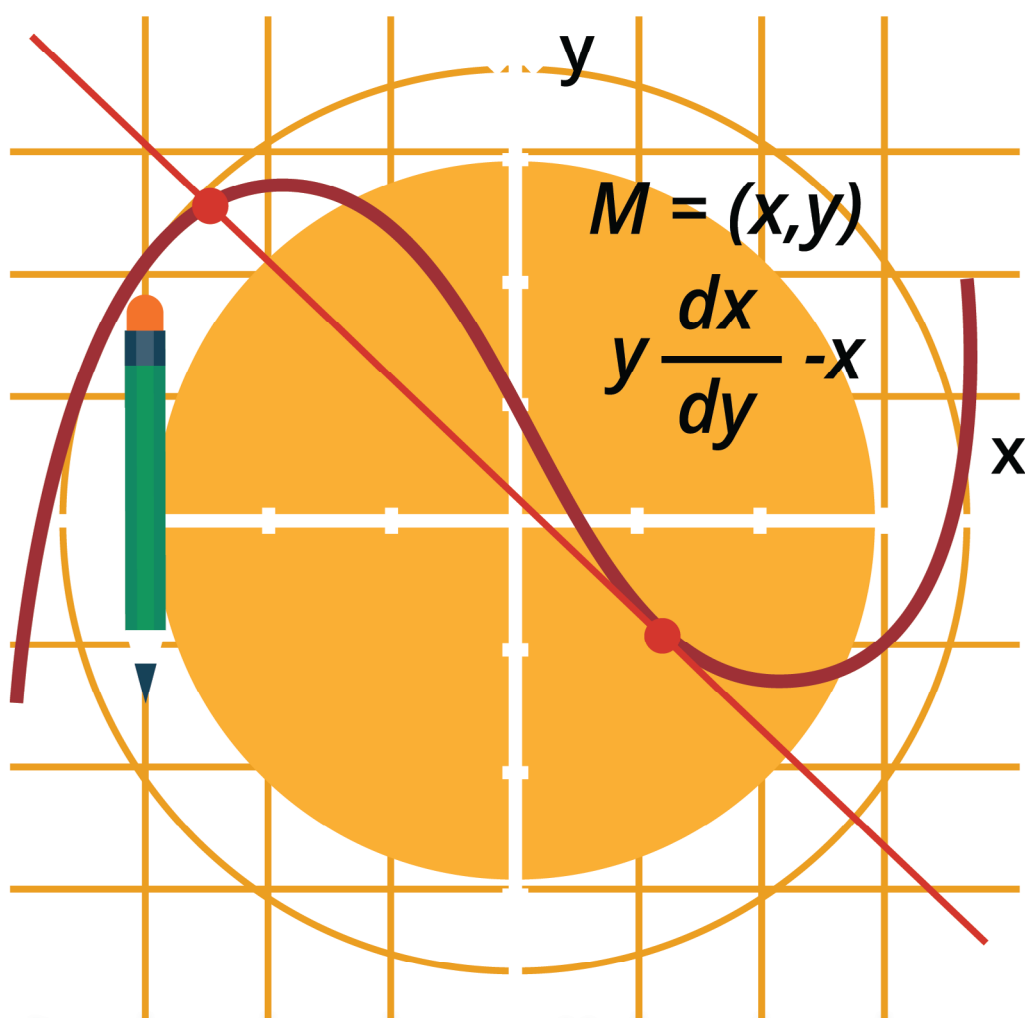


Vedantu's Tatva Practice Book

Mathematics

Volume 1

Class 12 **JEE**



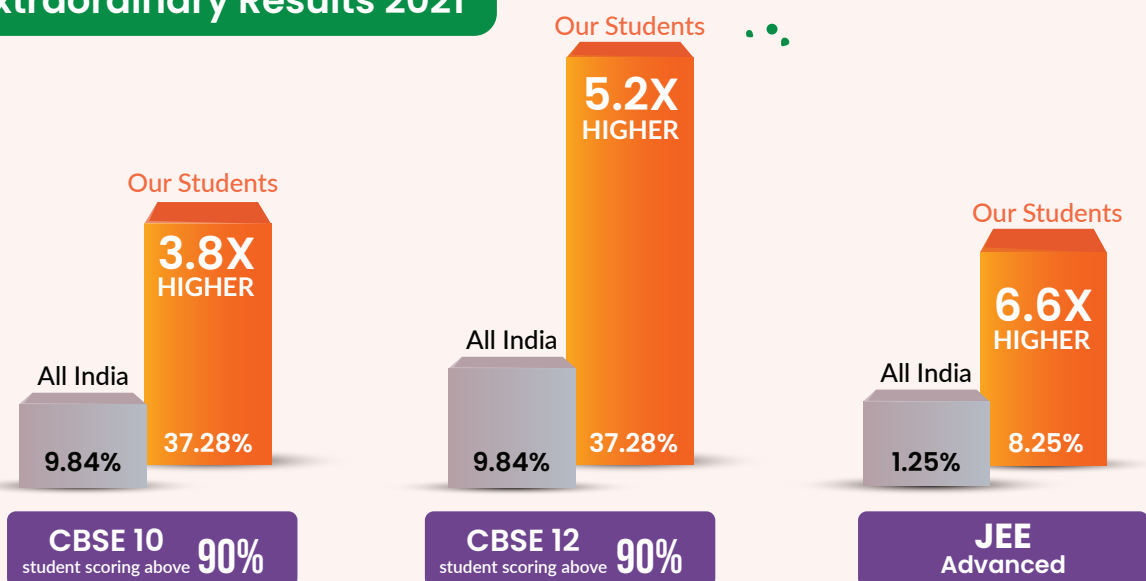
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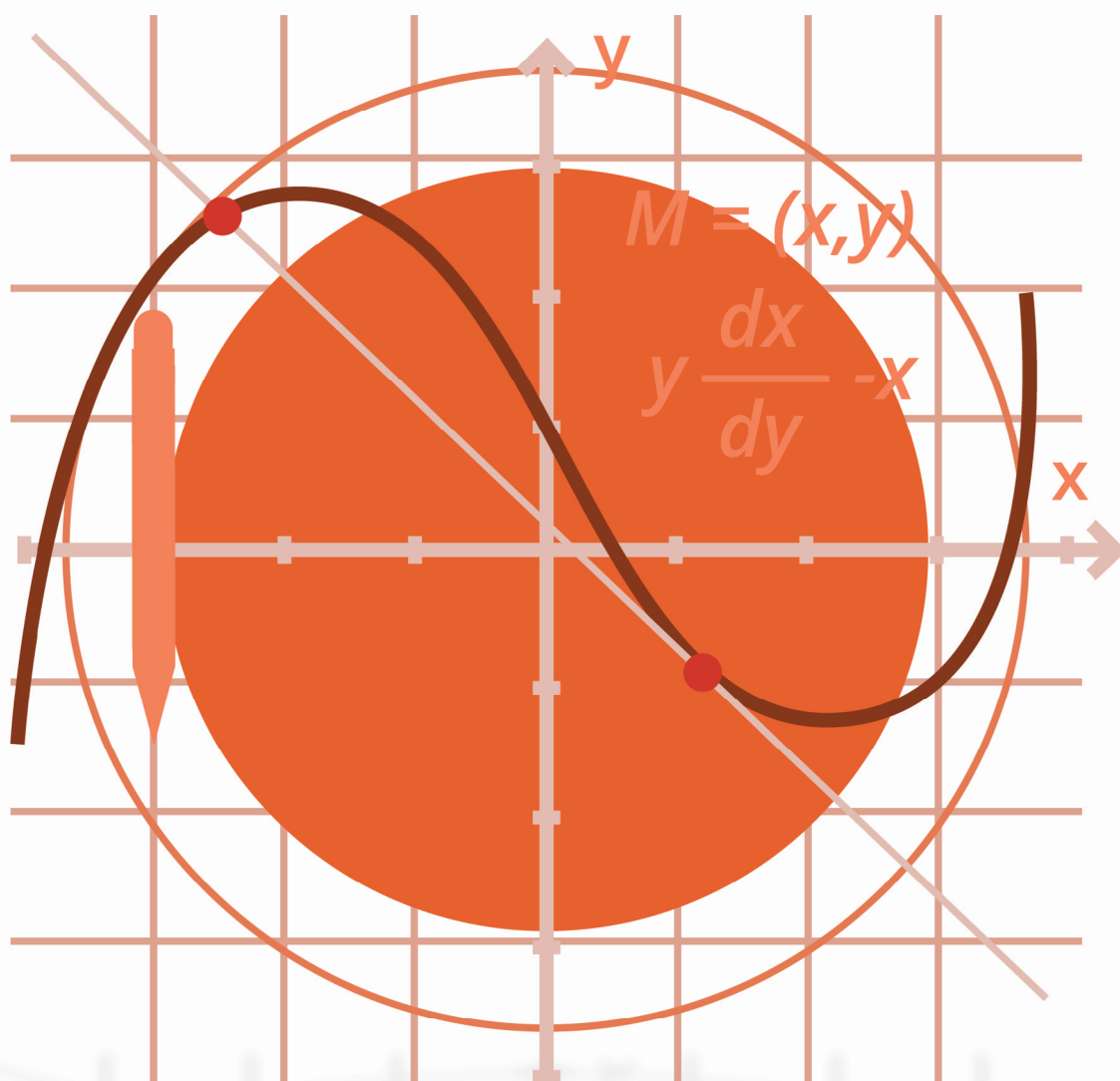


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Volume 1

Class 12 **JEE**



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How to use your Tatva Practice Book

4. Fundamental Principle of Counting :

If an event can occur in 'm' different ways following which another event can occur in 'n' different ways following which another event can occur in 'p' different ways then



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Permutations

1

Scan the QR Code in each chapter's theory section to view micro concept videos related to the chapter, on the Vedantu app

2

Solve all types of exercise questions based on the latest JEE pattern.

Exercise - 2:

1. If every element of a third order determinant of value Δ is multiplied by 5, then the value of new determinant is:

(JEE 2019)

- (a) Δ (b) 5Δ
(c) 25Δ (d) 125Δ

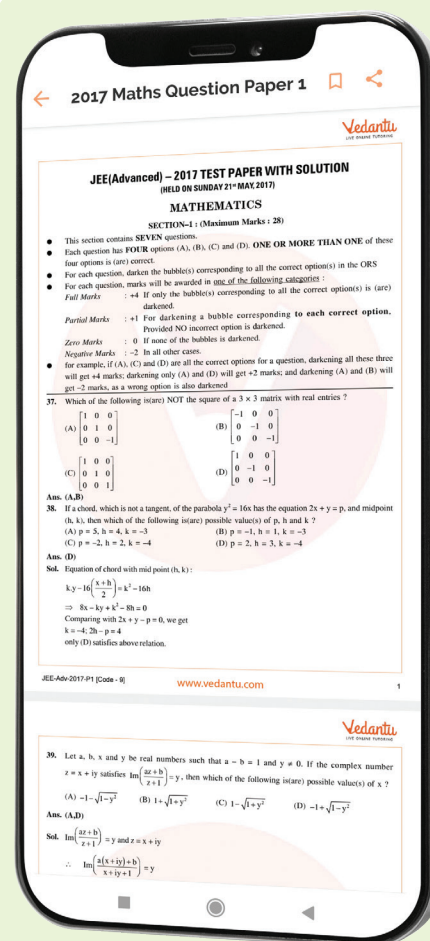
Answer Key

CHAPTER-1: DETERMINANTS
Exercise-1: Basic Objective



3

Scan the QR code in the Answers Section to view detailed solutions for all exercise questions.



For extra exam preparation content, visit the Vedantu app. You can find previous years' JEE papers with solutions and detailed analysis.

Founder's Message

Dear Student,

I am delighted to present to you a Ready Reckoner and an amazing book to guide you for your exams-'TATVA'. Tatva—which means the 'Core' is fully aligned with the culture, the mission, and the vision of Vedantu and therefore it gives me immense pleasure and joy to share this book with you. We at Vedantu have always believed in revolutionizing the teaching and learning process and always speedily progressed in the direction of bringing superior quality education and content to your table. Tatva is a step forward in this direction. This book is your guide, your practice guru, and your companion in moving towards your dreams. The book is a result of the consistent effort, diligence, and research by our experienced team of subject experts and teachers.

This book has been customized with curated content to suit the needs of JEE aspirants like you and guide you on the right path to cracking JEE and optimizing your efficiency. Tatva is a comprehensive amalgamation of important concepts, theories, derivations, definitions, solved examples, concept videos, practice questions, and important questions. We have ensured that high-quality content and the right ingredients are in place in this booklet to help you climb up the success ladder.

A few guiding points to optimally use Tatva with a planned approach:

Tatva equips you with Theory, Concept Videos, and Solved examples to help you revise concepts, mark your notes, walk you through the entire summary, and eventually makes you capable of clearing all your conceptual doubts all by yourself.

We suggest revision of theory followed by practice of solved examples.

Practice relevant questions daily after finishing Vedantu lectures and session assignments. We believe that a daily dose of Tatva will keep all your exam blues at bay.

Use the Tatva booklet to mark notes so that it always comes in handy for last-minute revision sessions before your exams. Notes should include key points of theory, solved examples, and some questions which you couldn't solve in the first attempt.

Exercise 1 and Exercise 2 of JEE Tatva deal with basic questions and those which can be asked or already asked in JEE Main. Similarly, Exercise 3 and Exercise 4 deal with JEE Advanced level questions. We recommend you solve basic JEE Main questions before moving to JEE Advanced level questions.

Before wrapping up, the practice mantra: "Don't practice until you get it right. Practice until you can't get it wrong."

We strongly believe in you and your capabilities. So believe in yourself because success is only one step away. Wishing that your talent shines bright. All the very best!

Anand Prakash

Founder and Academic Head, Vedantu



Anand Prakash Sir has been a pioneer in producing Top Ranks in JEE/NEET and Olympiads. He has personally taught and mentored AIR 1, 6, 7 (JEE Advanced), AIR-1, 7, 9(AIIMS), and thousands of more students who have successfully cleared these competitive exams in the last few years.

Credits

*“Happiness lies in the joy of achievement
and the thrill of creative effort.”*

—Franklin D. Roosevelt

Tatva is the brainchild of a group of creative Vedans who have strived tirelessly to weave success stories for you. We extend our heartfelt gratitude to the superb team of Vedans who give wings to the vision of Vedantu, starting with our leaders who have been guiding and encouraging us at every step of the way:

Vamsi Krishna Sir, Anand Prakash Sir and Pulkit Jain Sir

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Harish Rao, Neha Surana, Charubak Chakrabarti, Prashant Palande

Mathematics Team

We truly appreciate all the Master Teachers of Vedantu whose relentless efforts helped us translate this vision into reality. Our heartfelt gratitude to our creative content developers and the typesetting team, who have put in their hard work, insight, and eagerness to nurture and execute Tatva into ‘your ready handbook’ and bring a positive learning experience to you.

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The journey of bringing Tatva to life, from an idea to the book you are holding, would not have been possible without the extensive support of our diligent Operations Team, our amazing Academic Team, our dedicated team of Teachers, and our talented Tech Team.

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DETERMINANTS & MATRICES

DETERMINANTS & MATRICES



DETERMINANTS

1. INTRODUCTION TO DETERMINANTS

1.1 Definition

- (i) The determinant consisting two rows and two columns is

$$D = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}. \text{ Its value is given by :}$$

$$D = a_1 b_2 - a_2 b_1$$

- (ii) A determinant which consists of 3 rows and 3 columns is called a 3rd-order-determinant and is of the following form.

$$D = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

It's value is :

$$D = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31} - a_{23}a_{32}a_{11} - a_{12}a_{21}a_{33}$$

1.2 Minors and Cofactors

$$\text{Let } \Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

Here a_{ij} = Element in i^{th} row and j^{th} column of Δ .

Minor of a_{ij} :

It is defined as the value of the determinant obtained by eliminating the i^{th} row and j^{th} column of Δ .

We denote the minor of a_{ij} by M_{ij} .

$$\text{e.g. } M_{11} = \text{minor of } a_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} = a_{22}a_{33} - a_{32}a_{23}$$

Cofactor of a_{ij} :

Denoted by C_{ij}

Cofactor of a_{ij} (C_{ij}) = $(-1)^{i+j}$ Minor of a_{ij}

$$\text{e.g. Cofactor of } a_{11} (C_{11}) = (-1)^{1+1} M_{11} = M_{11}$$

1.3 Evaluation of Determinant

Value of any determinant can be obtained by adding product of all elements of a row (or column) to their corresponding cofactors.

$$\text{e.g. } \Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$\begin{aligned} \Delta &= a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} \\ &= a_{21}C_{21} + a_{22}C_{22} + a_{23}C_{23} \\ &= a_{31}C_{31} + a_{32}C_{32} + a_{33}C_{33} \\ &= a_{11}C_{11} + a_{21}C_{21} + a_{31}C_{31} \\ &= a_{12}C_{12} + a_{22}C_{22} + a_{32}C_{32} \\ &= a_{13}C_{13} + a_{23}C_{23} + a_{33}C_{33} \end{aligned}$$

2. PROPERTIES OF DETERMINANTS

- (i) The value of a determinant remains unaltered, if the rows and columns are interchanged,

$$D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

- (ii) If any two rows (or columns) of a determinant be interchanged, the value of determinant changes in sign only.

$$D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \text{ and } D' = \begin{vmatrix} a_2 & b_2 & c_2 \\ a_1 & b_1 & c_1 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

$$D' = -D.$$

- (iii) If a determinant has all the elements zero in any row (or column) then its value is zero.

$$D = \begin{vmatrix} 0 & 0 & 0 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$$



- (iv) If a determinant has any two rows (or columns) identical or proportional, then its value is zero.

$$D = \begin{vmatrix} a_1 & b_1 & c_1 \\ ka_1 & kb_1 & kc_1 \\ c_1 & c_2 & c_3 \end{vmatrix} = 0$$

- (v) If all the elements of any row (or column) are multiplied by the same number, then the determinant is multiplied by that number.

$$D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}; \text{ and } D' = \begin{vmatrix} Ka_1 & Kb_1 & Kc_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

$$\text{Then } D' = KD$$

- (vi) If each element of any row (or column) can be expressed as a sum of two terms then the determinant can be expressed as the sum of two determinants, i.e.

$$\begin{vmatrix} a_1 + x & b_1 + y & c_1 + z \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + \begin{vmatrix} x & y & z \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

- (vii) The value of determinant is not altered by adding to the elements of any row (or column) a constant multiple of the corresponding elements of any other row (or column).

e.g.

$$R_1 \rightarrow R_1 + mR_2 \text{ (change } R_1 \text{ as sum of } R_1 \text{ and } m(R_2)).$$

$$R_3 \rightarrow R_3 + nR_2 \text{ (change } R_3 \text{ as sum of } R_3 \text{ and } n(R_2)).$$

$$D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \text{ and}$$

$$D' = \begin{vmatrix} a_1 + ma_2 & b_1 + mb_2 & c_1 + mc_2 \\ a_2 & b_2 & c_2 \\ a_3 + na_2 & b_3 + nb_2 & c_3 + nc_2 \end{vmatrix}$$

$$\text{Then } D' = D.$$

3. CRAMER'S RULE

(Not in CBSE Syllabus)

- (i) Two Variables

$$\text{If } a_1x + b_1y = c_1 \quad \dots (i)$$

$$a_2x + b_2y = c_2 \quad \dots (ii)$$

$$\text{then } x = \frac{D_x}{D}, y = \frac{D_y}{D}$$

values of x, y are unique, if $D \neq 0$. where, $D = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$,

$$D_x = \begin{vmatrix} c_1 & b_1 \\ c_2 & b_2 \end{vmatrix}, \quad D_y = \begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}$$

Similarly 'n' equations in 'n' variables can be solved.

- (ii) Three Variables

$$\text{Let, } a_1x + b_1y + c_1z = d_1 \dots\dots\dots(i)$$

$$a_2x + b_2y + c_2z = d_2 \dots\dots\dots(ii)$$

$$a_3x + b_3y + c_3z = d_3 \dots\dots\dots(iii)$$

$$\text{Then, } x = \frac{D_x}{D}, y = \frac{D_y}{D}, z = \frac{D_z}{D}$$

$$\text{Where, } D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}; D_x = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix};$$

$$D_y = \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix} \text{ and } D_z = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}$$

Consistency of a System of Equations

- If $D \neq 0$ then the given system of equations are consistent and have unique solution.
- If $D = 0$ but at least one of D_x, D_y, D_z is not zero then the equations are inconsistent and have no solution.
- If $D = D_x = D_y = D_z = 0$ then the given system of equations are consistent and have infinite solution except the case of parallel planes when there is no solution.
- If $d_1 = d_2 = d_3 = 0$ then system of equation is called Homogenous system of equations.



- (v) Solution of Homogenous Equations is always consistent, as $x = 0 = y = z$ is always a solution. This is known as TRIVIAL solution.
- (vi) For Homogenous Equations, if $D \neq 0$. Then $x = 0 = y = z$ is the only solution.
- (vii) For Homogenous Equations, if $D = 0$, then there exists non zero solutions [NON TRIVIAL SOLUTIONS] also.

4. APPLICATION OF DETERMINANTS

Following examples of short hand writing of large expressions are:

- (i) Area of triangle whose vertices are $(x_r, y_r); r = 1, 2, 3$, is:

$$D = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

If $D = 0$ then the three points are collinear.

- (ii) Equation of straight line passing through (x_1, y_1) &

$$(x_2, y_2) \text{ is } \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$$

- (iii) The lines :

$$a_1x + b_1y + c_1 = 0 \dots\dots\dots(1)$$

$$a_2x + b_2y + c_2 = 0 \dots\dots\dots(2)$$

$$a_3x + b_3y + c_3 = 0 \dots\dots\dots(3)$$

are concurrent if, $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$

This is condition for the consistency of simultaneous linear equation in two variables.

- (iv) $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of straight lines if :

$$abc + 2fgh - af^2 - bg^2 - ch^2 = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$$

- (v) To find the variable $(x, y, z \text{ etc})$ in linear equations (Cramer's rule)

5. SOME MORE PROPERTIES OF DETERMINANT

- (i) Determinant of a skew-symmetric matrix of odd order is zero.

e.g. $D = \begin{vmatrix} 0 & 2 & 9 \\ -2 & 0 & \log_a b \\ -9 & \log_a \left(\frac{1}{b}\right) & 0 \end{vmatrix} = 0$

- (ii) Determinant of a skew-symmetric matrix of even order is always a perfect square.

e.g. $D = \begin{vmatrix} 0 & 5 \\ -5 & 0 \end{vmatrix}_{2 \times 2} = 25$

- (iii) Δ^{n-1} , where n is order of the determinant is equal to the determinant made from cofactors of elements of Δ .

e.g. $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}_{3 \times 3}^{3-1} = \begin{vmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{vmatrix}$

Where A_i 's are co-factors of a_i 's

- (iv) Determinant of a diagonal matrix is product of its diagonal elements

e.g. $D = \begin{vmatrix} 5 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 6 \end{vmatrix} = 5 \times 2 \times 6 = 60$

- (v) If a determinant considered as a polynomial becomes zero when $x = a$, then $x - a$ is factor of this. (This is an application of Factor Theorem)

$$D = \begin{vmatrix} x & a & a^2 \\ a & x & x^2 \\ a & x & a \end{vmatrix}$$

Because $D = 0$ when $x = a$ so $x - a$ is factor of D .

- (vi) The sum of the products of the elements of the i^{th} row/column with the co-factor of the corresponding elements of k^{th} row/column is zero provided $i \neq k$.



i.e. (i) $\sum_{j=1}^n a_{ij}C_{kj}=0$; if $i \neq k$

(ii) $\sum_{i=1}^n a_{ij}C_{ik}=0$ if $j \neq k$

(vii) $|AB| = |A||B|$

i.e. $A = \begin{vmatrix} 5 & -2 \\ -1 & -1 \end{vmatrix}$; $B = \begin{vmatrix} -2 & -3 \\ -4 & 1 \end{vmatrix}$; $AB = \begin{vmatrix} -2 & -17 \\ 6 & 2 \end{vmatrix}$

$|A| = -7$ $|B| = -14$

$|A||B| = -7 \times -14 = 98$,

$|AB| = -4 + 102 \Rightarrow |AB| = 98$.

(viii) Determinant of a triangular matrix is product of its diagonal elements only.

$D = \begin{vmatrix} 3 & 2 & 1 \\ 0 & 4 & 3 \\ 0 & 0 & -1 \end{vmatrix} = 3 \times 4 \times -1 = -12$

$D' = \begin{vmatrix} 5 & 0 & 0 \\ 4 & 9 & 0 \\ 3 & 2 & 1 \end{vmatrix} = 5 \times 9 \times 1 = 45$

(iii) $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(a+b+c)$

(iv) $\begin{vmatrix} 1 & 1 & 1 \\ a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(ab+bc+ca)$

7. MULTIPLICATION OF TWO DETERMINANTS

(a) $\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \times \begin{vmatrix} \ell_1 & m_1 \\ \ell_2 & m_2 \end{vmatrix} = \begin{vmatrix} a_1\ell_1 + b_1\ell_2 & a_1m_1 + b_1m_2 \\ a_2\ell_1 + b_2\ell_2 & a_2m_1 + b_2m_2 \end{vmatrix}$

(b) $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \times \begin{vmatrix} \ell_1 & m_1 & n_1 \\ \ell_2 & m_2 & n_2 \\ \ell_3 & m_3 & n_3 \end{vmatrix}$

$= \begin{vmatrix} a_1\ell_1 + b_1\ell_2 + c_1\ell_3 & a_1m_1 + b_1m_2 + c_1m_3 & a_1n_1 + b_1n_2 + c_1n_3 \\ a_2\ell_1 + b_2\ell_2 + c_2\ell_3 & a_2m_1 + b_2m_2 + c_2m_3 & a_2n_1 + b_2n_2 + c_2n_3 \\ a_3\ell_1 + b_3\ell_2 + c_3\ell_3 & a_3m_1 + b_3m_2 + c_3m_3 & a_3n_1 + b_3n_2 + c_3n_3 \end{vmatrix}$

6. SPECIAL DETERMINANTS

(i) Circulant Determinants :

The elements of the rows (or columns) are in cyclic arrangement

$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = -(a^3 + b^3 + c^3 - 3abc)$

$= -(a+b+c)(a^2 + b^2 + c^2 - ab - bc - ac)$

(ii) $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = (a-b)(b-c)(c-a)$

8. SUMMATION OF DETERMINANTS

Let $\Delta_r = \begin{vmatrix} f(r) & a & \ell \\ g(r) & b & m \\ h(r) & c & n \end{vmatrix}$

Where a, b, c, ℓ, m and n are constants independent of r , then

$\sum_{r=1}^n \Delta_r = \begin{vmatrix} \sum_{r=1}^n f(r) & a & \ell \\ \sum_{r=1}^n g(r) & b & m \\ \sum_{r=1}^n h(r) & c & n \end{vmatrix}$



Here functions of r can be the elements of only one row or column. None of the elements other than row or column should be dependent on r .

9. DIFFERENTIATION AND INTEGRATION OF DETERMINANT

$$\Delta(x) = \begin{vmatrix} f_1(x) & f_2(x) & f_3(x) \\ g_1(x) & g_2(x) & g_3(x) \\ h_1(x) & h_2(x) & h_3(x) \end{vmatrix}$$

Then

$$\Delta'(x) = \begin{vmatrix} f_1'(x) & f_2'(x) & f_3'(x) \\ g_1(x) & g_2(x) & g_3(x) \\ h_1(x) & h_2(x) & h_3(x) \end{vmatrix} + \begin{vmatrix} f_1(x) & f_2(x) & f_3(x) \\ g_1'(x) & g_2'(x) & g_3'(x) \\ h_1(x) & h_2(x) & h_3(x) \end{vmatrix} + \begin{vmatrix} f_1(x) & f_2(x) & f_3(x) \\ g_1(x) & g_2(x) & g_3(x) \\ h_1'(x) & h_2'(x) & h_3'(x) \end{vmatrix}$$

$$+ \begin{vmatrix} f_1(x) & f_2(x) & f_3(x) \\ g_1(x) & g_2(x) & g_3(x) \\ h_1(x) & h_2'(x) & h_3'(x) \end{vmatrix}$$

Integration of determinant

$$\text{If } \Delta(x) = \begin{vmatrix} f(x) & g(x) \\ \lambda_1 & \lambda_2 \end{vmatrix},$$

$$\text{then } \int_a^b \Delta(x) dx = \begin{vmatrix} \int_a^b f(x) dx & \int_a^b g(x) dx \\ \lambda_1 & \lambda_2 \end{vmatrix}.$$

Here $f(x)$ and $g(x)$ are functions of x and λ_1, λ_2 are constants.

NOTES:

This formula is only applicable if there is a variable only in one row or column, otherwise expand the determinant and then integrate.

$$\text{Example : If } f(x) = \begin{vmatrix} x^3 & \cos^2 x & 2x^4 \\ \tan^5 x & 1 & \sec 2x \\ \sin^3 x & x^4 & 5 \end{vmatrix} \text{ then } \int_{-\pi/2}^{\pi/2} f(x) dx =$$

- (a) 2 (b) -2 (c) 0 (d) none of these

Sol. We have

$$f(-x) = \begin{vmatrix} -x^3 & \cos^2 x & 2x^4 \\ -\tan^5 x & 1 & \sec 2x \\ -\sin^3 x & x^4 & 5 \end{vmatrix} = -f(x)$$

$$\therefore \int_{-\pi/2}^{\pi/2} f(x) dx = 0 \text{ (since } f(x) \text{ is an odd function)}$$

Example : If $f(x) = \begin{vmatrix} x^3 & \sin x \\ 1 & 2 \end{vmatrix}$, then $\int_{-a}^a f(x) dx$ is

- (a) 0 (b) $\frac{1}{2}$ (c) 3 (d) $-\frac{1}{2}$

$$\text{Sol. } \int_{-a}^a f(x) dx = \begin{vmatrix} \int_{-a}^a x^3 dx & \int_{-a}^a \sin x dx \\ 1 & 2 \end{vmatrix} = \begin{vmatrix} 0 & 0 \\ 1 & 1 \end{vmatrix} = 0$$

Hence (a) is correct answer.

MATRICES

1. INTRODUCTION TO MATRICES

A set of $(m \times n)$ numbers arranged in the form of an ordered set of m rows and n columns is called a matrix of order $m \times n$.

$$A = [a_{ij}]_{m \times n}$$

$$\text{or } A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ a_{31} & a_{32} & \dots & a_{3n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \text{ is a matrix of order } m \times n.$$

NOTES:

The matrix is not a number. It has no numerical value. But it is an arrangement of numbers.



2. TYPES OF MATRICES

2.1 Row Matrix

A matrix having only one row is called a row matrix or a row vector.

e.g. $A = [1 \ 2 \ -1 \ -2]$ is a row matrix of order 1×4 .

2.2 Column Matrix

A matrix having only one column is called a column matrix or a column vector.

e.g. $A = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$ and $B = \begin{bmatrix} 3 \\ 2 \\ 5 \\ 4 \end{bmatrix}$ are column matrices or order 3×1 and 4×1 respectively.

2.3 Square Matrix

A matrix in which the number of rows is equal to the number of column, say $(n \times n)$ is called a square matrix of order n .

e.g. the matrix $\begin{bmatrix} 2 & 1 & -1 \\ 3 & -2 & 5 \\ 1 & 5 & -3 \end{bmatrix}$ is square matrix of order 3.

Sum of diagonal elements of a square matrix is called its trace ($\text{tr}(A)$). Here $\text{tr}(A) = 2 - 2 - 3 = -3$

2.4 Diagonal Matrix

A square matrix is called a diagonal matrix if all the elements, except those in the leading diagonal, are zero.

$A = [a_{ij}]_{n \times n}$, $a_{ij} = 0$ for all $i \neq j$

2.5 Scalar Matrix

A diagonal matrix in which all the diagonal elements are equal is called the scalar matrix.

A square matrix $A = [a_{ij}]_{n \times n}$ is called a scalar matrix if.

- (i) $a_{ij} = 0$ for all $i \neq j$ and
- (ii) $a_{ii} = C$ for all $i \in \{1, 2, \dots, n\}$

2.6 Identity or Unit Matrix

A square matrix each of whose diagonal element is unity and each of whose non diagonal element is equal to zero is called an identity or unit matrix.

A square matrix $A = [a_{ij}]_{n \times n}$ is called identity or unit matrix if

- (i) $a_{ij} = 0$ for all $i \neq j$, and
- (ii) $a_{ii} = 1$ for all $i \in \{1, 2, \dots, n\}$

The identity matrix of order n is denoted by I_n .

e.g. $I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

2.7 Null Matrix

A matrix whose all elements are zero is called a null matrix or a zero matrix, represented by O .

e.g. $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$, $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

2.8 Upper Triangular Matrix

A square matrix $A = [a_{ij}]$ is called an upper triangular matrix if $a_{ij} = 0 \ \forall \ i > j$.

e.g. $A = \begin{bmatrix} 5 & 4 & 3 \\ 0 & 2 & 1 \\ 0 & 0 & 6 \end{bmatrix}$

2.9 Lower Triangular Matrix

A square matrix $A = [a_{ij}]$ is called lower triangular if $a_{ij} = 0 \ \forall \ i < j$.

e.g. $A = \begin{bmatrix} 2 & 0 & 0 \\ 3 & 2 & 0 \\ 4 & 5 & 3 \end{bmatrix}$

2.10 Singular Matrix

A square matrix with zero determinant is called a singular matrix.

3. EQUALITY OF MATRICES

Two matrices $A = [a_{ij}]_{m \times n}$ and $B = [b_{ij}]_{r \times s}$ are equal if

- (i) $m = r$, i.e., the number of rows in A equals the number of rows in B .
- (ii) $n = s$, i.e., the number of columns in A equals the number of columns in B .



(iii) $a_{ij} = b_{ij}$ for $i = 1, 2, \dots, m$ and $j = 1, 2, \dots, n$.

If two matrices A and B are equal, we write $A = B$, otherwise we write $A \neq B$.

4. SUM OF MATRICES

Let $A = [a_{ij}]$, $B = [b_{ij}]$ be matrices of the same order $m \times n$.

Then $C = A + B = [c_{ij}]$, is a matrix of order $m \times n$.

Where, $[c_{ij}] = [a_{ij} + b_{ij}]$

e.g. $A = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 5 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 7 & 3 & 2 \\ 5 & 1 & 9 \end{bmatrix}$

$$A + B = \begin{bmatrix} 1+7 & 2+3 & 4+2 \\ 0+5 & 5+1 & 3+9 \end{bmatrix} = \begin{bmatrix} 8 & 5 & 6 \\ 5 & 6 & 12 \end{bmatrix}$$

$$A - B = \begin{bmatrix} 1-7 & 2-3 & 4-2 \\ 0-5 & 5-1 & 3-9 \end{bmatrix} = \begin{bmatrix} -6 & -1 & 2 \\ -5 & 4 & -6 \end{bmatrix}$$

Properties of Matrix Addition

(i) **Matrix addition is commutative**

$$A + B = B + A$$

(ii) **Matrix addition is associative**

$$A + (B + C) = (A + B) + C.$$

5. SCALAR MULTIPLE OF A MATRIX

If A be a given matrix and k is any scalar number real or complex.

Then matrix kA is a matrix of same order, where all the elements of kA are k times of the corresponding elements of A.

e.g. If $A = \begin{bmatrix} 2 & 3 & 1 \\ 5 & 2 & 4 \end{bmatrix}$

$$\text{Then } 3A = \begin{bmatrix} 3 \cdot 2 & 3 \cdot 3 & 3 \cdot 1 \\ 3 \cdot 5 & 3 \cdot 2 & 3 \cdot 4 \end{bmatrix} = \begin{bmatrix} 6 & 9 & 3 \\ 15 & 6 & 12 \end{bmatrix}$$

Properties of Multiplication by a Scalar

If $A = [a_{ij}]$ and $B = [b_{ij}]$ are matrix of the same order and α and β are any scalars, then

(i) $\alpha(A + B) = \alpha A + \alpha B$

(ii) $(\alpha + \beta)A = \alpha A + \beta A$

(iii) $\alpha(\beta A) = (\alpha\beta)A$.

(iv) If A is a square matrix of order 'n'

$$\text{Then } |kA| = k^n |A|$$

6. MATRIX MULTIPLICATION

If $A = [a_{ij}]_{m \times p}$ and $B = [b_{jk}]_{p \times n}$

Then $A_{m \times p} \times B_{p \times n} = (AB)_{m \times n}$

or $C = AB = [c_{ik}]_{m \times n}$ where $c_{ik} = \sum_{j=1}^p a_{ij}b_{jk}$

For multiplication number of columns of first matrix should be equal to number of rows of second matrix.

i.e. $c_{ik} = a_{i1}b_{1k} + a_{i2}b_{2k} + \dots + a_{ip}b_{pk}$

In other words c_{ik} = Sum of the products of i^{th} row of A (having p elements) with k^{th} column of B (having p elements).

e.g. If $A = \begin{bmatrix} 1 & -2 & 3 \\ -4 & 2 & 5 \end{bmatrix}_{2 \times 3}$ and $B = \begin{bmatrix} 2 & 3 \\ 4 & 5 \\ 2 & 1 \end{bmatrix}_{3 \times 2}$

Compute AB and show that $AB \neq BA$. A is 2×3 type and B is 3×2 type and hence both AB and BA are defined because the number of columns in pre factor is equal to the number of rows in post factor.

Sol.

$$AB = \begin{bmatrix} 1 \cdot 2 - 2 \cdot 4 + 3 \cdot 2 & 1 \cdot 3 - 2 \cdot 5 + 3 \cdot 1 \\ -4 \cdot 2 + 2 \cdot 4 + 5 \cdot 2 & -4 \cdot 3 + 2 \cdot 5 + 5 \cdot 1 \end{bmatrix} = \begin{bmatrix} 0 & -4 \\ 10 & 3 \end{bmatrix}_{2 \times 2}$$

$$BA = \begin{bmatrix} 2 & 3 \\ 4 & 5 \\ 2 & 1 \end{bmatrix}_{3 \times 2} \begin{bmatrix} 1 & -2 & 3 \\ -4 & 2 & 5 \end{bmatrix}_{2 \times 3} = \begin{bmatrix} -10 & 2 & 21 \\ -16 & 2 & 37 \\ -2 & -2 & 11 \end{bmatrix}_{3 \times 3}$$

Hence $AB \neq BA$.

7. PROPERTIES OF MATRIX MULTIPLICATION

(i) **Multiplication of matrices is distributive with respect to a addition of matrices.**

$$A(B + C) = AB + AC \text{ and } (A + B)C = AC + BC$$

(ii) **Matrix multiplication is associative if conformability is assured.**

$$\text{i.e. } A(BC) = (AB)C.$$

(iii) **The multiplication of matrices is not always commutative.**
i.e. AB is not always equal to BA.



- (iv) **Multiplication of a matrix A by a null matrix conformable with A for multiplication is a null matrix i.e. $AO = O$.**

In particular if A be a square matrix and O be square null matrix of the same order, then $OA = AO = O$.

- (v) **If $AB = O$ then it does not necessarily mean that $A = O$ or $B = O$.**

$$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

None of the matrices on the left is a null matrix whereas their products is a null matrix.

- (vi) **Multiplication of matrix A by a unit matrix I :**
Let A be a $m \times n$ matrix.

$$\text{Then } AI_n = A \quad \text{and } I_m A = A.$$

- (vii) **If A and B are square matrices of order 'n'**

$$\text{Then } |AB| = |A| |B|$$

- (viii) **Positive Integral Powers of Matrix**

Let A be any square matrix of order n.

$$\text{Then } A^2 = A.A$$

$$A^3 = A.A.A$$

$$A^m = A.A.A \dots \text{m times}$$

All are square matrix of order n.

- (i) $A^m . A^n = (A.A.A \dots \text{m times}) (A.A.A \dots \text{n times})$
 $= A.A.A \dots (m+n) \text{ times}$
 $= A^{m+n}$

- (ii) $(A^m)^n = A^{mn}$

Also, we define $A^0 = I$

8. TRANSPOSE OF A MATRIX

If A be a given matrix of the order $m \times n$ then the matrix obtained by changing the rows of A into columns and columns of A into rows is called Transpose of matrix A and is denoted by A' or A^T . Hence the matrix A' is of order $n \times m$.

$$\text{e.g. } A = \begin{bmatrix} 3 & 4 \\ 2 & 1 \\ 5 & 9 \end{bmatrix}_{3 \times 2} \quad \text{then } A^T = A' = \begin{bmatrix} 3 & 2 & 5 \\ 4 & 1 & 9 \end{bmatrix}_{2 \times 3}$$

Properties of Transpose

- $(A')' = A$.
- $(kA)' = kA'$. k being a scalar.
- $(A + B)' = A' + B'$.
- $(AB)' = B'A'$.
- $(ABC)' = C'B'A'$.

9. SYMMETRIC AND SKEW SYMMETRIC MATRICES

- (i) A square matrix $A = [a_{ij}]$ will be called **symmetric** if $A^T = A$.
 i.e. every ij^{th} element = ji^{th} element.

$$\text{e.g. } A = \begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}_{3 \times 3}$$

- (ii) A square matrix $A = [a_{ij}]$ will be called **skew symmetric** if $A^T = -A$.

i.e. every ij^{th} element = $-(ji^{\text{th}} \text{ element})$.

- (iii) For any square matrix A, $A + A^T$ is symmetric and $A - A^T$ is skew - symmetric.

- (iv) Any square matrix can be uniquely expressed as a sum of a symmetric matrix and a skew-symmetric matrix.

$$\frac{1}{2}(A + A^T) \text{ is symmetric}$$

$$\text{and } \frac{1}{2}(A - A^T) \text{ is skew-symmetric}$$

- (v) Let A and B be symmetric matrices of the same order. Then the following hold:

- A^n is symmetric for all positive integers n.
- AB is symmetric if and only if $AB = BA$.
- $AB + BA$ is symmetric.
- $AB - BA$ is skew - symmetric

NOTES:

For a skew symmetric matrix :

$$a_{ii} = -a_{ii} \text{ for all values of } i$$

[$i = j$ when elements are diagonals].

DETERMINANTS & MATRICES



$$2a_{ii} = 0 \quad \therefore \quad a_{ii} = 0$$

Hence the diagonal elements of skew symmetric matrix are zero.

e.g. $\begin{bmatrix} 0 & h & g \\ -h & 0 & f \\ -g & -f & 0 \end{bmatrix}$ is a skew symmetric matrix

10. ADJOINT

If A is a square matrix, then transpose of a matrix made from cofactors of elements of A is called adjoint matrix of A. It's denoted by $\text{adj } A$.

Properties of Adjoint Matrix

- (i) $A \cdot (\text{Adj } A) = |A| I_n = (\text{adj } A) \cdot A$
- (ii) $|\text{adj } A| = |A|^{n-1}$
- (iii) $\text{adj}(\text{adj } A) = |A|^{n-2} A$
- (iv) $(\text{adj } A)^T = \text{adj}(A^T)$
- (v) $\text{adj}(AB) = (\text{adj } B) \cdot (\text{adj } A)$
- (iv) $\text{Adj}(A^{-1}) = (\text{adj } A)^{-1}$
- (vii) $|\text{adj}(\text{adj}(A))| = |A|^{(n-1)^2}$

11. INVERSE OF MATRIX (A)

11.1

A square matrix A of order n is said to be invertible or non-singular if there exists a square matrix B of order n such that

$$AB = I_n = BA$$

where I_n is the identity matrix of order n, B is called inverse of A and is denoted by A^{-1} .

$$A^{-1} = \frac{1}{|A|} \text{adj}(A)$$

11.2 Properties of Inverse Matrices

- (i) $(A^T)^{-1} = (A^{-1})^T$
- (ii) $(AB)^{-1} = B^{-1} A^{-1}$
- (iii) $(A^{-1})^{-1} = A$
- (iv) $|A^{-1}| = \frac{1}{|A|}$

$$(v) \quad (kA)^{-1} = \frac{1}{k} A^{-1} \text{ if } k \neq 0.$$

(vi) Let A, B, C be square matrix of the same order n. If A is non-singular matrix then

- (a) $AB = AC \Rightarrow B = C$ (left cancellation law)
- (b) $BA = CA \Rightarrow B = C$ (right cancellation law)

(vii) If A is non singular matrix such that A is symmetric then A^{-1} is also symmetric.

12. ELEMENTARY TRANSFORMATIONS

Any one of the following operations on a matrix is called an elementary transformation.

- (i) Interchanging any two rows (or column).
- (ii) Multiplication of the elements of any row (or column) by a non zero scalar quantity.
- (iii) Addition of constant multiple of the elements of any row (or column) to the corresponding element of any other row (or column).

Two matrices are said to be equivalent if one is obtained from the other by elementary transformation. The symbol $\approx$ is used for equivalence.

Method to Find Inverse by Elementary Transformations :

Row Transformation

- (i) A^{-1} exists if $|A| \neq 0$.
- (ii) To find A^{-1} by row transformation, then we write $IA = A$.
- (iii) Apply row transformations to the pre-factor I on L.H.S. & to A on R.H.S. such that A becomes a unit matrix.
- (iv) Now equation becomes $BA = I$, so $B = A^{-1}$

13. SOLUTION OF A SYSTEM OF LINEAR EQUATION BY MATRIX METHOD

Consider a system of linear equation

$$a_{11}x_1 + a_{12}x_2 + \dots a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots a_{2n}x_n = b_2$$

$$\dots \dots \dots$$

$$\dots \dots \dots$$

$$a_{n1}x_1 + a_{n2}x_2 + \dots a_{nn}x_n = b_n$$



We can express these equations as a single matrix equation.

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \dots \\ b_n \end{bmatrix}$$

$$A \quad X \quad B$$

Let $|A| \neq 0$ so that A^{-1} exists uniquely

pre-multiplying both sides of $AX = B$ by A^{-1} we get

$$\begin{aligned} A^{-1}(AX) &= A^{-1}B & \Rightarrow & (AA^{-1})X = A^{-1}B \\ \Rightarrow IX &= A^{-1}B & \Rightarrow & X = A^{-1}B \end{aligned}$$

Criterion of Consistency

Let $AX = B$ be a system of n linear equation in n variables.

- If $|A| \neq 0$ then the system of the equations is consistent and has a unique solution given by $X = A^{-1}B$.
- If $|A| = 0$ and $(\text{adj } A)B = O$ then the system of equations is consistent and has infinitely many solutions except the case of parallel planes when there is no solution.
- If $|A| = 0$ and $(\text{adj } A)B \neq O$ then the system of equations is inconsistent i.e. it has no solution.

Homogeneous Equation

The system of equations $AX = B$ is said to be homogeneous if the constants $b_1, b_2, b_3, \dots, b_n$ are all zero. That is if the matrix B is a zero matrix and the system is of the form

$$AX = O$$

Where O is the null matrix of order $n \times 1$.

- If $|A| \neq 0$ then its only solution $X = 0$ is called trivial solution. ($x = y = z = 0$)
- If $|A| = 0$ then $AX = O$ have both trivial and non trivial type solutions. In this case number of solutions will be Infinite.
- The condition for
 $a_1x + b_1y + c_1z = 0$;
 $a_2x + b_2y + c_2z = 0$ and
 $a_3x + b_3y + c_3z = 0$
 to have non-zero or non-trivial solutions is :

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$$

14. SOME OTHER TYPES OF MATRICES

- (a) **Orthogonal Matrix** : A square matrix A is called an orthogonal matrix if the product of the matrix A and its transpose A' is identity matrix.

$$AA' = I$$

NOTES:

- If $AA' = I$ then $A^{-1} = A'$
- If A and B are orthogonal then AB is also orthogonal.
- All above properties are defined for square matrix only.
- Elements of all 3 rows (or columns) of orthogonal matrix of order 3×3 represent unit vectors.

- (b) **Idempotent Matrix** : A matrix ' A ' such that $A^2 = A$ is called idempotent..

- Only square matrix can be idempotent matrix.
- Identity matrix (unit matrix) is also idempotent matrix.

Example: $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$$A^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

- Periodic Matrix** : A matrix ' A ' will be called a periodic matrix if $A^{k+1} = A$ where k is +ve integer and k is least positive integer for which $A^{k+1} = A$, then k is said to be the period of A .
- Nilpotent Matrix** : A matrix ' A ' will be called nilpotent matrix if $A^k = O$ (null matrix), k is least positive integer and k is called index of the nilpotent matrix.
- Involutory Matrix** : A matrix ' A ' will be called an involutory matrix if $A^2 = I$ (Unit Matrix). Unit matrix is also involutory matrix.



15. SOME IMPORTANT APPLICATIONS

15.1

Let $f(x) = a_0 + a_1x + a_2x^2 + \dots + a_mx^m$ be a polynomial in x where $a_0, a_1, a_2, \dots, a_m$ are real numbers, such that $a_0 \neq 0$. If A is a non-singular matrix such that $f(A) = 0$, then

$$A^{-1} = \frac{-1}{a_0} (a_1 + a_2A + a_3A^2 + \dots + a_mA^{m-1})$$

15.2

To find the inverse of a square matrix A , or to express A^{-1} in terms of A , the concept of a characteristic polynomial of a square matrix and the much known Cayley-Hamilton Theorem are useful, especially for 2×2 and 3×3 matrices.

15.3

If A is a square matrix and I is the corresponding unit matrix, then the polynomial $|A - xI|$ in x is called characteristic polynomial of A and the equation $|A - xI| = 0$ is called the characteristic equation of the matrix A .

15.4

(Cayley-Hamilton) Every square matrix satisfies its characteristic equation; that is, if A is a square matrix of order n and

$$f(x) = |A - xI| = a_0 + a_1x + a_2x^2 + \dots + a_nx^n = 0$$

is its characteristic equation, then

$$f(A) = a_0I_n + a_1A + a_2A^2 + \dots + a_nA^n = O$$

Also if $a_0 \neq 0$, then

$$A^{-1} = \frac{-1}{a_0} (a_1I + a_2A + a_3A^2 + \dots + a_nA^{n-1})$$

Note that A^{-1} exists if and only if the constant term of the characteristic of A is non-zero.



SOLVED EXAMPLES

Example – 1

Find minors and cofactors of elements of the following determinants

$$(i) \begin{vmatrix} 0 & 1 & 2 \\ 3 & 0 & 1 \\ 2 & 3 & 0 \end{vmatrix} \quad (ii) \begin{vmatrix} -1 & 0 & 4 \\ -2 & 1 & 3 \\ 0 & -4 & 2 \end{vmatrix}$$

Sol. (i) $D = \begin{vmatrix} 0 & 1 & 2 \\ 3 & 0 & 1 \\ 2 & 3 & 0 \end{vmatrix}$

Minors

$$M_{11} = \begin{vmatrix} 0 & 1 \\ 3 & 0 \end{vmatrix};$$

$$= 0 - 3 = -3$$

$$M_{12} = \begin{vmatrix} 3 & 1 \\ 2 & 0 \end{vmatrix};$$

$$= 0 - 2 = -2$$

$$M_{13} = \begin{vmatrix} 3 & 0 \\ 2 & 3 \end{vmatrix};$$

$$= 9 - 0 = 9$$

$$M_{21} = \begin{vmatrix} 1 & 2 \\ 3 & 0 \end{vmatrix};$$

$$= 0 - 6 = -6$$

$$M_{22} = \begin{vmatrix} 0 & 2 \\ 2 & 0 \end{vmatrix};$$

$$= 0 - 4 = -4$$

$$M_{23} = \begin{vmatrix} 0 & 1 \\ 2 & 3 \end{vmatrix};$$

$$= 0 - 2 = -2$$

Cofactors

$$A_{11} = (-1)^{1+1} M_{11}$$

$$= -3$$

$$A_{12} = (-1)^{1+2} M_{12}$$

$$= 2$$

$$A_{13} = (-1)^{1+3} M_{13}$$

$$= 9$$

$$A_{21} = (-1)^{2+1} M_{21}$$

$$= 6$$

$$A_{22} = (-1)^{2+2} M_{22}$$

$$= -4$$

$$A_{23} = (-1)^{2+3} M_{23}$$

$$= 2$$

$$M_{31} = \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix};$$

$$= 1 - 0 = 1$$

$$M_{32} = \begin{vmatrix} 0 & 2 \\ 3 & 1 \end{vmatrix};$$

$$= 0 - 6 = -6$$

$$M_{33} = \begin{vmatrix} 0 & 1 \\ 3 & 0 \end{vmatrix};$$

$$= 0 - 3 = -3$$

$$A_{31} = (-1)^{3+1} M_{31}$$

$$= 1$$

$$A_{32} = (-1)^{3+2} M_{32}$$

$$= 6$$

$$A_{33} = (-1)^{3+3} M_{33}$$

$$= -3$$

(ii) $D = \begin{vmatrix} -1 & 0 & 4 \\ -2 & 1 & 3 \\ 0 & -4 & 2 \end{vmatrix}$

Minors

$$M_{11} = \begin{vmatrix} 1 & 3 \\ -4 & 2 \end{vmatrix};$$

$$= 2 + 12 = 14$$

$$M_{12} = \begin{vmatrix} -2 & 3 \\ 0 & 2 \end{vmatrix};$$

$$= -4 - 0 = -4$$

$$M_{13} = \begin{vmatrix} -2 & 1 \\ 0 & -4 \end{vmatrix};$$

$$= 8 - 0 = 8$$

$$M_{21} = \begin{vmatrix} 0 & 4 \\ -4 & 2 \end{vmatrix};$$

$$= 0 + 16 = 16$$

$$M_{22} = \begin{vmatrix} -1 & 4 \\ 0 & 2 \end{vmatrix};$$

$$= -2 - 0 = -2$$

Cofactors

$$A_{11} = (-1)^{1+1} M_{11}$$

$$= 14$$

$$A_{12} = (-1)^{1+2} M_{12}$$

$$= 4$$

$$A_{13} = (-1)^{1+3} M_{13}$$

$$= 8$$

$$A_{21} = (-1)^{2+1} M_{21}$$

$$= -16$$

$$A_{22} = (-1)^{2+2} M_{22}$$

$$= -2$$



$$= -2 + 0 = -2 \quad = -2$$

$$M_{23} = \begin{vmatrix} -1 & 0 \\ 0 & -4 \end{vmatrix}; \quad A_{23} = (-1)^{2+3} M_{23}$$

$$= 4 - 0 = 4 \quad = -4$$

$$M_{31} = \begin{vmatrix} 0 & 4 \\ 1 & 3 \end{vmatrix}; \quad A_{31} = (-1)^{3+1} M_{31}$$

$$= 0 - 4 = -4 \quad = -4$$

$$M_{32} = \begin{vmatrix} -1 & 4 \\ -2 & 3 \end{vmatrix}; \quad A_{32} = (-1)^{3+2} M_{32}$$

$$= -3 + 8 = 5 \quad = -5$$

$$M_{33} = \begin{vmatrix} -1 & 0 \\ -2 & 1 \end{vmatrix}; \quad A_{33} = (-1)^{3+3} M_{33}$$

$$= -1 + 0 = -1 \quad = -1$$

Example – 2

Show that $\begin{vmatrix} x-y & y-z & z-x \\ y-z & z-x & x-y \\ z-x & x-y & y-z \end{vmatrix} = 0$

Sol. $D = \begin{vmatrix} x-y & y-z & z-x \\ y-z & z-x & x-y \\ z-x & x-y & y-z \end{vmatrix}$

$C_3 \rightarrow C_3 + C_2 + C_1$ gives

$$D = \begin{vmatrix} x-y & y-z & 0 \\ y-z & z-x & 0 \\ z-x & x-y & 0 \end{vmatrix}$$

$$= 0$$

[$\therefore$ all elements of C_3 are zero.]

Example – 3

Evaluate

$$(i) \begin{vmatrix} 16 & 29 & 35 \\ 50 & 100 & 110 \\ 82 & 158 & 180 \end{vmatrix}$$

$$(ii) \begin{vmatrix} 5 & 13 & 17 \\ 30 & 68 & 105 \\ 25 & 66 & 84 \end{vmatrix}$$

Sol. (i) $D = \begin{vmatrix} 16 & 29 & 35 \\ 50 & 100 & 110 \\ 82 & 158 & 180 \end{vmatrix}$

Using $\left(\frac{1}{10} R_2\right)$,

$$D = 10 \begin{vmatrix} 16 & 29 & 35 \\ 5 & 10 & 11 \\ 82 & 158 & 180 \end{vmatrix}$$

$R_1 = R_1 - 3R_2, R_3 \rightarrow R_3 - 16R_2$

$$D = 10 \begin{vmatrix} 1 & -1 & 2 \\ 5 & 10 & 11 \\ 2 & -2 & 4 \end{vmatrix} = 0$$

$\therefore (R_3 \equiv 2 \times R_1)$

$$(ii) D = \begin{vmatrix} 5 & 13 & 17 \\ 30 & 68 & 105 \\ 25 & 66 & 84 \end{vmatrix}$$

$R_2 \rightarrow R_2 - 6R_1, R_3 \rightarrow R_3 - 5R_1$

$$\therefore D = \begin{vmatrix} 5 & 13 & 17 \\ 0 & -10 & 3 \\ 0 & 1 & -1 \end{vmatrix}$$

$$= 5(10 - 3)$$

(Expanding along 1st column) = 35



Example – 4

Show that $\begin{vmatrix} 10 & 24 & 36 \\ 36 & 10 & 24 \\ 24 & 36 & 10 \end{vmatrix}$ is divisible by 35.

Sol. $D = \begin{vmatrix} 10 & 24 & 36 \\ 36 & 10 & 24 \\ 24 & 36 & 10 \end{vmatrix}$

$$R_1 \rightarrow R_1 + R_2 + R_3$$

$$\therefore D = \begin{vmatrix} 70 & 70 & 70 \\ 36 & 10 & 24 \\ 24 & 36 & 10 \end{vmatrix}$$

Using $\frac{1}{35} R_1$, we get

$$D = 35 \begin{vmatrix} 2 & 2 & 2 \\ 36 & 10 & 24 \\ 24 & 36 & 10 \end{vmatrix}$$

All elements in determinant are natural numbers

$\therefore$ Its value will be whole number

$\therefore$ D is multiple of 35 i.e. it is divisible by 35.

Example – 5

If x, y, z are different and $\Delta = \begin{vmatrix} x & x^2 & 1+x^3 \\ y & y^2 & 1+y^3 \\ z & z^2 & 1+z^3 \end{vmatrix} = 0$,

then show that $1 + xyz = 0$

Sol. We have

$$\Delta = \begin{vmatrix} x & x^2 & 1+x^3 \\ y & y^2 & 1+y^3 \\ z & z^2 & 1+z^3 \end{vmatrix}$$

$$= \begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{vmatrix} + \begin{vmatrix} x & x^2 & x^3 \\ y & y^2 & y^3 \\ z & z^2 & z^3 \end{vmatrix} \quad (\text{Using Property 6})$$

Take common x, y, z from R_1, R_2, R_3 resp.

$$= (-1)^2 \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} + xyz \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix}$$

(Using $C_3 \leftrightarrow C_2$ and then $C_1 \leftrightarrow C_2$)

$$= \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} (1 + xyz)$$

$$= (1 + xyz) \begin{vmatrix} 1 & x & x^2 \\ 0 & y-x & y^2-x^2 \\ 0 & z-x & z^2-x^2 \end{vmatrix}$$

(Using $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$)

Taking out common factor $(y-x)$ from R_2 and $(z-x)$ from R_3 , we get

$$\Delta = (1 + xyz)(y-x)(z-x) \begin{vmatrix} 1 & x & x^2 \\ 0 & 1 & y+x \\ 0 & 1 & z+x \end{vmatrix}$$

$$= (1 + xyz)(y-x)(z-x)(z-y) \quad (\text{on expanding along } C_1)$$

Since $\Delta = 0$ and x, y, z are all different, i.e., $x-y \neq 0$,

$y-z \neq 0, z-x \neq 0$, we get $1 + xyz = 0$

Example – 6

If l, m, n are $p^{\text{th}}, q^{\text{th}}$ and r^{th} terms of an arithmetic progression

respectively, prove that $\begin{vmatrix} l & p & 1 \\ m & q & 1 \\ n & r & 1 \end{vmatrix} = 0$

Sol. In arithmetic progression

$$T_1 = a$$

$$T_2 = a + d$$

(where d is constant difference)

$$T_3 = a + 2d$$

$$T_n = a + (n-1)d$$

$$\therefore l = T_p = a + (p-1)d = a + pd - d$$



$$m = T_q = a + (q-1)d = a + qd - d$$

$$n = T_r = a + (r-1)d = a + rd - d$$

$$\therefore \text{L.H.S.} = \begin{vmatrix} l & p & 1 \\ m & q & 1 \\ n & r & 1 \end{vmatrix}$$

$$= \begin{vmatrix} a+pd-d & p & 1 \\ a+qd-d & q & 1 \\ a+rd-d & r & 1 \end{vmatrix}$$

gives $C_1 \rightarrow C_1 - d \times C_2$

$$\text{L.H.S.} = \begin{vmatrix} a-d & p & 1 \\ a-d & q & 1 \\ a-d & r & 1 \end{vmatrix}$$

Using $\left(\frac{1}{a-d}\right) C_1$, we get

$$\text{L.H.S.} = (a-d) \begin{vmatrix} 1 & p & 1 \\ 1 & q & 1 \\ 1 & r & 1 \end{vmatrix}$$

$$= 0 \quad (\because C_1 \equiv C_3)$$

Example - 7

Without expanding the determinants, show that

$$(i) \begin{vmatrix} 1/x & x & yz \\ 1/y & y & zx \\ 1/z & z & xy \end{vmatrix} = 0 \quad (ii) \begin{vmatrix} 0 & b & c \\ -b & 0 & a \\ -c & -a & 0 \end{vmatrix} = 0$$

Sol. (i) $D = \begin{vmatrix} 1/x & x & yz \\ 1/y & y & zx \\ 1/z & z & xy \end{vmatrix}$

using xR_1, yR_2 and zR_3 , we get

$$D = \frac{1}{xyz} \begin{vmatrix} 1 & x^2 & xyz \\ 1 & y^2 & xyz \\ 1 & z^2 & xyz \end{vmatrix}$$

Using $\left(\frac{1}{xyz}\right) C_3$, we get

$$D = \frac{xyz}{xyz} \begin{vmatrix} 1 & x^2 & 1 \\ 1 & y^2 & 1 \\ 1 & z^2 & 1 \end{vmatrix}$$

$$= 0 \quad (\because C_1 \equiv C_3)$$

$$(ii) D = \begin{vmatrix} 0 & b & c \\ -b & 0 & a \\ -c & -a & 0 \end{vmatrix}$$

$$= (-1)^3 \begin{vmatrix} 0 & -b & -c \\ b & 0 & -a \\ c & a & 0 \end{vmatrix} \quad (\text{Taking } -1 \text{ common from } R_1, R_2 \text{ \& } R_3)$$

$$\text{So } D = - \begin{vmatrix} 0 & -b & -c \\ b & 0 & -a \\ c & a & 0 \end{vmatrix}$$

$$= - \begin{vmatrix} 0 & b & c \\ -b & 0 & a \\ -c & -a & 0 \end{vmatrix} \quad (\text{Interchanging rows with columns \& vice-versa}).$$

$$= -D$$

$$\therefore 2D = 0 \quad \Rightarrow D = 0.$$

Example - 8

Show that

$$\begin{vmatrix} 1 & x & yz \\ 1 & y & zx \\ 1 & z & xy \end{vmatrix} = (x-y)(y-z)(z-x)$$

Sol. $D = \begin{vmatrix} 1 & x & yz \\ 1 & y & zx \\ 1 & z & xy \end{vmatrix}$

$R_1 \rightarrow R_1 - R_3, R_2 \rightarrow R_2 - R_3$, gives



$$D = \begin{vmatrix} 0 & x-z & -y(x-z) \\ 0 & y-z & -x(y-z) \\ 1 & z & xy \end{vmatrix}$$

Using $\left(\frac{1}{z-x}\right)R_1$ and $\left(\frac{1}{y-z}\right)R_2$, we get

$$D = (z-x)(y-z) \begin{vmatrix} 0 & -1 & y \\ 0 & 1 & -x \\ 1 & z & xy \end{vmatrix}$$

$R_1 \rightarrow R_1 + R_2$ gives

$$D = (z-x)(y-z) \begin{vmatrix} 0 & 0 & y-x \\ 0 & 1 & -x \\ 1 & z & xy \end{vmatrix}$$

Expanding along R_1

$$\begin{aligned} \therefore D &= (z-x)(y-z)(y-x) \begin{vmatrix} 0 & 1 \\ 1 & z \end{vmatrix} \\ &= (z-x)(y-z)(y-x)(0-1) \\ &= (x-y)(y-z)(z-x) \\ &= \text{R.H.S.} \end{aligned}$$

Example - 9

Show that

$$\begin{vmatrix} a+b+c & -c & -b \\ -c & a+b+c & -a \\ -b & -a & a+b+c \end{vmatrix}$$

$$= 2(a+b)(b+c)(c+a)$$

$$\text{Sol. } D = \begin{vmatrix} a+b+c & -c & -b \\ -c & a+b+c & -a \\ -b & -a & a+b+c \end{vmatrix}$$

$C_1 \rightarrow C_1 + C_3; C_2 \rightarrow C_2 + C_3$ gives

$$D = \begin{vmatrix} a+c & -(b+c) & -b \\ -(a+c) & b+c & -a \\ a+c & b+c & a+b+c \end{vmatrix}$$

Using $\left(\frac{1}{a+c}\right)C_1$ and $\left(\frac{1}{b+c}\right)C_2$ we get

$$D = (a+c)(b+c) \begin{vmatrix} 1 & -1 & -b \\ -1 & 1 & -a \\ 1 & 1 & a+b+c \end{vmatrix}$$

$R_1 \rightarrow R_1 + R_2$ gives

$$D = (b+c)(c+a) \begin{vmatrix} 0 & 0 & -(a+b) \\ -1 & 1 & -a \\ 1 & 1 & a+b+c \end{vmatrix}$$

Expanding along R_1

$$\begin{aligned} &= (b+c)(c+a) \times [-(a+b)(-1-1)] \\ &= 2(a+b)(b+c)(c+a) \\ &= \text{R.H.S.} \end{aligned}$$

Example - 10

Show that

$$\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = abc \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) = abc + bc + ca + ab$$

Sol. Taking out factors a, b, c common from R_1, R_2 and R_3 , we get

$$\text{L.H.S.} = abc \begin{vmatrix} \frac{1}{a}+1 & \frac{1}{a} & \frac{1}{a} \\ \frac{1}{b} & \frac{1}{b}+1 & \frac{1}{b} \\ \frac{1}{c} & \frac{1}{c} & \frac{1}{c}+1 \end{vmatrix}$$

Applying $R_1 \rightarrow R_1 + R_2 + R_3$, we have

$$\Delta = abc \begin{vmatrix} 1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} & 1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} & 1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \\ \frac{1}{b} & \frac{1}{b}+1 & \frac{1}{b} \\ \frac{1}{c} & \frac{1}{c} & \frac{1}{c}+1 \end{vmatrix}$$



Taking $\left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)$ common from R_1

$$= abc \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) \begin{vmatrix} 1 & 1 & 1 \\ \frac{1}{b} & \frac{1}{b} + 1 & \frac{1}{b} \\ \frac{1}{c} & \frac{1}{c} & \frac{1}{c} + 1 \end{vmatrix}$$

Now applying $C_2 \rightarrow C_2 - C_1, C_3 \rightarrow C_3 - C_1$, we get

$$\Delta = abc \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) \begin{vmatrix} 1 & 0 & 0 \\ \frac{1}{b} & 1 & 0 \\ \frac{1}{c} & 0 & 1 \end{vmatrix}$$

Expanding along C_3

$$= abc \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) [1(1-0)]$$

$$= abc \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) = abc + bc + ca + ab = \text{R.H.S.}$$

Example – 11

Prove that

$$\Delta = \begin{vmatrix} a+bx & c+dx & p+qx \\ ax+b & cx+d & px+q \\ u & v & w \end{vmatrix} = (1-x^2) \begin{vmatrix} a & c & p \\ b & d & q \\ u & v & w \end{vmatrix}$$

Sol. Applying $R_1 \rightarrow R_1 - x R_2$ to Δ , we get

$$\Delta = \begin{vmatrix} a(1-x^2) & c(1-x)^2 & p(1-x^2) \\ ax+b & cx+d & px+q \\ u & v & w \end{vmatrix}$$

Taking $1-x^2$ common from R_1

$$= (1-x^2) \begin{vmatrix} a & c & p \\ ax+b & cx+d & px+q \\ u & v & w \end{vmatrix}$$

Applying $R_2 \rightarrow R_2 - x R_1$, we get

$$\Delta = (1-x^2) \begin{vmatrix} a & c & p \\ b & d & q \\ u & v & w \end{vmatrix}$$

= RHS

Example – 12

Find x , if

$$(i) \begin{vmatrix} 1 & x & x^2 \\ 1 & 2 & 4 \\ 4 & 6 & 9 \end{vmatrix} = 0 \quad (ii) \begin{vmatrix} 1 & 2x & 4x^2 \\ 1 & 4 & 16 \\ 1 & 1 & 1 \end{vmatrix} = 0$$

Sol. (i) $\begin{vmatrix} 1 & x & x^2 \\ 1 & 2 & 4 \\ 4 & 6 & 9 \end{vmatrix} = 0$

Expanding along R_1

$$\therefore 1(18-24) - x(9-16) + x^2(6-8) = 0$$

$$\therefore -6 + 7x - 2x^2 = 0$$

$$\therefore 2x^2 - 7x + 6 = 0$$

$$\therefore (2x-3)(x-2) = 0$$

$$\therefore 2x-3 = 0 \text{ or } x-2 = 0$$

$$\therefore x = \frac{3}{2} \text{ or } x = 2$$

(ii) Given $\begin{vmatrix} 1 & 2x & 4x^2 \\ 1 & 4 & 16 \\ 1 & 1 & 1 \end{vmatrix} = 0$

Method I: Here $a_{13} = (a_{12})^2$ (element in third column is equal to square of respective element in 2nd column)

We know, when two rows are identical then the determinant is zero.

$$\therefore \text{for } R_1 = R_2, 2x = 4 \text{ i.e. } x = 2$$

$$\therefore \text{for } R_1 = R_3, 2x = 1 \text{ i.e. } x = \frac{1}{2}$$

$$\therefore \text{solution is } x = \frac{1}{2}, 2$$



Method II: if we expand the given determinant by R_1 , we get

$$1(4-16)-2x(1-16)+4x^2(1-4)=0$$

$$\therefore -12-2x+32x-12x^2=0$$

$$\therefore -12x^2+30x-12=0$$

$$\therefore 4x^2-10x+4=0$$

$$\therefore (4x-2)(x-2)=0$$

$$\therefore 4x-2=0 \text{ or } x-2=0$$

$$\therefore x=\frac{1}{2} \text{ or } x=2$$

Example – 13

Solve the following equations using Cramer's Rule

$$x+2y-z=3, 3x-y+2z=1,$$

$$2x-2y+3z=2$$

Sol. The given equations are

$$x+2y-z=3$$

$$3x-y+2z=1$$

$$2x-2y+3z=2$$

$$D = \begin{vmatrix} 1 & 2 & -1 \\ 3 & -1 & 2 \\ 2 & -2 & 3 \end{vmatrix}$$

Expanding along R_1

$$= 1(-3+4) - 2(9-4) - 1(-6+2)$$

$$= 1 - 10 + 4 = -5$$

$$D_x = \begin{vmatrix} 3 & 2 & -1 \\ 1 & -1 & 2 \\ 2 & -2 & 3 \end{vmatrix}$$

Expanding along R_1

$$= 3(-3+4) - 2(3-4) - 1(-2+2)$$

$$= 5$$

$$D_y = \begin{vmatrix} 1 & 3 & -1 \\ 3 & +1 & 2 \\ 2 & 2 & 3 \end{vmatrix}$$

Expanding along R_1

$$= 1(3-4) - 3(9-4) - 1(6-2)$$

$$= -20$$

$$D_z = \begin{vmatrix} 1 & 2 & 3 \\ 3 & -1 & 1 \\ 2 & -2 & 2 \end{vmatrix}$$

$$= 1(-2+2) - 2(6-2) + 3(-6+2)$$

$$= -20$$

By Cramer's Rule

$$x = \frac{D_x}{D} = \frac{5}{-5} = -1$$

$$y = \frac{D_y}{D} = \frac{-20}{-5} = 4$$

$$z = \frac{D_z}{D} = \frac{-20}{-5} = 4$$

$$\therefore \text{Solution is } x=-1, y=4, z=4$$

Example – 14

Show that the following equations are consistent

$$2x+3y+1=0, x+2y+1=0, x+y=0$$

Sol. Given

$$2x+3y+1=0$$

$$x+2y+1=0$$

$$x+y=0$$

By condition of consistency

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$$

$$\text{Now, } \begin{vmatrix} 2 & 3 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 0 \end{vmatrix}$$

$$R_1 \rightarrow R_1 - R_2 \text{ gives}$$

$$= \begin{vmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 1 & 1 & 0 \end{vmatrix} = 0 \quad (\because R_1 \equiv R_3)$$

$\therefore$ The given equations are consistent.



Example – 15

Find k, if the following equations are consistent

$$(k-2)x + (k-1)y = 17,$$

$$(k-1)x + (k-2)y = 18,$$

$$x + y = 5$$

Sol. The given equations are

$$(k-2)x + (k-1)y - 17 = 0$$

$$(k-1)x + (k-2)y - 18 = 0$$

$$x + y - 5 = 0$$

∴ The equations are consistent

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} k-2 & k-1 & -17 \\ k-1 & k-2 & -18 \\ 1 & 1 & -5 \end{vmatrix} = 0$$

$R_1 \rightarrow R_1 - R_2$ gives

$$= \begin{vmatrix} -1 & 1 & 1 \\ k-1 & k-2 & -18 \\ 1 & 1 & -5 \end{vmatrix} = 0$$

$C_2 \rightarrow C_2 + C_1, C_3 \rightarrow C_3 + C_1$ gives

$$\begin{vmatrix} -1 & 0 & 0 \\ k-1 & 2k-3 & k-19 \\ 1 & 2 & -4 \end{vmatrix}$$

Expanding along R_1

$$\therefore -1(-8k + 12 - 2k + 38) = 0$$

$$\therefore 10k - 50 = 0$$

$$\therefore k = 5$$

Example – 16

Find k if the area of the triangle ABC is 35 sq. units, where A = (2, 6), B = (5, 4) and C = (k, 4)

Sol. Area = $\frac{1}{2} \begin{vmatrix} 2 & 6 & 1 \\ 5 & 4 & 1 \\ k & 4 & 1 \end{vmatrix} = 35$ sq. units

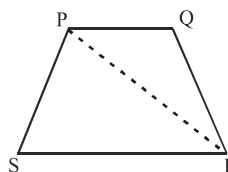
$$\Rightarrow \frac{1}{2}(k-5)2 = \pm 35$$

$$\Rightarrow k-5 = \pm 35$$

$$\Rightarrow k = 40, -30.$$

Example – 17

Find the area of the quadrilateral whose vertices are P(-3, 1), Q(1, -1), R(2, 1), S(0, 3).



Sol.

$$A(PQRS) = A(\Delta PQR) + A(\Delta PRS)$$

$$A(\Delta PQR) = \frac{1}{2} \begin{vmatrix} -3 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & 1 \end{vmatrix}$$

$$= \frac{1}{2} [-3(-1-1) - 1(1-2) + 1(1+2)]$$

$$= \frac{1}{2} (6 + 1 + 3)$$

$$= 5 \text{ Sq. units}$$

$$A(\Delta PRS) = \frac{1}{2} \begin{vmatrix} -3 & 1 & 1 \\ 2 & 1 & 1 \\ 0 & 3 & 1 \end{vmatrix}$$

$$= \frac{1}{2} [-3(1-3) - 1(2-0) + 1(6-0)]$$

$$= \frac{1}{2} [6 - 2 + 6]$$

$$= 5 \text{ sq. units}$$

$$\therefore A(PQRS) = 5 + 5 = 10 \text{ Sq. units.}$$



Example – 18

The sum of first and second numbers is greater than the third number by 5. The sum of first and third numbers is more than the second number by 7. The sum of second and third numbers is greater than the first number by 2. Find such three numbers.

Sol. Let the numbers be x, y, z respectively. We get the following equations.

$$x + y - z = 5$$

$$x - y + z = 7$$

$$-x + y + z = 2$$

$$\therefore D = \begin{vmatrix} 1 & 1 & -1 \\ 1 & -1 & 1 \\ -1 & 1 & 1 \end{vmatrix} = \begin{vmatrix} 2 & 0 & 0 \\ 1 & -1 & 1 \\ -1 & 1 & 1 \end{vmatrix}$$

Expanding along R_1

$$= 2(-1-1)$$

$$= -4$$

$$D_x = \begin{vmatrix} 5 & 1 & -1 \\ 7 & -1 & 1 \\ 2 & 1 & 1 \end{vmatrix} = \begin{vmatrix} 12 & 0 & 0 \\ 7 & -1 & 1 \\ 2 & 1 & 1 \end{vmatrix}$$

Expanding along R_1

$$= 12(-1-1)$$

$$= -24$$

$$D_y = \begin{vmatrix} 1 & 5 & -1 \\ 1 & 7 & 1 \\ -1 & 2 & 1 \end{vmatrix}$$

Expanding along R_1

$$= 1(7-2) - 5(1+1) - 1(2+7)$$

$$= -14$$

$$D_z = \begin{vmatrix} 1 & 1 & 5 \\ 1 & -1 & 7 \\ -1 & 1 & 2 \end{vmatrix}$$

Expanding along R_1

$$= 1(-2-7) - 1(+2+7) + 5(1-1)$$

$$= -18$$

By Cramer's Rule

$$x = \frac{D_x}{D} = \frac{-24}{-4} = 6$$

$$y = \frac{D_y}{D} = \frac{-14}{-4} = \frac{7}{2}$$

$$z = \frac{D_z}{D} = \frac{-18}{-4} = \frac{9}{2}$$

$\therefore$ The numbers are $6, \frac{7}{2}$ and $\frac{9}{2}$ respectively.

Example – 19

If $A = \begin{bmatrix} 2 & 2 \\ -3 & 1 \\ 4 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 6 & 2 \\ 1 & 3 \\ 0 & 4 \end{bmatrix}$, find matrix C such that

$A + B + C = 0$, where 0 is the zero matrix.

Sol. Given, $A + B + C = 0$

$$\therefore C = -[A + B]$$

$$A + B = \begin{bmatrix} 2 & 2 \\ -3 & 1 \\ 4 & 0 \end{bmatrix} + \begin{bmatrix} 6 & 2 \\ 1 & 3 \\ 0 & 4 \end{bmatrix}$$

$$C = -[A + B] = -\begin{bmatrix} 8 & 4 \\ -2 & 4 \\ 4 & 4 \end{bmatrix} = \begin{bmatrix} -8 & -4 \\ 2 & -4 \\ -4 & -4 \end{bmatrix}$$

Example – 20

Find matrices A and B , where

$$2A - B = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \text{ and } A + 3B = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

$$\text{Sol. Given } 2A - B = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \quad \dots (i)$$

$$A + 3B = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \quad \dots (ii)$$

From $3 \times (i) + (ii)$, we get



$$7A = \begin{bmatrix} 3 & -3 \\ 0 & 3 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 3+0 & -3+1 \\ 0-1 & 3+0 \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ -1 & 3 \end{bmatrix}$$

$$\therefore A = \frac{1}{7} \begin{bmatrix} 3 & -2 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 3/7 & -2/7 \\ -1/7 & 3/7 \end{bmatrix}$$

$$\therefore B = 2A - \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \quad \dots (i)$$

$$\therefore B = 2 \begin{bmatrix} 3/7 & -2/7 \\ -1/7 & 3/7 \end{bmatrix} - \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 6/7 & -4/7 \\ -2/7 & 6/7 \end{bmatrix} - \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -1/7 & 3/7 \\ -2/7 & -1/7 \end{bmatrix}$$

Example – 21

If $A_\alpha = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$, show that $A_\alpha \cdot A_\beta = A_{\alpha+\beta}$

Sol. $A_\alpha = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$

then $A_\beta = \begin{bmatrix} \cos \beta & \sin \beta \\ -\sin \beta & \cos \beta \end{bmatrix}$

$$A_\alpha \cdot A_\beta = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} \cos \beta & \sin \beta \\ -\sin \beta & \cos \beta \end{bmatrix}$$

$$= \begin{bmatrix} \cos \alpha \cos \beta - \sin \alpha \sin \beta & \cos \alpha \sin \beta + \sin \alpha \cos \beta \\ -\sin \alpha \cos \beta - \cos \alpha \sin \beta & -\sin \alpha \sin \beta + \cos \alpha \cos \beta \end{bmatrix}$$

$$= \begin{bmatrix} \cos(\alpha + \beta) & \sin(\alpha + \beta) \\ -\sin(\alpha + \beta) & \cos(\alpha + \beta) \end{bmatrix}$$

$$= A_{(\alpha+\beta)}$$

Example – 22

If $A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 \\ 3 & -2 \end{bmatrix}$, verify that $|AB| = |A| |B|$.

Sol. $AB = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & -2 \end{bmatrix}$

$$= \begin{bmatrix} 2+3 & 4-2 \\ 0+9 & 0-6 \end{bmatrix}$$

$$\therefore AB = \begin{bmatrix} 5 & 2 \\ 9 & -6 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 2 & 1 \\ 0 & 3 \end{vmatrix} = 6 - 0 = 6$$

$$|B| = \begin{vmatrix} 1 & 2 \\ 3 & -2 \end{vmatrix} = -2 - 6 = -8$$

$$|AB| = \begin{vmatrix} 5 & 2 \\ 9 & -6 \end{vmatrix} = -30 - 18 = -48$$

$$\text{Also, } |A| \cdot |B| = 6(-8) = -48$$

Hence, $|AB| = |A| \cdot |B|$ is verified.

Example – 23

If $A = \begin{bmatrix} 3 & -5 \\ -4 & 2 \end{bmatrix}$, show that $A^2 - 5A - 14I = 0$

Sol. $A = \begin{bmatrix} 3 & -5 \\ -4 & 2 \end{bmatrix}$

$$\therefore A^2 = A \cdot A = \begin{bmatrix} 3 & -5 \\ -4 & 2 \end{bmatrix} \begin{bmatrix} 3 & -5 \\ -4 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 9+20 & -15-10 \\ -12-8 & 20+4 \end{bmatrix} = \begin{bmatrix} 29 & -25 \\ -20 & 24 \end{bmatrix}$$

$$\therefore A^2 - 5A - 14I = \begin{bmatrix} 29 & -25 \\ -20 & 24 \end{bmatrix} - 5 \begin{bmatrix} 3 & -5 \\ -4 & 2 \end{bmatrix} - 14 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$



$$= \begin{bmatrix} 29 & -25 \\ -20 & 24 \end{bmatrix} - \begin{bmatrix} 15 & -25 \\ -20 & 10 \end{bmatrix} - \begin{bmatrix} 14 & 0 \\ 0 & 14 \end{bmatrix}$$

$$= \begin{bmatrix} 29-15-14 & -25+25+0 \\ -20+20-0 & 24-10-14 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\therefore A^2 - 4A + 3I = 0$$

Example – 24

If $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

show that $(A+B)(A-B) \neq A^2 - B^2$.

Sol. $(A+B)(A-B) = A^2 - AB + BA - B^2$

$$AB = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0+1 & 0+0 \\ 0+0 & -1+0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

And, $BA = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

$$= \begin{bmatrix} 0-1 & 0+0 \\ 0+0 & 1+0 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\therefore AB \neq BA$$

$$\therefore -AB + BA \neq 0$$

$$\therefore (A+B)(A-B) \neq A^2 - B^2$$

Example – 25

If $A = \begin{bmatrix} -3 & 2 \\ 2 & -4 \end{bmatrix}$, $B = \begin{bmatrix} 1 & x \\ y & 0 \end{bmatrix}$, and $(A+B)(A-B) = A^2 - B^2$,

find x and y

Sol. Condition given

$$(A+B)(A-B) = A^2 - B^2$$

$$\therefore A^2 - AB + BA - B^2 = A^2 - B^2$$

$$\therefore -AB + BA = 0$$

$$\therefore AB = BA$$

$$\therefore \begin{bmatrix} -3 & 2 \\ 2 & -4 \end{bmatrix} \begin{bmatrix} 1 & x \\ y & 0 \end{bmatrix} = \begin{bmatrix} 1 & x \\ y & 0 \end{bmatrix} \begin{bmatrix} -3 & 2 \\ 2 & -4 \end{bmatrix}$$

$$\therefore \begin{bmatrix} -3+2y & -3x+0 \\ 2-4y & 2x+0 \end{bmatrix} = \begin{bmatrix} -3+2x & 2-4x \\ -3y+0 & 2y+0 \end{bmatrix}$$

Comparing corresponding elements.

$$-3x = 2 - 4x \text{ and } -3y = 2 - 4y$$

$$\therefore x = 2 \quad y = 2$$

$$[\therefore x = y = 2]$$

Example – 26

If $A = \begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix}$, find A^3

Sol. $A = \begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix}$

$$\therefore A^2 = A \cdot A = \begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} 4-3 & -2+2 \\ 6-6 & -3+4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

$$\therefore A^3 = A \cdot A^2 = A \cdot I = A = \begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix}$$

Example – 27

If $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 2 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 1 & -1 & 1 \\ -3 & 2 & -1 \\ -2 & 1 & 0 \end{bmatrix}$,

show that AB and BA are both singular matrices.

Sol. $AB = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & -1 & 1 \\ -3 & 2 & -1 \\ -2 & 1 & 0 \end{bmatrix}$

$$= \begin{bmatrix} 1-6-6 & -1+4+3 & 1-2+0 \\ 2-12-12 & -2+8+6 & 2-4+0 \\ 1-6-6 & -1+4+3 & 1-2+0 \end{bmatrix}$$



$$= \begin{bmatrix} -11 & 6 & -1 \\ -22 & 12 & -2 \\ -11 & 6 & -1 \end{bmatrix}$$

$$|AB| = \begin{vmatrix} -11 & 6 & -1 \\ -22 & 12 & -2 \\ -11 & 6 & -1 \end{vmatrix} = 0 \quad (\because R_1 = R_3)$$

Similarly,

$$BA = \begin{bmatrix} 1 & -1 & 1 \\ -3 & 2 & -1 \\ -2 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 2 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 1-2+1 & 2-4+2 & 3-6+3 \\ -3+4-1 & -6+8-2 & -9+12-3 \\ -2+2+0 & -4+4+0 & -6+6+0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\therefore |BA| = 0 \quad (\because \text{it is zero matrix})$$

Hence, AB and BA both are singular.

Example – 28

Express the matrix $B = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$ as the sum of a

symmetric and a skew symmetric matrix.

Sol. Here

$$B' = \begin{bmatrix} 2 & -1 & 1 \\ -2 & 3 & -2 \\ -4 & 4 & -3 \end{bmatrix}$$

$$\text{Let } P = \frac{1}{2}(B+B') = \frac{1}{2} \begin{bmatrix} 4 & -3 & -3 \\ -3 & 6 & 2 \\ -3 & 2 & -6 \end{bmatrix} = \begin{bmatrix} 2 & \frac{-3}{2} & \frac{-3}{2} \\ \frac{-3}{2} & 3 & 1 \\ \frac{-3}{2} & 1 & -3 \end{bmatrix}$$

$$\text{Now } P' = \begin{bmatrix} 2 & \frac{-3}{2} & \frac{-3}{2} \\ \frac{-3}{2} & 3 & 1 \\ \frac{-3}{2} & 1 & -3 \end{bmatrix} = P$$

Thus $P = \frac{1}{2}(B+B')$ is a symmetric matrix.

Also, let

$$Q = \frac{1}{2}(B-B') = \frac{1}{2} \begin{bmatrix} 0 & -1 & -5 \\ 1 & 0 & 6 \\ 5 & -6 & 0 \end{bmatrix} = \begin{bmatrix} 0 & \frac{-1}{2} & \frac{-5}{2} \\ \frac{1}{2} & 0 & 3 \\ \frac{5}{2} & -3 & 0 \end{bmatrix}$$

$$\text{Then } Q' = \begin{bmatrix} 0 & \frac{1}{2} & \frac{5}{2} \\ \frac{-1}{2} & 0 & -3 \\ \frac{-5}{2} & 3 & 0 \end{bmatrix} = -Q$$

Thus $Q = \frac{1}{2}(B-B')$ is a skew symmetric matrix.

Now

$$P+Q = \begin{bmatrix} 2 & \frac{-3}{2} & \frac{-3}{2} \\ \frac{-3}{2} & 3 & 1 \\ \frac{-3}{2} & 1 & -3 \end{bmatrix} + \begin{bmatrix} 0 & \frac{-1}{2} & \frac{-5}{2} \\ \frac{1}{2} & 0 & 3 \\ \frac{5}{2} & -3 & 0 \end{bmatrix} = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix} = B$$

Thus, B is represented as the sum of a symmetric and a skew symmetric matrix.



Example – 29

The sum of three numbers is 6. If we multiply third number by 3 and add second number to it, we get 11. By adding first and third numbers, we get double of the second number. Represent it algebraically and find the numbers using matrix method.

Sol. Let first, second and third numbers be denoted by x , y and z , respectively. Then, according to given conditions, we have

$$x + y + z = 6$$

$$y + 3z = 11$$

$$x + z = 2y \text{ or } x - 2y + z = 0$$

This system can be written as $AX = B$, where

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 1 & -2 & 1 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } B = \begin{bmatrix} 6 \\ 11 \\ 0 \end{bmatrix}$$

Here $|A| = 1(1+6) - (0-3) + (0-1) = 9 \neq 0$. Now we find $\text{adj } A$

$$A_{11} = 1(1+6) = 7, A_{12} = -(0-3) = 3, A_{13} = -(0-1) = 1$$

$$A_{21} = -(1+2) = -3, A_{22} = 0, A_{23} = -(-2-1) = 3$$

$$A_{31} = (3-1) = 2, A_{32} = -(3-0) = -3, A_{33} = (1-0) = 1$$

$$\text{Hence } \text{adj } A = \begin{bmatrix} 7 & -3 & 2 \\ 3 & 0 & -3 \\ -1 & 3 & 1 \end{bmatrix}$$

$$\text{Thus } A^{-1} = \frac{1}{|A|} \text{adj}(A) = \frac{1}{9} \begin{bmatrix} 7 & -3 & 2 \\ 3 & 0 & -3 \\ -1 & 3 & 1 \end{bmatrix}$$

Since $X = A^{-1}B$

$$X = \frac{1}{9} \begin{bmatrix} 7 & -3 & 2 \\ 3 & 0 & -3 \\ -1 & 3 & 1 \end{bmatrix} \begin{bmatrix} 6 \\ 11 \\ 0 \end{bmatrix}$$

$$\text{or } \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{9} \begin{bmatrix} 42-33+0 \\ 18+0+0 \\ -6+33+0 \end{bmatrix} = \frac{1}{9} \begin{bmatrix} 9 \\ 18 \\ 27 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Thus $x = 1, y = 2, z = 3$

Example – 30

$$\text{Use product } \begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & -3 \\ 3 & -2 & 4 \end{bmatrix} \begin{bmatrix} -2 & 0 & 1 \\ 9 & 2 & -3 \\ 6 & 1 & -2 \end{bmatrix} \text{ to solve the}$$

system of equations

$$x - y + 2z = 1$$

$$2y - 3z = 1$$

$$3x - 2y + 4z = 2$$

$$\text{Sol. Consider the product } \begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & -3 \\ 3 & -2 & 4 \end{bmatrix} \begin{bmatrix} -2 & 0 & 1 \\ 9 & 2 & -3 \\ 6 & 1 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} -2-9+12 & 0-2+2 & 1+3-4 \\ 0+18-18 & 0+4-3 & 0-6+6 \\ -6-18+24 & 0-4+4 & 3+6-8 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{Hence } \begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & -3 \\ 3 & -2 & 4 \end{bmatrix}^{-1} = \begin{bmatrix} -2 & 0 & 1 \\ 9 & 2 & -3 \\ 6 & 1 & -2 \end{bmatrix}$$

Now, given system of equations can be written, in matrix form, as follows

$$\begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & -3 \\ 3 & -2 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

$$\text{or } \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & -3 \\ 3 & -2 & 4 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} -2 & 0 & 1 \\ 9 & 2 & -3 \\ 6 & 1 & -2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} -2+0+2 \\ 9+2-6 \\ 6+1-4 \end{bmatrix} = \begin{bmatrix} 0 \\ 5 \\ 3 \end{bmatrix}$$

Hence $x = 0, y = 5$ and $z = 3$



Example – 31

Given $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 4 & 1 \\ 2 & 3 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix}$.

Find P such that $BPA = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$.

Sol. Given $BPA = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$.

Pre-multiplying both sides by B^{-1}

$$B^{-1} BPA = B^{-1} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\Rightarrow IPA = B^{-1} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\Rightarrow PA = B^{-1} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \dots (i)$$

To find B^{-1} .

Now $B = \begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix}$

$$|B| = \begin{vmatrix} 2 & 3 \\ 3 & 4 \end{vmatrix} = 8 - 9 = -1 \neq 0. \text{ As } |B| \neq 0$$

so it is non-singular matrix and hence inverse of B exists.

$$\Rightarrow B^{-1} = \frac{\text{Adj.}B}{|B|} = \begin{bmatrix} -4 & 3 \\ 3 & -2 \end{bmatrix}$$

NOTES :

For a 2×2 matrix, adjoint can be obtained by swapping diagonal elements and changing the sign of non-diagonal elements.

Now from (i),

$$PA = \begin{bmatrix} -4 & 3 \\ 3 & -2 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\Rightarrow PA = \begin{bmatrix} -4 & 3 & -4 \\ 3 & -2 & 3 \end{bmatrix}$$

Post-multiplying both sides by A^{-1}

$$PAA^{-1} = \begin{bmatrix} -4 & 3 & -4 \\ 3 & -2 & 3 \end{bmatrix} A^{-1}$$

$$\Rightarrow PI = \begin{bmatrix} -4 & 3 & -4 \\ 3 & -2 & 3 \end{bmatrix} A^{-1}$$

$$\therefore P = \begin{bmatrix} -4 & 3 & -4 \\ 3 & -2 & 3 \end{bmatrix} A^{-1} \dots (ii)$$

For A^{-1} :

since $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 4 & 1 \\ 2 & 3 & 1 \end{bmatrix}$. Now $|A| = -1 \neq 0$

$\Rightarrow$ it is non-singular matrix and hence A^{-1} exists

$$\text{adj.}(A) = \begin{bmatrix} 1 & 2 & -3 \\ 0 & -1 & 1 \\ -2 & -1 & 2 \end{bmatrix}$$

$$A^{-1} = \frac{\text{Adj.}A}{|A|} = \begin{bmatrix} -1 & -2 & 3 \\ 0 & 1 & -1 \\ 2 & 1 & -2 \end{bmatrix}$$

Now From (ii),

$$P = \begin{bmatrix} -4 & 3 & -4 \\ 3 & -2 & 3 \end{bmatrix} \times \begin{bmatrix} -1 & -2 & 3 \\ 0 & 1 & -1 \\ 2 & 1 & -2 \end{bmatrix}$$

$$\Rightarrow P = \begin{bmatrix} -4 & 7 & -7 \\ 3 & -5 & 5 \end{bmatrix}$$



Example – 32

Solve the system of equations :

$$x + 2y + z = 2$$

$$2x - 3y + 4z = 1$$

$$3x + 6y + 3z = 6$$

Sol. 1st and 3rd equations are integral multiple of each other.
(dependent equations)

$$\Rightarrow D = D_1 = D_2 = D_3 = 0$$

$\Rightarrow$ infinite solutions

consider $x + 2y + z = 2$

$$2x - 3y + 4z = 1$$

Let $z = k$

$$\Rightarrow \begin{cases} x + 2y = 2 - k \\ 2x - 3y = 1 - 4k \end{cases}$$

$$\Rightarrow y = \frac{3+2k}{7} \text{ and } x = \frac{8-11k}{7}$$

$$\text{Hence : } \left[x = \frac{8-11k}{7}, y = \frac{3+2k}{7} \text{ and } z = k \right]$$

where k is an arbitrary constant.

Example – 33

Obtain the inverse of the following matrix using elementary operations

$$A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$$

Sol. Write $A = IA$, i.e.,

$$\begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

$$\text{or } \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 3 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \text{ (applying } R_1 \leftrightarrow R_2)$$

$$\text{or } \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & -5 & -8 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & -3 & 1 \end{bmatrix} A \text{ (applying } R_3 \rightarrow R_3 - 3R_1)$$

$$\text{or } \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & -5 & -8 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & -3 & 1 \end{bmatrix} A \text{ (applying } R_1 \rightarrow R_1 - 2R_2)$$

$$\text{or } \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 0 \\ 1 & 0 & 0 \\ 5 & -3 & 1 \end{bmatrix} A \text{ (applying } R_3 \rightarrow R_3 + 5R_2)$$

$$\text{or } \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 0 \\ 1 & 0 & 0 \\ \frac{5}{2} & \frac{-3}{2} & \frac{1}{2} \end{bmatrix} A \text{ (applying } R_3 \rightarrow \frac{1}{2} R_3)$$

$$\text{or } \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{-1}{2} & \frac{1}{2} \\ 1 & 0 & 0 \\ \frac{5}{2} & \frac{-3}{2} & \frac{1}{2} \end{bmatrix} A \text{ (applying } R_1 \rightarrow R_1 + R_3)$$

$$\text{or } \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{-1}{2} & \frac{1}{2} \\ -4 & 3 & -1 \\ \frac{5}{2} & \frac{-3}{2} & \frac{1}{2} \end{bmatrix} A \text{ (applying } R_2 \rightarrow R_2 - 2R_3)$$

$$\text{Hence } A^{-1} = \begin{bmatrix} \frac{1}{2} & \frac{-1}{2} & \frac{1}{2} \\ -4 & 3 & -1 \\ \frac{5}{2} & \frac{-3}{2} & \frac{1}{2} \end{bmatrix}$$



Example – 34

If M is a 3×3 matrix, where $M^T M = I$ and $\det(M) = 1$, then prove that $\det(M - I) = 0$.

Sol. $(M - I)^T = M^T - I = M^T - M^T M = M^T (I - M)$

$$\Rightarrow |M - I|^T = |M - I| = |M^T| |I - M|$$

$$= |I - M| \Rightarrow |M - I| = 0.$$

Alternate Method :

$$\det(M - I) = \det(M - I) \det(M^T)$$

$$= \det(M M^T - M^T)$$

$$= \det(I - M^T) = -\det(M^T - I)$$

$$= -\det(M - I)^T = -\det(M - I)$$

$$\Rightarrow \det(M - I) = 0.$$

Example – 35

If S is a skew-symmetric matrix of order n and $I + S$ is non-singular, then prove that

$A = (I - S)(I + S)^{-1}$ is an orthogonal matrix of order n .

Sol. $A^T = [(I + S)^T]^{-1} [I - S]^T$

$$= (I - S)^{-1} (I + S),$$

[since $S^T = -S$; S being skew symmetric].

$$\therefore A^T A = (I - S)^{-1} (I + S) (I - S) (I + S)^{-1}$$

$$= (I - S)^{-1} (I - S) (I + S) (I + S)^{-1},$$

$$\text{since } (I + S)(I - S) = (I - S)(I + S)$$

$$= I$$

$\therefore A$ is orthogonal, $I - S$ is a square matrix of order n .

$\therefore A = (I - S)(I + S)^{-1}$ is a square matrix of order n .

Example – 36

If A and B are n -rowed non-zero square matrix such that $AB = 0$, then show that both A and B are singular. If both A and B are singular and $AB = 0$, does it follow that $BA = 0$.

Justify your answer.

Sol. (i) $AB = 0$ and A non-singular implies

$$A^{-1} AB = A^{-1} (0) = 0$$

$$\Rightarrow (A^{-1} A) B = 0$$

$$\Rightarrow IB = 0 \Rightarrow B = 0 \text{ [Note true]}$$

(ii) $AB = 0$ and B non-singular implies

$$ABB^{-1} = 0(B)^{-1} = 0 \Rightarrow AI = 0$$

$$\Rightarrow A = 0 \text{ [Not true]}$$

$\therefore$ Both A and B are singular.

(iii) Consider the counter example

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \text{ and } B = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$AB = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\text{whereas } BA = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \neq 0$$

Example – 37

Let A and B be matrix of order n . Prove that if $(I - AB)$ is invertible, then $(I - BA)$ is also invertible and $(I - BA)^{-1} = I + B(I - AB)^{-1}A$.

Sol. $I - BA = B I B^{-1} - B A B B^{-1}$

$$= B(I - AB)B^{-1} \quad \dots(i)$$

$$\text{Hence, } |I - BA| = |B| |I - AB| |B^{-1}|$$

$$= |I - AB| |B| |B^{-1}|$$

$$= |I - AB| |B| |B^{-1}|$$

$$= |I - AB| \quad \dots(ii)$$

$$\text{Since } |B| |B^{-1}| = |B B^{-1}| = |I| = 1$$

If $I - AB$ is invertible, $|I - AB|$ has to be non-zero.

Hence, $|I - BA| \neq 0$ and therefore $I - BA$ is also invertible

$$\text{Now } (I - BA) \{I + B(I - AB)^{-1}A\}$$

$$= (I - BA) + (I - BA) B (I - AB)^{-1} A$$

$$= (I - BA) + \{B(I - AB)B^{-1}\} B (I - AB)^{-1} A$$

(Using (i))

$$= (I - BA) + B(I - AB)(I - AB)^{-1} A$$

$$= I - BA + BA = I$$

$$\text{Hence, } (I - BA)^{-1} = I + B(I - AB)^{-1} A$$



Example – 38

If $A = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$, prove that

$$A^n = \begin{bmatrix} \cos n\theta & \sin n\theta \\ -\sin n\theta & \cos n\theta \end{bmatrix} \text{ for all } n \in \mathbb{N}$$

Sol. Consider $A^n = \begin{bmatrix} \cos n\theta & \sin n\theta \\ -\sin n\theta & \cos n\theta \end{bmatrix}$

for $n = 1$, $A^1 = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$,

it is true as given

$\therefore A^n$ is true for $n = 1$

Let A^n is true for $n = r$, where $r \in \mathbb{N}$

$\therefore A^r = \begin{bmatrix} \cos r\theta & \sin r\theta \\ -\sin r\theta & \cos r\theta \end{bmatrix}$

then,

$$A^{r+1} = A \cdot A^r$$

$$= \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \cos r\theta & \sin r\theta \\ -\sin r\theta & \cos r\theta \end{bmatrix}$$

$$= \begin{bmatrix} \cos \theta \cos r\theta - \sin \theta \sin r\theta & \cos \theta \sin r\theta + \sin \theta \cos r\theta \\ -\sin \theta \cos r\theta - \cos \theta \sin r\theta & -\sin \theta \sin r\theta + \cos \theta \cos r\theta \end{bmatrix}$$

$$= \begin{bmatrix} \cos(\theta + r\theta) & \sin(\theta + r\theta) \\ -\sin(\theta + r\theta) & \cos(\theta + r\theta) \end{bmatrix}$$

$$= \begin{bmatrix} \cos(r+1)\theta & \sin(r+1)\theta \\ -\sin(r+1)\theta & \cos(r+1)\theta \end{bmatrix}$$

i.e. A^n is true for $n = r + 1$,

$\therefore A^n$ is true for $n = 1$

And A^n is true for $n = r + 1$, if it is true for $n = r$

$\therefore A^n$ is true for all $n \in \mathbb{N}$

$\therefore A^n = \begin{bmatrix} \cos n\theta & \sin n\theta \\ -\sin n\theta & \cos n\theta \end{bmatrix} \text{ for all } n \in \mathbb{N}.$

Example – 39

$$A = \begin{bmatrix} a & 0 & 1 \\ 1 & c & b \\ 1 & d & b \end{bmatrix}, B = \begin{bmatrix} a & 1 & 1 \\ 0 & d & c \\ f & g & h \end{bmatrix}, U = \begin{bmatrix} f \\ g \\ h \end{bmatrix}, V = \begin{bmatrix} a^2 \\ 0 \\ 0 \end{bmatrix}, ab \neq 1$$

If there is vector matrix X , such that $AX = U$ has infinitely many solutions, then prove that $BX = V$ cannot have unique solution. If $af \neq 0$ then prove that $BX = V$ has no solution.

Sol. $AX = U$ has infinite solutions

$$\Rightarrow |A| = 0$$

$$\begin{vmatrix} a & 0 & 1 \\ 1 & c & b \\ 1 & d & b \end{vmatrix} = 0$$

$$\Rightarrow ab = 1 \text{ or } c = d$$

and $|A_1| = \begin{vmatrix} a & 0 & f \\ 1 & c & g \\ 1 & d & h \end{vmatrix} = 0$

$\Rightarrow g = h$; [Here A_1 is actually D_1 for A : Cramer's Rule in Determinants section]

$$|A_2| = \begin{vmatrix} a & f & 1 \\ 1 & g & b \\ 1 & h & b \end{vmatrix} = 0 \Rightarrow g = h$$

$$\Rightarrow |A_3| = \begin{vmatrix} f & 0 & 1 \\ g & c & b \\ h & d & b \end{vmatrix} = 0 \Rightarrow g = h, c = d$$

$$\Rightarrow c = d \text{ and } g = h$$

So, for infinite solutions $c = d$ and $g = h$

$$BX = V$$

$$|B| = \begin{vmatrix} a & 1 & 1 \\ 0 & d & c \\ f & g & h \end{vmatrix} = 0 \text{ (Since } C_2 \text{ and } C_3 \text{ are equal)}$$

$$\Rightarrow BX = V \text{ has no unique solution}$$

and $|B_1| = \begin{vmatrix} a^2 & 1 & 1 \\ 0 & d & c \\ 0 & g & h \end{vmatrix} = 0$ (since $c = d, g = h$)



$$|B_2| = \begin{vmatrix} a & a^2 & 1 \\ 0 & 0 & c \\ f & 0 & h \end{vmatrix} = a^2cf = a^2df \quad (\text{since } c = d)$$

$$|B_3| = \begin{vmatrix} a & 1 & a^2 \\ 0 & d & 0 \\ f & g & 0 \end{vmatrix} = a^2df$$

Since if $adf \neq 0$ then $|B_2| = |B_3| \neq 0$. Hence no solution exists.

Example – 40

Prove that the inverse of $\begin{pmatrix} A & 0 \\ B & C \end{pmatrix}$ is $\begin{pmatrix} A^{-1} & 0 \\ -C^{-1}BA^{-1} & C^{-1} \end{pmatrix}$

where A, C are non-singular matrix and hence find the

inverse of $\begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{pmatrix}$

Sol. First part :

$$\text{As } \begin{pmatrix} A & 0 \\ B & C \end{pmatrix} \begin{pmatrix} A^{-1} & 0 \\ -C^{-1}BA^{-1} & C^{-1} \end{pmatrix}$$

$$\begin{pmatrix} AA^{-1} & 0 \\ BA^{-1} - CC^{-1}BA^{-1} & CC^{-1} \end{pmatrix} = \begin{pmatrix} I & 0 \\ 0 & I \end{pmatrix}$$

$$\text{and } \begin{pmatrix} A^{-1} & 0 \\ -C^{-1}BA^{-1} & C^{-1} \end{pmatrix} \begin{pmatrix} A & 0 \\ B & C \end{pmatrix}$$

$$\begin{pmatrix} A^{-1}A & 0 \\ -C^{-1}B + C^{-1}B & C^{-1}C \end{pmatrix} = \begin{pmatrix} I & 0 \\ 0 & I \end{pmatrix}$$

Hence $\begin{pmatrix} A^{-1} & 0 \\ C^{-1}BA^{-1} & C^{-1} \end{pmatrix}$ is the inverse of $\begin{pmatrix} A & 0 \\ B & C \end{pmatrix} = I$.

Second Part :

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} A & 0 \\ B & C \end{pmatrix}$$

$$\text{where } A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, B = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, C = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

$$\text{Inverse of } \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{pmatrix}$$

$$\text{since } C^{-1}BA^{-1} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$



EXERCISE - 1 : BASIC OBJECTIVE QUESTIONS

Properties of determinants

1. If $P = \begin{vmatrix} 2 & 1 & 0 \\ 3 & 1 & 2 \\ 5 & 2 & 3 \end{vmatrix}$, then $\begin{vmatrix} 2 & 2 & 0 \\ 9 & 6 & 6 \\ 5 & 4 & 3 \end{vmatrix}$ is equal to

- (a) $2P$ (b) $3P$
(c) $5P$ (d) $6P$

2. $\begin{vmatrix} 11 & 12 & 13 \\ 12 & 13 & 14 \\ 13 & 14 & 15 \end{vmatrix}$ is equal to

- (a) 1 (b) 0
(c) -1 (d) 67

3. If every element of a third order determinant of value Δ is multiplied by 5, then the value of new determinant is

- (a) Δ (b) 5Δ
(c) 25Δ (d) 125Δ

4. $\Delta = \begin{vmatrix} 18 & 40 & 89 \\ 40 & 89 & 198 \\ 89 & 198 & 440 \end{vmatrix}$ is equal to

- (a) 1 (b) -1
(c) zero (d) 2

5. The value of $\begin{vmatrix} 1 & \omega & \omega^2 \\ \omega & \omega^2 & 1 \\ \omega^2 & 1 & \omega \end{vmatrix}$, ω being a cube root of unity, is

- (a) 0 (b) 1
(c) ω^2 (d) ω

6. $\Delta = \begin{vmatrix} 0 & p-q & a-b \\ q-p & 0 & x-y \\ b-a & y-x & 0 \end{vmatrix}$ is equal to

- (a) 0 (b) $a+b$
(c) $x+y$ (d) $p+q$

7. If α, β & γ are the roots of the equation $x^3 + px + q = 0$ then the value of the determinant

$$\begin{vmatrix} \alpha & \beta & \gamma \\ \beta & \gamma & \alpha \\ \gamma & \alpha & \beta \end{vmatrix} =$$

- (a) p (b) q
(c) $p^2 - 2q$ (d) none

8. Given a, b, c are in A.P. Then determinant

$$\begin{vmatrix} x+1 & x+2 & x+a \\ x+2 & x+3 & x+b \\ x+3 & x+4 & x+c \end{vmatrix}$$
 in its simplified form is :

- (a) $x^3 + 3ax + 7c$ (b) 0
(c) 15 (d) $10x^2 + 5x + 2c$

9. If $a + b + c = 0$, one root of :

$$\begin{vmatrix} a-x & c & b \\ c & b-x & a \\ b & a & c-x \end{vmatrix} = 0$$
 is

- (a) $x = 1$ (b) $x = 2$
(c) $x = a^2 + b^2 + c^2$ (d) $x = 0$

10. The determinant $\begin{vmatrix} b_1 + c_1 & c_1 + a_1 & a_1 + b_1 \\ b_2 + c_2 & c_2 + a_2 & a_2 + b_2 \\ b_3 + c_3 & c_3 + a_3 & a_3 + b_3 \end{vmatrix} =$

(a) $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ (b) $2 \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$

(c) $3 \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ (d) none of these



11. If
$$\begin{vmatrix} r & 2r-1 & 3r-2 \\ \frac{n}{2} & n-1 & a \\ \frac{1}{2}n(n-1) & (n-1)^2 & \frac{1}{2}(n-1)(3n+4) \end{vmatrix},$$
 then the

value of $\sum_{r=1}^{n-1} \Delta r$:

- (a) depends only on a
- (b) depends only on n
- (c) depends both on a and n
- (d) is independent of both a and n.

Algebra of matrices

12. A matrix $A = [a_{ij}]$ of order 2×3 whose elements are such that $a_{ij} = i + j$ is -

- (a) $\begin{bmatrix} 2 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix}$
- (b) $\begin{bmatrix} 2 & 3 & 4 \\ 5 & 4 & 3 \end{bmatrix}$
- (c) $\begin{bmatrix} 2 & 3 & 4 \\ 5 & 5 & 3 \end{bmatrix}$
- (d) none of these

13. If A and B are 3×3 matrices, then $AB = O$ implies :

- (a) $A = O$ and $B = O$
- (b) $|A| = 0$ and $|B| = 0$
- (c) either $|A| = 0$ or $|B| = 0$
- (d) $A = O$ or $B = O$

14. If X and Y two matrices are such that

$X - Y = \begin{bmatrix} 3 & 2 \\ -1 & 0 \end{bmatrix}$ and $X + Y = \begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix}$ then Y is given by

- (a) $\begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix}$
- (b) $\begin{bmatrix} -1 & -2 \\ 3 & 4 \end{bmatrix}$
- (c) $\begin{bmatrix} -1 & -2 \\ 2 & 2 \end{bmatrix}$
- (d) None of these

15. If $A + B = \begin{bmatrix} 7 & 4 \\ 8 & 9 \end{bmatrix}$ and $A - B = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$ then the value of A is-

- (a) $\begin{bmatrix} 3 & 1 \\ 4 & 3 \end{bmatrix}$
- (b) $\begin{bmatrix} 4 & 3 \\ 4 & 6 \end{bmatrix}$
- (c) $\begin{bmatrix} 6 & 2 \\ 8 & 6 \end{bmatrix}$
- (d) $\begin{bmatrix} 7 & 6 \\ 8 & 6 \end{bmatrix}$

16. If $A = [a_{ij}]$ is a square matrix of order $n \times n$ and k is a scalar, then $|kA| =$

- (a) $k^n |A|$
- (b) $k |A|$
- (c) $kn^{-1} |A|$
- (d) none of these

17. If A, B, C are matrices of order 1×3 , 3×3 and 3×1 respectively, the order of ABC will be -

- (a) 3×3
- (b) 1×3
- (c) 1×1
- (d) 3×1

18. The order of $[xyz] \begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ is

- (a) 3×1
- (b) 1×1
- (c) 1×3
- (d) 3×3

19. If $[1 \times 2] \begin{bmatrix} 2 & 3 & 1 \\ 0 & 4 & 2 \\ 0 & 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ 1 \\ -1 \end{bmatrix} = O$, then the value of x is

- (a) -1
- (b) 0
- (c) 1
- (d) 2

20. If $\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$ then

- (a) $x = 2, y = 1$
- (b) $x = 1, y = 2$
- (c) $x = 3, y = 2$
- (d) $x = 2, y = 3$

21. If $A = \begin{bmatrix} -1 & 2 \\ 3 & -4 \end{bmatrix}$ then element a_{21} of A^2 is -

- (a) 22
- (b) -15
- (c) -10
- (d) 7



22. If $E(\theta) = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$, then value of $E(\alpha) \cdot E(\beta)$ is -

- (a) $E(0^\circ)$ (b) $E(90^\circ)$
(c) $E(\alpha + \beta)$ (d) $E(\alpha - \beta)$

23. The root of the equation $\begin{bmatrix} x & 1 & 2 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ -1 \\ 1 \end{bmatrix} = 0$ is

- (a) $1/3$ (b) $-1/3$
(c) 0 (d) 1

24. If $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, the $A^4 =$

- (a) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$
(c) $\begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

25. If $A = \begin{bmatrix} p & q \\ -q & p \end{bmatrix}$, $B = \begin{bmatrix} r & s \\ -s & r \end{bmatrix}$ then

- (a) $AB = BA$ (b) $AB \neq BA$
(c) $AB = -BA$ (d) none of these

26. If $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ and a and b are arbitrary constants then

- $(aI + bA)^2 =$
(a) $a^2I + abA$ (b) $a^2I + 2abA$
(c) $a^2I + b^2A$ (d) none of these

27. If $A = \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 4 \\ 1 & 6 \end{bmatrix}$ then $(AB)^T$ equals -

- (a) $\begin{bmatrix} 5 & 16 \\ 9 & 16 \end{bmatrix}$ (b) $\begin{bmatrix} 5 & 9 \\ 16 & 12 \end{bmatrix}$
(c) $\begin{bmatrix} 5 & 9 \\ 4 & 3 \end{bmatrix}$ (d) none of these

28. If $A = \begin{bmatrix} 2 & -1 \\ -7 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 1 \\ 7 & 2 \end{bmatrix}$ then $B^T A^T$ is equal to -

- (a) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$
(c) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ (d) $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

29. If $A = \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix}$; $B = \begin{bmatrix} 3 & 4 \\ 1 & 6 \end{bmatrix}$ then which of the following statements is true -

- (a) $AB = BA$ (b) $A^2 = B$
(c) $(AB)^T = \begin{bmatrix} 5 & 9 \\ 16 & 12 \end{bmatrix}$ (d) none of these

30. If $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, then which one of the following holds for all $n \geq 1$, by the principle of mathematical induction?

- (a) $A^n = 2^{n-1} A - (n-1)I$ (b) $A^n = nA - (n-1)I$
(c) $A^n = 2^{n-1} A + (n-1)I$ (d) $A^n = nA + (n-1)I$

Type of matrices

31. In the following, singular matrix is -

- (a) $\begin{bmatrix} 2 & 3 \\ 1 & 3 \end{bmatrix}$ (b) $\begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix}$
(c) $\begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}$ (d) $\begin{bmatrix} 2 & 6 \\ 4 & 12 \end{bmatrix}$

32. If $A = \begin{bmatrix} 1 & -3 & 2 \\ 2 & k & 5 \\ 4 & 2 & 1 \end{bmatrix}$ is a singular matrix, then k is equal to

- (a) -1 (b) 8
(c) 4 (d) -8

33. If $A = [a_{ij}]$ is a skew-symmetric matrix of order n , then $a_{ii} =$

- (a) 0 for some i (b) 0 for all $i = 1, 2, \dots, n$
(c) 1 for some i (d) 1 for all $i = 1, 2, \dots, n$



34. Matrix $\begin{bmatrix} 0 & 5 & -7 \\ -5 & 0 & 11 \\ 7 & -11 & 0 \end{bmatrix}$ is a -

- (a) diagonal matrix (b) upper triangular matrix
(c) skew-symmetric matrix (d) symmetric matrix

35. If A and B are square matrices of same order, then which of the following is skew-symmetric -

- (a) $\frac{A + A^T}{2}$ (b) $\frac{A^T + B^T}{2}$
(c) $\frac{A^T - B^T}{2}$ (d) $\frac{B - B^T}{2}$

36. If A is symmetric as well as skew symmetric matrix, then -

- (a) A is a diagonal matrix (b) A is a null matrix
(c) A is a unit matrix (d) A is a triangular matrix

37. If $A = \begin{bmatrix} -1 & 7 \\ 2 & 3 \end{bmatrix}$, then skew-symmetric part of A is -

- (a) $\begin{bmatrix} -1 & 9/2 \\ -9/2 & 3 \end{bmatrix}$ (b) $\begin{bmatrix} -0 & -5/2 \\ 5/2 & 0 \end{bmatrix}$
(c) $\begin{bmatrix} -1 & -9/2 \\ 9/2 & 3 \end{bmatrix}$ (d) $\begin{bmatrix} 0 & 5/2 \\ -5/2 & 0 \end{bmatrix}$

Adjoint of matrix and its properties

38. If $A = \begin{bmatrix} 1 & 3 & 5 \\ 3 & 5 & 1 \\ 5 & 1 & 3 \end{bmatrix}$, then adj. A is equal to -

- (a) $\begin{bmatrix} 14 & -4 & -22 \\ -4 & -22 & 14 \\ -22 & 14 & -4 \end{bmatrix}$ (b) $\begin{bmatrix} -14 & 4 & 22 \\ 4 & 22 & -14 \\ 22 & -14 & 4 \end{bmatrix}$
(c) $\begin{bmatrix} 14 & 4 & -22 \\ 4 & -22 & -14 \\ -22 & -14 & -4 \end{bmatrix}$ (d) none of these

39. If $A = \begin{bmatrix} 1 & 2 & 3 \\ 5 & 0 & 4 \\ 2 & 6 & 7 \end{bmatrix}$ then adj A is equal to -

- (a) $\begin{bmatrix} -24 & 4 & 8 \\ 4 & 1 & 2 \\ 8 & 11 & -11 \end{bmatrix}$ (b) $\begin{bmatrix} -24 & 4 & 8 \\ 4 & 1 & 11 \\ 30 & -2 & -10 \end{bmatrix}$

- (c) $\begin{bmatrix} -24 & 4 & 8 \\ -27 & 1 & 11 \\ 30 & -2 & -10 \end{bmatrix}$ (d) none of these

40. If $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 3 & 1 \\ 2 & 1 & 2 \end{bmatrix}$, then A (adj A) equals -

- (a) $\begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix}$ (b) $-\begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix}$
(c) $\begin{bmatrix} 0 & 0 & 9 \\ 0 & 9 & 0 \\ 9 & 0 & 0 \end{bmatrix}$ (d) none of these

41. If A is an invertible matrix of order n, then the determinant of Adj. A =

- (a) $|A|^n$ (b) $|A|^{n+1}$
(c) $|A|^{n-1}$ (d) $|A|^{n+2}$

Inverse of a matrix and Its properties

42. If A and B are invertible matrices of the order n, then $(AB)^{-1}$ is equal to

- (a) AB^{-1} (b) $A^{-1}B$
(c) $B^{-1}A^{-1}$ (d) $A^{-1}B^{-1}$

43. If $A^2 - A + I = 0$, then the inverse of A is

- (a) A (b) A + I
(c) I - A (d) A - I



44. If $A = \begin{bmatrix} 2 & 3 \\ 5 & -2 \end{bmatrix}$, then $19A^{-1}$ is equal to

- (a) A' (b) $2A$
(c) $\frac{1}{2}A$ (d) A

45. $\begin{bmatrix} -6 & 5 \\ -7 & 6 \end{bmatrix}^{-1} =$

- (a) $\begin{bmatrix} -6 & 5 \\ -7 & 6 \end{bmatrix}$ (b) $\begin{bmatrix} 6 & -5 \\ -7 & 6 \end{bmatrix}$
(c) $\begin{bmatrix} 6 & 5 \\ 7 & 6 \end{bmatrix}$ (d) $\begin{bmatrix} 6 & -5 \\ 7 & -6 \end{bmatrix}$

46. Inverse matrix of $\begin{bmatrix} 2 & -3 \\ -4 & 2 \end{bmatrix}$ is -

- (a) $-\frac{1}{8}\begin{bmatrix} 2 & 3 \\ 4 & 2 \end{bmatrix}$ (b) $-\frac{1}{8}\begin{bmatrix} 2 & 4 \\ 3 & 2 \end{bmatrix}$
(c) $\frac{1}{8}\begin{bmatrix} 2 & 3 \\ 4 & 2 \end{bmatrix}$ (d) $\begin{bmatrix} 2 & 3 \\ 4 & 2 \end{bmatrix}$

47. If $A = \begin{bmatrix} 0 & -1 & 2 \\ 2 & -2 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}$ and $M = AB$, then M^{-1} is

equal to -

- (a) $\begin{bmatrix} 2 & -2 \\ 2 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 1/3 & 1/3 \\ -1/3 & 1/6 \end{bmatrix}$
(c) $\begin{bmatrix} 1/3 & -1/3 \\ 1/3 & 1/6 \end{bmatrix}$ (d) $\begin{bmatrix} 1/3 & -1/3 \\ -1/3 & 1/6 \end{bmatrix}$

48. If $A = \begin{bmatrix} 1 & 2 \\ 3 & -5 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$ and X is a matrix such that $A = BX$, then X equals -

- (a) $\frac{1}{2}\begin{bmatrix} -2 & 4 \\ 3 & 5 \end{bmatrix}$ (b) $\frac{1}{2}\begin{bmatrix} 2 & 4 \\ 3 & -5 \end{bmatrix}$
(c) $\begin{bmatrix} 2 & 4 \\ 3 & -5 \end{bmatrix}$ (d) none of these

49. Matrix $\begin{bmatrix} \lambda & -1 & 4 \\ -3 & 0 & 1 \\ -1 & 1 & 2 \end{bmatrix}$ is not invertible if -

- (a) $\lambda = -15$ (b) $\lambda = -17$
(c) $\lambda = -16$ (d) $\lambda = -18$

50. If $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ then A^{-n} is equal to -

- (a) $\begin{bmatrix} 1 & 0 \\ n & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 0 \\ -n & -1 \end{bmatrix}$
(c) $\begin{bmatrix} 1 & 0 \\ -n & 1 \end{bmatrix}$ (d) none of these

51. $\begin{bmatrix} 1 & -\tan \theta/2 \\ \tan \theta/2 & 1 \end{bmatrix} \begin{bmatrix} 1 & \tan \theta/2 \\ -\tan \theta/2 & 1 \end{bmatrix}^{-1}$ is equal to

- (a) $\begin{bmatrix} \sin \theta & -\cos \theta \\ \cos \theta & \sin \theta \end{bmatrix}$ (b) $\begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$
(c) $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ (d) none of these

Consistency of simultaneous Equations

52. The system of linear equations $x + y + z = 2$, $2x + y - z = 3$, $3x + 2y + kz = 4$ has a unique solution if

- (a) $k \neq 0$ (b) $-1 < k < 1$
(c) $-2 < k < 2$ (d) $k = 0$



53. If the system of equations $x + y + z = 6$, $x + 2y + 3z = 10$ and $x + 2y + \lambda z = \mu$ has no solution, then the values of λ and μ are
- (a) $\lambda = 3, \mu = 10$ (b) $\lambda = 3, \mu \neq 10$
 (c) $\lambda \neq 3, \mu = 10$ (d) $\lambda \neq 3, \mu \neq 10$
54. Consider the system of equations $a_1x + b_1y + c_1z = 0$, $a_2x + b_2y + c_2z = 0$, $a_3x + b_3y + c_3z = 0$ if
- $$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$$
- , then the system has
- (a) more than two solutions
 (b) only non trivial solutions
 (c) no solution
 (d) only trivial solution $(0, 0, 0)$.
55. Consider the system of linear equations
- $$x_1 + 2x_2 + x_3 = 3$$
- $$2x_1 + 3x_2 + x_3 = 3$$
- $$3x_1 + 5x_2 + 2x_3 = 1$$
- The system has
- (a) Infinite number of solutions
 (b) Exactly 3 solutions
 (c) A unique solution
 (d) No solution
56. If the trivial solution is the only solution of the system of equations
- $$x - ky + z = 0$$
- $$kx + 3y - kz = 0$$
- $$3x + y - z = 0$$
- Then, the set of all values of k is
- (a) $\{2, -3\}$ (b) $R - \{2, -3\}$
 (c) $R - \{2\}$ (d) $R - \{-3\}$
57. The number of values of k , for which the system of equations
- $$(k + 1)x + 8y = 4k$$
- $$kx + (k + 3)y = 3k - 1$$
- has no solution, is
- (a) infinite (b) 1
 (c) 2 (d) 3
58. If a, b, c are non-zero real numbers and if the system of equations
- $$(a - 1)x = y + z,$$
- $$(b - 1)y = z + x,$$
- $$(c - 1)z = x + y,$$
- has a non-trivial solution, then $ab + bc + ca$ equals:
- (a) $a + b + c$ (b) abc
 (c) 1 (d) -1
59. With the help of matrices, the solution of the equations
- $$3x + y + 2z = 3,$$
- $$2x - 3y - z = -3,$$
- $$x + 2y + z = 4$$
- is given by
- (a) $x = 1, y = 2, z = -1$ (b) $x = -1, y = 2, z = 1$
 (c) $x = 1, y = -2, z = -1$ (d) $x = -1, y = -2, z = 1$
60. Solution of
- $$x + 3y - 2z = 0$$
- $$2x - y + 4z = 0$$
- $$x - 11y + 14z = 0$$
- is
- (a) $\frac{x}{8} = \frac{y}{-10} = \frac{z}{7} = \lambda$ (b) $\frac{x}{-10} = \frac{y}{8} = \frac{z}{7} = \lambda$
 (c) $\frac{x}{7} = \frac{y}{8} = \frac{z}{-10} = \lambda$ (d) None of these

Numerical Value Type Questions

61. Let $f(\theta) = \begin{vmatrix} \cos^2 \theta & \cos \theta \sin \theta & -\sin \theta \\ \cos \theta \sin \theta & \sin^2 \theta & \cos \theta \\ \sin \theta & -\cos \theta & 0 \end{vmatrix}$ then $f\left(\frac{\pi}{6}\right) =$



62. If $f(x) = \tan x$ and A, B, C are the angles of

$$\Delta \begin{vmatrix} f(A) & f(\pi/4) & f(\pi/4) \\ f(\pi/4) & f(B) & f(\pi/4) \\ f(\pi/4) & f(\pi/4) & f(C) \end{vmatrix}$$

then is equal to

63. $\begin{vmatrix} 109 & 102 & 95 \\ 6 & 13 & 20 \\ 1 & -6 & -13 \end{vmatrix}$ is equal to

64. If $\omega (\neq 1)$ is a cube root of unity, then

$$\begin{vmatrix} 1 & 1+i+\omega^2 & \omega^2 \\ 1-i & -1 & \omega^2-1 \\ -i & -i+\omega-1 & -1 \end{vmatrix} =$$

65. If $\begin{vmatrix} y+z & x & x \\ y & z+x & y \\ z & z & x+y \end{vmatrix} = k(xyz)$, then k is equal to

66. If $ax^3 + bx^2 + cx + d = \begin{vmatrix} 3x & x+1 & x-1 \\ x-3 & -2x & x+2 \\ x+3 & x-4 & 5x \end{vmatrix}$

be an identity in x, where a, b, c are constants, then the value of -d is

67. If l, m, n are the p^{th} , q^{th} and r^{th} term of a G.P. all positive, then

$$\begin{vmatrix} \log \ell & p & 1 \\ \log m & q & 1 \\ \log n & r & 1 \end{vmatrix} \text{ equals}$$

68. If $A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$ and $A^2 - 4A - nI = 0$, then -n is equal to -

69. If $\begin{bmatrix} 3 & 0 & 3 \\ 2 & 1 & 0 \\ 4 & 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 8 \\ 1 \\ 4 \end{bmatrix} + \begin{bmatrix} 2y \\ z \\ 3y \end{bmatrix}$, then find the value of

$$x + \frac{y}{2} + \frac{z}{3}$$

70. If p and q are real so that the system of equations $px + 4y + z = 0$, $2y + 3z = 1$ and $3x - qz = -2$ has infinite

solutions then $\sqrt{q^2 - p^2}$ is equal to -

71. The system of equations $\lambda x + y + z = 1$, $x + \lambda y + z = \lambda$ and $x + y + \lambda z = \lambda^2$ have no solution. Then the value of λ^4 is

72. The system of equations $kx + y + z = 1$, $x + ky + z = k$ and $x + y + kz = k^2$ have no solution if -k equals

73. Let a, b, c be any real numbers. Suppose that there are real numbers x, y, z not all zero such that $x = cy + bz$, $y = az + cx$ and $z = bx + ay$. Then $a^2 + b^2 + c^2 + 2abc$ is equal to

74. The number of values of k for which the linear equations $4x + ky + 2z = 0$, $kx + 4y + z = 0$ and $2x + 2y + z = 0$ possess a non-zero solution is

75. If $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ and $\det(A^n - I) = 1 - \lambda^n$, $n \in \mathbb{N}$, then λ is equal to



EXERCISE - 2 : PREVIOUS YEAR JEE MAIN QUESTIONS

- The system of linear equations

$$\begin{aligned} x + \lambda y - z &= 0 \\ \lambda x - y - z &= 0 \\ x + y - \lambda z &= 0 \end{aligned}$$

has a non-trivial solution for :

 - exactly one value of λ .
 - exactly two values of λ .
 - exactly three values of λ .
 - infinitely many values of λ .

(2016)
- If $A = \begin{bmatrix} 5a & -b \\ 3 & 2 \end{bmatrix}$ and $A \text{ adj } A = AA^T$, then $5a + b$ is equal to:

 - 5
 - 4
 - 13
 - 1

(2016)
- If $P = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$, $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $Q = PAP^T$, then $P^T Q^{2015} P$ is

 - $\begin{bmatrix} 0 & 2015 \\ 0 & 0 \end{bmatrix}$
 - $\begin{bmatrix} 2015 & 1 \\ 0 & 2015 \end{bmatrix}$
 - $\begin{bmatrix} 2015 & 0 \\ 1 & 2015 \end{bmatrix}$
 - $\begin{bmatrix} 1 & 2015 \\ 0 & 1 \end{bmatrix}$

(2016/Online Set-1)
- The number of distinct real roots of the equation,

$$\begin{vmatrix} \cos x & -\sin x & \sin x \\ \sin x & \cos x & \sin x \\ \sin x & \sin x & \cos x \end{vmatrix} = 0$$

in the intervals $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$ is:

 - 4
 - 3
 - 2
 - 1

(2016/Online Set-1)
- Let A be a 3×3 matrix such that $A^2 - 5A + 7I = O$.

Statement – I : $A^{-1} = \frac{1}{7} (5I - A)$.

Statement – II : The Polynomial $A^3 - 2A^2 - 3A + I$ can be reduced to $5(A - 4I)$. Then :

 - Statement-I is true, but Statement-II is false.
 - Statement-I is false, but Statement-II is true.
 - Both the statements are true.
 - Both the statements are false.
- If $A = \begin{bmatrix} -4 & -1 \\ 3 & 1 \end{bmatrix}$, then the determinant of the matrix $(A^{2016} - 2A^{2015} - A^{2014})$ is :

 - 2014
 - 175
 - 2016
 - 25

(2016/Online Set-2)
- If S is the set of distinct values of 'b' for which the following system of linear equations

$$\begin{aligned} x + y + z &= 1 \\ x + ay + z &= 1 \\ ax + by + z &= 0 \end{aligned}$$

has no solution, then S is :

 - an empty set
 - an infinite set
 - a finite set containing two or more elements
 - a singleton

(2017)
- Let ω be a complex number such that $2\omega + 1 = z$ where $z = \sqrt{-3}$. If

$$\begin{vmatrix} 1 & 1 & 1 \\ 1 & -\omega^2 - 1 & \omega^2 \\ 1 & \omega^2 & \omega^7 \end{vmatrix} = 3k$$

then k is equal to:

 - z
 - z
 - 1
 - 1



9. Let A be any 3×3 invertible matrix. Then which one of the following is **not** always true ? (2017)

- (a) $\text{adj } A = |A| \cdot A^{-1}$
 (b) $\text{adj } (\text{adj } A) = |A| \cdot A$
 (c) $\text{adj } (\text{adj } A) = |A|^2 \cdot (\text{adj } A)^{-1}$
 (d) $\text{adj } (\text{adj } (A)) = |A| \cdot (\text{adj } (A))^{-1}$

10. If $S = \left\{ x \in [0, 2\pi] : \begin{vmatrix} 0 & \cos x & -\sin x \\ \sin x & 0 & \cos x \\ \cos x & \sin x & 0 \end{vmatrix} = 0 \right\}$, then

$\sum_{x \in S} \tan\left(\frac{\pi}{3} + x\right)$ is equal to : (2017)

- (a) $4 + 2\sqrt{3}$ (b) $-2 + \sqrt{3}$
 (c) $-2 - \sqrt{3}$ (d) $-4 - 2\sqrt{3}$

11. If $A = \begin{bmatrix} 2 & -3 \\ -4 & 1 \end{bmatrix}$, then $\text{adj } (3A^2 + 12A)$ is equal to:

(2017)

- (a) $\begin{bmatrix} 72 & -84 \\ -63 & 51 \end{bmatrix}$ (b) $\begin{bmatrix} 51 & 63 \\ 84 & 72 \end{bmatrix}$
 (c) $\begin{bmatrix} 51 & 84 \\ 63 & 72 \end{bmatrix}$ (d) $\begin{bmatrix} 72 & -63 \\ -84 & 51 \end{bmatrix}$

12. For two 3×3 matrices A and B , let $A + B = 2B'$ and $3A + 2B = I_3$, where B' is the transpose of B and I_3 is 3×3 identity matrix. Then : (2017)

- (a) $5A + 10B = 2I_3$ (b) $10A + 5B = 3I_3$
 (c) $B + 2A = I_3$ (d) $3A + 6B = 2I_3$

13. The number of real values of λ for which the system of linear equations

$$2x + 4y - \lambda z = 0$$

$$4x + \lambda y + 2z = 0$$

$$\lambda x + 2y + 2z = 0$$

has infinitely many solutions, is (2017/Online Set-1)

- (a) 0 (b) 1
 (c) 2 (d) 3

14. If the system of linear equations

$$x + ky + 3z = 0$$

$$3x + ky - 2z = 0$$

$$2x + 4y - 3z = 0$$

has a non-zero solution (x, y, z) , then $\frac{xz}{y^2}$ is equal to :

(2018)

- (a) 30 (b) -10
 (c) 10 (d) -30

15. If $\begin{vmatrix} x-4 & 2x & 2x \\ 2x & x-4 & 2x \\ 2x & 2x & x-4 \end{vmatrix} = (A+Bx)(x-A)^2$, then the

ordered pair (A, B) is equal to :

(2018)

- (a) $(4, 5)$ (b) $(-4, -5)$
 (c) $(-4, 3)$ (d) $(-4, 5)$

16. Let A be matrix such that $A \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$ is a scalar matrix and $|3A| = 108$. Then A^2 equals : (2018/Online Set-1)

- (a) $\begin{bmatrix} 4 & -32 \\ 0 & 36 \end{bmatrix}$ (b) $\begin{bmatrix} 36 & 0 \\ -32 & 4 \end{bmatrix}$
 (c) $\begin{bmatrix} 4 & 0 \\ -32 & 36 \end{bmatrix}$ (d) $\begin{bmatrix} 36 & -32 \\ 0 & 4 \end{bmatrix}$

17. If $f(x) = \begin{vmatrix} \cos x & x & 1 \\ 2 \sin x & x^2 & 2x \\ \tan x & x & 1 \end{vmatrix}$, then $\lim_{x \rightarrow 0} \frac{f'(x)}{x}$

(2018/Online Set-1)

- (a) does not exist
 (b) exists and is equal to 2
 (c) exists and is equal to 0
 (d) exists and is equal to -2



18. Let S be the set of all real values of k for which the system of linear equations

$$x + y + z = 2$$

$$2x + y - z = 3$$

$$3x + 2y + kz = 4$$

has a unique solution. Then S is :

(2018/Online Set-1)

- (a) an empty set (b) equal to $\{0\}$
(c) equal to R (d) equal to $R - \{0\}$
19. Suppose A is any 3×3 non-singular matrix and $(A - 3I)(A - 5I) = O$, where $I = I_3$ and $O = O_3$.

If $\alpha A + \beta A^{-1} = 4I$, then $\alpha + \beta$ is equal to :

(2018/Online Set-2)

- (a) 8 (b) 7
(c) 13 (d) 12

20. If the system of linear equations

$$x + ay + z = 3$$

$$x + 2y + 2z = 6$$

$$x + 5y + 3z = b$$

has no solution, then :

(2018/Online Set-2)

- (a) $a = -1, b = 9$ (b) $a = -1, b \neq 9$
(c) $a \neq -1, b = 9$ (d) $a = 1, b \neq 9$

21. Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$ and $B = A^{20}$. Then the sum of the

elements of the first column of B is :

(2018/Online Set-3)

- (a) 210 (b) 211
(c) 231 (d) 251

22. The number of values of k for which the system of linear equations,

$$(k+2)x + 10y = k$$

$$kx + (k+3)y = k-1$$

has no solution, is :

(2018/Online Set-3)

- (a) 1 (b) 2
(c) 3 (d) infinitely many

23. The greatest value of $c \in R$ for which the system of linear equations

$$x - cy - cz = 0$$

$$cx - y + cz = 0$$

$$cx + cy - z = 0$$

has a non-trivial solution, is :

(2019-04-08/Shift-1)

- (a) -1 (b) $\frac{1}{2}$
(c) 2 (d) 0

24. Let the numbers 2, b, c be in an A.P. and

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & b & c \\ 4 & b^2 & c^2 \end{bmatrix}. \text{ If } \det(A) \in [2, 16], \text{ then } c \text{ lies in the}$$

interval :

(2019-04-08/Shift-2)

- (a) $[2, 3)$ (b) $(2 + 2^{3/4}, 4)$
(c) $[4, 6]$ (d) $[3, 2 + 2^{3/4}]$

25. If the system of linear equations

$$x - 2y + kz = 1$$

$$2x + y + z = 2$$

$$3x - y - kz = 3$$

has a solution (x, y, z) , $z \neq 0$, then (x, y) lies on the straight line whose equation is :

(2019-04-08/Shift-2)

- (a) $3x - 4y - 1 = 0$ (b) $4x - 3y - 4 = 0$
(c) $4x - 3y - 1 = 0$ (d) $3x - 4y - 4 = 0$

26. If $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \cdots \begin{bmatrix} 1 & n-1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 78 \\ 0 & 1 \end{bmatrix}$ then

the inverse of $\begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix}$ is

(2019-04-09/Shift-1)

- (a) $\begin{bmatrix} 1 & 0 \\ 12 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & -13 \\ 0 & 1 \end{bmatrix}$
(c) $\begin{bmatrix} 1 & -12 \\ 0 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 1 & 0 \\ 13 & 1 \end{bmatrix}$



27. The total number of matrices $A = \begin{pmatrix} 0 & 2y & 1 \\ 2x & y & -1 \\ 2x & -y & 1 \end{pmatrix}$

$(x, y \in \mathbb{R}, x \neq y)$ for which $A^T A = 3I_3$ is:

(2019-04-09/Shift-2)

- (a) 2 (b) 3
(c) 6 (d) 4

28. If the system of equations $2x + 3y - z = 0$, $x + ky - 2z = 0$ and $2x - y + z = 0$ has a non-trivial solution (x, y, z) , then

$\frac{x}{y} + \frac{y}{z} + \frac{z}{x} + k$ is equal to (2019-04-09/Shift-2)

- (a) $\frac{3}{4}$ (b) $\frac{1}{2}$
(c) $-\frac{1}{4}$ (d) -4

29. If $\Delta_1 = \begin{vmatrix} x & \sin \theta & \cos \theta \\ -\sin \theta & -x & 1 \\ \cos \theta & 1 & x \end{vmatrix}$ and

$$\Delta_2 = \begin{vmatrix} x & \sin 2\theta & \cos 2\theta \\ -\sin 2\theta & -x & 1 \\ \cos 2\theta & 1 & x \end{vmatrix}, x \neq 0$$

then for all $\Delta_1 = \theta \in \left(0, \frac{\pi}{2}\right)$: (2019-04-10/Shift-1)

- (a) $\Delta_1 - \Delta_2 = -2x^3$
(b) $\Delta_1 - \Delta_2 = x(\cos 2\theta - \cos 4\theta)$
(c) $\Delta_1 \times \Delta_2 = -2(x^3 + x - 1)$
(d) $\Delta_1 + \Delta_2 = -2x^3$

30. Let λ be a real number for which the system of linear equations:

$$x + y + z = 6$$

$$4x + \lambda y - \lambda z = \lambda - 2$$

$$3x + 2y - 4z = -5$$

has infinitely many solutions. Then λ is a root of the quadratic equation: (2019-04-10/Shift-2)

- (a) $\lambda^2 + 3\lambda - 4 = 0$ (b) $\lambda^2 - 3\lambda - 4 = 0$
(c) $\lambda^2 + \lambda - 6 = 0$ (d) $\lambda^2 - \lambda - 6 = 0$

31. The sum of the real roots of the equation

$$\begin{vmatrix} x & -6 & -1 \\ 2 & -3x & x-3 \\ -3 & 2x & x+2 \end{vmatrix} = 0, \text{ is equal to: (2019-04-10/Shift-2)}$$

- (a) 6 (b) 0
(c) 1 (d) -4

32. If A is a symmetric matrix and B is a skew-symmetric

matrix such that $A+B = \begin{bmatrix} 2 & 3 \\ 5 & -1 \end{bmatrix}$, then AB is equal to :

(2019-04-12/Shift-1)

- (a) $\begin{bmatrix} -4 & -1 \\ -1 & 4 \end{bmatrix}$ (b) $\begin{bmatrix} 4 & -2 \\ -1 & -4 \end{bmatrix}$
(c) $\begin{bmatrix} 4 & -2 \\ 1 & -4 \end{bmatrix}$ (d) $\begin{bmatrix} -4 & 2 \\ 1 & 4 \end{bmatrix}$

33. If $B = \begin{bmatrix} 5 & 2\alpha & 1 \\ 0 & 2 & 1 \\ \alpha & 3 & -1 \end{bmatrix}$ is the inverse of a 3×3 matrix A ,

then the sum of all values of α for which $\det(A) + 1 = 0$, is (2019-04-12/Shift-1)

- (a) 0 (b) -1
(c) 1 (d) 2



34. A value of $\theta \in \left(0, \frac{\pi}{3}\right)$, for which

$$\begin{vmatrix} 1 + \cos^2 \theta & \sin^2 \theta & 4 \cos 6\theta \\ \cos^2 \theta & 1 + \sin^2 \theta & 4 \cos 6\theta \\ \cos^2 \theta & \sin^2 \theta & 1 + 4 \cos 6\theta \end{vmatrix} = 0 \text{ is } \underline{\hspace{2cm}}.$$

(2019-04-12/Shift-2)

- (a) $\frac{\pi}{9}$ (b) $\frac{\pi}{18}$
(c) $\frac{7\pi}{24}$ (d) $\frac{7\pi}{36}$

35. The system of linear equations

$$x + y + z = 2$$

$$2x + 3y + 2z = 5$$

$$2x + 3y + (a^2 - 1)z = a + 1$$

(2019-01-09/Shift-1)

- (a) is inconsistent when $a = 4$
(b) has a unique solution for $|a| = \sqrt{3}$
(c) has infinitely many solutions for $a = 4$
(d) is inconsistent when $|a| = \sqrt{3}$

36. If $A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$, then the matrix A^{-50} when $\theta = \frac{\pi}{12}$

is equal to

(2019-01-09/Shift-1)

(a) $\begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}$ (b) $\begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$

(c) $\begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$ (d) $\begin{bmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}$

37. If the system of linear equations

$$x - 4y + 7z = g$$

$$3y - 5z = h$$

$$-2x + 5y - 9z = k$$

is consistent, then :

(2019-01-09/Shift-2)

- (a) $g + 2h + k = 0$ (b) $g + h + 2k = 0$
(c) $2g + h + k = 0$ (d) $g + h + k = 0$

38. If $A = \begin{bmatrix} e^t & e^{-t} \cos t & e^{-t} \sin t \\ e^t & -e^{-t} \cos t - e^{-t} \sin t & -e^{-t} \sin t + e^{-t} \cos t \\ e^t & 2e^{-t} \sin t & -2e^{-t} \cos t \end{bmatrix}$

then A is:

(2019-01-09/Shift-2)

- (a) invertible for all $t \in \mathbb{R}$
(b) invertible only if $t = \pi$
(c) not invertible for any $t \in \mathbb{R}$
(d) invertible only if $t = \frac{\pi}{2}$

39. If the system of equations

$$x + y + z = 5$$

$$x + 2y + 3z = 9$$

$$x + 3y + \alpha z = \beta$$

has infinitely many solutions, then $\beta - \alpha$ equals:

(2019-01-10/Shift-1)

40. The number of values of $\theta \in (0, \pi)$ for which the system of linear equations

$$x + 3y + 7z = 0$$

$$-x + 4y + 7z = 0$$

$(\sin 3\theta)x + (\cos 2\theta)y + 2z = 0$ has a non-trivial solution, is

(2019-01-10/Shift-2)

- (a) three (b) two
(c) four (d) one



41. Let $A = \begin{bmatrix} 2 & b & 1 \\ b & b^2+1 & b \\ 1 & b & 2 \end{bmatrix}$ where $b > 0$. Then the minimum

value of $\frac{\det(A)}{b}$ is: (2019-01-10/Shift-2)

- (a) $2\sqrt{3}$ (b) $-2\sqrt{3}$
(c) $-\sqrt{3}$ (d) $\sqrt{3}$

42. Let $a_1, a_2, a_3, \dots, a_{10}$ be in G.P. with $a_i > 0$ for $i = 1, 2, \dots, 10$ and S be the set of pairs (r, k) , $r, k \in \mathbb{N}$ (the set of natural numbers) for which

$$\begin{vmatrix} \log_e a_1^r a_2^k & \log_e a_2^r a_3^k & \log_e a_3^r a_4^k \\ \log_e a_4^r a_5^k & \log_e a_5^r a_6^k & \log_e a_6^r a_7^k \\ \log_e a_7^r a_8^k & \log_e a_8^r a_9^k & \log_e a_9^r a_{10}^k \end{vmatrix} = 0$$

Then the number of elements in S , is :

(2019-01-10/Shift-2)

- (a) 4 (b) infinitely many
(c) 2 (d) 10

43. If the system of linear equations

$$2x + 2y + 3z = a$$

$$3x - y + 5z = b$$

$$x - 3y + 2z = c$$

where, a, b, c are non-zero real numbers, has more than one solution, then :

(2019-01-11/Shift-1)

- (a) $b - c + a = 0$ (b) $b - c - a = 0$
(c) $a + b + c = 0$ (d) $b + c - a = 0$

44. If $\begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix}$

$$= (a+b+c)(x+a+b+c)^2, x \neq 0 \text{ and } a+b+c \neq 0,$$

then x is equal to : (2019-01-11/Shift-2)

- (a) abc (b) $-(a+b+c)$

- (c) $2(a+b+c)$ (d) $-2(a+b+c)$

45. Let A and B be two invertible matrices of order 3×3 . If $\det(ABA^T) = 8$ and $\det(AB^{-1}) = 8$ then $\det(BA^{-1}B^T)$ is equal to : (2019-01-11/Shift-2)

- (a) $\frac{1}{4}$ (b) 1

- (c) $\frac{1}{16}$ (d) 16

46. An ordered pair (α, β) for which the system of linear equations

$$(1+\alpha)x + \beta y + z = 2,$$

$$\alpha x + (1+\beta)y + z = 3,$$

$$\alpha x + \beta y + 2z = 2$$

has a unique solution, is: (2019-01-12/Shift-1)

- (a) $(2, 4)$ (b) $(-3, 1)$
(c) $(-4, 2)$ (d) $(1, -3)$

47. If $A = \begin{bmatrix} 1 & \sin \theta & 1 \\ -\sin \theta & 1 & \sin \theta \\ -1 & -\sin \theta & 1 \end{bmatrix}$; then for all

$\theta \in \left(\frac{3\pi}{4}, \frac{5\pi}{4}\right)$, $\det(A)$ lies in the interval :

(2019-01-12/Shift-2)

- (a) $\left(1, \frac{5}{2}\right]$ (b) $\left[\frac{5}{2}, 4\right)$

- (c) $\left(0, \frac{3}{2}\right]$ (d) $\left[\frac{3}{2}, 3\right)$



48. The set of all values of λ for which the system of linear equations

$$x - 2y - 2z = \lambda x$$

$$x + 2y + z = \lambda y$$

$$-x - y = \lambda z$$

has a non-trivial solution : (2019-01-12/Shift-2)

- (a) is a singleton
(b) contains exactly two elements
(c) is an empty set
(d) contains more than two elements

49. Let A be a 2×2 real matrix with entries from $\{0, 1\}$ and $|A| \neq 0$. Consider the following two statements :

(P) If $A \neq I_2$, then $|A| = -1$

(Q) If $|A| = 1$, then $\text{tr}(A) = 2$,

Where I_2 denotes 2×2 identity matrix and $\text{tr}(A)$ denotes the sum of the diagonal entries of A . Then :

(2020-09-02/Shift-1)

- (a) Both (P) and (Q) are false
(b) (P) is true and (Q) is false
(c) Both (P) and (Q) are true
(d) (P) is false and (Q) is true

50. Let S be the set of all $\lambda \in R$ for which the system of linear equations

$$2x - y + 2z = 2$$

$$x - 2y + \lambda z = -4$$

$$x + \lambda y + z = 4$$

has no solution. Then the set S (2020-09-02/Shift-1)

- (a) is an empty set.
(b) is a singleton.
(c) contains more than two elements.
(d) contains exactly two elements.

51. Let $A = \{X = (x, y, z)^T : PX = 0 \text{ and } x^2 + y^2 + z^2 = 1\}$, where

$$P = \begin{bmatrix} 1 & 2 & 1 \\ -2 & 3 & -4 \\ 1 & 9 & -1 \end{bmatrix}, \text{ then the set } A : (2020-09-02/Shift-2)$$

- (a) contains more than two elements
(b) is a singleton.
(c) contains exactly two elements
(d) is an empty set.

52. Let $a, b, c \in R$ be all non-zero and satisfy

$$a^3 + b^3 + c^3 = 2 \text{ If the matrix } A = \begin{pmatrix} a & b & c \\ b & c & a \\ c & a & b \end{pmatrix} \text{ satisfies}$$

$A^T A = I$, then a value of abc can be :

(2020-09-02/Shift-2)

- (a) $\frac{2}{3}$ (b) 3
(c) $a - b$ (d) $\frac{1}{3}$

53. If $\Delta = \begin{vmatrix} x-2 & 2x-3 & 3x-4 \\ 2x-3 & 3x-4 & 4x-5 \\ 3x-5 & 5x-8 & 10x-17 \end{vmatrix} = Ax^3 + Bx^2 + Cx + D$,

then $B + C$ is equal to : (2020-09-03/Shift-1)

- (a) 1 (b) -1
(c) -3 (d) 9

54. Let $A = \begin{bmatrix} x & 1 \\ 1 & 0 \end{bmatrix}$, $x \in R$ and $A^4 = [a_{ij}]$. If $a_{11} = 109$, then

a_{22} is equal to (2020-09-03/Shift-1)



55. Let A be a 3×3 matrix such that $\text{adj } A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 0 & 2 \\ 1 & -2 & -1 \end{bmatrix}$

and $B = \text{adj}(\text{adj } A)$. If $|A| = \lambda$ and $|(B^{-1})^T| = \mu$ then the ordered pair, $(|\lambda|, \mu)$ is equal to : (2020-09-03/Shift-2)

(a) $\left(9, \frac{1}{81}\right)$ (b) $\left(9, \frac{1}{9}\right)$

(c) $\left(3, \frac{1}{81}\right)$ (d) $(3, 81)$

56. Let S be the set of all integer solutions, (x, y, z) , of the system of equations

$$x - 2y + 5z = 0$$

$$-2x + 4y + z = 0$$

$$-7x + 14y + 9z = 0$$

such that $15 \leq x^2 + y^2 + z^2 \leq 150$. Then, the number of elements in the set S is equal to

(2020-09-03/Shift-2)

57. If $A = \begin{bmatrix} \cos \theta & i \sin \theta \\ i \sin \theta & \cos \theta \end{bmatrix}$, $\left(\theta = \frac{\pi}{24}\right)$ and $A^5 = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$,

where $i = \sqrt{-1}$ then which one of the following is not true ? (2020-09-04/Shift-1)

(a) $a^2 - d^2 = 0$ (b) $a^2 - c^2 = 1$

(c) $0 \leq a^2 + b^2 \leq 1$ (d) $a^2 - b^2 = \frac{1}{2}$

58. If the system of equations

$$x - 2y + 3z = 9$$

$$2x + y + z = b$$

$$x - 7y + az = 24$$

has infinitely many solutions, then $a - b$ is equal to.....

(2020-09-04/Shift-1)

59. Suppose the vectors x_1, x_2 and x_3 are the solutions of the system of linear equations, $Ax = b$ when the vector b on the right side is equal to b_1, b_2 and b_3 respectively.

$$x_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, x_2 = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}, x_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix},$$

$$b_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, b_2 = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} \text{ and } b_3 = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix},$$

If then the determinant of A is equal to:

(2020-09-04/Shift-2)

(a) 2 (b) $\frac{1}{2}$

(c) $\frac{3}{2}$ (d) 4

60. If the system of equations

$$x + y + z = 2$$

$$2x + 4y - z = 6$$

$$3x + 2y + \lambda z = \mu$$

has infinitely many solutions, then

(2020-09-04/Shift-2)

(a) $\lambda - 2\mu = -5$ (b) $2\lambda + \mu = 14$

(c) $\lambda + 2\mu = 14$ (d) $2\lambda - \mu = 5$

61. If the minimum and the maximum values of the function

$$f: \left[\frac{\pi}{4}, \frac{\pi}{2}\right] \rightarrow \mathbb{R}, \text{ defined by}$$

$$f(\theta) = \begin{vmatrix} -\sin^2 \theta & -1 - \sin^2 \theta & 1 \\ -\cos^2 \theta & -1 - \cos^2 \theta & 1 \\ 12 & 10 & -2 \end{vmatrix}$$

are m and M respectively, then the ordered pair (m, M) is equal to:

(2020-09-05/Shift-1)

(a) $(0, 4)$ (b) $(-4, 0)$

(c) $(-4, 4)$ (d) $(0, 2\sqrt{2})$



62. Let $\lambda \in \mathbb{R}$. The system of linear equations
- $$2x_1 - 4x_2 + \lambda x_3 = 1$$
- $$x_1 - 6x_2 + x_3 = 2$$
- $$\lambda x_1 - 10x_2 + 4x_3 = 3$$
- is inconsistent for: (2020-09-05/Shift-1)

- (a) exactly two values of λ
- (b) exactly one negative value of λ
- (c) every value of λ
- (d) exactly one positive value of λ

63. $a + x = b + y = c + z + 1$, where a, b, c, x, y, z are non-zero

distinct real numbers, then $\begin{vmatrix} x & a+y & x+a \\ y & b+y & y+b \\ z & c+y & z+c \end{vmatrix}$ is equal to:

(2020-09-05/Shift-2)

- (a) $y(a-b)$ (b) 0
- (c) $y(b-a)$ (d) $y(a-c)$

64. If the system of linear equations
- $$x + y + 3z = 0$$
- $$x + 3y + k^2z = 0$$
- $$3x + y + 3z = 0$$
- has a non-zero solution (x, y, z) for some $k \in \mathbb{R}$, then

$x + \left(\frac{y}{z}\right)$ is equal to: (2020-09-05/Shift-2)

(a) -9 (b) 9

(c) -3 (d) 3

65. The values of λ and μ for which the system of linear equations
- $$x + y + z = 2$$
- $$x + 2y + 3z = 5$$
- $$x + 3y + \lambda z = \mu$$
- has infinitely many solutions are, respectively:

(2020-09-06/Shift-1)

- (a) 6 and 8 (b) 5 and 8
- (c) 5 and 7 (d) 4 and 9

66. Let m and M be respectively the minimum and maximum

values of $\begin{vmatrix} \cos^2 x & 1 + \sin^2 x & \sin 2x \\ 1 + \cos^2 x & \sin^2 x & \sin 2x \\ \cos^2 x & \sin^2 x & 1 + \sin 2x \end{vmatrix}$

Then the ordered pair (m, M) is equal to:

(2020-09-06/Shift-1)

- (a) $(-3, -1)$ (b) $(-4, -1)$
- (c) $(1, 3)$ (d) $(-3, 3)$

67. Let $\theta = \frac{\pi}{5}$ and $A = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$. If $B = A + A^4$, then $\det(B)$: (2020-09-06/Shift-2)

- (a) is one (b) lies in $(1, 2)$
- (c) lies in $(2, 3)$ (d) is zero

68. The sum of distinct values of λ for which the system of equations

$$(\lambda - 1)x + (3\lambda + 1)y + 2\lambda z = 0$$

$$(\lambda - 1)x + (4\lambda - 2)y + (\lambda + 3)z = 0$$

$2x + (3\lambda + 1)y + 3(\lambda - 1)z = 0$, has non-zero solutions, is _____ (2020-09-06/Shift-2)

69. Let α be the root of the equation $x^2 + x + 1 = 0$ and the

matrix $A = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1 & 1 \\ 1 & \alpha & \alpha^2 \\ 1 & \alpha^2 & \alpha^4 \end{bmatrix}$, then the matrix A^{31} is equal

to, (2020-01-07/Shift-1)

- (a) A (b) A^2
- (c) A^3 (d) I_3



70. If the system of linear equations
- $$2x + 2ay + az = 0$$
- $$2x + 3by + bz = 0$$
- $$2x + 4cy + c = 0$$
- Where $a, b, c \in \mathbb{R}$ are non-zero and distinct; has non-zero solution, then **(2020-01-07/Shift-1)**
- (a) $a + b + c = 0$ (b) a, b, c are A.P.
 (c) $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in A.P. (d) a, b, c are in G.P.
71. $A = [a_{ij}]$ and $B = [b_{ij}]$ be two 3×3 real matrices such that $b_{ij} = (3)^{(i+j-2)} a_{ji}$, where $i, j = 1, 2, 3$. If the determinant of B is 81, then the determinant of A is : **(2020-01-07/Shift-2)**
- (a) $\frac{1}{9}$ (b) $\frac{1}{81}$
 (c) $\frac{1}{3}$ (d) 3
72. If system of linear equations
- $$x + y + z = 6$$
- $$x + 2y + 3z = 10$$
- $$3x + 2y + \lambda z = \mu$$
- has more than two solutions, then $\mu - \lambda^2$ is equal to _____ **(2020-01-07/Shift-2)**
73. For which of the following ordered pairs (μ, δ) , the system of linear equations
- $$x + 2y + 3z = 1$$
- $$3x + 4y + 5z = \mu$$
- $$4x + 4y + 4z = \delta$$
- is inconsistent? **(2020-01-08/Shift-1)**
- (a) (4, 6) (b) (3, 4)
 (c) (1, 0) (d) (4, 3)
74. The number of all 3×3 matrices A, with entries from the set $\{-1, 0, 1\}$ such that the sum of the diagonal elements of (AA^T) is 3, is _____. **(2020-01-08/Shift-1)**
75. If $A = \begin{bmatrix} 2 & 2 \\ 9 & 4 \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, then $10A^{-1}$ is equal to: **(2020-01-08/Shift-2)**
- (a) $6I - A$ (b) $A - 6I$
 (c) $4I - A$ (d) $A - 4I$
76. The system of linear equations
- $$\lambda x + 2y + 2z = 5$$
- $$2\lambda x + 3y + 5z = 8$$
- $$4x + \lambda y + 6z = 10$$
- has: **(2020-01-08/Shift-2)**
- (a) no solution when $\lambda = 2$
 (b) infinitely many solutions when $\lambda = 2$
 (c) no solution when $\lambda = 8$
 (d) a unique solution when $\lambda = -8$
77. If for some α and β in $\mathbb{R}$, the intersection of the following three planes
- $$x + 4y - 2z = 1$$
- $$x + 7y - 5z = \beta$$
- $$x + 5y + \alpha z = 5$$
- is a line in $\mathbb{R}^3$, then $\alpha + \beta$ is equal to: **(2020-01-09/Shift-1)**
- (a) 0 (b) 10
 (c) -10 (d) 2
78. If the matrices $A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 3 & 4 \\ 1 & -1 & 3 \end{bmatrix}$, $B = \text{adj } A$ and $C = 3A$, then $\frac{|\text{adj } B|}{|C|}$ is equal to: **(2020-01-09/Shift-1)**
- (a) 16 (b) 2
 (c) 8 (d) 72



79. Let $a - 2b + c = 1$. If $f(x) = \begin{vmatrix} x+a & x+2 & x+1 \\ x+b & x+3 & x+2 \\ x+c & x+4 & x+3 \end{vmatrix}$, then:

(2020-01-09/Shift-2)

- (a) $f(-50) = 501$ (b) $f(-50) = 10$
(c) $f(50) = 1$ (d) $f(50) = -501$

80. The following system of linear equations

$$7x + 6y - 2z = 0,$$

$$3x + 4y + 2z = 0,$$

$$x - 2y - 6z = 0,$$

(2020-01-09/Shift-2)

- (a) infinitely many solutions, (x, y, z) satisfying $y = 2z$
(b) infinitely many solutions (x, y, z) satisfying $x = 2z$
(c) no solution
(d) only the trivial solution

81. Let $A = [a_{ij}]$ be a 3×3 matrix,

$$\text{where } a_{ij} = \begin{cases} 1, & \text{if } i = j \\ -x, & \text{if } |i - j| = 1 \\ 2x + 1, & \text{otherwise} \end{cases}$$

Let a function $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = \det(A)$. Then the sum of maximum and minimum values of f on $\mathbb{R}$ is equal to:

(2021-07-20/Shift-1)

- (a) $\frac{20}{27}$ (b) $-\frac{88}{27}$
(c) $-\frac{20}{27}$ (d) $\frac{88}{27}$

82. Let $A = \begin{bmatrix} 2 & 3 \\ a & 0 \end{bmatrix}$, $a \in \mathbb{R}$ be written as $P + Q$, where P is a symmetric matrix and Q is a skew-symmetric matrix. If $\det(Q) = 9$, then the modulus of the sum of all possible values of determinant of P is equal to:

(2021-07-20/Shift-1)

- (a) 24 (b) 18
(c) 45 (d) 36

83. Let a, b, c, d be in arithmetic progression with common difference λ .

If $\begin{vmatrix} x+a-c & x+b & x+a \\ x-1 & x+c & x+b \\ x-b+d & x+d & x+c \end{vmatrix} = 2$, then value of λ^2 is equal

to _____.

(2021-07-20/Shift-1)

84. Let $A = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$ and $B = 7A^{20} - 20A^7 + 2I$, where

I is an identity matrix of order 3×3 . If $B = [b_{ij}]$, then b_{13} is equal to _____.

(2021-07-20/Shift-1)

85. The value of $k \in \mathbb{R}$, for which the following system of linear equations

$$3x - y + 4z = 3,$$

$$x + 2y - 3z = -2,$$

$$6x + 5y + kz = -3,$$

has infinitely many solutions, is: (2021-07-20/Shift-2)

- (a) 3 (b) -3
(c) 5 (d) -5

86. Let $A = \{a_{ij}\}$ be a 3×3 matrix,

$$\text{where } a_{ij} = \begin{cases} (-1)^{j-i} & \text{if } i < j, \\ 2 & \text{if } i = j, \\ (-1)^{i+j} & \text{if } i > j \end{cases} \text{ then } \det(3\text{Adj}(2A^{-1})) \text{ is}$$

equal to _____ ?

(2021-07-20/Shift-2)

87. Let $S = \left\{ n \in \mathbb{N} \mid \begin{vmatrix} 0 & i \\ 1 & 0 \end{vmatrix}^n \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \forall a, b, c, d \in \mathbb{R} \right\}$,

where $i = \sqrt{-1}$. Then the number of 2-digit numbers in the set S is _____ ? (2021-07-25/Shift-1)



88. The values of a and b , for which the system of equations

$$2x + 3y + 6z = 8$$

$$x + 2y + az = 5$$

$$3x + 5y + 9z = b$$

Has no solution, are ? (2021-07-25/Shift-1)

- (a) $a = 3, b = 13$ (b) $a \neq 3, b \neq 13$
(c) $a \neq 3, b = 3$ (d) $a = 3, b \neq 13$

89.
$$f(x) = \begin{vmatrix} \sin^2 x & -2 + \cos^2 x & \cos 2x \\ 2 + \sin^2 x & \cos^2 x & \cos 2x \\ \sin^2 x & \cos^2 x & 1 + \cos 2x \end{vmatrix}, x \in [0, \pi]$$

Then the maximum value of $f(x)$ is equal to _____.

(2021-07-27/Shift-1)

90. For real numbers α and β consider the following system of linear equations:

$$x + y - z = 2, x + 2y + \alpha z = 1, 2x - y + z = \beta.$$

If the system has infinite solutions, then $\alpha + \beta$ is equal to _____.

(2021-07-27/Shift-1)

91. Let $A = \begin{bmatrix} 1 & 2 \\ -1 & 4 \end{bmatrix}$. If $A^{-1} = \alpha I + \beta A$, $\alpha, \beta \in \mathbb{R}$, I is a 2×2 identity matrix, then $4(\alpha - \beta)$ is equal to:

(2021-07-27/Shift-1)

- (a) 5 (b) 4
(c) 2 (d) $\frac{8}{3}$

92. Let A and B be two 3×3 real matrices such that $(A^2 - B^2)$ is invertible matrix. If $A^5 = B^5$ and $A^3 B^2 = A^2 B^3$, then the value of the determinant of the matrix $A^3 + B^3$ is equal to:

(2021-07-27/Shift-2)

- (a) 0 (b) 2
(c) 1 (d) 4

93. If $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ and $M = A + A^2 + A^3 + \dots + A^{20}$, then

the sum of the all the elements of the matrix M is equal to _____.

(2021-07-27/Shift-2)

94. Let $A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. Then the number of 3×3 matrices B

with entries from the set $\{1, 2, 3, 4, 5\}$ and satisfying $AB = BA$ is _____.

(2021-07-22/Shift-2)

95. Let $A = [a_{ij}]$ be a real matrix of order 3×3 such that $a_{i1} + a_{i2} + a_{i3} = 1$, for $i = 1, 2, 3$. Then, the sum of all the entries of the matrix A^3 is equal to:

(2021-07-22/Shift-2)

- (a) 1 (b) 3
(c) 2 (d) 9

96. The value of λ and μ such that the system of equations $x + y + z = 6$, $3x + 5y + 5z = 26$, $x + 2y + \lambda z = \mu$ has no solution, are:

(2021-07-22/Shift-2)

- (a) $\lambda = 3, \mu \neq 10$ (b) $\lambda \neq 2, \mu = 10$
(c) $\lambda = 3, \mu = 5$ (d) $\lambda = 2, \mu \neq 10$

97. The number of distinct real roots of
$$\begin{vmatrix} \sin x & \cos x & \cos x \\ \cos x & \sin x & \cos x \\ \cos x & \cos x & \sin x \end{vmatrix} = 0$$
 in the interval $-\frac{\pi}{4} \leq x \leq \frac{\pi}{4}$ is:

(2021-07-25/Shift-2)

- (a) 1 (b) 2
(c) 3 (d) 4

98. If $P = \begin{bmatrix} 1 & 0 \\ \frac{1}{2} & 1 \end{bmatrix}$, then P^{50} is: (2021-07-25/Shift-2)

- (a) $\begin{bmatrix} 1 & 25 \\ 0 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 0 \\ 25 & 1 \end{bmatrix}$
(c) $\begin{bmatrix} 1 & 0 \\ 50 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 1 & 50 \\ 0 & 1 \end{bmatrix}$



99. Consider the system of linear equations

$$-x + y + 2z = 0$$

$$3x - ay + 5z = 1$$

$$2x - 2y - az = 7$$

Let S_1 be the set of all $a \in \mathbb{R}$ for which the system is inconsistent and S_2 be the set of all $a \in \mathbb{R}$ for which the system has infinitely many solutions. If $n(S_1)$ and $n(S_2)$ denote the number of elements in S_1 and S_2 respectively, then (2021-09-01/Shift-2)

(a) $n(S_1) = 0, n(S_2) = 2$

(b) $n(S_1) = 2, n(S_2) = 2$

(c) $n(S_1) = 2, n(S_2) = 0$

(d) $n(S_1) = 1, n(S_2) = 0$

100. Let $J_{n,m} = \int_0^{1/2} \frac{x^n}{x^m - 1} dx, \forall n > m \text{ and } n, m \in \mathbb{N}$. Consider

a matrix $A = [a_{ij}]_{3 \times 3}$ where $a_{ij} = \begin{cases} J_{6+i,3} - J_{i+3,3}, & i \leq j \\ 0, & i > j \end{cases}$.

Then $|\text{adj} A^{-1}|$ is (2021-09-01/Shift-2)

(a) $(15)^2 \times 2^{34}$ (b) $(105)^2 \times 2^{38}$

(c) $(15)^2 \times 2^{42}$ (d) $(105)^2 \times 2^{36}$

101. Two fair dice are thrown. The number on them are taken as λ and μ and a system of linear equations

$$x + y + z = 5$$

$$x + 2y + 3z = \mu$$

$$x + 3y + \lambda z = 1$$

Is constructed. If p is true probability that the system has a unique solution and q is the probability that the system has no solution, then: (2021-08-26/Shift-2)

(a) $p = \frac{5}{6}$ and $q = \frac{5}{36}$ (b) $p = \frac{1}{6}$ and $q = \frac{1}{36}$

(c) $p = \frac{1}{6}$ and $q = \frac{5}{36}$ (d) $p = \frac{5}{6}$ and $q = \frac{1}{36}$

102. Let A be a 3×3 real matrix.

If $\det(2\text{Adj}(2\text{Adj}(\text{Adj}(2A)))) = 2^{41}$, then the value of $\det(A^2)$ equals (2021-08-26/Shift-2)

103. Let $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{pmatrix}$. Then $A^{2025} - A^{2020}$ is equal to: (2021-08-26/Shift-2)

(a) A^5 (b) A^6

(c) $A^6 - A$ (d) $A^5 - A$

104. If the matrix $A = \begin{pmatrix} 0 & 2 \\ k & -1 \end{pmatrix}$ satisfies $A(A^3 + 3I) = 2I$, then the value of k is: (2021-08-27/Shift-1)

(a) 1 (b) -1

(c) $-\frac{1}{2}$ (d) $\frac{1}{2}$

105. If the system of linear equations

$$2x + y - z = 3$$

$$x - y - z = \alpha$$

$$3x + 3y + \beta z = 3$$

has infinitely many solutions, then $\alpha + \beta - \alpha\beta$ is equal to (2021-08-27/Shift-1)

106. Let $\theta \in \left(0, \frac{\pi}{2}\right)$. If the system of linear equations,

$$(1 + \cos^2 \theta)x + (\sin^2 \theta)y + (4 \sin 3\theta)z = 0$$

$$(\cos^2 \theta)x + (1 + \sin^2 \theta)y + (4 \sin 3\theta)z = 0$$

$$(\cos^2 \theta)x + (\sin^2 \theta)y + (1 + 4 \sin 3\theta)z = 0$$

Has a non-trivial solution, then the value of θ is: (2021-08-26/Shift-1)

(a) $\frac{4\pi}{9}$ (b) $\frac{\pi}{18}$

(c) $\frac{5\pi}{18}$ (d) $\frac{7\pi}{18}$



107. If $A = \begin{pmatrix} \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \\ -2 & \frac{1}{\sqrt{5}} \end{pmatrix}$, $B = \begin{pmatrix} 1 & 0 \\ i & 1 \end{pmatrix}$, $i = \sqrt{-1}$ and

$Q = A^T B A$, then the inverse of the matrix $A Q^{2021} A^T$ is equal to : (2021-08-26/Shift-1)

(a) $\begin{pmatrix} \frac{1}{\sqrt{5}} & -2021 \\ 2021 & \frac{1}{\sqrt{5}} \end{pmatrix}$ (b) $\begin{pmatrix} 1 & 0 \\ 2021i & 1 \end{pmatrix}$

(c) $\begin{pmatrix} 1 & 0 \\ -2021i & 1 \end{pmatrix}$ (d) $\begin{pmatrix} 1 & -2021i \\ 0 & 1 \end{pmatrix}$

108. Let $[\lambda]$ be the greatest integer less than or equal to λ . The set of all value of λ for which the system of linear equations $x + y + z = 4$, $3x + 2y + 5z = 3$, $9x + 4y + (28 + [\lambda])z = [\lambda]$ has a solution is (2021-08-27/Shift-2)

- (a) $(-\infty, -9) \cup [-8, \infty)$ (b) $(-\infty, -9) \cup (-9, \infty)$
(c) $[-9, -8]$ (d) $\mathbb{R}$

109. Let $A = \begin{pmatrix} [x+1] & [x+2] & [x+3] \\ [x] & [x+3] & [x+3] \\ [x] & [x+2] & [x+4] \end{pmatrix}$, where $[x]$ denotes

the greatest integer less than or equal to x . If $\det(A) = 192$, then the set of values of x is in the interval:

- (a) $[62, 63]$ (b) $[60, 61]$
(c) $[68, 69]$ (d) $[65, 66]$

110. If the following system of linear equations

$$2x + y + z = 5$$

$$x - y + z = 3$$

$$x + y + az = b$$

Has no solution, then ? (2021-08-31/Shift-1)

(a) $a \neq \frac{1}{3}, b = \frac{7}{3}$ (b) $a \neq -\frac{1}{3}, b = \frac{7}{3}$

(c) $a = \frac{1}{3}, b \neq \frac{7}{3}$ (d) $a = -\frac{1}{3}, b \neq \frac{7}{3}$

111. The number of elements in the set

$$\left\{ A = \begin{pmatrix} a & b \\ 0 & d \end{pmatrix}; a, b, d \in \{-1, 0, 1\} \text{ and } (I - A)^3 = I - A^3 \right\},$$

where I is 2×2 identity matrix, is ____

(2021-08-31/Shift-2)

112. If $\alpha + \beta + \gamma = 2\pi$, then the system of equations

$$x + (\cos \gamma)y + (\cos \beta)z = 0$$

$$(\cos \gamma)x + y + (\cos \alpha)z = 0$$

$$(\cos \beta)x + (\cos \alpha)y + z = 0$$

has: (2021-08-31/Shift-2)

- (a) exactly two solutions (b) a unique solution
(c) no solution (d) infinitely many solutions

113. Let A be the set of all points (α, β) such that the area of triangle formed by the points $(5, 6)$, $(3, 2)$ and (α, β) is 12 square units. Then the least possible length of a line segment joining the origin to a point in A , is:

(2021-08-31/Shift-2)

(a) $\frac{16}{\sqrt{5}}$ (b) $\frac{12}{\sqrt{5}}$

(c) $\frac{8}{\sqrt{5}}$ (d) $\frac{4}{\sqrt{5}}$



114. Consider the following system of equations:

$$x + 2y - 3z = a$$

$$2x + 6y - 11z = b$$

$$x - 2y + 7z = c,$$

where a, b and c are real constants. Then the system of equations: (2021-02-26/Shift-2)

- (a) has a unique solution when $5a = 2b + c$
- (b) has no solution for all a, b and c
- (c) has infinite number of solutions when $5a = 2b + c$
- (d) has a unique solution for all a, b and c

115. If the matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 3 & 0 & -1 \end{bmatrix}$ satisfies the equation

$$A^{20} + \alpha A^{19} + \beta A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ for some real numbers } \alpha$$

and β , then $\beta - \alpha$ is equal to _____.

(2021-02-26/Shift-2)

116. Let A be a symmetric matrix of order 2 with integer entries. If the sum of the diagonal elements of A^2 is 1, then the possible number of such matrices is :

(2021-02-26/Shift-1)

- (a) 12
- (b) 4
- (c) 6
- (d) 1

117. The value of $\begin{vmatrix} (a+1)(a+2) & a+2 & 1 \\ (a+2)(a+3) & a+3 & 1 \\ (a+3)(a+4) & a+4 & 1 \end{vmatrix}$ is :

(2021-02-26/Shift-1)

- (a) $(a+2)(a+3)(a+4)$
- (b) 0
- (c) -2
- (d) $(a+1)(a+2)(a+3)$

118. If $A = \begin{bmatrix} 0 & -\tan\left(\frac{\theta}{2}\right) \\ \tan\left(\frac{\theta}{2}\right) & 0 \end{bmatrix}$ and

$$(I_2 + A)(I_2 - A)^{-1} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}, \text{ then } 13(a^2 + b^2) \text{ is equal}$$

to _____. (2021-02-25/Shift-1)

119. Let θ , where θ and ϕ are real numbers such that $\theta + \phi = \frac{\pi}{2}$. If $\sin \theta = \frac{1}{2}$, then the value of $\sin \phi$ is _____. (2021-02-25/Shift-1)

120. If the system of equations

$$kx + y + 2z = 1$$

$$3x - y - 2z = 2$$

$$-2x - 2y - 4z = 3$$

has infinitely many solutions, then k is equal to _____.

(2021-02-25/Shift-1)

121. If for the matrix, $A = \begin{bmatrix} 1 & -\alpha \\ \alpha & \beta \end{bmatrix}$, $AA^T = I_2$, then the value

of $\alpha^4 + \beta^4$ is:

(2021-02-25/Shift-2)

- (a) 3
- (b) 1
- (c) 4
- (d) 2

122. The following system of linear equations

$$2x + 3y + 2z = 9$$

$$3x + 2y + 2z = 9$$

$$x - y + 4z = 8$$

(2021-02-25/Shift-2)

- (a) does not have any solution
- (b) has infinitely many solutions
- (c) has a unique solution
- (d) has a solution (α, β, γ) satisfying $\alpha + \beta^2 + \gamma^3 = 12$

123. Let A be a 3×3 matrix with $\det(A) = 4$. Let R_i denote the i^{th} row of A . If a matrix B is obtained by performing the operation $R_2 \rightarrow 2R_2 + 5R_3$ on $2A$, then $\det(B)$ is equal to:

(2021-02-25/Shift-2)

- (a) 80
- (b) 128
- (c) 64
- (d) 16



124. For the system of linear equations:

$$x - 2y = 1, x - y + kz = -2, ky + 4z = 6, k \in \mathbb{R}$$

consider the following statements :

- (A) The system has unique solution if $k \neq 2, k \neq -2$.
 (B) The system has unique solution if $k = -2$.
 (C) The system has unique solution if $k = 2$.
 (D) The system has no-solution if $k = 2$.
 (E) The system has infinite number of solutions if $k \neq -2$.

Which of the following statements are correct ?

(2021-02-24/Shift-2)

- (a) (B) and (E) only (b) (A) and (E) only
 (c) (A) and (D) only (d) (C) and (D) only

125. Let A and B be 3×3 real matrices such that A is symmetric matrix and B is skew-symmetric matrix. Then the system of linear equations $(A^2 B^2 - B^2 A^2)X = O$, where X is a 3×1 column matrix of unknown variables and O is a 3×1 null matrix, has

(2021-02-24/Shift-2)

- (a) a unique solution (b) exactly two solutions
 (c) no solution (d) infinitely many solutions

126. The system of linear equations

$$3x - 2y - kz = 10$$

$$2x - 4y - 2z = 6$$

$$x + 2y - z = 5m$$

is inconsistent if:

(2021-02-24/Shift-1)

- (a) $k = 3, m = \frac{4}{5}$ (b) $k \neq 3, m \in \mathbb{R}$
 (c) $k \neq 3, m \neq \frac{4}{5}$ (d) $k = 3, m \neq \frac{4}{5}$

127. Let M be any 3×3 matrix with entries from the set $\{0, 1, 2\}$. The maximum number of such matrices, for which the sum of diagonal elements of $M^T M$ is seven, is _____.

(2021-02-24/Shift-1)

128. Let $P = \begin{bmatrix} 3 & -1 & -2 \\ 2 & 0 & \alpha \\ 3 & -5 & 0 \end{bmatrix}$, where $\alpha \in \mathbb{R}$. Suppose

$Q = [q_{ij}]$ is a matrix satisfying $PQ = kI_3$ for some non-zero $k \in \mathbb{R}$. If $q_{23} = -\frac{k}{8}$ and $|Q| = \frac{k^2}{2}$, then $\alpha^2 + k^2$ is equal to _____.

(2021-02-24/Shift-1)

129. The solutions of the equation

$$\begin{vmatrix} 1 + \sin^2 x & \sin^2 x & \sin^2 x \\ \cos^2 x & 1 + \cos^2 x & \cos^2 x \\ 4 \sin 2x & 4 \sin 2x & 1 + 4 \sin 2x \end{vmatrix} = 0, (0 < x < \pi), \text{ are :}$$

(2021-03-18/Shift-1)

- (a) $\frac{7\pi}{12}, \frac{11\pi}{12}$ (b) $\frac{\pi}{12}, \frac{\pi}{6}$
 (c) $\frac{5\pi}{12}, \frac{7\pi}{12}$ (d) $\frac{\pi}{6}, \frac{5\pi}{6}$

130. Let $A + 2B = \begin{bmatrix} 1 & 2 & 0 \\ 6 & -3 & 3 \\ -5 & 3 & 1 \end{bmatrix}$ and $2A - B = \begin{bmatrix} 2 & -1 & 5 \\ 2 & -1 & 6 \\ 0 & 1 & 2 \end{bmatrix}$.

If $T_r(A)$ denotes the sum of all diagonal elements of the matrix A, then $T_r(A) - T_r(B)$ has value equal to :

(2021-03-18/Shift-1)

- (a) 3 (b) 1
 (c) 2 (d) 0

131. Let α, β, γ be the real roots of the equation $x^3 + ax^2 + bx + c = 0$, ($a, b, c \in \mathbb{R}$ and $a, b \neq 0$). If the system of equation (in u, v, w) given by $\alpha u + \beta v + \gamma w = 0$; $\beta u + \gamma v + \alpha w = 0$;

$\gamma u + \alpha v + \beta w = 0$ has non-trivial solution, then the value

of $\frac{a^2}{b}$ is : (2021-03-18/Shift-1)

- (a) 5 (b) 1
 (c) 3 (d) 0



132. Let the system of linear equations

$$4x + \lambda y + 2z = 0$$

$$2x - y + z = 0$$

$$\mu x + 2y + 3z = 0, \lambda, \mu \in \mathbb{R}.$$

has a non-trivial solution. Then which of the following is true ? (2021-03-18/Shift-2)

(a) $\lambda = 3, \mu \in \mathbb{R}$ (b) $\lambda = 2, \mu \in \mathbb{R}$

(c) $\mu = -6, \lambda \in \mathbb{R}$ (d) $\mu = 6, \lambda \in \mathbb{R}$

133. Define a relation R over a class of $n \times n$ real matrices A and B as "ARB iff there exists a non-singular matrix P such that $PAP^{-1} = B$ ".

Then which of the following is true?

(2021-03-18/Shift-2)

(a) R is reflexive, symmetric but not transitive

(b) R is an equivalence relation

(c) R is symmetric, transitive but not reflexive,

(d) R is reflexive, transitive but not symmetric

134. Let I be an identity matrix of order 2×2 and $P = \begin{bmatrix} 2 & -1 \\ 5 & -3 \end{bmatrix}$.

Then the value of $n \in \mathbb{N}$ for which $P^n = 5I - 8P$ is equal to _____. (2021-03-18/Shift-2)

135. Let $A = \begin{bmatrix} i & -i \\ -i & i \end{bmatrix}, i = \sqrt{-1}$. Then, the system of linear

equations $A^8 \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 8 \\ 64 \end{bmatrix}$ has: (2021-03-16/Shift-1)

(a) A unique solution

(b) Exactly two solutions

(c) Infinitely many solutions

(d) No solution

136. The total number of 3×3 matrices A having entries from the set $\{0, 1, 2, 3\}$ such that the sum of all the diagonal entries of AA^T is 9, is equal to _____. (2021-03-16/Shift-1)

137. Let $P = \begin{bmatrix} -30 & 20 & 56 \\ 90 & 140 & 112 \\ 120 & 60 & 14 \end{bmatrix}$ and $A = \begin{bmatrix} 2 & 7 & \omega^2 \\ -1 & -\omega & 1 \\ 0 & -\omega & -\omega+1 \end{bmatrix}$

where $\omega = \frac{-1+i\sqrt{3}}{2}$, and I_3 be the identity matrix of order

3. If the determinant of the matrix $(P^{-1}AP - I_3)^2$ is $\alpha\omega^2$, then the value of α is equal to _____. (2021-03-16/Shift-1)

138. The maximum value of

$$f(x) = \begin{vmatrix} \sin^2 x & 1 + \cos^2 x & \cos 2x \\ 1 + \sin^2 x & \cos^2 x & \cos 2x \\ \sin^2 x & \cos^2 x & \sin 2x \end{vmatrix}, x \in \mathbb{R} \text{ is}$$

(2021-03-16/Shift-2)

(a) $\sqrt{5}$

(b) 5

(c) $\frac{3}{4}$

(d) $\sqrt{7}$

139. Let $A = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$ and $B = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$ be two 2×1 matrices with

real numbers such that $A = XB$, where $X = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & -1 \\ 1 & k \end{bmatrix}$,

and $k \in \mathbb{R}$. If $a_1^2 + a_2^2 = \frac{2}{3}(b_1^2 + b_2^2)$ and

$(k^2 + 1)b_2^2 \neq -2b_1b_2$ then the value of k is ____.

(2021-03-16/Shift-2)

140. The system of equations $kx + y + z = 1, x + ky + z = k$ and $x + y + zx = k^2$ has no solution is k is equal to :

(2021-03-17/Shift-1)

(a) 1

(b) -2

(c) -1

(d) 0



141. If $A = \begin{pmatrix} 0 & \sin \alpha \\ \sin \alpha & 0 \end{pmatrix}$ and $\det \left(A^2 - \frac{1}{2}I \right) = 0$, then a possible value of α is : **(2021-03-17/Shift-1)**
- (a) $\frac{\pi}{4}$ (b) $\frac{\pi}{2}$
(c) $\frac{\pi}{3}$ (d) $\frac{\pi}{6}$
142. If $A = \begin{bmatrix} 2 & 3 \\ 0 & -1 \end{bmatrix}$, then the value of $\det(A^4) + \det(A^{10} - (\text{Adj}(2A))^{10})$ is equal to **(2021-03-17/Shift-1)**
143. If x, y, z are in arithmetic progression with common difference d , $x \neq 3d$, and the determinant of the matrix $\begin{bmatrix} 3 & 4\sqrt{2} & x \\ 4 & 5\sqrt{2} & y \\ 5 & k & z \end{bmatrix}$ is zero, then the value of k^2 is : **(2021-03-17/Shift-2)**
- (a) 36 (b) 12
(c) 6 (d) 72
144. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $B = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ such that $AB = B$ and $a + d = 2021$, then the value of $ad - bc$ is equal to **(2021-03-17/Shift-2)**
145. If $1, \log_{10}(4^x - 2)$ and $\log_{10}\left(4^x + \frac{18}{5}\right)$ are in arithmetic progression for a real number x , then the value of the determinant $\begin{vmatrix} 2\left(x - \frac{1}{2}\right) & x-1 & x^2 \\ 1 & 0 & x \\ x & 1 & 0 \end{vmatrix}$ is equal to : **(2021-03-17/Shift-2)**



EXERCISE - 3 : ADVANCED OBJECTIVE QUESTIONS

Objective Questions I [Only one correct option]

1. If $A = \text{diag}(d_1, d_2, d_3, \dots, d_n)$, then A^n is equal to

(a) $\text{diag}(d_1^{n-1}, d_2^{n-1}, d_3^{n-1}, \dots, d_n^{n-1})$

(b) $\text{diag}(d_1^n, d_2^n, d_3^n, \dots, d_n^n)$

(c) A

(d) none of these

2. If $f(x)$, $g(x)$ and $h(x)$ are three polynomials of degree 2, then

$$\phi(x) = \begin{vmatrix} f(x) & g(x) & h(x) \\ f'(x) & g'(x) & h'(x) \\ f''(x) & g''(x) & h''(x) \end{vmatrix} \text{ is a polynomial of degree}$$

(a) 2

(b) 3

(c) 4

(d) none of these

3. If n is not a multiple of 3 and 1, ω , ω^2 are the cube roots of unity, then

$$\Delta = \begin{vmatrix} 1 & \omega^n & \omega^{2n} \\ \omega^{2n} & 1 & \omega^n \\ \omega^n & \omega^{2n} & 1 \end{vmatrix} \text{ is equal to}$$

(a) 0

(b) ω

(c) ω^2

(d) 1

4. $D = \begin{vmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 1 & c \end{vmatrix} = 0$, then the value of $\frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c}$ is :

(a) -1

(b) 0

(c) 1

(d) none of these

5. If $\begin{vmatrix} x^2 - 2x + 3 & 7x + 2 & x + 4 \\ 2x + 7 & x^2 - x + 2 & 3x \\ 3 & 2x - 1 & x^2 - 4x + 7 \end{vmatrix} =$

$ax^6 + bx^5 + cx^4 + dx^3 + ex^2 + fx + g$ the value of g is

(a) 2

(b) 1

(c) -2

(d) none of these

6. Which one of the following is correct ? If A is non-singular matrix, then, :

(a) $\det(A^{-1}) = \det(A)$ (b) $\det(A^{-1}) = \frac{1}{\det(A)}$

(c) $\det(A^{-1}) = 1$

(d) none of these

7. If $A = \begin{bmatrix} 1 & \tan x \\ -\tan x & 1 \end{bmatrix}$, then the value of $|A^T A^{-1}|$ is

(a) $\cos 4x$

(b) $\sec^2 x$

(c) $-\cos 4x$

(d) 1

8. For what value of x , the matrix

$$\begin{bmatrix} 3-x & 2 & 2 \\ 2 & 4-x & 1 \\ -2 & -4 & -1-x \end{bmatrix} \text{ is singular.}$$

(a) $x = 1, 2$

(b) $x = 0, 2$

(c) $x = 0, 1$

(d) $x = 0, 3$

9. If m is a positive integer and

$$\Delta_r = \begin{vmatrix} 2r-1 & {}^m C_r & 1 \\ m^2-1 & 2^m & m+1 \\ \sin^2(m^2) & \sin^2(m) & \sin^2(m+1) \end{vmatrix}$$

Then the value of $\sum_{r=0}^m \Delta_r$ is

(a) 0

(b) $m^2 - 1$

(c) 2^m

(d) $2^m \sin^2(2^m)$

10. Let $D_r = \begin{vmatrix} a & 2^r & 2^{16}-1 \\ b & 3(4^r) & 2(4^{16}-1) \\ c & 7(8^r) & 4(8^{16}-1) \end{vmatrix}$, then the value of

$$\sum_{k=1}^{16} D_k \text{ is}$$

(a) 0

(b) $a + b + c$

(c) $ab + bc + ca$

(d) none of these



11. If $A = \begin{bmatrix} 4 & x+2 \\ 2x-3 & x+1 \end{bmatrix}$ is symmetric, then $x =$

- (a) 3 (b) 5
(c) 2 (d) 4

12. If $\begin{vmatrix} x & 3 & 6 \\ 3 & 6 & x \\ 6 & x & 3 \end{vmatrix} = \begin{vmatrix} 2 & x & 7 \\ x & 7 & 2 \\ 7 & 2 & x \end{vmatrix} = \begin{vmatrix} 4 & 5 & x \\ 5 & x & 4 \\ x & 4 & 5 \end{vmatrix} = 0$ then x is equal to

- (a) 9 (b) -9
(c) 0 (d) None of these

13. $\Delta = \begin{vmatrix} p & 2-i & i+1 \\ 2+i & q & 3+i \\ 1-i & 3-i & r \end{vmatrix}$ is always (where $p, q, r \in \mathbb{R}$)

- (a) real (b) imaginary
(c) zero (d) none of these

14. If $\Delta = \begin{vmatrix} 1 & a & bc \\ 1 & b & ca \\ 1 & c & ab \end{vmatrix} = \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}$, then

- (a) $\Delta = (a-b)(b-c)(c-a)$
(b) a, b, c are in G.P.
(c) b, c, a are in G.P.
(d) a, c, b are in G.P.

15. If $E(\theta) = \begin{bmatrix} \cos^2 \theta & \cos \theta \sin \theta \\ \cos \theta \sin \theta & \sin^2 \theta \end{bmatrix}$ and θ and ϕ differ by an

- odd multiple of $\pi/2$, then $E(\theta) \cdot E(\phi)$ is a
(a) Null matrix (b) Unit matrix
(c) Diagonal matrix (d) none of these.

16. If $U_n = \begin{vmatrix} n & 1 & 5 \\ n^2 & 2N+1 & 2N+1 \\ n^3 & 3N^2 & 3N+1 \end{vmatrix}$, then $\sum_{n=1}^N U_n$ is equal to

- (a) $2 \sum_{n=1}^N n$ (b) $2 \sum_{n=1}^N n^2$
(c) $\frac{1}{2} \sum_{n=1}^N n^2$ (d) 0

17. If $a^2 + b^2 + c^2 = -2$ and

$$f(x) = \begin{vmatrix} 1+a^2x & (1+b^2)x & (1+c^2)x \\ (1+a^2)x & 1+b^2x & (1+c^2)x \\ (1+a^2)x & (1+b^2)x & 1+c^2x \end{vmatrix}$$

Then $f(x)$ is a polynomial of degree :

- (a) 2 (b) 3
(c) 0 (d) 1

18. If $F(x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and

$$G(y) = \begin{bmatrix} \cos y & 0 & \sin y \\ 0 & 1 & 0 \\ -\sin y & 0 & \cos y \end{bmatrix}$$
 then $[F(x)G(y)]^{-1}$ is equal to

- (a) $F(-x)G(-y)$ (b) $F(x^{-1})G(y^{-1})$
(c) $G(-y)F(-x)$ (d) $G(y^{-1})F(x^{-1})$

19. If the system of linear equations

$$x + 2ay + az = 0$$

$$x + 3by + bz = 0$$

$$x + 4cy + cz = 0$$

has a non-zero solution, then a, b, c :

- (a) are in AP (b) are in GP
(c) are in HP (d) satisfy $a + 2b + 3c = 0$

20. The system of equations

$$\alpha x + y + z = \alpha - 1$$

$$x + \alpha y + z = \alpha - 1$$

$$x + y + \alpha z = \alpha - 1$$

has no solution, if α is :

- (a) 1 (b) not -2
(c) either -2 or 1 (d) -2

21. The system of equation $-2x + y + z = 1$,

$$x - 2y + z = -2, x + y + \lambda z = 4$$
 will have no solution if

- (a) $\lambda = -2$ (b) $\lambda = -1$
(c) $\lambda = 3$ (d) none of these



22. The system of equations $x + 2y + 3z = 4$, $2x + 3y + 4z = 5$, $3x + 4y + 5z = 6$ has

(a) Infinitely many solutions
(b) No solution
(c) A unique solution
(d) None of these

23. If $a, b, c > 0$ & $x, y, z \in \mathbb{R}$ then the determinant

$$\begin{vmatrix} (a^x + a^{-x})^2 & (a^x - a^{-x})^2 & 1 \\ (b^y + b^{-y})^2 & (b^y - b^{-y})^2 & 1 \\ (c^z + c^{-z})^2 & (c^z - c^{-z})^2 & 1 \end{vmatrix} =$$

(a) $a^x b^y c^z$ (b) $a^{-x} b^{-y} c^{-z}$
(c) $a^{2x} b^{2y} c^{2z}$ (d) zero

24. If $\begin{vmatrix} p & q-y & r-z \\ p-x & q & r-z \\ p-x & q-y & r \end{vmatrix} = 0$, then the value of $\frac{p}{x} + \frac{q}{y} + \frac{r}{z}$ is

(a) 0 (b) 1
(c) 2 (d) $4pqr$

25. Suppose $D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ and

$$D' = \begin{vmatrix} a_1 + pb_1 & b_1 + qc_1 & c_1 + ra_1 \\ a_2 + pb_2 & b_2 + qc_2 & c_2 + ra_2 \\ a_3 + pb_3 & b_3 + qc_3 & c_3 + ra_3 \end{vmatrix}. \text{ Then}$$

(a) $D' = D$ (b) $D' = D(1 - pqr)$
(c) $D' = D(1 + p + q + r)$ (d) $D' = D(1 + pqr)$

26. Let $ax^7 + bx^6 + cx^5 + dx^4 + ex^3 + fx^2 + gx + h$

$$= \begin{vmatrix} (x+1) & (x^2+2) & (x^2+x) \\ (x^2+x) & (x+1) & (x^2+2) \\ (x^2+2) & (x^2+x) & x+1 \end{vmatrix}. \text{ Then}$$

(a) $g = 3$ and $h = -5$ (b) $g = -3$ and $h = -5$
(c) $g = -3$ and $h = -9$ (d) None of these

Objective Questions II

[One or more than one correct option]

27. If $\begin{vmatrix} x^2 + x & x+1 & x-2 \\ 2x^2 + 3x - 1 & 3x & 3x-3 \\ x^2 + 2x + 3 & 2x-1 & 2x-1 \end{vmatrix} = Ax + B$, where A and B

are constants, then

(a) $A + B = 12$ (b) $A - B = 36$
(c) $A^2 + B^2 = 720$ (d) $A + 2B = 0$

28. Let A be a symmetric matrix such that $A^5 = O$ and $B = I + A + A^2 + A^3 + A^4$, then B is

(a) symmetric (b) singular
(c) non-singular (d) skew symmetric

29. The determinant $\Delta = \begin{vmatrix} a^2 + x^2 & ab & ac \\ ab & b^2 + x^2 & bc \\ ac & bc & c^2 + x^2 \end{vmatrix}$ is

divisible by

(a) x (b) x^2
(c) x^3 (d) x^4

30. If $f(\theta) = \begin{vmatrix} 1 & 1 & -1 \\ 1 & e^{i\theta} & 1 \\ 1 & -1 & -e^{-i\theta} \end{vmatrix}$ then

(a) $\int_{-\pi/2}^{\pi/2} f(\theta) d\theta = 2 \int_0^{\pi/2} f(\theta) d\theta$

(b) $f(\theta)$ is purely real
(c) $f(\pi/2) = 2$
(d) None of these

31. The value of the determinant

$$\begin{vmatrix} \sqrt{6} & 2i & 3+\sqrt{6} \\ \sqrt{12} & \sqrt{3}+\sqrt{8}i & 3\sqrt{2}+\sqrt{6}i \\ \sqrt{18} & \sqrt{2}+\sqrt{12}i & \sqrt{27}+2i \end{vmatrix}, \text{ where } i = \sqrt{-1}, \text{ is}$$

(a) complex number (b) real number
(c) irrational number (d) rational number



32. If $a > b > c$ and the system of equations $ax + by + cz = 0$, $bx + cy + az = 0$, $cx + ay + bz = 0$ has a non trivial solution, then both the roots of the quadratic equation $at^2 + bt + c = 0$ are

- (a) real (b) of opposite sign
(c) positive (d) complex

33. If $\Delta = \begin{vmatrix} e^x & \sin x & 1 \\ \cos x & \log_e(1+x^2) & 1 \\ x & x^2 & 1 \end{vmatrix} = a + bx + cx^2 \dots$ then

- (a) $a = 0$ (b) $a = 1$
(c) $b = -1$ (d) $b = -2$

34. If $f(x)$ and $g(x)$ are functions such that $f(x+y) = f(x)g(y) + g(x)f(y)$, then

$\begin{vmatrix} f(\alpha) & g(\alpha) & f(\alpha + \theta) \\ f(\beta) & g(\beta) & f(\beta + \theta) \\ f(\gamma) & g(\gamma) & f(\gamma + \theta) \end{vmatrix}$ is independent of

- (a) α (b) β
(c) γ (d) θ

35. $\begin{vmatrix} 1 & a & a^2 \\ \cos(xz - yz) & \cos xz & \cos(xz - yz) \\ \sin(xz - yz) & \sin xz & \sin(xz - yz) \end{vmatrix}$ depends on

- (a) x (b) y
(c) z (d) a

36. If $x \in \mathbb{N}$ and ${}^x C_1, {}^x C_2$ and ${}^x C_3, (i = 1, 2, 3)$ are binomial coefficients, then

$12 \begin{vmatrix} {}^x C_1 & {}^x C_2 & {}^x C_3 \\ {}^x C_1 & {}^x C_2 & {}^x C_3 \\ {}^x C_1 & {}^x C_2 & {}^x C_3 \end{vmatrix}$ is divisible by

- (a) x^3 (b) x^6
(c) x^9 (d) x^{12}

37. The digits A, B, C are such that the three digit numbers A 88, 6B8, 86C are divisible by 72, then the determinant

$\begin{vmatrix} A & 6 & 8 \\ 8 & B & 6 \\ 8 & 8 & C \end{vmatrix}$ is divisible by

- (a) 72 (b) 144
(c) 288 (d) 216

38. Let $f(a, b) = \begin{vmatrix} a & a^2 & 0 \\ 1 & (2a+b) & (a+b)^2 \\ 0 & 1 & (2a+3b) \end{vmatrix}$, then

- (a) $(a+b)$ is a factor of $f(a, b)$
(b) $(a+2b)$ is a factor of $f(a, b)$
(c) $(2a+b)$ is a factor of $f(a, b)$
(d) a is a factor of $f(a, b)$

39. a, b, c are non-zero real numbers. Then

$\begin{vmatrix} bc & ca & ab \\ ca & ab & bc \\ ab & bc & ca \end{vmatrix} = 0$, if

- (a) $\frac{1}{a} + \frac{1}{b\omega} + \frac{1}{c\omega^2} = 0$ (b) $\frac{1}{a} + \frac{1}{b\omega^2} + \frac{1}{c\omega} = 0$
(c) $\frac{1}{a\omega} + \frac{1}{b\omega^2} + \frac{1}{c} = 0$ (d) none of these

40. Let $f(x) = \begin{vmatrix} 1/x & \ln x & x^n \\ 1 & -1/n & (-1)^n \\ 1 & a & a^2 \end{vmatrix}$, then $\frac{d^n}{dx^n} f(x)$ at $x = 1$ is

- (a) independent of a (b) independent of n
(c) independent of a and n (d) zero

41. If $a^2 + b^2 + c^2 = 1$, then

$\begin{vmatrix} a^2 + (b^2 + c^2)\cos\phi & ab(1 - \cos\phi) & ac(1 - \cos\phi) \\ ba(1 - \cos\phi) & b^2 + (c^2 + a^2)\cos\phi & bc(1 - \cos\phi) \\ ca(1 - \cos\phi) & cb(1 - \cos\phi) & c^2 + (a^2 + b^2)\cos\phi \end{vmatrix}$

is independent of

- (a) a (b) b
(c) c (d) ϕ



42. Let A and B be two matrices different from I such that $AB = BA$ and $A^n - B^n$ is invertible for some positive integer n. If $A^n - B^n = A^{n+1} - B^{n+1} = A^{n+2} - B^{n+2}$, then

- (a) $I - A$ is singular
(b) $I - B$ is singular
(c) $A = B$
(d) $(I - A)(I - B)$ is non-singular

43. Let $f_1(x) = x + a$, $f_2(x) = x^2 + bx + c$ and

$$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ f_1(x_1) & f_1(x_2) & f_1(x_3) \\ f_2(x_1) & f_2(x_2) & f_2(x_3) \end{vmatrix}, \text{ then}$$

- (a) Δ is independent of a
(b) Δ is independent of b and c
(c) Δ is independent of x_1, x_2, x_3
(d) none of the above

44. Let $0 < \theta < \pi/2$ and

$$\Delta(x, \theta) = \begin{vmatrix} x & \tan \theta & \cot \theta \\ -\tan \theta & -x & 1 \\ \cot \theta & 1 & x \end{vmatrix}$$

then

- (a) $\Delta(0, \theta) = 0$
(b) $\Delta(x, \pi/4) = x^2 + 1$
(c) $\min_{0 < \theta < \pi/2} \Delta(1, \theta) = 2$
(d) $\Delta(x, \theta)$ is independent of x

45. Let A, B and C be 2×2 matrices with entries from the set of real numbers. Define * as follows :

$$A * B = \frac{1}{2} (AB' + A'B)$$

- (a) $A * B = B * A$
(b) $A * A = A^2$
(c) $A * (B + C) = A * B + A * C$
(d) $A * I = A + A'$

46. If $f(x) = \begin{vmatrix} 1 + \sin^2 x & \cos^2 x & 4 \sin 2x \\ \sin^2 x & 1 + \cos^2 x & 4 \sin 2x \\ \sin^2 x & \cos^2 x & 1 + 4 \sin 2x \end{vmatrix}$,

f(a) and f(b) be the least and greatest value of f(x), then

- (a) $f(a) = 2, f(b) = 6$ (b) $f(a) = -2, f(b) = 6$
(c) $f(a) = 2, f(b) = -6$ (d) period of f(x) is π

47. Eliminating a, b, c from $x = \frac{a}{b-c}, y = \frac{b}{c-a}, z = \frac{c}{a-b}$, we get

(a) $\begin{vmatrix} 1 & -x & x \\ -y & 1 & y \\ 1 & -z & z \end{vmatrix} = 0$ (b) $\begin{vmatrix} 1 & -x & x \\ 1 & 1 & -y \\ 1 & z & 1 \end{vmatrix} = 0$

(c) $\begin{vmatrix} 1 & -x & x \\ y & 1 & -y \\ -z & z & 1 \end{vmatrix} = 0$ (d) none of these

Numerical Value Type Questions

48. If $\begin{vmatrix} x & 3 & 6 \\ 3 & 6 & x \\ 6 & x & 3 \end{vmatrix} = \begin{vmatrix} 2 & x & 7 \\ x & 7 & 2 \\ 7 & 2 & x \end{vmatrix} = \begin{vmatrix} 4 & 5 & x \\ 5 & x & 4 \\ x & 4 & 5 \end{vmatrix} = 0$,

then x^2 is equal to

49. The value of the determinant $\begin{vmatrix} 1 & a & a^2 - bc \\ 1 & b & b^2 - ca \\ 1 & c & c^2 - ab \end{vmatrix}$ is

50. If $x \neq 0, y \neq 0, z \neq 0$ and $\begin{vmatrix} 1+x & 1 & 1 \\ 1+y & 1+2y & 1 \\ 1+z & 1+z & 1+3z \end{vmatrix} = 0$, then

$-(x^{-1} + y^{-1} + z^{-1})$ is equal to

51. If $f(x) = \begin{vmatrix} x & \sin x & \cos x \\ x^2 & \tan x & -x^3 \\ 2x & \sin 2x & 5x \end{vmatrix}$, then $\lim_{x \rightarrow 0} \frac{f'(x)}{x}$ is



Assertion & Reason

- (A) If both assertion and reason are correct and reason is the correct explanation of assertion.
 (B) If both assertion and reason are true but reason is not the correct explanation of assertion.
 (C) If assertion is true but reason is false.
 (D) If assertion is false but reason is true.

52. Assertion : If $\Delta(x) = \begin{vmatrix} f_1(x) & f_2(x) \\ g_1(x) & g_2(x) \end{vmatrix}$, then

$$\Delta'(x) \neq \begin{vmatrix} f_1'(x) & f_2'(x) \\ g_1'(x) & g_2'(x) \end{vmatrix}$$

Reason : $\frac{d}{dx} \{f(x)g(x)\} \neq \frac{d}{dx} f(x) \frac{d}{dx} g(x)$

- (a) A (b) B
 (c) C (d) D

53. Assertion : The system of equations possess a non trivial solution over the set of rationals $x + ky + 3z = 0$, $3x + ky - 2z = 0$, $2x + 3y - 4z = 0$, then the value of k is $31/2$,

Reason : For non trivial solution $\Delta = 0$.

- (a) A (b) B
 (c) C (d) D

54. Let $A(\theta) = \begin{pmatrix} \cos \theta + \sin \theta & \sqrt{2} \sin \theta \\ -\sqrt{2} \sin \theta & \cos \theta - \sin \theta \end{pmatrix}$

Assertion : $A(\pi/3)^3 = -I$

Reason : $A(\theta)A(\phi) = A(\theta + \phi)$

- (a) A (b) B
 (c) C (d) D

55. Assertion : $\begin{vmatrix} a^2 + x^2 & ab - cx & ac + bx \\ ab + cx & b^2 + x^2 & bc - ax \\ ac - bx & bc + ax & c^2 + x^2 \end{vmatrix} = \begin{vmatrix} x & c & -b \\ -c & x & a \\ b & -a & x \end{vmatrix}^2$

Reason : $\Delta^c = \Delta^{n-1}$ where n is order of determinant, and Δ^c is the determinant of cofactors of Δ .

- (a) A (b) B
 (c) C (d) D

56. Assertion : $f(x) = \begin{vmatrix} (1+x)^{21} & (1+x)^{22} & (1+c)^{23} \\ (1+x)^{31} & (1+x)^{32} & (1+x)^{33} \\ (1+x)^{41} & (1+x)^{42} & (1+x)^{43} \end{vmatrix}$, then

coefficient of x in $f(x)$ is zero.

Reason : If $F(x) = A_0 + A_1x + A_2x^2 + \dots + A_nx^n$, then $A_1 = F'(0)$, when dash denotes the differential coefficient.

- (a) A (b) B
 (c) C (d) D

57. Assertion : $\begin{vmatrix} \cos(\theta + \alpha) & \cos(\theta + \beta) & \cos(\theta + \gamma) \\ \sin(\theta + \alpha) & \sin(\theta + \beta) & \sin(\theta + \gamma) \\ \sin(\beta - \gamma) & \sin(\gamma - \alpha) & \sin(\alpha - \beta) \end{vmatrix}$ is

independent of θ .

Reason : If $f(\theta) = c$, then $f(\theta)$ is independent of θ .

- (a) A (b) B
 (c) C (d) D

58. Suppose a, b, c are distinct real numbers.

Let $f(x) = \begin{vmatrix} x^3 + a^3 & x^2 - a^2 & x + a \\ x^3 + b^3 & x^2 - b^2 & x + b \\ x^3 + c^3 & x^2 - c^2 & x + c \end{vmatrix}$

Assertion : $f(x)$ is a polynomial of degree 3.

Reason : a, b, c are zeroes of $f(x)$.

- (a) A (b) B
 (c) C (d) D

59. Suppose $X = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ satisfies the equation $X^2 - 4X + 3I = O$.

Assertion : If $a + d \neq 4$, then there are just two such matrix X .

Reason : There are infinite number of matrices X , satisfying $X^2 - 4X + 3I = O$.

- (a) A (b) B
 (c) C (d) D

60. Let $a_i, b_i, c_i \in \mathbb{N}$ for $i = 1, 2, 3$ and let

$$\Delta = \begin{vmatrix} \frac{1+a_1^3b_1^3}{1+a_1b_1} & \frac{1+a_1^3b_2^3}{1+a_1b_2} & \frac{1+a_1^3b_3^3}{1+a_1b_3} \\ \frac{1+a_2^3b_1^3}{1+a_2b_1} & \frac{1+a_2^3b_2^3}{1+a_2b_2} & \frac{1+a_2^3b_3^3}{1+a_2b_3} \\ \frac{1+a_3^3b_1^3}{1+a_3b_1} & \frac{1+a_3^3b_2^3}{1+a_3b_2} & \frac{1+a_3^3b_3^3}{1+a_3b_3} \end{vmatrix}$$

Assertion : $\Delta = 0$

Reason : Δ can be written as product of two determinants.

- (a) A (b) B
 (c) C (d) D



Match the Following

Each question has two columns. Four options are given representing matching of elements from Column-I and Column-II. Only one of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.

61. Match the following List-I and List-II

Column-I

Column-II

(P) If $A = \begin{bmatrix} 1 & \tan x \\ -\tan x & 1 \end{bmatrix}$, then the (1) 10

value of $|A^T A^{-1}|$ is

(Q) If x, y, z are cube root of unity and (2) 1

$$D = \begin{vmatrix} x^2 + y^2 & z^2 & z^2 \\ x^2 & y^2 + z^2 & x^2 \\ y^2 & y^2 & z^2 + x^2 \end{vmatrix}$$

then real part of ID is

(R) If any triangle the area $A_1 \leq \frac{b^2 + c^2}{\lambda}$, (3) 4

then largest possible numerical

quantity λ is

(S) The equation $x^4 - 4x + c$ has no real (4) 0

roots, then minimum integral value

of c^2 can be

Codes :

	P	Q	R	S
(A)	2	4	1	3
(B)	1	2	3	4
(C)	4	3	2	1
(D)	2	4	3	1

62. Match the following List-I and List-II

Column-I

Column-II

(A) If 2 is the root of the equation (P) e
 $|A - xI| = 0$, (where A is a non singular

matrix), $\frac{|A|}{2}$ a root of $|B - xI| = 0$,

then B can be

(B) If $e^{i\alpha}$ is the root of $|A - yI| = 0$ then (Q) $\text{adj}(A)$

a root of $|\bar{A} - xI| = 0$ is (where A is
 a non singular matrix)

(C) Let A_{ij} be a 2×2 non singular matrix (R) $\cos \alpha - i \sin \alpha$
 where $i, j \in \mathbb{N}$ and

$$B = \begin{vmatrix} |A_{11}| & |A_{12}| & \dots & |A_{1n}| \\ 0 & |A_{22}| & \dots & |A_{2n}| \\ 0 & 0 & |A_{33}| & \dots & |A_{3n}| \\ \dots & \dots & \dots & \dots & \dots \\ 0 & \dots & \dots & \dots & |A_{nn}| \end{vmatrix}$$

then $|B - \lambda I| = 0$ has root as

(D) Consider a matrix such that $\bar{A}' = A$ (S) $|A_{11}|$

then the equation $|A - xI| = 0$ can

have root as (where A is a non
 singular matrix)

The correct matching is

(a) A-Q; B-R; C-S; D-P

(b) A-R; B-Q; C-S; D-P

(c) A-Q; B-S; C-R; D-P

(d) A-Q; B-R; C-P; D-S



Paragraph Type Questions

Use the following passage, solve Q. 63 to Q. 67

Passage

Let $\Delta = 0$ and Δ^c denotes the determinant of cofactors, then $\Delta^c = \Delta^{n-1}$, where $n (> 0)$ is the order of Δ .

On the basis of above information, answer the following questions :

63. If a, b, c are the roots of the equation $x^3 - px^2 + r = 0$, then the value of

$$\begin{vmatrix} bc - a^2 & ca - b^2 & ab - c^2 \\ ca - b^2 & ab - c^2 & bc - a^2 \\ ab - c^2 & bc - a^2 & ca - b^2 \end{vmatrix} \text{ is}$$

- (a) p^2 (b) p^4
(c) p^6 (d) p^9

64. If $l_1, m_1, n_1; l_2, m_2, n_2; l_3, m_3, n_3$ are real quantities satisfying the six relations :

$$\ell_1^2 + m_1^2 + n_1^2 = \ell_2^2 + m_2^2 + n_2^2 = \ell_3^2 + m_3^2 + n_3^2 = 1$$

$$l_2 l_3 + m_2 m_3 + n_2 n_3 = l_3 l_1 + m_3 m_1 + n_3 n_1 =$$

$$l_1 l_2 + m_1 m_2 + n_1 n_2 = 0, \text{ then the value of}$$

$$\begin{vmatrix} \ell_1 & m_1 & n_1 \\ \ell_2 & m_2 & n_2 \\ \ell_3 & m_3 & n_3 \end{vmatrix} \text{ is}$$

- (a) 0 (b) ± 1
(c) ± 2 (d) ± 3

65. If a, b, c are the roots of the equation $x^3 - 3x^2 + 3x + 7 = 0$, then the value of

$$\begin{vmatrix} 2bc - a^2 & c^2 & b^2 \\ c^2 & 2ac - b^2 & a^2 \\ b^2 & a^2 & 2ab - c^2 \end{vmatrix} \text{ is}$$

- (a) 9 (b) 27
(c) 81 (d) 0

66. If $a^2 + b^2 + c^2 = \lambda^2$, then the value of

$$\begin{vmatrix} a^2 + \lambda^2 & ab + c\lambda & ca - b\lambda \\ ab - c\lambda & b^2 + \lambda^2 & bc + a\lambda \\ ac + b\lambda & bc - a\lambda & c^2 + \lambda^2 \end{vmatrix} \times \begin{vmatrix} \lambda & c & -b \\ -c & \lambda & a \\ b & -a & \lambda \end{vmatrix} \text{ is}$$

- (a) $8\lambda^6$ (b) $27\lambda^9$
(c) $8\lambda^9$ (d) $27\lambda^6$

67. Suppose $a, b, c \in \mathbb{R}$, $a + b + c > 0$, $A = bc - a^2$, $B = ca - b^2$ and $C = ab - c^2$ and

$$\begin{vmatrix} A & B & C \\ B & C & A \\ C & A & B \end{vmatrix} = 49, \text{ then } \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} \text{ equals}$$

- (a) -7 (b) 7
(c) -2401 (d) 2401

Use the following passage, solve Q. 68 to Q. 70

Passage

Elementary Transformation of a matrix :

The following operation on a matrix are called elementary operations (transformations)

1. The interchange of any two rows (or columns)
2. The multiplication of the elements of any row (or column) by any non zero number

The addition to the elements of any row (or column) the corresponding elements of any other row (or column) the corresponding elements of any other row (or column) multiplied by any number

Echelon form of matrix :

A matrix A is said to be in echelon form if

- (i) every row of A which has all its elements 0, occurs below row, which has a non-zero elements
- (ii) The first non zero element in each non-zero row is 1.
- (iii) the number of zeros before the first non zero elements in a row is less than the number of such zeroes in the next row.

[A row of matrix is said to be a zero row if all its elements are zero]

Note : Rank of a matrix does not change by application of any elementary operations.



For example $\begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 1 & 3 & 6 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ are echelon forms

The number of non-zero rows in the echelon form of a matrix is defined as its RANK.

For example we can reduce the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 6 & 10 \end{bmatrix}$ into

echelon form using following elementary row transformation.

$$(i) \quad R_2 \rightarrow R_2 - 2R_1 \text{ and } R_3 \rightarrow R_3 - 3R_1 \quad \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$(ii) \quad R_2 \rightarrow R_2 - 2R_1 \quad \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

This is the echelon form of matrix A

Number of non zero rows in the echelon form = 2

$\Rightarrow$ Rank of the matrix A is 2

68. Rank of the matrix $\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & -1 \\ 3 & 1 & 1 \end{bmatrix}$ is

- (a) 1 (b) 2
(c) 3 (d) 0

69. Rank of the matrix $\begin{bmatrix} 1 & 1 & 1 & -1 \\ 1 & 2 & 4 & 4 \\ 3 & 4 & 5 & 2 \end{bmatrix}$ is

- (a) 1 (b) 2
(c) 3 (d) 4

70. The echelon form of the matrix $\begin{bmatrix} 1 & 3 & 4 & 3 \\ 3 & 9 & 12 & 9 \\ 1 & 3 & 4 & 1 \end{bmatrix}$ is

$$(a) \begin{bmatrix} 1 & 3 & 4 & 3 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (b) \begin{bmatrix} 1 & 2 & 4 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -2 \end{bmatrix}$$

$$(c) \begin{bmatrix} 1 & 3 & 4 & 3 \\ 0 & 0 & 0 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (d) \begin{bmatrix} 1 & 3 & 4 & -\frac{3}{2} \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$



EXERCISE - 4 : PREVIOUS YEAR JEE ADVANCED QUESTIONS

Objective Questions I [Only one correct option]

1. The number of distinct real roots of

$$\begin{vmatrix} \sin x & \cos x & \cos x \\ \cos x & \sin x & \cos x \\ \cos x & \cos x & \sin x \end{vmatrix} = 0 \text{ in the interval } -\frac{\pi}{4} \leq x \leq \frac{\pi}{4} \text{ is}$$

(2001)

- (a) 0 (b) 2
(c) 1 (d) 3

2. The number of values of k for which the system of equations

$$(k+1)x + 8y = 4k$$

$$kx + (k+3)y = 3k - 1$$

has infinitely many solution, is

(2002)

- (a) 0 (b) 1
(c) 2 (d) infinite

3. If $A = \begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 5 & 1 \end{bmatrix}$, then value of α for which

$$A^2 = B, \text{ is}$$

(2003)

- (a) 1 (b) -1
(c) 4 (d) no real values

4. If the system of equations $x + ay = 0$, $az + y = 0$ and $ax + z = 0$ has infinite solutions, then the value of a is

(2003)

- (a) -1 (b) 1
(c) 0 (d) no real values

5. Given $2x - y - z = 2$, $x - 2y + z = -4$, $x + y + \lambda z = 4$ then the value of λ such that the given system of equation has no solution, is

(2004)

- (a) 3 (b) -2
(c) 0 (d) -3

6. If $A = \begin{bmatrix} \alpha & 2 \\ 2 & \alpha \end{bmatrix}$ and $|A^3| = 125$ then the value of α is

(2004)

- (a) ± 1 (b) ± 2
(c) ± 3 (d) ± 5

7. If $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & 4 \end{bmatrix}$, $6A^{-1} = A^2 + cA + dI$, then (c, d) is

(2005)

- (a) $(-11, 6)$ (b) $(-6, 11)$
(c) $(6, 11)$ (d) $(11, 6)$

8. If $P = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$, $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $Q = PAP^T$, then

$P^T(Q^{2005})P$ is equal to

(2005)

- (a) $\begin{bmatrix} 1 & \frac{\sqrt{3}}{2} \\ 0 & 2005 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 2005 \\ 0 & 1 \end{bmatrix}$
(c) $\begin{bmatrix} \frac{\sqrt{3}}{2} & 2005 \\ 2 & 0 \end{bmatrix}$ (d) $\begin{bmatrix} \frac{1}{2} & 2005 \\ \frac{\sqrt{3}}{2} & 1 \end{bmatrix}$

9. The number of 3×3 matrices A whose entries are either

0 or 1 and for which the system $A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ has exactly

two distinct solutions, is

(2010)

- (a) 0 (b) $2^9 - 1$
(c) 168 (d) 2



10. Let $\omega \neq 1$ be a cube root of unity and S be the set of all non-

singular matrices of the form $\begin{bmatrix} 1 & a & b \\ \omega & 1 & c \\ \omega^2 & \omega & 1 \end{bmatrix}$, where each of

a, b and c is either ω or ω^2 . Then, the number of distinct matrices in the set S is (2011)

- (a) 2 (b) 6
(c) 4 (d) 8

11. How many 3×3 matrices M with entries from $\{0, 1, 2\}$ are there, for which the sum of the diagonal entries of $M^T M$ is 5? (2017)

- (a) 126 (b) 198
(c) 162 (d) 135

12. Let $M = \begin{bmatrix} \sin^4 \theta & -1 - \sin^2 \theta \\ 1 + \cos^2 \theta & \cos^4 \theta \end{bmatrix} = \alpha I + \beta M^{-1}$, where

$\alpha = \alpha(\theta)$ and $\beta = \beta(\theta)$ are real numbers and I is the 2×2 identity matrix.

If α^* is the minimum of the set $\{\alpha(\theta) : \theta \in [0, 2\pi)\}$ and β^* is the minimum of the set $\{\beta(\theta) : \theta \in [0, 2\pi)\}$, then

the value of $\alpha^* + \beta^*$ is (2019)

- (a) $\frac{-37}{16}$ (b) $\frac{-29}{16}$
(c) $\frac{-31}{16}$ (d) $\frac{-17}{16}$

Objective Questions II

[One or more than one correct option]

13. Let M and N be two 3×3 non-singular skew-symmetric matrices such that $MN = NM$. If P^T denotes the transpose of P, then $M^2 N^2 (M^T N)^{-1} (MN^{-1})^T$ is equal to (2011)

- (a) M^2 (b) $-N^2$
(c) $-M^2$ (d) MN

14. If the adjoint of a 3×3 matrix P is $\begin{bmatrix} 1 & 4 & 4 \\ 2 & 1 & 7 \\ 1 & 1 & 3 \end{bmatrix}$, then the possible value(s) of the determinant of P is/are (2012)

- (a) -2 (b) -1
(c) 1 (d) 2

15. For 3×3 matrices M and N, which of the following statement(s) is (are) not correct? (2013)

- (a) $N^T M N$ is symmetric or skew-symmetric, according as M is symmetric or skew-symmetric
(b) $MN - NM$ is skew symmetric for all symmetric matrices M and N
(c) MN is symmetric for all symmetric matrices M and N
(d) $(\text{adj } M)(\text{adj } N) = \text{adj } (MN)$ for all invertible matrices M and N

16. Let M and N be two 3×3 matrices such that $MN = NM$. Further, if $M \neq N^2$ and $M^2 = N^4$, then (2014)

- (a) determinant of $(M^2 + MN^2)$ is 0
(b) there is a 3×3 non-zero matrix U such that $(M^2 + MN^2)U$ is the zero matrix
(c) determinant of $(M^2 + MN^2) \geq 1$
(d) for a 3×3 matrix U, if $(M^2 + MN^2)U$ equals the zero matrix then U is the zero matrix

17. Let M be a 2×2 symmetric matrix with integer entries. Then M is invertible if (2014)

- (a) the first column of M is the transpose of the second row of M
(b) the second row of M is the transpose of the first column of M
(c) M is a diagonal matrix with nonzero entries in the main diagonal
(d) the product of entries in the main diagonal of M is not the square of an integer

18. Which of the following values of α satisfy the equation

$$\begin{vmatrix} (1+\alpha)^2 & (1+2\alpha)^2 & (1+3\alpha)^2 \\ (2+\alpha)^2 & (2+2\alpha)^2 & (2+3\alpha)^2 \\ (3+\alpha)^2 & (3+2\alpha)^2 & (3+3\alpha)^2 \end{vmatrix} = -648\alpha \quad (2015)$$

- (a) -4 (b) 9
(c) -9 (d) 4



19. Let X and Y be two arbitrary, 3×3 , non-zero, skew-symmetric matrices and Z be an arbitrary 3×3 , non zero, symmetric matrix. Then which of the following matrices is (are) skew symmetric ? (2015)

(a) $Y^3Z^4 - Z^4Y^3$ (b) $X^{44} + Y^{44}$
(c) $X^4Z^3 - Z^3X^4$ (d) $X^{23} - Y^{23}$

20. Let $P = \begin{bmatrix} 3 & -1 & -2 \\ 2 & 0 & \alpha \\ 3 & -5 & 0 \end{bmatrix}$, where $\alpha \in \mathbb{R}$. Suppose $Q = [q_{ij}]$

is a matrix such that $PQ = kI$, where $k \in \mathbb{R}$, $k \neq 0$ and I is the identity matrix of order 3. If $q_{23} = -\frac{k}{8}$ and $\det$

$(Q) = \frac{k^2}{2}$, then (2016)

(a) $\alpha = 0, k = 8$ (b) $4\alpha - k + 8 = 0$
(c) $\det(P \operatorname{adj}(Q)) = 2^9$ (d) $\det(Q \operatorname{adj}(P)) = 2^{13}$

21. Which of the following is(are) **NOT** the square of a 3×3 matrix with real entries? (2017)

(a) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ (b) $\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$

22. Let S be the set of all column matrices $\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$ such that

$b_1, b_2, b_3 \in \mathbb{R}$ and the system of equations (in real variables)

$-x + 2y + 5z = b_1$

$2x - 4y + 3z = b_2$

$x - 2y + 2z = b_3$

has at least one solution. Then, which of the following system(s) (in real variables) has (have) at least one

solution for each $\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \in S$? (2018)

- (a) $x + 2y + 3z = b_1, 4y + 5z = b_2$ and $x + 2y + 6z = b_3$
(b) $x + y + 3z = b_1, 5x + 2y + 6z = b_2$ and $-2x - y - 3z = b_3$
(c) $-x + 2y - 5z = b_1, 2x - 4y + 10z = b_2$ and $x - 2y + 5z = b_3$
(d) $x + 2y + 5z = b_1, 2x + 3z = b_2$ and $x + 4y - 5z = b_3$

23. Let $M = \begin{bmatrix} 0 & 1 & a \\ 1 & 2 & 3 \\ 3 & b & 1 \end{bmatrix}$ and $\operatorname{adj} M = \begin{bmatrix} -1 & 1 & -1 \\ 8 & -6 & 2 \\ -5 & 3 & -1 \end{bmatrix}$ Where a

and b are real numbers. Which of the following options is /are correct? (2019)

(a) $\det(\operatorname{adj} M^2) = 81$
(b) $a + b = 3$

(c) If $M \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, then $\alpha - \beta + \gamma = 3$

(d) $(\operatorname{adj} M)^{-1} + \operatorname{adj} M^{-1} = -M$

24. Let $x \in \mathbb{R}$ and let

$P = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}, Q = \begin{bmatrix} 2 & x & x \\ 0 & 4 & 0 \\ x & x & 6 \end{bmatrix}$ and $R = PQP^{-1}$

Then which of the following is/are correct (2019)

(a) There exists a real number x such that $PQ = QP$

(b) $\det R = \det \begin{bmatrix} 2 & x & x \\ 0 & 4 & 0 \\ x & x & 5 \end{bmatrix} + 8$ for all $x \in \mathbb{R}$

(c) For $x = 1$ there exists a unit vector $\alpha \hat{i} + \beta \hat{j} + \gamma \hat{k}$ for

which are $R \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

(d) For $x = 0$ if $R \begin{bmatrix} 1 \\ a \\ b \end{bmatrix} = 6 \begin{bmatrix} 1 \\ a \\ b \end{bmatrix}$ then $a + b = 5$



25. Let

$$P_1 = I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, P_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, P_3 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$P_4 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}, P_5 = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, P_6 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$\text{and } X = \sum_{k=1}^6 P_k \begin{bmatrix} 2 & 1 & 3 \\ 1 & 0 & 2 \\ 3 & 2 & 1 \end{bmatrix} P_k^T.$$

Where P_k^T denotes the transpose of matrix P_k . Then which of the following options is/are correct? (2019)

- (a) X is a symmetric matrix
(b) The sum of diagonal entries of X is 18.

(c) if $X \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \alpha \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, then $\alpha=30$

(d) $X - 30I$ is an invertible matrix

26. Let M be a 3×3 invertible matrix with real entries and let I denote the 3×3 identity matrix. If $M^{-1} = \text{adj}(\text{adj } M)$, then which of the following statement is/are ALWAYS TRUE? (2020)

- (a) $M = I$ (b) $\det M = 1$
(c) $M^2 = I$ (d) $(\text{adj } M)^2 = I$

27. For any 3×3 matrix M, let $|M|$ denote the determinant of M.

$$\text{Let } E = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 8 & 13 & 18 \end{bmatrix}, P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\text{and } F = \begin{bmatrix} 1 & 3 & 2 \\ 8 & 18 & 13 \\ 2 & 4 & 3 \end{bmatrix}$$

If Q is a non-singular matrix of order 3×3 , then which of

the following statements is/are TRUE?

(2021)

(a) $F = PEP$ and $P^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(b) $|EQ + PFQ^{-1}| = |EQ| + |PFQ^{-1}|$

(c) $|(EF)^3| > |EF|^2$

(d) Sum of the diagonal entries of $P^{-1}EP + F$ is equal to the sum of diagonal entries of $E + P^{-1}FP$

28. For any 3×3 matrix M, let $|M|$ denote the determinant of M. Let I be the 3×3 identity matrix. Let E and F be two 3×3 matrices such that $(I - EF)$ is invertible. If $G = (I - EF)^{-1}$, then which of the following statements is (are) TRUE? (2021)

(a) $|FE| = |I - FE||FGE|$

(b) $(I - FE)(I + FGE) = I$

(c) $EFG = GEF$

(d) $(I - FE)(I - FGE) = I$

Numerical Value Type Questions

29. If matrix $A = \begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix}$ where a, b, c are real positive

numbers, $abc = 1$ and $A^T A = I$, then find value of

$a^3 + b^3 + c^3$. (2003)

30. Let k be a positive real number and let

$$A = \begin{bmatrix} 2k-1 & 2\sqrt{k} & 2\sqrt{k} \\ 2\sqrt{k} & 1 & -2k \\ -2\sqrt{k} & 2k & -1 \end{bmatrix} \text{ and } B = \begin{bmatrix} 0 & 2k-1 & \sqrt{k} \\ 1-2k & 0 & 2\sqrt{k} \\ -\sqrt{k} & -2\sqrt{k} & 0 \end{bmatrix}$$

If $\det(\text{adj } A) + \det(\text{adj } B) = 10^6$, then $[k]$ is equal to....

(2010)



31. The number of all possible values of θ , where $0 < \theta < \pi$, for which the system of equations

$$(y+z) \cos 3\theta = (xyz) \sin 3\theta$$

$$x \sin 3\theta = \frac{2 \cos 3\theta}{y} + \frac{2 \sin 3\theta}{z}$$

and $(xyz) \sin 3\theta = (y+2z) \cos 3\theta + y \sin 3\theta$ have a solution (x_0, y_0, z_0) with $y_0 z_0 \neq 0$, is (2010)

32. Let ω be the complex number $\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}$. Then the number of distinct complex number z satisfying

$$\begin{vmatrix} z+1 & \omega & \omega^2 \\ \omega & z+\omega^2 & 1 \\ \omega^2 & 1 & z+\omega \end{vmatrix} = 0 \text{ is equal to} \quad (2010)$$

33. Let M be a 3×3 matrix satisfying

$$M \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix}, M \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}, \text{ and } M \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 12 \end{bmatrix}$$

Then, the sum of the diagonal entries of M is (2011)

34. Let $z = \frac{-1 + \sqrt{3}i}{2}$, where $i = \sqrt{-1}$, and $r, s \in \{1, 2, 3\}$.

$$\text{Let } P = \begin{bmatrix} (-z)^r & z^{2s} \\ z^{2s} & z^r \end{bmatrix} \text{ and } I \text{ be the identity matrix of}$$

order 2. Then the total number of ordered pairs (r, s) for which $P^2 = -I$ is (2016)

35. The total number of distinct $x \in \mathbb{R}$ for which

$$\begin{vmatrix} x & x^2 & 1+x^3 \\ 2x & 4x^2 & 1+8x^3 \\ 3x & 9x^2 & 1+27x^3 \end{vmatrix} = 10 \text{ is} \quad (2016)$$

36. For a real number α , if the system

$$\begin{bmatrix} 1 & \alpha & \alpha^2 \\ \alpha & 1 & \alpha \\ \alpha^2 & \alpha & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ -z \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \text{ of linear equations, has}$$

infinitely many solutions, then $1 + \alpha + \alpha^2 =$ (2017)

37. The trace of a square matrix is defined to be the sum of its diagonal entries. If A is a 2×2 matrix such that the trace of A is 3 and the trace of A^3 is -18, then the value of the determinant of A is _____. (2020)

38. Let α, β and γ be real numbers such that the system of linear equations

$$x + 2y + 3z = \alpha$$

$$4x + 5y + 6z = \beta$$

$$7x + 8y + 9z = \gamma - 1$$

is consistent. Let $|M|$ represent the determinant of the matrix

$$M = \begin{bmatrix} \alpha & 2 & \gamma \\ \beta & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

The value of $|M|$ is _____. (2021)

Assertion & Reason

- (A) If ASSERTION is true, REASON is true, REASON is a correct explanation for ASSERTION.
(B) If ASSERTION is true, REASON is true, REASON is not a correct explanation for ASSERTION.
(C) If ASSERTION is true, REASON is false
(D) If ASSERTION is false, REASON is true

39. Consider the system of equations

$$x - 2y + 3z = -1, -x + y - 2z = k, x - 3y + 4z = 1$$

Assertion : The system of equations has no solution for $k \neq 3$.

Reason : The determinant $\begin{vmatrix} 1 & 3 & -1 \\ -1 & -2 & k \\ 1 & 4 & 1 \end{vmatrix} \neq 0$, for $k \neq 3$

(1997)

- (a) A (b) B
(c) C (d) D



Paragraph Type Questions

Using the following passage, solve Q.40 to Q.42

Passage

$A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix}$, If U_1, U_2 are U_3 are columns matrices

satisfying $AU_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $AU_2 = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}$ and $AU_3 = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$ and U is

3×3 matrix when columns are U_1, U_2, U_3 then answer the following questions.

40. The value of $|U|$ is (2006)

- (a) 3 (b) -3
(c) 3/2 (d) 2

41. The sum of the elements of U^{-1} is (2006)

- (a) -1 (b) 0
(c) 1 (d) 3

42. The value of $[3 \ 2 \ 0] U \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix}$ is (2006)

- (a) 5 (b) 5/2
(c) 4 (d) 3/2

Using the following passage, solve Q.43 to Q.45

Passage

Let p be an odd prime number and T_p be the following set of 2×2 matrices

$$T_p = \left\{ A = \begin{bmatrix} a & b \\ c & a \end{bmatrix}; a, b, c \in \{0, 1, 2, \dots, p-1\} \right\} \quad (2010)$$

43. The number of A in T_p such that A is either symmetric or skew-symmetric or both, and $\det(A)$ is divisible by p is

- (a) $(p-1)^2$ (b) $2(p-1)$
(c) $(p-1)^2 + 1$ (d) $2p-1$

44. The number of A in T_p such that the trace of A is not divisible by p but $\det(A)$ is divisible by p is

[Note : The trace of a matrix is the sum of its diagonal entries]

- (a) $(p-1)(p^2-p+1)$ (b) $p^3-(p-1)^2$
(c) $(p-1)^2$ (d) $(p-1)(p^2-2)$

45. The number of A in T_p such that $\det(A)$ is not divisible by p , is

- (a) $2p^2$ (b) p^3-5p
(c) p^3-3p (d) p^3-p^2

Find Answer Key and Detailed Solutions at the end of this book



DETERMINANTS & MATRICES



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RELATIONS , FUNCTIONS & INVERSE TRIGONOMETRIC FUNCTIONS

RELATIONS, FUNCTIONS & INVERSE TRIGONOMETRIC FUNCTIONS



RELATIONS

1. CARTESIAN PRODUCT OF SETS

Definition : Given two non-empty sets P & Q. The cartesian product $P \times Q$ is the set of all ordered pairs of elements from P & Q i.e.

$$P \times Q = \{(p, q); p \in P; q \in Q\}$$

2. RELATIONS

2.1 Definition : Let A & B be two non-empty sets. Then any subset 'R' of $A \times B$ is a relation from A to B.

If $(a, b) \in R$, then we write it as $a R b$ which is read as 'a is related to b' by the relation R, 'b' is also called image of 'a' under R.

2.2 Domain and range of a relation : If R is a relation from A to B, then the set of first elements in R is called domain & the set of second elements in R is called range of R. symbolically.

$$\text{Domain of } R = \{x : (x, y) \in R\}$$

$$\text{Range of } R = \{y : (x, y) \in R\}$$

The set B is called co-domain of relation R.

Note that $\text{range} \subset \text{co-domain}$.

NOTES:

Total number of relations that can be defined from a set A to a set B is the number of possible subsets of $A \times B$. If $n(A) = p$ and $n(B) = q$, then $n(A \times B) = pq$ and total number of relations is 2^{pq} .

2.3 Inverse of a relation : Let A, B be two sets and let R be a relation from a set A to set B. Then the inverse of R, denoted by R^{-1} , is a relation from B to A and is defined by

$$R^{-1} = \{(b, a) : (a, b) \in R\}$$

Clearly, $(a, b) \in R \Leftrightarrow (b, a) \in R^{-1}$

Also, $\text{Domain}(R) = \text{Range}(R^{-1})$ and $\text{Range}(R) = \text{Domain}(R^{-1})$.

3. TYPES OF RELATION

- (a) **Void Relation :** Let A be a non-empty set. Then $\phi \subseteq A \times A$ and so it is a relation on set A. This relation is called the void or empty relation on set A.
- (b) **Universal Relation :** Let A be a non-empty set. Then, $A \times A$ is known as the universal relation set A.
- (c) **Identity Relation :** Let A be a non-empty set. Then, $I_A = \{(a, a) : a \in A\}$ is called the identity relation on A.
- (d) **Reflexive Relation :** A relation R on a set A is said to be reflexive if every element of A is related to itself. Thus, R is reflexive $\Leftrightarrow (a, a) \in R$ for all $a \in A$.
- (e) **Symmetric Relation :** A relation R on a set A is said to be a symmetric relation iff $(a, b) \in R \Rightarrow (b, a) \in R$ for all $a, b \in A$.
i.e. $a R b \Rightarrow b R a$ for all $a, b \in A$.
- (f) **Antisymmetric Relation :** A relation R on set A is said to be antisymmetric relation iff $(a, b) \in R$ and $(b, a) \in R \Rightarrow a = b$ for all $a, b \in A$.
- (g) **Transitive Relation :**
We say that a relation R on a set A is transitive if whenever $a R b$ and $b R c$, then $a R c$.
It means that if a related to b and b related to c then a related to c for all $(a, b, c) \in A$.
- (h) **Equivalence Relation :** A relation R on a set A is said to be an equivalence relation on A iff
 - (i) it is reflexive
 - (ii) it is symmetric and
 - (iii) it is transitive

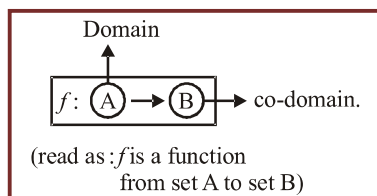
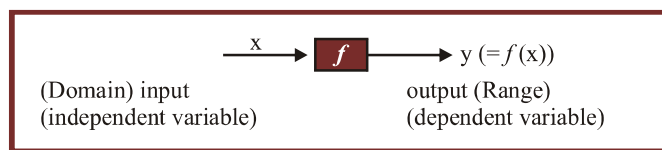
FUNCTIONS

1. DEFINITION

A relation 'f' from a set A to set B is said to be a function if every element of set A has one and only one image in set B.



Notations



2. DOMAIN, CO-DOMAIN AND RANGE OF A FUNCTION

Domain : When we define $y = f(x)$ with a formula and the domain is not stated explicitly, the domain is assumed to be the largest set of x -values for which the formula gives real y -values.

The domain of $y = f(x)$ is the set of all real x for which $f(x)$ is defined (real).

Rules for finding Domain :

- Expression under even root (i.e. square root, fourth root etc.) should be non-negative.
- Denominator $\neq 0$.
- $\log_a x$ is defined when $x > 0$, $a > 0$ and $a \neq 1$.
- If domain of $y = f(x)$ and $y = g(x)$ are D_1 and D_2 respectively, then the domain of $f(x) \pm g(x)$ or $f(x) \cdot g(x)$ is $D_1 \cap D_2$. While domain of

$$\frac{f(x)}{g(x)} \text{ is } D_1 \cap D_2 - \{x: g(x) = 0\}.$$

Range : The set of all f -images of elements of A is known as the range of f & denoted by $f(A)$.

$$\text{Range} = f(A) = \{f(x) : x \in A\};$$

$$f(A) \subseteq B \text{ \{Range} \subseteq \text{Co-domain}\}.$$

Rule for finding range :

First of all find the domain of $y = f(x)$

- If domain $\in$ finite number of points
 $\Rightarrow$ range $\in$ set of corresponding $f(x)$ values.
- If domain $\in \mathbb{R}$ or $\mathbb{R} - \{\text{some finite points}\}$
 Put $y = f(x)$

Then express x in terms of y . From this find y for x to be defined. (i.e., find the values of y for which x exists).

- If domain $\in$ a finite interval, find the least and greater value for range using monotonicity.

NOTES :

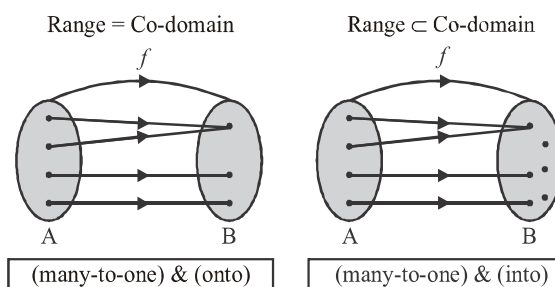
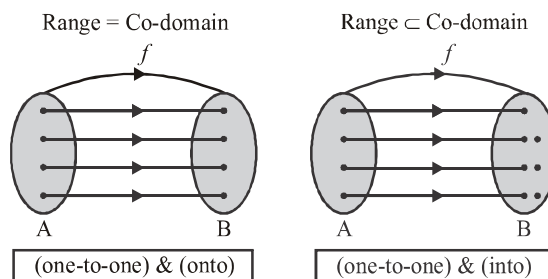
Two functions f & g are said to be equal (identical) iff

- Domain of f = Domain of g
- Co-Domain of f = Co-Domain of g
- $f(x) = g(x) \forall x \in \text{Domain}$.

3. CLASSIFICATION OF FUNCTION

Definition 1 : A function $f : X \rightarrow Y$ is defined to be one-one (or injective), if the images of distinct elements of X under f are distinct, i.e., for every $x_1, x_2 \in X$, $f(x_1) = f(x_2)$ implies $x_1 = x_2$. Otherwise, f is called many-one.

Definition 2 : A function $f : X \rightarrow Y$ is said to be onto (or surjective), if every element of Y is the image of some element of X under f , i.e., for every $y \in Y$, there exists an element x in X such that $f(x) = y$.





Methods to check one-one mapping

- Theoretically :** If $f(x_1) = f(x_2)$
 $\Rightarrow x_1 = x_2$ only, then $f(x)$ is one-one.
- Graphically :** A function is one-one, iff no line parallel to x-axis meets the graph of function at more than one point.
- By Calculus :** For checking whether $f(x)$ is One-One, find whether function is only increasing or only decreasing in their domain. If yes, then function is one-one,

i.e. if $f'(x) \geq 0, \forall x \in \text{domain}$

or if $f'(x) \leq 0, \forall x \in \text{domain}$,

then function is one-one.

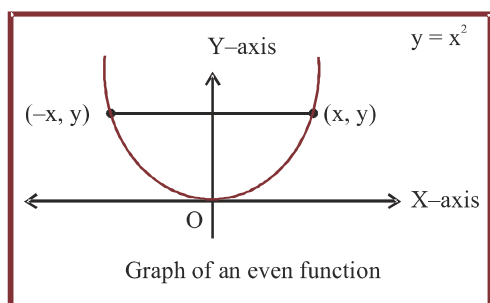
Methods to check into/onto mapping

Find the range of $f(x)$ and compare with co-domain. If range equals co-domain then function is onto, otherwise it is into.

4. EVEN AND ODD FUNCTIONS

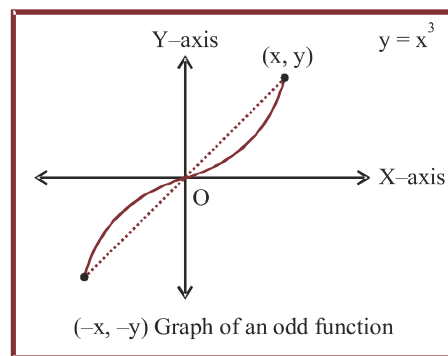
1. Even Function : $f(-x) = f(x), \forall x \in \text{Domain}$

The graph of an even function $y = f(x)$ is symmetric about the y-axis. i.e., (x, y) lies on the graph $\Leftrightarrow (-x, y)$ lies on the graph.



2. Odd Function : $f(-x) = -f(x), \forall x \in \text{Domain}$

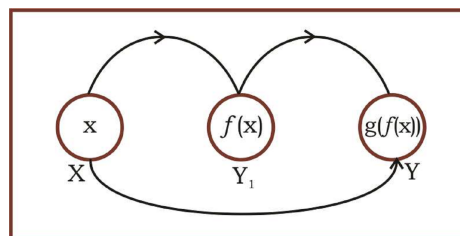
The graph of an odd function $y = f(x)$ is symmetric about origin i.e. if point (x, y) is on the graph of an odd function, then $(-x, -y)$ will also lie on the graph.



5. COMPOSITE FUNCTIONS

Let us consider two functions, $f: X \rightarrow Y_1$ and $g: Y_1 \rightarrow Y$. We define function $h: X \rightarrow Y$; such that

$$h(x) = g(f(x)) = (g \circ f)(x).$$



To obtain $h(x)$, we first take f -image of an element $x \in X$ so that $f(x) \in Y_1$, which is the domain of $g(x)$. Then take g -image of $f(x)$, i.e., $g(f(x))$ which would be an element of Y .

NOTES:

It should be noted that $g \circ f$ exists iff; the range of $f \subseteq \text{domain of } g$. Similarly, $f \circ g$ exists; iff; the range of $g \subseteq \text{domain of } f$.

6. INVERSE OF FUNCTION

6.1 Definition : Let $f: A \rightarrow B$ be a one-one and onto function, then there exists a unique function, $g: B \rightarrow A$ such that $f(x) = y \Leftrightarrow g(y) = x, \forall x \in A$ and $y \in B$. Then g is said to be inverse of f .

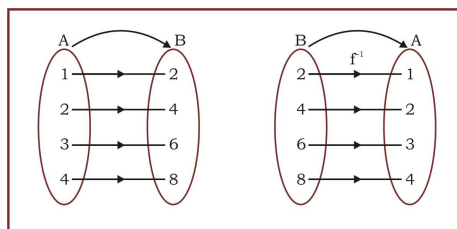
$$\text{Thus, } g = f^{-1}: B \rightarrow A = \{(f(x), x) | (x, f(x)) \in f\}$$



Let us consider one-one function with domain A and range B.

where $A = \{1, 2, 3, 4\}$ and $B = \{2, 4, 6, 8\}$ and $f: A \rightarrow B$ is given by $f(x) = 2x$, then write f and f^{-1} as a set of ordered pairs.

Here, member $y \in B$ arises from one and only one member $x \in A$.



So, $f = \{(1, 2), (2, 4), (3, 6), (4, 8)\}$

and $f^{-1} = \{(2, 1), (4, 2), (6, 3), (8, 4)\}$

NOTES:

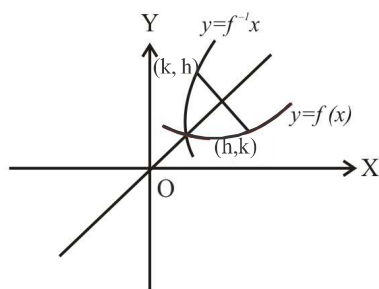
In above function, Domain of $f = \{1, 2, 3, 4\}$ = range of f^{-1}

Range of $f = \{2, 4, 6, 8\}$ = domain of f^{-1}

Which represents for a function to have its inverse, it must be one-one onto or bijective.

6.2 Graph of the inverse of an invertible function :

Let (h, k) be a point on the graph of the function f . Then (k, h) is the corresponding point on the graph of inverse of f i.e., f^{-1} .



The line segment joining the points (h, k) and (k, h) is bisected at right angle by the line $y = x$.

So that the two points play object-image role in the line $y = x$ as plane mirror.

It follows that the graph of $y = f(x)$ and its inverse written in form $y = g(x)$ are symmetrical about the line $y = x$.

6.3 Properties of inverse of a function

- The inverse of bijection is unique.
- The inverse of bijection is also bijection.
- If $f: A \rightarrow B$ is bijection and $g: B \rightarrow A$ is inverse of f , then $f \circ g = I_B$ and $g \circ f = I_A$.
Where, I_A and I_B are identity function on the sets A and B respectively.
- If $f: A \rightarrow B$ and $g: B \rightarrow C$ are two bijections, then $g \circ f: A \rightarrow C$ is bijection and $(g \circ f)^{-1} = (f^{-1} \circ g^{-1})$.
- In general, $f \circ g \neq g \circ f$ but if, $(f \circ g)(x) = x$ and $(g \circ f)(x) = x$, then $f^{-1} = g$ and $g^{-1} = f$.

7. FUNCTIONAL EQUATION

If x, y are independent variable then ;

- $f(xy) = f(x) + f(y) \Rightarrow f(x) = k \ln x$ or $f(x) = 0$
- $f(xy) = f(x) \cdot f(y) \Rightarrow f(x) = x^n, n \in \mathbb{R}$ or $f(x) = 0$
- $f(x + y) = f(x) \cdot f(y) \Rightarrow f(x) = a^{kx}$ or $f(x) = 0$
- $f(x)$ is continuous and takes rational values for all $x \Rightarrow f(x)$ is constant function.
- By considering a general n^{th} degree polynomial and writing the expression.

$$f(x) \cdot f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$$

$$\Rightarrow f(x) = \pm x^n + 1 = 1 \pm x^n$$

BINARY OPERATIONS

Definition 1 : A binary operations $*$ on a set A is a function $*$: $A \times A \rightarrow A$. We denote $*(a, b)$ by $a * b$.

Definition 2 : A binary operation $*$ on a set A is called commutative, if $a * b = b * a$, for every $a, b \in A$.

Definition 3 : A binary operation $*$: $A \times A \rightarrow A$ is said to be associative if $(a * b) * c = a * (b * c)$, $\forall a, b, c, \in A$.



Definition 4 : Given a binary operation $*$: $A \times A \rightarrow A$, an element $e \in A$, if it exists, is called identity for the operation $*$, if $a * e = a = e * a$, $\forall a \in A$.

INVERSE TRIGONOMETRIC FUNCTIONS

1. INTRODUCTION

Function	Domain	Range
1. $y = \sin^{-1} x$ iff $x = \sin y$	$-1 \leq x \leq 1$,	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
2. $y = \cos^{-1} x$ iff $x = \cos y$	$-1 \leq x \leq 1$	$[0, \pi]$
3. $y = \tan^{-1} x$ iff $x = \tan y$	$-\infty < x < \infty$	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
4. $y = \cot^{-1} x$ iff $x = \cot y$	$-\infty < x < \infty$	$(0, \pi)$
5. $y = \operatorname{cosec}^{-1} x$ iff $x = \operatorname{cosec} y$	$(-\infty, -1] \cup [1, \infty)$	$\left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right]$
6. $y = \sec^{-1} x$ iff $x = \sec y$	$(-\infty, -1] \cup [1, \infty)$	$\left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right]$

NOTES :

(i) $\sin^{-1} x$ & $\tan^{-1} x$ are increasing functions in their domain.

(ii) $\cos^{-1} x$ & $\cot^{-1} x$ are decreasing functions in their domain.

2. COMPOSITION OF TRIGONOMETRIC AND INVERSE TRIGONOMETRIC FUNCTIONS

2.1 (i) $\sin(\sin^{-1} x) = x$, for all $x \in [-1, 1]$

(ii) $\cos(\cos^{-1} x) = x$, for all $x \in [-1, 1]$

(iii) $\tan(\tan^{-1} x) = x$, for all $x \in \mathbb{R}$

(iv) $\operatorname{cosec}(\operatorname{cosec}^{-1} x) = x$, for all $x \in (-\infty, -1] \cup [1, \infty)$

(v) $\sec(\sec^{-1} x) = x$, for all $x \in (-\infty, -1] \cup [1, \infty)$

(vi) $\cot(\cot^{-1} x) = x$, for all $x \in \mathbb{R}$

Proof. We know that, if $f : A \rightarrow B$ is a bijection, then $f^{-1} : B \rightarrow A$ exists such that $f \circ f^{-1}(y) = f(f^{-1}(y)) = y$ for all $y \in B$.

Clearly, all these results are direct consequences of this property.

Aliter : Let $\theta \in [-\pi/2, \pi/2]$ and $x \in [-1, 1]$ such that $\sin \theta = x$ then, $\theta = \sin^{-1} x$

$\therefore x = \sin \theta = \sin(\sin^{-1} x)$

Hence, $\sin(\sin^{-1} x) = x$ for all $x \in [-1, 1]$

Similarly, we can prove other results.

2.2 $T^{-1}(T(x)) \neq x$ always

$T^{-1}(T(x)) = x$ when x lies in principal domain of T .

eg: $\sin^{-1}(\sin \theta) \neq \theta$, if $\theta \notin [-\pi/2, \pi/2]$. Infact, we have

$$\sin^{-1}(\sin \theta) = \begin{cases} -\pi - \theta, & \text{if } \theta \in [-3\pi/2, -\pi/2] \\ \theta, & \text{if } \theta \in [-\pi/2, \pi/2] \\ \pi - \theta, & \text{if } \theta \in [\pi/2, 3\pi/2] \\ -2\pi + \theta, & \text{if } \theta \in [3\pi/2, 5\pi/2] \end{cases} \text{ and so on.}$$

Similarly,

$$\cos^{-1}(\cos \theta) = \begin{cases} -\theta, & \text{if } \theta \in [-\pi, 0] \\ \theta, & \text{if } \theta \in [0, \pi] \\ 2\pi - \theta, & \text{if } \theta \in [\pi, 2\pi] \\ -2\pi + \theta, & \text{if } \theta \in [2\pi, 3\pi] \end{cases} \text{ and so on.}$$



$$\tan^{-1}(\tan \theta) = \begin{cases} \pi + \theta, & \text{if } \theta \in (-3\pi/2, -\pi/2) \\ \theta, & \text{if } \theta \in (-\pi/2, \pi/2) \\ \theta - \pi, & \text{if } \theta \in (\pi/2, 3\pi/2) \\ \theta - 2\pi, & \text{if } \theta \in (3\pi/2, 5\pi/2) \end{cases} \text{ and so on.}$$

Graph for $y = \sin^{-1}(\sin x)$

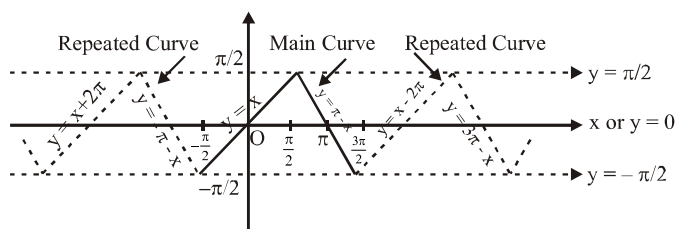
As, $y = \sin^{-1}(\sin x)$ is periodic with period 2π .

$\therefore$ to draw this graph we should draw the graph for one interval of length 2π and repeat for entire values of x .

As we know,

$$\sin^{-1}(\sin x) = \begin{cases} x, & -\frac{\pi}{2} \leq x \leq \frac{\pi}{2} \\ \pi - x, & \frac{\pi}{2} \leq x \leq \frac{3\pi}{2} \end{cases}$$

which is defined for the interval of length 2π , plotted as ;



Thus, the graph for $y = \sin^{-1}(\sin x)$, is a straight line up and a straight line down with slopes 1 and -1 respectively lying

between $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

Graph for $y = \cos^{-1}(\cos x)$

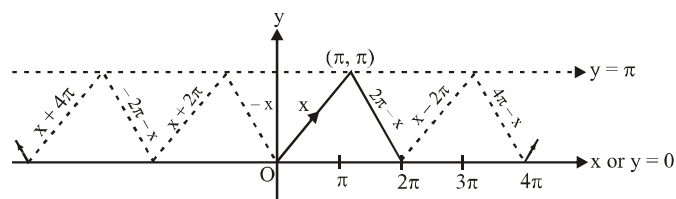
As, $y = \cos^{-1}(\cos x)$ is periodic with period 2π .

$\therefore$ to draw this graph we should draw the graph for one interval of length 2π and repeat for entire values of x of length 2π .

As we know;

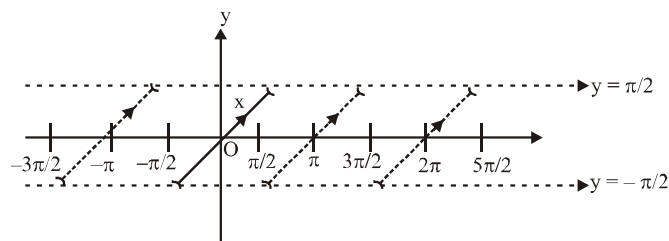
$$\cos^{-1}(\cos x) = \begin{cases} x; & 0 \leq x \leq \pi \\ 2\pi - x; & \pi \leq x \leq 2\pi, \end{cases}$$

Thus, it has been defined for $0 \leq x \leq 2\pi$ that has length 2π . So, its graph could be plotted as;



Thus, the curve $y = \cos^{-1}(\cos x)$.

Graph for $y = \tan^{-1}(\tan x)$



This is the curve for $y = \tan^{-1}(\tan x)$, where y is not defined

for $x \in (2n+1)\frac{\pi}{2}$.

3. PROPERTIES

3.1 PROPERTY – I

- (i) $\cos^{-1}(-x) = \pi - \cos^{-1}(x)$, for all $x \in [-1, 1]$
- (ii) $\sec^{-1}(-x) = \pi - \sec^{-1}x$, for all $x \in (-\infty, -1] \cup [1, \infty)$
- (iii) $\cot^{-1}(-x) = \pi - \cot^{-1}x$, for all $x \in \mathbb{R}$
- (iv) $\sin^{-1}(-x) = -\sin^{-1}(x)$, for all $x \in [-1, 1]$
- (v) $\tan^{-1}(-x) = -\tan^{-1}x$, for all $x \in \mathbb{R}$
- (vi) $\operatorname{cosec}^{-1}(-x) = -\operatorname{cosec}^{-1}x$, for all $x \in (-\infty, -1] \cup [1, \infty)$

Proof. (i) Clearly, $-x \in [-1, 1]$ for all $x \in [-1, 1]$

let $\cos^{-1}(-x) = \theta$... (i)

then, $-x = \cos \theta$

$\Rightarrow x = -\cos \theta$

$\Rightarrow x = \cos(\pi - \theta)$

$\Rightarrow \cos^{-1}x = \cos^{-1}(\cos(\pi - \theta))$

$\cos^{-1}x = \pi - \theta$ { $\because x \in (-1, 1)$ and $\pi - \theta \in [0, \pi]$ for all $\theta \in [0, \pi]$ }

$\Rightarrow \theta = \pi - \cos^{-1}x$... (ii)

from (i) and (ii), we get

$\cos^{-1}(-x) = \pi - \cos^{-1}x$

Similarly, we can prove other results.

(iv) Clearly, $-x \in [-1, 1]$ for all $x \in [-1, 1]$

let $\sin^{-1}(-x) = \theta$

then, $-x = \sin \theta$... (i)

$\Rightarrow x = -\sin \theta$

$\Rightarrow x = \sin(-\theta) \Rightarrow \sin^{-1}(x) = \sin^{-1}(\sin(-\theta))$



$\Rightarrow -\theta = \sin^{-1} x \quad \dots (ii)$
 $\{ \because x \in [-1, 1] \text{ and } -\theta \in [-\pi/2, \pi/2] \text{ for all } \theta \in [-\pi/2, \pi/2]$
 from (i) and (ii), we get
 $\sin^{-1}(-x) = -\sin^{-1}(x)$

3.2 PROPERTY – II

(i) $\sin^{-1}\left(\frac{1}{x}\right) = \operatorname{cosec}^{-1} x$, for all $x \in (-\infty, -1] \cup [1, \infty)$

Proof. Let, $\operatorname{cosec}^{-1} x = \theta \quad \dots (i)$
 then, $x = \operatorname{cosec} \theta$

$\Rightarrow \frac{1}{x} = \sin \theta \Rightarrow \sin^{-1}\left(\frac{1}{x}\right) = \sin^{-1}(\sin \theta) = \theta$
 $\{ \because x \in (-\infty, -1] \cup [1, \infty) \Rightarrow \frac{1}{x} \in [-1, 1] - \{0\}$
 $\operatorname{cosec}^{-1} x = \theta \Rightarrow \theta \in [-\pi/2, \pi/2] - \{0\}$

$\Rightarrow \theta = \sin^{-1}\left(\frac{1}{x}\right) \quad \dots (ii)$

from (i) and (ii); we get

$$\sin^{-1}\left(\frac{1}{x}\right) = \operatorname{cosec}^{-1} x$$

(ii) $\cos^{-1}\left(\frac{1}{x}\right) = \sec^{-1} x$, for all $x \in (-\infty, -1] \cup [1, \infty)$

(iii) $\tan^{-1}\left(\frac{1}{x}\right) = \begin{cases} \cot^{-1} x, & \text{for } x > 0 \\ -\pi + \cot^{-1} x, & \text{for } x < 0 \end{cases}$

Proof. Let $\cot^{-1} x = \theta$. Then $x \in \mathbb{R}$, $x \neq 0$ and $\theta \in (0, \pi) \quad \dots (i)$

Now two cases arises :

Case I : When $x > 0$

In this case, $\theta \in (0, \pi/2)$

$$\therefore \cot^{-1} x = \theta$$

$$\Rightarrow x = \cot \theta$$

$$\Rightarrow \frac{1}{x} = \tan \theta \Rightarrow \tan^{-1}\left(\frac{1}{x}\right) = \tan^{-1}(\tan \theta)$$

$$\theta = \tan^{-1}\left(\frac{1}{x}\right) \quad \dots (ii) \quad \{ \because \theta \in (0, \pi/2) \}$$

from (i) and (ii), we get

$$\tan^{-1}\left(\frac{1}{x}\right) = \cot^{-1} x, \text{ for all } x > 0.$$

Case II : When $x < 0$

In this case $\theta \in (\pi/2, \pi) \quad \{ \because x = \cot \theta < 0 \}$

$$\text{Now, } \frac{\pi}{2} < \theta < \pi$$

$$\Rightarrow -\frac{\pi}{2} < \theta - \pi < 0$$

$$\Rightarrow \theta - \pi \in (-\pi/2, 0)$$

$$\therefore \cot^{-1} x = \theta$$

$$\Rightarrow x = \cot \theta$$

$$\Rightarrow \frac{1}{x} = \tan \theta$$

$$\Rightarrow \frac{1}{x} = -\tan(\pi - \theta)$$

$$\Rightarrow \frac{1}{x} = \tan(\theta - \pi) \quad \{ \because \tan(\pi - \theta) = -\tan \theta \}$$

$$\Rightarrow \tan^{-1}\left(\frac{1}{x}\right) = \tan^{-1}(\tan(\theta - \pi))$$

$$\Rightarrow \tan^{-1}\left(\frac{1}{x}\right) = -\pi + \theta \quad \dots (iii) \quad \{ \because \theta - \pi \in (-\pi/2, 0) \}$$

from (i) and (iii), we get

$$\tan^{-1}\left(\frac{1}{x}\right) = -\pi + \cot^{-1} x, \text{ if } x < 0$$

Hence,

$$\tan^{-1}\left(\frac{1}{x}\right) = \begin{cases} \cot^{-1} x, & \text{for } x > 0 \\ -\pi + \cot^{-1} x, & \text{for } x < 0 \end{cases}$$



3.3 PROPERTY – III

- (i) $\sin^{-1} x + \cos^{-1} x = \pi/2$, for all $x \in [-1, 1]$
- (ii) $\tan^{-1} x + \cot^{-1} x = \pi/2$, for all $x \in \mathbb{R}$
- (iii) $\sec^{-1} x + \operatorname{cosec}^{-1} x = \frac{\pi}{2}$, for all $x \in (-\infty, -1] \cup [1, \infty)$

Proof. Let, $\sin^{-1} x = \theta$... (i)

then, $\theta \in [-\pi/2, \pi/2]$ ($\because x \in [-1, 1]$)

$$\Rightarrow -\pi/2 \leq \theta \leq \pi/2$$

$$\Rightarrow -\pi/2 \leq -\theta \leq \pi/2$$

$$\Rightarrow 0 \leq \frac{\pi}{2} - \theta \leq \pi$$

$$\Rightarrow \frac{\pi}{2} - \theta \in [0, \pi]$$

Now, $\sin^{-1} x = \theta$

$$\Rightarrow x = \sin \theta$$

$$\Rightarrow x = \cos\left(\frac{\pi}{2} - \theta\right) \Rightarrow \cos^{-1} x = \cos^{-1}\left(\cos\left(\frac{\pi}{2} - \theta\right)\right)$$

$$\Rightarrow \cos^{-1} x = \frac{\pi}{2} - \theta$$

{ $\because x \in [-1, 1]$ and $(\pi/2 - \theta) \in [0, \pi]$ }

$$\Rightarrow \theta + \cos^{-1} x = \pi/2 \quad \dots (ii)$$

from (i) and (ii), we get

$$\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$$

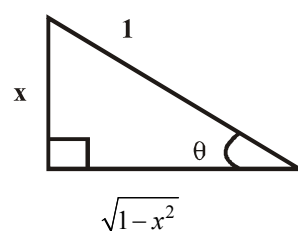
4. INTERCONVERSION

4.1 If $x > 0$

If $\sin^{-1} x = \theta$ { θ lies in I quadrant}

Then to convert $\sin^{-1} x$ to other inverse trigonometric

functions, $\sin \theta = \frac{x}{1} = \frac{p}{h}$



$$\text{So, } \cos \theta = \sqrt{1-x^2} \Rightarrow \theta = \cos^{-1}(\sqrt{1-x^2})$$

$$\Rightarrow \sin^{-1} x = \cos^{-1}(\sqrt{1-x^2})$$

$$\text{and } \tan \theta = \frac{x}{\sqrt{1-x^2}} \Rightarrow \theta = \tan^{-1}\left(\frac{x}{\sqrt{1-x^2}}\right)$$

Similarly, $\sin^{-1} x$ can be converted to any other inverse trigonometric function.

Similar procedure can be applied to convert any inverse trigonometric ratio to any other inverse trigonometric ratio.

4.2 when $x < 0$

We can convert these to positive number first.

eg. $\sin^{-1} x = \sin^{-1}(-(-x)) = -\sin^{-1}(-x)$ and

$$\cos^{-1}(x) = \cos^{-1}(-(-x)) = \pi - \cos^{-1}(-x).$$

Now $(-x)$ is positive and so procedure learnt in 4.1 can be applied to it.

5. SUM AND DIFFERENCE FORMULAE

$$(i) \quad \tan^{-1} x + \tan^{-1} y = \begin{cases} \tan^{-1}\left(\frac{x+y}{1-xy}\right), & x \geq 0, y \geq 0, xy < 1 \\ \pi + \tan^{-1}\left(\frac{x+y}{1-xy}\right), & x \geq 0, y \geq 0, xy > 1 \\ \frac{\pi}{2}, & x \geq 0, y \geq 0, xy = 1 \end{cases}$$

$$(ii) \quad \tan^{-1} x - \tan^{-1} y = \tan^{-1}\left(\frac{x-y}{1+xy}\right), \quad x \geq 0, y \geq 0$$

$$(iii) \quad \sin^{-1} x + \sin^{-1} y = \begin{cases} \sin^{-1}(x\sqrt{1-y^2} + y\sqrt{1-x^2}) & \text{if } x \geq 0, y \geq 0, x^2 + y^2 \leq 1 \\ \pi - \sin^{-1}(x\sqrt{1-y^2} + y\sqrt{1-x^2}) & \text{if } x \geq 0, y \geq 0, x^2 + y^2 > 1 \end{cases}$$

$$(iv) \quad \sin^{-1} x - \sin^{-1} y = \sin^{-1}(x\sqrt{1-y^2} - y\sqrt{1-x^2}), \quad \text{if } 0 \leq x, y \leq 1$$

$$(v) \quad \cos^{-1} x + \cos^{-1} y = \cos^{-1}[xy - \sqrt{(1-x^2)(1-y^2)}], \quad 0 \leq x, y \leq 1$$



$$(vi) \cos^{-1} x - \cos^{-1} y = \begin{cases} \cos^{-1} \left[xy + \sqrt{(1-x^2)(1-y^2)} \right], & 0 \leq x < y \leq 1 \\ -\cos^{-1} \left[xy + \sqrt{(1-x^2)(1-y^2)} \right], & 0 \leq y < x \leq 1 \end{cases}$$

6. SUMMATION OF SERIES

The formula to be used in such problems is

$$\tan^{-1} \left(\frac{x-y}{1+xy} \right) = \tan^{-1} x - \tan^{-1} y$$

So first convert tan inverse term to form given in L.H.S.

If series given is in some other inverse trigonometric function, then first convert it to tan inverse using interconversion.

7. SIMPLIFICATION

Terms involving inverse trigonometric ratios can be simplified using proper trigonometric substitutions. For example,

$$1. \quad \tan^{-1} \left(\frac{2x}{1-x^2} \right) = 2 \tan^{-1} x, \quad |x| < 1.$$

For this we use substitution $x = \tan \theta$ in LHS.

$$2. \quad \sin^{-1} \left(\frac{2x}{1+x^2} \right) = 2 \tan^{-1} x, \quad |x| \leq 1$$

Substitution used : $x = \tan \theta$

$$3. \quad \sin^{-1} \left(2x\sqrt{1-x^2} \right) = 2 \sin^{-1} x, \quad |x| \leq \frac{1}{\sqrt{2}}$$

Substitution used : $x = \sin \theta$



SOLVED EXAMPLES

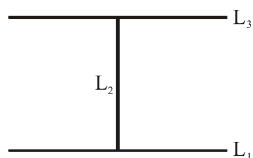
RELATIONS AND FUNCTIONS-II

Example -1

Let L be the set of all lines in a plane and R be the relation in L defined as $R = \{(L_1, L_2) : L_1 \text{ is perpendicular to } L_2\}$. Show that R is symmetric but neither reflexive nor transitive.

Sol. R is not reflexive: As a line L_1 can not be perpendicular to itself, i.e., $(L_1, L_1) \notin R$,

R is symmetric : As $(L_1, L_2) \in R$.



$$\Rightarrow L_1 \text{ is perpendicular to } L_2$$

$$\Rightarrow L_2 \text{ is perpendicular to } L_1$$

$$\Rightarrow (L_2, L_1) \in R.$$

R is not transitive: If $(L_1, L_2) \in R$ & $(L_2, L_3) \in R$

$\Rightarrow L_1$ is perpendicular to L_2 & L_2 is perpendicular to L_3 .
Then L_1 can not be perpendicular to L_3 as L_1 is parallel to L_3 .

$$\Rightarrow (L_1, L_3) \notin R.$$

Example -2

Show that the relation R in the set Z of integers given by

$$R = \{(a, b) : 2 \text{ divides } a - b\}$$

is an equivalence relation.

Sol. R is reflexive: As $2|a-a \quad \forall a \in Z$ ($|$ this symbol mean divide)

$$\Rightarrow (a, a) \in R,$$

R is symmetric : Let $(a, b) \in R$

$$\Rightarrow 2|a-b$$

$$\Rightarrow 2|-(b-a)$$

$$\Rightarrow (b, a) \in R$$

R is transitive : Let $(a, b) \in R$ & $(b, c) \in R$

$$\Rightarrow 2|a-b \text{ \& } 2|b-c$$

$$\Rightarrow 2|a-b + b-c$$

$$\Rightarrow 2|a-c \Rightarrow (a, c) \in R$$

Example -3

Let $f: X \rightarrow Y$ be a function. Define a relation R in X given by $R = \{(a, b) : f(a) = f(b)\}$. Examine whether R is an equivalence relation or not.

Sol. R is reflexive: Since $f(a) = f(a) \quad \forall a \in X$

$$\Rightarrow (a, a) \in R \Rightarrow R \text{ is reflexive}$$

R is symmetric : Let $(a, b) \in R$

$$\Rightarrow f(a) = f(b)$$

$$\Rightarrow f(b) = f(a)$$

$$\Rightarrow (b, a) \in R$$

$$\Rightarrow R \text{ is symmetric}$$

R is transitive : Let (a, b) & $(b, c) \in R$

$$\Rightarrow f(a) = f(b) \text{ \& } f(b) = f(c)$$

$$\Rightarrow f(a) = f(c)$$

$$\Rightarrow (a, c) \in R$$

$$\Rightarrow R \text{ is transitive}$$

Example -4

Let $A = \{2, 3, 4, 5, 6, 7, 8, 9\}$. Let R be the relation on A defined by

$$R = \{(x, y) : x \in A, y \in A \text{ and } x \text{ divides } y\}.$$

Find (i) R ; (ii) domain of R ; (iii) range of R ; (iv) R^{-1}

State whether or not R is (a) reflexive (b) symmetric (c) transitive.

Sol. Here, $x R y$ iff x divides y , therefore,

$$(i) \quad R = \{(2, 2), (2, 4), (2, 6), (2, 8), (3, 3), (3, 6), (3, 9), (4, 4), (4, 8), (5, 5), (6, 6), (7, 7), (8, 8), (9, 9)\}$$

$$(ii) \quad \text{Domain of } R = \{2, 3, 4, 5, 6, 7, 8, 9\} = A$$

$$(iii) \quad \text{Range of } R = \{2, 3, 4, 5, 6, 7, 8, 9\} = A$$

$$(iv) \quad R^{-1} = \{(y, x) : (x, y) \in R\} \\ = \{(2, 2), (4, 2), (6, 2), (8, 2), (3, 3), (6, 3), (9, 3), (4, 4), (8, 4), (5, 5), (6, 6), (7, 7), (8, 8), (9, 9)\}$$

In fact R^{-1} is $\{(y, x) : x, y \in A, y \text{ is divisible by } x\}$.



- (a) As (2, 2) (3, 3), (4, 4), (5, 5), (6, 6), (7, 7), (8, 8), (9, 9) belong to R, therefore, (R) is reflexive.
- (b) Here, R is not symmetric. We may observe that (2, 4) ∈ R but (4, 2) ∉ R. Infact, 'x divides y' does not imply 'y divides x' when x ≠ y.
- (c) As x divides y and 'y divides z' imply 'x divides z', therefore, the relation R is transitive.

Example - 5

Determine which of the following binary operations on the set R are associative and which are commutative.

(a) $a * b = 1 \quad \forall a, b \in R$

(b) $a * b = \frac{(a+b)}{2} \quad \forall a, b \in R$

Sol. (a) Clearly, by definition $a * b = b * a = 1$,
 $\forall a, b \in R$. Also $(a * b) * c = (1) * c = 1$ and

$a * (b * c) = a * (1) = 1, \quad \forall a, b, c \in R$. Hence R is both associative and commutative.

(b) $a * b = \frac{a+b}{2} = \frac{b+a}{2} = b * a$, shows that * is commutative. Further,

$$(a * b) * c = \left(\frac{a+b}{2} \right) * c$$

$$= \frac{\left(\frac{a+b}{2} \right) + c}{2} = \frac{a+b+2c}{4}$$

But $a * (b * c) = a * \left(\frac{b+c}{2} \right)$

$$= \frac{a + \frac{b+c}{2}}{2} = \frac{2a+b+c}{4} \neq \frac{a+b+2c}{4}$$

in general.

Hence * is not associative.

Example - 6

Find the domain of

$$f(x) = \frac{1}{\sqrt{\log_{1/2}(x^2 - 7x + 13)}}$$

Sol.

$$f(x) = \frac{1}{\sqrt{\log_{1/2}(x^2 - 7x + 13)}} \text{ exists if;}$$

$$\log_{1/2}(x^2 - 7x + 13) > 0$$

$$\Rightarrow (x^2 - 7x + 13) < 1 \quad \dots(i)$$

$$\text{and } x^2 - 7x + 13 > 0 \quad \dots(ii)$$

considering equation (ii), $x^2 - 7x + 13 > 0$, we have

$$\left(x^2 - 7x + \frac{49}{4} \right) + 13 - \frac{49}{4} > 0$$

$$\Rightarrow \left(x - \frac{7}{2} \right)^2 + \frac{3}{4} > 0$$

which is true for all $x \in R$

$$\text{as } \left(x - \frac{7}{2} \right)^2 \geq 0 \text{ for all } x. \quad \dots(a)$$

again taking (i), $x^2 - 7x + 13 < 1$

$$\begin{array}{c} + \quad | \quad + \\ 3 \quad \quad \quad 4 \end{array}$$

$$\Rightarrow x^2 - 7x + 12 < 0$$

$$\Rightarrow (x-3)(x-4) < 0$$

$$\Rightarrow 3 < x < 4 \quad \dots(b)$$

combining (a) and (b), we have

Hence domain of $f(x) \in (3, 4) \cup]3, 4[$

Example - 7

Find domain for $f(x) = \sin^{-1} \left(\frac{1+x^2}{2x} \right)$.

Sol. $f(x) = \sin^{-1} \left(\frac{1+x^2}{2x} \right)$ is defined for ;

$$-1 \leq \frac{1+x^2}{2x} \leq 1 \quad \text{or} \quad \left| \frac{1+x^2}{2x} \right| \leq 1$$

(since domain of $\sin^{-1} x = [-1, 1]$)

$$\Rightarrow |1+x^2| \leq |2x|$$

$$\Rightarrow 1+x^2 \leq |2x|, \quad \{\text{as } 1+x^2 > 0\}$$

$$\Rightarrow x^2 - 2|x| + 1 \leq 0$$

$$\Rightarrow |x|^2 - 2|x| + 1 \leq 0 \quad \{\text{as } x^2 = |x|^2\}$$

$$\Rightarrow (|x| - 1)^2 \leq 0$$



But $(|x| - 1)^2$ is either always positive or zero.

Thus, $(|x| - 1)^2 = 0$

$$\Rightarrow |x| = 1$$

$$\Rightarrow x = \pm 1$$

Thus, domain for $f(x)$ is $\{-1, 1\}$

$$\Rightarrow \frac{9}{4} \leq \left(\sin x - \frac{5}{2} \right)^2 \leq \frac{49}{4}$$

squaring both sides ... (ii)

$\therefore$ From Eqs (i) and (ii),

$$-10 \leq f(x) \leq 0$$

$\therefore$ Range of $f(x) \in [-10, 0]$.

Example -8

Find the range of the function :

$$f(x) = 3\sin x + 8\cos\left(x - \frac{\pi}{3}\right) + 5$$

Sol. Here $f(x) = 3\sin x + 8\cos\left(x - \frac{\pi}{3}\right) + 5$

$$= 3\sin x + 4(\cos x + \sqrt{3}\sin x) + 5$$

$$= (3 + 4\sqrt{3})\sin x + 4\cos x + 5.$$

Put $3 + 4\sqrt{3} = r \cos \theta$... (i) and $4 = r \sin \theta$... (ii)

squaring and adding (1) & (2), dividing (i) and (2)

$$r = \sqrt{73 + 24\sqrt{3}} \text{ and } \theta = \tan^{-1} \frac{4}{3 + 4\sqrt{3}}$$

$$\Rightarrow f(x) = \sqrt{73 + 24\sqrt{3}} \sin(x + \theta) + 5$$

$\Rightarrow$ Range of $f(x)$ is

$$\left[5 - \sqrt{73 + 24\sqrt{3}}, 5 + \sqrt{73 + 24\sqrt{3}} \right].$$

Example -9

The range of the function $\sin^2 x - 5 \sin x - 6$ is

Sol. Here, $f(x) = \sin^2 x - 5 \sin x - 6$

$$= \left(\sin^2 x - 5 \sin x + \frac{25}{4} \right) - 6 - \frac{25}{4}$$

$$= \left(\sin x - \frac{5}{2} \right)^2 - \frac{49}{4} \quad \dots (i)$$

$$\text{where } \frac{9}{4} \leq \left(\sin x - \frac{5}{2} \right)^2 \leq \frac{49}{4} \quad \dots (ii)$$

$$\left(\text{since } -1 \leq \sin x \leq 1 \Rightarrow \frac{-7}{2} \leq \sin x - \frac{5}{2} \leq \frac{-3}{2} \right)$$

Squaring Both sides 2

Example -10

Find the range of the function :

$$f(x) = \ln \sqrt{x^2 + 4x + 5}$$

Sol. Here $f(x) = \ln \sqrt{x^2 + 4x + 5} = \ln \sqrt{(x+2)^2 + 1}$

i.e. $x^2 + 4x + 5$ takes all values in $[1, \infty)$

since $(x+2)^2 + 1 \geq 1$

$\Rightarrow f(x)$ will take all values in $[0, \infty)$.

Hence range of $f(x)$ is $[0, \infty)$.

Example -11

Find the range of the function

$$f(x) = \tan \sqrt{\frac{\pi^2}{9} - x^2} \text{ is}$$

Sol. For $f(x)$ to be defined, $\frac{\pi^2}{9} - x^2 \geq 0$

$$\Rightarrow -\frac{\pi}{3} \leq x \leq \frac{\pi}{3}$$

$$\therefore \text{Domain of } f = \left[-\frac{\pi}{3}, \frac{\pi}{3} \right].$$

The greatest value of $f(x) = \tan \sqrt{\frac{\pi^2}{9} - 0}$, when $x = 0$

$$= \tan \frac{\pi}{3}$$

$$= \sqrt{3}$$

and the least value of $f(x) = \tan \sqrt{\frac{\pi^2}{9} - \frac{\pi^2}{9}}$, when $x = \frac{\pi}{3}$.

$$= \tan 0$$

$$= 0$$



$\therefore$ The greatest value of $f(x) = \sqrt{3}$ and the least value of $f(x) = 0$.

$\therefore$ Range of $f = [0, \sqrt{3}]$

Example - 12

Find the period of the function.

$$f(x) = \sin x + \{x\}$$

Sol. Here $f(x) = \sin x + \{x\}$

Period of $\sin x$ is 2π and that of $\{x\}$ is 1.

But the L.C.M. of 2π and 1 does not exist.

Hence $\sin x + \{x\}$ is not periodic.

Example - 13

Find the period of the function

$$f(x) = \tan \frac{x}{3} + \sin 2x$$

Sol. Here $f(x) = \tan x/3 + \sin 2x$.

Here $\tan(x/3)$ is periodic with period 3π and $\sin 2x$ is periodic with period π .

Hence $f(x)$ will be periodic with period 3π .

(Since L.C.M of 3π & π is 3π)

Example - 14

Find the period of the function

$$f(x) = |\sin x| + |\cos x|.$$

Sol. Here $f(x) = |\sin x| + |\cos x|$

Now, $|\sin x| = \sqrt{\sin^2 x} = \sqrt{\frac{1 - \cos 2x}{2}}$, which is periodic

with period π .

Similarly, $|\cos x|$ is periodic with period π .

Hence, according to rule of LCM, period of $f(x)$ must be π .

$$\text{But } \left| \sin \left(\frac{\pi}{2} + x \right) \right| = |\cos x| \text{ and } \left| \cos \left(\frac{\pi}{2} + x \right) \right| = |\sin x|$$

Since $\pi/2 < \pi$, period of $f(x)$ is $\pi/2$.

Example - 15

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = 3x - 2$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ be

defined by $g(x) = \frac{x+2}{3}$. Show that $f \circ g = I_{\mathbb{R}} = g \circ f$.

Sol. For all $x \in \mathbb{R}$, $(f \circ g)(x) = f(g(x))$

$$= f\left(\frac{x+2}{3}\right) = 3\left(\frac{x+2}{3}\right) - 2 = x = I_{\mathbb{R}}(x)$$

Hence $f \circ g = I_{\mathbb{R}}$

Again,

$$(g \circ f)(x) = g(f(x)) = g(3x - 2) = \frac{(3x - 2) + 2}{3} = x = I_{\mathbb{R}}(x)$$

$\therefore g \circ f = I_{\mathbb{R}}$.

Example - 16

Let $X = \{-2, -1, 0, 1, 2, 3\}$ and $Y = \{0, 1, 2, \dots, 10\}$ and $f: X \rightarrow Y$ be a function defined by $f(x) = x^2$ for all $x \in X$, find $f^{-1}(A)$ where $(A) = \{0, 1, 2, 4\}$.

Sol. Here, we have to find $f^{-1}(0)$, $f^{-1}(1)$, $f^{-1}(2)$ and $f^{-1}(4)$.

Now $f(x) = 0 \Rightarrow x^2 = 0 \Rightarrow x = 0 \Rightarrow f^{-1}(0) = \{0\}$,

$$f(x) = 1 \Rightarrow x^2 = 1 \Rightarrow x = -1, 1$$

$$\Rightarrow f^{-1}(1) = \{-1, 1\}, f(x) = 2 \Rightarrow x^2 = 2$$

$$\Rightarrow x = -\sqrt{2}, \sqrt{2} \text{ but none of these is in } X.$$

$$\Rightarrow f^{-1}(2) = \emptyset, f(x) = 4 \Rightarrow x^2 = 4 \Rightarrow x = -2, 2$$

$$\Rightarrow f^{-1}(4) = \{-2, 2\}.$$

Hence, $f^{-1}(A) = \{0, -1, 1, -2, 2\}$.

Example - 17

Let A be a non-empty set and $f: A \rightarrow A$, $g: A \rightarrow A$ be two functions such that $f \circ g = I_A = g \circ f$, show that f and g are bijections and that $g = f^{-1}$.

Sol. Consider $f: A \rightarrow A$, Let $y \in A$ be arbitrary. Since $f \circ g = I_A$, therefore, $(f \circ g)(y) = y$

$$\Rightarrow f(g(y)) = y$$

$$\Rightarrow f(t) = y, \text{ where } t = g(y) \in A.$$

This means that for $y \in A$, there exists $t \in A$ such that $f(t) = y$. Hence f is onto.

Let $x, y \in A$ such that $f(x) = f(y)$

$$\Rightarrow g(f(x)) = g(f(y))$$

($\because g$ is a function)

$$\Rightarrow (g \circ f)(x) = (g \circ f)(y)$$

$$\Rightarrow I_A(x) = I_A(y)$$



$$\Rightarrow x = y.$$

So, $f(x) = f(y) \Rightarrow x = y \Rightarrow f$ is one-one. Thus, we see that f is both one-one and onto i.e. f is a bijection. Similarly, we can show that g is a bijection.

Moreover, for all $x \in A$,

$$x = I_A(x) = (f \circ g)(x) = f(g(x))$$

$$\Rightarrow x = f(g(x)) \Rightarrow f^{-1}(x) = g(x)$$

$$\Rightarrow f^{-1} = g.$$

Example – 18

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \frac{e^x - e^{-x}}{2}$. Is $f(x)$

invertible? If so, find its inverse.

Sol. Let us check invertibility of $f(x)$:

(a) **One-one**: Here, $f'(x) = \frac{e^x + e^{-x}}{2}$

$$\Rightarrow f'(x) = \frac{e^{2x} + 1}{2e^x} \text{ which is strictly increasing as}$$

$$e^{2x} > 0 \text{ for all } x.$$

Thus, one-one.

(b) **Onto**: Let $y = f(x)$

$$\Rightarrow y = \frac{e^x - e^{-x}}{2} \text{ where } y \text{ is strictly monotonic.}$$

Hence, range of $f(x) = (f(-\infty), f(\infty))$

(Since domain of $f = (-\infty, \infty)$)

$$\Rightarrow \text{range of } f(x) = (-\infty, \infty)$$

So range of $f(x) = \text{co-domain}$.

Hence, $f(x)$ is one-one and onto.

$\Rightarrow f$ is invertible

(c) **To find f^{-1}** : $y = \frac{e^{2x} - 1}{2e^x}$

$$\Rightarrow e^{2x} - 2e^x y - 1 = 0$$

$$\Rightarrow e^x = \frac{2y \pm \sqrt{4y^2 + 4}}{2}$$

$$\Rightarrow x = \log(y \pm \sqrt{y^2 + 1})$$

$$\Rightarrow f^{-1}(y) = \log(y \pm \sqrt{y^2 + 1})$$

$$[\text{as } f(x) = y \Rightarrow x = f^{-1}(y)]$$

Since, $e^{f^{-1}(x)}$ is always positive.

So, neglecting negative sign.

$$\text{Hence, } f^{-1}(x) = \log(x + \sqrt{x^2 + 1})$$

Example – 19

Let $f: [1/2, \infty) \rightarrow [3/4, \infty)$, where $f(x) = x^2 - x + 1$. Find the inverse of $f(x)$. Hence, solve the equation

$$x^2 - x + 1 = \frac{1}{2} + \sqrt{x - \frac{3}{4}}.$$

Sol. (a) $f(x) = x^2 - x + 1$

$$\Rightarrow f(x) = \left(x - \frac{1}{2}\right)^2 + \frac{3}{4} > 0, \text{ which is clearly one-one}$$

and onto in given domain and co-domain.

$\Rightarrow f(x)$ is strictly increasing.

$\Rightarrow f(x)$ is one-one.

Also $f(x)$ is onto.

(b) Thus, its inverse can be obtained.

$$\text{Let } f(x) = y$$

$$\Rightarrow y = \left(x - \frac{1}{2}\right)^2 + \frac{3}{4}$$

$$\Rightarrow x - \frac{1}{2} = \pm \sqrt{y - \frac{3}{4}}$$

$$\Rightarrow x = \frac{1}{2} \pm \sqrt{y - \frac{3}{4}} \quad [f(x) = y \Rightarrow x = f^{-1}(y)]$$

$$\Rightarrow f^{-1}(y) = \frac{1}{2} + \sqrt{y - \frac{3}{4}}$$

[neglecting -ve sign as $x > 0$]

(as x is always +ve)

$$\Rightarrow f^{-1}(x) = \frac{1}{2} + \sqrt{x - \frac{3}{4}}$$

(c) To solve $x^2 - x + 1 = \frac{1}{2} + \sqrt{x - \frac{3}{4}}$, as $f(x) = f^{-1}(x)$ has only one solution in this case.

$$\text{ie, } f(x) = x$$

$$\Rightarrow x^2 - x + 1 = x$$

$$\Rightarrow x^2 - 2x + 1 = 0$$

$$\Rightarrow (x - 1)^2 = 0$$

$x = 1$ is the required solution.



Example – 20

If $f(x) = x^2 - 3x + 2$ be a real valued function of the real variable, find $f \circ f$.

Sol. We are given that the function $f: \mathbb{R} \rightarrow \mathbb{R}$, defined by $f(x) = x^2 - 3x + 2$ for all $x \in \mathbb{R}$.

$$\begin{aligned} \text{Now, } (f \circ f)(x) &= f(f(x)) = (f(x))^2 - 3f(x) + 2 \\ &= (x^2 - 3x + 2)^2 - 3(x^2 - 3x + 2) + 2 \\ &= x^4 + 9x^2 + 4 - 6x^3 - 12x + 4x^2 - 3x^2 + 9x - 6 + 2 \\ &= x^4 - 6x^3 + 10x^2 - 3x. \end{aligned}$$

Example – 21

Two functions are defined as under :

$$f(x) = \begin{cases} x+1, & x \leq 1 \\ 2x+1, & 1 < x \leq 2 \end{cases}$$

$$g(x) = \begin{cases} x^2, & -1 \leq x < 2 \\ x+2, & 2 \leq x \leq 3 \end{cases}$$

Find $f \circ g$ and $g \circ f$.

Sol. $(f \circ g)(x) = f(g(x)) = \begin{cases} g(x)+1, & g(x) \leq 1 \\ 2g(x)+1, & 1 < g(x) \leq 2 \end{cases}$

Let us consider $-1 \leq f(x) < 2$

$$\begin{aligned} \text{(i)} \quad x^2 &\leq 1, -1 \leq x < 2 \\ \Rightarrow -1 &\leq x \leq 1, -1 \leq x < 2 \\ \Rightarrow -1 &\leq x \leq 1 \\ \text{(ii)} \quad x+2 &\leq 1, 2 \leq x \leq 3 \\ \Rightarrow x &\leq -1, 2 \leq x \leq 3 \\ \Rightarrow x &= \phi. \end{aligned}$$

(so no value of x is possible)

Let us consider, $1 < g(x) \leq 2$.

$$\begin{aligned} \text{(iii)} \quad 1 < x^2 &\leq 2, -1 \leq x < 2 \\ \Rightarrow x &\in [-\sqrt{2}, -1) \cup (1, \sqrt{2}], -1 \leq x < 2 \\ \Rightarrow 1 < x &\leq \sqrt{2}. \\ \text{(iv)} \quad 1 < x+2 &\leq 2, 2 \leq x \leq 3 \\ \Rightarrow -1 < x &\leq 0, 2 \leq x \leq 3, x = \phi \end{aligned}$$

$$\text{Thus } f(g(x)) = \begin{cases} x^2+1, & -1 \leq x \leq 1 \\ 2x^2+1, & 1 < x \leq \sqrt{2} \end{cases}$$

Now, Let us consider $g \circ f$:

$$g \circ f = g(f(x)) = \begin{cases} f^2(x), & -1 \leq f(x) < 2 \\ f(x)+2, & 2 \leq f(x) \leq 3 \end{cases}$$

Let us consider $-1 \leq f(x) < 2$

$$\begin{aligned} \text{(i)} \quad -1 &\leq x+1 < 2, x \leq 1 \\ \Rightarrow -2 &\leq x < 1, x \leq 1 \\ \Rightarrow -2 &\leq x < 1 \\ \text{(ii)} \quad -1 &\leq 2x+1 < 2, 1 < x \leq 2 \\ \Rightarrow -1 &\leq x < 1/2, 1 < x \leq 2 \\ \Rightarrow x &= \phi. \end{aligned}$$

Let us consider $2 \leq f(x) \leq 3$

$$\begin{aligned} \text{(iii)} \quad 2 &\leq x+1 \leq 3, x \leq 1 \\ \Rightarrow 1 &\leq x \leq 2, x \leq 1 \\ \Rightarrow x &= 1 \\ \text{(iv)} \quad 2 &\leq 2x+1 \leq 3, 1 < x \leq 2 \\ \Rightarrow 1 &\leq 2x \leq 2, 1 < x \leq 2 \\ \Rightarrow 1/2 &\leq x \leq 1, 1 < x \leq 2 \\ \Rightarrow x &= \phi. \end{aligned}$$

$$g(f(x)) = \begin{cases} (x+1)^2, & -2 \leq x < 1 \\ x+3, & x = 1 \end{cases}$$

If we like we can also write $g(f(x)) = (x+1)^2, -2 \leq x \leq 1$.

Example – 22

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^3 + ax^2 + 3x + 100$. Then find the values of a for which f is a one-one function.

Sol. $f(x) = x^3 + ax^2 + 3x + 100$

$$\Rightarrow f'(x) = 3x^2 + 2ax + 3.$$

For $f(x)$ to be one-one. $f'(x) \geq 0$ or ≤ 0

But $f(x)$ is a quadratic expression and coefficient of $x^2 > 0$ so that $f'(x) \geq 0$

$$\begin{aligned} \Rightarrow D &\leq 0 \\ \Rightarrow 4a^2 - 36 &\leq 0 \\ \Rightarrow a^2 &\leq 9 \\ \Rightarrow -3 &\leq a \leq 3. \end{aligned}$$

Example – 23

Find whether the given function is even or odd ?

$$f(x) = \frac{x}{e^x - 1} + \frac{x}{2} + 1$$



Sol. We have

$$f(-x) = \frac{-x}{e^{-x}-1} - \frac{x}{2} + 1 = \frac{-e^x \cdot x}{1-e^x} - \frac{x}{2} + 1$$

$$= \frac{(e^x - 1 + 1)x}{(e^x - 1)} - \frac{x}{2} + 1$$

$$= x + \frac{x}{e^x - 1} - \frac{x}{2} + 1 = \frac{x}{e^x - 1} + \frac{x}{2} + 1 = f(x)$$

Hence $f(x)$ is an even function.

Example -24

If f is an even function, find the real values of x satisfying

the equation $f(x) = f\left(\frac{x+1}{x+2}\right)$.

Sol. Since, $f(x)$ is even, so $f(-x) = f(x)$

Thus, $x = \frac{x+1}{x+2}$ or $-x = \frac{x+1}{x+2}$

$$\Rightarrow x^2 + 2x = x + 1 \text{ or } -x^2 - 2x = x + 1$$

$$\Rightarrow x^2 + x - 1 = 0 \text{ or } -x^2 - 3x - 1 = 0$$

$$\Rightarrow x = \frac{-1 \pm \sqrt{5}}{2} \text{ or } x = \frac{-3 \pm \sqrt{5}}{2}$$

Thus, $x \in \left\{ \frac{-1+\sqrt{5}}{2}, \frac{-1-\sqrt{5}}{2}, \frac{-3+\sqrt{5}}{2}, \frac{-3-\sqrt{5}}{2} \right\}$

Example -25

Let $f(x) = \frac{9^x}{9^x + 3}$. Show $f(x) + f(1-x) = 1$, and hence evaluate

$$f\left(\frac{1}{1996}\right) + f\left(\frac{2}{1996}\right) + f\left(\frac{3}{1996}\right) + \dots + f\left(\frac{1995}{1996}\right).$$

Sol. $f(x) = \frac{9^x}{9^x + 3}$... (i)

and $f(1-x) = \frac{9^{1-x}}{9^{1-x} + 3}$

$$\Rightarrow f(1-x) = \frac{\frac{9}{9^x}}{\frac{9}{9^x} + 3} = \frac{9}{9 + 3 \cdot 9^x}$$

$$f(1-x) = \frac{9}{3(3+9^x)} \dots (ii)$$

Adding Eqs. (i) and (ii), we get

$$f(x) + f(1-x) = \frac{9^x}{9^x + 3} + \frac{9}{3(3+9^x)}$$

$$= \frac{3 \cdot 9^x + 9}{3(9^x + 3)} = \frac{3(9^x + 3)}{3(9^x + 3)}$$

$$\therefore f(x) + f(1-x) = 1 \dots (iii)$$

Now, putting $x = \frac{1}{1996}, \frac{2}{1996}, \frac{3}{1996}, \dots, \frac{998}{1996}$

in (Eq. (iii)), we get

$$f\left(\frac{1}{1996}\right) + f\left(\frac{1995}{1996}\right) = 1$$

$$\Rightarrow f\left(\frac{2}{1996}\right) + f\left(\frac{1994}{1996}\right) = 1$$

$$\Rightarrow f\left(\frac{3}{1996}\right) + f\left(\frac{1993}{1996}\right) = 1$$

$$\Rightarrow f\left(\frac{997}{1996}\right) + f\left(\frac{999}{1996}\right) = 1$$

$$\Rightarrow f\left(\frac{998}{1996}\right) + f\left(\frac{998}{1996}\right) = 1$$

or $f\left(\frac{998}{1996}\right) = \frac{1}{2}$

Adding all the above expressions, we get

$$f\left(\frac{1}{1996}\right) + f\left(\frac{2}{1996}\right) + \dots + f\left(\frac{1995}{1996}\right)$$

$$= (1 + 1 + 1 + \dots \cdot 997 \text{ times}) + \frac{1}{2}$$

$$= 997 + \frac{1}{2}$$

$$= 997.5.$$



INVERSE TRIGONOMETRIC FUNCTIONS

Example – 26

Prove that (i) $\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right) = -\frac{\pi}{3}$,

(ii) $\cos^{-1}\left(\cos\left(\frac{5\pi}{3}\right)\right) = \frac{\pi}{3}$.

Sol. (i) Let $\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right) = \theta$ so that $\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

(since range of $\sin^{-1} x$ is $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$)

$$\Rightarrow -\frac{\sqrt{3}}{2} = \sin \theta$$

$$\Rightarrow \sin \theta = -\sin \frac{\pi}{3} = \sin\left(-\frac{\pi}{3}\right) \text{ note that } -\frac{\pi}{3} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

$$\Rightarrow \theta = -\frac{\pi}{3} \Rightarrow \sin^{-1}\left(-\frac{\sqrt{3}}{2}\right) = -\frac{\pi}{3}.$$

$$(ii) \cos^{-1}\left(\cos\left(\frac{5\pi}{3}\right)\right) = \cos^{-1}\left(\cos\left(2\pi - \frac{\pi}{3}\right)\right)$$

$$= \cos^{-1}\left(\cos\left(\frac{\pi}{3}\right)\right) = \frac{\pi}{3},$$

note that $\frac{\pi}{3} \in [0, \pi] = \text{range of } \cos^{-1} x$.

Example – 27

Evaluate

$$(i) \sin\left(2\sin^{-1}\left(\frac{3}{5}\right)\right) \quad (ii) \sin\left(2\sin^{-1}\left(-\frac{4}{5}\right)\right)$$

$$(iii) \sin\left(2\cos^{-1}\left(-\frac{3}{5}\right)\right) \quad (iv) \sin\left(3\sin^{-1}\left(\frac{2}{5}\right)\right)$$

Sol. (i) $\sin\left(2\sin^{-1}\left(\frac{3}{5}\right)\right) = \sin 2\theta$, where $\theta = \sin^{-1}\left(\frac{3}{5}\right)$

$$= 2 \sin \theta \cos \theta$$

$$= 2 \sin\left(\sin^{-1}\left(\frac{3}{5}\right)\right) \cos\left(\sin^{-1}\left(\frac{3}{5}\right)\right)$$

$$= 2 \times \frac{3}{5} \sqrt{1 - \left(\frac{3}{5}\right)^2} \quad (\because \cos(\sin^{-1} x) = \sqrt{1-x^2} \text{ for } |x| \leq 1)$$

$$= 2 \times \frac{3}{5} \times \frac{4}{5} = \frac{24}{25}$$

$$(ii) \sin\left(2\sin^{-1}\left(-\frac{4}{5}\right)\right) = \sin\left(-2\sin^{-1}\frac{4}{5}\right)$$

$$(\because \sin^{-1}(-x) = -\sin^{-1} x)$$

$$= -\sin\left(2\sin^{-1}\left(\frac{4}{5}\right)\right)$$

$$= -\sin\left(2\sin^{-1}\left(\frac{4}{5}\right)\right)$$

$$= -\sin 2\theta, \text{ where } \theta = \sin^{-1}\left(\frac{4}{5}\right) = 2 \sin \theta \cos \theta$$

$$= -2 \sin\left(\sin^{-1}\left(\frac{4}{5}\right)\right) \cos\left(\sin^{-1}\left(\frac{4}{5}\right)\right)$$

$$= -2\left(\frac{4}{5}\right) \sqrt{1 - \left(\frac{4}{5}\right)^2}$$

$$(\because \cos(\sin^{-1} x) = \sqrt{1-x^2})$$

$$= -\frac{8}{5} \times \frac{3}{5} = -\frac{24}{25}$$

$$(iii) \sin\left(2\cos^{-1}\left(-\frac{3}{5}\right)\right) = \sin(2\theta), \text{ where } \theta = \cos^{-1}\left(-\frac{3}{5}\right)$$

$$= 2 \sin \theta \cos \theta$$

$$= 2 \sin\left(\cos^{-1}\left(-\frac{3}{5}\right)\right) \cos\left(\cos^{-1}\left(-\frac{3}{5}\right)\right)$$

$$= 2\sqrt{1 - \left(-\frac{3}{5}\right)^2} \left(-\frac{3}{5}\right) (\because \sin(\cos^{-1} x) = \sqrt{1-x^2} \text{ for } |x| \leq 1)$$

$$= 2\left(\frac{4}{5}\right)\left(-\frac{3}{5}\right) = -\frac{24}{25}.$$



$$\begin{aligned} \text{(iv)} \quad & \sin \left(3 \sin^{-1} \left(\frac{2}{5} \right) \right) = \sin 3\theta, \text{ where } \theta = \sin^{-1} \left(\frac{2}{5} \right) \\ & = 3 \sin \theta - 4 \sin^3 \theta \\ & = 3 \left(\frac{2}{5} \right) - 4 \left(\frac{2}{5} \right)^3 \\ & \left(\because \theta = \sin^{-1} \left(\frac{2}{5} \right), \therefore \sin \theta = \frac{2}{5} \right) \\ & = \frac{6}{5} - \frac{32}{125} = \frac{118}{125}. \end{aligned}$$

Example – 28

Prove that

$$\text{(i)} \quad \sin^{-1} \left(\frac{3}{5} \right) - \sin^{-1} \left(\frac{8}{17} \right) = \cos^{-1} \left(\frac{84}{85} \right)$$

$$\text{(ii)} \quad \sin^{-1} \left(\frac{3}{5} \right) + \sin^{-1} \left(\frac{8}{17} \right) = \sin^{-1} \left(\frac{77}{85} \right)$$

Sol. (i) Since $\sin^{-1} x$ is an increasing function in $[-1, 1]$ and $\frac{3}{5} > \frac{8}{17}$, therefore,

$$\sin^{-1} \left(\frac{3}{5} \right) > \sin^{-1} \left(\frac{8}{17} \right) \Rightarrow \sin^{-1} \left(\frac{3}{5} \right) - \sin^{-1} \left(\frac{8}{17} \right) > 0.$$

$$\Rightarrow \sin^{-1} \left(\frac{3}{5} \right) - \sin^{-1} \left(\frac{8}{17} \right) \in [0, \pi] = \text{range of } \cos^{-1} x$$

$$\left(\because 0 \leq \sin^{-1} \left(\frac{3}{5} \right) \leq \frac{\pi}{2} \text{ and } 0 \leq \sin^{-1} \left(\frac{8}{17} \right) \leq \frac{\pi}{2} \right)$$

$$\begin{aligned} \text{Now } & \cos \left\{ \sin^{-1} \left(\frac{3}{5} \right) - \sin^{-1} \left(\frac{8}{17} \right) \right\} \\ & = \cos \left(\sin^{-1} \left(\frac{3}{5} \right) \right) \cos \left(\sin^{-1} \left(\frac{8}{17} \right) \right) + \sin \left(\sin^{-1} \left(\frac{3}{5} \right) \right) \sin \left(\sin^{-1} \left(\frac{8}{17} \right) \right) \\ & = \sqrt{1 - \frac{9}{25}} \sqrt{1 - \frac{64}{289}} + \frac{3}{5} \times \frac{8}{17} = \frac{4}{5} \times \frac{15}{17} + \frac{3}{5} \times \frac{8}{17} = \frac{84}{85} \\ & \Rightarrow \sin^{-1} \left(\frac{3}{5} \right) - \sin^{-1} \left(\frac{8}{17} \right) = \cos^{-1} \left(\frac{84}{85} \right). \end{aligned}$$

(ii) We know that

$$0 \leq \sin^{-1} \frac{3}{5} \leq \frac{\pi}{2} \text{ and } 0 \leq \sin^{-1} \frac{8}{17} \leq \frac{\pi}{2}, \text{ therefore, } 0 \leq \sin^{-1} \left(\frac{3}{5} \right) +$$

$$\sin^{-1} \left(\frac{8}{17} \right) \leq \pi$$

We compute

$$\begin{aligned} & \cos \left(\sin^{-1} \left(\frac{3}{5} \right) + \sin^{-1} \left(\frac{8}{17} \right) \right) \\ & = \cos \left(\sin^{-1} \left(\frac{3}{5} \right) \right) \cos \left(\sin^{-1} \left(\frac{8}{17} \right) \right) \\ & \quad - \sin \left(\sin^{-1} \left(\frac{3}{5} \right) \right) \sin \left(\sin^{-1} \left(\frac{8}{17} \right) \right) \\ & = \sqrt{1 - \left(\frac{3}{5} \right)^2} \sqrt{1 - \left(\frac{8}{17} \right)^2} - \frac{3}{5} \times \frac{8}{17} = \frac{4}{5} \times \frac{15}{17} - \frac{3}{5} \times \frac{8}{17} = \frac{36}{85} \\ & \Rightarrow \sin^{-1} \left(\frac{3}{5} \right) + \sin^{-1} \left(\frac{8}{17} \right) = \cos^{-1} \left(\frac{36}{85} \right) = \sin^{-1} \left(\sqrt{1 - \left(\frac{36}{85} \right)^2} \right) \end{aligned}$$

$$\left(\because \cos^{-1} x = \sin^{-1} \sqrt{1 - x^2} \text{ for } 0 \leq x \leq 1 \right)$$

$$\Rightarrow \sin^{-1} \left(\frac{3}{5} \right) + \sin^{-1} \left(\frac{8}{17} \right) = \sin^{-1} \left(\frac{77}{85} \right), \text{ as desired.}$$

Example – 29

$$\text{Show that } \sin^{-1} \left(\frac{4}{5} \right) + \sin^{-1} \left(\frac{5}{13} \right) + \sin^{-1} \left(\frac{16}{65} \right) = \frac{\pi}{2}.$$

$$\text{Sol. Let } \theta = \sin^{-1} \left(\frac{4}{5} \right) + \sin^{-1} \left(\frac{5}{13} \right)$$

then $0 < \theta < \pi$

$$\left(\because 0 < \sin^{-1} \left(\frac{4}{5} \right) < \frac{\pi}{2}, 0 < \sin^{-1} \left(\frac{5}{13} \right) < \frac{\pi}{2} \right)$$

$$\text{Now } \cos \theta = \cos \left(\sin^{-1} \frac{4}{5} + \sin^{-1} \frac{5}{13} \right)$$

$$= \cos \left(\sin^{-1} \frac{4}{5} \right) \cos \left(\sin^{-1} \frac{5}{13} \right) - \sin \left(\sin^{-1} \frac{4}{5} \right) \sin \left(\sin^{-1} \frac{5}{13} \right)$$



$$\left(\sin^{-1} \frac{5}{13}\right)$$

$$(\because \cos(A+B) = \cos A \cos B - \sin A \sin B)$$

$$= \sqrt{1 - \left(\frac{4}{5}\right)^2} \sqrt{1 - \left(\frac{5}{13}\right)^2} - \frac{4}{5} \cdot \frac{5}{13}$$

$$(\because \cos(\sin^{-1} x) = \sqrt{1-x^2})$$

$$= \frac{3}{5} \cdot \frac{12}{13} - \frac{20}{65} = \frac{36-20}{65} = \frac{16}{65} \Rightarrow \theta = \cos^{-1}\left(\frac{16}{65}\right)$$

$$\text{Hence } \sin^{-1}\left(\frac{4}{5}\right) + \sin^{-1}\left(\frac{5}{13}\right) + \sin^{-1}\left(\frac{16}{65}\right)$$

$$= \left\{ \sin^{-1}\left(\frac{4}{5}\right) + \sin^{-1}\left(\frac{5}{13}\right) \right\} + \sin^{-1}\left(\frac{16}{65}\right)$$

$$= \cos^{-1}\left(\frac{16}{65}\right) + \sin^{-1}\left(\frac{16}{65}\right)$$

$$(\because \sin^{-1} t + \cos^{-1} t = \frac{\pi}{2} \text{ for } -1 \leq t \leq 1)$$

$$= \frac{\pi}{2}$$

Example – 30

$$\text{Prove that } \cot^{-1}(13) + \cot^{-1}(21) + \cot^{-1}(-8) = \pi.$$

$$\text{Sol. L.H.S.} = \cot^{-1}(13) + \cot^{-1}(21) + \cot^{-1}(-8)$$

$$= \tan^{-1}\left(\frac{1}{13}\right) + \tan^{-1}\left(\frac{1}{21}\right) + \pi - \cot^{-1} 8$$

$$(\because \cot^{-1}(-x) = \pi - \cot^{-1} x, x \in \mathbb{R})$$

$$= \tan^{-1}\left(\frac{\frac{1}{13} + \frac{1}{21}}{1 - \frac{1}{13} \cdot \frac{1}{21}}\right) + \pi - \tan^{-1}\left(\frac{1}{8}\right)$$

$$= \tan^{-1}\left(\frac{21+13}{13 \times 21 - 1}\right) + \pi - \tan^{-1}\left(\frac{1}{8}\right)$$

$$= \tan^{-1}\left(\frac{34}{272}\right) + \tan^{-1}\left(\frac{34}{272}\right) = \tan^{-1}\left(\frac{1}{8}\right) + \pi - \tan^{-1}\left(\frac{1}{8}\right)$$

$$= \pi = \text{R.H.S.}$$

Example – 31

$$\text{Prove that } 2 \tan^{-1}(-3) = -\frac{\pi}{2} + \tan^{-1}\left(-\frac{4}{3}\right).$$

$$\text{Sol. L.H.S.} = 2 \tan^{-1}(-3) = -2 \tan^{-1} 3 = -2 \cot^{-1}\left(\frac{1}{3}\right)$$

$$= -2 \left(\frac{\pi}{2} - \tan^{-1}\left(\frac{1}{3}\right) \right) = -\pi + 2 \tan^{-1}\left(\frac{1}{3}\right)$$

$$= -\pi + \tan^{-1} \left\{ \frac{2(1/3)}{1 - (1/3)^2} \right\}$$

$$\left(\because 2 \tan^{-1} x = \tan^{-1} \frac{2x}{1-x^2} \text{ for } |x| < 1 \right)$$

$$= -\pi + \tan^{-1} \frac{3}{4} = -\pi + \cot^{-1} \frac{4}{3} = -\pi + \frac{\pi}{2} - \tan^{-1}\left(\frac{4}{3}\right)$$

$$= -\frac{\pi}{2} - \tan^{-1}\left(\frac{4}{3}\right) = -\frac{\pi}{2} + \tan^{-1}\left(-\frac{4}{3}\right) = \text{R.H.S.}$$

Example – 32

$$\text{Find } x \text{ if } \sin^{-1} x + \sin^{-1} 2x = \frac{\pi}{3}.$$

$$\text{Sol. Given } \sin^{-1} x + \sin^{-1} 2x = \frac{\pi}{3} \Rightarrow \sin^{-1} 2x = \frac{\pi}{3} - \sin^{-1} x$$

$$\Rightarrow 2x = \sin\left(\frac{\pi}{3} - \sin^{-1} x\right)$$

$$\Rightarrow 2x = \sin \frac{\pi}{3} \cos(\sin^{-1} x) - \cos \frac{\pi}{3} \sin(\sin^{-1} x)$$

$$\Rightarrow 2x = \frac{\sqrt{3}}{2} \sqrt{1-x^2} - \frac{1}{2} x \Rightarrow 2x + \frac{x}{2} = \frac{\sqrt{3}}{2} \sqrt{1-x^2}$$

$$\Rightarrow \frac{5x}{2} = \frac{\sqrt{3}}{2} \sqrt{1-x^2}$$

On squaring both sides, we get

$$\frac{25x^2}{4} = \frac{3}{4} (1-x^2)$$

$$\Rightarrow 28x^2 = 3 \Rightarrow x^2 = \frac{3}{28} \Rightarrow x = \pm \sqrt{\frac{3}{28}}$$

$$\Rightarrow x = \sqrt{\frac{3}{28}}$$



($\therefore x = -\sqrt{\frac{3}{28}}$ does not satisfy the given equation)

Example – 33

If $\cos^{-1}\left(\frac{x}{a}\right) + \cos^{-1}\left(\frac{y}{b}\right) = \alpha$,

Prove that $\frac{x^2}{a^2} - \frac{2xy}{ab} \cos \alpha + \frac{y^2}{b^2} = \sin^2 \alpha$.

Sol. Given $\cos^{-1}\left(\frac{x}{a}\right) + \cos^{-1}\left(\frac{y}{b}\right) = \alpha$

$$\Rightarrow \cos^{-1}\left(\frac{y}{b}\right) = \alpha - \cos^{-1}\left(\frac{x}{a}\right)$$

$$\Rightarrow \frac{y}{b} = \cos\left(\alpha - \cos^{-1}\left(\frac{x}{a}\right)\right)$$

$$\Rightarrow \frac{y}{b} = \cos \alpha \cos\left(\frac{x}{a}\right) + \sin \alpha \sin\left(\cos^{-1}\left(\frac{x}{a}\right)\right)$$

$$\Rightarrow \frac{y}{b} - \frac{x}{a} \cos \alpha = \sin \alpha \sqrt{1 - \left(\frac{x}{a}\right)^2}$$

($\therefore \sin(\cos^{-1} x) = \sqrt{1-x^2}$ for $|x| \leq 1$)

$$\Rightarrow \left(\frac{y}{b} - \frac{x}{a} \cos \alpha\right)^2 = \sin^2 \alpha \left(1 - \frac{x^2}{a^2}\right)$$

$$\Rightarrow \frac{y^2}{b^2} + \frac{x^2}{a^2} \cos^2 \alpha - 2\left(\frac{y}{b}\right)\left(\frac{x}{a}\right) \cos \alpha = \sin^2 \alpha \left(1 - \frac{x^2}{a^2}\right)$$

$$\Rightarrow \frac{x^2}{a^2} (\cos^2 \alpha + \sin^2 \alpha) - \frac{2xy}{ab} \cos \alpha + \frac{y^2}{b^2} = \sin^2 \alpha$$

$$\Rightarrow \frac{x^2}{a^2} - \frac{2xy}{ab} \cos \alpha + \frac{y^2}{b^2} = \sin^2 \alpha$$

Example – 34

If $\cos^{-1} x + \cos^{-1} y + \cos^{-1} z = \pi$,

Prove that $x^2 + y^2 + z^2 + 2xyz = 1$.

Sol. Let $\cos^{-1} x = A$, $\cos^{-1} y = B$, $\cos^{-1} z = C$

so that $x = \cos A$, $y = \cos B$, $z = \cos C$ and $A + B + C = \pi$.

$\therefore x^2 + y^2 + z^2 + 2xyz = \cos^2 A + \cos^2 B + \cos^2 C + 2 \cos A \cos B \cos C$

$$= \frac{1 + \cos 2A}{2} + \frac{1 + \cos 2B}{2} + \frac{1 + \cos 2C}{2} + 2 \cos A \cos B \cos C$$

$$= \frac{3}{2} + \frac{1}{2} (\cos 2A + \cos 2B + \cos 2C) + 2 \cos A \cos B \cos C$$

$$= \frac{3}{2} + \frac{1}{2} (-1 - 4 \cos A \cos B \cos C) + 2 \cos A \cos B \cos C$$

(Using a result from conditional identities)

= 1, as required.

Example – 35

$$2 \sin^{-1} x = \cos^{-1}(1 - 2x^2), 0 \leq x \leq 1.$$

Sol. Let $\sin^{-1} x = \theta \Rightarrow x = \sin \theta$, $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$

$$\text{But } 0 \leq x \leq 1 \Rightarrow 0 \leq \theta \leq \frac{\pi}{2}$$

$$\text{Hence } \cos^{-1}(1 - 2x^2) = \cos^{-1}(1 - 2 \sin^2 \theta), 0 \leq 2\theta \leq \pi$$

$$= \cos^{-1}(\cos 2\theta), 0 \leq 2\theta \leq \pi = 2\theta$$

$$\Rightarrow \cos^{-1}(1 - 2x^2) = 2\theta = 2 \sin^{-1} x.$$

Example – 36

$$3 \cos^{-1} x = \cos^{-1}(4x^3 - 3x), \frac{1}{2} \leq x \leq 1.$$

Sol. Let $\cos^{-1} x = \theta$ so that

$x = \cos \theta$ and $0 \leq \theta \leq \pi$

As $x \in \left[\frac{1}{2}, 1\right]$, therefore, $\frac{1}{2} \leq x \leq 1$

$$\Rightarrow \cos \frac{\pi}{3} \leq \cos \theta \leq \cos 0$$

$$\Rightarrow \frac{\pi}{3} \geq \theta \geq 0$$

Note (that $\cos \theta$ is decreasing in $[0, \pi]$)

$$\Rightarrow 0 \leq 3\theta \leq \pi, \text{ i.e. } 3\theta \in [0, \pi]$$

$$\therefore \cos^{-1}(4x^3 - 3x)$$

$$= \cos^{-1}(4 \cos^3 \theta - 3 \cos \theta)$$

$$= \cos^{-1}(\cos 3\theta)$$

$$= 3\theta \quad (\text{Note that } 0 \leq 3\theta \leq \pi)$$

$$= 3 \cos^{-1} x.$$



Example – 37

Prove that $\tan^{-1} \left(\frac{x}{\sqrt{a^2 - x^2}} \right) = \sin^{-1} \left(\frac{x}{a} \right)$, $|x| < a$.

Sol. Let $\sin^{-1} \left(\frac{x}{a} \right) = \theta \Rightarrow x = a \sin \theta$, and $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$

($\because |x| < a$)

Now $\sqrt{a^2 - x^2} = \sqrt{a^2 - a^2 \sin^2 \theta} = \sqrt{a^2 \cos^2 \theta} = a \cos \theta$

$\left(\because -\frac{\pi}{2} < \theta < \frac{\pi}{2} \Rightarrow \cos \theta > 0 \right)$

Hence $\tan^{-1} \left(\frac{x}{\sqrt{a^2 - x^2}} \right) = \tan^{-1} \left(\frac{a \sin \theta}{a \cos \theta} \right)$

$= \tan^{-1} (\tan \theta) = \theta = \sin^{-1} \frac{x}{a}$.

Example – 38

Show that $\tan^{-1} \left(\frac{\sqrt{1+x^2} + 1}{x} \right) = \frac{\pi}{2} - \frac{1}{2} \tan^{-1} x$, $x > 0$

Sol. Let $\theta = \tan^{-1} x \Rightarrow x = \tan \theta$, $0 < \theta < \frac{\pi}{2}$

$\therefore \tan^{-1} \left(\frac{\sqrt{1+x^2} + 1}{x} \right) = \tan^{-1} \left(\frac{\sqrt{1+\tan^2 \theta} + 1}{\tan \theta} \right) = \tan^{-1} \left(\frac{\sec \theta + 1}{\tan \theta} \right)$

$= \tan^{-1} \left(\frac{\frac{1}{\cos \theta} + 1}{\frac{\sin \theta}{\cos \theta}} \right) = \tan^{-1} \left(\frac{1 + \cos \theta}{\sin \theta} \right) = \tan^{-1} \left(\frac{2 \cos^2 \frac{\theta}{2}}{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}} \right)$

$= \tan^{-1} \left(\cot \frac{\theta}{2} \right)$

$= \tan^{-1} \left(\tan \left(\frac{\pi}{2} - \frac{\theta}{2} \right) \right)$

$= \frac{\pi}{2} - \frac{\theta}{2} = \frac{\pi}{2} - \frac{1}{2} \tan^{-1} x$.

Example – 39

Prove that $\sin^2 \left(2 \tan^{-1} \sqrt{\frac{1+x}{1-x}} \right) = 1 - x^2$ where $-1 \leq x < 1$.

Sol. L.H.S. $\sin^2 \left(2 \tan^{-1} \sqrt{\frac{1+x}{1-x}} \right) = \sin^2 (2\theta)$,

where $\theta = \tan^{-1} \sqrt{\frac{1+x}{1-x}}$

ie. $\tan \theta = \sqrt{\frac{1+x}{1-x}}$

Thus L.H.S. $= (\sin 2\theta)^2 = \left(\frac{2 \tan \theta}{1 + \tan^2 \theta} \right)^2$

$= \left\{ \frac{2\sqrt{1+x} / \sqrt{1-x}}{1 + \left(\frac{1+x}{1-x} \right)} \right\}^2$

$= \frac{4(1+x)(1-x)}{(1-x+1+x)^2} = 1 - x^2 = \text{R.H.S.}$

Example – 40

Prove that $\tan^{-1} \left(\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}} \right) = \frac{\pi}{4} + \frac{1}{2} \cos^{-1} (x^2)$.

Sol. Let $x^2 = \cos 2\theta$ so that $0 \leq 2\theta \leq \frac{\pi}{2}$, i.e., $0 \leq \theta \leq \frac{\pi}{4}$.

Now, $\sqrt{1+x^2} = \sqrt{1+\cos 2\theta} = \sqrt{2 \cos^2 \theta} = \sqrt{2} \cos \theta$

and $\sqrt{1-x^2} = \sqrt{1-\cos 2\theta} = \sqrt{2 \sin^2 \theta} = \sqrt{2} \sin \theta$

Hence, $\tan^{-1} \left(\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}} \right) = \tan^{-1} \left(\frac{\sqrt{2} \cos \theta + \sqrt{2} \sin \theta}{\sqrt{2} \cos \theta - \sqrt{2} \sin \theta} \right)$

$= \tan^{-1} \left(\frac{\sqrt{2} \cos \theta + \sqrt{2} \sin \theta}{\sqrt{2} \cos \theta - \sqrt{2} \sin \theta} \right) = \tan^{-1} \left(\frac{1 + \tan \theta}{1 - \tan \theta} \right)$



$$= \tan^{-1} \left[\tan \left(\frac{\pi}{4} + \theta \right) \right] = \frac{\pi}{4} + \theta = \frac{\pi}{4} + \frac{1}{2} \cos^{-1}(x^2)$$

$$[\because \cos 2\theta = x^2 \Rightarrow 2\theta = \cos^{-1}(x^2) \Rightarrow \theta = \frac{1}{2} \cos^{-1}(x^2)]$$

Example – 41

Solve the equation $\tan^{-1} \left(\frac{x-1}{x-2} \right) + \tan^{-1} \left(\frac{x+1}{x+2} \right) = \frac{\pi}{4}$,
 $|x| < 1$.

Sol. Given equation is $\tan^{-1} \left(\frac{x-1}{x-2} \right) + \tan^{-1} \left(\frac{x+1}{x+2} \right) = \frac{\pi}{4}$

$$\Rightarrow \tan^{-1} \left(\frac{\frac{x-1}{x-2} + \frac{x+1}{x+2}}{1 - \left(\frac{x-1}{x-2} \right) \left(\frac{x+1}{x+2} \right)} \right) = \frac{\pi}{4}$$

$$\left\{ \begin{array}{l} \because \tan^{-1} x + \tan^{-1} y = \tan^{-1} \left(\frac{x+y}{1-xy} \right) \text{ for } xy < 1 \\ \text{and for } |x| < 1, \left(\frac{x-1}{x-2} \right) \left(\frac{x+1}{x+2} \right) = \frac{1-x^2}{4-x^2} < 1 \end{array} \right\}$$

$$\Rightarrow \frac{(x-1)(x+2) + (x+1)(x-2)}{(x-2)(x+2) - (x-1)(x+1)} = \tan \frac{\pi}{4}$$

$$\Rightarrow \frac{2x^2 - 4}{(x^2 - 4) - (x^2 - 1)} = 1 \Rightarrow 2x^2 - 4 = -3$$

$$\Rightarrow 2x^2 = 1 \Rightarrow x^2 = \frac{1}{2} \Rightarrow x = \pm \frac{1}{\sqrt{2}}$$

Example – 42

Solve the equation $\sin \{2 \cos^{-1}(\cot(2 \tan^{-1} x))\} = 0$.

Sol. Given equation is $\sin \{2 \cos^{-1}(\cot(2 \tan^{-1} x))\} = 0$

$$\Rightarrow 2 \cos^{-1}(\cot(2 \tan^{-1} x)) = n\pi, n \in \mathbb{I}$$

$$\Rightarrow \cos^{-1} \{ \cot(2 \tan^{-1} x) \} = \frac{n\pi}{2}, n \in \mathbb{I}$$

$$\Rightarrow \cos^{-1} \{ \cot(2 \tan^{-1} x) \} = 0, \frac{\pi}{2}, \pi$$

($\because \cos^{-1} x$ lies in $[0, \pi]$)

$$\Rightarrow \cot(2 \tan^{-1} x) = \cos 0, \cos \frac{\pi}{2}, \cos \pi$$

$$\Rightarrow \frac{1}{\tan(2 \tan^{-1} x)} = -1, 0, 1 \text{ (As } \tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta} \text{)}$$

$$\Rightarrow \frac{1-x^2}{2x} = -1, 0, 1$$

$$\Rightarrow \frac{1-x^2}{2x} = -1, \frac{1-x^2}{2x} = 0 \text{ or } \frac{1-x^2}{2x} = 1$$

$$\Rightarrow x^2 - 2x - 1 = 0, x^2 = 1 \text{ or } x^2 + 2x - 1 = 0$$

$$\Rightarrow x = 1 \pm \sqrt{2}, \pm 1 \text{ or } -1 \pm \sqrt{2}$$

Example – 43

Solve : $2 \tan^{-1}(\cos x) = \tan^{-1}(2 \operatorname{cosec} x)$, $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2} \right]$

Sol. Given that,

$$2 \tan^{-1}(\cos x) = \tan^{-1}(2 \operatorname{cosec} x)$$

$$\Rightarrow \tan^{-1} \left(\frac{2 \cos x}{1 - \cos^2 x} \right) = \tan^{-1}(2 \operatorname{cosec} x)$$

$$\Rightarrow \frac{2 \cos x}{1 - \cos^2 x} = 2 \operatorname{cosec} x$$

$$\Rightarrow \frac{2 \cos x}{\sin^2 x} = \frac{2}{\sin x}$$

$$\Rightarrow \frac{\cos x}{\sin x} = 1 \quad (\because \sin x \neq 0)$$

$$\Rightarrow \cot x = 1 \Rightarrow x = \frac{\pi}{4} \text{ as } x \in \left[-\frac{\pi}{2}, \frac{\pi}{2} \right]$$

Example – 44

Solve for x : $\sin^{-1}(1-x) - 2 \sin^{-1} x = \frac{\pi}{2}$.

Sol. We have $\sin^{-1}(1-x) - 2 \sin^{-1} x = \frac{\pi}{2}$

$$\Rightarrow \sin^{-1}(1-x) = \frac{\pi}{2} + 2 \sin^{-1} x$$

$$\Rightarrow 1-x = \sin \left(\frac{\pi}{2} + 2 \sin^{-1} x \right)$$

$$\Rightarrow 1-x = \cos(2 \sin^{-1} x)$$

$$\Rightarrow 1-x = 1 - 2 \sin^2(\sin^{-1} x)$$



$$\Rightarrow 1 - x = 1 - 2 [\sin(\sin^{-1} x)]^2$$

$$\Rightarrow 1 - x = 1 - 2x^2$$

$$\Rightarrow 2x^2 - x = 0$$

$$\Rightarrow x(2x - 1) = 0$$

$$\Rightarrow x = 0, \frac{1}{2}$$

$$\text{For } x = \frac{1}{2}, \sin^{-1}(1 - x) - 2 \sin^{-1} x$$

$$= \sin^{-1}\left(1 - \frac{1}{2}\right) - 2 \sin^{-1} \frac{1}{2}$$

$$= \sin^{-1} \frac{1}{2} - 2 \sin^{-1} \frac{1}{2}$$

$$= -\sin^{-1} \frac{1}{2} = \frac{-\pi}{6} \neq \text{R.H.S.}$$

$x = \frac{1}{2}$ is not a solution of given equation. Hence, $x = 0$ is the required solution.

Example - 45

If $\tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \pi$, prove that $x + y + z = xyz$.

Sol. Let $\tan^{-1} x = \alpha$, $\tan^{-1} y = \beta$ and $\tan^{-1} z = \gamma$

$$\Rightarrow x = \tan \alpha, y = \tan \beta \text{ and } z = \tan \gamma$$

Now, given that,

$$\tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \pi$$

$$\Rightarrow \alpha + \beta + \gamma = \pi$$

$$\Rightarrow \alpha + \beta = \pi - \gamma$$

$$\Rightarrow \tan(\alpha + \beta) = \tan(\pi - \gamma)$$

$$\Rightarrow \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} = -\tan \gamma$$

Cross multiply, we have

$$\Rightarrow \tan \alpha + \tan \beta = -\tan \gamma + \tan \alpha \tan \beta \tan \gamma$$

$$\Rightarrow \tan \alpha + \tan \beta + \tan \gamma = \tan \alpha \tan \beta \tan \gamma$$

$$\Rightarrow x + y + z = xyz. \text{ Hence, the result.}$$

Example - 46

Express each of the following in the simplest form :

$$(i) \tan^{-1} \left\{ \sqrt{\frac{1 - \cos x}{1 + \cos x}} \right\}, -\pi < x < \pi$$

$$(ii) \tan^{-1} \left(\frac{\cos x}{1 + \sin x} \right), -\frac{\pi}{2} < x < \frac{\pi}{2}$$

Sol. (i) We have,

$$\begin{aligned} \tan^{-1} \left\{ \sqrt{\frac{1 - \cos x}{1 + \cos x}} \right\} \\ &= \tan^{-1} \left\{ \sqrt{\frac{2 \sin^2 \frac{x}{2}}{2 \cos^2 \frac{x}{2}}} \right\} = \tan^{-1} \left\{ \sqrt{\tan^2 \frac{x}{2}} \right\} = \tan^{-1} \left(\left| \tan \frac{x}{2} \right| \right) \\ &= \begin{cases} \tan^{-1} \left(-\tan \frac{x}{2} \right), & \text{if } -\pi < x < 0 \\ \tan^{-1} \left(\tan \frac{x}{2} \right), & \text{if } 0 \leq x < \pi \end{cases} \\ &= \begin{cases} \tan^{-1} \left\{ \tan \left(-\frac{x}{2} \right) \right\} = -\frac{x}{2}, & \text{if } -\pi < x < 0 \\ \tan^{-1} \left\{ \tan \frac{x}{2} \right\} = \frac{x}{2}, & \text{if } 0 < x < \pi \end{cases} \end{aligned}$$

(ii) We have,

$$\begin{aligned} \tan^{-1} \left(\frac{\cos x}{1 + \sin x} \right) &= \tan^{-1} \left\{ \frac{\cos^2 \frac{x}{2} - \sin^2 \frac{x}{2}}{\cos^2 \frac{x}{2} + \sin^2 \frac{x}{2} + 2 \sin \frac{x}{2} \cos \frac{x}{2}} \right\} \\ &= \tan^{-1} \left\{ \frac{\left(\cos \frac{x}{2} - \sin \frac{x}{2} \right) \left(\cos \frac{x}{2} + \sin \frac{x}{2} \right)}{\left(\cos \frac{x}{2} + \sin \frac{x}{2} \right)^2} \right\} = \tan^{-1} \left\{ \frac{\cos \frac{x}{2} - \sin \frac{x}{2}}{\cos \frac{x}{2} + \sin \frac{x}{2}} \right\} \\ &= \tan^{-1} \left\{ \frac{1 - \tan \frac{x}{2}}{1 + \tan \frac{x}{2}} \right\} = \tan^{-1} \left\{ \tan \left(\frac{\pi}{4} - \frac{x}{2} \right) \right\} \\ &= \frac{\pi}{4} - \frac{x}{2} \\ &\left[\because -\frac{\pi}{2} < x < \frac{\pi}{2} \Rightarrow -\frac{\pi}{4} < -\frac{x}{2} < \frac{\pi}{4} \Rightarrow 0 < \frac{\pi}{4} - \frac{x}{2} < \frac{\pi}{2} \right] \end{aligned}$$

ALITER We have,

$$\begin{aligned} \tan^{-1} \left(\frac{\cos x}{1 + \sin x} \right) &= \tan^{-1} \left\{ \frac{\sin \left(\frac{\pi}{2} + x \right)}{1 - \cos \left(\frac{\pi}{2} + x \right)} \right\} \\ &= \tan^{-1} \left\{ \frac{2 \sin \left(\frac{\pi}{4} + \frac{x}{2} \right) \cos \left(\frac{\pi}{4} + \frac{x}{2} \right)}{2 \sin^2 \left(\frac{\pi}{4} + \frac{x}{2} \right)} \right\} = \tan^{-1} \left\{ \cot \left(\frac{\pi}{4} + \frac{x}{2} \right) \right\} \\ &= \tan^{-1} \left\{ \tan \left\{ \frac{\pi}{2} - \left(\frac{\pi}{4} + \frac{x}{2} \right) \right\} \right\} = \tan^{-1} \left\{ \tan \left(\frac{\pi}{4} - \frac{x}{2} \right) \right\} = \frac{\pi}{4} - \frac{x}{2} \end{aligned}$$



Example – 47

Write the following functions in the simplest form :

(i) $\tan^{-1} \left\{ \frac{x}{\sqrt{a^2 - x^2}} \right\}, -a < x < a$

(ii) $\tan^{-1} \left\{ \sqrt{\frac{a-x}{a+x}} \right\}, -a < x < a$

(iii) $\sin^{-1} \left\{ \frac{x}{\sqrt{x^2 + a^2}} \right\}$

Sol. (i) Putting $x = a \sin \theta$, we have

$$\begin{aligned} & \tan^{-1} \left\{ \frac{x}{\sqrt{a^2 - x^2}} \right\} \\ &= \tan^{-1} \left\{ \frac{a \sin \theta}{\sqrt{a^2 - a^2 \sin^2 \theta}} \right\} \\ &= \tan^{-1} \left\{ \frac{a \sin \theta}{a \cos \theta} \right\} = \tan^{-1} (\tan \theta) = \theta = \sin^{-1} \frac{x}{a} \end{aligned}$$

$$\left[\begin{aligned} \because x = a \sin \theta &\Rightarrow \sin \theta = \frac{x}{a} \\ &\Rightarrow \theta = \sin^{-1} \frac{x}{a} \end{aligned} \right]$$

$$\left(\text{since } -a < x < a \Rightarrow -1 \leq \sin \theta \leq 1 \Rightarrow -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2} \right)$$

(ii) Putting $x = a \cos \theta$, we have

$$\begin{aligned} & \tan^{-1} \sqrt{\frac{a-x}{a+x}} \\ &= \tan^{-1} \sqrt{\frac{a-a \cos \theta}{a+a \cos \theta}} \\ &= \tan^{-1} \sqrt{\frac{1-\cos \theta}{1+\cos \theta}} \\ &= \tan^{-1} \sqrt{\frac{2 \sin^2 \frac{\theta}{2}}{2 \cos^2 \frac{\theta}{2}}} = \tan^{-1} \left(\tan \frac{\theta}{2} \right) = \tan^{-1} \left(\tan \frac{\theta}{2} \right) \end{aligned}$$

$$\left[\because -a < x < a \Rightarrow 0 < \theta < \pi < \frac{\theta}{2} < \frac{\pi}{2} \right]$$

$$\left(\begin{aligned} & \text{since } -1 \leq \cos \theta \leq 1 \\ & \Rightarrow 0 \leq \theta \leq \pi \\ & \Rightarrow 0 \leq \frac{\theta}{2} \leq \frac{\pi}{2} \end{aligned} \right)$$

$$= \frac{\theta}{2} = \frac{1}{2} \cos^{-1} \frac{x}{a}$$

$$\left[\because x = a \cos \theta \Rightarrow \cos \theta = \frac{x}{a} \Rightarrow \theta = \cos^{-1} \frac{x}{a} \right]$$

(iii) Putting $x = a \tan \theta$, we have

$$\sin^{-1} \left\{ \frac{x}{\sqrt{x^2 + a^2}} \right\}$$

$$= \sin^{-1} \left\{ \frac{a \tan \theta}{\sqrt{a^2 \tan^2 \theta + a^2}} \right\}$$

$$= \sin^{-1} \left\{ \frac{a \tan \theta}{a \sec \theta} \right\}$$

$$= \sin^{-1} (\sin \theta)$$

$$= \theta = \tan^{-1} \frac{x}{a}$$

$$\left[\because x = a \tan \theta \Rightarrow \tan \theta = \frac{x}{a} \Rightarrow \theta = \tan^{-1} \frac{x}{a} \right]$$

Example – 48

Prove that :

$$2 \sin^{-1} x = \begin{cases} \sin^{-1} (2x \sqrt{1-x^2}) & , \text{if } -\frac{1}{\sqrt{2}} \leq x \leq \frac{1}{\sqrt{2}} \\ \pi - \sin^{-1} (2x \sqrt{1-x^2}) & , \text{if } \frac{1}{\sqrt{2}} \leq x \leq 1 \\ -\pi - \sin^{-1} (2x \sqrt{1-x^2}) & , \text{if } -1 \leq x \leq -\frac{1}{\sqrt{2}} \end{cases}$$

Sol. Let $\sin^{-1} x = \theta$. Then,

$$x = \sin \theta,$$

$$\Rightarrow \cos \theta = \sqrt{1-x^2}$$

$$[\because \cos \theta > 0 \text{ for } \theta \in [-\pi/2, \pi/2]]$$

$$\therefore \sin 2\theta = 2 \sin \theta \cos \theta \quad \dots (i)$$

$$\Rightarrow \sin 2\theta = 2x \sqrt{1-x^2}$$



CASE I: When $-\frac{1}{\sqrt{2}} \leq x \leq \frac{1}{\sqrt{2}}$

We have,

$$-\frac{1}{\sqrt{2}} \leq x \leq \frac{1}{\sqrt{2}}$$

$$\Rightarrow -\frac{\pi}{4} \leq \theta \leq \frac{\pi}{4}$$

$$\Rightarrow -\frac{\pi}{2} \leq 2\theta \leq \frac{\pi}{2}$$

Also, $-\frac{1}{\sqrt{2}} \leq x \leq \frac{1}{\sqrt{2}}$

$$\Rightarrow -1 \leq 2x \sqrt{1-x^2} \leq 1$$

$$\therefore \sin 2\theta = 2x \sqrt{1-x^2} \quad \dots (i)$$

$$\Rightarrow 2\theta = \sin^{-1}(2x \sqrt{1-x^2})$$

$$\Rightarrow 2 \sin^{-1} x = \sin^{-1}(2x \sqrt{1-x^2})$$

CASE II: When $\frac{1}{\sqrt{2}} \leq x \leq 1$:

We have,

$$\frac{1}{\sqrt{2}} \leq x \leq 1$$

$$\Rightarrow \frac{1}{\sqrt{2}} \leq \sin \theta \leq 1$$

$$\Rightarrow \frac{\pi}{4} \leq \theta \leq \frac{\pi}{2}$$

$$\Rightarrow \frac{\pi}{2} \leq 2\theta \leq \pi$$

$$\Rightarrow -\pi \leq -2\theta \leq -\frac{\pi}{2}$$

$$\Rightarrow 0 \leq \pi - 2\theta \leq \frac{\pi}{2}$$

Also, $\frac{1}{\sqrt{2}} \leq x \leq 1 \Rightarrow 0 \leq 2x \sqrt{1-x^2} < 1$

$$\therefore \sin 2\theta = 2x \sqrt{1-x^2} \quad \text{from (i)}$$

$$\Rightarrow \sin(\pi - 2\theta) = 2x \sqrt{1-x^2}$$

$$(\text{since } \sin(\pi - x) = \sin x)$$

$$\Rightarrow \pi - 2\theta = \sin^{-1}(2x \sqrt{1-x^2})$$

$$\Rightarrow \pi - 2 \sin^{-1} x = \sin^{-1}(2x \sqrt{1-x^2})$$

$$\Rightarrow 2 \sin^{-1} x = \pi - \sin^{-1}(2x \sqrt{1-x^2})$$

CASE III: When $-1 \leq x \leq -\frac{1}{\sqrt{2}}$

We have,

$$-1 \leq x \leq -\frac{1}{\sqrt{2}}$$

$$\Rightarrow -1 \leq \sin \theta \leq -\frac{1}{\sqrt{2}}$$

$$\Rightarrow -\frac{\pi}{2} \leq \theta \leq -\frac{\pi}{4}$$

$$\Rightarrow -\pi \leq 2\theta \leq -\frac{\pi}{2}$$

$$\Rightarrow 0 \leq \pi + 2\theta \leq \frac{\pi}{2}$$

Also, $-1 \leq x \leq -\frac{1}{\sqrt{2}} \Rightarrow -1 \leq 2x \sqrt{1-x^2} \leq 0$

$$\therefore \sin 2\theta = 2x \sqrt{1-x^2} \quad [\text{From (i)}]$$

$$\Rightarrow -\sin(\pi + 2\theta) = 2x \sqrt{1-x^2}$$

$$\Rightarrow \sin(-\pi - 2\theta) = 2x \sqrt{1-x^2}$$

$$\Rightarrow -\pi - 2\theta = \sin^{-1}(2x \sqrt{1-x^2})$$

$$\Rightarrow 2\theta = -\pi - \sin^{-1}(2x \sqrt{1-x^2})$$

$$\Rightarrow 2 \sin^{-1} x = -\pi - \sin^{-1}(2x \sqrt{1-x^2})$$

Example - 49

Prove that :

$$\tan^{-1} \left\{ \frac{\sqrt{1+\cos x} + \sqrt{1-\cos x}}{\sqrt{1+\cos x} - \sqrt{1-\cos x}} \right\} = \frac{\pi}{4} + \frac{x}{2}, 0 < x < \frac{\pi}{2}$$

Sol. We have,

$$\tan^{-1} \left\{ \frac{\sqrt{1+\cos x} + \sqrt{1-\cos x}}{\sqrt{1+\cos x} - \sqrt{1-\cos x}} \right\}$$



$$= \tan^{-1} \left\{ \frac{\sqrt{2 \cos^2 \frac{x}{2}} + \sqrt{2 \sin^2 \frac{x}{2}}}{\sqrt{2 \cos^2 \frac{x}{2}} - \sqrt{2 \sin^2 \frac{x}{2}}} \right\}$$

$$= \tan^{-1} \left\{ \frac{\cos \frac{x}{2} + \sin \frac{x}{2}}{\cos \frac{x}{2} - \sin \frac{x}{2}} \right\}$$

$$\left[\because 0 < \frac{x}{2} < \frac{\pi}{4} \therefore \cos \frac{x}{2} > 0, \sin \frac{x}{2} > 0 \right]$$

$$= \tan^{-1} \left\{ \frac{1 + \tan \frac{x}{2}}{1 - \tan \frac{x}{2}} \right\}$$

$$= \tan^{-1} \left\{ \tan \left(\frac{\pi}{4} + \frac{x}{2} \right) \right\}$$

$$= \frac{\pi}{4} + \frac{x}{2}$$

$$\left[\because 0 < x < \frac{\pi}{2} \therefore \frac{\pi}{4} < \frac{\pi}{4} + \frac{x}{2} < \frac{\pi}{2} \right]$$

Example – 50

Prove that :

$$\tan^{-1} \left\{ \frac{\sqrt{1 + \cos x} + \sqrt{1 - \cos x}}{\sqrt{1 + \cos x} - \sqrt{1 - \cos x}} \right\} = \frac{\pi}{4} - \frac{x}{2}, \text{ if } \pi < x < \frac{3\pi}{2}$$

Sol. We have,

$$\tan^{-1} \left\{ \frac{\sqrt{1 + \cos x} + \sqrt{1 - \cos x}}{\sqrt{1 + \cos x} - \sqrt{1 - \cos x}} \right\}$$

$$= \tan^{-1} \left\{ \frac{\sqrt{2 \cos^2 \frac{x}{2}} + \sqrt{2 \sin^2 \frac{x}{2}}}{\sqrt{2 \cos^2 \frac{x}{2}} - \sqrt{2 \sin^2 \frac{x}{2}}} \right\}$$

$$= \tan^{-1} \left\{ \frac{\sqrt{2} \left| \cos \frac{x}{2} \right| + \sqrt{2} \left| \sin \frac{x}{2} \right|}{\sqrt{2} \left| \cos \frac{x}{2} \right| - \sqrt{2} \left| \sin \frac{x}{2} \right|} \right\}$$

$$= \tan^{-1} \left\{ \frac{-\cos \frac{x}{2} + \sin \frac{x}{2}}{-\cos \frac{x}{2} - \sin \frac{x}{2}} \right\}$$

$$\left[\begin{aligned} \because \pi < x < \frac{3\pi}{2} &\Rightarrow -\frac{\pi}{2} \leq \frac{\pi}{4} - \frac{x}{2} < -\frac{\pi}{4} \\ &\Rightarrow -\frac{3\pi}{4} < -\frac{x}{2} < -\frac{\pi}{2} \\ &\Rightarrow \frac{\pi}{2} < \frac{x}{2} < \frac{3\pi}{4} \\ &\Rightarrow \pi < x < \frac{3\pi}{2} \end{aligned} \right]$$

$$= \tan^{-1} \left\{ \frac{\cos \frac{x}{2} - \sin \frac{x}{2}}{\cos \frac{x}{2} + \sin \frac{x}{2}} \right\}$$

$$= \tan^{-1} \left\{ \frac{1 - \tan \frac{x}{2}}{1 + \tan \frac{x}{2}} \right\}$$

$$= \tan^{-1} \left\{ \tan \left(\frac{\pi}{4} - \frac{x}{2} \right) \right\}$$

$$= \frac{\pi}{4} - \frac{x}{2}$$

$$\left[\because \pi < x < \frac{3\pi}{2} \therefore -\frac{\pi}{2} < \frac{\pi}{4} - \frac{x}{2} < -\frac{\pi}{4} \right]$$

$$\frac{-3\pi}{4} < \frac{-x}{2} < \frac{-\pi}{2}$$



EXERCISE - 1 : BASIC OBJECTIVE QUESTIONS

Relation and its Types

- Let $R = \{(3, 3), (6, 6), (9, 9), (12, 12), (6, 12), (3, 9), (3, 12), (3, 6)\}$ be a relation on the set $A = \{3, 6, 9, 12\}$. The relation is
 - reflexive and symmetric only
 - an equivalence relation
 - reflexive only
 - reflexive and transitive only
- Let W denotes the words in the English dictionary. Define the relation R by $R = \{(x, y) \in W \times W : \text{the words } x \text{ and } y \text{ have at least one letter in common}\}$. Then, R is
 - reflexive, symmetric and not transitive
 - reflexive, symmetric and transitive
 - reflexive, not symmetric and transitive
 - not reflexive, symmetric and transitive
- Let N be the set of natural numbers and a relation R on N be defined by

$$R = \{(x, y) \in N \times N : x^3 - 3x^2y - xy^2 + 3y^3 = 0\}.$$
 Then the relation R is:
 - reflexive and symmetric, but not transitive
 - reflexive but neither symmetric nor transitive
 - an equivalence relation
 - symmetric but neither reflexive nor transitive

Functions and its classifications

- Let $f(x) = \frac{\alpha x^2}{x+1}$, $x \neq -1$. The value of α for which $f(a) = a$, ($a \neq 0$) is
 - $1 - \frac{1}{a}$
 - $\frac{1}{a}$
 - $1 + \frac{1}{a}$
 - $\frac{1}{a} - 1$
- The domain of $f(x) = \frac{\log_2(x+3)}{x^2+3x+2}$ is
 - $R - \{-1, -2\}$
 - $(-2, +\infty)$
 - $R - \{-1, -2, -3\}$
 - $(-3, +\infty) - \{-1, -2\}$
- The range of the function $y = \log_3(5 + 4x - x^2)$ is
 - $(0, 2]$
 - $(-\infty, 2]$
 - $(0, 9]$
 - none of these
- If $e^x + e^{f(x)} = e$, then range of the function of f is
 - $(-\infty, 1]$
 - $(-\infty, 1)$
 - $(1, \infty)$
 - $[1, \infty)$
- Range of the function $f(x) = \frac{x^2 + x + 2}{x^2 + x + 1}$; $x \in R$ is
 - $(1, \infty)$
 - $\left(1, \frac{11}{7}\right)$
 - $\left(1, \frac{7}{3}\right]$
 - $\left(1, \frac{7}{5}\right)$
- The equation $2 \sin^2 \frac{x}{2} \cdot \cos^2 x = x + \frac{1}{x}$, $0 < x \leq \frac{\pi}{2}$ has
 - one real solution
 - no real solution
 - infinitely many real solutions
 - none of these
- Let $f(x) = \frac{x - [x]}{1 + x - [x]}$, $x \in R$, then the range of f is :
 - $[0, 1]$
 - $[0, 1/2]$
 - $[0, 1/2)$
 - $(0, 1)$
- The range of k for which $\|x-1|-5\| = k$ have four distinct solutions -
 - $[0, 5]$
 - $(-\infty, 5)$
 - $[0, \infty)$
 - $(0, 5)$
- The function $f(x) = \cos \left(\log \left(x + \sqrt{x^2 + 1} \right) \right)$ is :
 - even
 - odd
 - constant
 - None of these
- Let $f: R \rightarrow R$ be a function such that $f(x) = x^3 - 6x^2 + 11x - 6$. Then
 - f is one-one and into
 - f is one-one and onto
 - f is many-one and into
 - f is many-one and onto



14. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $f(x) = x^3 + x^2 + 3x + \sin x$. Then
 (a) f is one-one and into (b) f is many-one and into
 (c) f is one-one and onto (d) f is many-one and onto
15. Let f be a function from $\mathbb{R}$ to $\mathbb{R}$ given by $f(x) = 2x + |\cos x|$. Then f is
 (a) one-one and into (b) one-one and onto
 (c) many-one and into (d) many-one and onto
16. Let f be a function from $\mathbb{R}$ to $\mathbb{R}$ given by $f(x) = \frac{x^2 - 4}{x^2 + 1}$. Then $f(x)$ is.
 (a) one-one and into (b) one-one and onto
 (c) many-one and into (d) many-one and onto
17. $f(x) = x + \sqrt{x^2}$ is a function from $\mathbb{R} \rightarrow \mathbb{R}$. Then $f(x)$ is
 (a) injective (b) surjective
 (c) bijective (d) none of these
18. A function $f : A \rightarrow B$, where $A = \{x : -1 \leq x \leq 1\}$ and $B = \{y : 1 \leq y \leq 2\}$ is defined by the rule $y = f(x) = 1 + x^2$. Which of the following statement is true?
 (a) f is injective but not surjective
 (b) f is surjective but not injective
 (c) f is both injective and surjective
 (d) f is neither injective nor surjective
19. $f(x) = \begin{cases} x, & \text{if } x \text{ is rational} \\ 0, & \text{if } x \text{ is irrational} \end{cases}$ and $g(x) = \begin{cases} 0, & \text{if } x \text{ is rational} \\ x, & \text{if } x \text{ is irrational} \end{cases}$. Then, $f - g$ is
 (a) one-one and into (b) neither one-one nor onto
 (c) many one and onto (d) one-one and onto
20. If $f : \mathbb{R} \rightarrow \mathbb{S}$, define by $f(x) = \sin x - \sqrt{3} \cos x + 1$, is onto, then the interval of $\mathbb{S}$ is
 (a) $[0, 1]$ (b) $[-1, 1]$
 (c) $[0, 3]$ (d) $[-1, 3]$
21. If a function $f : [2, \infty) \rightarrow B$ defined by $f(x) = x^2 - 4x + 5$ is a bijection, then B is :
 (a) $\mathbb{R}$ (b) $[1, \infty)$
 (c) $[4, \infty)$ (d) $[5, \infty)$
22. Which of the following function has period π ?
 (a) $2 \cos\left(\frac{2\pi x}{3}\right) + 3 \sin\left(\frac{\pi x}{3}\right)$
 (b) $|\tan x| + \cos 2x$
 (c) $4 \cos\left(2\pi x + \frac{\pi}{2}\right) + 2 \sin\left(\pi x + \frac{\pi}{4}\right)$
 (d) none of the above
23. Let $f(x) = \cos 3x + \sin \sqrt{3}x$. Then $f(x)$ is
 (a) a periodic function of period 2π .
 (b) a periodic function of period $\sqrt{3}\pi$.
 (c) not a periodic function
 (d) none of these
24. The period of $\sin^2 \theta$ is
 (a) π^2 (b) π
 (c) π^3 (d) $\pi/2$
25. Which one is not periodic
 (a) $|\sin 3x| + \sin^2 x$ (b) $\cos \sqrt{x} + \cos^2 x$
 (c) $\cos 4x + \tan^2 x$ (d) $\cos^2 x + \sin x$

Composition of a function

26. Let $f(x)$ be a function defined on $[0, 1]$ such that

$$f(x) = \begin{cases} x & x \in \mathbb{Q} \\ 1-x, & x \notin \mathbb{Q} \end{cases}$$

Then for all $x \in [0, 1]$, $f \circ f(x)$ is

- (a) a constant (b) $1+x$
 (c) x (d) none of these
27. If $f(x) = \sqrt{2-x}$ and $g(x) = \sqrt{1-2x}$, then the domain of $f[g(x)]$ is
 (a) $(-\infty, 1/2]$ (b) $[1/2, \infty)$
 (c) $(-\infty, -3/2]$ (d) none of these
28. Let $f(x) = \sin x$ and $g(x) = \ln |x|$. If the ranges of the composition functions $f \circ g$ and $g \circ f$ are R_1 and R_2 respectively, then
 (a) $R_1 = \{u : -1 \leq u < 1\}$, $R_2 = \{v : -\infty < v < 0\}$
 (b) $R_1 = \{u : -\infty < u < 0\}$, $R_2 = \{v : -1 \leq v \leq 0\}$
 (c) $R_1 = \{u : -1 < u < 1\}$, $R_2 = \{v : -\infty < v < 0\}$
 (d) $R_1 = \{u : -1 \leq u \leq 1\}$, $R_2 = \{v : -\infty < v \leq 0\}$



29. If $g\{f(x)\} = |\sin x|$ and $f\{g(x)\} = (\sin \sqrt{x})^2$, then

- (a) $f(x) = \sin^2 x, g(x) = \sqrt{x}$
- (b) $f(x) = \sin x, g(x) = |x|$
- (c) $f(x) = x^2, g(x) = \sin \sqrt{x}$
- (d) f and g cannot be determined

30. If $f(x) = \sin^2 x$, $g(x) = \sqrt{x}$ and $h(x) = \cos^{-1} x$, $0 \leq x \leq 1$, then -

- (a) $h \circ g \circ f(x) = g \circ f \circ h(x)$
- (b) $g \circ f \circ h(x) = f \circ h \circ g(x)$
- (c) $f \circ h \circ g(x) = h \circ g \circ f(x)$
- (d) None of these

31. If $f(g(x)) = |\cos x|$, $g(f(x)) = \cos^2 \sqrt{x}$, then -

- (a) $f(x)$ is a periodic function and $g(x)$ is a non-periodic function.
- (b) $f(x)$ is a non-periodic function and $g(x)$ is a periodic function.
- (c) Both $f(x)$ and $g(x)$ are periodic functions
- (d) Neither $f(x)$ nor $g(x)$ is a periodic function

32. Consider the functions $f(x) = \sqrt{x}$ and $g(x) = 7x + b$. If the function $y = f \circ g(x)$ passes through (4, 6) then the value of b is

- (a) 8
- (b) -8
- (c) -25
- (d) $4 - 7\sqrt{6}$

Inverse of a Function

33. The inverse of the function $f(x) = \frac{a^x - a^{-x}}{a^x + a^{-x}}$ is

(where codomain of $f(x)$ is $(-1, 1)$)

- (a) $\frac{1}{2} \log_a \left(\frac{1-x}{1+x} \right)$
- (b) $\frac{1}{2} \log_a \left(\frac{1+x}{1-x} \right)$
- (c) $\log_a \left(\frac{1+x}{1-x} \right)$
- (d) none of these

34. Let $f: [-1, \infty) \rightarrow \mathbb{R}$ be given by $f(x) = (x+1)^2 - 1$, $x \geq -1$. Then $f^{-1}(x)$, is :

- (a) $-1 + \sqrt{x+1}$
- (b) $-1 - \sqrt{x+1}$
- (c) does not exist because f is not one-one
- (d) does not exist because f is not onto

35. The inverse of the function $y = [1 - (x-3)^4]^{1/7}$ is

- (a) $3 + (1-x^7)^{1/4}$
- (b) $3 - (1-x^7)^{1/4}$
- (c) $3 - (1+x^7)^{1/4}$
- (d) none of these

Functional Equations

36. If $3f(x) + 5f\left(\frac{1}{x}\right) = \frac{1}{x} - 3$, $\forall x(x \neq 0) \in \mathbb{R}$, then $f(x)$ is equal to :

- (a) $\frac{1}{16} \left(\frac{3}{x} + 5x - 6 \right)$
- (b) $\frac{1}{16} \left(-\frac{3}{x} + 5x - 6 \right)$
- (c) $\frac{1}{16} \left(\frac{3}{x} - 5x - 6 \right)$
- (d) none of these

37. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function given by $f(x+y) = f(x) + f(y)$ for all $x, y \in \mathbb{R}$ such that $f(1) = a$. Then, $f(x) =$

- (a) a^x
- (b) ax
- (c) x^a
- (d) $a+x$

38. Let f be a real valued function satisfying $f(x+y) = f(x)f(y)$ for all $x, y \in \mathbb{R}$ such that $f(1) = a$. Then, $f(x) =$

- (a) a^x
- (b) ax
- (c) x^a
- (d) none of these

39. If $a f(x+1) + b f\left(\frac{1}{x+1}\right) = x$, $x \neq -1$, $a \neq -b$, then $f(1)$ is

equal to

- (a) $a+b$
- (b) $a^2 - b^2$
- (c) $\frac{1}{a+b}$
- (d) $f(1) = 0$



Simplification problems of ITF

Properties of ITF

40. If $\alpha = \tan^{-1} \left(\tan \frac{5\pi}{4} \right)$ and $\beta = \tan^{-1} \left(-\tan \frac{2\pi}{3} \right)$, then

- (a) $4\alpha = 3\beta$ (b) $3\alpha = 4\beta$
(c) $\alpha - \beta = \frac{7\pi}{12}$ (d) none of these

41. Which one of the following is correct?

- (a) $\tan 1 > \tan^{-1} 1$ (b) $\tan 1 < \tan^{-1} 1$
(c) $\tan 1 = \tan^{-1} 1$ (d) None of the above

42. The value of $\sin \left[\frac{\pi}{2} - \sin^{-1} \left(-\frac{\sqrt{3}}{2} \right) \right]$ is :

- (a) $\frac{\sqrt{3}}{2}$ (b) $-\frac{\sqrt{3}}{2}$
(c) $\frac{1}{2}$ (d) $-\frac{1}{2}$

43. $\cot^{-1} (\sqrt{\cos \alpha}) - \tan^{-1} (\sqrt{\cos \alpha}) = x$, then $\sin x$ is equal to

- (a) $\tan^2 \left(\frac{\alpha}{2} \right)$ (b) $\cot^2 \left(\frac{\alpha}{2} \right)$
(c) $\tan \alpha$ (d) $\cot \left(\frac{\alpha}{2} \right)$

44. If $\tan(\cos^{-1} x) = \sin \left(\cot^{-1} \frac{1}{2} \right)$, then x is equal to :

- (a) $\pm \frac{5}{3}$ (b) $\pm \frac{\sqrt{5}}{3}$
(c) $\pm \frac{5}{\sqrt{3}}$ (d) None of these

45. $\cos [\tan^{-1} \{ \sin (\cot^{-1} x) \}]$ is equal to :

- (a) $\sqrt{\frac{x^2+2}{x^2+3}}$ (b) $\sqrt{\frac{x^2+2}{x^2+1}}$
(c) $\sqrt{\frac{x^2+1}{x^2+2}}$ (d) None of these

46. If $\sin^{-1} x = \frac{\pi}{5}$, for some $x \in (-1, 1)$, then the value of $\cos^{-1} x$ is :

- (a) $\frac{3\pi}{10}$ (b) $\frac{5\pi}{10}$
(c) $\frac{7\pi}{10}$ (d) $\frac{9\pi}{10}$

47. The value of $\cot^{-1} \frac{3}{4} + \sin^{-1} \frac{5}{13}$ is :

- (a) $\sin^{-1} \frac{63}{65}$ (b) $\sin^{-1} \frac{12}{13}$
(c) $\sin^{-1} \frac{65}{68}$ (d) $\sin^{-1} \frac{5}{12}$

48. If $\cos^{-1} x - \cos^{-1} \frac{y}{2} = \alpha$, then $4x^2 - 4xy \cos \alpha + y^2$ is equal to

- (a) $-4 \sin^2 \alpha$ (b) $4 \sin^2 \alpha$
(c) 4 (d) $2 \sin 2\alpha$

49. If $\tan^{-1} \frac{x-1}{x+2} + \tan^{-1} \frac{x+1}{x+2} = \frac{\pi}{4}$, then x is equal to :

- (a) $\frac{1}{\sqrt{2}}$ (b) $-\frac{1}{\sqrt{2}}$
(c) $\sqrt{\frac{5}{2}}$ (d) $\pm \frac{1}{2}$

50. If $2 \tan^{-1} (\cos x) = \tan^{-1} (2 \operatorname{cosec} x)$, then the value of x is :

- (a) $\frac{3\pi}{4}$ (b) $\frac{\pi}{4}$
(c) $\frac{\pi}{3}$ (d) None of these

51. The equation $2 \cos^{-1} x + \sin^{-1} x = \frac{11\pi}{6}$ has :

- (a) no solution (b) only one solution
(c) two solutions (d) three solutions



ITF- Domain & Range

52. If $\tan^{-1}\left(\frac{a}{x}\right) + \tan^{-1}\left(\frac{b}{x}\right) = \frac{\pi}{2}$, then x is equal to :

- (a) $\sqrt{ab}$ (b) $\sqrt{2ab}$
(c) $2ab$ (d) ab

53. If $\cos^{-1} x > \sin^{-1} x$, then :

- (a) $x < 0$ (b) $-1 < x < 0$
(c) $0 \leq x < \frac{1}{\sqrt{2}}$ (d) $-1 \leq x < \frac{1}{\sqrt{2}}$

54. Set of values of x satisfying $\cos^{-1} \sqrt{x} > \sin^{-1} \sqrt{x}$

- (a) $\left(0, \frac{1}{2}\right)$ (b) $\left[0, \frac{1}{2}\right)$
(c) $\left(\frac{1}{2}, 1\right)$ (d) $\left[\frac{1}{2}, 1\right]$

55. The value of $\cos(2 \cos^{-1} x + \sin^{-1} x)$ at $x = \frac{1}{5}$ is :

- (a) 1 (b) 3
(c) 0 (d) $-\frac{2\sqrt{6}}{5}$

56. The value of $\sum_{r=0}^{\infty} \tan^{-1}\left(\frac{1}{1+r+r^2}\right)$ is equal to :

- (a) $\frac{\pi}{2}$ (b) $\frac{3\pi}{4}$
(c) $\frac{\pi}{4}$ (d) None of these

57. If $\tan^{-1} \frac{1}{1+2} + \tan^{-1} \frac{1}{1+(2)(3)} + \tan^{-1} \frac{1}{1+(3)(4)}$

$$+ \dots + \tan^{-1} \frac{1}{1+n(n+1)} = \tan^{-1} \theta, \text{ then } \theta =$$

- (a) $\frac{n}{n+1}$ (b) $\frac{n+1}{n+2}$
(c) $\frac{n}{n+2}$ (d) $\frac{n-1}{n+2}$

58. The domain of $\sin^{-1} \left[\log_3 \left(\frac{x}{3} \right) \right]$ is

- (a) $[1, 9]$ (b) $[-1, 9]$
(c) $[-9, 1]$ (d) $[-9, -1]$

59. The domain of the function $f(x) = \frac{\sin^{-1}(x-3)}{\sqrt{9-x^2}}$ is

- (a) $[1, 2]$ (b) $[2, 3]$
(c) $[2, 3]$ (d) $[1, 2]$

60. The largest interval lying in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ for which the function $f(x) = 4^{-x^2} + \cos^{-1}\left(\frac{x}{2} - 1\right) + \log(\cos x)$ is defined, is

- (a) $[0, \pi]$ (b) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
(c) $\left[-\frac{\pi}{4}, \frac{\pi}{2}\right)$ (d) $\left[0, \frac{\pi}{2}\right)$

61. If $\theta = \sin^{-1} x + \cos^{-1} x - \tan^{-1} x$, $x \geq 0$, then the smallest interval in which θ lies, is given by :

- (a) $\frac{\pi}{2} \leq \theta \leq \frac{3\pi}{4}$ (b) $-\frac{\pi}{4} \leq \theta \leq 0$
(c) $0 \leq \theta \leq \frac{\pi}{4}$ (d) $\frac{\pi}{4} \leq \theta \leq \frac{\pi}{2}$

62. Range of $f(x) = \sin^{-1} x + \tan^{-1} x + \sec^{-1} x$ is

- (a) $\left(\frac{\pi}{4}, \frac{3\pi}{4}\right)$ (b) $\left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$
(c) $\left\{\frac{\pi}{4}, \frac{3\pi}{4}\right\}$ (d) None of these

63. Range of $f(x) = \sin^{-1} x + \cot^{-1} x + \tan^{-1} x$ is

- (a) $[0, \pi]$ (b) $\left[\frac{\pi}{2}, \pi\right]$
(c) $\left[\frac{\pi}{4}, \pi\right]$ (d) $[-\pi, \pi]$



Numerical Value Type Questions

64. Let $n(A) = 4$ and $n(B) = 6$. Then the number of one-one functions from A to B is
65. The period of the function $f(x) = \sin^4 x + \cos^4 x$ is π/k . Then the value of k is
66. The period of the function $f(x) = |\sin 4x| + |\cos 4x|$ is π/k . Then the value of k is
67. Let $[x]$ denote the greatest integer $\leq x$. If $f(x) = [x]$ and $g(x) = |x|$, then the value of $f\left(g\left(\frac{8}{5}\right)\right) - g\left(f\left(-\frac{8}{5}\right)\right)$ is
68. If $f: \mathbb{R} \rightarrow \mathbb{R}$ is given by
- $$f(x) = \begin{cases} -1, & \text{when } x \text{ is rational} \\ 1, & \text{when } x \text{ is irrational} \end{cases}$$
- then $(f \circ f)(1 - \sqrt{3})$ is equal to
69. The number of real solutions of the equation $e^x = x$ is
70. The number of real solutions of the equation $\log_{0.5} x = |x|$ is
71. The number of real solutions of the equation $\sin(e^x) = 5^x + 5^{-x}$ is
72. If f is a real valued function such that $f(x+y) = f(x) + f(y)$ and $f(1) = 5$, then the value of $f(100)$ is
73. If $2f(x+1) + f\left(\frac{1}{x+1}\right) = 2x$ and $x \neq -1$, then $f(2)$ is equal to $k/6$. Then the value of k is.
74. If $f(x) = ax^2 + bx + c$ satisfies the identity $f(x+1) - f(x) = 8x + 3$ for all $x \in \mathbb{R}$. Then, $a + b =$
75. $\sec^2(\tan^{-1} 2) + \operatorname{cosec}^2(\cot^{-1} 3)$ is equal to
76. If $\sum_{i=1}^{20} \sin^{-1} x_i = 10\pi$, then $\sum_{i=1}^{20} x_i$ is equal to :
77. If $\cos^{-1} \frac{3}{5} - \sin^{-1} \frac{4}{5} = \cos^{-1} x$, then x is equal to
78. If $k \leq \sin^{-1} x + \cos^{-1} x + \tan^{-1} x \leq K$ and $k + K = m\pi$. Then the value of m is.
79. The value of $\tan^{-1}(1) + \tan^{-1}(0) + \tan^{-1}(2) + \tan^{-1}(3)$ is equal to $k\pi$. Then the value of k is
80. If $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, then the value of $\tan^{-1}\left(\frac{\tan x}{4}\right) + \tan^{-1}\left(\frac{3 \sin 2x}{5 + 3 \cos 2x}\right)$ is kx . Then the value of k is



EXERCISE - 2 : PREVIOUS YEAR JEE MAIN QUESTIONS

1. If $f(x) = 2\tan^{-1}x + \sin^{-1}\left(\frac{2x}{1+x^2}\right)$, $x > 1$, then $f(5)$ is equal to
(2015/Online Set-1)
- (a) $\frac{\pi}{2}$ (b) $\tan^{-1}\left(\frac{65}{156}\right)$
(c) $4\tan^{-1}(5)$ (d) π
2. If $f(x) + 2f\left(\frac{1}{x}\right) = 3x$, $x \neq 0$, and
 $S = \{x \in \mathbb{R} : f(x) = f(-x)\}$; then S :
(2016)
- (a) contains exactly one element
(b) contains exactly two elements.
(c) contains more than two elements.
(d) is an empty set.
3. For $x \in \mathbb{R}$, $x \neq 0$, $x \neq 1$, let $f_0(x) = \frac{1}{1-x}$
and $f_{n+1}(x) = f_0(f_n(x))$, $n = 0, 1, 2, \dots$. Then the value
of $f_{100}(3) + \frac{8}{3}f_1\left(\frac{2}{3}\right) + f_2\left(\frac{3}{2}\right)$ is equal to:
(2016/Online Set-1)
- (a) $\frac{8}{3}$ (b) $\frac{5}{3}$
(c) $\frac{4}{3}$ (d) $\frac{1}{3}$
4. The function $f: \mathbb{R} \rightarrow \left[-\frac{1}{2}, \frac{1}{2}\right]$ defined as $f(x) = \frac{x}{1+x^2}$,
is:
(2017)
- (a) invertible
(b) injective but not surjective
(c) surjective but not injective
(d) neither injective nor surjective.
5. Let $f(x) = 2^{10} \cdot x + 1$ and $g(x) = 3^{10} \cdot x - 1$.
If $(f \circ g)(x) = x$, then x is equal to : (2017/Online Set-1)
- (a) $\frac{3^{10}-1}{3^{10}-2^{-10}}$ (b) $\frac{2^{10}-1}{2^{10}-3^{-10}}$
(c) $\frac{1-3^{-10}}{2^{10}-3^{-10}}$ (d) $\frac{1-2^{-10}}{3^{10}-2^{-10}}$
6. The value of $\tan^{-1}\left|\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}}\right|$, $|x| < \frac{1}{2}$, $x \neq 0$, is
equal to : (2017/Online Set-1)
- (a) $\frac{\pi}{4} + \frac{1}{2}\cos^{-1}x^2$ (b) $\frac{\pi}{4} + \cos^{-1}x^2$
(c) $\frac{\pi}{4} - \frac{1}{2}\cos^{-1}x^2$ (d) $\frac{\pi}{4} - \cos^{-1}x^2$
7. A value of x satisfying the equation
 $\sin[\cot^{-1}(1+x)] = \cos[\tan^{-1}x]$, is :
(2017/Online Set-2)
- (a) $-\frac{1}{2}$ (b) -1
(c) 0 (d) $\frac{1}{2}$
8. The function $f: \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(x) = x - 5\left[\frac{x}{5}\right]$,
where $\mathbb{N}$ is the set of natural numbers and $[x]$ denotes the
greatest integer less than or equal to x , is :
(2017/Online Set-2)
- (a) one-one and onto.
(b) one-one but not onto.
(c) onto but not one-one.
(d) neither one-one nor onto.



9. Let $S = \{(\lambda, \mu) \in \mathbb{R} \times \mathbb{R} : f(t) = \lfloor |\lambda| e^{|t|} - \mu \rfloor$
 $\sin(2|t|), t \in \mathbb{R}$ is a differentiable function}. Then S is a
 subset of : **(2018/Online Set-1)**
- (a) $\mathbb{R} \times [0, \infty)$ (b) $[0, \infty) \times \mathbb{R}$
 (c) $\mathbb{R} \times (-\infty, 0)$ (d) $(-\infty, 0) \times \mathbb{R}$
10. Consider the following two binary relations on the set
 $A = \{a, b, c\}$:
 $R_1 = \{(c, a), (b, b), (a, c), (c, c), (b, c), (a, a)\}$ and
 $R_2 = \{(a, b), (b, a), (c, c), (c, a), (a, a), (b, b), (a, c)\}$. Then :
(2018/Online Set-1)
- (a) both R_1 and R_2 are not symmetric.
 (b) R_1 is not symmetric but it is transitive.
 (c) R_2 is symmetric but it is not transitive.
 (d) both R_1 and R_2 are transitive.
11. Let $f: A \rightarrow B$ be a function defined as $f(x) = \frac{x-1}{x-2}$, where
 $A = \mathbb{R} - \{2\}$ and $B = \mathbb{R} - \{1\}$. Then f is :
(2018/Online Set-2)
- (a) Invertible and $f^{-1}(y) = \frac{3y-1}{y-1}$
 (b) Invertible and $f^{-1}(y) = \frac{2y-1}{y-1}$
 (c) Invertible and $f^{-1}(y) = \frac{2y+1}{y-1}$
 (d) Not invertible
12. If the function f defined as $f(x) = \frac{1}{x} - \frac{k-1}{e^{2x}-1}, x \neq 0$, is
 continuous at $x=0$, then the ordered pair $(k, f(0))$ is equal
 to : **(2018/Online Set-2)**
- (a) $(3, 2)$ (b) $(3, 1)$
 (c) $(2, 1)$ (d) $\left(\frac{1}{3}, 2\right)$
13. If $f(x) = \sin^{-1}\left(\frac{2 \times 3^x}{1+9^x}\right)$, then $f'\left(-\frac{1}{2}\right)$ equals :
(2018/Online Set-2)
- (a) $-\sqrt{3} \log_e \sqrt{3}$ (b) $\sqrt{3} \log_e \sqrt{3}$
 (c) $-\sqrt{3} \log_e 3$ (d) $\sqrt{3} \log_e 3$
14. Let N denote the set of all natural numbers. Define two
 binary relations on N as
 $R_1 = \{(x, y) \in N \times N : 2x + y = 10\}$ and
 $R_2 = \{(x, y) \in N \times N : x + 2y = 10\}$. Then :
(2018/Online Set-3)
- (a) Range of R_1 is $\{2, 4, 8\}$.
 (b) Range of R_2 is $\{1, 2, 3, 4\}$.
 (c) Both R_1 and R_2 are symmetric relations.
 (d) Both R_1 and R_2 are transitive relations.
15. If $\alpha = \cos^{-1}\left(\frac{3}{5}\right), \beta = \tan^{-1}\left(\frac{1}{3}\right)$, where $0 < \alpha, \beta < \frac{\pi}{2}$,
 then $\alpha - \beta$ is equal to : **(2019-04-08/Shift-1)**
- (a) $\tan^{-1}\left(\frac{9}{5\sqrt{10}}\right)$ (b) $\cos^{-1}\left(\frac{9}{5\sqrt{10}}\right)$
 (c) $\tan^{-1}\left(\frac{9}{14}\right)$ (d) $\sin^{-1}\left(\frac{9}{5\sqrt{10}}\right)$
16. If $f(x) = \log_e\left(\frac{1-x}{1+x}\right), |x| < 1$, then $f\left(\frac{2x}{1+x^2}\right)$ is equal
 to : **(2019-04-08/Shift-1)**
- (a) $2f(x)$ (b) $2f(x^2)$
 (c) $(f(x))^2$ (d) $-2f(x)$
17. The sum of the solutions of the equation
 $|\sqrt{x}-2| + \sqrt{x}(\sqrt{x}-4) + 2 = 0, (x > 0)$ is equal to:
(2019-04-08/Shift-1)
18. Let $\sum_{k=1}^{10} f(a+k) = 16(2^{10}-1)$, where the function f
 satisfies $f(x+y) = f(x)f(y)$ for all natural numbers x ,
 y and $f(1) = 2$. Then the natural number 'a' is:
(2019-04-09/Shift-1)



19. Let $f(x) = x^2$, $x \in \mathbb{R}$. For any $A \subseteq \mathbb{R}$, define $g(A) = \{x \in \mathbb{R} : f(x) \in A\}$. If $S = [0, 4]$, then which one of the following statements is not true ?

(2019-04-10/Shift-1)

- (a) $g(f(S)) \neq S$ (b) $f(g(S)) = S$
(c) $g(f(S)) = g(S)$ (d) $f(g(S)) \neq f(S)$

20. Let $f(x) = \log_e(\sin x)$, $(0 < x < \pi)$ and $g(x) = \sin^{-1}(e^{-x})$ ($x \geq 0$). If α is a positive real number such that $a = (f \circ g)'(\alpha)$ and $b = (f \circ g)(\alpha)$, then:

(2019-04-10/Shift-2)

- (a) $a\alpha^2 + b\alpha + a = 0$ (b) $a\alpha^2 - b\alpha - a = 1$
(c) $a\alpha^2 - b\alpha - a = 0$ (d) $a\alpha^2 + b\alpha - a = -2a^2$

21. If $\cos^{-1} x - \cos^{-1} \frac{y}{2} = \alpha$, where

$$-1 \leq x \leq 1, -3 \leq y \leq 2, x \leq \frac{y}{2}, \text{ then for all}$$

$x, y, 4x^2 - 4xy \cos \alpha + y^2$ is equal to:

(2019-04-10/Shift-2)

- (a) $4 \sin^2 \alpha$ (b) $2 \sin^2 \alpha$
(c) $4 \sin^2 \alpha - 2x^2 y^2$ (d) $4 \cos^2 \alpha + 2x^2 y^2$

22. For $x \in \left(0, \frac{3}{2}\right)$ let $f(x) = \sqrt{x}$, $g(x) = \tan x$ and

$h(x) = \frac{1-x^2}{1+x^2}$. If $\phi(x) = ((h \circ f) \circ g)(x)$, then $\phi\left(\frac{\pi}{3}\right)$ is equal to

(2019-04-12/Shift-1)

- (a) $\tan \frac{\pi}{12}$ (b) $\tan \frac{11\pi}{12}$
(c) $\tan \frac{7\pi}{12}$ (d) $\tan \frac{5\pi}{12}$

23. The value of $\sin^{-1}\left(\frac{12}{13}\right) - \sin^{-1}\left(\frac{3}{5}\right)$ is equal to

(2019-04-12/Shift-1)

- (a) $\pi - \sin^{-1}\left(\frac{63}{65}\right)$ (b) $\frac{\pi}{2} - \sin^{-1}\left(\frac{56}{65}\right)$
(c) $\frac{\pi}{2} - \cos^{-1}\left(\frac{9}{65}\right)$ (d) $\pi - \cos^{-1}\left(\frac{33}{65}\right)$

24. If $[x]$ denotes the greatest integer $\leq x$, then the system of linear equations $[\sin \theta]x + [-\cos \theta]y = 0$, $[\cot \theta]x + y = 0$

(2019-04-12/Shift-2)

- (a) have infinitely many solutions if $\theta \in \left(\frac{\pi}{2}, \frac{2\pi}{3}\right)$ and

has a unique solution if $\theta \in \left(\pi, \frac{7\pi}{6}\right)$

- (b) has a unique solution if $\theta \in \left(\frac{\pi}{2}, \frac{2\pi}{3}\right) \cup \left(\pi, \frac{7\pi}{6}\right)$

- (c) has a unique solution if $\theta \in \left(\frac{\pi}{2}, \frac{2\pi}{3}\right)$ and have

infinitely many solutions if $\theta \in \left(\pi, \frac{7\pi}{6}\right)$

- (d) have infinitely many solutions if

$$\theta \in \left(\frac{\pi}{2}, \frac{2\pi}{3}\right) \cup \left(\pi, \frac{7\pi}{6}\right)$$

25. If $\cos^{-1}\left(\frac{2}{3x}\right) + \cos^{-1}\left(\frac{3}{4x}\right) = \frac{\pi}{2}$ ($x > \frac{3}{4}$), then x is equal to

(2019-01-09/Shift-1)

- (a) $\frac{\sqrt{145}}{12}$ (b) $\frac{\sqrt{145}}{10}$
(c) $\frac{\sqrt{146}}{12}$ (d) $\frac{\sqrt{145}}{11}$

26. If $x = \sin^{-1}(\sin 10)$ and $y = \cos^{-1}(\cos 10)$, then $y - x$ is equal to:

(2019-01-09/Shift-2)

- (a) 0 (b) 10
(c) 7π (d) π

27. The value of $\cot\left(\sum_{n=1}^{19} \cot^{-1}\left(1 + \sum_{p=1}^n 2p\right)\right)$ is:

(2019-01-10/Shift-2)

- (a) $\frac{21}{19}$ (b) $\frac{19}{21}$
(c) $\frac{22}{23}$ (d) $\frac{23}{22}$



28. All x satisfying the inequality $(\cot^{-1} x)^2 - 7(\cot^{-1} x) + 10 > 0$, lie in the interval :
(2019-01-11/Shift-2)
- (a) $(-\infty, \cot 5) \cup (\cot 4, \cot 2)$
(b) $(\cot 2, \infty)$
(c) $(-\infty, \cot 5) \cup (\cot 2, \infty)$
(d) $(\cot 5, \cot 4)$
29. Considering only the principal values of inverse functions, the set $A = \left\{ x \geq 0 : \tan^{-1}(2x) + \tan^{-1}(3x) = \frac{\pi}{4} \right\}$
(2019-01-12/Shift-1)
- (a) contains two elements
(b) contains more than two elements
(c) is a singleton
(d) is an empty set
30. The domain of the function $f(x) = \sin^{-1} \left(\frac{|x|+5}{x^2+1} \right)$ is $(-\infty, -a] \cup [a, \infty)$. Then a is equal to :
(2020-09-02/Shift-1)
- (a) $\frac{\sqrt{17}-1}{2}$ (b) $\frac{\sqrt{17}}{2}$
(c) $\frac{1+\sqrt{17}}{2}$ (d) $\frac{\sqrt{17}}{2} + 1$
31. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function which satisfies $f(x+y) = f(x) + f(y) \quad \forall x, y \in \mathbb{R}$. If $f(1) = 2$ and $g(n) = \sum_{k=1}^{(n-1)} f(k)$, $n \in \mathbb{N}$ then the value of n , for which $g(n) = 20$, is :
(2020-09-02/Shift-2)
- (a) 9 (b) 5
(c) 4 (d) 20
32. $2\pi - \left(\sin^{-1} \frac{4}{5} + \sin^{-1} \frac{5}{13} + \sin^{-1} \frac{16}{65} \right)$ is equal to :
(2020-09-03/Shift-1)
- (a) $\frac{5\pi}{4}$ (b) $\frac{3\pi}{2}$
(c) $\frac{7\pi}{4}$ (d) $\frac{\pi}{2}$
33. Let R_1 and R_2 be two relations defined as follows :
 $R_1 = \{(a, b) \in \mathbb{R}^2 : a^2 + b^2 \in \mathbb{Q}\}$ and
 $R_2 = \{(a, b) \in \mathbb{R}^2 : a^2 + b^2 \notin \mathbb{Q}\}$, where $\mathbb{Q}$ is the set of all rational numbers. Then :
(2020-09-03/Shift-2)
- (a) R_1 is transitive but R_2 is not transitive
(b) R_1 and R_2 are both transitive
(c) R_2 is transitive but R_1 is not transitive
(d) Neither R_1 nor R_2 is transitive
34. Let $A = \{a, b, c\}$ and $B = \{1, 2, 3, 4\}$. Then the number of elements in the set $C = \{f : A \rightarrow B \mid 2 \in f(A) \text{ and } f \text{ is not one-one}\}$ is :
(2020-09-05/Shift-2)
35. For a suitably chosen real constant a , let a function, $f : \mathbb{R} - \{-a\} \rightarrow \mathbb{R}$ be defined by $f(x) = \frac{a-x}{a+x}$. Further suppose that for any real number $x \neq -a$ and $f(x) \neq -a$, $(f \circ f)(x) = x$. Then $f\left(-\frac{1}{2}\right)$ is equal to:
(2020-09-06/Shift-2)
- (a) -3 (b) 3
(c) $\frac{1}{3}$ (d) $-\frac{1}{3}$
36. If $g(x) = x^2 + x - 1$ and $(g \circ f)(x) = 4x^2 - 10x + 5$ then, $f\left(\frac{5}{4}\right)$ is equal to :
(2020-01-07/Shift-1)
- (a) $-\frac{3}{2}$ (b) $-\frac{1}{2}$
(c) $\frac{1}{2}$ (d) $\frac{3}{2}$
37. The inverse function of $f(x) = \frac{8^{2x} - 8^{-2x}}{8^{2x} + 8^{-2x}}$, $x \in (-1, 1)$, is :
(2020-01-08/Shift-1)
- (a) $\frac{1}{4}(\log_8 e) \log_e \left(\frac{1-x}{1+x} \right)$ (b) $\frac{1}{4}(\log_8 e) \log_e \left(\frac{1+x}{1-x} \right)$
(c) $\frac{1}{4} \log_e \left(\frac{1+x}{1-x} \right)$ (d) $\frac{1}{4} \log_e \left(\frac{1-x}{1+x} \right)$



38. Let $f(x) = \sin^{-1} x$ and $g(x) = \frac{x^2 - x - 2}{2x^2 - x - 6}$. If

$g(2) = \lim_{x \rightarrow 2} g(x)$ then the domain of the function fg is

(26-02-2021/Shift-2)

(a) $(-\infty, -1] \cup [2, \infty)$ (b) $(-\infty, -2] \cup \left[-\frac{4}{3}, \infty\right)$

(c) $(-\infty, -2] \cup [-1, \infty)$ (d) $(-\infty, -2] \cup \left[-\frac{3}{2}, \infty\right)$

39. Let $A = \{1, 2, 3, \dots, 10\}$ and $f : A \rightarrow A$ be defined as

$$f(k) = \begin{cases} k+1 & \text{if } k \text{ is odd} \\ k & \text{if } k \text{ is even} \end{cases}$$

Then the number of possible functions $g : A \rightarrow A$ such that $gof = f$ is

(26-02-2021/Shift-2)

(a) 10^5 (b) ${}^{10}C_5$

(c) $5!$ (d) 5^5

40. Let $R = \{(P, Q) | P \text{ and } Q \text{ are at the same distance from the origin}\}$ be a relation, then the equivalence class of $(1, -1)$ is the set:

(26-02-2021/Shift-1)

(a) $S = \{(x, y) | x^2 + y^2 = 4\}$

(b) $S = \{(x, y) | x^2 + y^2 = 1\}$

(c) $S = \{(x, y) | x^2 + y^2 = 2\}$

(d) $S = \{(x, y) | x^2 + y^2 = \sqrt{2}\}$

41. Let $f, g : \mathbb{N} \rightarrow \mathbb{N}$ such that

$f(n+1) = f(n) + f(1) \forall n \in \mathbb{N}$ and g be any arbitrary function. Which of the following statements is **NOT** true?

(25-02-2021/Shift-1)

(a) If g is onto, then fg is one-one

(b) f is one-one

(c) If f is onto, then $f(n) = n \forall n \in \mathbb{N}$

(d) If fg is one-one, then g is one-one

42. A possible value of $\tan\left(\frac{1}{4}\sin^{-1}\frac{\sqrt{63}}{8}\right)$ is :

(24-02-2021/Shift-2)

(a) $2\sqrt{2} - 1$ (b) $\frac{1}{\sqrt{7}}$

(c) $\frac{1}{2\sqrt{2}}$ (d) $\sqrt{7} - 1$

43. If $a + \alpha = 1, b + \beta = 2$ and

$af(x) + \alpha f\left(\frac{1}{x}\right) = bx + \frac{\beta}{x}, x \neq 0$, then the value of the

expression $\frac{f(x) + f\left(\frac{1}{x}\right)}{x + \frac{1}{x}}$ is _____.

(24-02-2021/Shift-2)

44. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = 2x - 1$ and

$g : \mathbb{R} - \{1\} \rightarrow \mathbb{R}$ be defined as $g(x) = \frac{x - \frac{1}{2}}{x - 1}$. Then the

composition function $f(g(x))$ is (24-02-2021/Shift-2)

(a) One-one but not onto

(b) Both one-one and onto

(c) Neither one-one nor onto

(d) Onto but not one-one

45. Let $f : \mathbb{R} - \{3\} \rightarrow \mathbb{R} - \{1\}$ be defined by $f(x) = \frac{x-2}{x-3}$.

Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be given as $g(x) = 2x - 3$. Then, the sum

of all the values of x for which $f^{-1}(x) + g^{-1}(x) = \frac{13}{2}$ is equal to.

(18-03-2021/Shift-2)

(a) 2 (b) 7

(c) 5 (d) 3



46. Let $A = \{2, 3, 4, 5, \dots, 30\}$ and $'\sim'$ be an equivalence relation on $A \times A$, defined by $(a, b) \sim (c, d)$, if and only if $ad = bc$. Then the number of ordered pairs which satisfy this equivalence relation with ordered pair $(4, 3)$ is equal to
(16-03-2021/Shift-2)
- (a) 6 (b) 5
(c) 8 (d) 7
47. The inverse of $y = 5^{\log x}$ is : (17-03-2021/Shift-1)
- (a) $x = y^{\log 5}$ (b) $x = y^{\frac{1}{\log 5}}$
(c) $x = 5^{\log y}$ (d) $x = 5^{\frac{1}{\log y}}$
48. If $\cot^{-1}(\alpha) = \cot^{-1} 2 + \cot^{-1} 8 + \cot^{-1} 18 + \cot^{-1} 32 + \dots$ upto 100 terms, then α is (17-03-2021/Shift-1)
- (a) 1.03 (b) 1.02
(c) 1.01 (d) 1.00
49. The sum of possible values of x for $\tan^{-1}(x+1) + \cot^{-1}\left(\frac{1}{x-1}\right) = \tan^{-1}\left(\frac{8}{31}\right)$ is : (17-03-2021/Shift-1)
- (a) $-\frac{32}{4}$ (b) $-\frac{31}{4}$
(c) $-\frac{30}{4}$ (d) $-\frac{33}{4}$
50. If $f(x) = \sin\left(\cos^{-1}\left(\frac{1-2^{2x}}{1+2^{2x}}\right)\right)$ and its first derivative with respect to x is $-\frac{b}{a} \log_e 2$ when $x = 1$, where a and b are integers, then the minimum value of $|a^2 - b^2|$ is (17-03-2021/Shift-1)
51. The number of solutions of the equation $x + 2 \tan x = \frac{\pi}{2}$ in the interval $[0, 2\pi]$ is : (17-03-2021/Shift-2)
- (a) 4 (b) 5
(c) 3 (d) 2
52. Let $[x]$ denote the greatest integer $\leq x$, where $x \in \mathbb{R}$. If the domain of the real valued function $f(x) = \sqrt{\frac{[x]-2}{[x]-3}}$ is $(-\infty, a) \cup [b, c) \cup [4, \infty)$, $a < b < c$, then the value of $a + b + c$ is: (20-07-2021/Shift-1)
- (a) -3 (b) 1
(c) -2 (d) 8
53. Let $f : \mathbb{R} - \left\{\frac{\alpha}{6}\right\} \rightarrow \mathbb{R}$ be defined by $f(x) = \frac{5x+3}{6x-\alpha}$. Then the value of α for which $(f \circ f)(x) = x$, for all $x \in \mathbb{R} - \left\{\frac{\alpha}{6}\right\}$, is ? (20-07-2021/Shift-2)
- (a) No such α exists (b) 5
(c) 6 (d) 8
54. Let $g : \mathbb{N} \rightarrow \mathbb{N}$ be defined as $g(3n+1) = 3n+2$,
 $g(3n+2) = 3n+3$,
 $g(3n+3) = 3n+1$, for all $n \geq 0$.
Then which of the following statements is true ? (25-07-2021/Shift-1)
- (a) $g \circ g \circ g = g$
(b) There exists an onto function $f : \mathbb{N} \rightarrow \mathbb{N}$ such that $\text{fog} = f$
(c) There exists a one-one function $f : \mathbb{N} \rightarrow \mathbb{N}$ such that $\text{fog} = f$
(d) There exists a function $f : \mathbb{N} \rightarrow \mathbb{N}$ such that $\text{gof} = f$
55. Let $S = \{1, 2, 3, 4, 5, 6, 7\}$. Then the number of possible functions $f : S \rightarrow S$ such that $f(m.n) = f(m).f(n)$ for every $m, n \in S$ and $m, n \in S$ is equal to ____ (27-07-2021/Shift-1)



56. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined as

$$f(x+y) + f(x-y) = 2f(x)f(y), f\left(\frac{1}{2}\right) = -1. \text{ Then, the}$$

value of $\sum_{k=1}^{20} \frac{1}{\sin(k)\sin(k+f(k))}$ is equal to:

(27-07-2021/Shift-2)

(a) $\operatorname{cosec}^2(1)\operatorname{cosec}(21)\sin(20)$

(b) $\sec^2(1)\sec(21)\cos(20)$

(c) $\operatorname{cosec}^2(21)\cos(20)\cos(2)$

(d) $\sec^2(21)\sin(20)\sin(2)$

57. Let $[x]$ denote the greatest integer less than or equal to x . Then, the values of $x \in \mathbb{R}$ satisfying the equation $[e^x]^2 + [e^x + 1] - 3 = 0$ lie in the interval:

(22-07-2021/Shift-2)

(a) $[\ln 2, \ln 3]$ (b) $\left[0, \frac{1}{e}\right]$

(c) $[0, \ln 2]$ (d) $[1, e]$

58. Consider function $f : A \rightarrow B$ and

$g : B \rightarrow C$ ($A, B, C \subseteq \mathbb{R}$) such that $(g \circ f)^{-1}$ exists, then:

(25-07-2021/Shift-2)

(a) f and g both are one-one

(b) f is onto and g is one-one

(c) f is one-one and g is onto

(d) f and g both are onto

59. The range of the function

$$f(x) = \log_{\sqrt{3}} \left(3 + \cos\left(\frac{3\pi}{4} + x\right) + \cos\left(\frac{\pi}{4} + x\right) \right)$$

$$+ \cos\left(\frac{\pi}{4} - x\right) - \cos\left(\frac{3\pi}{4} - x\right)$$

(01-09-2021/Shift-2)

(a) $\left[\frac{1}{\sqrt{5}}, \sqrt{5}\right]$ (b) $(0, \sqrt{5})$

(c) $[0, 2]$ (d) $[-2, 2]$

60. If be a polynomial of degree 3 such that $f(k) = -\frac{2}{k}$ for

$k = 2, 3, 4, 5$. Then the value $52 - 10f(10)$ is equal to

(01-09-2021/Shift-2)

61. The domain of the function $\operatorname{cosec}^{-1}\left(\frac{1+x}{x}\right)$ is :

(26-08-2021/Shift-2)

(a) $\left[-\frac{1}{2}, \infty\right) - \{0\}$ (b) $\left(-\frac{1}{2}, \infty\right) - \{0\}$

(c) $\left[-\frac{1}{2}, 0\right) \cup [1, \infty)$ (d) $\left(-1, -\frac{1}{2}\right] \cup (0, \infty)$

62. Let z be the set of all integers,

$$A = \{(x, y) \in \mathbb{Z} \times \mathbb{Z}; (x-2)^2 + y^2 \leq 4\}$$

$$B = \{(x, y) \in \mathbb{Z} \times \mathbb{Z}; x^2 + y^2 \leq 4\} \text{ and}$$

$$C = \{(x, y) \in \mathbb{Z} \times \mathbb{Z}; (x-2)^2 + (y-2)^2 \leq 4\}$$

If the total number of relation from $A \cap B$ to $A \cap C$ is 2^p , then the value of p is :

(27-08-2021/Shift-2)

(a) 16 (b) 25

(c) 49 (d) 9

63. Let M and m respectively be the maximum and minimum values of the function $f(x) = \tan^{-1}(\sin x + \cos x)$ in

$$\left[0, \frac{\pi}{2}\right]. \text{ Then the value of } \tan(M-m) \text{ is equal to:}$$

(27-08-2021/Shift-2)

(a) $3 - 2\sqrt{2}$ (b) $3 + 2\sqrt{2}$

(c) $2 - \sqrt{3}$ (d) $2 + \sqrt{3}$



64. Which of the following is not correct for relation R on the set of real numbers? (31-08-2021/Shift-1)
- (a) $(x, y) \in R \Leftrightarrow |x - y| \leq 1$ is reflexive and symmetric.
- (b) $(x, y) \in R \Leftrightarrow 0 < |x - y| \leq 1$ is symmetric but not transitive.
- (c) $(x, y) \in R \Leftrightarrow |x| - |y| \leq 1$ is reflexive but not symmetric.
- (d) $(x, y) \in R \Leftrightarrow 0 < |x| - |y| \leq 1$ is neither transitive nor symmetric.
65. Let $f : \mathbb{N} \rightarrow \mathbb{N}$ be a function such that $f(m + n) = f(m) + f(n)$ for every $m, n \in \mathbb{N}$. If $f(6) = 18$, then $f(2) \cdot f(3)$ is equal to: (31-08-2021/Shift-2)
- (a) 36 (b) 6
- (c) 18 (d) 54
66. If the domain of the function $f(x) = \frac{\cos^{-1} \sqrt{x^2 - x + 1}}{\sqrt{\sin^{-1} \left(\frac{2x - 1}{2} \right)}}$ is the interval $(\alpha, \beta]$, then $\alpha + \beta$ is equal to: (22-07-2021/Shift-2)
- (a) 2 (b) $\frac{3}{2}$
- (c) $\frac{1}{2}$ (d) 1
67. The number of real roots of the equation $e^{4x} - e^{3x} - 4e^{2x} - e^x + 1 = 0$ is equal to _____. (27-07-2021/Shift-2)
68. If $x^2 + 9y^2 - 4x + 3 = 0$, $x, y \in \mathbb{R}$, then x and y respectively lie in the intervals: (27-08-2021/Shift--1)
- (a) $[1, 3]$ and $\left[-\frac{1}{3}, \frac{1}{3}\right]$
- (b) $\left[-\frac{1}{3}, \frac{1}{3}\right]$ and $[1, 3]$
- (c) $[1, 3]$ and $[1, 3]$
- (d) $\left[-\frac{1}{3}, \frac{1}{3}\right]$ and $\left[-\frac{1}{3}, \frac{1}{3}\right]$
69. The number of solutions of the equation $32^{\tan^2 x} + 32^{\sec^2 x} = 81, 0 \leq x \leq \frac{\pi}{4}$ is: (31-08-2021/Shift-2)
- (a) 0 (b) 1
- (c) 2 (d) 3
70. If the functions are defined as $f(x) = \sqrt{x}$ and $g(x) = \sqrt{1 - x}$, then what is the common domain of the following functions $f + g, f - g, \frac{f}{g}, \frac{g}{f}, g - f$ where $(f \pm g)(x) = f(x) \pm g(x), (f/g)(x) = \frac{f(x)}{g(x)}$ (18-03-2021/Shift-1)
- (a) $0 \leq x < 1$ (b) $0 < x \leq 1$
- (c) $0 \leq x \leq 1$ (d) $0 < x < 1$
71. The real valued function $f(x) = \frac{\operatorname{cosec}^{-1} x}{\sqrt{x - [x]}}$, where $[x]$ denotes the greatest integer less than or equal to x , is defined for all x belonging to: (18-03-2021/Shift-1)
- (a) all integers except 0, -1, 1
- (b) all reals except the interval $[-1, 1]$
- (c) all reals except integers
- (d) all non-integers except the interval $[-1, 1]$
72. Let x denote the total number of one - one functions from a set A with 3 elements to a set B with 5 elements and y denote the total number of one one functions from the set A to the set $A \times B$. Then: (25-02-2021/Shift-2)
- (a) $y = 91x$ (b) $2y = 91x$
- (c) $y = 273x$ (d) $2y = 273x$
73. A function $f(x)$ is given by $f(x) = \frac{5^x}{5^x + 5}$, then the sum of the series $f\left(\frac{1}{20}\right) + f\left(\frac{2}{20}\right) + f\left(\frac{3}{20}\right) + \dots + f\left(\frac{39}{20}\right)$ (25-02-2021/Shift-2)
- (a) $\frac{49}{2}$ (b) $\frac{29}{2}$
- (c) $\frac{39}{2}$ (d) $\frac{19}{2}$



74. Let f be any function defined on $\mathbb{R}$ and let it satisfy the condition $|f(x) - f(y)| \leq |x - y|^2, \forall (x, y) \in \mathbb{R}$

If $f(0) = 1$, then :

(26-02-2021/Shift-1)

(a) $f(x) > 0, \forall x \in \mathbb{R}$

(b) $f(x) = 0, \forall x \in \mathbb{R}$

(c) $f(x) < 0, \forall x \in \mathbb{R}$

(d) $f(x)$ can take any value in $\mathbb{R}$

75. $\operatorname{cosec}\left[2\cot^{-1}(5) + \cos^{-1}\left(\frac{4}{5}\right)\right]$ is equal to:

(25-02-2021/Shift-2)

(a) $\frac{65}{56}$

(b) $\frac{65}{33}$

(c) $\frac{75}{56}$

(d) $\frac{56}{33}$

76. If $\frac{\sin^{-1} x}{a} = \frac{\cos^{-1} x}{b} = \frac{\tan^{-1} y}{c}; 0 < x < 1$, then the value

of $\cos\left(\frac{\pi c}{a+b}\right)$ is:

(26-02-2021/Shift-1)

(a) $\frac{1-y^2}{1+y^2}$

(b) $\frac{1-y^2}{y\sqrt{y}}$

(c) $\frac{1-y^2}{2y}$

(d) $1-y^2$

77. If $0 < a, b < 1$, and $\tan^{-1} a + \tan^{-1} b = \frac{\pi}{4}$, then the value of

$(a+b) - \left(\frac{a^2+b^2}{2}\right) + \left(\frac{a^3+b^3}{3}\right) - \left(\frac{a^4+b^4}{4}\right) + \dots$ is:

(26-02-2021/Shift-2)

(a) $\log_e\left(\frac{e}{2}\right)$

(b) $\log_e 2$

(c) $e^2 - 1$

(d) e

78. Given that the inverse trigonometric functions take principal values only. Then, the number of real values of x

which satisfy $\sin^{-1}\left(\frac{3x}{5}\right) + \sin^{-1}\left(\frac{4x}{5}\right) = \sin^{-1} x$ is equal to

(16-03-2021/Shift-2)

(a) 1

(b) 3

(c) 2

(d) 0

79. The number of solutions of the equation

$\sin^{-1}\left[x^2 + \frac{1}{3}\right] + \cos^{-1}\left[x^2 - \frac{2}{3}\right] = x^2$, for $x \in [-1, 1]$, and

$[x]$ denotes the greatest integer less than or equal to x , is:

(17-03-2021/Shift-2)

(a) 4

(b) 0

(c) Infinite

(d) 2

80. The number of real roots of the equation

$\tan^{-1}\sqrt{x(x+1)} + \sin^{-1}\sqrt{x^2+x+1} = \frac{\pi}{4}$ is:

(20-07-2021/Shift-1)

(a) 0

(b) 4

(c) 1

(d) 2



EXERCISE - 3 : ADVANCED OBJECTIVE QUESTIONS

Objective Questions I [Only one correct option]

1. The domain of the function $f(x) = \sqrt[4]{\log_3 \left(\frac{1}{|\cos x|} \right)}$ is :

- (a) $(-\infty, \infty)$
 (b) $(-\infty, \infty) - \{n\pi | n \in \mathbb{I}\}$

(c) $(-\infty, \infty) - \left\{ (2n+1)\frac{\pi}{2} : n \in \mathbb{I} \right\}$

(d) none of the above

2. The range of the function $f(x) = \cos [x]$, for $-\pi/2 < x < \pi/2$ contains.

- (a) $\{-1, 1, 0\}$ (b) $\{\cos 1, 1, \cos 2\}$
 (c) $\{\cos 1, -\cos 1, 1\}$ (d) $[-1, 1]$

3. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by

$$f(x) = \frac{x^2 - 8}{x^2 + 2}, \text{ then } f \text{ is :}$$

- (a) one-one but not onto (b) one-one and onto
 (c) onto but not one-one (d) neither one-one nor onto

4. Let $f: \mathbb{R} - \{n\} \rightarrow \mathbb{R}$ be a function defined by

$$f(x) = \frac{x-m}{x-n}, \text{ where } m \neq n. \text{ This function is-}$$

- (a) one-one onto (b) one-one into
 (c) many-one onto (d) many one into

5. Let $A = (x_1, x_2, \dots, x_8)$, $B = (y_1, y_2, y_3)$, the total no. of functions $f: A \rightarrow B$ that are onto and there are exactly four elements (x) in A such that $f(x) = y_3$, is equal to

- (a) $16 \times {}^8C_4$ (b) $14 \times {}^8C_4$
 (c) $16 \times {}^4C_4$ (d) None of these

6. If $f(x+y) = f(x) \cdot f(y)$ for all real x, y and $f(0) \neq 0$ then the

$$\text{function } g(x) = \frac{f(x)}{1 + (f(x))^2} \text{ is}$$

- (a) even function (b) odd function
 (c) odd if $f(x) > 0$ (d) neither even nor odd

7. Let $f: (-\infty, 2] \rightarrow (-\infty, 4]$ be a function defined by $f(x) = 4x - x^2$. Then $f^{-1}(x)$ is :

- (a) $2 - \sqrt{4-x}$ (b) $2 + \sqrt{4-x}$
 (c) $\sqrt{4-x}$ (d) $\sqrt{4+x}$

8. Let $f: \mathbb{R} \rightarrow \mathbb{R}$, $g: \mathbb{R} \rightarrow \mathbb{R}$ be two functions given by $f(x) = 2x - 3$, $g(x) = x^3 + 5$. Then $(f \circ g)^{-1}(x)$ is equal to :

- (a) $\left(\frac{x-7}{2} \right)^{1/3}$ (b) $\left(\frac{x+7}{2} \right)^{1/2}$
 (c) $\left(x - \frac{7}{2} \right)^{1/3}$ (d) $\left(\frac{x-2}{7} \right)^{1/3}$

9. Let $f: \left[-\frac{\pi}{3}, \frac{2\pi}{3} \right]$ be a function defined as

$$f(x) = \sqrt{3} \sin x - \cos x + 2. \text{ The } f^{-1}(x) \text{ is given by}$$

- (a) $\sin^{-1} \left(\frac{x-2}{2} \right) - \frac{\pi}{6}$ (b) $\sin^{-1} \left(\frac{x-2}{2} \right) + \frac{\pi}{6}$
 (c) $\frac{2\pi}{3} + \cos^{-1} \left(\frac{x-2}{2} \right)$ (d) none of these

10. If the function $f: [1, \infty) \rightarrow [1, \infty)$ is defined by $f(x) = 2^{x(x-1)}$, then $f^{-1}(x)$ is

- (a) $\left(\frac{1}{2} \right)^{x(x-1)}$ (b) $\left(\frac{1}{2} \right) \left[1 + \sqrt{1 + 4 \log_2 x} \right]$
 (c) $\left(\frac{1}{2} \right) \left[1 - \sqrt{1 + 4 \log_2 x} \right]$ (d) not defined

11. If the function $f: \mathbb{R} \rightarrow \mathbb{R}$ be such that $f(x) = x - [x]$, where $[x]$ denotes the greatest integer less than or equal to x , then $f^{-1}(x)$ is

- (a) $\frac{1}{x - [x]}$ (b) $[x] - x$
 (c) not defined (d) none of these

12. The range of the function $f(x) = |x-1| + |x-2|$, $-1 \leq x \leq 3$, is

- (a) $[1, 3]$ (b) $[1, 5]$
 (c) $[3, 5]$ (d) none of these



13. The domain of the function $f(x) = \sqrt{\left(\frac{1}{\sin x} - 1\right)}$ is :

- (a) $\left\{2n\pi, 2n\pi + \frac{\pi}{2}\right\}, \forall n \in \mathbb{I}$
 (b) $(2n\pi, (2n+1)\pi) \forall n \in \mathbb{I}$
 (c) $((2n-1)\pi, 2n\pi) \forall n \in \mathbb{I}$
 (d) None of the above

14. $f(x) = \frac{\sqrt{|\tan x| + \tan x}}{\sqrt{3x}}$ is defined for :

- (a) $\mathbb{R}^+$ (b) $\mathbb{R}^+ - \left\{\frac{1}{3}\right\}$
 (c) $\mathbb{R}^+ - \left\{n\pi + \frac{\pi}{2} \mid n \in \mathbb{W}\right\}$ (d) none of these

15. The domain of the function $f(x) = x^{\frac{1}{\log x}}$ is :

- (a) $(0, \infty) - \{1\}$ (b) $(0, \infty)$
 (c) $[0, \infty)$ (d) $[0, \infty) - \{1\}$

16. The minimum value of $2^{(x^2-3)^3+27}$ is :

- (a) 1 (b) 2
 (c) 2^{27} (d) None of these

17. Let $f(x) = \min\{x, x^2\}$, for every $x \in \mathbb{R}$. Then :

- (a) $f(x) = \begin{cases} x, & x \geq 1 \\ x^2, & 0 \leq x < 1 \\ x, & x < 0 \end{cases}$ (b) $f(x) = \begin{cases} x^2, & x \geq 1 \\ x, & x < 1 \end{cases}$
 (c) $f(x) = \begin{cases} x, & x \geq 1 \\ x^2, & x < 1 \end{cases}$ (d) $f(x) = \begin{cases} x^2, & x \geq 1 \\ x, & 0 \leq x < 1 \\ x^2, & x < 0 \end{cases}$

18. The domain of definition of

$$f(x) = \log_2 \left(-\log_{1/2} \left(1 + \frac{1}{x^{1/4}} \right) - 1 \right) \text{ is}$$

- (a) $(0, 1)$ (b) $(0, 1]$
 (c) $[1, \infty)$ (d) $(1, \infty)$

19. The domain of the function

$$f(x) = \log_3 \left[-\log_{1/2} \left(1 + \frac{1}{x^{1/5}} \right) - 1 \right] \text{ is}$$

- (a) $(-\infty, 1)$ (b) $(0, 1)$
 (c) $(1, \infty)$ (d) none of these

20. Let f be a real valued function defined by

$$f(x) = \frac{e^x - e^{-|x|}}{e^x + e^{|x|}} \text{ then range of } f(x) \text{ is :}$$

- (a) $\mathbb{R}$ (b) $[0, 1]$
 (c) $[0, 1)$ (d) $[0, 1/2)$

21. The number of pairs, (x, y) , $x, y \in \mathbb{R}$, satisfying

$$4x^2 - 4x + 2 = \sin^2 y \text{ and } x^2 + y^2 \leq 3 \text{ are}$$

- (a) 0 (b) 4
 (c) 2 (d) infinite

22. If $f(x) = \frac{x-1}{x+1}$, then $f(2x)$ is :

- (a) $\frac{f(x)+1}{f(x)+3}$ (b) $\frac{3f(x)+1}{f(x)+3}$
 (c) $\frac{f(x)+3}{f(x)+1}$ (d) $\frac{f(x)+3}{3f(x)+1}$

23. If $f(x) = \begin{cases} |x|, & x \leq 1 \\ 2-x, & x > 1 \end{cases}$, then $f(f(x))$ is equal to

- (a) $\begin{cases} 2-|x|, & x < -1 \\ |x|, & -1 \leq x \leq 1 \\ |2-x|, & x > 1 \end{cases}$ (b) $\begin{cases} |x|, & x < -1 \\ 2-|x|, & -1 \leq x \leq 1 \\ |2-x|, & x > 1 \end{cases}$

- (c) $\begin{cases} |2-x|, & x < -1 \\ |x|, & -1 \leq x \leq 1 \\ 2-|x|, & x > 1 \end{cases}$ (d) none of these



24. If $f\left(x + \frac{y}{8}, x - \frac{y}{8}\right) = xy$, then $f(m, n) + f(n, m) = 0$

- (a) only when $m = n$ (b) only when $m \neq n$
(c) only when $m = -n$ (d) for all m and n

25. If $2 < x^2 < 3$ then the number of positive roots of

$\left\{\frac{1}{x}\right\} = \{x^2\}$, $\{.\}$ denotes the fractional part of x , is :

- (a) 0 (b) 1
(c) 2 (d) 3

26. If $[x]$ denotes the greatest integer $\leq x$, then

$\left[\frac{2}{3}\right] + \left[\frac{2}{3} + \frac{1}{99}\right] + \left[\frac{2}{3} + \frac{2}{99}\right] + \dots + \left[\frac{2}{3} + \frac{98}{99}\right]$ is equal to

- (a) 99 (b) 98
(c) 66 (d) 65

27. $f(x) = |x - 1|$, $f: \mathbb{R}^+ \rightarrow \mathbb{R}$ and $g(x) = e^x$, $g: [-1, \infty) \rightarrow \mathbb{R}$ If the function $f \circ g(x)$ is defined, then its domain and range respectively are.

- (a) $(0, \infty) \& [0, \infty)$ (b) $[-1, \infty) \& [0, \infty)$

- (c) $[-1, \infty) \& \left[1 - \frac{1}{e}, \infty\right)$ (d) $[-1, \infty) \& \left[\frac{1}{e} - 1, \infty\right)$

28. The number of positive integers satisfying the equation $x + \log_{10}(2^x + 1) = x \log_{10} 5 + \log_{10} 6$ is

- (a) 0 (b) 1
(c) 2 (d) infinite

29. A certain polynomial $P(x)$, $x \in \mathbb{R}$ when divided by $x - a$, $x - b$, $x - c$ leaves remainder a , b , c respectively. The remainder when $P(x)$ is divided by $(x - a)(x - b)(x - c)$ is (a , b , c and distinct).

- (a) 0 (b) x
(c) $ax + b - c$ (d) $ax^2 + bx + c$

30. Complete solution set of the equation

$|x^2 - 1 + \cos x| = |x^2 - 1| + |\cos x|$
belonging to $(-2\pi, \pi)$ is

(a) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \cup (-1, 1)$

(b) $\left[-\frac{3\pi}{2}, \frac{\pi}{2}\right] \cup [-1, 1] \cup \left[\frac{\pi}{2}, \pi\right)$

(c) $\left[-\frac{3\pi}{2}, -\frac{\pi}{2}\right] \cup \left[\frac{\pi}{2}, \pi\right)$

(d) $\left(-2\pi, -\frac{3\pi}{2}\right] \cup \left[-\frac{\pi}{2}, -1\right] \cup \left[1, \frac{\pi}{2}\right)$

31. The number points (x, y) , where curves $|y| = \ln |x|$ and $(x - 1)^2 + y^2 - 4 = 0$ cut each other, is

- (a) 2 (b) 3
(c) 1 (d) 6

32. Let f be a function satisfying

$$2f(xy) = (f(x))^y + (f(y))^x$$

and $f(1) = k \neq 1$, then $\sum_{r=1}^n f(r)$ is equal to :

- (a) $k^n - 1$ (b) k^n
(c) $k^n + 1$ (d) None of these

33. If $f(x) + 2f(1 - x) = x^2 + 2$, $\forall x \in \mathbb{R}$, then $f(x)$ is given as:

- (a) $\frac{(x-1)^2}{3}$ (b) $\frac{(x-2)^2}{3}$
(c) $x^2 - 1$ (d) $x^2 - 2$

34. The function $f(x) = \sqrt{\cos(\sin x)} + \sin^{-1}\left(\frac{1+x^2}{2x}\right)$ is defined for :

- (a) $x \in \{-1, 1\}$ (b) $x \in [-1, 1]$
(c) $x \in \mathbb{R}$ (d) $x \in (-1, 1)$

35. If $f(x) = \cos^{-1}\left(\frac{2-|x|}{4}\right) + [\log(3-x)]^{-1}$, then its domain is :

- (a) $[-2, 6]$ (b) $[-6, 2) \cup (2, 3)$
(c) $[-6, 2]$ (d) $[-2, 2) \cup (2, 3]$



36. If $1 < x < \sqrt{2}$, then number of solutions of the equation $\tan^{-1}(x-1) + \tan^{-1} x + \tan^{-1}(x+1) = \tan^{-1} 3x$, is/are
 (a) 0 (b) 1
 (c) 2 (d) 3
37. The complete solution set of $\sin^{-1}(\sin 5) > x^2 - 4x$ is
 (a) $|x-2| < \sqrt{9-2\pi}$ (b) $|x-2| > \sqrt{9-2\pi}$
 (c) $|x| < \sqrt{9-2\pi}$ (d) $|x| > \sqrt{9-2\pi}$
38. $\sum_{m=1}^n \tan^{-1} \frac{2m}{m^4 + m^2 + 2} =$
 (a) $\tan^{-1}(n^2 + n + 1)$ (b) $\tan^{-1}(n^2 - n + 1)$
 (c) $\tan^{-1} \frac{n^2 + n}{n^2 + n + 2}$ (d) None of these
39. The domain of the function $f(x) = \frac{1}{\sqrt{x^{12} - x^9 + x^4 - x + 1}}$ is
 (a) $(-\infty, -1)$ (b) $(1, \infty)$
 (c) $(-1, 1)$ (d) $(-\infty, \infty)$
40. If $f(x) = \lim_{t \rightarrow \infty} \frac{(1 + \sin \pi x)^t - 1}{(1 + \sin \pi x)^t + 1}$, then range of $f(x)$ is
 (a) $\{-1, 1\}$ (b) $\{0, 1\}$
 (c) $\{-1, 1\}$ (d) $\{-1, 0, 1\}$
41. Range of the function f defined by $f(x) = \left[\frac{1}{\sin \{x\}} \right]$ (where $[.]$ and $\{.\}$ respectively denote the greatest integer and the fractional part functions) is.
 (a) I , the set of integers
 (b) N , the set of natural numbers.
 (c) W , the set of whole numbers
 (d) $\{2, 3, 4, \dots\}$
42. Let f be a real valued function with domain R satisfying $f(x+k) = 1 + [2 - 5f(x) + 10(f(x))^2 - 10(f(x))^3 + 5(f(x))^4 - (f(x))^5]^{1/5}$ for all real x and some positive constant k , then the period of the function $f(x)$ is :
 (a) k (b) $2k$
 (c) non periodic (d) none of these
43. Let f be a real valued function with domain R satisfying $0 < f(x) \leq \frac{1}{2}$ and for some fixed a ,
 $f(x+a) = \frac{1}{2} - \sqrt{f(x) - (f(x))^2} \quad \forall x \in R$
 then the period of the function $f(x)$ is :
 (a) a (b) $2a$
 (c) non periodic (d) none of these
44. Let $f(x) = \max \{1 + \sin x, 1 - \cos x, 1\} \quad \forall x \in [0, 2\pi]$ and $g(x) = \max \{1, |x-1|\} \quad \forall x \in R$. Then $g \circ f$ is :
 (a) 2 (b) 1
 (c) $\begin{cases} 1 + \sin x & x \leq 0 \\ 1 - \cos x & x \geq 0 \end{cases}$ (d) None of these
45. Find all possible values of x satisfying :
 $\frac{[x]}{[x-2]} - \frac{[x-2]}{[x]} = \frac{8\{x\} + 12}{[x-2][x]}$
 (where $[.]$ denotes the greatest function $\{.\}$ is fractional part).
 (a) $\left\{4, \frac{11}{2}\right\}$ (b) $\{4, 5\}$
 (c) $\left\{5, \frac{11}{2}\right\}$ (d) $\left\{\frac{11}{2}, \frac{15}{4}\right\}$
46. Total number of solutions of the equation $\sin \pi x = |I_n| x|$ are :
 (a) 8 (b) 10
 (c) 9 (d) 6
47. The number of roots of the question $1 + \log_2(1-x) = 2^{-x}$ is :
 (a) 0 (b) 1
 (c) 2 (d) many
48. If $2f(x-1) - f\left(\frac{1-x}{x}\right) = x$, then $f(x)$ is :
 (a) $\frac{1}{3} \left[2(1+x) + \frac{1}{1+x} \right]$ (b) $2(x-1) - \frac{1-x}{x}$
 (c) $x^2 + \frac{1}{x^2} + 3$ (d) None of these



49. Let $f_1(n) = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}$, then

$f_1(1) + f_1(2) + f_1(3) + \dots + f_1(n)$ is equal to :

- (a) $n f_1(n) - 1$ (b) $(n+1) f_1(n) - n$
(c) $(n+1) f_1(n) + n$ (d) $n f_1(n) + n$

50. If $f(x) = \frac{4^x}{4^x + 2}$, then

$f\left(\frac{1}{1997}\right) + f\left(\frac{2}{1997}\right) + \dots + f\left(\frac{1996}{1997}\right)$ is equal to

- (a) 1997 (b) 998
(c) 0 (d) none of these

Objective Questions II [One or more than one correct option]

51. If f is an even function defined on the interval $[-5, 5]$, then the real values of x satisfying the equation

$$f(x) = f\left(\frac{x+1}{x+2}\right), \text{ are}$$

- (a) $\frac{-1 \pm \sqrt{5}}{2}$ (b) $\frac{-3 \pm \sqrt{5}}{2}$
(c) $\frac{-2 \pm \sqrt{5}}{2}$ (d) none of these

52. Let $R = \{(x, y) : x, y \in \mathbb{R}, x^2 + y^2 \leq 25\}$ and

$$R' = \left\{ (x, y) : x, y \in \mathbb{R}, y \geq \frac{4}{9}x^2 \right\} \text{ then is}$$

- (a) $\text{dom } R \cap R' = [-3, 3]$
(b) $\text{Range } R \cap R' \supset [0, 4]$
(c) $\text{Range } R \cap R' = [0, 5]$
(d) $R \cap R'$ defines a function

53. Let $f(x)$ and $g(x)$ be two real valued function given by, $f(x) = -\ln x$ and $g(x) = e^{-x}$. Let $h(x) = f(x) - x$ and $m(x) = g(x) - x$. Further more let the number of solutions of $h(x) = 0$ and $m(x) = 0$ be a and b , then.

- (a) $a \neq b$ (b) $a = b$
(c) $a = 1$ and $b = 1$ (d) None of these

54. Let $f(x)$ be invertible function and let $f^{-1}(x)$ be its inverse. Let equation $f\{f^{-1}(x)\} = f^{-1}(x)$ has two real roots α and β (within domain of $f(x)$), then

- (a) $f(x) = x$, also have same two real roots.
(b) $f^{-1}(x) = x$, also have same two real roots.
(c) $f(x) = f^{-1}(x)$, also have same two real roots.
(d) Area formed by $(0, 0)$, $(\alpha, f(\alpha))$ and $(\beta, f(\beta))$ is 1 unit.

55. 2π is fundamental period of the function

- (a) $\frac{(1 + \sin x)}{\cos x (1 + \operatorname{cosec} x)}$ (b) $|\sin x| + |\cos x|$
(c) $\sin 2x + \cos 3x$ (d) $\cos(\sin x) + \cos(\cos x)$

56. Which of the following functions are periodic ?

- (a) $f(x) = \operatorname{sgn}(e^{-x})$
(b) $f(x) = \begin{cases} 1 & \text{if } x \text{ is a rational number} \\ 0 & \text{if } x \text{ is an irrational number} \end{cases}$

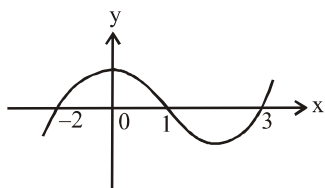
(c) $f(x) = \sqrt{\frac{8}{1 + \cos x} + \frac{8}{1 - \cos x}}$

(d) $f(x) = \left[x + \frac{1}{2} \right] + \left[x - \frac{1}{2} \right] + 2[-x]$

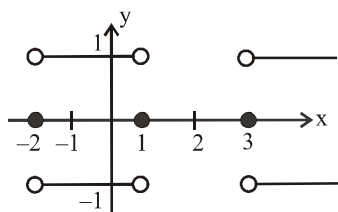
(where $[]$ denotes greatest integer function)



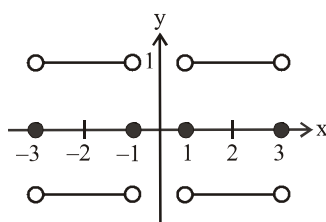
57. The graph of the function $y = f(x)$ is as shown in the figure. Then which one of the following graphs are correct?



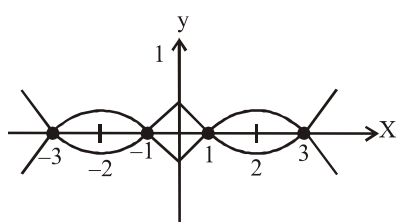
(a) $|y| = \text{sgn}(f(x))$



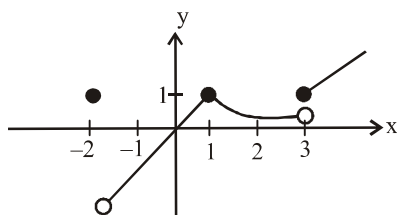
(b) $|y| = \text{sgn}(-f(|x|))$



(c) $|y| = |f(x)|$



(d) $y = x^{\text{sgn}(f(x))}$



58. Let $f(x)$ be defined on $[-\pi, \pi]$ and is given by,

$$f(x) = \begin{cases} \sin x & -\pi \leq x \leq 0 \\ \cos x & 0 < x \leq \pi \end{cases}$$

Let $g(x) = f(|x|) + |f(x)|$, $\forall x \in [-\pi, \pi]$, then

- (a) $g(x) = 0$, has no real roots
 (b) $g(x) = 0$, has infinitely many real roots
 (c) $g(x) = 0$
 (d) limit does not exist at $x = 0$

Numerical Value Type Questions

59. The number of integer values of m for which $f(x) = x^3 - mx^2 + 3x - 11$ invertible is
60. If $f(1) = 2$ and $f(x+y) = f(x)f(y)$ for all natural numbers x, y , the natural number a for which $\sum_{k=1}^n f(a+k) = 16(2^n - 1)$, is

61. The function $f(x) = \frac{x+5}{\sqrt{x^2+1}}$ takes exactly k integer values, then k must be

62. Let S be the set of points (x, y) given by $S = \{(x, y); x^2 + y^2 - 10x + 16 = 0\}$

and $f: S \rightarrow \mathbb{R}$ be given by $f(x, y) = \frac{y}{x}$

If range of f is $\left[-\frac{3}{k}, \frac{3}{k}\right]$ where $k > 0$. then k must be

Assertion & Reason

- (A) If ASSERTION is true, REASON is true, REASON is a correct explanation for ASSERTION.
 (B) If ASSERTION is true, REASON is true, REASON is not a correct explanation for ASSERTION.
 (C) If ASSERTION is true, REASON is false.
 (D) If ASSERTION is false, REASON is true.

63. **Assertion :** A function $y = f(x)$ is defined by $x^2 - \arccos y = \pi$, then domain of $f(x)$ is $\mathbb{R}$.

Reason : $\cos^{-1} y \in [0, \pi]$.

- (a) A (b) B
 (c) C (d) D



64. **Assertion :**

$$\operatorname{cosec}^{-1} \frac{3}{2} + \cos^{-1} \frac{2}{3} - 2 \cot^{-1} \frac{1}{7} - \cot^{-1} 7 = \cot^{-1} 7.$$

Reason : $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$, $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$,

$\operatorname{cosec}^{-1} x = \sin^{-1} \frac{1}{x}$, and for $x > 0$, $\cot^{-1} x = \tan^{-1} \frac{1}{x}$

- (a) A (b) B
(c) C (d) D

65. **Assertion :** If a is twice the tangent of the arithmetic mean of $\sin^{-1} x$ and $\cos^{-1} x$, b is the geometric mean of $\tan x$ and $\cot x$, then $x^2 - ax + b = 0 \Rightarrow x = 1$

Reason : $\tan \left(\frac{\sin^{-1} x + \cos^{-1} x}{2} \right) = 1$

- (a) A (b) B
(c) C (d) D

66. **Assertion :** $\sin^{-1} \left\{ x - \frac{x^2}{2} + \frac{x^3}{4} - \dots \right\}$

$= \frac{\pi}{2} - \cos^{-1} \left\{ x^2 - \frac{x^4}{2} + \frac{x^6}{4} - \dots \right\}$ for $0 < |x| < \sqrt{2}$ has a unique solution.

Reason : $\tan^{-1} \sqrt{x(x+1)} + \sin^{-1} \sqrt{x^2+x+1} = \frac{\pi}{2}$ has no solution for $-\sqrt{2} < x < 0$.

- (a) A (b) B
(c) C (d) D

67. **Assertion :** Let $f(x)$ be a function satisfying $f(x-1) + f(x+1) = \sqrt{2} f(x)$ for all $x \in \mathbb{R}$. Then $f(x)$ is periodic with period 8.

Reason : For every natural number n there exists a periodic function with period n .

- (a) A (b) B
(c) C (d) D

68. **Assertion :** $\sin^{-1} \left[\tan \left\{ \tan^{-1} x + \tan^{-1} (1-x) \right\} \right] = \frac{\pi}{2}$ has no non-zero integral solution.

Reason : The greatest and least values of $(\sin^{-1} x)^3 + (\cos^{-1} x)^3$ are $\frac{7\pi^3}{8}$ and $\frac{\pi^3}{32}$ respectively.

- (a) A (b) B
(c) C (d) D

Match the Following

Each question has two columns. Four options are given representing matching of elements from Column-I and Column-II. Only one of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.

69. Match the column.

Column-I

Column-II

- | | |
|--|-------|
| (A) The number of possible values of k if fundamental period of $\sin^{-1}(\sin kx)$ is $\frac{\pi}{2}$, is | (P) 1 |
| (B) Numbers of elements in the domain of $f(x) = \tan^{-1} x + \sin^{-1} x + \sec^{-1} x$ is | (Q) 2 |
| (C) Period of the function $f(x) = \sin \left(\frac{\pi x}{2} \right) \cdot \cos \left(\frac{\pi x}{2} \right)$ is | (R) 3 |
| (D) If the range of the function $f(x) = \cos^{-1}[5x]$ is $\{a, b, c\}$ & $a + b + c = \frac{\lambda\pi}{2}$, then λ is equal to | (S) 4 |

(where $[.]$ denotes greatest integer)

- (a) $A \rightarrow Q, B \rightarrow Q, C \rightarrow Q, D \rightarrow R$
(b) $A \rightarrow P, B \rightarrow Q, C \rightarrow Q, D \rightarrow R$
(c) $A \rightarrow Q, B \rightarrow P, C \rightarrow R, D \rightarrow R$
(d) $A \rightarrow Q, B \rightarrow Q, C \rightarrow Q, D \rightarrow P$



70. Functions in column I can take values of column II

Column-I	Column-II
(A) $x + 2\sqrt{x}$ can be	(P) 38
(B) $\frac{x-1}{x+1}$ can be	(Q) 0
(C) $2x^3 - 9x^2 + 12x + 6$ can be	(R) $\frac{3}{5}$
(D) $\left[[x] - \frac{x}{2} \right]$ can be	(S) -1

where $[.]$ is G.I.F. Correct matching is

- (a) $A \rightarrow Q, R; B \rightarrow P, Q, R, S; C \rightarrow P, Q, R; D \rightarrow P, Q, S$
 (b) $A \rightarrow P, Q, R; B \rightarrow P, Q, R, S; C \rightarrow P, Q, R, S; D \rightarrow P, Q, S$
 (c) $A \rightarrow P, Q, R; B \rightarrow P, Q, R; C \rightarrow P, Q, R; D \rightarrow P, Q$
 (d) $A \rightarrow P, Q; B \rightarrow P, Q; C \rightarrow P, Q, R, S; D \rightarrow P, Q$

71. Column-I Column-II

(A) Let $X = \{a_1, a_2, \dots, a_6\}$ and $Y = \{b_1, b_2, b_3\}$. The number of functions f from X to Y such that it is onto and there are exactly three elements x in X such that $f(x) = b_1$, is greater than	(P) 2
(B) The number of real solutions for x, y if $y = \sin x $ and $y = \sin^{-1}(\sin x)$ where $x \in [-2\pi, 2\pi]$, is	(Q) 5
(C) If a, b and c are distinct positive real numbers such that $a + b + c = 1$, then $\frac{(1+a)(1+b)(1+c)}{(1-a)(1-b)(1-c)}$ can be	(R) 120
(D) The period of the function $[6x + 7] + \cos \pi x - 6x$, where $[.]$ denotes the greatest integer function, is	(S) 80 (T) 10

The correct matching is

- (a) $A \rightarrow P, Q, S, T; B \rightarrow Q, C \rightarrow R, S, T; D \rightarrow P$
 (b) $A \rightarrow P, Q; B \rightarrow Q; C \rightarrow R, S; D \rightarrow P$
 (c) $A \rightarrow P; B \rightarrow Q; C \rightarrow R, S; D \rightarrow R, S$
 (d) $A \rightarrow P; B \rightarrow T; C \rightarrow R, S; D \rightarrow Q$

72. Match of the column.

Column-I	Column-II
(A) $\log_{10} 5 \cdot \log_{10} 20 + (\log_{10} 2)^2 =$	(P) 4
(B) Least value of x satisfying $ 2x - x - 4 = x + 4$ is	(Q) -4
(C) If $\cos^{-1} x = \frac{k}{2} \sin^{-1} \sqrt{\frac{1-x}{2}}$, for all $x \in (-1, 1)$, then k is equal to	(R) 1
(D) If $f: [0, 2] \rightarrow [2, 0]$ is bijective function defined by $f(x) = ax^2 + bx + c$, where a, b, c are non-zero real numbers then $f(2)$ is equal to	(S) 0

The correct matching is

- (a) $A \rightarrow R, B \rightarrow Q, C \rightarrow P, D \rightarrow S$
 (b) $A \rightarrow P, B \rightarrow Q, C \rightarrow R, D \rightarrow S$
 (c) $A \rightarrow Q, B \rightarrow R, C \rightarrow P, D \rightarrow S$
 (d) $A \rightarrow R, B \rightarrow Q, C \rightarrow S, D \rightarrow P$

73. Column II contain the ranges of the functions given in Column I.

Column I	Column II
(A) $y = \frac{e^x}{1 + [x]}; x \geq 0$ where $[.]$ denotes greatest integer function	(P) $\left[\frac{3}{4}, \infty \right)$
(B) $\cot^{-1}(2x - x^2)$	(Q) $[1, \infty)$
(C) $4^x - 2^x + 1$	(R) $[0, \infty)$
(D) $\ln(1 + x^2)$	(S) $\left[\frac{\pi}{4}, \pi \right)$

The correct matching is

- (a) $A \rightarrow (Q); B \rightarrow (S); C \rightarrow (P); D \rightarrow (R)$
 (b) $A \rightarrow (S); B \rightarrow (Q); C \rightarrow (P); D \rightarrow (R)$
 (c) $A \rightarrow (Q); B \rightarrow (S); C \rightarrow (R); D \rightarrow (P)$
 (d) $A \rightarrow (R); B \rightarrow (S); C \rightarrow (P); D \rightarrow (Q)$



Using the following passage, solve Q.74 to Q.76

Passage –1

A function f from a set X to Y is called onto, if for every $y \in Y$ there exist $x \in X$ such that $f(x) = y$. Unless the contrary is specified, a real function is onto if it takes all real values, otherwise it is called into function. Thus, if X and Y are finite sets, then f cannot be onto If Y contains more element than X .

74. The polynomial function $P(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_n$, where $a_0 \neq 0$, is onto, for
- (a) all positive integers n
 - (b) all even positive integers n
 - (c) all odd positive integers n
 - (d) no positive integer
75. Which of the following is not true ?
- (a) A one-one function from the set $\{a, b, c\}$ to $\{\alpha, \beta, \gamma\}$ is onto also.
 - (b) An onto function from an infinite set to a finite set cannot be one-one.
 - (c) An onto function is always invertible.
 - (d) The function $\tan x$ and $\cot x$ are onto
76. The function $f(x) = \frac{x^2 + 2x + c}{x^2 + 4x + 3c}$ is onto, if
- (a) $0 < c < 2$
 - (b) $0 < c < 4$
 - (c) $-\frac{1}{2} < c < \frac{1}{2}$
 - (d) $0 < c < 1$

Using the following passage, solve Q.77 to Q.79

Passage –2

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ is a function satisfying $f(2-x) = f(2+x)$ and $f(20-x) = f(x)$, $\forall x \in \mathbb{R}$. For this function f answer the following.

77. If $f(0) = 5$, then minimum possible number of values of x satisfying $f(x) = 5$, for $x \in [0, 170]$ is.
- (a) 21
 - (b) 12
 - (c) 11
 - (d) 22
78. Graph of $y = f(x)$ is
- (a) symmetrical about $x = 18$
 - (b) symmetrical about $x = 5$
 - (c) symmetrical about $x = 8$
 - (d) symmetrical about $x = 20$
79. If $f(2) \neq f(6)$, then
- (a) fundamental period of $f(x)$ is 1
 - (b) fundamental period of $f(x)$ may be 1
 - (c) period of $f(x)$ can't be 1
 - (d) fundamental period of $f(x)$ is 8

Text

80. Find the range of values of t for which

$$2 \sin t = \frac{1-2x+5x^2}{3x^2-2x-1}, \quad t \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$



EXERCISE - 4 : PREVIOUS YEAR JEE ADVANCED QUESTIONS

Objective Questions I [Only one correct option]

1. The domain of definition of the function $y(x)$ is given by the equation $2^x + 2^y = 2$, is (2000)
 (a) $0 < x \leq 1$ (b) $0 \leq x \leq 1$
 (c) $-\infty < x \leq 0$ (d) $-\infty < x < 1$
2. Let $g(x) = 1 + x - [x]$ and $f(x) = \begin{cases} -1, & x < 0 \\ 0, & x = 0 \\ 1, & x > 0 \end{cases}$, then for all x , $f[g(x)]$ is equal to (2001)
 (a) x (b) 1
 (c) $f(x)$ (d) $g(x)$
3. If $f: [1, \infty) \rightarrow [2, \infty)$ is given by $f(x) = x + \frac{1}{x}$ then $f^{-1}(x)$ equals. (2001)
 (a) $\frac{x + \sqrt{x^2 - 4}}{2}$ (b) $\frac{x}{1 + x^2}$
 (c) $\frac{x - \sqrt{x^2 - 4}}{4}$ (d) $1 + \sqrt{x^2 - 4}$
4. The domain of definition of $f(x) = \frac{\log_2(x+3)}{x^2 + 3x + 2}$ is (2001)
 (a) $\mathbb{R} \setminus \{-1, -2\}$ (b) $(-2, \infty)$
 (c) $\mathbb{R} \setminus \{-1, -2, -3\}$ (d) $(-3, \infty) \setminus \{-1, -2\}$
5. Let $f(x) = (1 + b^2)x^2 + 2bx + 1$ and let $m(b)$ be the minimum value of $f(x)$. As b varies, the range of $m(b)$ is (2001)
 (a) $[0, 1]$ (b) $\left[0, \frac{1}{2}\right]$
 (c) $\left[\frac{1}{2}, 1\right]$ (d) $(0, 1]$
6. Let $E = \{1, 2, 3, 4\}$ and $F = \{1, 2\}$, Then, the number of onto functions from E to F is (2001)
 (a) 14 (b) 16
 (c) 12 (d) 8
7. Let $f(x) = \frac{\alpha x}{x+1}$, $x \neq -1$. Then, for what value of α is $f[f(x)] = x$? (2001)
 (a) $\sqrt{2}$ (b) $-\sqrt{2}$
 (c) 1 (d) -1
8. Suppose $f(x) = (x+1)^2$ for $x \geq -1$. If $g(x)$ is the function whose graph is reflection of the graph of $f(x)$ with respect to the line $y = x$, then $g(x)$ equals (2002)
 (a) $-\sqrt{x} - 1, x \geq 0$ (b) $\frac{1}{(x+1)^2}, x > -1$
 (c) $\sqrt{x+1}, x \geq -1$ (d) $\sqrt{x} - 1, x \geq 0$
9. Let function $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = 2x + \sin x$ for $x \in \mathbb{R}$. Then f is (2002)
 (a) one-to-one and onto
 (b) one-to-one but NOT onto
 (c) onto but NOT one-to-one
 (d) neither one-to-one nor onto
10. If $f: [0, \infty) \rightarrow [0, \infty)$ and $f(x) = \frac{x}{1+x}$, then f is (2003)
 (a) one-one and onto
 (b) one-one but not onto
 (c) onto but not one-one
 (d) neither one-one nor onto
11. Range of the function $f(x) = \frac{x^2 + x + 2}{x^2 + x + 1}$; $x \in \mathbb{R}$ is (2003)
 (a) $(1, \infty)$ (b) $\left(1, \frac{11}{7}\right)$
 (c) $\left(1, \frac{7}{3}\right)$ (d) $\left(1, \frac{7}{5}\right)$



12. Domain of definition of the function

$$f(x) = \sqrt{\sin^{-1}(2x) + \frac{\pi}{6}} \text{ for real valued } x, \text{ is } \quad (2003)$$

(a) $\left[-\frac{1}{4}, \frac{1}{2}\right]$ (b) $\left[-\frac{1}{2}, \frac{1}{2}\right]$

(c) $\left(-\frac{1}{2}, \frac{1}{9}\right)$ (d) $\left[-\frac{1}{4}, \frac{1}{4}\right]$

13. If $f(x) = \sin x + \cos x$, $g(x) = x^2 - 1$, then $g(f(x))$ is invertible in the domain. (2004)

(a) $\left[0, \frac{\pi}{2}\right]$ (b) $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$

(c) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ (d) $[0, \pi]$

14.
$$f(x) = \begin{cases} x, & \text{if } x \text{ is rational} \\ 0, & \text{if } x \text{ is irrational} \end{cases}$$

$$g(x) = \begin{cases} 0, & \text{if } x \text{ is rational} \\ x, & \text{if } x \text{ is irrational} \end{cases}$$

Then $f-g$ is. (2005)

- (a) neither one-one nor onto
(b) one-one and onto
(c) one-one and into
(d) many one and onto

15. Suppose X and Y are two sets and $f: X \rightarrow Y$ is a function. For a subset A of X , define $f(A)$ to be the subset $\{f(a) : a \in A\}$ of Y . For a subset B of Y , define $f^{-1}(B)$ to be the subset $\{x \in X : f(x) \in B\}$ of X . Then which of the following statements is true? (2005)

- (a) $f^{-1}(f(A)) = A$ for every $A \subset X$
(b) $f^{-1}(f(A)) = A$ for every $A \subset X$ if and only if $f(X) = Y$
(c) $f(f^{-1}(B)) = B$ for every $B \subset Y$
(d) $f(f^{-1}(B)) = B$ for every $B \subset Y$ if and only if $f(X) = Y$

16. Let $S = \{1, 2, 3, 4\}$. The total number of unordered pairs of disjoint subsets of S is equal to (2010)

- (a) 25 (b) 34
(c) 42 (d) 41

17. Let $f(x) = x^2$ and $g(x) = \sin x$ for all $x \in \mathbb{R}$. Then, the set of all x satisfying $(f \circ g \circ g \circ f)(x) = (g \circ g \circ f)(x)$, where $(f \circ g)(x) = f(g(x))$, is (2011)

- (a) $\pm\sqrt{n\pi}, n \in \{0, 1, 2, \dots\}$
(b) $\pm\sqrt{n\pi}, n \in \{1, 2, \dots\}$
(c) $\frac{\pi}{2} + 2n\pi, n \in \{\dots, -2, -1, 0, 1, 2, \dots\}$
(d) $2n\pi, n \in \{\dots, -2, -1, 0, 1, 2, \dots\}$

18. The function $f: [0, 3] \rightarrow [1, 29]$, defined by $f(x) = 2x^3 - 15x^2 + 36x + 1$, is (2012)

- (a) one-one and onto
(b) onto but not one-one
(c) one-one but not onto
(d) neither one-one nor onto

19. For any positive integer n , define $f_n: (0, \infty) \rightarrow \mathbb{R}$ as

$$f_n(x) = \sum_{j=1}^n \tan^{-1} \left(\frac{1}{1 + (x+j)(x+j-1)} \right)$$

for all $x \in (0, \infty)$ (Here, the inverse trigonometric function

$\tan^{-1}x$ assumes value in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.) Then, which of the

following statement(s) is (are) TRUE? (2018)

- (a) $\sum_{j=1}^5 \tan^2(f_j(0)) = 55$
(b) $\sum_{j=1}^{10} (1 + f'_j(0)) \sec^2(f_j(0)) = 10$
(c) For any fixed positive integer n , $\lim_{x \rightarrow \infty} \tan(f_n(x)) = \frac{1}{n}$
(d) For any fixed positive integer n , $\lim_{x \rightarrow \infty} \sec^2(f_n(x)) = 1$

20. If the function $f: \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f(x) = |x|(x - \sin x)$, then which of the following statements is TRUE? (2020)

- (a) f is one-one, but **NOT** onto
(b) f is onto, but **NOT** one-one
(c) f is **BOTH** one-one and onto
(d) f is **NEITHER** one-one **NOR** onto



Objective Questions II [One or more than one correct option]

Numerical Value Type Questions

21. Let $f: (-1, 1) \rightarrow \mathbb{R}$ be such that $f(\cos 4\theta) = \frac{2}{2 - \sec^2 \theta}$ for $\theta \in \left(0, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$. Then, the value(s) of $f\left(\frac{1}{3}\right)$ is/are (2012)

- (a) $1 - \sqrt{\frac{3}{2}}$ (b) $1 + \sqrt{\frac{3}{2}}$
(c) $1 - \sqrt{\frac{2}{3}}$ (d) $1 + \sqrt{\frac{2}{3}}$

22. Let $f: \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \rightarrow \mathbb{R}$ be given by

$$f(x) = (\log(\sec x + \tan x))^3.$$

Then

- (a) $f(x)$ is an odd function
(b) $f(x)$ is a one-one function
(c) $f(x)$ is an onto function
(d) $f(x)$ is an even function
23. If $\alpha = 3 \sin^{-1}\left(\frac{6}{11}\right)$ and $\beta = 3 \cos^{-1}\left(\frac{4}{9}\right)$, where the inverse trigonometric functions take only the principal values, then the correct option(s) is (are) (2015)
- (a) $\cos \beta > 0$ (b) $\sin \beta < 0$
(c) $\cos(\alpha + \beta) > 0$ (d) $\cos \alpha < 0$
24. For any positive integer n , let $S_n: (0, \infty) \rightarrow \mathbb{R}$ be defined

$$\text{by } S_n(x) = \sum_{k=1}^n \cot^{-1}\left(\frac{1+k(k+1)x^2}{x}\right),$$

Where for any $x \in \mathbb{R}$, $\cot^{-1} x \in (0, \pi)$ and

$\tan^{-1}(x) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Then which of the following statements is (are) TRUE? (2021)

- (a) $S_{10}(x) = \frac{\pi}{2} - \tan^{-1}\left(\frac{1+11x^2}{10x}\right)$, for all $x > 0$
(b) $\lim_{n \rightarrow \infty} \cot(S_n(x)) = x$, for all $x > 0$
(c) The equation $S_3(x) = \frac{\pi}{4}$ has a root in $(0, \infty)$
(d) $\tan(S_n(x)) \leq \frac{1}{2}$, for all $n \geq 1$ and $x > 0$

25. The number of real solutions of the equation $\sin^{-1}\left(\sum_{i=1}^{\infty} x^{i+1} - x \sum_{i=1}^{\infty} \left(\frac{x}{2}\right)^i\right) = \frac{\pi}{2} - \cos^{-1}\left(\sum_{i=1}^{\infty} \left(-\frac{x}{2}\right)^i - \sum_{i=1}^{\infty} (-x)^i\right)$ lying in the interval $\left(-\frac{1}{2}, \frac{1}{2}\right)$ is _____. (2018)
- (Here, the inverse trigonometric functions $\sin^{-1} x$ and $\cos^{-1} x$ assume values in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ and $[0, \pi]$, respectively)
26. Let $f: [0, 4\pi] \rightarrow [0, \pi]$ be defined by $f(x) = \cos^{-1}(\cos x)$. The number of points $x \in [0, 4\pi]$ satisfying the equation $f(x) = \frac{10-x}{10}$ is (2014)
27. The value of $\sec^{-1}\left(\frac{1}{4} \sum_{k=0}^{10} \sec\left(\frac{7\pi}{12} + \frac{k\pi}{2}\right) \sec\left(\frac{7\pi}{12} + \frac{(k+1)\pi}{2}\right)\right)$ in the interval $\left[-\frac{\pi}{4}, \frac{3\pi}{4}\right]$ equals (2019)
28. Let the function $f: [0, 1] \rightarrow \mathbb{R}$ be defined by $f(x) = \frac{4^x}{4^x + 2}$. Then the value of $f\left(\frac{1}{40}\right) + f\left(\frac{2}{40}\right) + f\left(\frac{3}{40}\right) + \dots + f\left(\frac{39}{40}\right) - f\left(\frac{1}{2}\right)$ is..... (2020)

Assertion & Reason

For the following questions choose the correct answer from the codes (A), (B), (C) and (D) defined as follows.

(A) Statement I is true, Statement II is also true; Statement II is the correct explanation of Statement I.

(B) Statement I is true, Statement II is also true; Statement II is not the correct explanation of Statement I.

(C) Statement I is true, Statement II is false.

(D) Statement I is false, Statement II is true.

29. Let $f(x) = 2 + \cos x$ for all real x .
Statement I : For each real t , there exists a point c in $[t, t + \pi]$ such that $f'(c) = 0$.
Because
Statement II : $f(t) = f(t + 2\pi)$ for each real t . (2007)
- (a) A (b) B
(c) C (d) D



30. **Statement I :** The curve $y = -\frac{x^2}{2} + x + 1$ is symmetric with respect to the line $x = 1$.

Because

Statement II : A parabola is symmetric about its axis.

(2007)

- (a) A (b) B
(c) C (d) D

Match the Columns

Each question has two columns. Four options are given representing matching of elements from Column-I and Column-II. Only one of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.

31. Let $f(x) = \frac{x^2 - 6x + 5}{x^2 - 5x + 6}$. (2007)

Column I

(A) If $-1 < x < 1$, then $f(x)$ satisfies

(B) If $1 < x < 2$, then $f(x)$ satisfies

(C) If $3 < x < 5$, then $f(x)$ satisfies

(D) If $x > 5$, then $f(x)$ satisfies

Column II

(P) $0 < f(x) < 1$

(Q) $f(x) < 0$

(R) $f(x) > 0$

(S) $f(x) < 1$

The correct matching is

- (a) A-P, R, S; B-Q; C-Q, S; D-P, R
(b) A-P, R, S; B-Q, S; C-Q, S; D-P, R, S
(c) A-P; B-Q, S; C-Q; D-P, R, S
(d) A-R, S; B-S; C-Q, S; D-P, R

32. Let $E_1 = \left\{ x \in \mathbb{R} : x \neq 1 \text{ and } \frac{x}{x-1} > 0 \right\}$ and

$$E_2 = \left\{ x \in E_1 : \sin^{-1} \left(\log_e \left(\frac{x}{x-1} \right) \right) \text{ is a real number} \right\}.$$

(Here, the inverse trigonometric function $\sin^{-1}x$ assumes

values in $\left[-\frac{\pi}{2}, \frac{\pi}{2} \right]$)

Let $f : E_1 \rightarrow \mathbb{R}$ be the function defined by

$$f(x) = \log_e \left(\frac{x}{x-1} \right)$$

and $g : E_2 \rightarrow \mathbb{R}$ be the function defined by

$$g(x) = \sin^{-1} \left(\log_e \left(\frac{x}{x-1} \right) \right) \quad (2018)$$

Columns A

- (A) The range of f is
(B) The range of g contains
(C) The domain of f contains
(D) The domain of g is

Column B

- (P) $\left(-\infty, \frac{1}{1-e} \right] \cup \left[\frac{e}{e-1}, \infty \right)$
(Q) $(0, 1)$
(R) $\left[-\frac{1}{2}, \frac{1}{2} \right]$
(S) $(-\infty, 0) \cup (0, \infty)$
(T) $\left(-\infty, \frac{e}{e-1} \right]$

The correct matching is

- (a) A-S; B-Q; C-P, D-P
(b) A-S; B-Q; C-P, D-S
(c) A-Q; B-Q; C-P, D-P
(d) A-S; B-P; C-S, D-P

Using the following passage, solve Q.33 to Q.35

Passage

If a continuous function ' f ' defined on the real line $\mathbb{R}$, assumes positive and negative values in $\mathbb{R}$, then the equation $f(x) = 0$ has a root in $\mathbb{R}$. For example, if it is known that a continuous function f on $\mathbb{R}$ is positive at some point and its minimum values is negative, then the equation $f(x) = 0$ has a root in $\mathbb{R}$. Consider $f(x) = ke^x - x$ for all real x where k is real constant. (2007)

33. The line $y = x$ meets $y = ke^x$ for $k \leq 0$ at
(a) no point (b) one point
(c) two points (d) more than two points
34. The positive value of k for which $ke^x - x = 0$ has only one root is
(a) $\frac{1}{e}$ (b) 1
(c) e (d) $\log_e 2$
35. For $k > 0$, the set of all values of k for which $ke^x - x = 0$ has two distinct roots, is
(a) $\left(0, \frac{1}{e} \right)$ (b) $\left(\frac{1}{e}, 1 \right)$
(c) $\left(\frac{1}{e}, \infty \right)$ (d) $(0, 1)$



Note



Find Answer Key and Detailed Solutions at the end of this book



RELATIONS , FUNCTIONS & INVERSE TRIGONOMETRIC FUNCTIONS



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03

CONTINUITY, DIFFERENTIABILITY & DIFFERENTIATION

CONTINUITY, DIFFERENTIABILITY & DIFFERENTIATION

CONTINUITY

1. DEFINITION

A function $f(x)$ is said to be continuous at $x = a$; where $a \in \text{domain of } f(x)$, if

$$\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = f(a)$$

i.e., LHL = RHL = value of a function at $x = a$

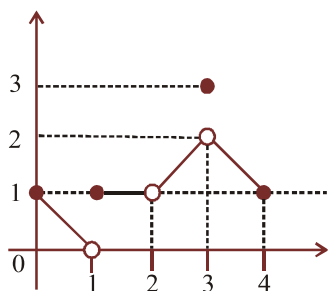
or $\lim_{x \rightarrow a} f(x) = f(a)$

1.1 Reasons of discontinuity

If $f(x)$ is not continuous at $x = a$, we say that $f(x)$ is discontinuous at $x = a$.

There are following possibilities of discontinuity :

1. $\lim_{x \rightarrow a^-} f(x)$ and $\lim_{x \rightarrow a^+} f(x)$ exist but they are not equal.
2. $\lim_{x \rightarrow a^-} f(x)$ and $\lim_{x \rightarrow a^+} f(x)$ exists and are equal but not equal to $f(a)$.
3. $f(a)$ is not defined.
4. At least one of the limits does not exist. Geometrically, the graph of the function will exhibit a break at the point of discontinuity.



The graph as shown is discontinuous at $x = 1, 2$ and 3 .

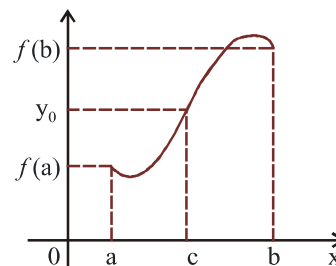
2. PROPERTIES OF CONTINUOUS FUNCTIONS

Let $f(x)$ and $g(x)$ be continuous functions at $x = a$. Then,

1. $cf(x)$ is continuous at $x = a$, where c is any constant.
2. $f(x) \pm g(x)$ is continuous at $x = a$.
3. $f(x) \cdot g(x)$ is continuous at $x = a$.
4. $f(x)/g(x)$ is continuous at $x = a$, provided $g(a) \neq 0$.
5. If $f(x)$ is continuous on $[a, b]$ such that $f(a)$ and $f(b)$ are of opposite signs, then there exists at least one solution of equation $f(x) = 0$ in the open interval (a, b) .

3. THE INTERMEDIATE VALUE THEOREM

Suppose $f(x)$ is continuous on an interval I , and a and b are any two points of I . Then if y_0 is a number between $f(a)$ and $f(b)$, there exists a number c between a and b such that $f(c) = y_0$.



The Function f , being continuous on (a, b) takes on every value between $f(a)$ and $f(b)$

NOTES :

That a function f which is continuous in $[a, b]$ possesses the following properties :

- (i) If $f(a)$ and $f(b)$ possess opposite signs, then there exists at least one solution of the equation $f(x) = 0$ in the open interval (a, b) .
- (ii) If K is any real number between $f(a)$ and $f(b)$, then there exists at least one solution of the equation $f(x) = K$ in the open interval (a, b) .



4. CONTINUITY IN AN INTERVAL

- (a) A function f is said to be continuous in (a, b) if f is continuous at each and every point $\in (a, b)$.
- (b) A function f is said to be continuous in a closed interval $[a, b]$ if:
- f is continuous in the open interval (a, b) and
 - f is right continuous at 'a' i.e. $\lim_{x \rightarrow a^+} f(x) = f(a) = \text{a finite quantity}$.
 - f is left continuous at 'b'; i.e. $\lim_{x \rightarrow b^-} f(x) = f(b) = \text{a finite quantity}$.

5. CONTINUOUS OF COMPOSITE FUNCTIONS

Theorem 5.1

Suppose $a < c < b, f: [a, b] \rightarrow R$ is a function and $\lim_{x \rightarrow c} f(x) = l$. Further suppose that $g: R \rightarrow R$ is continuous at l . Then $\lim_{x \rightarrow c} (g \circ f)(x)$ exists and is equal to $g(l)$.

Proof : Write $p = g(l)$ and $\varepsilon > 0$. Then there exists $\delta > 0$ such that $y \in (l - \delta, l + \delta) \Rightarrow |g(y) - p| < \varepsilon$ (since g is continuous at l). Since $f(x) \rightarrow l$ as $x \rightarrow c$, corresponding to this $\delta > 0$ there exists $\eta > 0$ such that

$$\begin{aligned} x \in (c - \eta, c + \eta) &\Rightarrow |f(x) - l| < \delta \\ &\Rightarrow 1 - \delta < f(x) < 1 + \delta \\ &\Rightarrow |g(f(x)) - p| < \varepsilon \\ &\Rightarrow |(g \circ f)(x) - p| < \varepsilon \end{aligned}$$

Hence

$$\lim_{x \rightarrow c} (g \circ f)(x) = p = g(l)$$

Theorem 5.2

Suppose $a < c < b$ and $f: [a, b] \rightarrow R$ and $g: R \rightarrow R$ are, respectively, continuous at c and $f(c)$. Then $g \circ f: [a, b] \rightarrow R$ is continuous at c .

Proof : By Theorem 5.1

$$\lim_{x \rightarrow c} (g \circ f)(x) = g(f(c)) = (g \circ f)(c)$$

Hence $g \circ f$ is continuous at c .

Note : If $c = a$ or b , in a similar way as Theorem we can show that $g \circ f$ is continuous at a or b .

Theorem 5.3

Suppose $f: [a, b] \rightarrow R$ is continuous, $f([a, b]) \subset [p, q]$ and $g: [p, q] \rightarrow R$ is continuous. Then $g \circ f: [a, b] \rightarrow R$ is continuous.

Proof : Suppose $c \in [a, b]$. Since f is continuous on $[a, b]$ and g is continuous on $[p, q]$ it follows that $g \circ f$ is continuous at c (by Theorem and the note). Hence $g \circ f$ is continuous on $[a, b]$.

Theorem 5.4

Suppose $\alpha \in R, a < c < b, f: [a, b] \rightarrow [0, \infty)$ and

$$\lim_{x \rightarrow c} f(x) = l, \text{ then } \lim_{x \rightarrow c} f^\alpha(x) = l^\alpha.$$

Proof : Since $f(x) \geq 0$ on $[a, b]$, by Theorem it follows that

$$\lim_{x \rightarrow c} f(x) = l \geq 0. \text{ Write } g(x) = x^\alpha \forall x \geq 0. \text{ By Example we}$$

get g is continuous at l . Therefore by Theorem it follows that

$$\lim_{x \rightarrow c} f^\alpha(x) = \lim_{x \rightarrow c} (f(x))^\alpha = \lim_{x \rightarrow c} g(f(x)) = g(l) = l^\alpha$$

6. A LIST OF CONTINUOUS FUNCTIONS

Function $f(x)$	Interval in which $f(x)$ is continuous
1. constant c	$(-\infty, \infty)$
2. x^n, n is an integer ≥ 0	$(-\infty, \infty)$
3. x^{-n}, n is a positive integer	$(-\infty, \infty) - \{0\}$
4. $ x - a $	$(-\infty, \infty)$
5. $P(x) = a_0x^n + a_1x^{n-1} + \dots + a_n$	$(-\infty, \infty)$
6. $\frac{p(x)}{q(x)}$, where $p(x)$ and $q(x)$ are polynomial in x	$(-\infty, \infty) - \{x; q(x) = 0\}$
7. $\sin x$	$(-\infty, \infty)$
8. $\cos x$	$(-\infty, \infty)$
9. $\tan x$	$(-\infty, \infty) - \left\{ (2n+1)\frac{\pi}{2} : n \in I \right\}$
10. $\cot x$	$(-\infty, \infty) - \{n\pi : n \in I\}$
11. $\sec x$	$(-\infty, \infty) - \{(2n+1)\frac{\pi}{2} : n \in I\}$





12. $\operatorname{cosec} x$ $(-\infty, \infty) - \{n\pi : n \in \mathbb{I}\}$
13. e^x $(-\infty, \infty)$
14. $\log_e x$ $(0, \infty)$

7. TYPES OF DISCONTINUITIES

Type-1 : (Removable type of discontinuities)

In case, $\lim_{x \rightarrow c} f(x)$ exists but is not equal to $f(c)$ then the function is said to have a **removable discontinuity or discontinuity of the first kind**. In this case, we can redefine the function such that $\lim_{x \rightarrow c} f(x) = f(c)$ and make it continuous at $x = c$. Removable type of discontinuity can be further classified as :

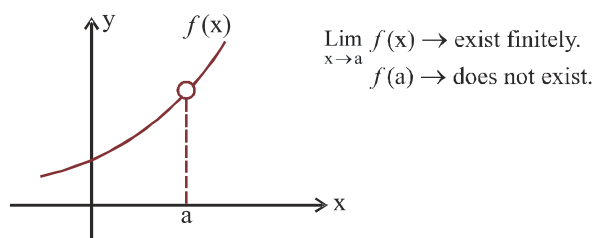
(a) Missing Point Discontinuity :

Where $\lim_{x \rightarrow a} f(x)$ exists finitely but $f(a)$ is not defined.

E.g. $f(x) = \frac{(1-x)(9-x^2)}{(1-x)}$ has a missing point discontinuity

at $x = 1$, and

$f(x) = \frac{\sin x}{x}$ has a missing point discontinuity at $x = 0$.



missing point discontinuity at $x = a$

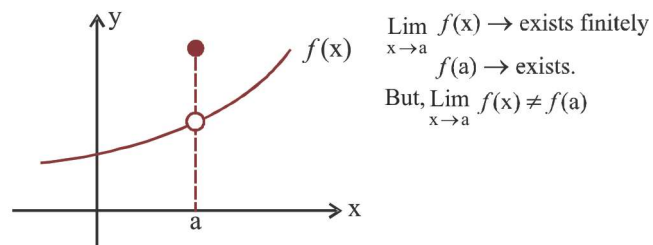
(b) Isolated Point Discontinuity :

Where $\lim_{x \rightarrow a} f(x)$ exists & $f(a)$ also exists but;

$\lim_{x \rightarrow a} f(x) \neq f(a)$.

e.g. $f(x) = \frac{x^2 - 16}{x - 4}$, $x \neq 4$ and $f(4) = 9$ has an isolated point discontinuity at $x = 4$.

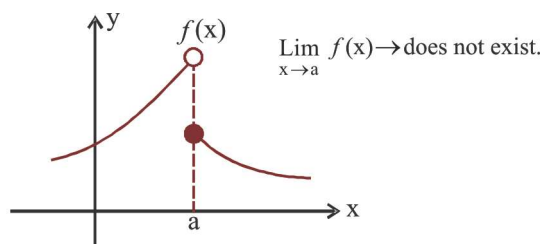
Similarly $f(x) = [x] + [-x] = \begin{cases} 0 & \text{if } x \in \mathbb{I} \\ -1 & \text{if } x \notin \mathbb{I} \end{cases}$ has an isolated point discontinuity at all $x \in \mathbb{I}$.



Isolated point discontinuity at $x = a$

Type-2 : (Non-Removable type of discontinuities)

In case, $\lim_{x \rightarrow a} f(x)$ does not exist, then it is not possible to make the function continuous by redefining it. Such discontinuities are known as **non-removable discontinuity or discontinuity of the 2nd kind**. Non-removable type of discontinuity can be further classified as :



non-removable discontinuity at $x = a$

(a) Finite Discontinuity :

e.g., $f(x) = x - [x]$ at all integral x ; $f(x) = \tan^{-1} \frac{1}{x}$ at $x = 0$ and

$f(x) = \frac{1}{1 + 2^x}$ at $x = 0$ (note that $f(0^+) = 0$; $f(0^-) = 1$)

(b) Infinite Discontinuity :

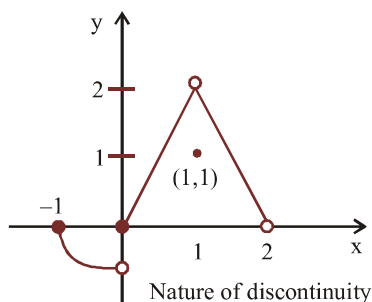
e.g., $f(x) = \frac{1}{x-4}$ or $g(x) = \frac{1}{(x-4)^2}$ at $x = 4$; $f(x) = 2^{\tan x}$

at $x = \frac{\pi}{2}$ and $f(x) = \frac{\cos x}{x}$ at $x = 0$.

(c) Oscillatory Discontinuity :

e.g., $f(x) = \sin \frac{1}{x}$ at $x = 0$.

In all these cases the value of $f(a)$ of the function at $x = a$ (point of discontinuity) may or may not exist but $\lim_{x \rightarrow a}$ does not exist.



From the adjacent graph note that

- f is continuous at $x = -1$
- f has isolated discontinuity at $x = 1$
- f has missing point discontinuity at $x = 2$
- f has non-removable (finite type) discontinuity at the origin.

NOTES :

- (a) In case of dis-continuity of the second kind the non-negative difference between the value of the RHL at $x = a$ and LHL at $x = a$ is called the **jump of discontinuity**. A function having a finite number of jumps in a given interval I is called a piece wise continuous or sectionally continuous function in this interval.
- (b) All Polynomials, Trigonometrical functions, exponential and Logarithmic functions are continuous in their domains.
- (c) If $f(x)$ is continuous and $g(x)$ is discontinuous at $x = a$ then the product function $\phi(x) = f(x) \cdot g(x)$ is not necessarily be discontinuous at $x = a$. e.g.

$$f(x) = x \text{ and } g(x) = \begin{cases} \sin \frac{\pi}{x} & x \neq 0 \\ 0 & x = 0 \end{cases}$$

- (d) If $f(x)$ and $g(x)$ both are discontinuous at $x = a$ then the product function $\phi(x) = f(x) \cdot g(x)$ is not necessarily be discontinuous at $x = a$. e.g.

$$f(x) = -g(x) = \begin{cases} 1 & x \geq 0 \\ -1 & x < 0 \end{cases}$$

- (e) Point functions are to be treated as discontinuous eg.

$$f(x) = \sqrt{1-x} + \sqrt{x-1} \text{ is not continuous at } x = 1.$$

- (f) A continuous function whose domain is closed must have a range also in closed interval.

- (g) If f is continuous at $x = a$ and g is continuous at $x = f(a)$ then the composite $g[f(x)]$ is continuous at

$$x = a \text{ E.g. } f(x) = \frac{x \sin x}{x^2 + 2} \text{ and } g(x) = |x| \text{ are continuous at } x$$

$$= 0, \text{ hence the composite } (g \circ f)(x) = \left| \frac{x \sin x}{x^2 + 2} \right| \text{ will also be continuous at } x = 0.$$

DIFFERENTIABILITY

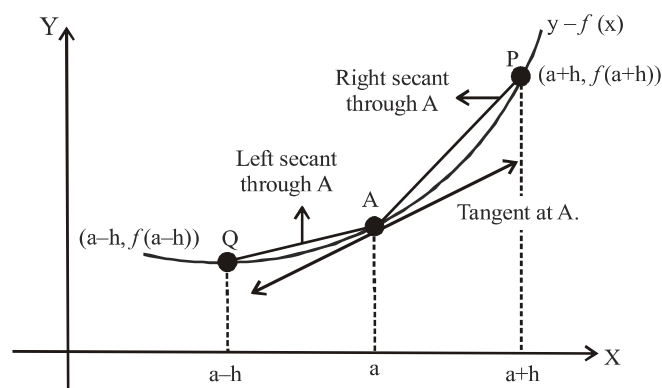
8. DEFINITION

Let $f(x)$ be a real valued function defined on an open interval (a, b) where $c \in (a, b)$. Then $f(x)$ is said to be differentiable or derivable at $x = c$,

$$\text{iff, } \lim_{x \rightarrow c} \frac{f(x) - f(c)}{(x - c)} \text{ exists finitely.}$$

This limit is called the derivative or differentiable coefficient of the function $f(x)$ at $x = c$, and is denoted by

$$f'(c) \text{ or } \frac{d}{dx} (f(x))_{x=c}.$$



- Slope of Right hand secant = $\frac{f(a+h) - f(a)}{h}$ as

$h \rightarrow 0, P \rightarrow A$ and secant (AP) $\rightarrow$ tangent at A

$$\Rightarrow \text{Right hand derivative} = \lim_{h \rightarrow 0} \left(\frac{f(a+h) - f(a)}{h} \right)$$





= Slope of tangent at A (when approached from right) $f'(a^+)$.

• Slope of Left hand secant = $\frac{f(a-h) - f(a)}{-h}$ as $h \rightarrow 0$, $Q \rightarrow A$ and secant AQ $\rightarrow$ tangent at A

$\Rightarrow$ Left hand derivative = $\lim_{h \rightarrow 0} \left(\frac{f(a-h) - f(a)}{-h} \right)$

= Slope of tangent at A (when approached from left) $f'(a^-)$.

Thus, $f(x)$ is differentiable at $x = c$.

$\Leftrightarrow \lim_{x \rightarrow c} \frac{f(x) - f(c)}{(x-c)}$ exists finitely

$\Leftrightarrow \lim_{x \rightarrow c^-} \frac{f(x) - f(c)}{(x-c)} = \lim_{x \rightarrow c^+} \frac{f(x) - f(c)}{(x-c)}$

$\Leftrightarrow \lim_{h \rightarrow 0} \frac{f(c-h) - f(c)}{-h} = \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h}$

Hence, $\lim_{x \rightarrow c^-} \frac{f(x) - f(c)}{(x-c)} = \lim_{h \rightarrow 0} \frac{f(c-h) - f(c)}{-h}$ is called

the **left hand derivative** of $f(x)$ at $x = c$ and is denoted by $f'(c^-)$ or $Lf'(c)$.

While, $\lim_{x \rightarrow c^+} \frac{f(x) - f(c)}{(x-c)} = \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h}$ is called

the **right hand derivative** of $f(x)$ at $x = c$ and is denoted by $f'(c^+)$ or $Rf'(c)$.

If $f'(c^-) \neq f'(c^+)$, we say that $f(x)$ is not differentiable at $x = c$.

9. DIFFERENTIABILITY IN A SET

1. A function $f(x)$ defined on an open interval (a, b) is said to be differentiable or derivable in open interval (a, b) , if it is differentiable at each point of (a, b) .
2. A function $f(x)$ defined on closed interval $[a, b]$ is said to be differentiable or derivable. "If f is derivable in the open interval (a, b) and also the end points a and b , then f is said to be derivable in the closed interval $[a, b]$ ".

i.e., $\lim_{x \rightarrow a^+} \frac{f(x) - f(a)}{x-a}$ and $\lim_{x \rightarrow b^-} \frac{f(x) - f(b)}{x-b}$, both exist.

A function f is said to be a differentiable function if it is differentiable at every point of its domain.

NOTES:

1. If $f(x)$ and $g(x)$ are derivable at $x = a$ then the functions $f(x) + g(x)$, $f(x) - g(x)$, $f(x) \cdot g(x)$ will also be derivable at $x = a$ and if $g(a) \neq 0$ then the function $f(x)/g(x)$ will also be derivable at $x = a$.
2. If $f(x)$ is differentiable at $x = a$ and $g(x)$ is not differentiable at $x = a$, then the product function $F(x) = f(x) \cdot g(x)$ can still be differentiable at $x = a$. E.g. $f(x) = x$ and $g(x) = |x|$.
3. If $f(x)$ and $g(x)$ both are not differentiable at $x = a$ then the product function; $F(x) = f(x) \cdot g(x)$ can still be differentiable at $x = a$. E.g., $f(x) = |x|$ and $g(x) = |x|$.
4. If $f(x)$ and $g(x)$ both are not differentiable at $x = a$ then the sum function $F(x) = f(x) + g(x)$ may be a differentiable function. E.g., $f(x) = |x|$ and $g(x) = -|x|$.

10. RELATION B/W CONTINUITY & DIFFERENTIABILITY

In the previous section we have discussed that if a function is differentiable at a point, then it should be continuous at that point and a discontinuous function cannot be differentiable. This fact is proved in the following theorem.

Theorem : If a function is differentiable at a point, it is necessarily continuous at that point. But the converse is not necessarily true,

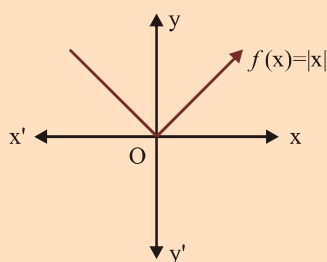
or $f(x)$ is differentiable at $x = c$

$\Rightarrow f(x)$ is continuous at $x = c$.

NOTES:

Converse : The converse of the above theorem is not necessarily true i.e., a function may be continuous at a point but may not be differentiable at that point.

e.g., The function $f(x) = |x|$ is continuous at $x = 0$ but it is not differentiable at $x = 0$, as shown in the figure.



The figure shows that sharp edge at $x = 0$ hence, function is not differentiable but continuous at $x = 0$.

NOTES :

(a) Let $f^{'+}(a) = p$ & $f'^-(a) = q$ where p & q are finite then:

(i) $p = q \Rightarrow f$ is derivable at $x = a$
 $\Rightarrow f$ is continuous at $x = a$.

(ii) $p \neq q \Rightarrow f$ is not derivable at $x = a$.

It is very important to note that f may be still continuous at $x = a$.

In short, for a function f :

Differentiable $\Rightarrow$ Continuous;

Not Differentiable $\nRightarrow$ Not Continuous
(i.e., function may be continuous)

But,

Not Continuous $\Rightarrow$ Not Differentiable.

(b) If a function f is not differentiable but is continuous at $x = a$ it geometrically implies a sharp corner at $x = a$.

Theorem 2 : Let f and g be real functions such that fg is defined if g is continuous at $x = a$ and f is continuous at $g(a)$, show that fg is continuous at $x = a$.

Derivative at a Point

The derivative of a function at a point $x = a$ is defined by

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \text{ (provided the limit exists and is finite)}$$

The above definition of derivative is also called derivative by first principle.

(1) Geometrical meaning of derivatives at a point : Consider the curve $y = f(x)$. Let $f(x)$ be differentiable at $x = c$. Let $P(c, f(c))$ be a point on the curve and $Q(x, f(x))$ be a neighbouring point on the curve. Then,

Slope of the chord $PQ = \frac{f(x) - f(c)}{x - c}$. Taking limit as

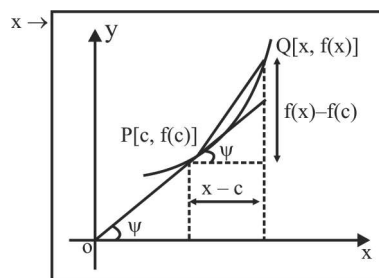
$Q \rightarrow P$, i.e., we get

$$\lim_{Q \rightarrow P} (\text{Slope of the chord } PQ) = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} \quad \text{..(i)}$$

As $Q \rightarrow P$, chord PQ becomes tangent at P .

Therefore from (i), we have

$$\text{slope of the tangent at } P = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} = \left(\frac{df(x)}{dx} \right)_{x=c}$$



NOTES :

Thus, the derivatives of a function at point $x = c$ is the slope of the tangent to curve $y = f(x)$ at point $(c, f(c))$.

(2) Physical interpretation at a point : Let a particle moves in a straight line OX starting from O towards X . Clearly, the position of the particle at any instant would depend upon the time elapsed. In other words, the distance of the particle from O will be some function f of time t .

Let at any time $t = t_0$, the particle be at P and after a further

DIFFERENTIATION

11. DEFINITION

The rate of change of one quantity with respect to some another quantity has a great importance. For example, the rate of change of displacement of a particle with respect to time is called its velocity and the rate of change of velocity is called its acceleration.

The rate of change of a quantity 'y' with respect to another quantity 'x' is called the derivative or differential coefficient of y with respect to x.



time h , it is at Q so that $OP = f(t_0)$ and $OQ = f(t_0 + h)$.
hence, the average speed of the particle during the journey

from P to Q is $\frac{PQ}{h}$, i.e., $\frac{f(t_0 + h) - f(t_0)}{h} = f(t_0, h)$. Taking

the limit of $f(t_0, h)$ as $h \rightarrow 0$, we get its instantaneous

speed to be $\lim_{h \rightarrow 0} \frac{f(t_0 + h) - f(t_0)}{h}$, which is simply $f'(t_0)$.

Thus, if $f(t)$ gives the distance of a moving particle at time t , then the derivative of f at $t = t_0$ represents the instantaneous speed of the particle at the point P , i.e., at time $t = t_0$.

Important Tips

* $\frac{dy}{dx}$ is $\frac{d}{dx}(y)$ in which $\frac{d}{dx}$ is simply a symbol of operation and 'd' divided by dx .

* If $f'(x_0) = \infty$, the function is said to have an infinite derivative at the point x_0 . In this case the line tangent to the curve of $y = f(x)$ at the point x_0 is perpendicular to the x -axis.

(a) Let us consider a function $y = f(x)$ defined in a certain interval. It has a definite value for each value of the independent variable x in this interval.

Now, the ratio of the increment of the function to the increment in the independent variable,

$$\frac{\Delta y}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

Now, as $\Delta x \rightarrow 0$, $\Delta y \rightarrow 0$ and $\frac{\Delta y}{\Delta x} \rightarrow$ finite quantity, then

derivative $f'(x)$ exists and is denoted by y' or $f'(x)$ or $\frac{dy}{dx}$

$$\text{Thus, } f'(x) = \lim_{\Delta x \rightarrow 0} \left(\frac{\Delta y}{\Delta x} \right) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

(if it exists)

for the limit to exist,

$$\lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{f(x - h) - f(x)}{-h}$$

(Right Hand derivative)

(Left Hand derivative)

(b) The derivative of a given function f at a point $x = a$ of its domain is defined as :

Limit $\frac{f(a + h) - f(a)}{h}$, provided the limit exists & is denoted by $f'(a)$.

Note that alternatively, we can define

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}, \text{ provided the limit exists.}$$

This method is called first principle of finding the derivative of $f(x)$.

12. DERIVATIVE OF STANDARD FUNCTION

$$(i) \quad \frac{d}{dx}(x^n) = n \cdot x^{n-1}; x \in \mathbb{R}, n \in \mathbb{R}, x > 0$$

$$(ii) \quad \frac{d}{dx}(e^x) = e^x$$

$$(iii) \quad \frac{d}{dx}(a^x) = a^x \cdot \ln a \quad (a > 0)$$

$$(iv) \quad \frac{d}{dx}(\ln|x|) = \frac{1}{x}$$

$$(v) \quad \frac{d}{dx}(\log_a|x|) = \frac{1}{x} \log_a e$$

$$(vi) \quad \frac{d}{dx}(\sin x) = \cos x$$

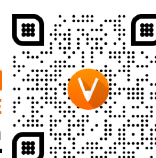
$$(vii) \quad \frac{d}{dx}(\cos x) = -\sin x$$

$$(viii) \quad \frac{d}{dx}(\tan x) = \sec^2 x$$

$$(ix) \quad \frac{d}{dx}(\sec x) = \sec x \cdot \tan x$$

$$(x) \quad \frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cdot \cot x$$

$$(xi) \quad \frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$$





$$(xii) \quad \frac{d}{dx}(\text{constant}) = 0$$

$$(xiii) \quad \frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}, \quad -1 < x < 1$$

$$(xiv) \quad \frac{d}{dx}(\cos^{-1} x) = \frac{-1}{\sqrt{1-x^2}}, \quad -1 < x < 1$$

$$(xv) \quad \frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}, \quad x \in \mathbb{R}$$

$$(xvi) \quad \frac{d}{dx}(\cot^{-1} x) = \frac{-1}{1+x^2}, \quad x \in \mathbb{R}$$

$$(xvii) \quad \frac{d}{dx}(\sec^{-1} x) = \frac{1}{|x|\sqrt{x^2-1}}, \quad |x| > 1$$

$$(xviii) \quad \frac{d}{dx}(\operatorname{cosec}^{-1} x) = \frac{-1}{|x|\sqrt{x^2-1}}, \quad |x| > 1$$

(xix) Results :

If the inverse functions f & g are defined by

$y = f(x)$ & $x = g(y)$. Then $g(f(x)) = x$.

$$\Rightarrow g'(f(x)) \cdot f'(x) = 1.$$

This result can also be written as, if $\frac{dy}{dx}$ exists & $\frac{dy}{dx} \neq 0$, then

$$\frac{dx}{dy} = 1 / \left(\frac{dy}{dx} \right) \text{ or } \frac{dy}{dx} \cdot \frac{dx}{dy} = 1 \text{ or } \frac{dy}{dx} = 1 / \left(\frac{dx}{dy} \right) \left[\frac{dx}{dy} \neq 0 \right]$$

13. THEOREMS ON DERIVATIVES

If u and v are derivable functions of x , then,

$$(i) \quad \text{Term by term differentiation : } \frac{d}{dx}(u \pm v) = \frac{du}{dx} \pm \frac{dv}{dx}$$

$$(ii) \quad \text{Multiplication by a constant } \frac{d}{dx}(K u) = K \frac{du}{dx}, \text{ where } K \text{ is any constant}$$

$$(iii) \quad \text{“Product Rule” } \frac{d}{dx}(u \cdot v) = u \frac{dv}{dx} + v \frac{du}{dx} \text{ known as}$$

In general,

(a) If $u_1, u_2, u_3, u_4, \dots, u_n$ are the functions of x , then

$$\begin{aligned} & \frac{d}{dx}(u_1 \cdot u_2 \cdot u_3 \cdot u_4 \dots u_n) \\ &= \left(\frac{du_1}{dx} \right) (u_2 u_3 u_4 \dots u_n) + \left(\frac{du_2}{dx} \right) (u_1 u_3 u_4 \dots u_n) \\ &+ \left(\frac{du_3}{dx} \right) (u_1 u_2 u_4 \dots u_n) + \left(\frac{du_4}{dx} \right) (u_1 u_2 u_3 u_5 \dots u_n) \\ &+ \dots + \left(\frac{du_n}{dx} \right) (u_1 u_2 u_3 \dots u_{n-1}) \end{aligned}$$

$$(iv) \quad \text{“Quotient Rule” } \frac{d}{dx} \left(\frac{u}{v} \right) = \frac{v \left(\frac{du}{dx} \right) - u \left(\frac{dv}{dx} \right)}{v^2} \text{ where } v \neq 0$$

known as

(b) **Chain Rule :** If $y = f(u)$, $u = g(w)$, $w = h(x)$

$$\text{then } \frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dw} \cdot \frac{dw}{dx}$$

$$\text{or } \frac{dy}{dx} = f'(u) \cdot g'(w) \cdot h'(x)$$

NOTES :

In general if $y = f(u)$ then $\frac{dy}{dx} = f'(u) \cdot \frac{du}{dx}$.

14. METHODS OF DIFFERENTIATION

14.1 Derivative by using Trigonometrical Substitution

Using trigonometrical transformations before differentiation shorten the work considerably. Some important results are given below :

$$(i) \quad \sin 2x = 2 \sin x \cos x = \frac{2 \tan x}{1 + \tan^2 x}$$

$$(ii) \quad \cos 2x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x = \frac{1 - \tan^2 x}{1 + \tan^2 x}$$

$$(iii) \quad \tan 2x = \frac{2 \tan x}{1 - \tan^2 x}, \tan^2 x = \frac{1 - \cos 2x}{1 + \cos 2x}$$

$$(iv) \quad \sin 3x = 3 \sin x - 4 \sin^3 x$$





(v) $\cos 3x = 4 \cos^3 x - 3 \cos x$

(vi) $\tan 3x = \frac{3 \tan x - \tan^3 x}{1 - 3 \tan^2 x}$

(vii) $\tan\left(\frac{\pi}{4} + x\right) = \frac{1 + \tan x}{1 - \tan x}$

(viii) $\tan\left(\frac{\pi}{4} - x\right) = \frac{1 - \tan x}{1 + \tan x}$

(ix) $\sqrt{1 \pm \sin x} = \left| \cos \frac{x}{2} \pm \sin \frac{x}{2} \right|$

(x) $\tan^{-1} x \pm \tan^{-1} y = \tan^{-1} \left(\frac{x \pm y}{1 \mp xy} \right)$

(xi) $\sin^{-1} x \pm \sin^{-1} y = \sin^{-1} \left\{ x \sqrt{1 - y^2} \pm y \sqrt{1 - x^2} \right\}$

(xii) $\cos^{-1} x \pm \cos^{-1} y = \cos^{-1} \left\{ xy \mp \sqrt{1 - x^2} \sqrt{1 - y^2} \right\}$

(xiii) $\sin^{-1} x + \cos^{-1} x = \tan^{-1} x + \cot^{-1} x = \sec^{-1} x + \operatorname{cosec}^{-1} x = \pi/2$

(xiv) $\sin^{-1} x = \operatorname{cosec}^{-1}(1/x); \cos^{-1} x = \sec^{-1}(1/x); \tan^{-1} x = \cot^{-1}(1/x)$

NOTES :

Some standard substitutions :

Expressions Substitutions

$\sqrt{a^2 - x^2}$ $x = a \sin \theta$ or $a \cos \theta$

$\sqrt{a^2 + x^2}$ $x = a \tan \theta$ or $a \cot \theta$

$\sqrt{x^2 - a^2}$ $x = a \sec \theta$ or $a \operatorname{cosec} \theta$

$\sqrt{\frac{a+x}{a-x}}$ or $\sqrt{\frac{a-x}{a+x}}$ $x = a \cos \theta$ or $a \cos 2\theta$

$\sqrt{(a-x)(x-b)}$ or $x = a \cos^2 \theta + b \sin^2 \theta$

$\sqrt{\frac{a-x}{x-b}}$ or $\sqrt{\frac{x-b}{a-x}}$

$\sqrt{(x-a)(x-b)}$ or $x = a \sec^2 \theta - b \tan^2 \theta$

$\sqrt{\frac{x-a}{x-b}}$ or $\sqrt{\frac{x-b}{x-a}}$

$\sqrt{2ax - x^2}$ $x = a(1 - \cos \theta)$

14.2 Logarithmic Differentiation

To find the derivative of :

If $y = \{f_1(x)\}^{f_2(x)}$ or $y = f_1(x) \cdot f_2(x) \cdot f_3(x) \dots$

or $y = \frac{f_1(x) \cdot f_2(x) \cdot f_3(x) \dots}{g_1(x) \cdot g_2(x) \cdot g_3(x) \dots}$

then it is convenient to take the logarithm of the function first and then differentiate. This is called derivative of the logarithmic function.

Important Notes (Alternate methods)

1. If $y = \{f(x)\}^{g(x)} = e^{g(x) \ln f(x)}$ ((variable)^{variable}) $\therefore x = e^{\ln x}$

$\therefore \frac{dy}{dx} = e^{g(x) \ln f(x)} \cdot \left\{ g(x) \cdot \frac{d}{dx} \ln f(x) + \ln f(x) \cdot \frac{d}{dx} g(x) \right\}$

$= \{f(x)\}^{g(x)} \cdot \left\{ g(x) \cdot \frac{f'(x)}{f(x)} + \ln f(x) \cdot g'(x) \right\}$

2. If $y = \{f(x)\}^{g(x)}$

$\therefore \frac{dy}{dx} =$ Derivative of y treating $f(x)$ as constant + Derivative of y treating $g(x)$ as constant

$= \{f(x)\}^{g(x)} \cdot \ln f(x) \cdot \frac{d}{dx} g(x) + g(x) \cdot \{f(x)\}^{g(x)-1} \cdot \frac{d}{dx} f(x)$

$= \{f(x)\}^{g(x)} \cdot \ln f(x) \cdot g'(x) + g(x) \cdot \{f(x)\}^{g(x)-1} \cdot f'(x)$

14.3 Implicit Differentiation : $\phi(x, y) = 0$

- (i) In order to find dy/dx in the case of implicit function, we differentiate each term w.r.t. x , regarding y as a function of x & then collect terms in dy/dx together on one side to finally find dy/dx .



- (ii) In answers of dy/dx in the case of implicit function, both x & y are present.

Alternate Method : If $f(x, y) = 0$

$$\text{then } \frac{dy}{dx} = -\frac{\left(\frac{\partial f}{\partial x}\right)}{\left(\frac{\partial f}{\partial y}\right)} = -\frac{\text{diff. of } f \text{ w.r.t. } x \text{ treating } y \text{ as constant}}{\text{diff. of } f \text{ w.r.t. } y \text{ treating } x \text{ as constant}}$$

14.4 Parametric Differentiation

If $y = f(t)$ & $x = g(t)$ where t is a Parameter, then

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} \quad \dots(1)$$

NOTES :

- $\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$
- $\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dt} \left(\frac{dy}{dx} \right) \cdot \frac{dt}{dx} \left(\because \frac{dy}{dx} \text{ in terms of } t \right)$
 $= \frac{d}{dt} \left(\frac{f'(t)}{g'(t)} \right) \cdot \frac{1}{f'(t)} \quad \{\text{From (1)}\}$
 $= \frac{f''(t)g'(t) - g''(t)f'(t)}{\{f'(t)\}^3}$

14.5 Derivative of a Function w.r.t. another Function

Let $y = f(x)$; $z = g(x)$ then $\frac{dy}{dz} = \frac{dy/dx}{dz/dx} = \frac{f'(x)}{g'(x)}$

14.6 Derivative of Infinite Series

If taking out one or more than one terms from an infinite series, it remains unchanged. Such that

(A) If $y = \sqrt{f(x)} + \sqrt{f(x)} + \sqrt{f(x)} + \dots \infty$

then $y = \sqrt{f(x) + y} \Rightarrow (y^2 - y) = f(x)$

Differentiating both sides w.r.t. x , we get $(2y - 1) \frac{dy}{dx} = f'(x)$

(B) If $y = \{f(x)\}^{\{f(x)\}^{\{f(x)\}^{\dots \infty}}}$ then $y = \{f(x)\}^y \Rightarrow y = e^{y \ln f(x)}$

Differentiating both sides w.r.t. x , we get

$$\frac{dy}{dx} = \frac{y \{f(x)\}^{y-1} \cdot f'(x)}{1 - \{f(x)\}^y \cdot \ln f(x)} = \frac{y^2 f'(x)}{f(x) \{1 - y \ln f(x)\}}$$

15. DERIVATIVE OF ORDER TWO & THREE

Let a function $y = f(x)$ be defined on an open interval (a, b) . It's derivative, if it exists on (a, b) , is a certain function $f'(x)$ [or (dy/dx) or y'] & is called the first derivative of y w.r.t. x . If it happens that the first derivative has a derivative on (a, b) then this derivative is called the second derivative of y w.r.t. x & is denoted by $f''(x)$ or (d^2y/dx^2) or y'' .

Similarly, the 3rd order derivative of y w.r.t. x , if it exists, is

defined by $\frac{d^3y}{dx^3} = \frac{d}{dx} \left(\frac{d^2y}{dx^2} \right)$ it is also denoted by $f'''(x)$ or y''' .

Some Standard Results :

- $\frac{d^n}{dx^n} (ax + b)^m = \frac{m!}{(m-n)!} \cdot a^n \cdot (ax + b)^{m-n}, m \geq n.$
- $\frac{d^n}{dx^n} x^n = n!$
- $\frac{d^n}{dx^n} (e^{mx}) = m^n \cdot e^{mx}, m \in \mathbb{R}$
- $\frac{d^n}{dx^n} (\sin(ax + b)) = a^n \sin\left(ax + b + \frac{n\pi}{2}\right), n \in \mathbb{N}$
- $\frac{d^n}{dx^n} (\cos(ax + b)) = a^n \cos\left(ax + b + \frac{n\pi}{2}\right), n \in \mathbb{N}$
- $\frac{d^n}{dx^n} \{e^{ax} \sin(bx + c)\} = r^n \cdot e^{ax} \cdot \sin(bx + c + n\phi), n \in \mathbb{N}$
 where $r = \sqrt{a^2 + b^2}, \phi = \tan^{-1}(b/a).$
- $\frac{d^n}{dx^n} \{e^{ax} \cdot \cos(bx + c)\} = r^n \cdot e^{ax} \cdot \cos(bx + c + n\phi), n \in \mathbb{N}$
 where $r = \sqrt{a^2 + b^2}, \phi = \tan^{-1}(b/a).$





16. DIFFERENTIATION OF DETERMINANTS

$$\text{If } F(X) = \begin{vmatrix} f(x) & g(x) & h(x) \\ \ell(x) & m(x) & n(x) \\ u(x) & v(x) & w(x) \end{vmatrix},$$

where $f, g, h, \ell, m, n, u, v, w$ are differentiable function of x then

$$F'(x) = \begin{vmatrix} f'(x) & g'(x) & h'(x) \\ \ell(x) & m(x) & n(x) \\ u(x) & v(x) & w(x) \end{vmatrix} + \begin{vmatrix} f(x) & g(x) & h(x) \\ \ell'(x) & m'(x) & n'(x) \\ u(x) & v(x) & w(x) \end{vmatrix} + \begin{vmatrix} f(x) & g(x) & h(x) \\ \ell(x) & m(x) & n(x) \\ u'(x) & v'(x) & w'(x) \end{vmatrix}$$

$$+ \begin{vmatrix} f(x) & g(x) & h(x) \\ \ell(x) & m(x) & n(x) \\ u(x) & v(x) & w(x) \end{vmatrix}$$

17. L' HOSPITAL'S RULE

If $f(x)$ & $g(x)$ are functions of x such that :

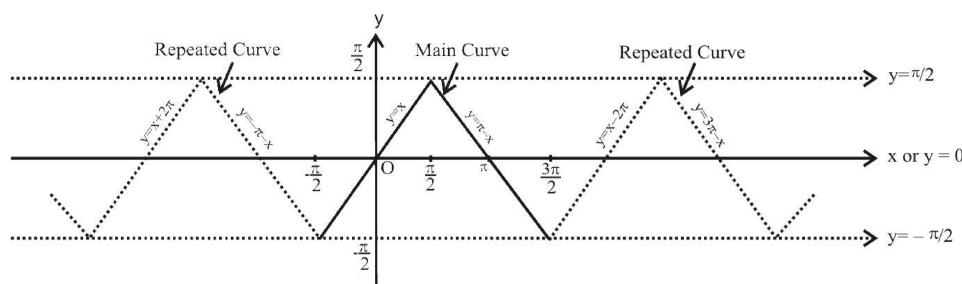
- (i) $\lim_{x \rightarrow a} f(x) = 0 = \lim_{x \rightarrow a} g(x)$ or $\lim_{x \rightarrow a} f(x) = \infty = \lim_{x \rightarrow a} g(x)$ and
- (ii) Both $f(x)$ & $g(x)$ are continuous at $x = a$ and
- (iii) Both $f(x)$ & $g(x)$ are differentiable at $x = a$ and
- (iv) Both $f'(x)$ & $g'(x)$ are continuous at $x = a$, Then

$$\text{Limit}_{x \rightarrow a} \frac{f(x)}{g(x)} = \text{Limit}_{x \rightarrow a} \frac{f'(x)}{g'(x)} = \text{Limit}_{x \rightarrow a} \frac{f''(x)}{g''(x)} \text{ \& so on till}$$

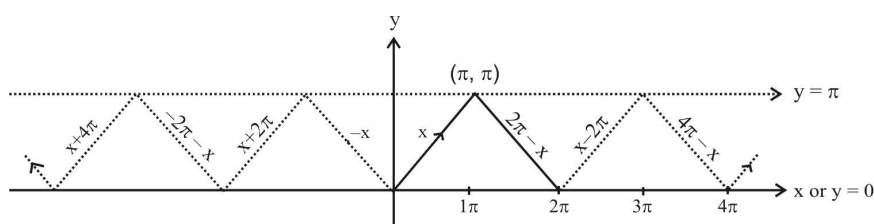
indeterminant form vanishes..

18. ANALYSIS & GRAPHS OF SOME USEFUL FUNCTION

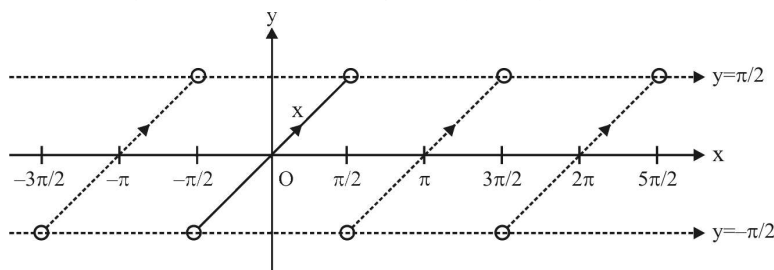
(i) $y = \sin^{-1}(\sin x)$ $x \in \mathbb{R}; y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$



(ii) $y = \cos^{-1}(\cos x)$ $x \in \mathbb{R}; y \in [0, \pi]$



(iii) $y = \tan^{-1}(\tan x)$ $x \in \mathbb{R} - \left\{x : x = (2n+1)\frac{\pi}{2}, n \in \mathbb{Z}\right\}; y \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$





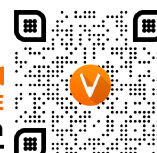
19. LEIBNITZ THEOREM

19.1 If u and v are functions of x such that their n^{th} derivatives exist, then the n^{th} derivative of their product is given by

$$(uv)_n = u_n v + {}^nC_1 u_{n-1} v_1 + {}^nC_2 u_{n-2} v_2 + \dots + {}^nC_r u_{n-r} v_r + \dots + u v_n$$

where u_r and v_r represent r^{th} derivatives of u and v respectively.

$$\mathbf{19.2} \quad \frac{d}{dx} \left[\int_{g(x)}^{h(x)} f(t) dt \right] = \frac{d}{dx} [h(x)] \cdot f[h(x)] - \frac{d}{dx} [g(x)] \cdot f[g(x)]$$





SOLVED EXAMPLES

Example – 1

If f is a real-valued function satisfying the relation

$$f(x) + 2f\left(\frac{1}{x}\right) = 3x \text{ for all real } x \neq 0, \text{ then } \lim_{x \rightarrow 0} (\sin x) f(x)$$

is equal to

- (a) 1 (b) 2
(c) 0 (d) ∞

Ans. (b)

Sol. We have $f(x) + 2f\left(\frac{1}{x}\right) = 3x$... (1)

Replacing x with $1/x$, we have

$$2f(x) + f\left(\frac{1}{x}\right) = \frac{3}{x}$$
 ... (2)

From Eqs. (1) and (2) we get

$$f(x) = \frac{2}{x} - x$$

Therefore

$$\begin{aligned} \lim_{x \rightarrow 0} (\sin x) f(x) &= \lim_{x \rightarrow 0} \left(\frac{2 \sin x}{x} - x \sin x \right) \\ &= 2(1) - 0 = 2. \end{aligned}$$

Example – 2

Show that $f(x) = \begin{cases} 5x - 4, & \text{when } 0 < x \leq 1 \\ 4x^3 - 3x, & \text{when } 1 < x < 2 \end{cases}$

is continuous at $x = 1$.

Sol. We have,

$$(\text{LHL at } x = 1) = \lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} 5x - 4$$

$$\begin{aligned} [\because f(x) &= 5x - 4, \text{ when } x \leq 1] \\ &= 5 \times 1 - 4 = 1, \end{aligned}$$

$$(\text{RHL at } x = 1) = \lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} 4x^3 - 3x$$

$$[\because f(x) = 4x^3 - 3x, x > 1]$$

$$= 4(1)^3 - 3(1) = 1,$$

$$\text{and, } f(1) = 5 \times 1 - 4 = 1$$

$$[\because f(x) = 5x - 4, \text{ where } x \leq 1]$$

$$\therefore \lim_{x \rightarrow 1^-} f(x) = f(1) = \lim_{x \rightarrow 1^+} f(x)$$

So, $f(x)$ is continuous at $x = 1$.

Example – 3

Test the continuity of the function $f(x)$ at the origin :

$$f(x) = \begin{cases} \frac{|x|}{x}, & x \neq 0 \\ 1, & x = 0 \end{cases}$$

Sol. We have,

$$(\text{LHL at } x = 0) = \lim_{x \rightarrow 0^-} f(x) = \lim_{h \rightarrow 0} f(0 - h) = \lim_{h \rightarrow 0} f(-h)$$

$$= \lim_{h \rightarrow 0} \frac{-h}{-h} = \lim_{h \rightarrow 0} \frac{h}{-h} = \lim_{h \rightarrow 0} -1 = -1$$

$$\text{and, } (\text{RHL at } x = 0) = \lim_{x \rightarrow 0^+} f(x) = \lim_{h \rightarrow 0} f(0 + h) = \lim_{h \rightarrow 0} f(h)$$

$$= \lim_{h \rightarrow 0} \frac{h}{h} = \lim_{h \rightarrow 0} \frac{h}{h} = \lim_{h \rightarrow 0} 1 = 1$$

Thus, $f(x)$ is not continuous at the origin.

Example – 4

Discuss the continuity of the function of given by

$$f(x) = |x - 1| + |x - 2| \text{ at } x = 1 \text{ and } x = 2.$$

Sol. We have,

$$f(x) = |x - 1| + |x - 2|$$



$$\Rightarrow f(x) = \begin{cases} -(x-1) - (x-2), & \text{if } x < 1 \\ (x-1) - (x-2), & \text{if } 1 \leq x < 2 \\ (x-1) + (x-2), & \text{if } x \geq 2 \end{cases}$$

$$\Rightarrow f(x) = \begin{cases} -2x+3, & \text{if } x < 1 \\ 1, & \text{if } 1 \leq x < 2 \\ 2x-3, & \text{if } x \geq 2 \end{cases}$$

Continuity at $x = 1$:

We have,

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (-2x+3) = -2 \times 1 + 3 = 1$$

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} 1 = 1$$

and, $f(1) = 1$.

$$\therefore \lim_{x \rightarrow 1^-} f(x) = f(1) = \lim_{x \rightarrow 1^+} f(x)$$

So, $f(x)$ is continuous at $x = 1$.

Continuity at $x = 2$

We have,

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} 1 = 1$$

$$\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} (2x-3) = 2 \times 2 - 3 = 1$$

and, $f(2) = 2 \times 2 - 3 = 1$.

$$\therefore \lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x) = f(2)$$

So, $f(x)$ is continuous at $x = 2$.

Example - 5

Examine the function $f(t)$ given by

$$f(t) = \begin{cases} \frac{\cos t}{\pi/2 - t}; & t \neq \pi/2 \\ 1 & ; t = \pi/2 \end{cases}$$

for continuity at $t = \pi/2$

Sol. We have,

$$(\text{LHL at } t = \pi/2) = \lim_{t \rightarrow \pi/2^-} f(t)$$

$$= \lim_{h \rightarrow 0} f(\pi/2 - h) = \lim_{h \rightarrow 0} \frac{\cos(\pi/2 - h)}{\pi/2 - (\pi/2 - h)} = \lim_{h \rightarrow 0} \frac{\sin h}{h} = 1$$

$$\text{and, (RHL at } t = \pi/2) = \lim_{t \rightarrow \pi/2^+} f(t)$$

$$= \lim_{h \rightarrow 0} f(\pi/2 + h) = \lim_{h \rightarrow 0} \frac{\cos(\pi/2 + h)}{\pi/2 - (\pi/2 + h)}$$

$$= \lim_{h \rightarrow 0} \frac{-\sin h}{-h} = \lim_{h \rightarrow 0} \frac{\sin h}{h} = 1$$

Also, $f(\pi/2) = 1$.

$$\therefore \lim_{t \rightarrow \pi/2^-} f(t) = \lim_{t \rightarrow \pi/2^+} f(t) = f(\pi/2)$$

So, $f(t)$ is continuous at $t = \pi/2$.

Example - 6

Show that the function $f(x)$ given by

$$f(x) = \begin{cases} \frac{e^{1/x} - 1}{e^{1/x} + 1}, & \text{when } x \neq 0 \\ 0 & , \text{ when } x = 0 \end{cases} \text{ is discontinuous at } x=0.$$

Sol. We have,

$$(\text{LHL at } x=0) = \lim_{x \rightarrow 0^-} f(x) = \lim_{h \rightarrow 0} f(0-h) = \lim_{h \rightarrow 0} f(-h)$$

$$= \lim_{h \rightarrow 0} \frac{e^{-1/h} - 1}{e^{-1/h} + 1} = \lim_{h \rightarrow 0} \frac{\frac{1}{e^{1/h}} - 1}{\frac{1}{e^{1/h}} + 1} = \frac{0-1}{0+1} = -1$$

$$\left[\because \lim_{h \rightarrow 0} \frac{1}{e^{1/h}} = 0 \right]$$

$$\text{and, (RHL at } x=0) = \lim_{x \rightarrow 0^+} f(x) = \lim_{h \rightarrow 0} f(0+h) = \lim_{h \rightarrow 0} f(h)$$

$$= \lim_{h \rightarrow 0} \frac{e^{1/h} - 1}{e^{1/h} + 1} = \lim_{h \rightarrow 0} \frac{1 - 1/e^{1/h}}{1 + 1/e^{1/h}} = \frac{1-0}{1+0} = 1$$

So, $f(x)$ is not continuous at $x = 0$ as $\text{LHL} \neq \text{RHL}$ and has a discontinuity of first kind at $x = 0$.

Example - 7

Show that the function $f(x) = |\sin x + \cos x|$ is continuous at $x = \pi$.

Sol. Let $g(x) = \sin x + \cos x$ and $h(x) = |x|$. Then, $f(x) = \text{hog}(x)$. In order to prove that $f(x)$ is continuous at $x = \pi$. It is sufficient to prove that $g(x)$ is continuous at $x = \pi$ and $h(x)$ is continuous at $y = g(\pi) = \sin \pi + \cos \pi = -1$.



Now,

$$\lim_{x \rightarrow \pi} g(x) = \lim_{x \rightarrow \pi} (\sin x + \cos x) = \sin \pi + \cos \pi = -1$$

and $g(\pi) = -1$

$$\therefore \lim_{x \rightarrow \pi} g(x) = g(\pi)$$

LHL at $x = \pi$

$$\lim_{x \rightarrow \pi} f(x) = \lim_{x \rightarrow \pi} |\sin \pi + \cos \pi| = +1$$

RHL at $x = \pi$

$$\lim_{x \rightarrow \pi} |\sin \pi + \cos \pi| = +1$$

Hence continuous

Example - 8

Let $f(x+y) = f(x) + f(y)$ for all $x, y \in \mathbb{R}$. If $f(x)$ is continuous at $x = 0$, show that $f(x)$ is continuous at all x .

Sol. Since $f(x)$ is continuous at $x = 0$. Therefore,

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x) = f(0)$$

$$\Rightarrow \lim_{h \rightarrow 0} f(0-h) = \lim_{h \rightarrow 0} f(0+h) = f(0)$$

$$\Rightarrow \lim_{h \rightarrow 0} f(0+(-h)) = \lim_{h \rightarrow 0} f(0+h) = f(0)$$

$$\Rightarrow \lim_{h \rightarrow 0} [f(0) + f(-h)] = \lim_{h \rightarrow 0} [f(0) + f(h)] = f(0)$$

[Using : $f(x+y) = f(x) + f(y)$]

$$\Rightarrow f(0) + \lim_{h \rightarrow 0} f(-h) = f(0) + \lim_{h \rightarrow 0} f(h) = f(0)$$

$$\Rightarrow \lim_{h \rightarrow 0} f(-h) = \lim_{h \rightarrow 0} f(h) = 0 \quad \dots (i)$$

Let a be any real number. Then,

$$\lim_{x \rightarrow a^-} f(x) = \lim_{h \rightarrow 0} f(a-h) = \lim_{h \rightarrow 0} f(a+(-h))$$

$$\Rightarrow \lim_{x \rightarrow a^-} f(x) = \lim_{h \rightarrow 0} [f(a) + f(-h)]$$

[$\because f(x+y) = f(x) + f(y)$]

$$\Rightarrow \lim_{x \rightarrow a^-} f(x) = f(a) + \lim_{h \rightarrow 0} f(-h)$$

$$\Rightarrow \lim_{x \rightarrow a^-} f(x) = f(a) + 0 \quad \text{[Using (i)]}$$

$$\Rightarrow \lim_{x \rightarrow a^-} f(x) = f(a). \quad \text{and,}$$

$$\lim_{x \rightarrow a^+} f(x) = \lim_{h \rightarrow 0} f(a+h)$$

$$\Rightarrow \lim_{x \rightarrow a^+} f(x) = \lim_{h \rightarrow 0} [f(a) + f(h)]$$

[$\because f(x+y) = f(x) + f(y)$]

$$\Rightarrow \lim_{x \rightarrow a^+} f(x) = f(a) + \lim_{h \rightarrow 0} f(h)$$

$$\Rightarrow \lim_{x \rightarrow a^+} f(x) = f(a) + 0 = f(a) \quad \text{[Using (i)]}$$

Thus, we have

$$\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = f(a)$$

$\therefore f(x)$ is continuous at $x = a$.

Since a is an arbitrary real number. So, $f(x)$ is continuous at all $x \in \mathbb{R}$.

Example - 9

If f is a real-valued function defined for all $x \neq 0, 1$ and satisfying the relation

$$f(x) + f\left(\frac{1}{1-x}\right) = \frac{2}{x} - \frac{2}{1-x}. \quad \text{Then } \lim_{x \rightarrow 2} f(x) \text{ is } \underline{\hspace{2cm}}.$$

Sol. Given relation is

$$f(x) + f\left(\frac{1}{1-x}\right) = \frac{2}{x} - \frac{2}{1-x} \quad \dots (1)$$

Replacing x with $\frac{1}{1-x}$ in above equation Eq. we have

$$f\left(\frac{1}{1-x}\right) + f\left(\frac{x-1}{x}\right) = 2(1-x) + \frac{2(1-x)}{x} \quad \dots (2)$$

Again replacing x with $\frac{1}{1-x}$ in Eq. (1), we get

$$f\left(\frac{x-1}{x}\right) + f(x) = -2x - \frac{2x}{1-x} \quad \dots (3)$$

Now adding Eqs. (1) and (3) and subtracting Eq. (2) gives

$$2f(x) = \left(\frac{2}{x} - \frac{2}{1-x} - 2x - \frac{2x}{1-x}\right) - 2(1-x) - \frac{2(1-x)}{x}$$



$$\begin{aligned}
 &= \left(\frac{2}{x} - \frac{2(1-x)}{x} \right) + \left(\frac{-2}{1-x} - \frac{2x}{1-x} \right) - 2x - 2(1-x) \\
 &= \frac{2x}{x} - \frac{2(1+x)}{(1-x)} - 2 \\
 &= 2 + \frac{2(x+1)}{x-1} - 2 \\
 &= \frac{2(x+1)}{x-1}
 \end{aligned}$$

Therefore

$$f(x) = \frac{x+1}{x-1}$$

Taking limit we get

$$\lim_{x \rightarrow 2} \left(\frac{x+1}{x-1} \right) = \frac{2+1}{2-1} = 3$$

Example – 10

Prove that the greatest integer function $[x]$ is continuous at all points except at integer points.

Sol. Let $f(x) = [x]$ be the greatest integer function. Let k be any integer. Then,

$$f(x) = [x] = \begin{cases} k-1, & \text{if } k-1 \leq x < k \\ k, & \text{if } k \leq x < k+1 \end{cases} \quad [\text{By def.}]$$

Now (LHL at $x=k$)

$$\begin{aligned}
 &= \lim_{x \rightarrow k^-} f(x) = \lim_{h \rightarrow 0} f(k-h) = \lim_{h \rightarrow 0} [k-h] \\
 &= \lim_{h \rightarrow 0} (k-1) = k-1
 \end{aligned}$$

$$[\because k-1 \leq k-h < k \therefore [k-h] = k-1]$$

and, (RHL at $x=k$)

$$\begin{aligned}
 &= \lim_{x \rightarrow k^+} f(x) = \lim_{h \rightarrow 0} f(k+h) = \lim_{h \rightarrow 0} [k+h] \\
 &= \lim_{h \rightarrow 0} k = k
 \end{aligned}$$

$$[\because k \leq k+h < k+1 \therefore [k+h] = k]$$

$$\therefore \lim_{x \rightarrow k^-} f(x) \neq \lim_{x \rightarrow k^+} f(x)$$

So, $f(x)$ is not continuous at $x=k$.

Since k is an arbitrary integer. Therefore, $f(x)$ is not continuous at integer points.

Let a be any real number other than an integer. Then, there exists an integer k such that $k-1 < a < k$.

Now, (LHL at $x=a$)

$$\lim_{x \rightarrow a^-} f(x) = \lim_{h \rightarrow 0} f(a-h) = \lim_{h \rightarrow 0} [a-h]$$

$$= \lim_{h \rightarrow 0} k-1 = k-1$$

$$[\because k-1 < a-h < k \therefore [a-h] = k-1]$$

(RHL at $x=a$)

$$= \lim_{x \rightarrow a^+} f(x) = \lim_{h \rightarrow 0} f(a+h)$$

$$= \lim_{h \rightarrow 0} [a+h] = \lim_{h \rightarrow 0} (k-1) = k-1$$

$$[\because k-1 < a+h < k \therefore [a+h] = k-1]$$

and, $f(a) = k-1$

$$[\because k-1 < a < k \therefore [a] = k-1]$$

Thus,

$$\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = f(a)$$

So, $f(x)$ is continuous at $x=a$. Since a is an arbitrary real number, other than an integer. Therefore, $f(x)$ is continuous at all real points except integer points.

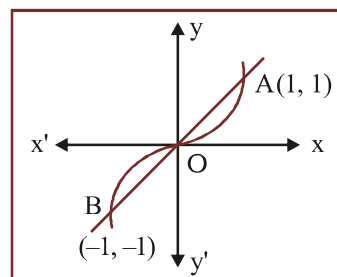
Example – 11

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x) = \max \{x, x^3\}$. Show that the set of points $\{-1, 0, 1\}$, $f(x)$ is not differentiable.

Sol. $f(x) = \max \{x, x^3\}$ considering the graph separately, $y = x^3$ and $y = x$.

$$\text{Now, } f(x) = \begin{cases} x & \text{in } (-\infty, -1] \\ x^3 & \text{in } [-1, 0] \\ x & \text{in } [0, 1] \\ x^3 & \text{in } [1, \infty) \end{cases}$$

The point of consideration are





$$f'(-1^-) = 1 \text{ and } f'(-1^+) = 3$$

$$f'(-0^-) = 0 \text{ and } f'(0^+) = 1$$

$$f'(1^-) = 1 \text{ and } f'(1^+) = 3$$

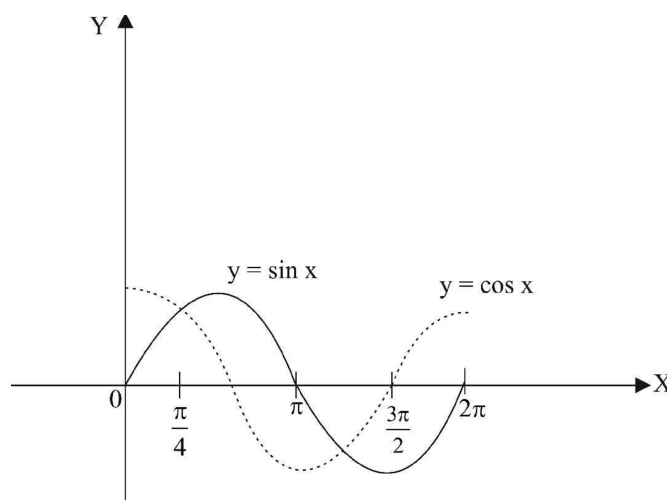
Hence, f is not differentiable at $-1, 0, 1$.

Example – 12

Find the number of points of non-differentiability of $f(x) = \max. \{\sin x, \cos x, 0\}$ in $(0, 2n\pi)$. Where $n \in \mathbb{N}$

Sol. Here, we know $\sin x$ and $\cos x$ are periodic with period 2π . Thus we could sketch the curve as; (In the interval 0 to 2π)

Which shows



$$y = \max. \{\sin x, \cos x, 0\}$$

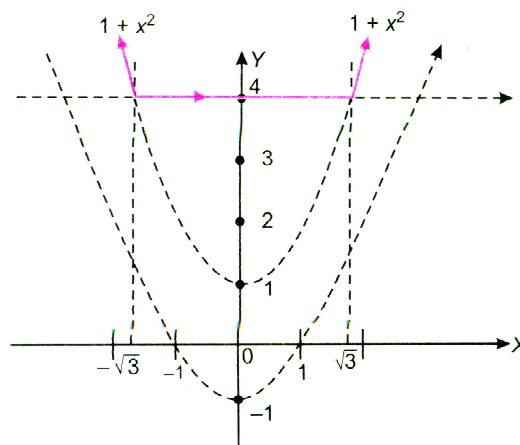
$$= \begin{cases} \cos x, & 0 < x < \frac{\pi}{4} \text{ or } \frac{3\pi}{2} < x < 2\pi \\ 0, & \pi < x < \frac{3\pi}{2} \\ \sin x, & \frac{\pi}{4} < x < \pi \end{cases}$$

Clearly, $y = \max. \{\sin x, \cos x, 0\}$ is not differentiable at 3 points when $x = (0, 2\pi)$.

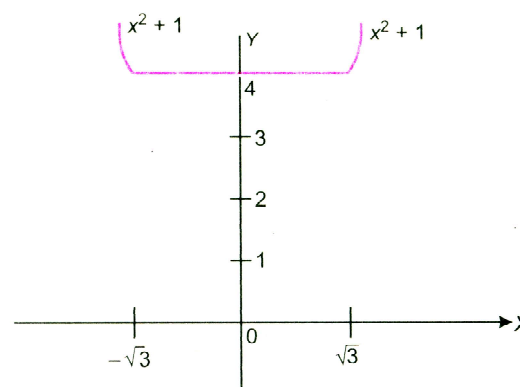
Example – 13

Let $f(x) = \text{maximum} \{4, 1+x^2, x^2-1\} \forall x \in \mathbb{R}$. Then find the total number of points, where $f(x)$ is not differentiable.

Sol. Using functions for sketching maximum $\{4, 1+x^2, x^2-1\}$ as



OR



Thus, from above graph we can simply say,

$f(x)$ is not differentiable at $x = \pm \sqrt{3}$.

And it could be defined as :

$$f(x) = \begin{cases} 4, & -\sqrt{3} \leq x \leq \sqrt{3} \\ x^2 + 1, & x \leq -\sqrt{3} \text{ or } x \geq \sqrt{3} \end{cases}$$

Example – 14

Discuss the differentiability of $f(x) = |x-1| + |x-2|$.

Sol. We have,

$$f(x) = |x-1| + |x-2|$$

$$\Rightarrow f(x) = \begin{cases} -(x-1) - (x-2) & \text{for } x < 1 \\ x-1 - (x-2) & \text{for } 1 \leq x < 2 \\ (x-1) + (x-2) & \text{for } x \geq 2 \end{cases}$$



$$\Rightarrow f(x) = \begin{cases} -2x+3, & x < 1 \\ 1, & 1 \leq x < 2 \\ 2x-3, & x \geq 2 \end{cases}$$

When $x < 1$, we have

$f(x) = -2x + 3$ which, being a polynomial function is continuous and differentiable.

When $1 \leq x < 2$, we have

$f(x) = 1$ which, being a constant function, is differentiable on $(1, 2)$.

When $x \geq 2$, we have

$f(x) = 2x - 3$ which, being a polynomial function, is differentiable for all $x > 2$. Thus, the possible points of non-differentiability of $f(x)$ are $x = 1$ and $x = 2$.

Now,

$$(\text{LHD at } x = 1) = \lim_{x \rightarrow 1^-} \frac{f(x) - f(1)}{x - 1}$$

$$\Rightarrow (\text{LHD at } x = 1) = \lim_{x \rightarrow 1^-} \frac{(-2x+3)-1}{x-1}$$

$$[\because f(x) = -2x + 3 \text{ for } x < 1]$$

$$\Rightarrow (\text{LHD at } x = 1) = \lim_{x \rightarrow 1^-} \frac{-2(x-1)}{x-1} = -2$$

$$(\text{RHD at } x = 1) = \lim_{x \rightarrow 1^+} \frac{f(x) - f(1)}{x - 1}$$

$$\Rightarrow (\text{RHD at } x = 1) = \lim_{x \rightarrow 1^+} \frac{1-1}{x-1} = 0$$

$$[\because f(x) = 1 \text{ for } 1 \leq x < 2]$$

$$\therefore (\text{LHD at } x = 1) \neq (\text{RHD at } x = 1)$$

So, $f(x)$ is not differentiable at $x = 1$.

$$(\text{LHD at } x = 2) = \lim_{x \rightarrow 2^-} \frac{f(x) - f(2)}{x - 2}$$

$$\Rightarrow (\text{LHD at } x = 2) = \lim_{x \rightarrow 2^-} \frac{1 - (2 \times 2 - 3)}{x - 2}$$

$$\left[\begin{array}{l} \because f(x) = 1 \text{ for } 1 \leq x < 2 \\ \text{and } f(2) = 2 \times 2 - 3 \end{array} \right]$$

$$\Rightarrow (\text{LHD at } x = 2) = \lim_{x \rightarrow 2^-} \frac{1-1}{x-2} = 0.$$

$$(\text{RHD at } x = 2) = \lim_{x \rightarrow 2^+} \frac{f(x) - f(2)}{x - 2}$$

$$\Rightarrow (\text{RHD at } x = 2) = \lim_{x \rightarrow 2^+} \frac{(2x-3) - (2 \times 2 - 3)}{x - 2}$$

$$[\because f(x) = 2x - 3 \text{ for } x \geq 2]$$

$$\Rightarrow (\text{RHD at } x = 2) = \lim_{x \rightarrow 2^+} \frac{2x-4}{x-2} = \lim_{x \rightarrow 2^+} \frac{2(x-2)}{x-2} = 2$$

$$\therefore (\text{LHD at } x = 2) \neq (\text{RHD at } x = 2)$$

So, $f(x)$ not differentiable at $x = 2$.

Remark It should be noted that the function $f(x)$ given by

$$f(x) = |x - a_1| + |x - a_2| + |x - a_3| + \dots + |x - a_n|$$

is not differentiable at $x = a_1, a_2, a_3, \dots, a_n$.

Example - 15

If $\sqrt{1-x^{2n}} + \sqrt{1-y^{2n}} = a(x^n - y^n)$ then prove that

$$\frac{dy}{dx} = \frac{x^{n-1}}{y^{n-1}} \sqrt{\frac{1-y^{2n}}{1-x^{2n}}}$$

$$\text{Sol. We have } \sqrt{1-x^{2n}} + \sqrt{1-y^{2n}} = a(x^n - y^n) \dots (1)$$

$$\left. \begin{array}{l} \text{Putting } x^n = \sin \theta \Rightarrow \theta = \sin^{-1} x^n \\ \text{and } y^n = \sin \phi \Rightarrow \phi = \sin^{-1} y^n \end{array} \right\} \dots (2)$$

then (1), becomes $\cos \theta + \cos \phi = a(\sin \theta - \sin \phi)$

$$\Rightarrow 2 \cos\left(\frac{\theta+\phi}{2}\right) \cos\left(\frac{\theta-\phi}{2}\right) = a.2 \cos\left(\frac{\theta+\phi}{2}\right) \sin\left(\frac{\theta-\phi}{2}\right)$$

$$\Rightarrow \cot\left(\frac{\theta+\phi}{2}\right) = a \Rightarrow \theta - \phi = 2 \cot^{-1} a$$

$$\Rightarrow \sin^{-1} x^n - \sin^{-1} y^n = 2 \cot^{-1} a \quad \{\text{from (2)}\}$$

Differentiating both sides w.r.t. x , we get

$$\frac{1}{\sqrt{1-x^{2n}}} \cdot nx^{n-1} - \frac{1}{\sqrt{1-y^{2n}}} \cdot ny^{n-1} \frac{dy}{dx} = 0$$

$$\Rightarrow \frac{dy}{dx} = \frac{x^{n-1}}{y^{n-1}} \sqrt{\frac{1-y^{2n}}{1-x^{2n}}}$$



Example – 16

If $y = \frac{\sqrt{a^2 + x^2} + \sqrt{a^2 - x^2}}{\sqrt{a^2 + x^2} - \sqrt{a^2 - x^2}}$, show that

$$\frac{dy}{dx} = \frac{-2a^2}{x^3} \left\{ 1 + \frac{a^2}{\sqrt{a^4 - x^4}} \right\}.$$

Sol. We have,

$$\begin{aligned} y &= \frac{\sqrt{a^2 + x^2} + \sqrt{a^2 - x^2}}{\sqrt{a^2 + x^2} - \sqrt{a^2 - x^2}} \\ &= \frac{\sqrt{a^2 + x^2} + \sqrt{a^2 - x^2}}{\sqrt{a^2 + x^2} - \sqrt{a^2 - x^2}} \cdot \frac{\sqrt{a^2 + x^2} + \sqrt{a^2 - x^2}}{\sqrt{a^2 + x^2} + \sqrt{a^2 - x^2}} \\ \Rightarrow y &= \frac{[\sqrt{a^2 + x^2} + \sqrt{a^2 - x^2}]^2}{(a^2 + x^2) - (a^2 - x^2)} \\ &= \frac{a^2 + x^2 + a^2 - x^2 + 2\sqrt{a^2 + x^2}\sqrt{a^2 - x^2}}{2x^2} \\ \Rightarrow y &= \frac{2a^2 + 2\sqrt{a^4 - x^4}}{2x^2} \\ \Rightarrow y &= \frac{a^2}{x^2} + \frac{\sqrt{a^4 - x^4}}{x^2} \\ \Rightarrow y &= a^2 x^{-2} + \sqrt{a^4 - x^4} x^{-2} \\ \Rightarrow \frac{dy}{dx} &= a^2 \frac{d}{dx}(x^{-2}) + \frac{d}{dx} \left\{ \sqrt{a^4 - x^4} x^{-2} \right\} \\ \Rightarrow \frac{dy}{dx} &= -2a^2 x^{-3} + (-2)x^{-3} \sqrt{a^4 - x^4} \\ &\quad + (x^{-2}) \frac{1}{2} (a^4 - x^4)^{-1/2} \frac{d}{dx} (a^4 - x^4) \\ \Rightarrow \frac{dy}{dx} &= \frac{-2a^2}{x^3} - \frac{2}{x^4} \sqrt{a^4 - x^4} + \frac{1}{2x^2 \sqrt{a^4 - x^4}} (-4x^3) \\ \Rightarrow \frac{dy}{dx} &= \frac{-2a^2}{x^3} - \frac{2}{x^3} \sqrt{a^4 - x^4} - \frac{2x}{\sqrt{a^4 - x^4}} \\ \Rightarrow \frac{dy}{dx} &= \frac{-2a^2}{x^3} - 2 \left\{ \frac{\sqrt{a^4 - x^4}}{x^3} + \frac{x}{\sqrt{a^4 - x^4}} \right\} \\ \Rightarrow \frac{dy}{dx} &= \frac{-2a^2}{x^3} - 2 \left\{ \frac{a^4 - x^4 + x^4}{x^3 \sqrt{a^4 - x^4}} \right\} \end{aligned}$$

$$\Rightarrow \frac{dy}{dx} = \frac{-2a^2}{x^3} - \frac{2a^4}{x^3 \sqrt{a^4 - x^4}} = \frac{-2a^2}{x^3} \left\{ 1 + \frac{a^2}{\sqrt{a^4 - x^4}} \right\}$$

Example – 17

If $x\sqrt{1+y} + y\sqrt{1+x} = 0$, prove that $\frac{dy}{dx} = -\frac{1}{(x+1)^2}$

Sol. We have,

$$\begin{aligned} x\sqrt{1+y} + y\sqrt{1+x} &= 0 \\ \Rightarrow x\sqrt{1+y} &= -y\sqrt{1+x} \\ \Rightarrow x^2(1+y) &= y^2(1+x) \quad [\text{On squaring both sides}] \\ \Rightarrow x^2 - y^2 &= y^2 x - x^2 y \\ \Rightarrow (x+y)(x-y) &= -xy(x-y) \\ \Rightarrow x+y &= -xy \quad [\because x-y \neq 0, \text{ as } y=x \text{ does not satisfy the given equation}] \\ \Rightarrow x &= -y - xy \\ \Rightarrow y(1+x) &= -x \\ \Rightarrow y &= -\frac{x}{1+x} \\ \Rightarrow \frac{dy}{dx} &= -\left\{ \frac{(1+x) \cdot 1 - x(0+1)}{(1+x)^2} \right\} \\ \Rightarrow \frac{dy}{dx} &= -\frac{1}{(1+x)^2} \end{aligned}$$

Example – 18

If $y = a^{x^a x^{\infty}}$, prove that $\frac{dy}{dx} = \frac{y^2 \log y}{x(1 - y \log x \cdot \log y)}$.

Sol. The given series may be written as

$$\begin{aligned} y &= a^{(x^y)} \\ \Rightarrow \log y &= x^y \log a \quad [\text{Taking log of both sides}] \\ \Rightarrow \log(\log y) &= y \log x + \log(\log a) \\ \Rightarrow \frac{1}{\log y} \frac{d}{dx}(\log y) &= \frac{dy}{dx} \cdot \log x + y \cdot \frac{d}{dx}(\log x) + 0 \\ &[\text{Differentiating both sides with respect to } x] \\ \Rightarrow \frac{1}{\log y} \cdot \frac{1}{y} \frac{dy}{dx} &= \frac{dy}{dx} \cdot \log x + y \cdot \frac{1}{x} \end{aligned}$$



$$\Rightarrow \frac{dy}{dx} \left\{ \frac{1}{y \log y} - \log x \right\} = \frac{y}{x}$$

$$\Rightarrow \frac{dy}{dx} \left\{ \frac{1 - y \log y \cdot \log x}{y \log y} \right\} = \frac{y}{x}$$

$$\Rightarrow \frac{dy}{dx} = \frac{y^2 \log y}{x \{1 - y \log y \cdot \log x\}}$$

Example – 19

If $y = e^{x+e^x+e^{x+\dots} \text{ to } \infty}$, show that $\frac{dy}{dx} = \frac{y}{1-y}$

Sol. The given function may be written as

$$y = e^{x+y}$$

$$\Rightarrow \log y = (x+y) \cdot \log e \text{ [Taking log of both sides]}$$

$$\Rightarrow \log y = x + y [\because \log e = 1]$$

$$\Rightarrow \frac{1}{y} \frac{dy}{dx} = 1 + \frac{dy}{dx} \text{ [Differentiating with respect to } x]$$

$$\Rightarrow \frac{dy}{dx} \left(\frac{1}{y} - 1 \right) = 1 \Rightarrow \frac{dy}{dx} = \frac{y}{1-y}$$

Example – 20

Show that the function $f(x) = \begin{cases} x-1, & \text{if } x < 2 \\ 2x-3, & \text{if } x \geq 2 \end{cases}$ is not differentiable at $x=2$.

Sol. We have,

$$(\text{LHD at } x=2) = \lim_{x \rightarrow 2^-} \frac{f(x) - f(2)}{x - 2}$$

$$\Rightarrow (\text{LHD at } x=2) = \lim_{x \rightarrow 2^-} \frac{(x-1) - (4-3)}{x-2}$$

$[\because f(x) = x-1 \text{ for } x < 2]$

$$\Rightarrow (\text{LHD at } x=2) = \lim_{x \rightarrow 2^-} \frac{x-2}{x-2} = \lim_{x \rightarrow 2^-} 1 = 1$$

$$\text{and, } (\text{RHD at } x=2) = \lim_{x \rightarrow 2^+} \frac{f(x) - f(2)}{x - 2}$$

$$\Rightarrow (\text{RHD at } x=2) = \lim_{x \rightarrow 2^+} \frac{(2x-3) - (4-3)}{x-2}$$

$$[\because f(x) = 2x-3 \text{ for } x \geq 2]$$

$$\Rightarrow (\text{RHD at } x=2) = \lim_{x \rightarrow 2^+} \frac{2x-4}{x-2} = \lim_{x \rightarrow 2^+} 2 = 2$$

$$\Rightarrow (\text{LHD at } x=2) \neq (\text{RHD at } x=2).$$

So, $f(x)$ is not differentiable at $x=2$.

Example – 21

Discuss the differentiability of

$$f(x) = \begin{cases} xe^{-\left(\frac{1}{x} + \frac{1}{x}\right)} & , x \neq 0 \\ 0 & , x = 0 \end{cases} \text{ at } x=0.$$

Sol. We have,

$$f(x) = \begin{cases} xe^{-\left(\frac{1}{x} + \frac{1}{x}\right)} = xe^{-2/x}, & x \geq 0 \\ xe^{-\left(\frac{-1}{x} + \frac{1}{x}\right)} = x & , x < 0 \\ 0 & , x = 0 \end{cases}$$

Now,

$$(\text{LHD at } x=0) = \lim_{x \rightarrow 0^-} \frac{f(x) - f(0)}{x - 0}$$

$$\Rightarrow (\text{LHD at } x=0) = \lim_{x \rightarrow 0^-} \frac{x-0}{x-0} = 1$$

$$[\because f(x) = x \text{ for } x < 0 \text{ and } f(0) = 0]$$

and,

$$(\text{RHD at } x=0) = \lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x - 0}$$

$$\Rightarrow (\text{RHD at } x=0) = \lim_{x \rightarrow 0^+} \frac{xe^{-2/x} - 0}{x}$$

$$\left[\because f(x) = xe^{-2/x} \text{ for } x > 0 \text{ and } f(0) = 0 \right]$$

$$\Rightarrow (\text{RHD at } x=0) = \lim_{x \rightarrow 0^+} e^{-2/x} = 0.$$

$$\frac{-2}{x} \text{ tends to } -\infty \text{ and } e^{-\infty} = 0$$

$$\Rightarrow (\text{LHD at } x=0) \neq (\text{RHD at } x=0)$$

So, $f(x)$ is not differentiable at $x=0$.



Example – 22

Show that the function $f(x)$ is continuous at $x = 0$ but its derivative does not exist at $x = 0$

$$\text{if } f(x) = \begin{cases} x \sin(\log x^2); & x \neq 0 \\ 0 & ; x = 0 \end{cases}$$

Sol. $\text{LHL} = \lim_{h \rightarrow 0} f(0-h) = \lim_{h \rightarrow 0} f(-h) \sin \log(-h)^2$

$$= -\lim_{h \rightarrow 0} h \sin \log h^2$$

As $h \rightarrow 0$, $\log h^2 \rightarrow -\infty$.

Hence $\sin \log h^2$ oscillates between -1 and $+1$.

$$\text{L.H.L} = -\lim_{h \rightarrow 0} (h) \times \lim_{h \rightarrow 0} (\sin \log h^2)$$

$$= -0 \times (\text{number between } -1 \text{ and } +1) = 0$$

$$\text{R.H.L} = \lim_{h \rightarrow 0} f(0+h)$$

$$= \lim_{h \rightarrow 0} h \sin \log h^2 = \lim_{h \rightarrow 0} h \lim_{h \rightarrow 0} \sin \log h^2$$

$$= 0 \times (\text{oscillating between } -1 \text{ and } +1) = 0$$

$$f(0) = 0 \quad (\text{Given})$$

$$\Rightarrow \text{L.H.L.} = \text{R.H.L.} = f(0)$$

Hence $f(x)$ is continuous at $x = 0$.

Test for differentiability :

$$\text{Lf}'(0) = \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{-h \sin \log(-h)^2 - 0}{-h}$$

$$= \lim_{h \rightarrow 0} \sin(\log h^2)$$

As the expression oscillates between -1 and $+1$, the limit does not exist.

$\Rightarrow$ Left hand derivative is not defined.

Hence the function is not differentiable at $x = 0$

Example – 23

Find the value of the constant λ so that the function given below is continuous at $x = -1$.

$$f(x) = \begin{cases} \frac{x^2 - 2x - 3}{x + 1}, & x \neq -1 \\ \lambda & , x = -1 \end{cases}$$

Sol. Since $f(x)$ is continuous at $x = -1$. Therefore,

$$\lim_{x \rightarrow -1} f(x) = f(-1)$$

$$\Rightarrow \lim_{x \rightarrow -1} \frac{x^2 - 2x - 3}{x + 1} = \lambda \quad [\because f(-1) = \lambda]$$

$$\Rightarrow \lim_{x \rightarrow -1} \frac{(x-3)(x+1)}{x+1} = \lambda \Rightarrow \lim_{x \rightarrow -1} (x-3) = \lambda \Rightarrow -4 = \lambda$$

So, $f(x)$ is continuous at $x = -1$, if $\lambda = -4$.

Example – 24

If the function $f(x)$ defined by

$$f(x) = \begin{cases} \frac{\log(1+ax) - \log(1-bx)}{x}, & \text{if } x \neq 0 \\ k & , \text{if } x = 0 \end{cases}$$

is continuous at $x = 0$, find k .

Sol. Since $f(x)$ is continuous at $x = 0$. Therefore,

$$\lim_{x \rightarrow 0} f(x) = f(0)$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{\log(1+ax) - \log(1-bx)}{x} = k \quad [\because f(0) = k]$$

$$\Rightarrow \lim_{x \rightarrow 0} \left[\frac{\log(1+ax)}{x} - \frac{\log(1-bx)}{x} \right] = k$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{\log(1+ax)}{x} - \lim_{x \rightarrow 0} \frac{\log(1-bx)}{x} = k$$

$$\Rightarrow a \lim_{x \rightarrow 0} \frac{\log(1+ax)}{ax} - (-b) \lim_{x \rightarrow 0} \frac{\log(1-bx)}{(-b)x} = k$$

$$\Rightarrow a(1) - (-b)(1) = k$$

$$\left[\text{Using : } \lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1 \right]$$

$$\Rightarrow a + b = k$$

Thus, $f(x)$ is continuous at $x = 0$, if $k = a + b$.



Example – 25

$$\text{Let } f(x) = \begin{cases} \frac{1 - \cos 4x}{x^2} & , \text{ if } x < 0 \\ a & , \text{ if } x = 0 \\ \frac{\sqrt{x}}{\sqrt{16 + \sqrt{x}} - 4} & , \text{ if } x > 0 \end{cases}$$

Determine the value of a so that $f(x)$ is continuous at $x = 0$.

Sol. For $f(x)$ to be continuous at $x = 0$, we must have

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x) = f(0)$$

$$\Rightarrow \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x) = a \quad \dots (i)$$

$$\text{Now, } \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0} \frac{1 - \cos 4x}{x^2}$$

$$\left[\because f(x) = \frac{1 - \cos 4x}{x^2} \text{ for } x < 0 \right]$$

$$\Rightarrow \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0} \frac{2 \sin^2 2x}{x^2}$$

$$\Rightarrow \lim_{x \rightarrow 0^-} f(x) = 2 \lim_{x \rightarrow 0} \left(\frac{\sin 2x}{x} \right)^2$$

$$\Rightarrow \lim_{x \rightarrow 0^-} f(x) = 2 \times 4 \cdot \lim_{x \rightarrow 0} \left(\frac{\sin 2x}{2x} \right)^2 = 8(1)^2 = 8 \quad \dots (ii)$$

$$\text{and, } \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0} \frac{\sqrt{x}}{\sqrt{16 + \sqrt{x}} - 4}$$

$$\left[\because f(x) = \frac{\sqrt{x}}{\sqrt{16 + \sqrt{x}} - 4} \text{ for } x > 0 \right]$$

$$\Rightarrow \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0} \frac{\sqrt{x}}{16 + \sqrt{x} - 16} \cdot (\sqrt{16 + \sqrt{x}} + 4)$$

Rationalizing Denominator

$$\Rightarrow \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0} (\sqrt{16 + \sqrt{x}} + 4) = 4 + 4 = 8 \quad \dots (iii)$$

From (i), (ii) and (iii), we get $a = 8$.

Example – 26

Determine the value of the constant m so that the function

$$f(x) = \begin{cases} m(x^2 - 2x), & \text{if } x < 0 \\ \cos x, & \text{if } x \geq 0 \end{cases} \text{ is continuous}$$

Sol. When $x < 0$, we have

$f(x) = m(x^2 - 2x)$, which being a polynomial is continuous at each $x < 0$.

When $x > 0$, we have

$f(x) = \cos x$, which being a cosine function is continuous at each $x > 0$.

So, consider the point $x = 0$.

We have,

$$(\text{LHL at } x = 0) = \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0} (x^2 - 2x) = 0, \text{ for all values of } m$$

$$\text{and } (\text{RHL at } x = 0) = \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0} \cos x = 1$$

Clearly, $\lim_{x \rightarrow 0^-} f(x) \neq \lim_{x \rightarrow 0^+} f(x)$ for all values of m .

So, $f(x)$ cannot be made continuous for any value of m .

In other words, the value of m does not exist for which $f(x)$ can be made continuous.

Example – 27

$$\text{If } f(x) = \begin{cases} 1 & , \text{ if } x \leq 3 \\ ax + b & , \text{ if } 3 < x < 5 \\ 7 & , \text{ if } 5 \leq x \end{cases}$$

Determine the values of a and b so that $f(x)$ is continuous.

Sol. The given function is a constant function for all $x < 3$ and for all $x > 5$ so it is continuous for all $x < 3$ and for all $x > 5$. We know that a polynomial function is continuous. So, the given function is continuous for all $x \in (3, 5)$. Thus, $f(x)$ is continuous at each $x \in \mathbb{R}$ except possibly at $x = 3$ and $x = 5$.

At, $x = 3$, we have

$$\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3} 1 = 1, \lim_{x \rightarrow 3^+} f(x) = \lim_{x \rightarrow 3} ax + b = 3a + b, \text{ and, } f(3) = 1$$

For $f(x)$ be continuous at $x = 3$, we must have

$$\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^+} f(x) = f(3)$$

$$\Rightarrow 1 = 3a + b \quad \dots (i)$$



At $x = 5$, we have

$$\lim_{x \rightarrow 5^-} f(x) = \lim_{x \rightarrow 5^-} ax + b = 5a + b;$$

$$\lim_{x \rightarrow 5^+} f(x) = \lim_{x \rightarrow 5^+} 7 = 7, \text{ and, } f(5) = 7$$

For $f(x)$ to be continuous at $x = 5$, we must have

$$\lim_{x \rightarrow 5^-} f(x) = \lim_{x \rightarrow 5^+} f(x) = f(5)$$

$$\Rightarrow 5a + b = 7 \quad \dots\dots (ii)$$

Solving (i) and (ii), we get $a = 3$, $b = -8$

Example – 28

Find the values of a & b so that the function is continuous for $0 \leq x \leq \pi$

$$f(x) = \begin{cases} x + a\sqrt{2} \sin x, & 0 \leq x < \frac{\pi}{4} \\ 2x \cot x + b, & \frac{\pi}{4} \leq x \leq \frac{\pi}{2} \\ a \cos 2x - b \sin x, & \frac{\pi}{2} < x \leq \pi \end{cases}$$

Sol. Since, $f(x)$ is continuous for $0 \leq x \leq \pi$

$$\therefore \text{RHL} \left(\text{at } x = \frac{\pi}{4} \right) = \text{LHL} \left(\text{at } x = \frac{\pi}{4} \right)$$

$$\Rightarrow \left(2 \cdot \frac{\pi}{4} \cot \frac{\pi}{4} + b \right) = \left(\frac{\pi}{4} + a\sqrt{2} \cdot \sin \frac{\pi}{4} \right)$$

$$\Rightarrow \frac{\pi}{2} + b = \frac{\pi}{4} + a$$

$$\Rightarrow a - b = \frac{\pi}{4} \quad \dots(i)$$

$$\text{Also, } \text{RHL} \left(\text{at } x = \frac{\pi}{2} \right) = \text{LHL} \left(\text{at } x = \frac{\pi}{2} \right)$$

$$\Rightarrow \left(a \cos \frac{2\pi}{2} - b \sin \frac{\pi}{2} \right) = \left(2 \cdot \frac{\pi}{2} \cdot \cot \frac{\pi}{2} + b \right)$$

$$\Rightarrow -a - b = b$$

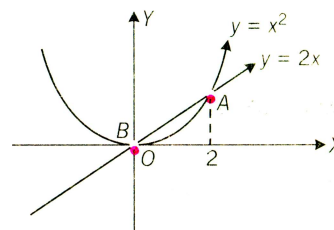
$$\Rightarrow a + 2b = 0 \quad \dots(ii)$$

$$\text{From eqs. (i) and (ii), } a = \frac{\pi}{6} \text{ and } b = -\frac{\pi}{12}$$

Example – 29

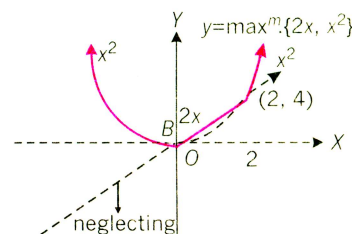
Draw graph for $y = \max. \{2x, x^2\}$ and discuss the continuity and differentiability.

Sol. Here, to draw, $y = \max. \{2x, x^2\}$



Firstly plot $y = 2x$ and $y = x^2$ on graph and put $2x = x^2 \Rightarrow x = 0, 2$ (i.e., their point of intersection).

Now, since $y = \max. \{2x, x^2\}$ we have to neglect the curve below point of intersections thus, the required graph is, as shown.



Thus, from the given graph $y = \max. \{2x, x^2\}$ we can say $y = \max. \{2x, x^2\}$ is continuous for all $x \in \mathbb{R}$.

But $y = \max. \{2x, x^2\}$ is differentiable for all $x \in \mathbb{R} - \{0, 2\}$

NOTES :

One must remember the formula we can write,

$$\max. \{f(x), g(x)\} = \frac{f(x) + g(x)}{2} + \left| \frac{f(x) - g(x)}{2} \right|$$

$$\min. \{f(x), g(x)\} = \frac{f(x) + g(x)}{2} - \left| \frac{f(x) - g(x)}{2} \right|$$

Example – 30

Differentiate the following functions with respect to x :

$$(i) x^{x^x} \quad (ii) (x^x)^x$$

Sol. (i) Let $y = x^{x^x}$. Then,

$$y = e^{x^x \cdot \log x}$$



On differentiating both sides with respect to x , we get

$$\frac{dy}{dx} = e^{x^x \cdot \log x} \cdot \frac{d}{dx}(x^x \cdot \log x)$$

$$\Rightarrow \frac{dy}{dx} = x^{x^x} \cdot \frac{d}{dx}(e^{x \log x} \cdot \log x)$$

$$\Rightarrow \frac{dy}{dx} = x^{x^x} \left\{ \log x \cdot \frac{d}{dx}(e^{x \log x}) + e^{x \log x} \cdot \frac{d}{dx}(\log x) \right\}$$

$$\Rightarrow \frac{dy}{dx} = x^{x^x} \left\{ \log x \cdot e^x \log x \cdot \frac{d}{dx}(x \log x) + e^{x \log x} \cdot \frac{1}{x} \right\}$$

$$\Rightarrow \frac{dy}{dx} = x^{x^x} \left\{ \log x \cdot x^x \left(x \cdot \frac{1}{x} + \log x \right) + x^x \cdot \frac{1}{x} \right\}$$

$$\Rightarrow \frac{dy}{dt} = x^{x^x} \left\{ x^x (1 + \log x) \cdot \log x + \frac{x^x}{x} \right\}$$

$$\Rightarrow \frac{dy}{dt} = x^{x^x} \cdot x^x \left\{ (1 + \log x) \cdot \log x + \frac{1}{x} \right\}$$

(ii) Let $y = (x^x)^x$. Then,

$$y = x^{x \cdot x} = x^{x^2}$$

$$\log y = x^2 \log x$$

$$\Rightarrow y = e^{x^2 \cdot \log x}$$

On differentiating both sides with respect to x , we get

$$\frac{dy}{dx} = e^{x^2 \cdot \log x} \cdot \frac{d}{dx}(x^2 \cdot \log x)$$

$$\Rightarrow \frac{dy}{dx} = e^{x^2 \cdot \log x} \left(\log x \cdot \frac{d}{dx}(x^2) + x^2 \cdot \frac{d}{dx}(\log x) \right)$$

$$\Rightarrow \frac{dy}{dx} = x^{x^2} \left(\log x \cdot 2x + x^2 \cdot \frac{1}{x} \right)$$

$$\left[\because e^{x^2 \cdot \log x} = x^{x^2} \right]$$

$$\Rightarrow \frac{dy}{dx} = x^{x^2} (2x \cdot \log x + x)$$

$$\Rightarrow \frac{dy}{dx} = x \cdot x^{x^2} (2 \log x + 1).$$

Example – 31

Differentiate : $(\log x)^x + x^{\log x}$ with respect to x .

Sol. Let $y = (\log x)^x + x^{\log x}$. Then,

$$y = e^{\log(\log x)^x} + e^{\log(x^{\log x})}$$

$$\Rightarrow y = e^{x \log(\log x)} + e^{x \log x \cdot \log x}$$

On differentiating both sides with respect to x , we get

$$\frac{dy}{dx} = e^{x \log(\log x)} \cdot \frac{d}{dx}\{x \log(\log x)\} + e^{(\log x)^2} \cdot \frac{d}{dx}(\log x)^2$$

$$\Rightarrow \frac{dy}{dx} = (\log x)^x \left\{ \log(\log x) \cdot \frac{d}{dx}(x) + x \cdot \frac{d}{dx}(\log(\log x)) \right\} + x^{\log x}$$

$$\left\{ 2(\log x) \cdot \frac{d}{dx}(\log x) \right\}$$

$$\Rightarrow \frac{dy}{dx} = (\log x)^x \left\{ \log(\log x) + x \cdot \frac{1}{\log x} \cdot \frac{1}{x} \right\} + x^{\log x} \left\{ 2 \log x \cdot \frac{1}{x} \right\}$$

$$\Rightarrow \frac{dy}{dx} = (\log x)^x \left\{ \log(\log x) + \frac{1}{\log x} \right\} + x^{\log x} \left\{ \frac{2 \log x}{x} \right\}.$$

Example – 32

Differentiate the following functions with respect to x :

$$x^{\cot x} + \frac{2x^2 - 3}{x^2 + x + 2}$$

Sol. Let $y = x^{\cot x} + \frac{2x^2 - 3}{x^2 + x + 2}$. Then,

$$y = e^{\cot x \cdot \log x} + \frac{2x^2 - 3}{x^2 + x + 2}$$

$$\left[\because x^{\cot x} = e^{\log x^{\cot x}} = e^{\cot x \cdot \log x} \right]$$

On differentiating both sides with respect to x , we get



$$\frac{dy}{dx} = \frac{d}{dx} (e^{\cot x \cdot \log x}) + \frac{d}{dx} \left(\frac{2x^2 - 3}{x^2 + x + 2} \right)$$

$$\Rightarrow -\frac{1}{2} \tan \frac{x}{2} - \frac{1}{4} \tan \frac{x}{4} - \frac{1}{8} \tan \frac{x}{8} \dots = \cot x - \frac{1}{x}$$

$$\frac{dy}{dx} = e^{\cot x \cdot \log x} \frac{d}{dx} (\cot x \cdot \log x)$$

Differentiating both sides with respect to x , we get

$$-\frac{1}{2^2} \sec^2 \frac{x}{2} - \frac{1}{4^2} \sec^2 \frac{x}{4} - \frac{1}{8^2} \sec^2 \frac{x}{8} \dots = -\operatorname{cosec}^2 x + \frac{1}{x^2}$$

$$+ \frac{(x^2 + x + 2) \frac{d}{dx} (2x^2 - 3) - (2x^2 - 3) \frac{d}{dx} (x^2 + x + 2)}{(x^2 + x + 2)^2}$$

$$\frac{1}{2^2} \sec^2 \frac{x}{2} + \frac{1}{4^2} \sec^2 \frac{x}{4} + \frac{1}{8^2} \sec^2 \frac{x}{8} \dots = \operatorname{cosec}^2 x - \frac{1}{x^2}$$

$$\frac{dy}{dx} = x^{\cot x} \left\{ \log x \cdot \frac{d}{dx} (\cot x) + \cot x \cdot \frac{d}{dx} (\log x) \right\}$$

Example – 34

$$+ \frac{(x^2 + x + 2)(4x) - (2x^2 - 3)(2x + 1)}{(x^2 + x + 2)^2}$$

If $y = f\left(\frac{2x-1}{x^2+1}\right)$ and $f'(x) = \sin x^2$, find $\frac{dy}{dx}$.

$$\frac{dy}{dx} = x^{\cot x} \left\{ -\operatorname{cosec}^2 x \cdot \log x + \frac{\cot x}{x} \right\} + \frac{2x^2 + 14x + 3}{(x^2 + x + 2)^2}$$

Sol. Let $z = \frac{2x-1}{x^2+1}$. Then,

$$y = f(z)$$

$$\Rightarrow \frac{dy}{dx} = \frac{d}{dx} \{f(z)\} = \frac{d}{dz} \{f(z)\} \cdot \frac{dz}{dx}$$

$$\Rightarrow \frac{dy}{dx} = f'(z) \frac{d}{dx} \left(\frac{2x-1}{x^2+1} \right)$$

$$\Rightarrow \frac{dy}{dx} = f'(z) \left\{ \frac{2(x^2+1) - (2x-1)2x}{(x^2+1)^2} \right\}$$

$$\Rightarrow \frac{dy}{dx} = \sin z^2 \frac{2(x^2+1) - (4x^2-2x)}{(x^2+1)^2}$$

$$\left[\begin{array}{l} \because f'(x) = \sin x^2 \\ \therefore f'(z) = \sin z^2 \end{array} \right]$$

$$\Rightarrow \frac{dy}{dx} = 2 \sin \left(\frac{2x-1}{x^2+1} \right) \left\{ \frac{1+x-x^2}{(x^2+1)^2} \right\}$$

Example – 33

Given that $\cos \frac{x}{2} \cdot \cos \frac{x}{4} \cdot \cos \frac{x}{8} \dots = \frac{\sin x}{x}$, prove that

$$\frac{1}{2^2} \sec^2 \frac{x}{2} + \frac{1}{4^2} \sec^2 \frac{x}{4} + \dots = \operatorname{cosec}^2 x - \frac{1}{x^2}$$

Sol. We have,

$$\cos \frac{x}{2} \cdot \cos \frac{x}{4} \cdot \cos \frac{x}{8} \dots = \frac{\sin x}{x}$$

Taking log on both sides, we get

$$\log \cos \frac{x}{2} + \log \cos \frac{x}{4} + \log \cos \frac{x}{8} \dots = \log \sin x - \log x$$

Differentiating both sides with respect to x , we get

$$-\frac{1}{2} \frac{\sin \frac{x}{2}}{\cos \frac{x}{2}} - \frac{1}{4} \frac{\sin \frac{x}{4}}{\cos \frac{x}{4}} - \frac{1}{8} \frac{\sin \frac{x}{8}}{\cos \frac{x}{8}} \dots = \frac{\cos x}{\sin x} - \frac{1}{x}$$



Example – 35

If $x = \sqrt{a^{\sin^{-1} t}}$, $y = \sqrt{a^{\cos^{-1} t}}$, $a > 0$ and $-1 < t < 1$, show

that $\frac{dy}{dx} = -\frac{y}{x}$

Sol. We have,

$$x = \sqrt{a^{\sin^{-1} t}} \text{ and } y = \sqrt{a^{\cos^{-1} t}}$$

$$\Rightarrow \frac{dx}{dt} = \frac{1}{2} \left(a^{\sin^{-1} t} \right)^{-1/2} \frac{d}{dt} \left(a^{\sin^{-1} t} \right) \text{ and } \frac{dy}{dx} = \frac{1}{2} \left(a^{\cos^{-1} t} \right)^{-1/2} \frac{d}{dt} \left(a^{\cos^{-1} t} \right)$$

$$\Rightarrow \frac{dx}{dt} = \frac{1}{2} \left(a^{\sin^{-1} t} \right)^{-1/2} \left(a^{\sin^{-1} t} \log_e a \right) \cdot \frac{d}{dt} (\sin^{-1} t) \text{ and ,}$$

$$\frac{dy}{dt} = \frac{1}{2} \left(a^{\cos^{-1} t} \right)^{-1/2} \left(a^{\cos^{-1} t} \log_e a \right) \cdot \frac{d}{dt} (\cos^{-1} t)$$

$$\Rightarrow \frac{dx}{dt} = \frac{1}{2} \left(a^{\sin^{-1} t} \right)^{1/2} (\log_e a) \times \frac{1}{\sqrt{1-t^2}} = \frac{x \log_e a}{2\sqrt{1-t^2}} \text{ and ,}$$

$$\frac{dy}{dt} = \frac{1}{2} \left(a^{\cos^{-1} t} \right)^{1/2} (\log_e a) \times \frac{-1}{\sqrt{1-t^2}} = \frac{-y \log_e a}{2\sqrt{1-t^2}}$$

$$\therefore \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$

$$\Rightarrow \frac{dy}{dx} = \frac{-y \log_e a}{2\sqrt{1-t^2}} \times \frac{2\sqrt{1-t^2}}{x \log_e a} = \frac{-y}{x}$$



EXERCISE - 1 : BASIC OBJECTIVE QUESTIONS

Continuity of the function

1. The function $f(x) = \begin{cases} \frac{1}{4^x - 1} & x \neq 0 \\ 0 & x = 0 \end{cases}$ is continuous

- (a) everywhere except at $x = 0$ and $x = 1$
- (b) nowhere
- (c) everywhere
- (d) everywhere except at $x = 0$

2. The function $f(x) = (1+x)^{\cot x}$ is not defined at $x = 0$. The value of $f(0)$ so that $f(x)$ becomes continuous at $x = 0$ is

- (a) 1
- (b) 0
- (c) e
- (d) none of these

3. If $f(x) = \begin{cases} x, & \text{when } x \text{ is rational} \\ 1-x, & \text{when } x \text{ is irrational, then} \end{cases}$

- (a) $f(x)$ is continuous for all real x
- (b) $f(x)$ is discontinuous for all real x
- (c) $f(x)$ is continuous only at $x = 1/2$
- (d) $f(x)$ is discontinuous only at $x = 1/2$

Differentiability of the function

4. Let $f(x) = [n + p \sin x]$, $x \in (0, \pi)$, $n \in \mathbb{I}$, p is a prime number and $[x]$ denotes the greatest integer less than or equal to x . The number of points at which $f(x)$ is not differentiable is

- (a) $p - 1$
- (b) p
- (c) $2p + 1$
- (d) $2p - 1$

5. The set of points where the function

$$f(x) = [x] + |1 - x|, -1 \leq x \leq 3$$

where $[.]$ denotes the greatest integer function, is not differentiable, is

- (a) $\{-1, 0, 1, 2, 3\}$
- (b) $\{-1, 0, 2\}$
- (c) $\{0, 1, 2, 3\}$
- (d) $\{-1, 0, 1, 2\}$

6. The set of points where the function $f(x) = |x-1| e^x$ is differentiable is

- (a) $\mathbb{R}$
- (b) $\mathbb{R} - \{1\}$
- (c) $\mathbb{R} - \{-1\}$
- (d) $\mathbb{R} - \{0\}$

7. The function $f(x) = \text{maximum} \left\{ \sqrt{x(2-x)}, 2-x \right\}$ is non-differentiable at x equal to :

- (a) 1
- (b) 0, 2
- (c) 0, 1
- (d) 1, 2

Continuity/ Differentiability of the function

8. The function $f(x) = \sin^{-1}(\cos x)$ is

- (a) discontinuous at $x = 0$
- (b) continuous at $x = 0$
- (c) differentiable at $x = 0$
- (d) None of these

9. If $f(x) = \begin{cases} \frac{|x+2|}{\tan^{-1}(x+2)} & x \neq -2 \\ 2 & x = -2 \end{cases}$ then $f(x)$ is

- (a) continuous at $x = -2$
- (b) not continuous at $x = -2$
- (c) differentiable at $x = -2$
- (d) continuous but not diff. at $x = -2$

10. If $f(x) = \begin{cases} \frac{x(3e^{1/x} + 4)}{2 - e^{1/x}} & x \neq 0 \\ 0 & x = 0 \end{cases}$ then

- (a) $\lim_{x \rightarrow 0} f(x) = 1$
- (b) $f(x)$ is continuous at $x = 0$
- (c) $f(x)$ is differentiable at $x = 0$
- (d) None of these

11. If $f(x) = x + \frac{x}{1+x} + \frac{x}{(1+x)^2} + \dots$ to ∞ , then at $x = 0$, $f(x)$

- (a) $\lim_{x \rightarrow 0} f(x)$ does not exist
- (b) is discontinuous
- (c) is continuous but not differentiable
- (d) is differentiable



12. If $f(x) = x^2|x|\operatorname{sgn} x$, then
 (a) f is derivable at $x = 0$
 (b) f is continuous but not derivable at $x = 0$
 (c) LHD at $x = 0$ is 1
 (d) RHD at $x = 0$ is 1
13. The function $f(x) = 1 + |\sin x|$ is
 (a) continuous no where
 (b) continuous every where and no differentiable at $x = 0$
 (c) differentiable no where
 (d) differentiable at $x = 0$

14. For the function $f(x) = \begin{cases} |x-3|, & x \geq 1 \\ \frac{x^2}{4} - \frac{3x}{2} + \frac{13}{4}, & x < 1 \end{cases}$ which of the following is incorrect?
 (a) continuous at $x = 1$ (b) continuous at $x = 3$
 (c) derivable at $x = 1$ (d) derivable at $x = 3$

15. If $f(x) = \begin{cases} 3^x, & -1 \leq x \leq 1 \\ 4-x, & 1 < x \leq 4 \end{cases}$, then $f(x)$ is
 (a) continuous as well as differentiable at $x = 1$
 (b) continuous but not differentiable at $x = 1$
 (c) differentiable but not continuous at $x = 1$
 (d) none of the above

16. A function is defined as follows :

$$f(x) = \begin{cases} x^3; & x^2 < 1 \\ x; & x^2 \geq 1 \end{cases} \text{ The function is}$$

- (a) discontinuous at $x = 1$
 (b) differentiable at $x = 1$
 (c) continuous but not differentiable at $x = 1$
 (d) none of these

17. If $f(x) = \begin{cases} 4, & -3 < x < -1 \\ 5+x, & -1 \leq x < 0 \\ 5-x, & 0 \leq x < 2 \\ x^2+x-3, & 2 \leq x < 3 \end{cases}$, then $f(x)$ is

- (a) differentiable but not continuous in $(-3, 3)$
 (b) continuous but not differentiable in $(-3, 3)$
 (c) continuous as well as differentiable in $(-3, 3)$
 (d) neither continuous nor differentiable in $(-3, 3)$

Defferentiation of logarithmic functions

18. Derivative of $x^6 + 6^x$ with respect to x is
 (a) $12x$ (b) $x+4$
 (c) $6x^5 + 6^x \log 6$ (d) $6x^5 + x6^{x-1}$
19. If $x^y = e^{x-y}$, then $\frac{dy}{dx}$ is equal to
 (a) $(1 + \log x)^{-1}$ (b) $(1 + \log x)^{-2}$
 (c) $\log x \cdot (1 + \log x)^{-2}$ (d) None of these
20. If $y = \log_a x + \log_x a + \log_x x + \log_a a$, then $\frac{dy}{dx}$ is equal to
 (a) $\frac{1}{x} + x \log a$ (b) $\frac{\log a}{x} + \frac{x}{\log a}$
 (c) $\frac{1}{x \log a} + x \log a$ (d) $\frac{1}{x \log a} - \frac{\log a}{x(\log x)^2}$
21. If $f(x) = \log |x|$, $x \neq 0$ then $f'(x)$ equals
 (a) $\frac{1}{|x|}$ (b) $\frac{1}{x}$
 (c) $-\frac{1}{x}$ (d) None of these
22. The derivative of $y = x^{\ln x}$ is
 (a) $x^{\ln x} \ln x$ (b) $x^{\ln x-1} \ln x$
 (c) $2x^{\ln x-1} \ln x$ (d) $x^{\ln x-2}$
23. If $y = \{f(x)\}^{\phi(x)}$, then $\frac{dy}{dx}$ is
 (a) $e^{\phi(x) \log f(x)} \left\{ \frac{\phi(x) df(x)}{f(x) dx} + \log f(x) \cdot \frac{d\phi(x)}{dx} \right\}$
 (b) $\frac{\phi(x)}{f(x)} \left(\frac{df(x)}{dx} \right) + \frac{d\phi(x)}{dx} \log f(x)$
 (c) $e^{\phi(x) \log f(x)} \left\{ \phi(x) \frac{f'(x)}{f(x)} + \phi'(x) \log f(x) \right\}$
 (d) None of these



Parametric differentiation

Various rule of solving differentiation

24. If $x = \frac{1-t^2}{1+t^2}$ and $y = \frac{2t}{1+t^2}$, then $\frac{dy}{dx}$ is equal to

(a) $-\frac{y}{x}$ (b) $\frac{y}{x}$

(c) $-\frac{x}{y}$ (d) $\frac{x}{y}$

25. If $y = e^{\sin^{-1} x}$ and $u = \log x$, then $\frac{dy}{du}$ is

(a) $\frac{e^{\sin^{-1} x}}{\sqrt{1-x^2}}$ (b) $x e^{\sin^{-1} x}$

(c) $\frac{x e^{\sin^{-1} x}}{\sqrt{1-x^2}}$ (d) $\frac{e^{\sin^{-1} x}}{x}$

26. If $x = a \cos^4 \theta$, $y = a \sin^4 \theta$, then $\frac{dy}{dx}$ at $\theta = \frac{3\pi}{4}$ is

(a) -1 (b) 1
(c) $-a^2$ (d) a^2

Differentiation of function w.r.t functions

27. The derivative of e^{x^3} with respect to $\log x$ is

(a) e^{x^3} (b) $3x^2 2e^{x^3}$
(c) $3x^3 e^{x^3}$ (d) $3x^2 e^{x^3} + 3x^2$

28. The derivative of $\log_{10} x$ with respect to x^2 is

(a) $\frac{1}{2x^2} \log_e 10$ (b) $\frac{2}{x^2} \log_{10} e$
(c) $\frac{1}{2x^2} \log_{10} e$ (d) None of these

29. The derivative of $\sin^2 x$ with respect to $\cos^2 x$ is

(a) $\tan^2 x$ (b) $\tan x$
(c) $-\tan x$ (d) None of these

30. If $y = \log_{\cos x} \sin x$, then $\frac{dy}{dx}$ is equal to

(a) $(\cot x \log \cos x + \tan x \log \sin x) / (\log \cos x)^2$
(b) $(\tan x \log \cos x + \cot x \log \sin x) / (\log \cos x)^2$
(c) $(\cot x \log \cos x + \tan x \log \sin x) / (\log \sin x)^2$
(d) None of these

31. If $y = f\left(\frac{2x-1}{x^2+1}\right)$ and $f'(x) = \sin x^2$, then $\frac{dy}{dx}$ is equal to

(a) $\sin\left(\frac{2x-1}{x^2+1}\right)^2 \left\{ \frac{x^2+2x+2}{(x^2+1)^2} \right\}$

(b) $\sin\left(\frac{2x-1}{x^2+1}\right)^2 \left\{ \frac{2+2x-2x^2}{(x^2+1)^2} \right\}$

(c) $\sin\left(\frac{2x-1}{x^2+1}\right)^2 \left\{ \frac{2+2x-x^2}{(x^2+1)} \right\}$

(d) None of these

32. If $y = 2^x \cdot 3^{2x-1}$, then $\frac{dy}{dx}$ is equal to

(a) $(\log 2)(\log 3)$ (b) $(\log 18)$
(c) $(\log 18^2) y^2$ (d) $(\log 18) y$

33. If $y = (1+x^2) \tan^{-1} x - x$, then $\frac{dy}{dx}$ is equal to

(a) $\tan^{-1} x$ (b) $2x \tan^{-1} x$
(c) $2x \tan^{-1} x - 1$ (d) $\frac{2x}{\tan^{-1} x}$

34. If $y = (1+x)(1+x^2)(1+x^4) \dots \left(1+x^{2^n}\right)$, then $\frac{dy}{dx}$ at $x=0$ is

(a) -1 (b) 1
(c) 0 (d) None of these



Derivatives of inverse trigonometric functions

35. If $f(x) = 2 \tan^{-1} x + \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right)$, then

- (a) $f'(-2) = \frac{4}{5}$
 (b) $f'(-1) = -1$
 (c) $f'(x) = 0$ for all $x < 0$
 (d) None of these

36. If $y = \left[\tan^{-1} \frac{1}{1+x+x^2} + \tan^{-1} \frac{1}{x^2+3x+3} + \tan^{-1} \frac{1}{x^2+5x+7} + \dots \text{upto } n \text{ terms} \right]$ then $y'(0)$ equals

- (a) $\frac{-1}{n^2+1}$ (b) $\frac{-n^2}{n^2+1}$
 (c) $\frac{n^2}{n^2+1}$ (d) None of these

37. If $y = \cos^{-1} \left(\frac{2 \cos x - 3 \sin x}{\sqrt{13}} \right)$; $0 < x < \frac{\pi}{2}$, then $\frac{dy}{dx}$ is

- (a) zero (b) constant = 1
 (c) constant $\neq 1$ (d) none of these

Differentiation of infinite series

38. If $y = \sqrt{\sin x + \sqrt{\sin x + \sqrt{\sin x + \dots \infty}}}$, then $\frac{dy}{dx}$ is equal to

- (a) $\frac{\cos x}{2y-1}$ (b) $\frac{-\cos x}{2y-1}$
 (c) $\frac{\sin x}{1-2y}$ (d) $\frac{-\sin x}{1-2y}$

39. If $y = x^{x^{x^{\dots \infty}}}$, then $x(1-y \log x) \frac{dy}{dx}$

- (a) x^2 (b) y^2
 (c) xy^2 (d) xy

40. If $y = x + \frac{1}{x + \frac{1}{x + \frac{1}{x + \dots \infty}}}$, then find $\frac{dy}{dx}$

- (a) $\frac{y}{2y-x}$ (b) $\frac{x}{2x-y}$
 (c) $\frac{y}{y-x}$ (d) $\frac{x}{x-y}$

41. For $|x| < 1$, let $y = 1 + x + x^2 + \dots$ to ∞ , then $\frac{dy}{dx}$ equal to

- (a) $\frac{x}{y}$ (b) $\frac{x^2}{y^2}$
 (c) $\frac{x}{y^2}$ (d) $xy^2 + y$

Higher order Derivative problems

42. If $x = a \sin \theta$ and $y = b \cos \theta$, then $\frac{d^2y}{dx^2}$ is equal to

- (a) $\frac{a}{b^2} \sec^2 \theta$ (b) $-\frac{b}{a} \sec^2 \theta$
 (c) $\frac{b}{a^2} \sec^3 \theta$ (d) $-\frac{b}{a^2} \sec^3 \theta$

43. If f be a polynomial then, the second derivative of $f(e^x)$ is

- (a) $f''(e^x)$ (b) $f'''(e^x) e^x + f''(e^x)$
 (c) $f'''(e^x) e^{2x} + f''(e^x)$ (d) $f'''(e^x) e^{2x} + f'(e^x) e^x$

44. If $x^2 + y^2 = 1$, then

- (a) $yy'' - 2(y')^2 + 1 = 0$ (b) $yy'' + (y')^2 + 1 = 0$
 (c) $yy'' + (y')^2 - 1 = 0$ (d) $yy'' + 2(y')^2 + 1 = 0$

45. If $\sqrt{x+y} + \sqrt{y-x} = c$ then $\frac{d^2y}{dx^2}$ is

- (a) $\frac{2x}{c^2}$ (b) $\frac{2}{c^3}$
 (c) $-\frac{2}{c^2}$ (d) $\frac{2}{c^2}$



46. If $y^2 = ax^2 + bx + c$ where a, b, c are constants, then $y^3 \frac{d^2y}{dx^2}$, is equal to
- (a) a constant
(b) a function of x
(c) a function of y
(d) a function of x and y both
47. Let $x = \sin(lnt)$ and $y = \cos(lnt)$ then $\frac{d^2y}{dx^2}$ is
- (a) $-\frac{1}{y^2}$ (b) $-\frac{1}{y^3}$
(c) $\frac{1}{y^2}$ (d) $\frac{1}{y^3}$
48. If $y = a \cos(\log x) + b \sin(\log x)$ where a, b are parameters, the $x^2 y'' + xy'$ is equal to
- (a) y (b) $-y$
(c) $2y$ (d) $-2y$
49. If $y = A \cos nx + B \sin nx$, then $\frac{d^2y}{dx^2} =$
- (a) $-n^2 y$ (b) $-y$
(c) $n^2 y$ (d) none of these
53. Let $f(x) = \frac{(x+1)^2(x-1)}{(x-2)^3}$, then $f'(0)$ is
54. If $t \in (0, \frac{1}{2})$ and $x = \sin^{-1}(3t - 4t^3)$ and $y = \cos^{-1}(\sqrt{1-t^2})$, then $\frac{dy}{dx}$ is equal to
55. If $u = f(x^2)$, $v = g(x^3)$, $f'(x) = \sin x$ and $g'(x) = \cos x$ then $\frac{du}{dx} = k \sin(x^2)/x \cos(x^3)$, then find the value of k
56. Find the derivative of $f(\tan x)$ with respect to $g(\sec x)$ at $x = \pi/4$, if $f'(1) = 2$, $g'(\sqrt{2}) = 4$.
57. If $f(x) = \sqrt{1 + \cos^2(x^2)}$, then the value of $f'\left(\frac{\sqrt{\pi}}{2}\right)$ is
58. If $y = |\cos x| + |\sin x|$ then $\frac{dy}{dx}$ at $x = \frac{2\pi}{3}$ is
59. If Differentiation of $\log(\sin x^2)$ with respect to x is $k \cdot x \cot(x^2)$, then the value of k is
60. If $f(x) = \log_x(\log_e x)$, then $f'(x)$ at $x = e$ is
61. If $y = \tan^{-1} \left[\frac{\sin x + \cos x}{\cos x - \sin x} \right]$, then $\frac{dy}{dx}$ is equal to
62. Let $f(x) = \sin^{-1} \left(\frac{2x}{1+x^2} \right)$, find $f'(1/2)$
63. Find the derivative of $y = \tan^{-1} \left(\frac{2x}{1-x^2} \right)$ at $x = 0$
64. If $y = \frac{\sin x}{1 + \frac{\cos x}{1 + \frac{\sin x}{1 + \frac{\cos x}{1 + \frac{\sin x}{\dots \infty}}}}}$ then $y'(0)$ is

Numerical Value Type Questions

50. Let $f(x) = \frac{1 - \tan x}{4x - \pi}$, $x \neq \frac{\pi}{4}$, $x \in \left[0, \frac{\pi}{2}\right]$.
If $f(x)$ is continuous in $\left[0, \frac{\pi}{2}\right]$, then $f\left(\frac{\pi}{4}\right)$ is
51. If $y = \frac{a + bx^{3/2}}{x^{5/4}}$ and $y' = 0$ at $x = 5$, then the ratio $a : b$ is equal to
52. If $2^x + 2^y = 2^{x+y}$, then the value of $\frac{dy}{dx}$ at $x = y = 1$, is
65. If $e^y + xy = e$, then the value of $\frac{d^2y}{dx^2}$ for $x = 0$, is



EXERCISE - 2 : PREVIOUS YEAR JEE MAINS QUESTIONS

1. If the function.

$g(x) = \begin{cases} k\sqrt{x+1} & , 0 \leq x \leq 3 \\ mx+2 & , 3 < x \leq 5 \end{cases}$ is differentiable, then the value of $k+m$ is: (2015)

(a) $\frac{10}{3}$ (b) 4

(c) 2 (d) $\frac{16}{5}$

2. For $x \in \mathbb{R}$, $f(x) = |\log 2 - \sin x|$ and $g(x) = f(f(x))$, then :

(2016)

(a) $g'(0) = \cos(\log 2)$

(b) $g'(0) = -\cos(\log 2)$

(c) g is differentiable at $x=0$ and $g'(0) = -\sin(\log 2)$

(d) g is not differentiable at $x=0$

3. If $y = \left[x + \sqrt{x^2 - 1} \right]^{15} + \left[x - \sqrt{x^2 - 1} \right]^{15}$,

then $(x^2 - 1) \frac{d^2 y}{dx^2} + x \frac{dy}{dx}$ is equal to :

(2017/Online Set-1)

(a) $125y$ (b) $224y^2$

(c) $225y^2$ (d) $225y$

4. If $2x = y^{\frac{1}{5}} + y^{-\frac{1}{5}}$ and $(x^2 - 1) \frac{d^2 y}{dx^2} + \lambda x \frac{dy}{dx} + ky = 0$, then

$\lambda + k$ is equal to : (2017/Online Set-2)

(a) -23 (b) -24

(c) 26 (d) -26

5. Let f be a polynomial function such that

$f(3x) = f'(x) \cdot f''(x)$, for all $x \in \mathbb{R}$. Then :

(2017/Online Set-2)

(a) $f(2) + f'(2) = 28$ (b) $f''(2) - f'(2) = 0$

(c) $f''(2) - f(2) = 4$ (d) $f(2) - f'(2) + f''(2) = 10$

6. The value of k for which the function

$$f(x) = \begin{cases} \left(\frac{4}{5} \right)^{\frac{\tan 4x}{\tan 5x}}, & 0 < x < \frac{\pi}{2} \\ k + \frac{2}{5}, & x = \frac{\pi}{2} \end{cases}$$

is continuous at $x = \frac{\pi}{2}$, is : (2017/Online Set-2)

(a) $\frac{17}{20}$ (b) $\frac{2}{5}$

(c) $\frac{3}{5}$ (d) $-\frac{2}{5}$

7. Let $S = \{t \in \mathbb{R} : f(x) = |x - \pi| \cdot (e^{|x|} - 1) \sin |x| \text{ is not differentiable at } t\}$. Then the set S is equal to: (2018)

(a) $\{0, \pi\}$ (b) $\emptyset$ (an empty set)

(c) $\{0\}$ (d) $\{\pi\}$

8. Let $f(x) = \begin{cases} (x-1)^{\frac{1}{2-x}}, & x > 1, x \neq 2 \\ k, & x = 2 \end{cases}$

The value of k for which f is continuous at $x = 2$ is:

(2018/Online Set-2)

(a) 1 (b) e

(c) e^{-1} (d) e^2

9. If $2y = \left(\cot^{-1} \left(\frac{\sqrt{3} \cos x + \sin x}{\cos x - \sqrt{3} \sin x} \right) \right)^2, x \in \left(0, \frac{\pi}{2} \right)$

then $\frac{dy}{dx}$ is equal to: (2019-04-08/Shift-1)

(a) $\frac{\pi}{6} - x$ (b) $x - \frac{\pi}{6}$

(c) $\frac{\pi}{3} - x$ (d) $2x - \frac{\pi}{3}$



10. Let $f: [-1, 3] \rightarrow \mathbb{R}$ be defined as

$$f(x) = \begin{cases} |x| + [x], & -1 \leq x < 1 \\ x + |x|, & 1 \leq x < 2 \\ x + [x], & 2 \leq x \leq 3, \end{cases}$$

where $[t]$ denotes the greatest integer less than or equal to t . Then f is discontinuous at: (2019-04-08/Shift-2)

- (a) only one point (b) only two points
(c) only three points (d) four or more points
11. If $f(1) = 1$, $f'(1) = 3$, then the derivative of $f(f(x)) + (f(x))^2$ at $x = 1$ is: (2019-04-08/Shift-2)
- (a) 33 (b) 12
(c) 9 (d) 15

12. If the function f defined on $\left(\frac{\pi}{6}, \frac{\pi}{3}\right)$ by

$$f(x) = \begin{cases} \frac{\sqrt{2} \cos x - 1}{\cot x - 1}, & x \neq \frac{\pi}{4} \\ k, & x = \frac{\pi}{4} \end{cases}$$

is continuous, then k is equal to: (2019-04-09/Shift-1)

- (a) 2 (b) $\frac{1}{2}$
(c) 1 (d) $\frac{1}{\sqrt{2}}$
13. If $f(x) = [x] - \left[\frac{x}{4}\right]$, $x \in \mathbb{R}$, where $[x]$ denotes the greatest integer function, then: (2019-04-09/Shift-2)
- (a) f is continuous at $x = 4$.
(b) $\lim_{x \rightarrow 4^+} f(x)$ exists but $\lim_{x \rightarrow 4^-} f(x)$ does not exist
(c) Both $\lim_{x \rightarrow 4^-} f(x)$ and $\lim_{x \rightarrow 4^+} f(x)$ exist but are not equal.
(d) $\lim_{x \rightarrow 4^-} f(x)$ exists but $\lim_{x \rightarrow 4^+} f(x)$ does not exist

14. If the function

$$f(x) = \begin{cases} a|\pi - x| + 1, & x \leq 5 \\ b|x - \pi| + 3, & x > 5 \end{cases}$$

is continuous at $x = 5$, then the value of $a-b$ is:

(2019-04-09/Shift-2)

- (a) $\frac{2}{\pi+5}$ (b) $\frac{-2}{\pi+5}$
(c) $\frac{2}{\pi-5}$ (d) $\frac{2}{5-\pi}$

15. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be differentiable at $c \in \mathbb{R}$ and $f'(c) = 0$. If $g(x) = |f(x)|$, then at $x = c$, g is: (2019-04-10/Shift-1)

- (a) not differentiable if $f'(c) = 0$
(b) differentiable if $f'(c) \neq 0$
(c) differentiable if $f'(c) = 0$
(d) not differentiable

16. If $f(x) = \begin{cases} \frac{\sin(p+1)x + \sin x}{x}, & x < 0 \\ q, & x = 0 \\ \frac{\sqrt{x+x^2} - \sqrt{x}}{x^{3/2}}, & x > 0 \end{cases}$

is continuous at $x = 0$, then the ordered pair (p, q) is equal to: (2019-04-10/Shift-1)

- (a) $\left(-\frac{3}{2}, -\frac{1}{2}\right)$ (b) $\left(-\frac{1}{2}, \frac{3}{2}\right)$
(c) $\left(-\frac{3}{2}, \frac{1}{2}\right)$ (d) $\left(\frac{5}{2}, \frac{1}{2}\right)$

17. If $e^y + xy = e$, the ordered pair $\left(\frac{dy}{dx}, \frac{d^2y}{dx^2}\right)$ at $x = 0$ is equal to: (2019-04-12/Shift-1)

- (a) $\left(\frac{1}{e}, -\frac{1}{e^2}\right)$ (b) $\left(-\frac{1}{e}, \frac{1}{e^2}\right)$
(c) $\left(\frac{1}{e}, \frac{1}{e^2}\right)$ (d) $\left(-\frac{1}{e}, -\frac{1}{e^2}\right)$

18. The derivative of $\tan^{-1}\left(\frac{\sin x - \cos x}{\sin x + \cos x}\right)$ with respect

to $\frac{x}{2}$ where $x \in \left(0, \frac{\pi}{2}\right)$ is _____. (2019-04-12/Shift-2)

- (a) 1 (b) $\frac{2}{3}$
(c) $\frac{1}{2}$ (d) 2



19. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined as

$$f(x) = \begin{cases} 5, & \text{if } x \leq 1 \\ a + bx, & \text{if } 1 < x < 3 \\ b + 5x, & \text{if } 3 \leq x < 5 \\ 30, & \text{if } x \geq 5 \end{cases}$$

Then, f is:

(2019-01-09/Shift-1)

- (a) continuous if $a = 5$ and $b = 5$
- (b) continuous if $a = -5$ and $b = 10$
- (c) continuous if $a = 0$ and $b = 5$
- (d) not continuous for any values of a and b

20. Let f be a differentiable function from $\mathbb{R}$ to $\mathbb{R}$ such that

$$|f(x) - f(y)| \leq 2|x - y|^{3/2}, \text{ for all } x, y \in \mathbb{R}.$$

If $f(0) = 1$ then $\int_0^1 (f(x))^2 dx$ is equal to

(2019-01-09/Shift-2)

- (a) 1
- (b) 2
- (c) $\frac{1}{2}$
- (d) 0

21. If $x = 3 \tan t$ and $y = 3 \sec t$, then the value of $\frac{d^2y}{dx^2}$ at

$$t = \frac{\pi}{4}, \text{ is:}$$

(2019-01-09/Shift-2)

- (a) $\frac{1}{3\sqrt{2}}$
- (b) $\frac{1}{6\sqrt{2}}$
- (c) $\frac{3}{2\sqrt{2}}$
- (d) $\frac{1}{6}$

22. Let $f(x) = \begin{cases} \max\{|x|, x^2\} & |x| \leq 2 \\ 8 - 2|x| & 2 < |x| \leq 4 \end{cases}$

Let S be the set of points in the interval $(-4, 4)$ at which f is not differentiable. Then S :

(2019-01-10/Shift-1)

- (a) is an empty set
- (b) equals $\{-2, -1, 0, 1, 2\}$
- (c) equals $\{-2, -1, 1, 2\}$
- (d) equals $\{-2, 2\}$

23. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function such that

$$f(x) = x^3 + x^2 f'(1) + x f''(2) + f'''(3), x \in \mathbb{R}.$$

then $f(2)$ equals:

(2019-01-10/Shift-1)

- (a) -4
- (b) 30
- (c) -2
- (d) 8

24. Let $f: (-1, 1) \rightarrow \mathbb{R}$ be a function defined by

$$f(x) = \max\{-|x|, -\sqrt{1-x^2}\}.$$

If K be the set of all points at which f is not differentiable, then K has exactly:

(2019-01-10/Shift-2)

- (a) five elements
- (b) one element
- (c) three elements
- (d) two elements

25. If $x \log_e (\log_e x) - x^2 + y^2 = 4 (y > 0)$, then $\frac{dy}{dx}$ at $x = e$

is equal to (2019-01-11/Shift-1)

$$(a) \frac{(1+2e)}{2\sqrt{4+e^2}} \quad (b) \frac{(2e-1)}{2\sqrt{4+e^2}}$$

$$(c) \frac{(1+2e)}{\sqrt{4+e^2}} \quad (d) \frac{e}{\sqrt{4+e^2}}$$

26. Let k be the set of all real values of x where the function $f(x) = \sin|x| - |x| + 2(x - \pi) \cos|x|$ is not differentiable.

Then the set k is equal to: (2019-01-11/Shift-2)

- (a) ϕ {an empty set}
- (b) $\{\pi\}$
- (c) $\{0\}$
- (d) $\{0, \pi\}$

27. For $x > 1$, If $(2x)^{2y} = 4e^{2x-2y}$, then $(1 + \log_e 2x)^2 \frac{dy}{dx}$ is

equal to _____. (2019-01-12/Shift-1)

$$(a) \frac{x \log_e 2x - \log_e 2}{x} \quad (b) \log_e 2x$$

$$(c) \frac{x \log_e 2x + \log_e 2}{x} \quad (d) x \log_e 2x$$

28. If a function $f(x)$ defined by

$$f(x) = \begin{cases} ae^x + be^{-x} & -1 \leq x < 1 \\ cx^2 & 1 \leq x \leq 3 \\ ax^2 + 2cx & 3 < x \leq 4 \end{cases} \text{ be continuous for}$$

some $a, b, c \in \mathbb{R}$ and $f'(0) + f'(2) = e$, then the value of a is:

(2020-09-02/Shift-1)

$$(a) \frac{1}{e^2 - 3e + 13} \quad (b) \frac{e}{e^2 - 3e - 13}$$

$$(c) \frac{e}{e^2 + 3e + 13} \quad (d) \frac{e}{e^2 - 3e + 13}$$



29. If $y = \sum_{k=1}^6 k \cos^{-1} \left\{ \frac{3}{5} \cos kx - \frac{4}{5} \sin kx \right\}$,
then $\frac{dy}{dx}$ at $x = 0$ is (2020-09-02/Shift-2)
30. If $y^2 + \log_e (\cos^2 x) = y$, $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, then
(a) $|y'(0)| + |y''(0)| = 1$
(b) $y''(0) = 0$
(c) $|y'(0)| + |y''(0)| = 3$
(d) $|y''(0)| = 2$ (2020-09-03/Shift-1)
31. If $(a + \sqrt{2} b \cos x)(a - \sqrt{2} b \cos y) = a^2 - b^2$, where
 $a > b > 0$, then $\frac{dx}{dy}$ at $\left(\frac{\pi}{4}, \frac{\pi}{4}\right)$ is: (2020-09-04/Shift-1)
- (a) $\frac{a+b}{a-b}$ (b) $\frac{a-2b}{a+2b}$
(c) $\frac{a-b}{a+b}$ (d) $\frac{2a+b}{2a-b}$
32. Suppose a differentiable function $f(x)$ satisfies the identity
 $f(x+y) = f(x) + f(y) + xy^2 + x^2y$, for all real x and y . If
 $\lim_{x \rightarrow 0} \frac{f(x)}{x} = 1$, then $f'(3)$ is equal to
(2020-09-04/Shift-1)
33. The function $f(x) = \begin{cases} \frac{\pi}{4} + \tan^{-1} x, & |x| \leq 1 \\ \frac{1}{2}(|x| - 1), & |x| > 1 \end{cases}$ is :
(a) both continuous and differentiable on $\mathbb{R} - \{-1\}$
(b) continuous on $\mathbb{R} - \{-1\}$ and differentiable on $\mathbb{R} - \{-1, 1\}$
(c) continuous on $\mathbb{R} - \{1\}$ and differentiable on $\mathbb{R} - \{-1, 1\}$
(d) both continuous and differentiable on $\mathbb{R} - \{1\}$ (2020-09-04/Shift-2)
34. If the function $f(x) = \begin{cases} k_1(x-\pi)^2 - 1, & x \leq \pi \\ k_2 \cos x, & x > \pi \end{cases}$ is twice
differentiable, then the ordered pair (k_1, k_2) is equal to:
(a) $(1, 1)$ (b) $(1, 0)$
(c) $\left(\frac{1}{2}, -1\right)$ (d) $\left(\frac{1}{2}, 1\right)$ (2020-09-05/Shift-1)
35. Let $f(x) = x \cdot \left[\frac{x}{2}\right]$, for $-10 < x < 10$, where $[t]$ denotes the
greatest integer function. Then the number of points of
discontinuity of f is equal to
(2020-09-05/Shift-1)
36. The derivative of $\tan^{-1} \left(\frac{\sqrt{1+x^2}-1}{x} \right)$ with respect to
 $\tan^{-1} \left(\frac{2x\sqrt{1-x^2}}{1-2x^2} \right)$ at $x = \frac{1}{2}$ is: (2020-09-05/Shift-2)
- (a) $\frac{2\sqrt{3}}{3}$ (b) $\frac{2\sqrt{3}}{5}$
(c) $\frac{\sqrt{3}}{12}$ (d) $\frac{\sqrt{3}}{10}$
37. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined as
$$f(x) = \begin{cases} x^5 \sin\left(\frac{1}{x}\right) + 5x^2, & x < 0 \\ 0, & x = 0 \\ x^5 \cos\left(\frac{1}{x}\right) + \lambda x^2, & x > 0 \end{cases}$$

The value of λ for which $f''(0)$ exists, is _____.
(2020-09-06/Shift-1)
38. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x) = \max\{x, x^2\}$.
Let S denote the set of all points in $\mathbb{R}$, where f is not
differentiable. Then (2020-09-06/Shift-2)
- (a) $\{0, 1\}$ (b) f (an empty set)
(c) $\{1\}$ (d) $\{0\}$



39. If $y(\alpha) = \sqrt{2\left(\frac{\tan \alpha + \cot \alpha}{1 + \tan^2 \alpha}\right) + \frac{1}{\sin^2 \alpha}}$

where $\alpha \in \left(\frac{3\pi}{4}, \pi\right)$, then $\frac{dy}{d\alpha}$ at $\alpha = \frac{5\pi}{6}$ is

(2020-01-07/Shift-1)

(a) $-\frac{1}{4}$

(b) $\frac{4}{3}$

(c) 4

(d) -4

40. Let $x^k + y^k = a^k$, ($a, k > 0$) and $\frac{dy}{dx} + \left(\frac{y}{x}\right)^{\frac{1}{3}} = 0$ then k is

(2020-01-07/Shift-1)

(a) $\frac{1}{3}$

(b) $\frac{3}{2}$

(c) $\frac{2}{3}$

(d) $\frac{4}{3}$

41. Let S be the set of points where the function, $f(x) = |2 - |x - 3||$, $x \in \mathbb{R}$, is not differentiable. Then, the value of $\sum_{x \in S} f(f(x))$ is equal to

(2020-01-07/Shift-1)

42. Let $y = y(x)$ be a function of x satisfying

$$y\sqrt{1-x^2} = k - x\sqrt{1-y^2} \text{ where } k \text{ is a constant and}$$

$$y\left(\frac{1}{2}\right) = -\frac{1}{4}. \text{ Then } \frac{dy}{dx} \text{ at } x = \frac{1}{2}, \text{ is equal to :}$$

(2020-01-07/Shift-2)

(a) $-\frac{\sqrt{5}}{2}$

(b) $\frac{\sqrt{5}}{2}$

(c) $-\frac{\sqrt{5}}{4}$

(d) $\frac{2}{\sqrt{5}}$

43. If the function f defined on $\left(-\frac{1}{3}, \frac{1}{3}\right)$ by

$$f(x) = \begin{cases} \left(\frac{1}{x}\right) \log_e \left(\frac{1+3x}{1-2x}\right), & \text{when } x \neq 0 \\ k, & \text{when } x = 0 \end{cases}$$

is continuous, then k is equal to _____.

(2020-01-07/Shift-2)

44. If $f(x) = \begin{cases} \frac{\sin(a+2)x + \sin x}{x}, & x < 0 \\ b, & x = 0 \\ \frac{(x+3x^2)^{\frac{1}{3}} - x^{\frac{1}{3}}}{x^{\frac{4}{3}}}, & x > 0 \end{cases}$

is continuous at $x = 0$ then $a + 2b$ is equal to:

(2020-01-09/Shift-1)

(a) -2

(b) 1

(c) 0

(d) -1

45. Let f and g be differentiable functions on $\mathbb{R}$, such that $f \circ g$ is the identity function. If for some $a, b \in \mathbb{R}$, $g'(a) = 5$ and $g(a) = b$, then $f'(b)$ is equal to :

(2020-01-09/Shift-2)

(a) $\frac{2}{5}$

(b) 5

(c) 1

(d) $\frac{1}{5}$

46. Let $[t]$ denotes the greatest integer $\leq t$ and $\lim_{x \rightarrow 0} x \left[\frac{4}{x} \right] = A$.

Then the function, $f(x) = [x^2] \sin \pi x$ is discontinuous, when x is equal to

(2020-01-09/Shift-2)

(a) $\sqrt{A+1}$

(b) $\sqrt{A}$

(c) $\sqrt{A+5}$

(d) $\sqrt{A+21}$



47. If $f: \mathbb{R} \rightarrow \mathbb{R}$ is a function defined by $f(x) = [x-1] \cos\left(\frac{2x-1}{2}\right)\pi$ where $[.]$ denotes the greatest integer function, then f is

(24-02-2021/Shift-1)

- (a) Discontinuous at all integral values of x except at
(b) Discontinuous only at $x = 1$
(c) Continuous only at $x = 1$
(d) Continuous for every real x

48. The number of points, at which the function $f(x) = |2x+1| - 3|x+2| + |x^2+x-2|$, $x \in \mathbb{R}$ is not differentiable, is _____.

(25-02-2021/Shift-1)

49. A function f is defined on $[-3, 3]$ as

$$f(x) = \begin{cases} \min\{|x|, 2-x^2\}, & -2 \leq x \leq 2 \\ [x], & 2 < |x| \leq 3 \end{cases} \text{ where } [x]$$

denotes the greatest integer $\leq x$. The number of points, where f is not differentiable $(-3, 3)$ is _____.

(25-02-2021/Shift-2)

50. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined as

$$f(x) = \begin{cases} 2 \sin\left(-\frac{\pi x}{2}\right), & \text{if } x < -1 \\ |ax^2 + x + b|, & \text{if } -1 \leq x \leq 1 \\ \sin(\pi x), & \text{if } x > 1 \end{cases}$$

If $f(x)$ is continuous on $\mathbb{R}$, then $a + b$ equals:

(26-02-2021/Shift-2)

- (a) -3
(b) 1
(c) 3
(d) -1

51. Let the function $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ be defined as:

$$f(x) = \begin{cases} x+2, & x < 0 \\ x^2, & x \geq 0 \end{cases} \text{ and } g(x) = \begin{cases} x^3, & x < 1 \\ 3x-2, & x \geq 1 \end{cases}$$

Then, the number of points in $\mathbb{R}$ where $(f \circ g)(x)$ is NOT differentiable is equal to

(16-03-2021/Shift-1)

- (a) 0
(b) 2
(c) 3
(d) 1

52. Let $\alpha \in \mathbb{R}$ be such that the function

$$f(x) = \begin{cases} \frac{\cos^{-1}(1-\{x\}^2) \sin^{-1}(1-\{x\})}{\{x\} - \{x\}^3}, & x \neq 0 \\ \alpha, & x = 0 \end{cases} \text{ is}$$

continuous at $x = 0$, where $\{x\} = x - [x]$, $[x]$ is the greatest integer less than or equal to x . Then

(16-03-2021/Shift-2)

- (a) $\alpha = 0$
(b) $\alpha = \frac{\pi}{\sqrt{2}}$
(c) no such α exists
(d) $\alpha = \frac{\pi}{4}$

53. Let $f: S \rightarrow S$ where $S = (0, \infty)$ be a twice differentiable function such that $f(x+1) = xf(x)$. If $g: S \rightarrow \mathbb{R}$ be defined as $g(x) = \log_e f(x)$, then the value of

$|g''(5) - g''(1)|$ is equal to:

(16-03-2021/Shift-2)

- (a) 1
(b) $\frac{197}{144}$
(c) $\frac{187}{144}$
(d) $\frac{205}{144}$

54. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ be defined as

$$f(x) = \begin{cases} x+a, & x < 0 \\ |x-1|, & x \geq 0 \end{cases} \text{ and } g(x) = \begin{cases} x+1, & x < 0 \\ (x-1)^2 + b, & x \geq 0 \end{cases}$$

where a, b are non-negative real numbers. If $(g \circ f)(x)$ is continuous for all $x \in \mathbb{R}$, then $a + b$ is equal to.

(16-03-2021/Shift-2)



55. If the function $f(x) = \frac{\cos(\sin x) - \cos x}{x^4}$ is continuous at each point in its domain and $f(0) = \frac{1}{k}$, then k is

(17-03-2021/Shift-1)

56. If $f(x) = \begin{cases} \frac{1}{|x|} & ; |x| \geq 1 \\ ax^2 + b & ; |x| < 1 \end{cases}$ is differentiable at every point of the domain, then the value of a and b are respectively :

(18-03-2021/Shift-1)

- (a) $\frac{5}{2}, -\frac{3}{2}$ (b) $\frac{1}{2}, -\frac{3}{2}$
(c) $-\frac{1}{2}, \frac{3}{2}$ (d) $\frac{1}{2}, \frac{1}{2}$

57. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined as

$$f(x) = \begin{cases} \frac{\sin(a+1)x + \sin 2x}{2x}, & \text{if } x < 0 \\ b & \text{if } x = 0 \\ \frac{\sqrt{x+bx^3} - \sqrt{x}}{bx^{5/2}}, & \text{if } x > 0 \end{cases}$$

If f is continuous at $x = 0$, then the value of $a + b$ is equal to

(18-03-2021/Shift-2)

- (a) $-\frac{5}{2}$ (b) -3
(c) $-\frac{3}{2}$ (d) -2

58. Let a function $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined as

$$f(x) = \begin{cases} \sin x - e^x & \text{if } x \leq 0 \\ a + [-x] & \text{if } 0 < x < 1 \\ 2x - b & \text{if } x \geq 1 \end{cases}$$

where $[x]$ is the greatest integer less than or equal to x .

If f is continuous on $\mathbb{R}$, then $(a + b)$ is equal to:

(20-07-2021/Shift-1)

- (a) 5 (b) 3
(c) 2 (d) 4

59. Let a function $g: [0, 4] \rightarrow \mathbb{R}$ be defined as,

$$g(x) = \begin{cases} \max\{t^3 - 6t^2 + 9t - 3\}, & 0 \leq t \leq x, 0 \leq x \leq 3 \\ 4 - x, & 3 < x \leq 4 \end{cases},$$

then the number of points in the interval $(0, 4)$

where $g(x)$ (20-07-2021/Shift-2)

60. Let $r: \mathbb{R} \rightarrow \mathbb{R}$ be function defined as

$$f(x) = \begin{cases} 3\left(1 - \frac{|x|}{2}\right) & \text{if } |x| \leq 2 \\ 0 & \text{if } |x| > 2 \end{cases}$$

Let $g: \mathbb{R} \rightarrow \mathbb{R}$ be given by $g(x) = f(x+2) - f(x-2)$

If n and m denote the number of points in $\mathbb{R}$ where g is not continuous and not differentiable, respectively, then $n + m$ is equal to

(22-07-2021/Shift-2)

61. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined as

$$f(x) = \begin{cases} \frac{\lambda|x^2 - 5x + 6|}{\mu(5x - x^2 - 6)} & ; x < 2 \\ e^{\frac{\tan(x-2)}{x-[x]}} & ; x > 2 \\ \mu & ; x = 2 \end{cases}$$

Where $[x]$ is the greatest integer less than or equal to x .

If f is continuous at $x = 2$, then $\lambda + \mu$ is equal to?

(25-07-2021/Shift-1)

- (a) $e(e-2)$ (b) $2e-1$
(c) $e(-e+1)$ (d) 1



62. Consider the function $f(x) = \begin{cases} \frac{P(x)}{\sin(x-2)} & ; x \neq 2 \\ 7 & ; x = 2 \end{cases}$

where $P(x)$ is a polynomial such that $P''(x)$ is always a constant and $P(3) = 9$. If $f(x)$ is continuous at $x = 2$, then $P(5)$ is equal to **(25-07-2021/Shift-2)**

63. Let $f : [0, \infty) \rightarrow [0, 3]$ be a function defined by

$$f(x) = \begin{cases} \max\{\sin t : 0 \leq t \leq x\}, & 0 \leq x \leq \pi \\ 2 + \cos x, & x > \pi \end{cases}$$

Then which of the following is true?

(27-07-2021/Shift-2)

- (a) f is differentiable everywhere in $(0, \infty)$
- (b) f is continuous everywhere but not differentiable exactly at two points in $(0, \infty)$
- (c) f is not continuous exactly at two points in $(0, \infty)$
- (d) f is continuous everywhere but not differentiable exactly at one point in $(0, \infty)$

64. Let $f : \left(-\frac{\pi}{4}, \frac{\pi}{4}\right) \rightarrow \mathbb{R}$ be defined as

$$f(x) = \begin{cases} (1 + |\sin x|)^{\frac{3a}{|\sin x|}}, & -\frac{\pi}{4} < x < 0 \\ b, & x = 0 \\ e^{\cot 4x / \cot 2x}, & 0 < x < \frac{\pi}{4} \end{cases}$$

If f is continuous at $x = 0$, the value of $6a + b^2$ is equal to : **(27-07-2021/Shift-1)**

- (a) e
- (b) $1 + e$
- (c) $1 - e$
- (d) $e - 1$

65. Let $f : [0, 3] \rightarrow \mathbb{R}$ be defined by

$$f(x) = \min\{x - [x], 1 + [x] - x\}$$

where $[x]$ is the greatest integer less than or equal to x .

Let P denote the set containing all $x \in [0, 3]$ where f is discontinuous, and Q denote the set containing all $x \in (0, 3)$ where f is not differentiable. Then the sum of number of elements in P and Q is equal to _____.

(27-07-2021/Shift-1)

66. Let $f(x) = \cos\left(2 \tan^{-1} \sin\left(\cot^{-1} \sqrt{\frac{1-x}{x}}\right)\right)$, $0 < x < 1$.

Then

(26-08-2021/Shift-1)

- (a) $(1+x)^2 f'(x) + 2(f(x))^2 = 0$
- (b) $(1-x)^2 f'(x) + 2(f(x))^2 = 0$
- (c) $(1+x)^2 f'(x) - 2(f(x))^2 = 0$
- (d) $(1-x)^2 f'(x) - 2(f(x))^2 = 0$

67. If $y = y(x)$ is an implicit function of x such that

$$\log_e(x+y) = 4xy, \text{ then } \frac{d^2y}{dx^2} \text{ at } x = 0 \text{ is equal to}$$

(26-08-2021/Shift-1)

68. Let $a, b \in \mathbb{R}$, $b \neq 0$. Define a function

$$f(x) = \begin{cases} a \sin \frac{\pi}{2}(x-1) & \text{for } x \leq 0 \\ \frac{\tan 2x - \sin 2x}{bx^3} & \text{for } x > 0 \end{cases}$$

If f is continuous at $x = 0$, then $10 - ab$ is equal to

(26-08-2021/Shift-1)



69. Let $[t]$ denote the greatest integer less than or equal to t .

Let $f(x) = x - [x]$, $g(x) = 1 - x + [x]$ and

$h(x) = \min\{f(x), g(x)\}$, $x \in [-2, 2]$. Then h is:

(26-08-2021/Shift-2)

- (a) not continuous at exactly four points in $[-2, 2]$
- (b) not continuous at exactly three points in $[-2, 2]$
- (c) continuous in $[-2, 2]$ but not differentiable at exactly three point in $(-2, 2)$
- (d) continuous in $[-2, 2]$ but not differentiable at more than four points in $(-2, 2)$

70. If $y^{\frac{1}{4}} + y^{-\frac{1}{4}} = 2x$, and $(x^2 - 1)\frac{d^2y}{dx^2} + \alpha x \frac{dy}{dx} + \beta y = 0$,

then $|\alpha - \beta|$ is equal to _____. (27-08-2021/Shift-1)

71. If $y(x) = \cot^{-1}\left(\frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}}\right)$, $x \in \left(\frac{\pi}{2}, \pi\right)$,

then $\frac{dy}{dx}$ at $x = \frac{5\pi}{6}$ is: (27-08-2021/Shift-2)

- (a) 0
- (b) -1
- (c) $\frac{1}{2}$
- (d) $-\frac{1}{2}$

72. If the function $f(x) = \begin{cases} \frac{1}{x} \log_e \left(\frac{1+\frac{x}{a}}{1-\frac{x}{b}} \right), & x < 0 \\ k, & x = 0 \\ \frac{\cos^2 x - \sin^2 x - 1}{\sqrt{x^2 + 1} - 1}, & x > 0 \end{cases}$ is

continuous at $x = 0$, then $\frac{1}{a} + \frac{1}{b} + \frac{4}{k}$ is equal to ?

(31-08-2021/Shift-1)

- (a) -4
- (b) -5
- (c) 4
- (d) 5

73. The function $f(x) = |x^2 - 2x - 3|e^{|9x^2 - 12x + 4|}$ is not differentiable at exactly ? (31-08-2021/Shift-1)

- (a) Two points
- (b) One points
- (c) Four points
- (d) Three

74. Let $[t]$ denote the greatest integer $\leq t$. The number of points where the function

$$f(x) = [x]|x^2 - 1| + \sin\left(\frac{\pi}{[x] + 3}\right) - [x + 1], x \in (-2, 2)$$

is not continuous is _____. (01-09-2021/Shift-2)

75. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined as

$$f(x) = \begin{cases} \frac{x^3}{(1 - \cos 2x)^2} \log_e \left(\frac{1 + 2xe^{-2x}}{(1 - xe^{-x})^2} \right) & x \neq 0 \\ \alpha & x = 0 \end{cases}$$

If f is continuous at $x = 0$, then α is equal to :

(22-07-2021/Shift-2)

- (a) 1
- (b) 0
- (c) 3
- (d) 2



EXERCISE - 3 : ADVANCED OBJECTIVE QUESTIONS

Objective Questions I [Only one correct option]

- Let $U(x)$ and $V(x)$ are differentiable functions such that $\frac{U(x)}{V(x)} = 7$. If $\frac{U'(x)}{V'(x)} = p$ and $\left(\frac{V(x)}{U(x)}\right)' = q$, then $\frac{p+q}{p-q}$ has the value equal to
 (a) 1 (b) 0
 (c) 7 (d) -7
- Suppose the function $f(x) - f(2x)$ has the derivative 5 at $x = 1$ and derivative 7 at $x = 2$. The derivative of the function $f(x) - f(4x)$ at $x = 1$, has the value equal to
 (a) 19 (b) 9
 (c) 17 (d) 14
- If f is a periodic function, then
 (a) f' and f'' are also periodic
 (b) f' is periodic but f'' is not periodic
 (c) f'' is periodic but f' is not periodic
 (d) None of these
- If $y = f(x)$ is an odd differentiable function defined on $(-\infty, \infty)$ such that $f'(3) = -2$, then $f'(-3)$ equals
 (a) 4 (b) 2
 (c) -2 (d) 0
- If $f(x) = \log |2x|$, $x \neq 0$, then $f'(x)$ is equal to
 (a) $\frac{1}{x}$ (b) $-\frac{1}{x}$
 (c) $\frac{1}{|x|}$ (d) None of these
- Let $f(x)$ be a polynomial function of second degree. If $f(1) = f(-1)$ and a, b, c are in AP, then $f'(a), f'(b)$ and $f'(c)$ are in
 (a) AP
 (b) GP
 (c) HP
 (d) Arithmetico-Geometric progression
- If $x^2 + y^2 = t + \frac{1}{t}$ and $x^4 + y^4 = t^2 + \frac{1}{t^2}$, then $\frac{dy}{dx}$ is equal to
 (a) $\frac{y}{x}$ (b) $-\frac{y}{x}$
 (c) $\frac{x}{y}$ (d) $-\frac{x}{y}$
- If $f(x) = |x-3|$ and $\phi(x) = (f \circ f)(x)$, then for $x > 10$, $\phi'(x)$ is equal to
 (a) 1 (b) 0
 (c) -1 (d) None of these
- Let $f(x) = \sin x$, $g(x) = 2x$ and $h(x) = \cos x$.
 If $\phi(x) = [g \circ (f \circ h)](x)$, then $\phi''\left(\frac{\pi}{4}\right)$ is equal to
 (a) 4 (b) 0
 (c) -4 (d) None of these
- If $f(x) = |x-1|$ and $g(x) = f[f\{f(x)\}]$, then for $x > 2$, $g'(x)$ is equal to
 (a) -1 if $2 < x < 3$ (b) 1 if $2 \leq x < 3$
 (c) 1 for all $x > 2$ (d) None of these
- Let $f(x) = 2/(x+1)$ and $g(x) = 3x$. It is given that $(f \circ g)(x_0) = (g \circ f)(x_0)$. Then $(g \circ f)'(x_0)$ equals
 (a) -32 (b) $\frac{32}{3}$
 (c) $\frac{-32}{9}$ (d) $\frac{-32}{3}$
- Let $f(x)$ be a polynomial of degree 3 such that $f(3) = 1$, $f'(3) = -1$, $f''(3) = 0$ and $f'''(3) = 12$. Then the value of $f'(1)$ is
 (a) 12 (b) 23
 (c) -13 (d) None of these



13. If $f(x) = \sin\left(\frac{\pi}{2}[x] - x^5\right)$, $1 < x < 2$ and $[x]$ denotes the greatest integer less than or equal to x , then $f'\left(\sqrt{\frac{\pi}{2}}\right)$ is equal to
- (a) $5\left(\frac{\pi}{2}\right)^{4/5}$ (b) $-5\left(\frac{\pi}{2}\right)^{4/5}$
 (c) 0 (d) None of these
14. If $f(x) = \sqrt{x^2 - 10x + 25}$, then the derivative of $f(x)$ on the interval $[0, 7]$ is
- (a) 1 (b) -1
 (c) 0 (d) none of these
15. If $y = \begin{vmatrix} \sin x & \cos x & \sin x \\ \cos x & -\sin x & \cos x \\ x & 1 & 1 \end{vmatrix}$, then $\frac{dy}{dx}$ is equal to
- (a) 1 (b) -1
 (c) 0 (d) None of these
16. If $\Delta_1 = \begin{vmatrix} x & b & b \\ a & x & b \\ a & a & x \end{vmatrix}$ and $\Delta_2 = \begin{vmatrix} x & b \\ a & x \end{vmatrix}$ are given, then
- (a) $\Delta_1 = 3(\Delta_2)^2$ (b) $\frac{d}{dx}\Delta_1 = 3\Delta_2$
 (c) $\frac{d}{dx}\Delta_1 = 3(\Delta_2)^2$ (d) $\Delta_1 = 3(\Delta_2)^{3/2}$
17. If $f(x) = \cos^2 x + \cos^2\left(x + \frac{\pi}{3}\right) - \cos x \cos\left(x + \frac{\pi}{3}\right)$ then $f'(x)$ is
- (a) 0 (b) 1
 (c) 2 (d) 3
18. The function $f(x) = (\sin 2x)^{\tan^2 2x}$ is not defined at $x = \pi/4$. The value of $f(\pi/4)$ so that f is continuous at $x = \pi/4$ is
- (a) $\sqrt{e}$ (b) $1/\sqrt{e}$
 (c) 2 (d) None of these
19. Let $f(x) = a + b|x| + c|x|^4$, where a, b and c are real constants. Then $f(x)$ is differentiable at $x = 0$ if
- (a) $a = 0$ (b) $b = 0$
 (c) $c = 0$ (d) None of these
20. The derivative of $f(\tan x)$ w.r.t. $g(\sec x)$ at $x = \frac{\pi}{4}$, where $f'(1) = 2$ and $g'(\sqrt{2}) = 4$, is
- (a) $\frac{1}{\sqrt{2}}$ (b) $\sqrt{2}$
 (c) 1 (d) None of these
21. Let $y = x^3 - 8x + 7$ and $x = f(t)$. If $\frac{dy}{dt} = 2$ and $x = 3$ at $t = 0$, then $\frac{dx}{dt}$ at $t = 0$ is given by
- (a) 1 (b) $\frac{19}{2}$
 (c) $\frac{2}{19}$ (d) None of these
22. $f(x), g(x), h(x)$ are functions having non-zero derivatives. The derivative of $f(x)$ w.r.t $g(x)$ is $\alpha(x)$ and derivative of $g(x)$ w.r.t $h(x)$ is $\beta(x)$. Then derivative of $h(x)$ w.r.t $f(x) =$
- (a) $\alpha(x) \cdot \beta(x)$ (b) $\frac{\alpha(x)}{\beta(x)}$
 (c) $\frac{1}{\alpha(x)\beta(x)}$ (d) $\frac{\beta(x)}{\alpha(x)}$
23. If f is an even function such that $\lim_{h \rightarrow 0^+} \frac{f(h) - f(0)}{h}$ has some finite non-zero value, then
- (a) f is continuous and derivable at $x = 0$
 (b) f is continuous but not derivable at $x = 0$
 (c) f may be discontinuous at $x = 0$
 (d) None of these



24. If $\sin y = x \sin (a + y)$, then $\frac{dy}{dx}$ is
- (a) $\frac{\sin a}{\sin^2 (a + y)}$ (b) $\frac{\sin^2 (a + y)}{\sin a}$
- (c) $\sin a \sin^2 (a + y)$ (d) $\frac{\sin^2 (a - y)}{\sin a}$
25. Let $f(x) = \alpha(x) \beta(x) \gamma(x)$ for all real x , where $\alpha(x)$, $\beta(x)$ and $\gamma(x)$ are differentiable functions of x . If $f'(2) = 18f(2)$, $\alpha'(2) = 3\alpha(2)$, $\beta'(2) = -4\beta(2)$ and $\gamma'(2) = k\gamma(2)$, then the value of k is
- (a) 14 (b) 16
- (c) 19 (d) None of these
26. If $y = \frac{ax + b}{x^2 + c}$, where a, b, c are constants then $(2xy' + y)y''$ is equal to
- (a) $3(xy'' + y')y''$ (b) $3(xy' + y'')y''$
- (c) $3(xy'' + y')y'$ (d) None of these
27. Function $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfies the functional equation
- $$f(x - y) = \frac{f(x)}{f(y)}$$
- If $f'(0) = p$ and $f'(a) = q$, then $f'(-a)$ is
- (a) $\frac{p^2}{q}$ (b) $\frac{q}{p}$
- (c) $\frac{p}{q}$ (d) q
28. Let $f(x) = x^n$, $n \in \mathbb{W}$. The number of values of n for which $f'(p + q) = f'(p) + f'(q)$ is valid for all +ve p & q is
- (a) 0 (b) 1
- (c) 2 (d) None of these

29. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function such that
- $$f\left(\frac{x+y}{3}\right) = \frac{f(x) + f(y)}{3}, f(0) = 0 \text{ and } f'(0) = 3. \text{ Then}$$
- (a) $f(x)$ is a quadratic function
- (b) $f(x)$ is continuous but not differentiable
- (c) $f(x)$ is differentiable in $\mathbb{R}$
- (d) $f(x)$ is bounded in $\mathbb{R}$
30. If $f_r(x), g_r(x), h_r(x), r = 1, 2, 3$ are polynomials in x such that $f_r(a) = g_r(a) = h_r(a), r = 1, 2, 3$ and
- $$F(x) = \begin{vmatrix} f_1(x) & f_2(x) & f_3(x) \\ g_1(x) & g_2(x) & g_3(x) \\ h_1(x) & h_2(x) & h_3(x) \end{vmatrix}, \text{ then } F'(a) \text{ is equal to}$$
- (a) a (b) $-a$
- (c) 0 (d) None of these.
31. Let f & g be differentiable functions satisfying $g'(a) = 2, g(a) = b$ & $f \circ g = I$ (Identity function). Then $f'(b)$ is equal to
- (a) 2 (b) $\frac{2}{3}$
- (c) $\frac{1}{2}$ (d) None
32. Let $f(x) = a[x] + b e^{[x]} + c |x|^2$, where a, b and c are real constants. where $[x]$ denotes greatest integer $\leq x$. If $f(x)$ is differentiable at $x = 0$, then
- (a) $b = 0, c = 0, a \in \mathbb{R}$ (b) $a = 0, c = 0, b \in \mathbb{R}$
- (c) $a = 0, b = 0, c \in \mathbb{R}$ (d) None of these
33. If $f(x) = \begin{cases} x(3e^{1/x} + 4) & , x \neq 0 \\ 0 & , x = 0 \end{cases}$, then $f(x)$ is
- (a) continuous as well as differentiable at $x = 0$
- (b) continuous but not differentiable at $x = 0$
- (c) differentiable but not continuous at $x = 0$
- (d) None of these



34. If $y = e^{\tan x}$, then $\cos^2 x \frac{d^2 y}{dx^2} =$

- (a) $(1 - \sin 2x) \frac{dy}{dx}$ (b) $-(1 + \sin 2x) \frac{dy}{dx}$
(c) $(1 + \sin 2x) \frac{dy}{dx}$ (d) None of these

35. Suppose $f(x) = e^{ax} + e^{bx}$, where $a \neq b$, and that $f''(x) - 2f'(x) - 15f(x) = 0$ for all x . Then the product ab is

- (a) 25 (b) -15
(c) 9 (d) -9

36. If $(\sin y)^{\sin(\pi x/2)} + \frac{\sqrt{3}}{2} \sec^{-1}(2x) + 2^x \tan(\log(x+2)) = 0$ then dy/dx at $x = -1$ is

- (a) $\frac{3}{\sqrt{\pi^2 - 3}}$ (b) $\frac{1}{\pi\sqrt{\pi^2 - 3}}$
(c) $\frac{3}{\pi\sqrt{\pi^2 - 3}}$ (d) $\frac{3\pi}{\sqrt{\pi^2 - 3}}$

37. If $ax^2 + 2hxy + by^2 = 1$, then $\frac{d^2 y}{dx^2}$ is equal to

- (a) $\frac{ab - h^2}{(hx + by)^3}$ (b) $\frac{h^2 - ab}{(hx + by)^3}$
(c) $\frac{h^2 + ab}{(hx + by)^3}$ (d) None of these

38. If $y^2 = P(x)$, a polynomial of degree $n \geq 3$, then

- $2 \frac{d}{dx} \left(y^3 \frac{d^2 y}{dx^2} \right)$
(a) $-P(x) \cdot P'''(x)$ (b) $P(x) \cdot P'''(x)$
(c) $P(x) \cdot P''(x)$ (d) None of these

39. Let $f(x) = e^x - e^{-x} - 2 \sin x - \frac{2}{3} x^3$, then the least value

- of n for which $\left| \frac{d^n f(x)}{dx^n} \right|_{x=0}$ is non-zero
(a) 4 (b) 5
(c) 7 (d) 3

40. People living at Mars, instead of the usual definition of derivative $Df(x)$, define a new kind of derivative, $D^*f(x)$ by the formula

$$D^*f(x) = \lim_{h \rightarrow 0} \frac{f^2(x+h) - f^2(x)}{h} \text{ where } f^2(x) \text{ means } [f(x)]^2.$$

If $f(x) = x \ln x$ then

$D^*f(x)|_{x=e}$ has the value

- (a) e (b) $2e$
(c) $4e$ (d) $8e$

41. If $y = k \sin px$, then the value of the determinant

$$\begin{vmatrix} y & y_1 & y_2 \\ y_3 & y_4 & y_5 \\ y_6 & y_7 & y_8 \end{vmatrix} \text{ is equal to}$$

- (a) 1 (b) 0
(c) -1 (d) None of these.

where y_n denotes n th derivative of y w.r.t. x .

42. If $f(x) = \begin{vmatrix} x^n & n! & 2 \\ \cos x & \cos \frac{n\pi}{2} & 4 \\ \sin x & \sin \frac{n\pi}{2} & 8 \end{vmatrix}$ then the value of

$$\frac{d^n}{dx^n} [f(x)]_{x=0} \text{ is}$$

- (a) 0 (b) 1
(c) -1 (d) None of these

43. A non zero polynomial with real coefficients has the property that $f(x) = f'(x) \cdot f''(x)$. The leading coefficient of $f(x)$ is

- (a) $\frac{1}{6}$ (b) $\frac{1}{9}$
(c) $\frac{1}{12}$ (d) $\frac{1}{18}$



44. If $0 < x < 1$, then

$$\frac{1}{1+x} + \frac{2x}{1+x^2} + \frac{4x^3}{1+x^4} + \frac{8x^7}{1+x^8} + \dots =$$

(a) $\frac{1}{1-x}$ (b) $\frac{x}{1-x}$

(c) $\frac{x}{1+x}$ (d) $\frac{1-x}{1+x}$

45. If $y = \left(\frac{ax+b}{cx+d} \right)$, then $2 \frac{dy}{dx} \cdot \frac{d^3y}{dx^3}$ is equal to

(a) $\left(\frac{d^2y}{dx^2} \right)^2$ (b) $3 \frac{d^2y}{dx^2}$

(c) $3 \left(\frac{d^2y}{dx^2} \right)^2$ (d) $3 \frac{d^2x}{dy^2}$

Objective Questions II [One or more than one correct option]

46. The function $f(x) = \max. \{ (1-x), (1+x), 2 \}$, $x \in (-\infty, \infty)$, is

- (a) continuous at all points
- (b) differentiable at all points
- (c) differentiable at all points except at $x = 1$ and $x = -1$.
- (d) continuous at all points except at $x = 1$ and $x = -1$, where it is discontinuous.

47. If $f(x) = \frac{1}{[\sin x]}$, where $[.]$ denotes the greatest function, then

(a) Domain of $f(x)$ is $(2n\pi + \pi, 2n\pi + 2\pi) \cup \left\{ 2n\pi + \frac{\pi}{2} \right\}$

where $n \in \mathbb{I}$

- (b) $f(x)$ is continuous when $x \in (2n\pi + \pi, 2n\pi + 2\pi)$
- (c) $f(x)$ is differentiable at $x = \pi/2$
- (d) None of these

48. Let $[x]$ denotes the greatest integer less than or equal to x . If $f(x) = [x \sin \pi x]$, then $f(x)$ is

- (a) continuous at $x = 0$ (b) continuous in $(-1, 0)$
- (c) differentiable at $x = 1$ (d) differentiable in $(-1, 1)$

49. Let $y = \sqrt{(\sin x + \sin 2x + \sin 3x)^2 + (\cos x + \cos 2x + \cos 3x)^2}$ then which of the following is correct?

(a) $\frac{dy}{dx}$ when $x = \frac{\pi}{2}$ is -2

(b) value of y when $x = \frac{\pi}{5}$ is $\frac{3+\sqrt{5}}{2}$

(c) value of y when $x = \frac{\pi}{12}$ is $\frac{\sqrt{1} + \sqrt{2} + \sqrt{3}}{2}$

(d) y simplifies to $(1 + 2 \cos x)$ in $[0, \pi]$

50. Let $f(x) = \frac{1-x^{n+1}}{1-x}$ and $g(x) = 1 - \frac{2}{x} + \frac{3}{x^2} - \dots + (-1)^n \frac{n+1}{x^n}$.

Then the constant term in $f'(x) \times g(x)$ is equal to

(a) $\frac{n(n^2-1)}{6}$ when n is even

(b) $\frac{n(n+1)}{2}$ when n is odd

(c) $-\frac{n}{2}(n+1)$ when n is even

(d) $\frac{n(n-1)}{2}$ when n is odd

51. If $F(x) = f(x)g(x)$ and $f'(x)g'(x) = c$, then

(a) $F' = c \left[\frac{f}{f'} + \frac{g}{g'} \right]$ (b) $\frac{F''}{F} = \frac{f''}{f} + \frac{g''}{g} + \frac{2c}{fg}$

(c) $\frac{F'''}{F} = \frac{f'''}{f} + \frac{g'''}{g}$ (d) $\frac{F'''}{F''} = \frac{f'''}{f''} + \frac{g'''}{g''}$

52. Suppose f and g are functions having second derivatives f'' and g'' everywhere, if $f(x) \cdot g(x) = 1$ for all x and f' and g'

are never zero, then $\frac{f''(x)}{f'(x)} - \frac{g''(x)}{g'(x)}$ equals

(a) $-2 \frac{f'(x)}{f(x)}$ (b) $\frac{2g'(x)}{g(x)}$

(c) $-\frac{f'(x)}{f(x)}$ (d) $2 \frac{f'(x)}{f(x)}$



Numerical Value Type Questions

53. If $f(x) = \begin{cases} \frac{8^x - 4^x - 2^x + 1^x}{x^2}, & x > 0 \\ e^x \sin x + \pi x + \lambda \ln 4, & x \leq 0 \end{cases}$ is continuous at

$x = 0$, then the value of $1000 e^\lambda$ must be

54. Let $f(x) = \frac{\cos^{-1}(1 - \{x\}) \cdot \sin^{-1}(1 - \{x\})}{\sqrt{2\{x\}} \cdot (1 - \{x\})}$, then the value

of $\frac{2008\sqrt{2}}{\pi} \lim_{x \rightarrow 0^+} f(x)$ must be (where $\{x\}$ denotes the fractional part of x).

55. If $f(x) = \begin{cases} \frac{36^x - 9^x - 4^x + 1}{\sqrt{2} - \sqrt{1 + \cos x}}, & x \neq 0 \\ \lambda, & x = 0 \end{cases}$ is continuous at

$x = 0$, then $\lambda = \sqrt{\mu} \ln 2 \cdot \ln 3$ then the value of μ must be

56. The function given by

$$f(x) = \begin{cases} \frac{\sqrt{\pi} - \sqrt{\cos^{-1} x}}{\sqrt{x+1}}, & x \neq -1 \\ \frac{1}{\sqrt{\lambda\pi}}, & x = -1 \end{cases}$$

The value of λ for which the function $f(x)$ is continuous at $x = -1$ from the right, must be

57. Let $P(x)$ be a polynomial of degree 4 such that $P(1) = P(3) = P(5) = P'(7) = 0$. If the real number $x \neq 1, 3, 5$ is such that $P(x) = 0$ can be expressed as $x = p/q$ where 'p' and 'q' are relatively prime, then find $(p+q)$.

58. If $f(x) = \frac{1}{\sin x - \sin a} - \frac{1}{(x-a) \cos x}$ then

$$\frac{d}{da} \lim_{x \rightarrow a} f(x) = \frac{-1}{k} \sec a - \sec a \tan^2 a.$$

The numerical quantity k should be equal to

59. If the third derivative of $\frac{x^4}{(x-1)(x-2)}$ is $\frac{-k}{(x-2)^4} + \frac{6}{(x-1)^4}$ then the numerical quantity k must be equal to

Match the Following

Each question has two columns. Four options are given representing matching of elements from Column-I and Column-II. Only one of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.

60. Column-I

Column-II

(A) If the function

(P) 6

$$f(x) = \begin{cases} \frac{\sin 3x}{x}, & x \neq 0 \\ \frac{K}{2}, & x = 0 \end{cases}$$

is continuous at $x = 0$, then $k =$

(B) If $f(x) = \frac{x^2 - 10x + 25}{x^2 - 7x + 10}$ for $x \neq 5$ (Q) $2 \log |a|$

& it is continuous at $x = 5$ then $f(5) =$

(C) If $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by (R) 3

$$f(x) = \begin{cases} a^2 \cos^2 x + b^2 \sin^2 x & (x \leq 0) \\ e^{ax+b} & (x > 0) \end{cases}$$

is continuous function then $b =$

(S) $\log |a|$

(T) 0

The correct matching is

(a) A - P; B - T; C - Q

(b) A - T; B - P; C - Q

(c) A - Q; B - T; C - P

(d) A - P; B - Q; C - T



61. Column - I Column - II

(A) If $f'(x) = \sqrt{3x^2 + 6}$ & $y = f(x^3)$ (P) -2

then at $x = 1$, $\frac{dy}{dx} =$

(B) If f be a differential function (Q) -1
such that

$$f(xy) = f(x) + f(y); \forall x, y \in \mathbb{R}$$

$$\text{then } f(e) + f(1/e) =$$

(C) If f be a twicedifferential function (R) 0
such that

$$f''(x) = -f(x) \text{ \& } f'(x) = g(x);$$

$$\text{If } h(x) = (f(x))^2 + (g(x))^2 \text{ \& } h(5) = 9$$

$$\text{then } h(10) =$$

(D) $y = \tan^{-1}(\cot x) + \cot^{-1}(\tan x)$, (S) 9

$$\frac{\pi}{2} < x < \pi \text{ then } \frac{dy}{dx} =$$

The correct matching is

(a) A-S; B-R; C-S; D-P

(b) A-R; B-S; C-S; D-P

(c) A-S; B-S; C-R; D-P

(d) A-S; B-R; C-P; D-S

Paragraph Type Questions

Using the following passage, solve Q.62 to Q.64

Passage

A curve is represented parametrically by the equations

$$x = f(t) = a^{t \ln(b^t)} \text{ and } y = g(t) = b^{-t \ln(a^t)} \text{ a, b} > 0 \text{ and } a \neq 1, b \neq 1 \text{ where } t \in \mathbb{R}.$$

62. Which of the following is not a correct expression for

$$\frac{dy}{dx} ?$$

$$(a) \frac{-1}{f(t)^2}$$

$$(b) -(g(t))^2$$

$$(c) \frac{-g(t)}{f(t)}$$

$$(d) \frac{-f(t)}{g(t)}$$

63. The value of $\frac{d^2y}{dx^2}$ at the point where $f(t) = g(t)$ is

$$(a) 0$$

$$(b) \frac{1}{2}$$

$$(c) 1$$

$$(d) 2$$

64. The value of $\frac{f(t)}{f'(t)} \cdot \frac{f''(-t)}{f'(-t)} + \frac{f(-t)}{f'(-t)} \cdot \frac{f''(t)}{f'(t)} \forall t \in \mathbb{R}$, is equal to

$$(a) -2$$

$$(b) 2$$

$$(c) -4$$

$$(d) 4$$

Text

65. Differentiate the following functions with respect to x :

(i) $\log(\sec x + \tan x)$ (ii) $e^{x \sin x}$



EXERCISE - 4 : PREVIOUS YEAR JEE ADVANCED QUESTIONS

Objective Questions I [Only one correct option]

- Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be any function. Define $g: \mathbb{R} \rightarrow \mathbb{R}$ by $g(x) = |f(x)|$ for all x . Then g is : (2000)
 - onto if f is onto
 - one-one if f is one-one
 - continuous if f is continuous
 - differentiable if f is differentiable
- Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x) = \max\{x, x^3\}$. The set of all points where $f(x)$ is not differentiable is : (2001)
 - $\{-1, 1\}$
 - $\{-1, 0\}$
 - $\{0, 1\}$
 - $\{-1, 0, 1\}$
- The left hand derivative of $f(x) = [x] \sin(\pi x)$ at $x = k$, k is an integer is (2001)
 - $(-1)^k(k-1)\pi$
 - $(-1)^{k-1}(k-1)\pi$
 - $(-1)^k k\pi$
 - $(-1)^{k-1} k\pi$
- Which of the following functions is differentiable at $x = 0$? (2001)
 - $(\cos |x|) + |x|$
 - $\cos(|x|) - |x|$
 - $\sin(|x|) + |x|$
 - $\sin(|x|) - |x|$
- The domain of the derivative of the functions $f(x) = \begin{cases} \tan^{-1} x, & \text{if } |x| \leq 1 \\ \frac{1}{2}(|x| - 1), & \text{if } |x| > 1 \end{cases}$ is (2002)
 - $\mathbb{R} - \{0\}$
 - $\mathbb{R} - \{1\}$
 - $\mathbb{R} - \{-1\}$
 - $\mathbb{R} - \{-1, 1\}$
- If y is a function of x and $\log(x+y) - 2xy = 0$, then the value of $y'(0)$ is equal to (2004)
 - 1
 - 1
 - 2
 - 0
- If $y = y(x)$ and it follows the relation $x \cos y + y \cos x = \pi$, then $y''(0)$ (2005)
 - 1
 - π
 - $-\pi$
 - 1
- Let $f(x) = ||x| - 1|$, then points where $f(x)$, is not differentiable is/(are) : (2005)
 - 0
 - 1
 - ± 1
 - $0, \pm 1$
- $\frac{d^2x}{dy^2}$ equals (2007)
 - $\left(\frac{d^2y}{dx^2}\right)^{-1}$
 - $-\left(\frac{d^2y}{dx^2}\right)^{-1} \left(\frac{dy}{dx}\right)^{-3}$
 - $\left(\frac{d^2y}{dx^2}\right) \left(\frac{dy}{dx}\right)^{-2}$
 - $-\left(\frac{d^2y}{dx^2}\right) \left(\frac{dy}{dx}\right)^{-3}$
- Let $g(x) = \log f(x)$ where $f(x)$ is a twice differentiable positive function on $(0, \infty)$ such that $f(x+1) = x f(x)$. Then, for $N = 1, 2, 3, \dots$, (2008)

$$g''\left(N + \frac{1}{2}\right) - g''\left(\frac{1}{2}\right) =$$
 - $-4 \left\{ 1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N-1)^2} \right\}$
 - $4 \left\{ 1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N-1)^2} \right\}$
 - $-4 \left\{ 1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N+1)^2} \right\}$
 - $4 \left\{ 1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N+1)^2} \right\}$
- Let f be a real-valued function defined on the interval $(-1, 1)$ such that $e^{-x}f(x) = 2 + \int_0^x \sqrt{t^4 + 1} dt$, for all $x \in (-1, 1)$ and let f^{-1} be the inverse function of f . Then $(f^{-1})'(2)$ is equal to (2010)
 - 1
 - $\frac{1}{3}$
 - $\frac{1}{2}$
 - $\frac{1}{e}$



12. Let $f(x) = \begin{cases} x^2 \left| \cos \frac{\pi}{x} \right|, & x \neq 0, x \in \mathbb{R}, \\ 0, & x = 0 \end{cases}$ then f is (2012)

- (a) differentiable both at $x=0$ and $x=2$
 (b) differentiable at $x=0$ but not differentiable at $x=2$
 (c) not differentiable at $x=0$ but differentiable at $x=2$
 (d) differentiable neither at $x=0$ nor at $x=2$

13. Let $f_1: \mathbb{R} \rightarrow \mathbb{R}$, $f_2: [0, \infty) \rightarrow \mathbb{R}$, $f_3: \mathbb{R} \rightarrow \mathbb{R}$ and $f_4: \mathbb{R} \rightarrow [0, \infty)$ be defined by

$$f_1(x) = \begin{cases} |x| & \text{if } x < 0, \\ e^x & \text{if } x \geq 0; \end{cases}$$

$$f_2(x) = x^2;$$

$$f_3(x) = \begin{cases} \sin x & \text{if } x < 0, \\ x & \text{if } x \geq 0 \end{cases}$$

and

$$f_4(x) = \begin{cases} f_2(f_1(x)) & \text{if } x < 0, \\ f_2(f_1(x)) - 1 & \text{if } x \geq 0. \end{cases} \quad (2014)$$

List I

List II

- | | | | |
|----|-------------------|----|--------------------------------|
| P. | f_4 is | 1. | onto but not one-one |
| Q. | f_3 is | 2. | neither continuous nor one-one |
| R. | f_2 of f_1 is | 3. | differentiable but not one-one |
| S. | f_2 is | 4. | continuous and one-one |

	P	Q	R	S
(A)	3	1	4	2
(B)	1	3	4	2
(C)	3	1	2	4
(D)	1	3	2	4

Objective Questions II [One or more than one correct option]

14. If $f(x) = \min \{1, x^2, x^3\}$, then (2006)
- (a) $f(x)$ is continuous $\forall x \in \mathbb{R}$
 (b) $f(x) > 0, \forall x > 1$
 (c) $f(x)$ is continuous but not differentiable $\forall x \in \mathbb{R}$
 (d) $f(x)$ is not differentiable at two points.

15. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $f(x+y) = f(x) + f(y)$, $\forall x, y \in \mathbb{R}$. If $f(x)$ is differentiable at $x=0$, then (2011)
- (a) $f(x)$ is differentiable only in a finite interval containing zero
 (b) $f(x)$ is continuous $\forall x \in \mathbb{R}$
 (c) $f'(x)$ is constant $\forall x \in \mathbb{R}$
 (d) $f(x)$ is differentiable except at finitely many points

16. If $f(x) = \begin{cases} -x - \frac{\pi}{2}, & x \leq -\frac{\pi}{2} \\ -\cos x, & -\frac{\pi}{2} < x \leq 0, \\ x-1, & 0 < x \leq 1 \\ \ln x, & x > 1 \end{cases}$ then (2011)

- (a) $f(x)$ is continuous at $x = -\frac{\pi}{2}$
 (b) $f(x)$ is not differentiable at $x=0$
 (c) $f(x)$ is differentiable at $x=1$

- (d) $f(x)$ is differentiable at $x = -\frac{3}{2}$

17. For every integer n , let a_n and b_n be real numbers. Let function $f: \mathbb{R} \rightarrow \mathbb{R}$ be given by

$$f(x) = \begin{cases} a_n + \sin \pi x, & \text{for } x \in [2n, 2n+1] \\ b_n + \cos \pi x, & \text{for } x \in (2n-1, 2n) \end{cases} \quad \text{for all integers } n.$$

If f is continuous, then which of the following hold(s) for all n ? (2012)

- (a) $a_{n-1} - b_{n-1} = 0$ (b) $a_n - b_n = 1$
 (c) $a_n - b_{n+1} = 1$ (d) $a_{n-1} - b_n = -1$

18. Let $f: [a, b] \rightarrow [1, \infty)$ be a continuous function and let $g: \mathbb{R} \rightarrow \mathbb{R}$ be defined as

$$g(x) = \begin{cases} 0 & \text{if } x < a, \\ \int_a^x f(t) dt & \text{if } a \leq x \leq b, \\ \int_a^b f(t) dt & \text{if } x > b \end{cases}$$

- Then (2014)
- (a) $g(x)$ is continuous but not differentiable at a
 (b) $g(x)$ is differentiable on $\mathbb{R}$
 (c) $g(x)$ is continuous but not differentiable at b
 (d) $g(x)$ is continuous and differentiable at either a or b but not both.



19. Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function with $g(0) = 0$, $g'(0) = 0$ and $g'(1) \neq 0$.

$$\text{Let } f(x) = \begin{cases} \frac{x}{|x|} g(x), & x \neq 0 \\ 0, & x = 0 \end{cases} \text{ and } h(x) = e^{|x|} \quad (2015)$$

- (a) f is differentiable at $x = 0$
 (b) h is differentiable at $x = 0$
 (c) $f \circ h$ is differentiable at $x = 0$
 (d) $h \circ f$ is differentiable at $x = 0$
20. Let $a, b \in \mathbb{R}$ and $f : \mathbb{R} \rightarrow \mathbb{R}$, be defined by $f(x) = a \cos(|x^2 - x|) + b|x| \sin(|x^3 + x|)$. Then f is (2016)
- (a) differentiable at $x = 0$ if $a = 0$ and $b = 1$
 (b) differentiable at $x = 1$ if $a = 1$ and $b = 0$
 (c) **NOT** differentiable at $x = 0$ if $a = 1$ and $b = 0$
 (d) **NOT** differentiable at $x = 1$ if $a = 1$ and $b = 1$
21. Let $f : \mathbb{R} \rightarrow \mathbb{R}$, $g : \mathbb{R} \rightarrow \mathbb{R}$, and $h : \mathbb{R} \rightarrow \mathbb{R}$, be differentiable functions such that $f(x) = x^3 + 3x + 2$, $g(f(x)) = x$ and $h(g(g(x))) = x$ for all $x \in \mathbb{R}$. Then (2016)

- (a) $g'(2) = \frac{1}{15}$ (b) $h'(1) = 666$
 (c) $h(0) = 16$ (d) $h(g(3)) = 36$

22. Let the function $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = x^3 - x^2 + (x-1) \sin x$ and let $g : \mathbb{R} \rightarrow \mathbb{R}$ be an arbitrary function. Let $fg : \mathbb{R} \rightarrow \mathbb{R}$ be the product function defined by $(fg)(x) = f(x)g(x)$. Then which of the following statements is/are TRUE? (2020)

- (a) If g is continuous at $x = 1$, then fg is differentiable at $x = 1$
 (b) If fg is differentiable at $x = 1$, then g is continuous at $x = 1$
 (c) If g is differentiable at $x = 1$, then fg is differentiable at $x = 1$
 (d) If fg is differentiable at $x = 1$, then g is differentiable at $x = 1$

23. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ be functions satisfying $f(x+y) = f(x) + f(y) + f(x)f(y)$ and $f(x) = xg(x)$ for all $x, y \in \mathbb{R}$. If $\lim_{x \rightarrow 0} g(x) = 1$, then which of the following statements is/are TRUE? (2020)

- (a) f is differentiable at every $x \in \mathbb{R}$
 (b) If $g(0) = 1$, then g is differentiable at every $x \in \mathbb{R}$
 (c) The derivative $f'(1)$ is equal to 1
 (d) The derivative $f'(0)$ is equal to 1

Numerical Value Type Questions

24. If two functions ' f ' and ' g ' satisfying given conditions for $\forall x, y \in \mathbb{R}$, $f(x-y) = f(x)g(y) - f(y)g(x)$ and $g(x-y) = g(x)g(y) + f(x)f(y)$. If right hand derivative at $x = 0$ exists for $f(x)$ then find the derivative of $g(x)$ at $x = 0$ (2005)
25. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ be respectively given by $f(x) = |x| + 1$ and $g(x) = x^2 + 1$. Define $h : \mathbb{R} \rightarrow \mathbb{R}$ by

$$h(x) = \begin{cases} \max \{f(x), g(x)\} & \text{if } x \leq 0, \\ \min \{f(x), g(x)\} & \text{if } x \geq 0. \end{cases} \quad (2014)$$

The number of points at which $h(x)$ is not differentiable is

26. Let the functions $f : (-1, 1) \rightarrow \mathbb{R}$ and $g : (-1, 1) \rightarrow (-1, 1)$ be defined by $f(x) = |2x - 1| + |2x + 1|$ and $g(x) = x - [x]$, where $[x]$ denotes the greatest integer less than or equal to x . Let $f \circ g : (-1, 1) \rightarrow \mathbb{R}$ be the composite function defined by $(f \circ g)(x) = f(g(x))$. Suppose c is the number of points in the interval $(-1, 1)$ at which $f \circ g$ is not continuous, and suppose d is the number of points in the interval $(-1, 1)$ at which $f \circ g$ is not differentiable. Then the value of $c + d$ is _____. (2020)

Match the Following

Each question has two columns. Four options are given representing matching of elements from Column-I and Column-II. For each question, choose the option corresponding to the correct matching.

27. In the following, $[x]$ denotes the greatest integer less than or equal to x . (2007)

Column I

- (A) $x|x|$
 (B) $\sqrt{|x|}$
 (C) $x + [x]$

Column II

- (P) continuous in $(-1, 1)$
 (Q) differentiable in $(-1, 1)$
 (R) strictly increasing $(-1, 1)$
 (S) not differentiable at least at one point in $(-1, 1)$

The correct matching is

- (a) A-P, Q, R; B-P, S; C-R, S; D-P, Q
 (b) A-P, S, R; B-P, Q, R; C-R, S; D-P, Q
 (c) A-P, Q, R; B-P, S; C-P, Q; D-R, S
 (d) A-P, Q; B-P, S; C-R, S; D-P, Q, R



Text

28. Let $\alpha \in \mathbb{R}$. Prove that a function $f: \mathbb{R} \rightarrow \mathbb{R}$ is differentiable at α if and only if there is a function $g: \mathbb{R} \rightarrow \mathbb{R}$ which is continuous at α and satisfies $f(x) - f(\alpha) = g(x)(x - \alpha)$ for all $x \in \mathbb{R}$. (2001)

29.
$$f(x) = \begin{cases} b \sin^{-1}\left(\frac{x+c}{2}\right), & -\frac{1}{2} < x < 0 \\ \frac{1}{2}, & x = 0 \\ \frac{e^{ax/2} - 1}{x}, & 0 < x < \frac{1}{2} \end{cases}$$

If $f(x)$ is differentiable at $x = 0$ and $|c| < \frac{1}{2}$, then find the value of a and prove that $64b^2 = (4 - c^2)$. (2004)

30. If $f: [-1, 1] \rightarrow \mathbb{R}$ and $f(0) = 0$ then $f'(0) = \lim_{n \rightarrow \infty} n f\left(\frac{1}{n}\right)$

Find the value of $\lim_{n \rightarrow \infty} \frac{2}{\pi} (n+1) \cos^{-1}\left(\frac{1}{n}\right) - n$

Given that $0 < \left| \lim_{n \rightarrow \infty} \cos^{-1}\left(\frac{1}{n}\right) \right| < \frac{\pi}{2}$ (2004)

Find Answer Key and Detailed Solutions at the end of this book



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04

APPLICATIONS OF DERIVATIVES

APPLICATIONS OF DERIVATIVES



1. DERIVATIVE AS RATE OF CHANGE

In various fields of applied mathematics one has the quest to know the rate at which one variable is changing, with respect to other. The rate of change naturally refers to time. But we can have rate of change with respect to other variables also.

An economist may want to study how the investment changes with respect to variations in interest rates.

A physician may want to know, how small changes in dosage can affect the body's response to a drug.

A physicist may want to know the rate of change of distance with respect to time.

All questions of the above type can be interpreted and represented using derivatives.

Definition :

The **average rate of change** of a function $f(x)$ with respect to x over an interval $[a, a+h]$ is defined as $\frac{f(a+h) - f(a)}{h}$.

Definition :

The **instantaneous rate of change** of f with respect to x is defined as

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}, \text{ provided the limit exists.}$$

NOTES :

To use the word 'instantaneous', x may not be representing time. We usually use the word 'rate of change' to mean 'instantaneous rate of change'.

2. EQUATIONS OF TANGENT & NORMAL

- (I) The value of the derivative at $P(x_1, y_1)$ gives the slope of the tangent to the curve at P . Symbolically

$$f'(x_1) = \left. \frac{dy}{dx} \right|_{(x_1, y_1)} = \text{Slope of tangent at}$$

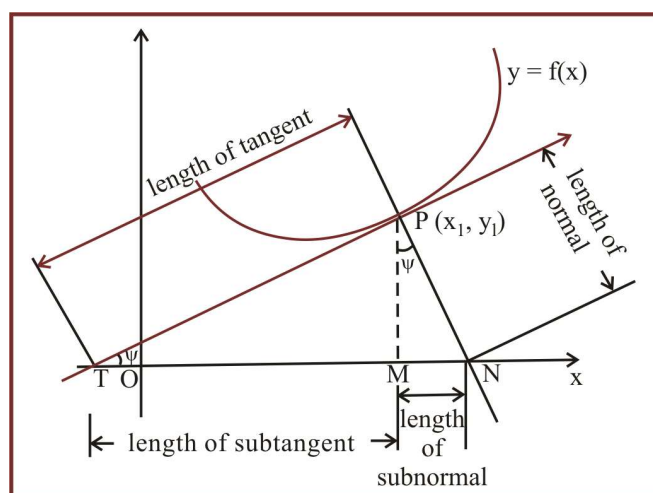
$P(x_1, y_1) = m$ (say).

- (II) Equation of tangent at (x_1, y_1) is ;

$$(y - y_1) = \left(\frac{dy}{dx} \right)_{(x_1, y_1)} \times (x - x_1)$$

- (III) Equation of normal at (x_1, y_1) is ;

$$(y - y_1) = \left(\frac{-1}{\frac{dy}{dx}} \right)_{(x_1, y_1)} \times (x - x_1)$$



NOTES :

1. The point $P(x_1, y_1)$ will satisfy the equation of the curve & the equation of tangent & normal line.
2. If the tangent at any point P on the curve is parallel to X -axis then $dy/dx = 0$ at the point P .
3. If the tangent at any point on the curve is parallel to Y -axis, then $dy/dx = \infty$ or $dx/dy = 0$.
4. If the tangent at any point on the curve is equally inclined to both the axes then $dy/dx = \pm 1$.
5. If the tangent at any point makes equal intercept on the coordinate axes then $dy/dx = \pm 1$.
6. Tangent to a curve at the point $P(x_1, y_1)$ can be drawn even though dy/dx at P does not exist. e.g. $x = 0$ is a tangent to $y = x^{2/3}$ at $(0, 0)$.
7. If a curve passing through the origin be given by a rational integral algebraic equation, the equation of the tangent (or tangents) at the origin is obtained by equating to zero the terms of the lowest degree in the equation. e.g. If the equation of a curve be $x^2 - y^2 + x^3 + 3x^2y - y^3 = 0$, the tangents at the origin are given by $x^2 - y^2 = 0$ i.e. $x + y = 0$ and $x - y = 0$.





(IV) (a) Length of the tangent (PT) = $\frac{y_1 \sqrt{1 + [f'(x_1)]^2}}{f'(x_1)}$

(b) Length of Subtangent (MT) = $\frac{y_1}{f'(x_1)}$

(c) Length of Normal (PN) = $y_1 \sqrt{1 + [f'(x_1)]^2}$

(d) Length of Subnormal (MN) = $y_1 f'(x_1)$

(V) Differential :

The differential of a function is equal to its derivative multiplied by the differential of the independent variable. Thus if, $y = \tan x$ then $dy = \sec^2 x \, dx$.

In general $dy = f'(x) \, dx$.

NOTES :

$d(c) = 0$ where 'c' is a constant.

$$d(u + v - w) = du + dv - dw$$

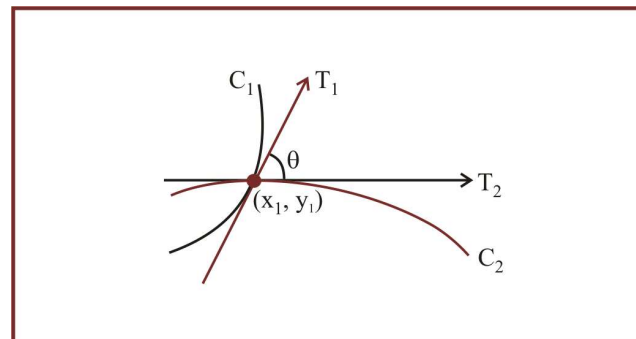
$$d(uv) = u \, dv + v \, du$$

* The relation $dy = f'(x) \, dx$ can be written as

$\frac{dy}{dx} = f'(x)$; thus the quotient of the differentials of 'y' and 'x' is equal to the derivative of 'y' w.r.t. 'x'.

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

where m_1 & m_2 are the slopes of tangents at the intersection point (x_1, y_1) .



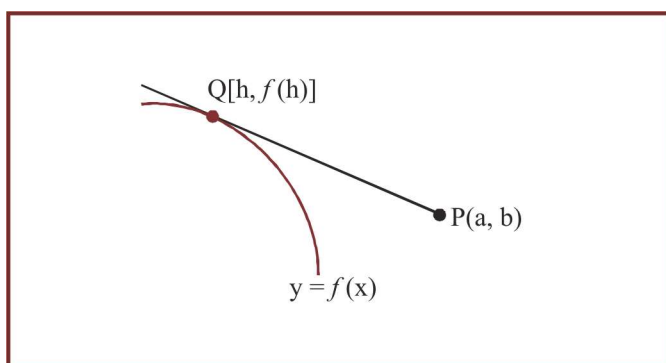
NOTES :

- The angle is defined between two curves if the curves are intersecting. This can be ensured by finding their point of intersection or by graphically.
- If the curves intersect at more than one point then angle between curves is found out with respect to the point of intersection.
- Two curves are said to be **orthogonal** if angle between them at each point of intersection is right angle i.e. $m_1 m_2 = -1$.

3. TANGENT FROM AN EXTERNAL POINT

Given a point P (a, b) which does not lie on the curve $y = f(x)$, then the equation of possible tangents to the curve $y = f(x)$, passing through (a, b) can be found by solving for the point of contact Q.

And equation of tangent is $y - b = \frac{f(h) - b}{h - a} (x - a)$



4. ANGLE BETWEEN THE CURVES

Angle between two intersecting curves is defined as the acute angle between their tangents or the normals at the point of intersection of two curves.

5. SHORTEST DISTANCE BETWEEN TWO CURVES

Shortest distance between two non-intersecting differentiable curves is always along their common normal. (Wherever defined)

6. ERRORS AND APPROXIMATIONS

(a) Errors

Let $y = f(x)$

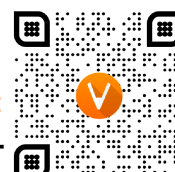
From definition of derivative, $\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \frac{dy}{dx}$

$$\frac{\delta y}{\delta x} = \frac{dy}{dx} \text{ approximately}$$

$$\text{or } \delta y = \left(\frac{dy}{dx} \right) \cdot \delta x \text{ approximately}$$

Definition :

- δx is known as absolute error in x.
- $\frac{\delta x}{x}$ is known as relative error in x.



(iii) $\frac{\delta x}{x} \times 100$ is known as percentage error in x .

NOTES:

δx and δy are known as differentials.

(b) Approximations

From definition of derivative,

$\therefore$ Derivative of $f(x)$ at $(x=a) = f'(a)$

$$\text{or } f'(a) = \lim_{\delta x \rightarrow 0} \frac{f(a + \delta x) - f(a)}{\delta x}$$

$$\text{or } \frac{f(a + \delta x) - f(a)}{\delta x} \rightarrow f'(a) \quad (\text{approximately})$$

$$f(a + \delta x) = f(a) + \delta x f'(a) \quad (\text{approximately})$$

7. DEFINITIONS

1. A function $f(x)$ is called an **Increasing Function** at a point $x = a$ if in a sufficiently small neighbourhood around $x = a$ we have

$$f(a + h) > f(a)$$

$$f(a - h) < f(a)$$

Similarly **Decreasing Function** if

$$f(a + h) < f(a)$$

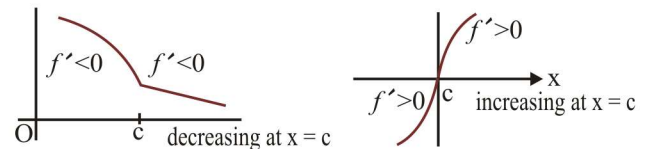
$$f(a - h) > f(a)$$

Above statements hold true irrespective of whether f is non derivable or even discontinuous at $x = a$

2. A differentiable function is called increasing in an interval (a, b) if it is increasing at every point within the interval (but not necessarily at the end points). A function decreasing in an interval (a, b) is similarly defined.
3. A function which in a given interval is increasing or decreasing is called "**Monotonic**" in that interval.
4. Tests for increasing and decreasing of a function at a point : If the derivative $f'(x)$ is positive at a point $x = a$, then the function $f(x)$ at this point is increasing. If it is negative, then the function is decreasing.

NOTES:

Even if $f'(a)$ is not defined, f can still be increasing or decreasing. (Look at the cases below).



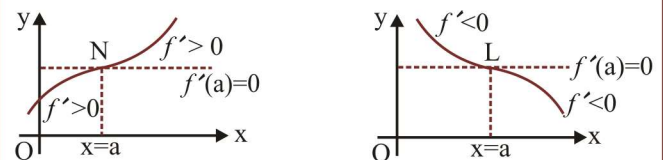
Note : $f'(c)$ is not defined in both the cases.

NOTES:

If $f'(a) = 0$, then for $x = a$ the function may be still increasing or it may be decreasing as shown. It has to be identified by a separate rule.

e.g. $f(x) = x^3$ is increasing at every point.

Note that, $dy/dx = 3x^2$.



NOTES:

1. If a function is invertible it has to be either increasing or decreasing.
2. If a function is continuous, the intervals in which it rises and falls may be separated by points at which its derivative fails to exist.
3. If f is increasing in $[a, b]$ and is continuous then $f(b)$ is the greatest and $f(a)$ is the least value of f in $[a, b]$. Similarly if f is decreasing in $[a, b]$ then $f(a)$ is the greatest value and $f(b)$ is the least value.

5. (a) ROLLE'S Theorem :

Let $f(x)$ be a function of x subject to the following conditions :

- (i) $f(x)$ is a continuous function of x in the closed interval of $a \leq x \leq b$.
- (ii) $f'(x)$ exists for every point in the open interval $a < x < b$.
- (iii) $f(a) = f(b)$.

Then there exists at least one point $x = c$ such that $a < c < b$ where $f'(c) = 0$.

(b) LMVT Theorem :

Let $f(x)$ be a function of x subject to the following conditions :





- (i) $f(x)$ is a continuous function of x in the closed interval of $a \leq x \leq b$.
- (ii) $f'(x)$ exists for every point in the open interval $a < x < b$.

Then there exists at least one point $x = c$ such that

$$a < c < b \text{ where } f'(c) = \frac{f(b) - f(a)}{b - a}$$

Geometrically, the slope of the secant line joining the curve at $x = a$ & $x = b$ is equal to the slope of the tangent line drawn to the curve at $x = c$.

Note the following : Rolles theorem is a special case of LMVT since

$$f(a) = f(b) \Rightarrow f'(c) = \frac{f(b) - f(a)}{b - a} = 0$$

NOTES :

Physical Interpretation of LMVT :

Now $[f(b) - f(a)]$ is the change in the function f as x changes

from a to b so that $\frac{f(b) - f(a)}{b - a}$ is the average rate of change

of the function over the interval $[a, b]$. Also $f'(c)$ is the actual rate of change of the function for $x = c$. Thus, the theorem states that the average rate of change of a function over an interval is also the actual rate of change of the function at some point of the interval. In particular, for instance, the average velocity of a particle over an interval of time is equal to the velocity at some instant belonging to the interval.

This interpretation of the theorem justifies the name "Mean Value" for the theorem.

- (c) Application of rolles theorem for isolating the real roots of an equation $f(x) = 0$

Suppose a & b are two real numbers such that ;

- (i) $f(x)$ & its first derivative $f'(x)$ are continuous for $a \leq x \leq b$.
- (ii) $f(a)$ & $f(b)$ have opposite signs.
- (iii) $f'(x)$ is different from zero for all values of x between a & b .

Then there is one & only one real root of the equation $f(x) = 0$ between a & b .

8. HOW MAXIMA & MINIMA ARE CLASSIFIED

1. Maxima & Minima

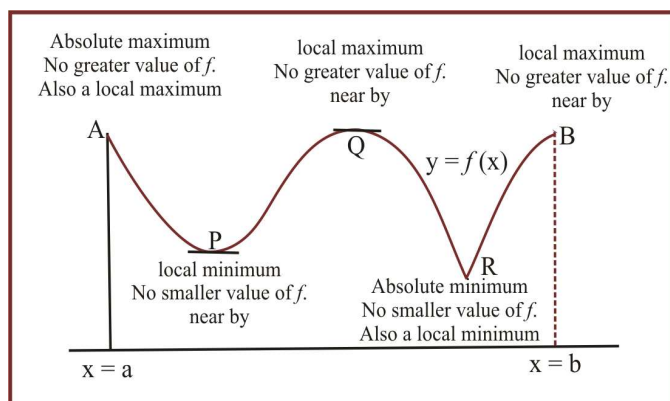
A function $f(x)$ is said to have a local maximum at $x = a$ if $f(a)$ is greater than every other value assumed by $f(x)$ in the immediate neighbourhood of $x = a$. Symbolically

$$\left. \begin{array}{l} f(a) > f(a+h) \\ f(a) > f(a-h) \end{array} \right\} \Rightarrow x=a \text{ gives maxima}$$

for a sufficiently small positive h .

Similarly, a function $f(x)$ is said to have a local minimum value at $x = b$ if $f(b)$ is least than every other value assumed by $f(x)$ in the immediate neighbourhood at $x = b$. Symbolically if

$$\left. \begin{array}{l} f(b) < f(b+h) \\ f(b) < f(b-h) \end{array} \right\} \Rightarrow x=b \text{ gives minima for a sufficiently small positive } h.$$



NOTES :

- The local maximum & local minimum values of a function are also known as local/relative maxima or local/relative minima as these are the greatest & least values of the function relative to some neighbourhood of the point in question.
- The term 'extremum' is used both for maxima or a minima.
- A local maximum (local minimum) value of a function may not be the greatest (least) value in a finite interval.
- A function can have several local maximum & local minimum values & a local minimum value may even be greater than a local maximum value.
- Maxima & minima of a continuous function occur alternately & between two consecutive maxima there is a minima & vice versa.



2. A necessary condition for maxima & minima

If $f(x)$ is a maxima or minima at $x = c$ & if $f'(c)$ exists then $f'(c) = 0$.

NOTES :

- The set of values of x for which $f'(x) = 0$ are often called as stationary points. The rate of change of function is zero at a stationary point.
- In case $f'(c)$ does not exist $f(c)$ may be a maxima or a minima & in this case left hand and right hand derivatives are of opposite signs.
- The greatest (global maxima) and the least (global minima) values of a function f in an interval $[a, b]$ are $f(a)$ or $f(b)$ or are given by the values of x which are critical points.
- Critical points** are those where :
 - $\frac{dy}{dx} = 0$, if it exists; (ii) or it fails to exist

3. Sufficient condition for extreme values

First Derivative Test

$$\left. \begin{array}{l} f'(c-h) > 0 \\ f'(c+h) < 0 \end{array} \right\} \Rightarrow x = c \text{ is a point of local maxima,}$$

where h is a sufficiently small positive quantity

$$\text{Similarly } \left. \begin{array}{l} f'(c-h) < 0 \\ f'(c+h) > 0 \end{array} \right\} \Rightarrow x = c \text{ is a point of local minima,}$$

where h is a sufficiently small positive quantity

Note: $f'(c)$ in both the cases may or may not exist. If it exists, then $f'(c) = 0$.

NOTES :

If $f'(x)$ does not change sign i.e. has the same sign in a certain complete neighbourhood of c , then $f(x)$ is either strictly increasing or decreasing throughout this neighbourhood implying that $f(c)$ is not an extreme value of f .

4. Use of second order derivative in ascertaining the maxima or minima

- $f'(c)$ is a minima of the function f , if $f'(c) = 0$ & $f''(c) > 0$.

- $f'(c)$ is a maxima of the function f , if $f'(c) = 0$ & $f''(c) < 0$.

NOTES :

If $f''(c) = 0$ then the test fails. Revert back to the first order derivative check for ascertaining the maxima or minima.

5. Summary-working rule

First : When possible, draw a figure to illustrate the problem & label those parts that are important in the problem. Constants & variables should be clearly distinguished.

Second : Write an equation for the quantity that is to be maximised or minimised. If this quantity is denoted by 'y', it must be expressed in terms of a single independent variable x . This may require some algebraic manipulations.

Third : If $y = f(x)$ is a quantity to be maximum or minimum, find those values of x for which $dy/dx = f'(x) = 0$.

Fourth : Test each values of x for which $f'(x) = 0$ to determine whether it provides a maxima or minima or neither. The usual tests are :

- If d^2y/dx^2 is positive when $dy/dx = 0$
 $\Rightarrow y$ is minima.
 If d^2y/dx^2 is negative when $dy/dx = 0$
 $\Rightarrow y$ is maxima.
 If $d^2y/dx^2 = 0$ when $dy/dx = 0$, the test fails.

- If $\frac{dy}{dx}$ is $\begin{array}{ll} \text{positive} & \text{for } x < x_0 \\ \text{zero} & \text{for } x = x_0 \\ \text{negative} & \text{for } x > x_0 \end{array} \Rightarrow \text{a maxima occurs at } x = x_0$.

But if dy/dx changes sign from negative to zero to positive as x advances through x_0 , there is a minima. If dy/dx does not change sign, neither a maxima nor a minima. Such points are called **INFLECTION POINTS**.

Fifth : If the function $y = f(x)$ is defined for only a limited range of values $a \leq x \leq b$ then examine $x = a$ & $x = b$ for possible extreme values.

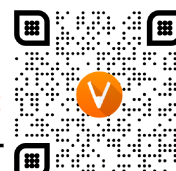
Sixth : If the derivative fails to exist at some point, examine this point as possible maxima or minima.

(In general, check at all Critical Points).

NOTES :

- If the sum of two positive numbers x and y is constant then their product is maximum if they are equal, i.e. $x + y = c$, $x > 0$, $y > 0$, then

$$xy = \frac{1}{4}[(x+y)^2 - (x-y)^2]$$





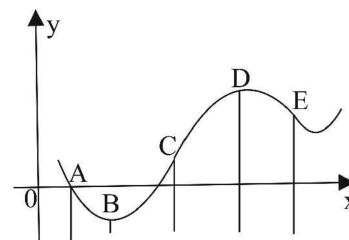
- If the product of two positive numbers is constant then their sum is least if they are equal.
i.e. $(x+y)^2 = (x-y)^2 + 4xy$

9. USEFUL FORMULAE OF MENSURATION TO REMEMBER

- Volume of a cuboid = lwh .
- Surface area of a cuboid = $2(lb + bh + hl)$.
- Volume of a prism = area of the base $\times$ height.
- Lateral surface of a prism = perimeter of the base $\times$ height.
- Total surface of a prism = lateral surface + 2 area of the base
(Note that lateral surfaces of a prism are all rectangles).
- Volume of a pyramid = $\frac{1}{3}$ area of the base $\times$ height.
- Curved surface of a pyramid = $\frac{1}{2}$ (perimeter of the base) $\times$ slant height.
(Note that slant surfaces of a pyramid are triangles).
- Volume of a cone = $\frac{1}{3}\pi r^2 h$.
- Curved surface of a cylinder = $2\pi rh$.
- Total surface of a cylinder = $2\pi rh + 2\pi r^2$.
- Volume of a sphere = $\frac{4}{3}\pi r^3$.
- Surface area of a sphere = $4\pi r^2$.
- Area of a circular sector = $\frac{1}{2}r^2\theta$, where θ is in radians.

10. SIGNIFICANCE OF THE SIGN OF 2ND ORDER DERIVATIVE AND POINTS OF INFLECTION

The sign of the 2nd order derivative determines the concavity of the curve. Such point such as C & E on the graph where the concavity of the curve changes are called the points of inflection. From the graph we find that if :



- $\frac{d^2y}{dx^2} > 0 \Rightarrow$ concave upwards
- $\frac{d^2y}{dx^2} < 0 \Rightarrow$ concave downwards.

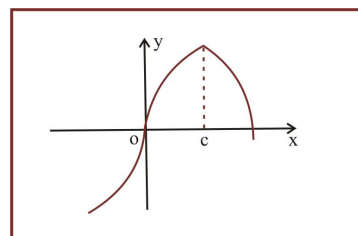
At the point of inflection we find that $\frac{d^2y}{dx^2} = 0$ and $\frac{d^2y}{dx^2}$ changes sign.

Inflection points can also occur if $\frac{d^2y}{dx^2}$ fails to exist (but changes its sign). For example, consider the graph of the function defined as,

$$f(x) = \begin{cases} x^{3/5} & \text{for } x \in (-\infty, 1) \\ 2 - x^2 & \text{for } x \in (1, \infty) \end{cases}$$

NOTES:

The graph below exhibits two critical points one is a point of local maximum ($x = c$) & the other a point of inflection ($x = 0$). This implies that not every Critical Point is a point of extrema.





SOLVED EXAMPLES

Example – 1

If the function $f(x) = 2x^3 - 9ax^2 + 12a^2x + 1$, where $a > 0$, attains its maximum and minimum at p and q respectively such that $p^2 = q$, then a equals

- (a) 1 (b) 2
(c) $\frac{1}{2}$ (d) 3

Ans. (b)

Sol. For maximum and minima $f'(x) = 0$

$$\Rightarrow 6x^2 - 18ax + 12a^2 = 0$$

$$\Rightarrow x = a, 2a$$

$$\text{Also, } f''(x) = 12x - 18a$$

$$f''(a) < 0 \Rightarrow \text{max at } 'a'$$

$$f''(2a) > 0 \Rightarrow \text{min at } '2a'$$

$$\text{So, } p = a \text{ and } q = 2a$$

$$\text{Given } p^2 = q$$

$$\Rightarrow a^2 = 2a \Rightarrow a^2 - 2a = 0$$

$$\Rightarrow a(a-2) = 0 \Rightarrow a = 0, a = 2$$

Example – 2

The real number x when added to its inverse gives the minimum value of the sum at x equal to

- (a) 1 (b) -1
(c) -2 (d) 2

Ans. (a)

Sol. $f(x) = x + \frac{1}{x}$

$$f'(x) = 1 - \frac{1}{x^2} \text{ and } f''(x) = \frac{2}{x^3}$$

$$\text{Now } f'(x) = 0$$

$$\Rightarrow x = \pm 1$$

$$\therefore f''(1) > 0$$

$$\Rightarrow x = 1 \text{ is point of minima.}$$

Example – 3

A function $y = f(x)$ has a second order derivative $f'' = 6(x-1)$. If its graph passes through the point $(2, 1)$ and at that point the tangent to the graph is $y = 3x - 5$, then the function is

- (a) $(x-1)^2$ (b) $(x-1)^3$
(c) $(x+1)^3$ (d) $(x+1)^2$

Ans. (b)

Sol. Given $f''(x) = 6(x-1)$

$$\Rightarrow f'(x) = \frac{6(x-1)^2}{2} + c$$

$$\Rightarrow 3 = 3 + c \left[\because \text{tangent at } x = 2 \text{ is } y = 3x + 5 \right]$$

$$\Rightarrow c = 0 \Rightarrow f'(2) = 3$$

$$\text{so } f'(x) = 3(x-1)^2$$

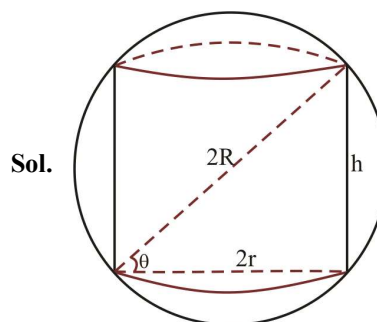
$$\Rightarrow f(x) = (x-1)^3 + c_1 \text{ as curve passes through } (2, 1)$$

$$\Rightarrow 1 = (2-1)^3 + c_1$$

$$\Rightarrow c_1 = 0 \text{ hence } f(x) = (x-1)^3$$

Example – 4

Find the maximum surface area of a cylinder that can be inscribed in a given sphere of radius R .



Sol.

Let r be the radius and h be the height of cylinder. Consider



the right triangle shown in the figure.

$$2r = 2R \cos \theta \text{ and } h = 2R \sin \theta$$

Surface area of the cylinder = $2\pi rh + 2\pi r^2$

$$\Rightarrow S(\theta) = 4\pi R^2 \sin \theta \cos \theta + 2\pi R^2 \cos^2 \theta$$

$$\Rightarrow S(\theta) = 2\pi R^2 \sin 2\theta + 2\pi R^2 \cos^2 \theta$$

$$\Rightarrow S'(\theta) = 4\pi R^2 \cos 2\theta - 2\pi R^2 \sin 2\theta$$

$$S'(\theta) = 0 \Rightarrow 2 \cos 2\theta - \sin 2\theta = 0$$

$$\Rightarrow \tan 2\theta = 2 \Rightarrow \theta = \theta_0 = \frac{1}{2} \tan^{-1} 2$$

$$S''(\theta_0) = -8\pi R^2 \sin 2\theta - 4\pi R^2 \cos 2\theta$$

$$S''(\theta) = -8\pi R^2 \left(\frac{2}{\sqrt{5}} \right) - 4\pi R^2 \left(\frac{1}{\sqrt{5}} \right) < 0$$

Hence surface area is maximum for $\theta = \theta_0 = \frac{1}{2} \tan^{-1} 2$

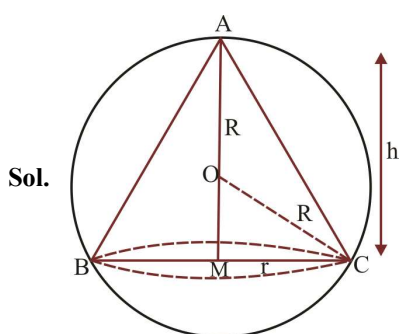
$$S_{\max} = 2\pi R^2 \sin 2\theta_0 + 2\pi R^2 \cos^2 \theta_0$$

$$\Rightarrow S_{\max} = 2\pi R^2 \left(\frac{2}{\sqrt{5}} \right) + 2\pi R^2 \left(\frac{1+1/\sqrt{5}}{2} \right)$$

$$\Rightarrow S_{\max} = \pi R^2 (1 + \sqrt{5})$$

Example - 5

Find the semi-vertical angle of the cone of maximum curved surface area that can be inscribed in a given sphere of radius R .



Let h be the height of cone and r be the radius of the cone.

Consider the right $\triangle OMC$ where O is the centre of sphere and AM is perpendicular to the base BC of cone.

$$OM = h - R, OC = R, MC = r$$

$$R^2 = (h - R)^2 + r^2 \quad \dots (i)$$

$$\text{and } r^2 + h^2 = R^2 \quad \dots (ii)$$

where l is the slant height of cone.

Curve surface area = $C = \pi r l$

Using (i) and (ii), express C in terms of h only.

$$C = \pi r \sqrt{r^2 + h^2} \Rightarrow C = \pi \sqrt{2hR - h^2} \sqrt{2hR}$$

We will maximise C^2 .

$$\text{Let } C^2 = f(h) = 2\pi^2 h R (2hR - h^2) = 2\pi^2 R (2h^2 R - h^3)$$

$$\Rightarrow f'(h) = 2\pi^2 R (4hR - 3h^2)$$

$$f'(h) = 0 \Rightarrow 4hR - 3h^2 = 0 \Rightarrow h(4R - 3h) = 0$$

$$\Rightarrow h = 4R/3.$$

$$f''(h) = 2\pi^2 R (4R - 6h)$$

$$f''\left(\frac{4R}{3}\right) = 2\pi^2 R (4R - 8R) < 0$$

Hence curved surface area is maximum for $h = \frac{4R}{3}$

Using (i), we get :

$$r^2 = 2hR - h^2 = \frac{8R^2}{9} \Rightarrow r = \frac{2\sqrt{2}}{3} R$$

$$\text{Semi-vertical angle } = \theta = \tan^{-1} r/h = \tan^{-1} 1/\sqrt{2}.$$

Example - 6

If f and g are differentiable functions in $[0, 1]$ satisfying $f(0) = 2 = g(1)$, $g(0) = 0$ and $f(1) = 6$, then for some $c \in]0, 1[$:

- (a) $f'(c) = 2g'(c)$ (b) $2f'(c) = g'(c)$
(c) $2f'(c) = 3g'(c)$ (d) $f'(c) = g'(c)$

Ans. (a)

Sol. By LMVT

$$f'(c) = \frac{f(1) - f(0)}{1 - 0} = \frac{6 - 2}{1} = 4$$

$$g'(c) = \frac{g(1) - g(0)}{1 - 0} = \frac{2 - 0}{1} \Rightarrow 2$$

$$\Rightarrow f'(c) = 2g'(c)$$

Example - 7

If $2a + 3b + 6c = 0$ ($a, b, c, \in \mathbb{R}$), then the quadratic equation $ax^2 + bx + c = 0$ has

- (a) at least one root in $(0, 1)$ (b) at least one root in $[2, 3]$
(c) at least one root in $[4, 5]$ (d) none of the above

Ans. (a)



Sol. Let us consider $f(x) = \frac{ax^3}{3} + \frac{bx^2}{2} + cx$

$$\therefore f(0)=0 \text{ and } f(1)=\frac{a}{3}+\frac{b}{2}+c$$

$$= \frac{2a+3b+6c}{6}=0 \text{ (given).}$$

As $f(0)=f(1)=0$ and $f(x)$ is continuous and differentiable also in $[0,1]$.

$\therefore$ By Rolle's theorem $f'(x)=0$

$\Rightarrow ax^2+bx+c=0$ has at least one root in the interval $(0,1)$.

Example – 8

Find the approximate value of $(0.007)^{1/3}$.

Sol. Let $f(x) = (x)^{1/3}$

$$\text{Now, } f(x + \delta x) - f(x) = f'(x) \cdot \delta x = \frac{\delta x}{3x^{2/3}}$$

we may write, $0.007 = 0.008 - 0.001$

Taking $x = 0.008$ and $\delta x = -0.001$, we have

$$f(0.007) - f(0.008) = -\frac{0.001}{3(0.008)^{2/3}}$$

$$\text{or } f(0.007) - (0.008)^{1/3} = -\frac{0.001}{3(0.2)^2} \text{ or}$$

$$f(0.007) = 0.2 - \frac{0.001}{3(0.04)} = 0.2 - \frac{1}{120} = \frac{23}{120}$$

$$\text{Hence } (0.007)^{1/3} = \frac{23}{120}.$$

Example – 9

Discuss concavity and convexity and find points of inflexion of $y = x^2 e^{-x}$.

Sol. Let $f(x) = x^2 e^{-x}$.

Differentiate w.r.t. x to get :

$$\begin{aligned} f'(x) &= e^{-x}(2x) + (-e^{-x})x^2 \\ &= xe^{-x}[2-x] \end{aligned}$$

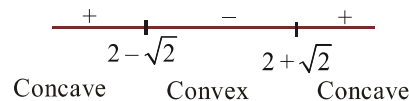
Differentiate again w.r.t. x to get :

$$\begin{aligned} f''(x) &= (2-2x)e^{-x} + (2x-x^2)(-e^{-x}) \\ &= e^{-x}(2-2x-2x+x^2) \end{aligned}$$

$$= e^{-x}(x^2 - 4x + 2)$$

$$= e^{-x}(x - (2 - \sqrt{2}))(x - (2 + \sqrt{2}))$$

See the figure and observe how the sign of $f''(x)$ changes.



Sign of $f''(x)$ is changing at $x = 2 \pm \sqrt{2}$.

Therefore points of inflexion of $f(x)$ are $x = 2 \pm \sqrt{2}$.

$$f''(x) \geq 0 \quad \forall x \in [-\infty, 2 - \sqrt{2}] \cup [2 + \sqrt{2}, \infty]$$

Therefore $f(x)$ is “Concave upward”

$$\forall x \in (-\infty, 2 - \sqrt{2}] \cup [2 + \sqrt{2}, \infty)$$

Similarly we can observe

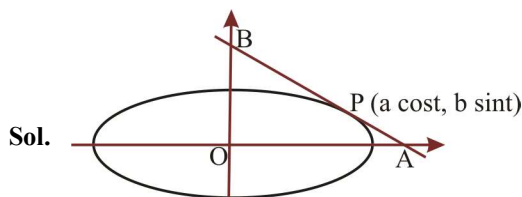
$$f''(x) \leq 0 \quad \forall x \in [2 - \sqrt{2}, 2 + \sqrt{2}]$$

Therefore $f(x)$ is “Convex downwards”

$$\forall x \in [2 - \sqrt{2}, 2 + \sqrt{2}]$$

Example – 10

Prove that the minimum intercept made by axes on the tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $a + b$. Also find the ratio in which the point of contact divides this intercept.



Sol.

Intercept made by the axes on the tangent is the length of the portion of the tangent intercepted between the axes.

Consider a point P on the ellipse whose coordinates are

$$x = a \cos t, y = b \sin t \text{ (where } t \text{ is the parameter)}$$

$$\frac{dx}{dt} = -a \sin t$$

$$\frac{dy}{dt} = b \cos t$$

The equation of the tangent is :



$$y - b \sin t = \frac{b \cos t}{-a \sin t} (x - a \cos t)$$

$$\Rightarrow \frac{x}{a} \cos t + \frac{y}{b} \sin t = 1$$

$$\Rightarrow OA = \frac{a}{\cos t}, OB = \frac{b}{\sin t}$$

$$\text{Length of intercept} = l = AB = \sqrt{\frac{a^2}{\cos^2 t} + \frac{b^2}{\sin^2 t}}$$

We will minimise l^2 .

$$\text{Let } l^2 = f(t) = a^2 \sec^2 t + b^2 \csc^2 t$$

$$\Rightarrow f'(t) = 2a^2 \sec^2 t \tan t - 2b^2 \csc^2 t \cot t$$

$$f'(t) = 0 \Rightarrow a^2 \sin^4 t = b^2 \cos^4 t$$

$$\Rightarrow t = \tan^{-1} \sqrt{b/a}$$

$$f''(t) = 2a^2 (\sec^4 t + 2 \tan^2 t \sec^2 t)$$

$$+ 2b^2 (\csc^4 t + 2 \cot^2 t \csc^2 t), \text{ which is positive.}$$

$$\text{Hence } f(t) \text{ is minimum for } \tan t = \sqrt{\frac{b}{a}}.$$

$$\Rightarrow l_{\min} = \sqrt{a^2(1 + b/a) + b^2(1 + a/b)}$$

$$\Rightarrow l_{\min} = a + b$$

$$PA^2 = \left(a \cos t - \frac{a}{\cos t}\right)^2 + b^2 \sin^2 t$$

$$= \frac{a^2 \sin^4 t}{\cos^2 t} + b^2 \sin^2 t$$

$$= (a^2 \tan^2 t + b^2) \sin^2 t$$

$$= (ab + b^2) \frac{b}{a + b} = b^2 \Rightarrow PA = b$$

$$\text{Hence } \frac{PA}{PB} = \frac{b}{a} \Rightarrow P \text{ divides } AB \text{ in the ratio } b : a$$

Example – 11

Find the equation of tangent to the curve $x^{2/3} + y^{2/3} = a^{2/3}$ at (x_0, y_0) . Hence prove that the length of the portion of tangent intercepted between the axes is constant.

Sol. Method 1 :

$$x^{2/3} + y^{2/3} = a^{2/3} \quad \text{Differentiating wrt } x,$$

$$\frac{2}{3} x^{-1/3} + \frac{2}{3} y^{-1/3} \frac{dy}{dx} = 0$$

$$\Rightarrow \left. \frac{dy}{dx} \right|_{(x_0, y_0)} = - \left(\frac{x_0}{y_0} \right)^{-1/3}$$

$$\Rightarrow \left. \frac{dy}{dx} \right|_{(x_0, y_0)} = - \left(\frac{y_0}{x_0} \right)^{1/3}$$

$$\Rightarrow \text{equation is } y - y_0 = - \left(\frac{y_0}{x_0} \right)^{1/3} (x - x_0)$$

$$\Rightarrow x_0^{1/3} y - y_0 x_0^{1/3} = -x y_0^{1/3} + x_0 y_0^{1/3}$$

$$\Rightarrow x y_0^{1/3} + y x_0^{1/3} = x_0 y_0^{1/3} + y_0 x_0^{1/3}$$

$$\Rightarrow \frac{x y_0^{1/3}}{x_0^{1/3} y_0^{1/3}} + \frac{y x_0^{1/3}}{x_0^{1/3} y_0^{1/3}} = x_0^{2/3} + y_0^{2/3}$$

$$\Rightarrow \text{equation of tangent is : } \frac{x}{x_0^{1/3}} + \frac{y}{y_0^{1/3}} = a^{2/3}$$

Length intercepted between the axes :

$$\text{length} = \sqrt{(\text{x intercept})^2 + (\text{y intercept})^2}$$

$$\text{x intercept} = x_0^{1/3} a^{2/3}$$

$$\text{y intercept} = y_0^{1/3} a^{2/3}$$

$$= \sqrt{(x_0^{1/3} a^{2/3})^2 + (y_0^{1/3} a^{2/3})^2}$$

$$= \sqrt{x_0^{2/3} a^{4/3} + y_0^{2/3} a^{4/3}}$$

$$= a^{2/3} \sqrt{x_0^{2/3} + y_0^{2/3}}$$

$$= a^{2/3} \sqrt{a^{2/3}}$$

$$= a \text{ i.e. constant.}$$

Method 2 :

Express the equation in parametric form

$$x = a \sin^3 t, \quad y = a \cos^3 t$$

$$\frac{dx}{dt} = 3a \sin^2 t \cos t, \quad \frac{dy}{dt} = -3a \cos^2 t \sin t$$

Equation of tangent is :

$$(y - a \cos^3 t) = \frac{-3a \cos^2 t \sin t}{3a \sin^2 t \cos t} (x - a \sin^3 t)$$

$$\Rightarrow y \sin t - a \sin t \cos^3 t = -x \cos t + a \sin^3 t \cos t$$

$$\Rightarrow x \cos t + y \sin t = a \sin t \cos t$$

$$\Rightarrow \frac{x}{\sin t} + \frac{y}{\cos t} = a$$

in terms of (x_0, y_0) equation is :

$$\frac{x}{(x_0/a)^{1/3}} + \frac{y}{(y_0/a)^{1/3}} = a$$

Length of tangent intercepted between axes



$$= \sqrt{(x_{\text{int}})^2 + (y_{\text{int}})^2}$$

$$= \sqrt{a^2 \sin^2 t + a^2 \cos^2 t} = a \text{ which is constant}$$

NOTES :

1. The parametric form is very useful in these type of problems.
2. Equation of tangent can also be obtained by substituting $b = a$ and $m = 2/3$ in the result

$$\frac{x}{a} \left(\frac{x_0}{a} \right)^{m-1} + \frac{y}{b} \left(\frac{y_0}{b} \right)^{m-1} = 1.$$

Example – 12

For the curve $xy = c^2$, prove that

- (i) the intercept between the axes on the tangent at any point is bisected at the point of contact.
- (ii) the tangent at any point makes with the co-ordinate axes a triangle of constant area.

Sol. Let the equation of the curve in parametric form be $x = ct, y = c/t$

$$\frac{dx}{dt} = c$$

$$\frac{dy}{dt} = \frac{-c}{t^2}$$

Let the point of contact be $(ct, c/t)$

Equation of tangent is :

$$y - c/t = \frac{-c/t^2}{c} (x - ct)$$

$$\Rightarrow t^2 y - ct = -x + ct$$

$$\Rightarrow x + t^2 y = 2ct \quad \dots\dots(i)$$

- (i) Let the tangent cut the x and y axes at A and B respectively.

$$\text{Writing the equations as : } \frac{x}{2ct} + \frac{y}{2c/t} = 1$$

$$\Rightarrow x_{\text{intercept}} = 2ct, y_{\text{intercept}} = 2c/t$$

$$\Rightarrow A \equiv (2ct, 0) \text{ and } B \equiv \left(0, \frac{2c}{t}\right)$$

$$\text{mid point of } AB \equiv \left(\frac{2ct+0}{2}, \frac{0+2c/t}{2}\right) \equiv (ct, c/t)$$

Hence, the point of contact bisects AB.

- (ii) If O is the origin,

Area of triangle $\Delta OAB = 1/2 (OA) (OB)$

$$= \frac{1}{2} (2ct) \left(\frac{2c}{t}\right)$$

$$= 2c^2$$

i.e. constant for all tangents because it is independent of t.

Example – 13

Find critical points of $f(x) = x^{2/3}(2x-1)$.

Sol. $f(x) = 2x^{5/3} - x^{2/3}$

Differentiate w.r.t. x to get,

$$f'(x) = \frac{10}{3} x^{2/3} - \frac{2}{3} x^{-1/3} = \frac{2}{3} \frac{(5x-1)}{x^{1/3}}.$$

For critical points,

$$f'(x) = 0 \text{ or } f'(x) \text{ is not defined.}$$

Put $f'(x) = 0$ to get $x = \frac{1}{5}$.

$f'(x)$ is not defined when denominator = 0.

$$\Rightarrow x^{1/3} = 0 \quad \Rightarrow x = 0$$

Now we can say that $x = 0$ and $x = \frac{1}{5}$ are critical points as

$$f(x) \text{ exists at both } x = 0 \text{ and } x = \frac{1}{5}.$$

$$\Rightarrow \text{Critical points of } f(x) \text{ are } x = 0, x = \frac{1}{5}.$$

Example – 14

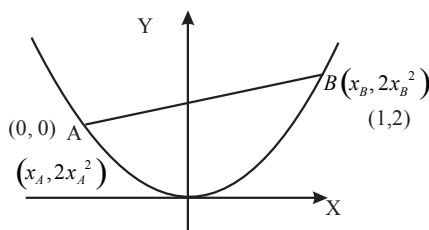
The ends A and B of a rod of length $\sqrt{5}$ are sliding along the curve $y = 2x^2$. Let x_A and x_B be the x-coordinate of the ends. At the moment when A is at (0, 0) and B is at (1, 2),

find the value of the derivative $\frac{dx_B}{dx_A}$.

Sol. We have $y = 2x^2$

$$(AB)^2 = (x_B - x_A)^2 + (2x_B^2 - 2x_A^2)^2 = 5 \left(AB \right) = \sqrt{5}$$

or $(x_B - x_A)^2 + 4(x_B^2 - x_A^2)^2 = 5$



Differentiating w.r.t. x_A and denoting $\frac{dx_B}{dx_A} = D$

$$2(x_B - x_A)(D - 1) + 8(x_B^2 - x_A^2)(2x_B D - 2x_A) = 0$$

Put $x_A = 0, x_B = 1$

$$2(1 - 0)(D - 1) + 8(1 - 0)(2D - 0) = 0$$

$$2D - 2 + 16D = 0 \Rightarrow D = 1/9 \Rightarrow \frac{dx_B}{dx_A} = \frac{1}{9}$$

Example – 15

The equation of the tangent to the curve $y = x + \frac{4}{x^2}$, that is parallel to the x-axis, is

- (a) $y = 0$ (b) $y = 1$
(c) $y = 2$ (d) $y = 3$

Ans. (d)

Sol. Tangent is parallel to x-axis

$$\Rightarrow \frac{dy}{dx} = 0 \Rightarrow 1 - \frac{8}{x^3} = 0 \Rightarrow x = 2 \Rightarrow y = 3$$

Example – 16

For $0 < x \leq \frac{\pi}{2}$, show that $x - \frac{x^3}{6} < \sin x < x$.

Sol. Let $f(x) = \sin x - x$

$$f'(x) = \cos x - 1 = -(1 - \cos x) = -2 \sin^2 x/2 < 0$$

$\therefore f(x)$ is a decreasing function

for $x > 0$

$$\therefore f(x) < f(0) \Rightarrow \sin x - x < 0 \quad (\because f(0) = 0)$$

$$\Rightarrow \sin x < x \quad \dots\dots(1)$$

$$\text{Now let } g(x) = x - \frac{x^3}{6} - \sin x \quad \therefore g'(x) = 1 - \frac{x^2}{2} - \cos x$$

To find sign of $g'(x)$ we consider $\phi(x) = 1 - \frac{x^2}{2} - \cos x$

$$\therefore \phi'(x) = -x + \sin x < 0 \quad [\text{From (1)}]$$

$$\therefore \phi(x) \text{ is a decreasing function} \Rightarrow g'(x) < 0$$

$$\Rightarrow g(x) \text{ is a decreasing function} \quad \because x > 0$$

$$\Rightarrow g(x) < g(0)$$

$$\Rightarrow x - \frac{x^3}{6} - \sin x < 0 \quad (\because g(0) = 0)$$

$$\Rightarrow x - \frac{x^3}{6} < \sin x \quad \dots\dots(2)$$

Combining (1) and (2) we get $x - \frac{x^3}{6} < \sin x < x$.

Example – 17

Show that $x/(1+x) < \log(1+x) < x$ for $x > 0$.

Sol. Let $f(x) = \log(1+x) - \frac{x}{1+x}$

$$f'(x) = \frac{1}{1+x} - \frac{(1+x) - x}{(1+x)^2}$$

$$f'(x) = \frac{x}{(1+x)^2} > 0 \text{ for } x > 0$$

$$\Rightarrow f(x) \text{ is increasing.}$$

Hence $x > 0 \Rightarrow f(x) > f(0)$ by the definition of the increasing function.

$$\Rightarrow \log(1+x) - \frac{x}{1+x} > \log(1+0) - \frac{0}{1+0}$$

$$\Rightarrow \log(1+x) - \frac{x}{1+x} > 0$$

$$\Rightarrow \log(1+x) > \frac{x}{1+x} \quad \dots\dots(i)$$

Now, let $g(x) = x - \log(1+x)$

$$g'(x) = 1 - \frac{1}{1+x} = \frac{x}{1+x} > 0 \text{ for } x > 0$$

$$\Rightarrow g(x) \text{ is increasing.}$$

Hence $x > 0 \Rightarrow g(x) > g(0)$

$$\Rightarrow x - \log(1+x) > 0 - \log(1+0)$$



$$\Rightarrow x - \log(1+x) > 0$$

$$\Rightarrow x > \log(1+x) \quad \dots (ii)$$

Combining (i) and (ii), we get :

$$\frac{x}{1+x} < \log(1+x) < x$$

Example – 18

Angle between the tangents to the curve $y = x^2 - 5x + 6$ at the points (2, 0) and (3, 0) is

- (a) $\pi/2$ (b) $\pi/3$
(c) $\pi/6$ (d) $\pi/4$

Ans. (a)

Sol. Given equation $y = x^2 - 5x + 6$, given point (2, 0), (3, 0)

$$\therefore \frac{dy}{dx} = 2x - 5$$

$$\text{say } m_1 = \left(\frac{dy}{dx} \right)_{x=2, y=0} = 4 - 5 = -1$$

$$\text{and } m_2 = \left(\frac{dy}{dx} \right)_{x=3, y=0} = 6 - 5 = 1$$

$$\text{since } m_1 m_2 = -1$$

$$\Rightarrow \text{tangents are at right angle i.e. } \frac{\pi}{2}$$

Example – 19

Determine the absolute extrema for the following function and interval.

$$g(t) = 2t^3 + 3t^2 - 12t + 4 \text{ on } [0, 2]$$

Sol. Differentiate w.r.t. t

$$g'(t) = 6t^2 + 6t - 12 = 6(t+2)(t-1)$$

Note that this problem is almost identical to the first problem. The only difference is the interval that we were working on.

The first step is to again find the critical points. From the first example we know these are $t = -2$ and $t = 1$. At this point it's important to recall that we only want the critical points that actually fall in the interval in question. This means that we only want $t = 1$ since $t = -2$ falls outside the interval so reject it.

Now for absolute maxima

We have,

$$\text{Max } \{g(1), g(0), g(2)\}$$

$$\text{i.e., Max } \{-3, 4, 8\}$$

On comparing all these values we get $g(t)$ has absolute max. as 8 at $t = 2$ and similarly absolute minimum of $g(t)$ is -3 at $t = 1$.

Example – 20

Let f be differentiable for all x .

If $f(1) = -2$ and $f'(x) \geq 2$ for $x \in [1, 6]$, then

- (a) $f(6) < 8$ (b) $f(6) \geq 8$
(c) $f(6) = 5$ (d) $f(6) < 5$

Ans. (b)

Sol. Using LMVT,

$$\frac{f(6) - f(1)}{6 - 1} = f'(c) \text{ for some } c \in (1, 6)$$

$$\Rightarrow \frac{f(6) - (-2)}{5} \geq 2$$

$$\Rightarrow f(6) \geq 8$$

Example – 21

Find points of local maximum and local minimum of $f(x) = x^{2/3}(2x - 1)$.

Sol. Let $f(x) = 2x^{5/3} - x^{2/3}$

Differentiate w.r.t. x to get :

$$f'(x) = 2 \left(\frac{5}{3} \right) x^{2/3} - \frac{2}{3} x^{-1/3} = \frac{2}{3} \frac{(5x - 1)}{x^{1/3}}$$

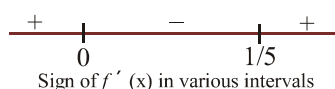
By taking $f'(x) = 0$ or $f'(x)$ is not defined.

Critical points of $f(x)$ are $x = \frac{1}{5}$ and $x = 0$.

Using the following figure, we can determine how sign of

$f'(x)$ is changing at $x = 0$ and $x = \frac{1}{5}$.

from figure,



$x = 0$ is point of local maximum as sign of $f'(x)$ changes from positive to negative and $x = \frac{1}{e}$ is a point of local minimum as sign of $f'(x)$ is changing from negative to positive.

Example – 22

Find the local maximum and local minimum values of the function $y = x^x$.

Sol. Let $f(x) = y = x^x$

$$\Rightarrow \log y = x \log x$$

$$\Rightarrow \frac{1}{y} \frac{dy}{dx} = x \frac{1}{x} + \log x$$

$$\Rightarrow \frac{dy}{dx} = x^x (1 + \log x)$$

$$f'(x) = 0 \Rightarrow x^x (1 + \log x) = 0$$

$$\Rightarrow \log x = -1 \Rightarrow x = e^{-1} = 1/e.$$

Method I : (First Derivative Test)

$$f'(x) = x^x (1 + \log x)$$

$$f'(x) = x^x \log x$$

$$x < 1/e \Rightarrow \log x < -1$$

$$\Rightarrow f'(x) < 0$$

$$x > 1/e \Rightarrow \log x > -1$$

$$\Rightarrow f'(x) > 0$$

The sign of $f'(x)$ changes from -ve to +ve around $x = 1/e$.

In other words, $f(x)$ changes from decreasing to increasing at $x = 1/e$.

Hence $x = 1/e$ is a point of local minimum.

$$\text{Local minimum value} = (1/e)^{1/e} = e^{-1/e}.$$

Method II : (Second Derivative Test)

$$f''(x) = (1 + \log x) \frac{d}{dx} x^x + x^x \left(\frac{1}{x} \right)$$

$$= x^x (1 + \log x)^2 + x^{x-1}$$

$$f''(1/e) = 0 + (e)^{(e-1)/e} > 0.$$

Hence $x = 1/e$ is a point of local minimum.

$$\text{Local minimum value is } (1/e)^{1/e} = e^{-1/e}.$$

Example – 23

The function $g(x) = \frac{x}{2} + \frac{2}{x}$ has a local minimum at

- (a) $x = 2$ (b) $x = -2$
(c) $x = 0$ (d) $x = 1$

Ans. (a)

Sol. Let $g(x) = \frac{x}{2} + \frac{2}{x}$

$$\therefore g'(x) = \frac{1}{2} - \frac{2}{x^2}$$

$$\text{for maxima and minima } g'(x) = 0 \Rightarrow x = \pm 2$$

$$\text{Again } g''(x) = \frac{4}{x^3} > 0 \text{ for } x = 2$$

$$< 0 \text{ for } x = -2 \therefore x = 2 \text{ is point of minima}$$

Example – 24

Suppose the cubic $x^3 - px + q$ has three distinct real roots where $p > 0$ and $q > 0$. Then which one of the following holds?

- (a) The cubic has maxima at both $\sqrt{\frac{p}{3}}$ and $-\sqrt{\frac{p}{3}}$
(b) The cubic has minima at $\sqrt{\frac{p}{3}}$ and maxima at $-\sqrt{\frac{p}{3}}$
(c) The cubic has minima at $-\sqrt{\frac{p}{3}}$ and maxima at $\sqrt{\frac{p}{3}}$
(d) The cubic has minima at both $\sqrt{\frac{p}{3}}$ and $-\sqrt{\frac{p}{3}}$

Ans. (b)

Sol. Let $f(x) = x^3 - px + q$



Now $f'(x) = 0$, i.e. $3x^2 - p = 0$

$$\Rightarrow x = -\sqrt{\frac{p}{3}}, \sqrt{\frac{p}{3}}$$

Also, $f''(x) = 6x \Rightarrow f''\left(-\sqrt{\frac{p}{3}}\right) < 0$ and $f''\left(\sqrt{\frac{p}{3}}\right) > 0$

Thus maxima at $-\sqrt{\frac{p}{3}}$ and minima at $\sqrt{\frac{p}{3}}$

Example – 25

Given $P(x) = x^4 + ax^3 + bx^2 + cx + d$ such that $x = 0$ is the only real root of $P'(x) = 0$. If $P(-1) < P(1)$, then in the interval $[-1, 1]$

- (a) $P(-1)$ is the minimum and $P(1)$ is the maximum of P
- (b) $P(-1)$ is not minimum but $P(1)$ is the maximum of P
- (c) $P(-1)$ is the minimum and $P(1)$ is not the maximum of P
- (d) neither $P(-1)$ is the minimum nor $P(1)$ is the maximum of P

Ans. (b)

Sol. $P(x) = x^4 + ax^3 + bx^2 + cx + d$

$$P'(x) = 4x^3 + 3ax^2 + 2bx + c$$

$$P'(0) = 0 \Rightarrow c = 0$$

$$\text{Now, } P'(x) = x(4x^2 + 3ax + 2b)$$

As $P'(x) = 0$ has no real roots except

$x = 0$, we have

Discriminant of $4x^2 + 3ax + 2b$ is less than zero.

$$\text{i.e., } (3a)^2 - (4)(4)(2b) < 0$$

$$\text{then } 4x^2 + 3ax + 2b > 0 \forall x \in R$$

(If $a > 0, b^2 - 4ac < 0$ then $ax^2 + bx + c > 0 \forall x \in R$)

So $P'(x) < 0$ if $x \in [-1, 0]$ i.e., decreasing

and $P'(x) > 0$ if $x \in [0, 1]$ i.e., increasing

Max. of $P(x) = P(1)$

But minimum of $P(x)$ doesn't occur at $x = -1$, i.e., $P(-1)$ is not the minimum.

Example – 26

For $x \in \left(0, \frac{5\pi}{2}\right)$, define $f(x) = \int_0^x \sqrt{t} \sin t \, dt$. Then, f has

- (a) local minimum at π and 2π
- (b) local minimum at π and local maximum at 2π
- (c) local maximum at π and local minimum at 2π
- (d) local maximum at π and 2π

Ans. (c)

Sol. $f'(x) = \sqrt{x} \sin x, f'(x) = 0$

$$\Rightarrow x = 0 \text{ or } \sin x = 0$$

$$\Rightarrow x = 2\pi, \pi \left(\because x \in \left(0, \frac{5\pi}{2}\right) \right)$$

$$f''(x) = \sqrt{x} \cos x + \frac{1}{2\sqrt{x}} \sin x = \frac{1}{2\sqrt{x}} (2x \cos x + \sin x)$$

$$f''(\pi) < 0 \text{ and } f''(2\pi) > 0$$

$\Rightarrow$ Local maxima at $x = \pi$ and local minima at $x = 2\pi$

Example – 27

Let $f: R \rightarrow R$ be defined by

$$f(x) = \begin{cases} k - 2x, & \text{if } x \leq -1 \\ 2x + 3, & \text{if } x > -1 \end{cases}$$

If f has a local minimum at $x = -1$, then a possible value of k is

- (a) 1
- (b) 0
- (c) $-\frac{1}{2}$
- (d) -1

Ans. (d)

Sol. $\lim_{x \rightarrow -1^+} f(x) = 1$

$$\text{As } f(-1) = k + 2$$

As f has a local minimum at $x = -1$

$$f(-1^+) \geq f(-1) \leq f(-1^-) \Rightarrow 1 \geq k + 2$$

$$\Rightarrow k + 2 \leq 1 \quad \therefore k \leq -1$$

Thus $k = -1$ is a possible value.

Example – 28

Let $a, b \in \mathbb{R}$ be such that the function f given by $f(x) = \log |x| + bx^2 + ax$, $x \neq 0$ has extreme values at $x = -1$ and $x = 2$.

Statement I f has local maximum at $x = -1$ and $x = 2$.

Statement II $a = \frac{1}{2}$ and $b = \frac{-1}{4}$.

- (a) Statement I is false, Statement II is true.
(b) Statement I is true, Statement II is true;
Statement II is a correct explanation for Statement I.
(c) Statement I is true, Statement II is true;
Statement II is not a correct explanation for Statement I.
(d) Statement I is true, Statement II is false.

Ans. (c)

Sol. Given $f(x) = \ln|x| + bx^2 + ax$

$$\therefore f'(x) = \frac{1}{x} + 2bx + a$$

$$\text{at } x = -1, f'(-1) = -1 - 2b + a = 0$$

$$\Rightarrow a - 2b = 1 \quad \dots(i)$$

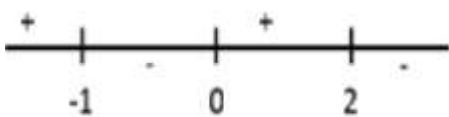
$$\text{at } x = 2, f'(-2) = \frac{1}{2} + 4b + a = 0$$

$$\Rightarrow a + 4b = -\frac{1}{2} \quad \dots(\text{ii})$$

Solving (i) and (ii) we get,

$$a = \frac{1}{2}, b = -\frac{1}{4}.$$

$$\Rightarrow f'(x) = \frac{1}{x} - \frac{x}{2} + \frac{1}{2} = \frac{2-x^2+x}{2x} = \frac{-(x+1)(x-2)}{2x}$$



$$\Rightarrow \text{maxima as } x = -1.2$$

Hence both statement are true but statement II. is not correct explanation of statement I.

Example – 29

The normal to the curve $x = a (\cos \theta + \theta \sin \theta)$,
 $y = a (\sin \theta - \theta \cos \theta)$ at any point θ is such that

- (a) it makes angle $\frac{\pi}{2} + \theta$ with x-axis
(b) it passes through the origin
(c) it is a constant distance from the origin
(d) it passes through $\left(a\frac{\pi}{2}, -a\right)$

Ans. (a,c)

Sol. $\frac{dy}{dx} = \frac{dy}{dx} \cdot \frac{d\theta}{dx} = \tan \theta = \text{Slope of tangent}$

$\therefore$ Slope of normal to the curve

$$= -\cot \theta \left(= \tan \left(\frac{\pi}{2} + \theta \right) \right)$$

Now, equation of normal to the curve

$$[y - a(\sin \theta - \theta \cos \theta)] = -\frac{\cos \theta}{\sin \theta} (x - a(\cos \theta + \theta \sin \theta))$$

$$\Rightarrow x \cos \theta + y \sin \theta = a(1)$$

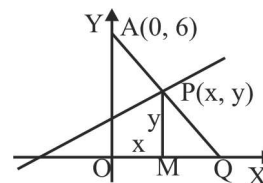
Now, distance from $(0,0)$ to $x \cos \theta + y \sin \theta = a$ is

$$\text{distance (d)} = \left| \frac{(0 + 0 - a)}{1} \right|$$

$\therefore$ distance is constant $= |a|$.

Example – 30

A point P (x, y) moves along the line whose equation is $x - 2y + 4 = 0$ in such a way that y increases at the rate of 3 units/sec. The point A (0, 6) is joined to P and the segment AP is prolonged to meet the x -axis in a point Q. Find how fast the distance from the origin to Q is changing when P reaches the point (4, 4).



Sol. The rate of change of y is given and it is desired to find the rate of change of OQ , which we denote by z . If MP is perpendicular to the x -axis, $MP = y$ and $OM = x$.

The triangles OAQ and MPQ are similar, hence

$$\frac{z}{6} = \frac{z-x}{y} \Rightarrow yz = 6z - 6x \Rightarrow z = \frac{6x}{6-y}$$



Substituting the value of x from the equation of the given line, we have

$$z = \frac{12(y-2)}{6-y}$$

$$\frac{dz}{dt} = \frac{48}{(6-y)^2} \frac{dy}{dt}$$

Setting $y = 4$ and $\frac{dy}{dt} = 3$, we obtain $\frac{dz}{dt} = 36$ that is, z is increasing at the rate of 36 units/sec.

Example – 31

The maximum distance from origin of a point on the curve

$$x = a \sin t - b \sin \left(\frac{at}{b} \right)$$

$$y = a \cos t - b \cos \left(\frac{at}{b} \right), \text{ both } a, b > 0, \text{ is}$$

- (a) $a - b$ (b) $a + b$
 (c) $\sqrt{a^2 + b^2}$ (d) $\sqrt{a^2 - b^2}$

Ans. (b)

Sol. Let $A(0, 0)$ and $B(x, y)$

$$\therefore AB^2 = x^2 + y^2$$

$$\Rightarrow AB = \sqrt{a^2(\sin^2 t + \cos^2 t) + b^2 \left(\sin^2 \left(\frac{at}{b} \right) + \cos^2 \left(\frac{at}{b} \right) \right) - 2ab \cos \left(t - \frac{at}{b} \right)}$$

$$= \sqrt{a^2 + b^2 - 2ab \cos \alpha} = \sqrt{a^2 + b^2 + 2ab}$$

($\because$ expression will take max value when $\cos \alpha = -1$)
 $= (a + b)$

Example – 32

The greatest value of $f(x) = (x+1)^{1/3} - (x-1)^{1/3}$ on $[0, 1]$ is

- (a) 1 (b) 2
 (c) 3 (d) $\frac{1}{3}$

Ans. (b)

Sol. We have

$$f(x) = (x+1)^{1/3} - (x-1)^{1/3}$$

$$\therefore f'(x) = \left[\frac{1}{(x+1)^{2/3}} - \frac{1}{(x-1)^{2/3}} \right] \cdot \frac{1}{3} = \frac{(x-1)^{2/3} - (x+1)^{2/3}}{3(x^2-1)^{2/3}}$$

for critical points : $f'(x) = 0$ or not defined.

Clearly, $f'(x)$ does not exist at $x = \pm 1$

$$\text{Now, } f'(x) = 0 \Rightarrow (x-1)^{2/3} = (x+1)^{2/3} \Rightarrow x = 0$$

Clearly, so $x = 0, \pm 1$ are critical point in $[0, 1]$.

$$f(0) = 2 \text{ and } f(1) = 2^{1/3}$$

Hence greatest value = 2

Example – 33

If x is real, the maximum value of $\frac{3x^2 + 9x + 17}{3x^2 + 9x + 7}$ is

- (a) $\frac{1}{4}$ (b) 41
 (c) 1 (d) $\frac{17}{7}$

Ans. (b)

Sol. For the range of the expression

$$\frac{3x^2 + 9x + 17}{3x^2 + 9x + 7} = y = \frac{ax^2 + bx + c}{px^2 + qx + r},$$

[find the solution of the inequality $Ay^2 + By + K \geq 0$

Where $A = q^2 - 4pr = -3$, $B = 4ar + 4pc - 2bq = 126$

$$K = b^2 - 4ac = -123$$

i.e., solve $3y^2 - 126y - 123 \geq 0$

$$\Rightarrow 3y^2 - 126y + 123 \leq 0 \Rightarrow y^2 - 42y + 41 \leq 0$$

$$\Rightarrow (y - 1)(y - 42) \leq 0 \Rightarrow 1 \leq y \leq 42$$

$\Rightarrow$ Maximum value of y is 42

Example – 34

If p and q are positive real numbers such that $p^2 + q^2 = 1$, then the maximum value of $(p + q)$ is

- (a) $\frac{1}{2}$ (b) $\frac{1}{\sqrt{2}}$
 (c) $\sqrt{2}$ (d) 2

Ans. (c)

Sol. 1st solution :

Let $p = \cos \theta$, $q = \sin \theta$ where $0 \leq \theta \leq \frac{\pi}{2}$

$$p + q = \cos \theta + \sin \theta$$

$\Rightarrow$ maximum value of $(p + q) = \sqrt{2}$



IInd solution :

By using A.M $\geq$ G.M, $\frac{p^2+q^2}{2} \geq pq \Rightarrow pq \leq \frac{1}{2}$

$$(p+q)^2 = p^2 + q^2 + 2pq \Rightarrow (p+q) \leq \sqrt{2}$$

Example – 35

Let f be a function defined by $f(x) = \begin{cases} \frac{\tan x}{x}, & x \neq 0 \\ 1, & x = 0 \end{cases}$

Statement I $x = 0$ is point of minima of f .

Statement II $f'(0) = 0$

- (a) Statement I is false, Statement II is true.
 (b) Statement I is true, Statement II is true;
 Statement II is correct explanation for Statement I.
 (c) Statement I is true, Statement II is true;
 Statement II is not a correct explanation for Statement I.
 (d) Statement I is true, Statement II is false.

Ans. (c)

Sol. $f(x) = \begin{cases} \frac{\tan x}{x}, & x \neq 0 \\ 1, & x = 0 \end{cases}$

In right neighbourhood of '0'

$$\tan x > x \Rightarrow \frac{\tan x}{x} > 1$$

In left neighbourhood of '0'

$$\tan x < x \Rightarrow \frac{\tan x}{x} > 1 (\because \tan x < 0)$$

$$\text{at } x = 0, f(x) = 1$$

$\Rightarrow x = 0$ is point of minima

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{\frac{\tan h}{h} - 1}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\tan h - h}{h^2} = 0$$

hence $f'(0) = 0$

$\Rightarrow$ statement I is true and statement II is true.

Example – 36

If $f(\theta) = \begin{vmatrix} 1 & \cos \theta & 1 \\ -\sin \theta & 1 & -\cos \theta \\ -1 & \sin \theta & 1 \end{vmatrix}$ and A and B are

respectively the maximum and the minimum values of $f(\theta)$, then (A, B) is equal to:

- (a) (3, -1) (b) $(4, 2 - \sqrt{2})$
 (c) $(2 + \sqrt{2}, 2 - \sqrt{2})$ (d) $(2 + \sqrt{2}, -1)$

Ans. (c)

Sol. $f(\theta) = \begin{vmatrix} 1 & \cos \theta & 1 \\ -\sin \theta & 1 & -\cos \theta \\ -1 & \sin \theta & 1 \end{vmatrix}$

$$\Rightarrow (1 + \sin \theta \cos \theta) - \cos \theta.(-\sin \theta - \cos \theta) + (-\sin^2 \theta + 1)$$

$$\Rightarrow f(\theta) = 2 + \sin 2\theta + \cos 2\theta$$

$$\Rightarrow f(\theta)_{\min} = 2 - \sqrt{2}$$

$$\Rightarrow f(\theta)_{\max} = 2 + \sqrt{2}$$

Example – 37

Find the interval in which

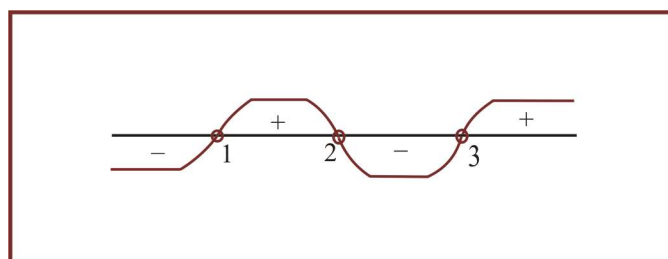
$f(x) = x^4 - 8x^3 + 22x^2 - 24x + 5$ is increasing.

Sol. Given $f(x) = x^4 - 8x^3 + 22x^2 - 24x + 5$

$$\therefore f'(x) = 4x^3 - 24x^2 + 44x - 24$$

$$= 4(x^3 - 6x^2 + 11x - 6)$$

$$= 4(x-1)(x-2)(x-3)$$



For increasing function $f'(x) > 0$

$$\text{or } 4(x-1)(x-2)(x-3) > 0$$

$$\text{or } (x-1)(x-2)(x-3) > 0$$

$$\therefore x \in (1, 2) \cup (3, \infty)$$

**Example – 38**

Find the interval in which $f(x) = x - 2 \sin x$, $0 \leq x \leq 2\pi$ is increasing

Sol. Given $f(x) = x - 2 \sin x$

$$\therefore f'(x) = 1 - 2 \cos x$$

$$f'(x) > 0 \text{ or } 1 - 2 \cos x > 0 \quad \therefore \cos x < \frac{1}{2}$$

$$\text{or } -\cos x > -\frac{1}{2}$$

$$\text{or } \cos(\pi + x) > \cos \frac{2\pi}{3}$$

$$\text{or } 2n\pi - \frac{2\pi}{3} < \pi + x < 2n\pi + \frac{2\pi}{3}, n \in \mathbb{I}$$

$$\text{or } 2n\pi - \frac{5\pi}{3} < x < 2n\pi - \frac{\pi}{3}$$

$$\text{For } n = 1, \frac{\pi}{3} < x < \frac{5\pi}{3} \text{ which is true } (\because 0 \leq x \leq 2\pi)$$

$$\text{Hence, } x \in \left(\frac{\pi}{3}, \frac{5\pi}{3}\right)$$

Example – 39

Find the intervals of monotonicity of the function

$$f(x) = \frac{|x-1|}{x^2}.$$

Sol. The given function $f(x)$ can be written as :

$$f(x) = \frac{|x-1|}{x^2} = \begin{cases} \frac{1-x}{x^2} & ; x < 1, x \neq 0 \\ \frac{x-1}{x^2} & ; x \geq 1 \end{cases}$$

Consider $x < 1$

$$f'(x) = \frac{-2}{x^3} + \frac{1}{x^2} = \frac{x-2}{x^3}$$

$$\text{For increasing, } f'(x) > 0 \Rightarrow \frac{x-2}{x^3} > 0$$

$$\Rightarrow x(x-2) > 0 \quad [\text{as } x^2 \text{ is positive}]$$

$$\Rightarrow x \in (-\infty, 0) \cup (2, \infty).$$

Combining with $x < 1$, we get $f(x)$ is increasing in $x < 0$ and decreasing in $x \in (0, 1)$... (i)

Consider $x \geq 1$

$$f'(x) = \frac{-1}{x^2} + \frac{2}{x^3} = \frac{2-x}{x^3}$$

$$\begin{array}{c} + \quad - \quad + \quad - \\ | \quad | \quad | \quad | \\ 0 \quad 1 \quad 2 \end{array}$$

For increasing $f'(x) > 0$

$$\Rightarrow (2-x) > 0 \quad [\text{as } x^3 \text{ is positive}]$$

$$\Rightarrow (x-2) < 0.$$

$$\Rightarrow x < 2.$$

Combining with $x > 1$, $f(x)$ is increasing in $x \in (1, 2)$ and decreasing in $x \in (2, \infty)$... (ii)

Combining (i) and (ii), we get :

$f(x)$ is strictly increasing on $x \in (-\infty, 0) \cup (1, 2)$ and strictly decreasing on $x \in (0, 1) \cup (2, \infty)$.

Example – 40

The function $f(x) = \log(x-2)^2 - x^2 + 4x + 1$ increases on the interval

$$(a) (1, 2)$$

$$(b) (2, 3)$$

$$(c) (5/2, 3)$$

$$(d) (2, 4)$$

Ans. (b,c)

Sol. $f(x) = 2 \log(x-2) - x^2 + 4x + 1$

$$\Rightarrow f'(x) = \frac{2}{x-2} - 2x + 4$$

$$\Rightarrow f'(x) = 2 \left[\frac{1-(x-2)^2}{x-2} \right] = -2 \frac{(x-1)(x-3)}{x-2}$$

$$\Rightarrow f'(x) = -\frac{2(x-1)(x-3)(x-2)}{(x-2)^2}$$

$$\therefore f'(x) > 0 \Rightarrow -2(x-1)(x-3)(x-2) > 0$$

$$\Rightarrow (x-1)(x-2)(x-3) < 0$$

$$\Rightarrow x \in (-\infty, 1) \cup (2, 3).$$

$$\begin{array}{c} - \quad + \quad - \quad + \\ -\infty \quad 1 \quad 2 \quad 3 \quad \infty \end{array}$$

**Example – 41**

A function is matched below against an interval where it is supposed to be increasing. Which of the following pairs is incorrectly matched ?

Interval	Function
(a) $(-\infty, -4)$	$x^3 + 6x^2 + 6$
(b) $\left(-\infty, \frac{1}{3}\right]$	$3x^2 - 2x + 1$
(c) $[2, \infty)$	$2x^3 - 3x^2 - 12x + 6$
(d) $(-\infty, \infty)$	$x^3 - 3x^2 + 3x + 3$

Ans. (b)

Sol. For function to be increasing, $f'(x) > 0$

(a) $f'(x) = 3x(x+4) \Rightarrow$ increasing in $(-\infty, -4) \cup (0, \infty)$

(b) $f'(x) = 2(3x-1) \Rightarrow$ decreasing in $(-\infty, \frac{1}{3})$

(c) $f'(x) = 6(x+1)(x-2) \Rightarrow$ increasing in $(-\infty, -1) \cup (2, \infty)$

(d) $f'(x) = 3(x-1)^2 \Rightarrow$ increasing in $(-\infty, \infty)$

so (b) match is incorrect.

Example – 42

The function $f(x) = \tan^{-1}(\sin x + \cos x)$ is an increasing function in

(a) $\left(0, \frac{\pi}{2}\right)$	(b) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
(c) $\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$	(d) $\left(-\frac{\pi}{2}, \frac{\pi}{4}\right)$

Ans. (d)

Sol. $f'(x) = \frac{1}{1 + (\sin x + \cos x)^2} \cdot (\cos x - \sin x)$

$$f'(x) = \frac{\cos x - \sin x}{2 + \sin 2x}$$

If $f'(x) > 0$ then $f'(x)$ is increasing function

For $-\frac{\pi}{2} < x < \frac{\pi}{4}$, $\cos x > \sin x$

Hence $y = f'(x)$ is increasing in $\left(-\frac{\pi}{2}, \frac{\pi}{4}\right)$

Example – 43

The function $f(x) = \cot^{-1} x + x$ increases in the interval

- (a) $(1, \infty)$ (b) $(-1, \infty)$
(c) $(-\infty, \infty)$ (d) $(0, \infty)$

Ans. (c)

Sol. $f(x) = \cot^{-1} x + x$

$$f'(x) = \frac{-1}{1+x^2} + 1 = \frac{x^2}{1+x^2} > 0 \forall x \in R$$

Example – 44

A spherical balloon is filled with 4500π cu m of helium gas. If a leak in the balloon causes the gas to escape at the rate of 72π cu m/min, then the rate (in m/min) at which the radius of the balloon decreases 49 min after the leakage began is

- (a) $\frac{9}{7}$ (b) $\frac{7}{9}$
(c) $\frac{2}{9}$ (d) $\frac{9}{2}$

Ans. (c)

Sol. $\frac{dv}{dt} = -72\pi \text{ m}^3/\text{min}, v_0 = 4500\pi$

$$v = \frac{4}{3}\pi r^3 \therefore \frac{dv}{dt} = \frac{4}{3}\pi \times 3r^2 \times \frac{dr}{dt}$$

After 49 min, $v = v_0 + 49 \cdot \frac{dv}{dt} = 4500\pi - 49 \times 72\pi$

$$= 4500\pi - 3528\pi = 972\pi$$

$$\Rightarrow 972\pi = \frac{4}{3}\pi r^3 \Rightarrow r^3 = 243 \times 3 = 36 \Rightarrow r = 9$$

$$\therefore 72\pi = 4\pi \times 81 \times \frac{dr}{dt} = -\frac{18}{81} = -\frac{2}{9}$$

Thus, radius decreases at a rate of $\frac{2}{9}$ m/min

Example – 45

A point on the parabola $y^2 = 18x$ at which the ordinate increases at twice the rate of the abscissa, is

- (a) $(2, 4)$ (b) $(2, -4)$
(c) $\left(-\frac{9}{8}, \frac{9}{2}\right)$ (d) $\left(\frac{9}{8}, \frac{9}{2}\right)$

Ans. (d)



Sol. $y^2 = 18x \Rightarrow 2y \frac{dy}{dx} = 18 \Rightarrow \frac{dy}{dx} = \frac{9}{y}$

Given $\frac{dy}{dx} = 2 \Rightarrow \frac{9}{y} = 2 \Rightarrow y = \frac{9}{2} \Rightarrow x = \frac{9}{8}$

Example – 46

If the volume of a spherical ball is increasing at the rate of 4π cc/sec, then the rate of increase of its radius (in cm/sec), when the volume is 288π cc, is:

- (a) $\frac{1}{6}$ (b) $\frac{1}{9}$
(c) $\frac{1}{36}$ (d) $\frac{1}{24}$

Ans. (c)

Sol. $\frac{dV}{dt} = 4\pi \text{ cc/sec}$

we know $V = \frac{4}{3}\pi r^3$

$$\frac{dV}{dt} = \frac{4}{3}\pi 3r^2 \frac{dr}{dt} = 4\pi r^2 \frac{dr}{dt}$$

when $V = 288\pi \text{ cc}$

$$\Rightarrow r^3 = 216$$

$$\Rightarrow r = 6$$

$$\frac{dV}{dt} = 4\pi r^2 \cdot \frac{dr}{dt}$$

$$\Rightarrow 4\pi = 4\pi \times 36 \times \frac{dr}{dt}$$

$$\Rightarrow \frac{dr}{dt} = \frac{1}{36}$$

Example – 47

The period T of a simple pendulum is

$$T = 2\pi\sqrt{\frac{l}{g}}$$

Find the maximum error in T due to possible errors upto 1% in l and 2.5% in g.

Sol. Since $T = 2\pi\sqrt{\frac{l}{g}} = 2\pi\left(\frac{l}{g}\right)^{1/2}$

Taking logarithm on both sides, we get

$$\ln T = \ln 2\pi + \frac{1}{2} \ln l - \frac{1}{2} \ln g$$

Differentiating both sides, we get

$$\frac{dT}{T} = 0 + \frac{1}{2} \cdot \frac{dl}{l} - \frac{1}{2} \cdot \frac{dg}{g}$$

or $\left(\frac{dT}{T} \times 100\right) = \frac{1}{2} \left(\frac{dl}{l} \times 100\right) - \frac{1}{2} \left(\frac{dg}{g} \times 100\right)$

$$\left(\frac{dT}{T} \times 100\right) = \frac{1}{2} (1 \pm 2.5) \left(\because \frac{dl}{l} \times 100 = 1 \text{ and } \frac{dg}{g} \times 100 = 2.5\right)$$

$\therefore$ Maximum error in T = 1.75%.

Example – 48

If the Rolle's theorem holds for the function $f(x) = 2x^3 + ax^2 + bx$ in the interval $[-1, 1]$ for the point

$c = \frac{1}{2}$, then the value of $2a + b$ is

- (a) 1 (b) -1
(c) 2 (d) -2

Ans. (b)

Sol. $f(x) = 2x^3 + ax^2 + bx$

Given Rolle's theorem is applicable

$$\Rightarrow f(-1) = f(1)$$

$$\Rightarrow -2 + a - b = 2 + a + b$$

$$\Rightarrow b = -2$$

$$f'(x) = 6x^2 + 2ax + b$$

$$= 6x^2 + 2ax - 2$$

$$f'\left(\frac{1}{2}\right) = 0$$

$$\Rightarrow a = \frac{1}{2}$$

$$\Rightarrow 2a + b = -1$$

**Example – 49**

Find the equation of the tangent to $\frac{x^m}{a^m} + \frac{y^m}{b^m} = 1$ at the point (x_0, y_0) .

Sol. $\frac{x^m}{a^m} + \frac{y^m}{b^m} = 1$ Differentiating wrt x ,

$$\Rightarrow \frac{mx^{m-1}}{a^m} + \frac{my^{m-1}}{b^m} \frac{dy}{dx} = 0$$

$$\Rightarrow \frac{dy}{dx} = -\frac{b^m}{a^m} \left(\frac{x}{y}\right)^{m-1}$$

$\Rightarrow$ at the given point (x_0, y_0) , slope of tangent is

$$\left. \frac{dy}{dx} \right|_{(x_0, y_0)} = -\left(\frac{b}{a}\right)^m \left(\frac{x_0}{y_0}\right)^{m-1}$$

$\Rightarrow$ the equation of tangent is

$$y - y_0 = -\left(\frac{b}{a}\right)^m \left(\frac{x_0}{y_0}\right)^{m-1} (x - x_0)$$

$$a^m y y_0^{m-1} - a^m y_0^m = -b^m x x_0^{m-1} + b^m x_0^m$$

$$a^m y y_0^{m-1} + b^m x x_0^{m-1} = a^m y_0^m + b^m x_0^m$$

using the equation of given curve, the right side can be replaced by $a^m b^m$.

$$\therefore a^m y y_0^{m-1} + b^m x x_0^{m-1} = a^m b^m$$

$\Rightarrow$ the equation of tangent is

$$\frac{x}{a} \left(\frac{x_0}{a}\right)^{m-1} + \frac{y}{b} \left(\frac{y_0}{b}\right)^{m-1} = 1$$

Example – 50

Find the equation of the tangent to $x^3 = ay^2$ at the point A (at^2, at^3) . Find also the Point where this tangent meets the curve again.

Sol. Equation of tangent to : $x = at^2, y = at^3$ is

$$\frac{dx}{dt} = 2at, \frac{dy}{dt} = 3at^2$$

$$y - at^3 = \frac{3at^2}{2at} (x - at^2)$$

$$\Rightarrow 2y - 2at^3 = 3tx - 3at^3$$

$$\text{i.e. } 3tx - 2y - at^3 = 0$$

Let B (at_1^2, at_1^3) be the point where it again meets the curve.

$\Rightarrow$ slope of tangent at A = slope of AB

$$\frac{3at^2}{2at} = \frac{a(t^3 - t_1^3)}{a(t^2 - t_1^2)}$$

$$\Rightarrow \frac{3t}{2} = \frac{t^2 + t_1^2 + t t_1}{t + t_1}$$

$$\Rightarrow 3t^2 + 3 t t_1 = 2t^2 + 2t_1^2 + 2 t t_1$$

$$\Rightarrow 2t_1^2 - t t_1 - t^2 = 0$$

$$\Rightarrow (t_1 - t)(2t_1 + t) = 0$$

$$\Rightarrow t_1 = t \text{ or } t_1 = -t/2$$

The relevant value is $t_1 = -t/2$

Hence the meeting point B is

$$= \left[a \left(\frac{-t}{2} \right)^2, a \left(\frac{-t}{2} \right)^3 \right] = \left[\frac{at^2}{4}, \frac{-at^3}{8} \right]$$

Example – 51

The normal to the curve $x = a(1 + \cos \theta)$, $y = a \sin \theta$ at θ always passes through the fixed point

(a) $(a, 0)$

(b) $(0, a)$

(c) $(0, 0)$

(d) (a, a)

Ans. (a)

Sol. $\frac{dx}{d\theta} = -a \sin \theta$ and $\frac{dy}{d\theta} = a \cos \theta \Rightarrow \frac{dy}{dx} = -\cot \theta$

$\therefore$ slope of normal at $\theta = \tan \theta$

$\therefore$ the equation of normal at θ is

$$y - a \sin \theta = \tan \theta (x - a - a \cos \theta)$$

$$\Rightarrow x \sin \theta - y \cos \theta = a \sin \theta$$

$$\Rightarrow y = (x - a) \tan \theta$$

which always passes through $(a, 0)$

Example – 52

Two ships A and B are sailing straight away from a fixed point O along routes such that $\angle AOB$ is always 120° . At a certain instance, $OA = 8$ km, $OB = 6$ km and the ship A is sailing at the rate of 20 km/hr while the ship B sailing at the rate of 30 km/hr. Then the distance between A and B is changing at the rate (in km/hr):

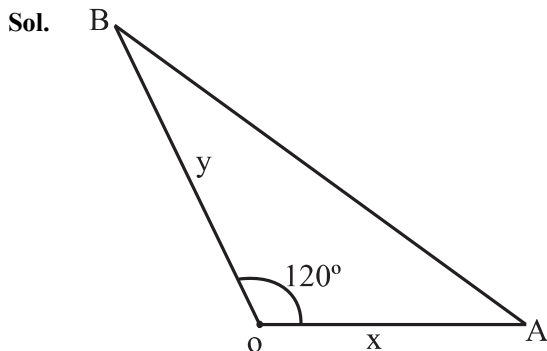
(a) $\frac{260}{\sqrt{37}}$

(b) $\frac{260}{37}$

(c) $\frac{80}{\sqrt{37}}$

(d) $\frac{80}{37}$

Ans. (a)



Let $OA = x$ and $OB = y$

$$\frac{dx}{dt} = 20 \text{ km/hr}, \quad \frac{dy}{dt} = 30 \text{ km/hr}.$$

When $OA = 8$, $OB = 6$

Applying cosine formula in $\triangle AOB$.

$$\cos 120^\circ = \frac{x^2 + y^2 - (AB)^2}{2xy} - \frac{1}{8} = \frac{64 + 36 - (AB)^2}{2 \times 8 \times 6}$$

$$\Rightarrow -48 = 64 + 36 - (AB)^2$$

$$\Rightarrow AB = 2\sqrt{37}$$

Again applying cosine formula in $\triangle AOB$

When $OA = x$ and $OB = y$

$$\Rightarrow -\frac{1}{2} = \frac{x^2 + y^2 - (AB)^2}{2xy}$$

$$\Rightarrow (AB)^2 = x^2 + y^2 + xy$$

$AB =$ distance between A and B = Z (let)

$$Z^2 = x^2 + y^2 + xy$$

differentiate w.r.t. "t"

$$2Z \cdot \frac{dz}{dt} = 2x \cdot \frac{dx}{dt} + 2y \cdot \frac{dy}{dt} + x \frac{dy}{dt} + y \frac{dx}{dt}$$

$$\Rightarrow 2 \times 2\sqrt{37} \frac{dz}{dt} = 16 \times 20 + 12 \times 30 + 240 + 120$$

$$\Rightarrow 4\sqrt{37} \frac{dz}{dt} = 1040$$

$$\Rightarrow \frac{dz}{dt} = \frac{260}{\sqrt{37}} \text{ km/hr}$$

Example – 53

If $2a + 3b + 6c = 0$, $a, b, c \in \mathbb{R}$ then show that the equation $ax^2 + bx + c = 0$ has at least one root between 0 and 1.

Sol. Given $2a + 3b + 6c = 0$

$$\text{or } \frac{a}{3} + \frac{b}{2} + c = 0 \quad \dots (i)$$

$$\text{Let } f'(x) = ax^2 + bx + c$$

On integrating both sides, we get

$$f(x) = \frac{ax^3}{3} + \frac{bx^2}{2} + cx + k$$

$$\text{Now, } f(1) = \frac{a}{3} + \frac{b}{2} + c + k \quad [\text{From (i)}]$$

$$= 0 + k = k$$

$$\text{and } f(0) = 0 + 0 + 0 + k = k$$

Since $f(x)$ is a polynomial of three degree, it is continuous and differentiable and $f(0) = f(1)$, then by Rolle's theorem

$f'(x) = 0$ i.e., $ax^2 + bx + c = 0$ has at least one real root between 0 and 1.

Example – 54

If $f(x) = (x-1)(x-2)(x-3)$ and $a=0$, $b=4$, find 'c' using Lagrange's mean value theorem.

Sol. We have $f(x) = (x-1)(x-2)(x-3) = x^3 - 6x^2 + 11x - 6$

$$\therefore f(a) = f(0) = (0-1)(0-2)(0-3) = -6$$

$$\text{and } f(b) = f(4) = (4-1)(4-2)(4-3) = 6$$

$$\therefore \frac{f(b) - f(a)}{b - a} = \frac{6 - (-6)}{4 - 0} = \frac{12}{4} = 3 \quad \dots (1)$$

$$\text{Also } f'(x) = 3x^2 - 12x + 11$$

$$\text{gives } f'(c) = 3c^2 - 12c + 11$$

$$\text{From LMVT, } \frac{f(b) - f(a)}{b - a} = f'(c) \quad \dots (2)$$

$$\Rightarrow 3 = 3c^2 - 12c + 11 \quad \{\text{From (1) and (2)}\}$$

$$\Rightarrow 3c^2 - 12c + 8 = 0$$

$$\therefore c = \frac{12 \pm \sqrt{144 - 96}}{6} = 2 \pm \frac{2\sqrt{3}}{3}$$

As both of these values of c lie in the open interval $(0, 4)$. Hence both of these are required values of c .

**Example – 55**

A value of c for which conclusion of Mean Value Theorem holds for the function $f(x) = \log_e x$ on the interval $[1, 3]$, is

(a) $\log_3 e$

(b) $\log_e 3$

(c) $2 \log_3 e$

(d) $\frac{1}{2} \log_e 3$

Ans. (c)

Sol. By LMVT

$$f'(c) = \frac{f(b) - f(a)}{b - a} = \frac{f(3) - f(1)}{3 - 1}$$

$$f'(c) = \frac{\log_e 3 - \log_e 1}{2} = \frac{1}{2} \log_e 3$$

$$\Rightarrow \frac{1}{c} = \frac{1}{2} \log_e 3 = \frac{1}{2 \log_3 e} \therefore c = 2 \log_3 e$$



EXERCISE - 1 : BASIC OBJECTIVE QUESTIONS

Derivative as rate Measure

- Gas is being pumped into a spherical balloon at the rate of $30 \text{ ft}^3/\text{min}$. Then, the rate at which the radius increases when it reaches the value 15 ft , is
 - $\frac{1}{30\pi} \text{ ft/min}$
 - $\frac{1}{15\pi} \text{ ft/min}$
 - $\frac{1}{20} \text{ ft/min}$
 - $\frac{1}{15} \text{ ft/min}$
- The position of a point in time 't' is given by $x = a + bt - ct^2$, $y = at + bt^2$. Its acceleration at time 't' is
 - $b - c$
 - $b + c$
 - $2b - 2c$
 - $2\sqrt{b^2 + c^2}$
- A spherical iron ball 10 cm in radius is coated with a layer of ice of uniform thickness that melts at a rate of $50 \text{ cm}^3/\text{min}$. When the thickness of ice is 5 cm , then the rate at which the thickness of ice decreases, is
 - $\frac{1}{18\pi} \text{ cm/min}$
 - $\frac{1}{36\pi} \text{ cm/min}$
 - $\frac{5}{6\pi} \text{ cm/min}$
 - $\frac{1}{54\pi} \text{ cm/min}$
- The rate of change of the surface area of a sphere of radius r , when the radius is increasing at the rate of 2 cm/s is proportional to
 - $\frac{1}{r}$
 - $\frac{1}{r^2}$
 - r
 - r^2
- For what values of x is the rate of increase of $x^3 - 5x^2 + 5x + 8$ is twice the rate of increase of x ?
 - $-3, -\frac{1}{3}$
 - $-3, \frac{1}{3}$
 - $3, -\frac{1}{3}$
 - $3, \frac{1}{3}$
- If a particle moving along a line follows the law $s = \sqrt{1+t}$, then the acceleration is proportional to
 - square of the velocity
 - cube of the displacement
 - cube of the velocity
 - square of the displacement
- If a particle is moving such that the velocity acquired is proportional to the square root of the distance covered, then its acceleration is
 - a constant
 - $\propto s^2$
 - $\propto \frac{1}{s^2}$
 - $\propto \frac{1}{s}$

Errors and Approximations

- If $y = x^n$, then the ratio of relative errors in y and x is
 - $1 : 1$
 - $2 : 1$
 - $1 : n$
 - $n : 1$
- If the ratio of base radius and height of a cone is $1 : 2$ and percentage error in radius is $\lambda\%$, then the error in its volume is
 - $\lambda\%$
 - $2\lambda\%$
 - $3\lambda\%$
 - none of these
- The height of a cylinder is equal to the radius. If an error of $\alpha\%$ is made in the height, then percentage error in its volume is
 - $\alpha\%$
 - $2\alpha\%$
 - $3\alpha\%$
 - none of these

Equation of Tangents and Normals

- For the curve $y = 3 \sin \theta \cos \theta$, $x = e^\theta \sin \theta$, $0 \leq \theta \leq \pi$, the tangent is parallel to x-axis when θ is:
 - $\frac{3\pi}{4}$
 - $\frac{\pi}{2}$
 - $\frac{\pi}{4}$
 - $\frac{\pi}{6}$



12. The curve $y - e^{xy} + x = 0$ has a vertical tangent at
 (a) (1, 1) (b) (0, 1)
 (c) (1, 0) (d) no point
13. If the line $ax + by + c = 0$ is a tangent to the curve $xy = 4$, then the possible answer is
 (a) $a > 0, b > 0$ (b) $a > 0, b < 0$
 (c) $a < 0, b > 0$ (d) none of these
14. The tangent to the curve $5x^2 + y^2 = 1$ at $\left(\frac{1}{3}, -\frac{2}{3}\right)$ passes through the point
 (a) (0, 0) (b) (1, -1)
 (c) (-1, 1) (d) none of these
15. The equation of the tangent to the curve $y = \sqrt{9 - 2x^2}$ at the point where the ordinate and the abscissa are equal, is
 (a) $2x + y - 3\sqrt{3} = 0$ (b) $2x + y + 3\sqrt{3} = 0$
 (c) $2x - y - 3\sqrt{3} = 0$ (d) none of these
16. The tangent to the curve $x^2 + y^2 = 25$ is parallel to the line $3x - 4y = 7$ at the point
 (a) (-3, -4) (b) (3, -4)
 (c) (3, 4) (d) none of these
17. If the tangent at each point of the curve $y = \frac{2}{3}x^3 - 2ax^2 + 2x + 5$ makes an acute angle with the positive direction of x-axis, then
 (a) $a \geq 1$ (b) $-1 \leq a \leq 1$
 (c) $a \leq -1$ (d) none of these
18. The equation of the tangent to the curve $(1 + x^2)y = 2 - x$, where it crosses the x-axis, is
 (a) $x + 5y = 2$ (b) $x - 5y = 2$
 (c) $5x - y = 2$ (d) $5x + y - 2 = 0$
19. The intercepts on x-axis made by tangents to the curve, $y = \int_0^x |t| dt, x \in \mathbb{R}$, which are parallel to the line $y = 2x$, are equal to
 (a) ± 1 (b) ± 2
 (c) ± 3 (d) ± 4

Length of tangent, normal, subtangent and subnormal

20. The length of subtangent to the curve $x^2y^2 = a^4$ at the point $(-a, a)$ is
 (a) $3a$ (b) $2a$
 (c) a (d) $4a$
21. For the parabola $y^2 = 4ax$, the ratio of the sub-tangent to the abscissa is
 (a) $1 : 1$ (b) $2 : 1$
 (c) $1 : 2$ (d) $3 : 1$
22. The length of subtangent to the curve $x^2y^2 = a^4$ at the point $(-a, a)$ is
 (a) $3a$ (b) $2a$
 (c) a (d) $4a$
23. The product of the lengths of subtangent and subnormal at any point of a curve is
 (a) square of the abscissae (b) square of the ordinate
 (c) constant (d) None of these

Angle of intersection between the curves

24. The curves $x^3 + pxy^2 = -2$ and $3x^2y - y^3 = 2$ are orthogonal for
 (a) $p = 3$ (b) $p = -3$
 (c) no value of p (d) $p = \pm 3$
25. The two tangents to the curve $ax^2 + 2hxy + by^2 = 1, a > 0$ at the points where it crosses x-axis, are
 (a) parallel (b) perpendicular
 (c) inclined at an angle $\frac{\pi}{4}$ (d) none of these
26. The lines $y = -\frac{3}{2}x$ and $y = -\frac{2}{5}x$ intersect the curve $3x^2 + 4xy + 5y^2 - 4 = 0$ at the points P and Q respectively. The tangents drawn to the curve at P and Q
 (a) intersect each other at angle of 45°
 (b) are parallel to each other
 (c) are perpendicular to each other
 (d) none of these
27. The angle between the curves $y = \sin x$ and $y = \cos x$ is
 (a) $\tan^{-1}(2\sqrt{2})$ (b) $\tan^{-1}(3\sqrt{2})$
 (c) $\tan^{-1}(3\sqrt{3})$ (d) $\tan^{-1}(5\sqrt{2})$



28. The angle between the tangents to the curve $y^2 = 2ax$ at the points where $x = \frac{a}{2}$, is
- (a) $\pi/6$ (b) $\pi/4$
(c) $\pi/3$ (d) $\pi/2$
29. The angle between the tangents at those points on the curve $y = (x+1)(x-3)$ where it meets x-axis, is
- (a) $\tan^{-1}\left(\frac{15}{8}\right)$ (b) $\tan^{-1}\left(\frac{8}{15}\right)$
(c) $\frac{\pi}{4}$ (d) none of these
30. The angle at which the curves $y = \sin x$ and $y = \cos x$ intersect in $[0, \pi]$, is
- (a) $\tan^{-1} 2\sqrt{2}$ (b) $\tan^{-1} \sqrt{2}$
(c) $\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$ (d) none of these
31. The two curves $x^3 - 3xy^2 + 2 = 0$ and $3x^2y - y^3 - 2 = 0$
- (a) cut at right angles (b) touch each other
(c) cut at an angle $\frac{\pi}{3}$ (d) cut at an angle $\frac{\pi}{4}$
32. The two curves $y = 3^x$ and $y = 5^x$ intersect at an angle
- (a) $\tan^{-1}\left(\frac{\log 5 - \log 3}{1 + \log 3 \cdot \log 5}\right)$ (b) $\tan^{-1}\left(\frac{\log 3 + \log 5}{1 - \log 3 \cdot \log 5}\right)$
(c) $\tan^{-1}\left(\frac{\log 3 + \log 5}{1 + \log 3 \cdot \log 5}\right)$ (d) none of these
33. The angle of intersection of the curve $y = x^2$ & $6y = 7 - x^3$ at $(1, 1)$ is
- (a) $\pi/5$ (b) $\pi/4$
(c) $\pi/3$ (d) $\pi/2$
34. The function $f(x) = 2x^2 - \log |x|$ monotonically decreases for
- (a) $x \in (-\infty, -1/2] \cup (0, 1/2]$
(b) $x \in (-\infty, 1/2]$
(c) $x \in [-1/2, 0) \cup [1/2, \infty)$
(d) none of these
35. The interval in which the function x^3 increases less rapidly than $6x^2 + 15x + 5$ is :
- (a) $(-\infty, -1)$ (b) $(-5, 1)$
(c) $(-1, 5)$ (d) $(5, \infty)$
36. The function $y = \frac{2x^2 - 1}{x^4}$ is
- (a) a decreasing function for all $x \in \mathbb{R} - \{0\}$
(b) a increasing function for all $x \in \mathbb{R} - \{0\}$
(c) increasing for $x > 0$
(d) none of these
37. The function $f(x) = \frac{\sin x}{x}$ is decreasing in the interval
- (a) $\left(-\frac{\pi}{2}, 0\right)$ (b) $\left(0, \frac{\pi}{2}\right)$
(c) $(0, \pi)$ (d) none of these
38. If $f(x) = \frac{1}{x+1} - \log(1+x)$, $x > 0$, then f is
- (a) an increasing function
(b) a decreasing function
(c) both increasing and decreasing function
(d) None of the above
39. Let $f(x) = \int_1^x e^x (x-1)(x-2) dx$. Then, f decreases in the interval
- (a) $(-\infty, 2)$ (b) $(-2, -1)$
(c) $(1, 2)$ (d) $(2, \infty)$
40. If $f(x) = x^3 + 4x^2 + \lambda x + 1$ is a strictly decreasing function of x in the largest possible interval $[-2, -2/3]$ then
- (a) $\lambda = 4$ (b) $\lambda = 2$
(c) $\lambda = -1$ (d) λ has no real value
41. The length of the longest interval, in which the function $3 \sin x - 4 \sin^3 x$ is increasing, is
- (a) $\frac{\pi}{3}$ (b) $\frac{\pi}{2}$
(c) $\frac{3\pi}{2}$ (d) π

Increasing and Decreasing Functions

34. The function $f(x) = 2x^2 - \log |x|$ monotonically decreases for
- (a) $x \in (-\infty, -1/2] \cup (0, 1/2]$
(b) $x \in (-\infty, 1/2]$
(c) $x \in [-1/2, 0) \cup [1/2, \infty)$
(d) none of these



42. The function $f(x) = x + \cos x$ is
 (a) always increasing
 (b) always decreasing
 (c) increasing for certain range of x
 (d) None of the above
43. How many real solutions does the equation $x^7 + 14x^5 + 16x^3 + 30x - 560 = 0$ have?
 (a) 5 (b) 7
 (c) 1 (d) 3
- Maxima and minima**
44. The function $f(x) = 2x^3 - 3x^2 - 12x + 4$ has
 (a) no maxima and minima
 (b) one maxima and one minima
 (c) two maxima
 (d) two minima
45. The greatest value of $f(x) = (x+1)^{1/3} - (x-1)^{1/3}$ on $[0, 1]$ is :
 (a) 1 (b) 2
 (c) 3 (d) $2^{1/3}$
46. The function $f(x) = x^2(x-2)^2$
 (a) decreases on $(0, 1) \cup (2, \infty)$
 (b) increase on $(-\infty, 0) \cup (1, 2)$
 (c) has a local maximum value 0
 (d) has a local maximum value 1
47. The maximum value of the function $y = x(x-1)^2$, $0 \leq x \leq 2$ is
 (a) 0 (b) $4/27$
 (c) -4 (d) none of these
48. The point in the interval $[0, 2\pi]$, where $f(x) = e^x \sin x$ has maximum slope, is
 (a) 0 (b) $\frac{\pi}{2}$
 (c) 2π (d) $\frac{3\pi}{2}$
49. The minimum value of x^x is attained (where x is positive real number) when x is equal to :
 (a) e (b) e^{-1}
 (c) 1 (d) e^2
50. The maximum value of $x^3 - 3x$ in the interval $[0, 2]$, is
 (a) -2 (b) 0
 (c) 2 (d) None of these
51. If $A > 0$, $B > 0$ and $A + B = \frac{\pi}{3}$, then the maximum value of $\tan A \tan B$ is
 (a) $\frac{1}{\sqrt{3}}$ (b) $\frac{1}{3}$
 (c) 3 (d) $\sqrt{3}$
52. The maximum slope of the curve $y = -x^3 + 3x^2 + 9x - 27$ is
 (a) 0 (b) 12
 (c) 16 (d) 32
53. The function $f(x) = \int_1^x \{2(t-1)(t-2)^3 + 3(t-1)^2(t-2)^2\} dt$ attains its local maximum value at $x =$
 (a) 1 (b) 2
 (c) 3 (d) 4
54. The maximum area of the rectangle that can be inscribed in a circle of radius r , is
 (a) πr^2 (b) r^2
 (c) $\pi r^2/4$ (d) $2r^2$
55. A triangular park is enclosed on two sides by a fence and on the third side by a straight river bank. The two sides having fence are of same length x . The maximum area enclosed by the park is
 (a) $\frac{3}{2}x^2$ (b) $\sqrt{\frac{x^3}{8}}$
 (c) $\frac{1}{2}x^2$ (d) πx^2
56. The greatest and the least value of the function, $f(x) = \sqrt{1-2x+x^2} - \sqrt{1+2x+x^2}$, $x \in (-\infty, \infty)$ are
 (a) 2, -2 (b) 2, -1
 (c) 2, 0 (d) none
57. The function $f(x) = 2x^3 - 15x^2 + 36x + 4$ has local maxima at
 (a) $x = 2$ (b) $x = 4$
 (c) $x = 0$ (d) $x = 3$



58. The maximum value of xy subject to $x + y = 8$, is
 (a) 8 (b) 16
 (c) 20 (d) 24
59. Let $f(x) = (1+b^2)x^2 + 2bx + 1$ and $m(b)$ the minimum value of $f(x)$ for a given b . As b varies, the range of $m(b)$ is
 (a) $[0, 1]$ (b) $(0, 1/2]$
 (c) $\left[\frac{1}{2}, 1\right]$ (d) $(0, 1]$
60. $f(x) = 1 + [\cos x]x$, in $0 \leq x \leq \frac{\pi}{2}$
 (a) has a minimum value 0
 (b) has a maximum value 2
 (c) is continuous in $\left[0, \frac{\pi}{2}\right]$
 (d) is not differentiable at $x = \frac{\pi}{2}$
61. The minimum value of $2^{(x^2-3)^3+27}$, is
 (a) 2^{27} (b) 2
 (c) 1 (d) 4
62. Area of the greatest rectangle that can be inscribed in the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is
 (a) ab (b) $2ab$
 (c) a/b (d) $\sqrt{ab}$
63. The difference between the greatest and least values of the function, $f(x) = \cos x + \frac{1}{2} \cos 2x - \frac{1}{3} \cos 3x$ is :
 (a) $4/3$ (b) 1
 (c) $9/4$ (d) $1/6$
64. A line is drawn through the point $(1, 2)$ to meet the coordinate axes at P and Q such that it forms a ΔOPQ , where O is the origin, if the area of the ΔOPQ is least, then the slope of the line PQ is
 (a) $-\frac{1}{4}$ (b) -4
 (c) -2 (d) $-\frac{1}{2}$

Numerical Value Type Questions

65. The radius of the base of a cone is increasing at the rate of 3 cm/minute and the altitude is decreasing at the rate of 4 cm/minute. The rate of change of lateral surface when the radius = 7 cm and altitude = 24 cm, in cm^2/min is:
66. A ladder 10 metres long rests with one end against a vertical wall, the other end on the floor. The lower end moves away from the wall at the rate of 2 metres/minute. The rate at which the upper end falls when its base is 6 metres away from the wall, in M/min is :
67. If the distance 's' metres travelled by a particle in t seconds is given by $s = t^3 - 3t^2$, then the velocity of the particle when the acceleration is zero in m/s is
68. An object is moving in the clockwise direction around the unit circle $x^2 + y^2 = 1$. As it passes through the point $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$, its y -coordinate is decreasing at the rate of 3 unit per second. The rate at which the x -coordinate changes at this point is (in unit per second)
69. If $V = \frac{4}{3}\pi r^3$, at what rate in cubic units is V increasing when $r = 10$ and $\frac{dr}{dt} = 0.01$?
70. Side of an equilateral triangle expands at the rate of 2 cm/s. The rate of increase of its area when each side is 10 cm, in cm^2/sec is:
71. The radius of a sphere is changing at the rate of 0.1 cm/s. The rate of change of its surface area when the radius is 200 cm, in cm^2/sec is:
72. The surface area of a sphere when its volume is increasing at the same rate as its radius, in sq. unit is :
73. The surface area of a cube is increasing at the rate of $2 \text{ cm}^2/\text{s}$. When its edge is 90 cm, the volume is increasing at the rate of (in cm^3/sec)
74. The sides of an equilateral triangle are increasing at the rate of 2 cm/s. The rate at which the area increases, when the side is 10 cm, in cm^2/s is:



75. The distance moved by the particle in time t is given by $x = t^3 - 12t^2 + 6t + 8$. At the instant when its acceleration is zero, the velocity is
76. The circumference of a circle is measured as 28 cm with an error of 0.01 cm. The percentage error in the area is
77. The triangle formed by the tangent to the curve $f(x) = x^2 + bx - b$ at the point $(1, 1)$ and the coordinate axes, lies in the first quadrant. If its area is 2, then the value of b is
78. If the normal to the curve $y = f(x)$ at the point $(3, 4)$ makes an angle $\frac{3\pi}{4}$ with the positive x -axis, then $f'(3)$ is equal to
79. Find the shortest distance between the line $y = x - 2$ and the parabola $y = x^2 + 3x + 2$.
80. If $f(x)$ is differentiable in the interval $[2, 5]$, where $f(2) = \frac{1}{5}$ and $f(5) = \frac{1}{2}$, then there exists a number c , $2 < c < 5$ for which $f'(c)$ is equal to



EXERCISE - 2 : PREVIOUS YEAR JEE MAINS QUESTIONS

1. The normal to the curve, $x^2 + 2xy - 3y^2 = 0$, at $(1, 1)$: (2015)
 (a) meets the curve again in the third quadrant.
 (b) meets the curve again in the fourth quadrant.
 (c) does not meet the curve again.
 (d) meets the curve again in the second quadrant.
2. Let $f(x)$ be a polynomial of degree four having extreme value at $x = 1$ and $x = 2$. If $\lim_{x \rightarrow 0} \left[1 + \frac{f(x)}{x^2} \right] = 3$, then $f(2)$ is equal to: (2015)
 (a) 0 (b) 4
 (c) -8 (d) -4
3. If Rolle's theorem holds for the function $f(x) = 2x^3 + bx^2 + cx$, $x \in [-1, 1]$, at the point $x = \frac{1}{2}$, then $2b + c$ equals : (2015/Online Set-1)
 (a) 1 (b) 2
 (c) -1 (d) -3
4. Let k and K be the minimum and the maximum values of the function $f(x) = \frac{(1+x)^{0.6}}{1+x^{0.6}}$ in $[0, 1]$ respectively, then the ordered pair (k, K) is equal to: (2015/Online Set-2)
 (a) $(2^{-0.4}, 1)$ (b) $(2^{-0.4}, 2^{0.6})$
 (c) $(2^{-0.6}, 1)$ (d) $(1, 2^{0.6})$
5. A wire of length 2 units is cut into two parts which are bent respectively to form a square of side $=x$ unit and a circle of radius $=r$ units. If the sum of the areas of the square and the circle so formed is minimum, then : (2016)
 (a) $(4 - \pi)x = \pi r$ (b) $x = 2r$
 (c) $2x = r$ (d) $2x = (\pi + 4)r$
6. Consider $f(x) = \tan^{-1} \left(\sqrt{\frac{1+\sin x}{1-\sin x}} \right)$, $x \in \left(0, \frac{\pi}{2} \right)$.
 A normal to $y = f(x)$ at $x = \frac{\pi}{6}$ also passes through the point : (2016)
 (a) $\left(0, \frac{2\pi}{3} \right)$ (b) $\left(\frac{\pi}{6}, 0 \right)$
 (c) $\left(\frac{\pi}{4}, 0 \right)$ (d) $(0, 0)$
7. If the tangent at a point P, with parameter t , on the curve $x = 4t^2 + 3$, $y = 8t^3 - 1$, $t \in \mathbb{R}$, meets the curve again at a point Q, then the coordinates of Q are : (2016/Online Set-1)
 (a) $(t^2 + 3, -t^3 - 1)$ (b) $(4t^2 + 3, -8t^3 - 1)$
 (c) $(t^2 + 3, t^3 - 1)$ (d) $(16t^2 + 3, -64t^3 - 1)$
8. The minimum distance of a point on the curve $y = x^2 - 4$ from the origin is : (2016/Online Set-1)
 (a) $\frac{\sqrt{19}}{2}$ (b) $\sqrt{\frac{15}{2}}$
 (c) $\frac{\sqrt{15}}{2}$ (d) $\sqrt{\frac{19}{2}}$
9. The normal to the curve $y(x-2)(x-3) = x+6$ at the point where the curve intersects the y-axis passes through the point: (2017)
 (a) $\left(-\frac{1}{2}, -\frac{1}{2} \right)$ (b) $\left(\frac{1}{2}, \frac{1}{2} \right)$
 (c) $\left(\frac{1}{2}, -\frac{1}{3} \right)$ (d) $\left(\frac{1}{2}, \frac{1}{3} \right)$
10. Twenty meters of wire is available for fencing off a flower-bed in the form of a circular sector. Then the maximum area (in sq. m) of the flower-bed, is: (2017)
 (a) 12.5 (b) 10
 (c) 25 (d) 30
11. The tangent at the point $(2, -2)$ to the curve, $x^2y^2 - 2x = 4$ $(1 - y)$ does not pass through the point : (2017/Online Set-1)
 (a) $\left(4, \frac{1}{3} \right)$ (b) $(8, 5)$
 (c) $(-4, -9)$ (d) $(-2, -7)$



12. The function f defined by $f(x) = x^3 - 3x^2 + 5x + 7$, is
(2017/Online Set-2)
- increasing in $\mathbb{R}$.
 - decreasing in $\mathbb{R}$.
 - decreasing in $(0, \infty)$ and increasing in $(-\infty, 0)$.
 - increasing in $(0, \infty)$ and decreasing in $(-\infty, 0)$.
13. If the curves $y^2 = 6x$, $9x^2 + by^2 = 16$ intersect each other at right angles, then the value of b is (2018)
- $\frac{9}{2}$
 - 6
 - $\frac{7}{2}$
 - 4
14. Let $f(x) = x^2 + \frac{1}{x^2}$ and $g(x) = x - \frac{1}{x}$,
 $x \in \mathbb{R} - \{-1, 0, 1\}$. If $h(x) = \frac{f(x)}{g(x)}$, then the local minimum value of $h(x)$ is: (2018)
- $2\sqrt{2}$
 - 3
 - 3
 - $-2\sqrt{2}$
15. If a right circular cone, having maximum volume, is inscribed in a sphere of radius 3 cm, then the curved surface area (in cm^2) of this cone is : (2018/Online Set-1)
- $6\sqrt{2}\pi$
 - $6\sqrt{3}\pi$
 - $8\sqrt{2}\pi$
 - $8\sqrt{3}\pi$
16. Let $f(x)$ be a polynomial of degree 4 having extreme values at $x = 1$ and $x = 2$. If $\lim_{x \rightarrow 0} \left(\frac{f(x)}{x^2} + 1 \right) = 3$ then $f(-1)$ is equal to: (2018/Online Set-2)
- $\frac{9}{2}$
 - $\frac{5}{2}$
 - $\frac{3}{2}$
 - $\frac{1}{2}$
17. Let M and m be respectively the absolute maximum and the absolute minimum values of the function, $f(x) = 2x^3 - 9x^2 + 12x + 5$ in the interval $[0, 3]$. Then $M - m$ is equal to : (2018/Online Set-3)
- 5
 - 9
 - 4
 - 1
18. The shortest distance between the line $y = x$ and the curve $y^2 = x - 2$ is: (2019-04-08/Shift-1)
- 2
 - $\frac{7}{8}$
 - $\frac{7}{4\sqrt{2}}$
 - $\frac{11}{4\sqrt{2}}$
19. If S_1 and S_2 are respectively the sets of local minimum and local maximum points of the function, $f(x) = 9x^4 + 12x^3 - 36x^2 + 25$, $x \in \mathbb{R}$ then (2019-04-08/Shift-1)
- $S_1 = \{-2\}$; $S_2 = \{0, 1\}$
 - $S_1 = \{-2, 0\}$; $S_2 = \{1\}$
 - $S_1 = \{-2, 1\}$; $S_2 = \{0\}$
 - $S_1 = \{-1\}$; $S_2 = \{0, 2\}$
20. Let $f: [0, 2] \rightarrow \mathbb{R}$ be a twice differentiable function such that $f'(x) > 0$, for all $x \in (0, 2)$. If $\phi(x) = f(x) + f(2 - x)$, then $\phi(x)$ is: (2019-04-08/Shift-1)
- increasing on $(0, 1)$ and decreasing on $(1, 2)$.
 - decreasing on $(0, 2)$
 - decreasing on $(0, 1)$ and increasing on $(1, 2)$.
 - increasing on $(0, 2)$
21. The height of a right circular cylinder of maximum volume inscribed in a sphere of radius 3 is : (2019-04-08/Shift-2)
- $\sqrt{6}$
 - $\frac{2}{3}\sqrt{3}$
 - $2\sqrt{3}$
 - $\sqrt{3}$
22. If the tangent to the curve, $y = x^3 + ax - b$ at the point $(1, -5)$ is perpendicular to the line, $-x + y + 4 = 0$, then which one of the following points lies on the curve? (2019-04-09/Shift-1)
- $(-2, 1)$
 - $(-2, 2)$
 - $(2, -1)$
 - $(2, -2)$



23. Let S be the set of all values of x for which the tangent to the curve $y = f(x) = x^3 - x^2 - 2x$ at (x, y) is parallel to the line segment joining the points $(1, f(1))$ and $(-1, f(-1))$ then S is equal to:
(2019-04-09/Shift-1)
- (a) $\left\{\frac{1}{3}, 1\right\}$ (b) $\left\{-\frac{1}{3}, -1\right\}$
(c) $\left\{\frac{1}{3}, -1\right\}$ (d) $\left\{-\frac{1}{3}, 1\right\}$
24. If $f(x)$ is a non-zero polynomial of degree four, having local extreme points at $x = -1, 0, 1$ then the set $S = \{x \in R : f(x) = f(0)\}$ contains exactly k real values, then k is
(2019-04-09/Shift-1)
25. A water tank has the shape of an inverted right circular cone, whose semi-vertical angle is $\tan^{-1}\left(\frac{1}{2}\right)$. Water is poured into it at a constant rate of 5 cubic meter per minute. Then the rate (in m/min.), at which the level of water is rising at the instant when the depth of water in the tank is 10m; is:
(2019-04-09/Shift-2)
- (a) $\frac{1}{15\pi}$ (b) $\frac{1}{10\pi}$
(c) $\frac{2}{\pi}$ (d) $\frac{1}{5\pi}$
26. Let $f(x) = e^x - x$ and $g(x) = x^2 - x, \forall x \in R$. Then the set of all $x \in R$, where the function $h(x) = (f \circ g)(x)$ is increasing, is:
(2019-04-10/Shift-1)
- (a) $\left[-1, \frac{-1}{2}\right] \cup \left[\frac{1}{2}, \infty\right)$ (b) $\left[0, \frac{1}{2}\right] \cup [1, \infty)$
(c) $[0, \infty)$ (d) $\left[-\frac{1}{2}, 0\right] \cup [1, \infty)$
27. If the tangent to the curve $y = \frac{x}{x^2 - 3}, x \in R (x \neq \pm\sqrt{3})$, at a point $(\alpha, \beta) \neq (0, 0)$ on it is parallel to the line $2x + 6y - 11 = 0$ then:
(2019-04-10/Shift-2)
- (a) $|6\alpha + 2\beta| = 19$ (b) $|6\alpha + 2\beta| = 9$
(c) $|2\alpha + 6\beta| = 19$ (d) $|2\alpha + 6\beta| = 9$
28. A spherical iron ball of radius 10 cm is coated with a layer of ice of uniform thickness that melts at a rate of $50\text{cm}^3/\text{min}$. When the thickness of the ice is 5 cm, then the rate at which the thickness (in cm/min) of the ice decreases, is:
(2019-04-10/Shift-2)
- (a) $\frac{1}{18\pi}$ (b) $\frac{1}{36\pi}$
(c) $\frac{5}{6\pi}$ (d) $\frac{1}{9\pi}$
29. Let $a_1, a_2, a_3, \dots$ be an A.P. with $a_6 = 2$ then the common difference of this A.P., which maximises the product $a_1 \cdot a_4 \cdot a_5$ is:
(2019-04-10/Shift-2)
- (a) $\frac{3}{2}$ (b) $\frac{8}{5}$
(c) $\frac{6}{5}$ (d) $\frac{2}{3}$
30. A 2 m ladder leans against a vertical wall. If the top of the ladder begins to slide down the wall at the rate 25 cm/sec., then the rate (in cm/sec.) at which the bottom of the ladder slides away from the wall on the horizontal ground when the top of the ladder is 1 m above the ground is:
(2019-04-12/Shift-1)
- (a) $25\sqrt{3}$ (b) $\frac{25}{\sqrt{3}}$
(c) $\frac{25}{3}$ (d) 25



31. The maximum volume (in cu.m) of the right circular cone having slant height 3 m is: (2019-01-09/Shift-1)
- (a) 6π (b) $3\sqrt{3}\pi$
- (c) $\frac{4}{3}\pi$ (d) $2\sqrt{3}\pi$
32. If θ denotes the acute angle between the curves, $y = 10 - x^2$ and $y = 2 + x^2$ at a point of their intersection, then $|\tan \theta|$ is equal to: (2019-01-09/Shift-1)
- (a) $\frac{4}{9}$ (b) $\frac{8}{15}$
- (c) $\frac{7}{17}$ (d) $\frac{8}{17}$
33. The shortest distance between the point $\left(\frac{3}{2}, 0\right)$ and the curve $y = \sqrt{x}, (x > 0)$, is: (2019-01-10/Shift-1)
- (a) $\frac{\sqrt{5}}{2}$ (b) $\frac{\sqrt{3}}{2}$
- (c) $\frac{3}{2}$ (d) $\frac{5}{4}$
34. The tangent to the curve, $y = xe^{x^2}$ passing through the point (1, e) also passes through the point: (2019-01-10/Shift-2)
- (a) (2, 3e) (b) $\left(\frac{4}{3}, 2e\right)$
- (c) $\left(\frac{5}{3}, 2e\right)$ (d) (3, 6e)
35. A helicopter is flying along the curve given by $y - x^{3/2} = 7, (x \geq 0)$. A soldier positioned at the point $\left(\frac{1}{2}, 7\right)$ wants to shoot down the helicopter when it is nearest to him. Then this nearest distance is: (2019-01-10/Shift-2)
- (a) $\frac{\sqrt{5}}{6}$ (b) $\frac{1}{3}\sqrt{\frac{7}{3}}$
- (c) $\frac{1}{6}\sqrt{\frac{7}{3}}$ (d) $\frac{1}{2}$
36. The maximum value of the function $f(x) = 3x^3 - 18x^2 + 27x - 40$ on the set $S = \{x \in R : x^2 + 30 \leq 11x\}$ is: (2019-01-11/Shift-1)
37. Let $f(x) = \frac{x}{\sqrt{a^2 + x^2}} - \frac{d - x}{\sqrt{b^2 + (d - x)^2}}, x \in R$, where a, b and d are non-zero real constants. Then: (2019-01-11/Shift-2)
- (a) f is an increasing function of x
- (b) f is a decreasing function of x
- (c) f' is not a continuous function of x
- (d) f is neither increasing nor decreasing function of x
38. If the function f given by $f(x) = x^3 - 3(a - 2)x^2 + 3ax + 7, f(0) = 7$ for some $a \in R$ is increasing in (0, 1] and decreasing in [1, 5], then a root of the equation, $\frac{f(x) - 14}{(x - 1)^2} = 0 (x \neq 1)$ (2019-01-12/Shift-2)
- (a) -7 (b) 5
- (c) 7 (d) 6
39. Let P (h, k) be a point on the curve $y = x^2 + 7x + 2$, nearest to the line, $y = 3x - 3$. Then the equation of the normal to the curve at P is: (2020-09-02/Shift-1)
- (a) $x + 3y - 62 = 0$ (b) $x - 3y - 11 = 0$
- (c) $x - 3y + 22 = 0$ (d) $x + 3y + 26 = 0$
40. If p(x) be a polynomial of degree three that has a local maximum value 8 at $x = 1$ and a local minimum value 4 at $x = 2$; then p(0) is equal to: (2020-09-02/Shift-1)
- (a) 12 (b) -12
- (c) -24 (d) 6
41. If the tangent to the curve $y = x + \sin y$ at a point (a, b) is parallel to the line joining $\left(0, \frac{3}{2}\right)$ and $\left(\frac{1}{2}, 2\right)$, then: (2020-09-02/Shift-1)
- (a) $b = \frac{\pi}{2} + a$ (b) $|a + b| = 1$
- (c) $|b - a| = 1$ (d) $b = a$



42. The equation of the normal to the curve $y = (1+x)^{2y} + \cos^2(\sin^{-1} x)$ at $x=0$ is :
(2020-09-02/Shift-2)
- (a) $y+4x=2$ (b) $2y+x=4$
(c) $x+4y=8$ (d) $y=4x+2$
43. Let $f: (-1, \infty) \rightarrow R$ be defined by $f(0) = 1$ and $f(x) = \frac{1}{x} \log_e(1+x), x \neq 0$. Then the function f :
(2020-09-02/Shift-2)
- (a) increases in $(-1, \infty)$
(b) decreases in $(-1, 0)$ and increases in $(0, \infty)$
(c) increases in $(-1, 0)$ and decreases in $(0, \infty)$
(d) decreases in $(-1, \infty)$.
44. The function, $f(x) = (3x-7)x^{2/3}, x \in R$ is increasing for all x lying in :
(2020-09-03/Shift-1)
- (a) $\left(-\infty, -\frac{14}{15}\right) \cup (0, \infty)$
(b) $\left(-\infty, \frac{14}{15}\right)$
(c) $(-\infty, 0) \cup \left(\frac{14}{15}, \infty\right)$
(d) $(-\infty, 0) \cup \left(\frac{3}{7}, \infty\right)$
45. Suppose $f(x)$ is a polynomial of degree four, having critical points at $-1, 0, 1$. If $T = \{x \in R \mid f'(x) = f'(0)\}$, then the sum of squares of all the element of T is :
(2020-09-03/Shift-2)
- (a) 6 (b) 2
(c) 8 (d) 4
46. If the surface area of a cube is increasing at a rate of $3.6 \text{ cm}^2/\text{sec}$, retaining its shape; then the rate of change of its volume (in cm^3/sec), when the length of a side of the cube is 10 cm , is :
(2020-09-03/Shift-2)
- (a) 9 (b) 10
(c) 18 (d) 20
47. The area (in sq. units) of the largest rectangle ABCD whose vertices A and B lie on the x -axis and vertices C and D lie on the parabola, $y = x^2 - 1$ below the x -axis, is:
(2020-09-04/Shift-2)
- (a) $\frac{2}{3\sqrt{3}}$ (b) $\frac{4}{3}$
(c) $\frac{1}{3\sqrt{3}}$ (d) $\frac{4}{3\sqrt{3}}$
48. If $x=1$ is a critical point of the function $f(x) = (3x^2+ax-2-a)e^x$, then: (2020-09-05/Shift-2)
- (a) $x=1$ is a local minima and $x = -\frac{2}{3}$ is a local maxima of f .
(b) $x=1$ is a local maxima and $x = -\frac{2}{3}$ is a local minima of f .
(c) $x=1$ and $x = -\frac{2}{3}$ are local minima of f .
(d) $x=1$ and $x = -\frac{2}{3}$ are local maxima of f .
49. Which of the following points lies on the tangent to the curve $x^4 e^y + 2\sqrt{y+1} = 3$ at the point $(1,0)$?
(2020-09-05/Shift-2)
- (a) (2,6) (b) (2,2)
(c) (-2,6) (d) (-2,4)
50. The position of a moving car at time t is given by $f(t) = at^2 + bt + c, t > 0$, where a, b and c are real numbers greater than 1. Then the average speed of the car over the time interval $[t_1, t_2]$ is attained at the point:
(2020-09-06/Shift-1)
- (a) $[t_1 + t_2] / 2$ (b) $2a(t_1 + t_2) + b$
(c) $(t_2 - t_1) / 2$ (d) $a(t_2 - t_1) + b$
51. Let AD and BC be two vertical poles at A and B respectively on a horizontal ground. If $AD = 8 \text{ m}$, $BC = 11 \text{ m}$ and $AB = 10 \text{ m}$; then the distance (in meters) of a point M on AB from the point A such that $MD^2 + MC^2$ is minimum is _____.
(2020-09-06/Shift-1)



52. The set of all real values of λ for which the function $f(x) = (1 - \cos^2 x)(\lambda + \sin x)$, $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$,

has exactly one maxima and exactly one minima, is:

(2020-09-06/Shift-2)

- (a) $\left(-\frac{3}{2}, \frac{3}{2}\right) - \{0\}$ (b) $\left(-\frac{1}{2}, \frac{1}{2}\right) - \{0\}$
(c) $\left(-\frac{3}{2}, \frac{3}{2}\right)$ (d) $\left(-\frac{1}{2}, \frac{1}{2}\right)$

53. If the tangent to the curve, $y = f(x) = x \log_e x$, ($x > 0$) at a point $(c, f(c))$ is parallel to the line-segment joining the points $(1, 0)$ and (e, e) , then c is equal to :

(2020-09-06/Shift-2)

- (a) $e^{\left(\frac{1}{1-e}\right)}$ (b) $\frac{e-1}{e}$
(c) $\frac{1}{e-1}$ (d) $e^{\left(\frac{1}{e-1}\right)}$

54. For all twice differentiable functions $f: \mathbb{R} \rightarrow \mathbb{R}$, with $f(0) = f(1) = f'(0) = 0$,

(2020-09-06/Shift-2)

- (a) $f''(x) = 0$, at every point $x \in (0, 1)$
(b) $f''(x) \neq 0$, at every point $x \in (0, 1)$
(c) $f''(x) = 0$, for some $x \in (0, 1)$
(d) $f''(0) = 0$

55. Let the function, $f: [-7, 0] \rightarrow \mathbb{R}$ be continuous on $[-7, 0]$ and differentiable on $(-7, 0)$. If $f(-7) = -3$ and $f'(x) \leq 2$ for all $x \in (-7, 0)$, then for all such functions f , $f(-1) + f(0)$ lies in the interval:

(7-1-2020/Shift-1)

- (a) $[-6, 20]$ (b) $(-\infty, 20]$
(c) $(-\infty, 11]$ (d) $[-3, 11]$

56. Let $f(x)$ be a polynomial of degree 5 such that $x = \pm 1$ are its critical points. If $\lim_{x \rightarrow 0} \left(2 + \frac{f(x)}{x^3}\right) = 4$, then which one

of the following is not true?

(7-1-2020/Shift-2)

- (a) $f(1) - 4f(-1) = 4$
(b) $x = 1$ is a point of maxima and $x = -1$ is a point of minimum of f .
(c) f is an odd function.
(d) $x = 1$ is a point of minima and $x = -1$ is a point of maxima of f .

57. The value of c in Lagrange's mean value theorem for the function $f(x) = x^3 - 4x^2 + 8x + 11$, where $x \in [0, 1]$ is :

(7-1-2020/Shift-2)

- (a) $\frac{4 - \sqrt{7}}{3}$ (b) $\frac{2}{3}$
(c) $\frac{\sqrt{7} - 2}{3}$ (d) $\frac{4 - \sqrt{5}}{3}$

58. If c is a point at which Rolle's theorem holds for the function, $f(x) = \log_e \left(\frac{x^2 + \alpha}{7x}\right)$ in the interval $[3, 4]$,

where $\alpha \in \mathbb{R}$ then $f''(c)$ is equal to:

(8-1-2020/Shift-1)

- (a) $-\frac{1}{24}$ (b) $-\frac{1}{12}$
(c) $\frac{\sqrt{3}}{7}$ (d) $\frac{1}{12}$

59. Let $f(x) = x \cos^{-1}(\sin(-|x|))$, $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$,

then which of the following is true? (8-1-2020/Shift-1)

- (a) $f'(0) = -\frac{\pi}{2}$
(b) f' is decreasing in $\left(-\frac{\pi}{2}, 0\right)$ and increasing in $\left(0, \frac{\pi}{2}\right)$
(c) f is not differentiable at $x = 0$
(d) f' is increasing in $\left(-\frac{\pi}{2}, 0\right)$ and decreasing in $\left(0, \frac{\pi}{2}\right)$

60. Let the normal at a P on the curve $y^2 - 3x^2 + y + 10 = 0$ intersect the y-axis at $\left(0, \frac{3}{2}\right)$. If m is the slope of the tangent at P to the curve, then $|m|$ is equal to _____.

(8-1-2020/Shift-1)



61. The length of the perpendicular from the origin, on the normal to the curve, $x^2 + 2xy - 3y^2 = 0$ at the point (2,2) is: (8-1-2020/Shift-2)
- (a) 2 (b) $2\sqrt{2}$
(c) $4\sqrt{2}$ (d) $\sqrt{2}$
62. Let $f(x)$ be a polynomial of degree 3 such that $f(-1) = 10$, $f(1) = -6$, $f(x)$ has a critical point at $x = -1$ and $f'(x)$ has a critical point at $x = 1$. Then the local minima at $x =$ _____ (8-1-2020/Shift-2)
63. A spherical iron ball of 10 cm radius is coated with a layer of ice of uniform thickness that melts at the rate of 50 cm³/min. When the thickness of ice is 5 cm, then the rate (in cm/min.) at which the thickness of ice decreases, is: (9-1-2020/Shift-1)
- (a) $\frac{5}{6\pi}$ (b) $\frac{1}{54\pi}$
(c) $\frac{1}{36\pi}$ (d) $\frac{1}{19\pi}$
64. Let f be any function continuous on $[a, b]$ and twice differentiable on (a, b) . If for all $x \in (a, b)$, $f'(x) > 0$ and $f''(x) < 0$, then for any $c \in (a, b)$, $\frac{f(c) - f(a)}{f(b) - f(c)}$ is greater than: (9-1-2020/Shift-1)
- (a) $\frac{b-c}{c-a}$ (b) 1
(c) $\frac{c-a}{b-c}$ (d) $\frac{b+a}{b-a}$
65. Let a function $f : [0, 5] \rightarrow \mathbb{R}$, be continuous, $f(1) = 3$ and F be defined as: $F(x) = \int_1^x t^2 g(t) dt$, where $g(t) = \int_1^t f(u) du$. Then for the function F , the point $x = 1$ is (9-1-2020/Shift-2)
- (a) a point of inflection (b) a point of local maxima
(c) a point of local minima (d) not a critical point
66. The function $f(x) = \frac{4x^3 - 3x^2}{6} - 2\sin x + (2x - 1)\cos x$ (24-02-2021/Shift-1)
- (a) Decreases in $\left(-\infty, \frac{1}{2}\right]$ (b) Increases in $\left(-\infty, \frac{1}{2}\right]$
(c) Increases in $\left[\frac{1}{2}, \infty\right)$ (d) Decreases in $\left[\frac{1}{2}, \infty\right)$
67. The minimum value of α for which the equation $\frac{4}{\sin x} + \frac{1}{1 - \sin x} = \alpha$ has at least one solution in $\left(0, \frac{\pi}{2}\right)$ is _____. (24-02-2021/Shift-1)
68. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined as
- $$f(x) = \begin{cases} -55x, & \text{if } x < -5 \\ 2x^3 - 3x^2 - 120x, & \text{if } -5 \leq x \leq 4 \\ 2x^3 - 3x^2 - 36x - 336, & \text{if } x > 4, \end{cases}$$
- Let $A = \{x \in \mathbb{R} : f \text{ is increasing}\}$. Then A is equal to (24-02-2021/Shift-2)
- (a) $(-\infty, -5) \cup (4, \infty)$ (b) $(-5, \infty)$
(c) $(-5, -4) \cup (4, \infty)$ (d) $(-\infty, -5) \cup (-4, \infty)$
69. If P is a point on the parabola $y = x^2 + 4$ which is closest to the straight line $y = 4x - 1$, then the co-ordinates of P are (24-02-2021/Shift-2)
- (a) (3, 13) (b) (2, 8)
(c) (-2, 8) (d) (1, 5)
70. If the curve $y = ax^2 + bx + c$, $x \in \mathbb{R}$, passes through the point (1, 2) and the tangent line to this curve at origin is $y = x$, then the possible values of a, b, c are : (24-02-2021/Shift-2)
- (a) $a = 1, b = 1, c = 0$ (b) $a = \frac{1}{2}, b = \frac{1}{2}, c = 1$
(c) $a = -1, b = 1, c = 1$ (d) $a = 1, b = 0, c = 1$



71. If the curves, $\frac{x^2}{a} + \frac{y^2}{b} = 1$ and $\frac{x^2}{c} + \frac{y^2}{d} = 1$ intersect each other at an angle of 90° , then which of the following relations is TRUE? (25-02-2021/Shift-1)
- (a) $ab = \frac{c+d}{a+b}$ (b) $a-c = b+d$
- (c) $a+b = c+d$ (d) $a-b = c-d$
72. If Rolle's theorem holds for the function $f(x) = x^3 - ax^2 + bx - 4$, $x \in [1, 2]$ with $f'\left(\frac{4}{3}\right) = 0$, then ordered pair (a, b) is equal to: (25-02-2021/Shift-1)
- (a) (5, 8) (b) (-5, 8)
- (c) (5, -8) (d) (-5, -8)
73. Let $f(x)$ be a polynomial of degree 6 in x , in which the coefficient of x^6 is unity and it has extrema at $x = -1$ and $x = 1$. If $\lim_{x \rightarrow 0} \frac{f(x)}{x^3} = 1$, then $5 \cdot f(2)$ is equal to _____. (25-02-2021/Shift-1)
74. The shortest distance between the line $x - y = 1$ and the curve $x^2 = 2y$ is: (25-02-2021/Shift-2)
- (a) $\frac{1}{2\sqrt{2}}$ (b) $\frac{1}{2}$
- (c) $\frac{1}{\sqrt{2}}$ (d) 0
75. If the curves $x = y^4$ and $xy = k$ cut at right angles, then $(4k)^6$ is equal to: (25-02-2021/Shift-2)
76. The maximum slope of the curve $y = \frac{1}{2}x^4 - 5x^3 + 18x^2 - 19x$ occurs at the point: (26-02-2021/Shift-1)
- (a) $\left(3, \frac{21}{2}\right)$ (b) (0, 0)
- (c) (2, 9) (d) (2, 2)
77. The triangle of maximum area that can be inscribed in a given circle of radius 'r' is (26-02-2021/Shift-2)
- (a) A right angle triangle having two of its sides of length $2r$ and r .
- (b) An equilateral triangle of height $\frac{2r}{3}$.
- (c) An isosceles triangle with base equal to $2r$.
- (d) An equilateral triangle having each of its side of length $\sqrt{3}r$.
78. Let a be an integer such that all the real roots of the polynomial $2x^5 + 5x^4 + 10x^3 + 10x^2 + 10x + 10$ lie in the interval $(a, a+1)$. (26-02-2021/Shift-2)
- Then, $|a|$ is equal to _____.
79. The range of $a \in \mathbb{R}$ for which the function $f(x) = (4a-3)(x + \log_e 5) + 2(a-7)\cot\left(\frac{x}{2}\right)\sin^2\left(\frac{x}{2}\right)$, $x \neq 2n\pi$, $n \in \mathbb{N}$ has critical points is: (16-03-2021/Shift-1)
- (a) $(-\infty, -1]$ (b) $(-3, 1)$
- (c) $\left[-\frac{4}{3}, 2\right]$ (d) $[1, \infty)$
80. Let f be a real valued function, defined on $\mathbb{R} - \{-1, 1\}$ and given by $f(x) = 3\log_e \left| \frac{x-1}{x+1} \right| - \frac{2}{x-1}$
- Then in which of the following intervals, function $f(x)$ is increasing? (16-03-2021/Shift-2)
- (a) $(-\infty, \infty) - \{-1, 1\}$
- (b) $\{-\infty, -1\} \cup \left(\left[\frac{1}{2}, \infty\right) - \{1\}\right)$
- (c) $\left(-1, \frac{1}{2}\right]$
- (d) $\left(-\infty, \frac{1}{2}\right] - \{-1\}$



81. Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} \left(2 - \sin\left(\frac{1}{x}\right)\right)|x|, & x \neq 0 \\ 0, & x = 0 \end{cases} \text{ Then } f \text{ is :}$$

(17-03-2021/Shift-2)

- (a) monotonic on $(-\infty, 0) \cup (0, \infty)$
- (b) not monotonic on $(-\infty, 0)$ and $(0, \infty)$
- (c) monotonic on $(-\infty, 0)$ only
- (d) monotonic on $(0, \infty)$ only

82. Let $f : [-1, 1] \rightarrow \mathbb{R}$ be defined as $f(x) = ax^2 + bx + c$ for all $x \in [-1, 1]$, where $f''(x)$ is $\frac{1}{2}$. If $f(x) \leq \alpha$, $x \in [-1, 1]$, then the least value of α is equal to

(17-03-2021/Shift-2)

83. Let a tangent be drawn to the ellipse $\frac{x^2}{27} + y^2 = 1$ at

$(3\sqrt{3}\cos\theta, \sin\theta)$ where $\theta \in \left(0, \frac{\pi}{2}\right)$. Then the value of

θ such that the sum of intercepts on axes made by this tangent is minimum is equal to: (18-03-2021/Shift-2)

- (a) $\frac{\pi}{3}$
- (b) $\frac{\pi}{6}$
- (c) $\frac{\pi}{8}$
- (d) $\frac{\pi}{4}$

84. Let 'a' be a real number such that the function $f(x) = ax^2 + 6x - 15, x \in \mathbb{R}$ is increasing in $\left(-\infty, \frac{3}{4}\right)$ and decreasing in $\left(\frac{3}{4}, \infty\right)$. Then the function $g(x) = ax^2 - 6x + 15, x \in \mathbb{R}$ has a: (20-07-2021/Shift-1)

- (a) local minimum at $x = -\frac{3}{4}$

(b) local maximum at $x = \frac{3}{4}$

(c) local minimum at . .

(d) local maximum at $x = -\frac{3}{4}$

85. The sum of all the local minimum values of the twice differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = x^3 - 3x^2 - \frac{3f''(2)}{2}x + f''(1) \text{ is?}$$

(20-07-2021/Shift-2)

- (a) -22
- (b) 0
- (c) -27
- (d) 5

86. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined as

$$f(x) = \begin{cases} -\frac{4}{3}x^3 + 2x^2 + 3x, & x > 0 \\ 3xe^x, & x \leq 0 \end{cases}$$

Then f is increasing function in the interval.

(22-07-2021/Shift-2)

- (a) $\left(-1, \frac{3}{2}\right)$
- (b) $\left(\frac{-1}{2}, 2\right)$
- (c) $(0, 2)$
- (d) $(-3, -1)$

87. Let

$$f(x) = 3\sin^4 x + 10\sin^3 x + 6\sin^2 x - 3, x \in \left[-\frac{\pi}{6}, \frac{\pi}{2}\right].$$

Then, f is?

(25-07-2021/Shift-1)

- (a) Increasing in $\left(-\frac{\pi}{6}, 0\right)$
- (b) Decreasing in $\left(0, \frac{\pi}{2}\right)$
- (c) Decreasing in $\left(-\frac{\pi}{6}, 0\right)$
- (d) Increasing in $\left(-\frac{\pi}{6}, \frac{\pi}{2}\right)$



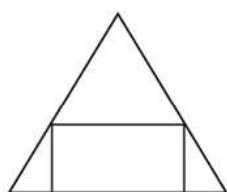
88. The number of real roots of the equation $e^{6x} - e^{4x} - 2e^{3x} - 12e^{2x} + e^x + 1 = 0$ is ?

(25-07-2021/Shift-1)

- (a) 1 (b) 6
(c) 4 (d) 2

89. If a rectangle is inscribed in an equilateral triangle of side length $2\sqrt{2}$ as shown in the figure, then the square of the largest area of such a rectangle is -

(25-07-2021/Shift-2)



90. Let $f : (a, b) \rightarrow \mathbb{R}$ be twice differentiable function such

that $f(x) = \int_a^x g(t) dt$ for a differentiable function $g(x)$.

If $f(x) = 0$ has exactly five distinct roots in (a, b) , then

$g(x)g'(x) = 0$ has at least: (27-07-2021/Shift-2)

- (a) seven roots in (a, b) (b) five roots in (a, b)
(c) three roots in (a, b) (d) twelve roots in (a, b)

91. A wire of length 36 m is cut into two pieces, one of the pieces is bent to form a square and the other is bent to form a circle. If the sum of the areas of the two figures is minimum, and the circumference of the circle is K (meter),

then $\left(\frac{4}{\pi} + 1\right)K$ is equal to _____.

(26-08-2021/Shift-1)

92. The local maximum value of the function

$f(x) = \left(\frac{2}{x}\right)^{x^2}$, $x > 0$ is: (26-08-2021/Shift-2)

- (a) $(e)^{\frac{2}{e}}$ (b) $\left(\frac{4}{\sqrt{e}}\right)^{\frac{e}{4}}$
(c) $(2\sqrt{e})^{\frac{1}{e}}$ (d) 1

93. A wire of length 20 m is to be cut into two pieces. One of the pieces is to be made into a square and the other into a regular hexagon. Then the length of the side (in meters) of the hexagon, so that the combined area of the square and the hexagon is minimum, is: (27-08-2021/Shift-1)

- (a) $\frac{10}{3+2\sqrt{3}}$ (b) $\frac{5}{3+\sqrt{3}}$
(c) $\frac{10}{2+3\sqrt{3}}$ (d) $\frac{5}{2+\sqrt{3}}$

94. The number of distinct real roots of the equation is

$$3x^4 + 4x^3 - 12x^2 + 4 = 0 \text{ _____}.$$

(27-08-2021/Shift-1)

95. A box open from top is made from a rectangular sheet of dimension x from each of the four corners and folding up the flaps. If the volume of the box is maximum, then x is equal to: (27-08-2021/Shift-2)

- (a) $\frac{a+b-\sqrt{a^2+b^2-ab}}{6}$
(b) $\frac{a+b+\sqrt{a^2+b^2-ab}}{6}$
(c) $\frac{a+b-\sqrt{a^2+b^2-ab}}{12}$
(d) $\frac{a+b-\sqrt{a^2+b^2+ab}}{6}$

96. The number of real roots of the equation $e^{4x} + 2e^{3x} - e^x - 6 = 0$ is? (31-08-2021/Shift-1)

- (a) 0 (b) 1
(c) 4 (d) 2

97. If ' R ' is the least value of ' a ' such that the function $f(x) = x^2 + ax + 1$ is increasing on $[1, 2]$ and ' S ' is the greatest value of ' a ' such that the function $f(x) = x^2 + ax + 1$ is decreasing on $[1, 2]$, then the value $|R - S|$ is _____? (31-08-2021/Shift-1)



98. Let f be any continuous function on $[0, 2]$ and twice differentiable on $(0, 2)$. If $f(0) = 0, f(1) = 1$ and $f(2) = 2$, then: (31-08-2021/Shift-2)
- (a) $f''(x) > 0$ for all $x \in (0, 2)$
(b) $f'(x) = 0$ for some $x \in [0, 2]$
(c) $f''(x) = 0$ for all $x \in (0, 2)$
(d) $f''(x) = 0$ for some $x \in (0, 2)$
99. An angle of intersection of the curves $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and $x^2 + y^2 = ab, a < b$ is (31-08-2021/Shift-2)
- (a) $\tan^{-1}\left(\frac{a-b}{2\sqrt{ab}}\right)$ (b) $\tan^{-1}\left(\frac{a+b}{\sqrt{ab}}\right)$
(c) $\tan^{-1}\left(\frac{a-b}{\sqrt{ab}}\right)$ (d) $\tan^{-1}(2\sqrt{ab})$
100. Let $f(x)$ be a cubic polynomial with $f(1) = -10, f(-1) = 6$, and has a local minima at $x = -1$. Then $f(3)$ is equal to ____ (31-08-2021/Shift-2)



EXERCISE - 3 : ADVANCED OBJECTIVE QUESTIONS

Objective Questions I [Only one correct option]

- The two curves $y^2 = 4x$ and $x^2 + y^2 - 6x + 1 = 0$ at the point (1,2)
 - intersect orthogonally
 - intersect at an angle $\frac{\pi}{3}$
 - touch each other
 - none of these
- The function 'f' is defined by $f(x) = x^p (1-x)^q$ for all $x \in \mathbb{R}$, where p,q are positive integers, has a local maximum value, for x equal to :
 - $\frac{pq}{p+q}$
 - 1
 - 0
 - $\frac{p}{p+q}$
- The triangle formed by the tangent to the parabola $y = x^2$ at the point with abscissa x_1 , the y-axis and the straight line $y = x_1^2$ has the greatest area where $x_1 \in [1, 3]$. Then x_1 equals:
 - 3
 - 2
 - 1
 - none
- Let f be a differentiable function with $f(2) = 3$ and $f'(2) = 5$, and let g be the function defined by $g(x) = x f(x)$. y-intercept of the tangent line to the graph of 'g' at point with abscissa 2, is
 - 20
 - 8
 - 20
 - 18
- If $px^2 + qx + r = 0$, p, q, r $\in \mathbb{R}$ has no real zero and the line $y + 2 = 0$ is tangent to $f(x) = px^2 + qx + r$ then
 - $p + q + r > 0$
 - $p - q + r > 0$
 - $r < 0$
 - None of these
- If $P(x) = a_0 + a_1 x^2 + a_2 x^4 + \dots + a_n x^{2n}$ be a polynomial in $x \in \mathbb{R}$ with $0 < a_1 < a_2 < \dots < a_n$, then P(x) has
 - no point of minima
 - only one point of minima
 - only two points of minima
 - none of these
- Consider the following statements S and R :
 S : Both $\sin x$ and $\cos x$ are decreasing functions in the interval $\left(\frac{\pi}{2}, \pi\right)$
 R : If a differentiable function decreases in an interval (a, b), then its derivative also decreases in (a, b). Which of the following is true ?
 - Both S and R are wrong.
 - Both S and R are correct, but R is not the correct explanation for S.
 - S is correct and R is the correct explanation for S.
 - S is correct and R is wrong.
- If the function $f(x)$ increases in the interval (a, b) then the function $\phi(x) = [f(x)]^2$.
 - Increases in (a, b)
 - decreases in (a, b)
 - we cannot say that $\phi(x)$ increases or decreases in (a, b)
 - none of these
- If at any point on a curve the sub-tangent and sub-normal are equal, then the length of the normal is equal to
 - $\sqrt{2}$ ordinate
 - ordinate
 - $\sqrt{2}$ ordinate
 - none of these
- A curve passes through the point (2, 0) and the slope of the tangent at any point (x, y) is $x^2 - 2x$ for all values of x then $3y_{\text{local max}}$ is equal to
 - 4
 - 3
 - 1
 - 2
- The radius of a right circular cylinder increases at a constant rate. Its altitude is a linear function of the radius and increases three times as fast as radius. When the radius is 1 cm the altitude is 6 cm. When the radius is 6 cm, the volume is increasing at the rate of 1 cu cm/sec. When the radius is 36 cm, the volume is increasing at a rate of n cu cm/sec. The value of 'n' is equal to
 - 12
 - 22
 - 30
 - 33



12. Slope of tangent to the curve $y = 2e^x \sin\left(\frac{\pi}{4} - \frac{x}{2}\right) \cos\left(\frac{\pi}{4} - \frac{x}{2}\right)$, where $0 \leq x \leq 2\pi$ is minimum at $x =$
- (a) 0 (b) π
(c) 2π (d) none of these
13. Let $f(x) = \begin{cases} x^3 - x^2 + 10x - 5, & x \leq 1 \\ -2x + \log_2(b^2 - 2), & x > 1 \end{cases}$ the set of values of b for which $f(x)$ have greatest value at $x = 1$ is given by :
- (a) $1 \leq b \leq 2$
(b) $b = \{1, 2\}$
(c) $b \in (-\infty, -1)$
(d) $[-\sqrt{130}, -\sqrt{2}] \cup (\sqrt{2}, \sqrt{130}]$
14. A curve is represented parametrically by the equation $x = t + e^{at}$ and $y = -t + e^{at}$ when $t \in \mathbb{R}$ and $a > 0$. If the curve touches the axis of x at the point A, then the coordinates of the point A are
- (a) (1, 0) (b) $(1/e, 0)$
(c) (e, 0) (d) $(2e, 0)$
15. If $ax + \frac{b}{x} \geq c$ for all positive x , where $a, b, c > 0$, then
- (a) $ab < \frac{c^2}{4}$ (b) $ab \geq \frac{c^2}{4}$
(c) $ab \geq \frac{c}{4}$ (d) none of these
16. If $f(x)$ is a differentiable function and $\phi(x)$ is twice differentiable function and α and β are roots of the equation $f(x) = 0$ and $\phi'(x) = 0$ respectively, then which of the following statement is true ? ($\alpha < \beta$).
- (a) there exists exactly one root of the equation $\phi'(x).f''(x) + \phi''(x).f'(x) = 0$ and (α, β)
(b) there exists at least one root of the equation $\phi'(x).f''(x) + \phi''(x).f'(x) = 0$ and (α, β)
(c) there exists odd number of roots of the equation $\phi'(x).f''(x) + \phi''(x).f'(x) = 0$ and (α, β)
(d) None of these
17. The sub-normal at any point of the curve $x^2y^2 = a^2(x^2 - a^2)$ varies as
- (a) (abscissa)⁻³ (b) (abscissa)³
(c) (ordinate)⁻³ (d) none of these
18. The sub-tangent at any point of the curve $x^m y^n = a^{m+n}$ varies as
- (a) (abscissa)² (b) (abscissa)³
(c) abscissa (d) ordinate
19. The length of the perpendicular from the origin to the normal of curve $x = a(\cos \theta + \theta \sin \theta)$, $y = a(\sin \theta - \theta \cos \theta)$ at any point θ is
- (a) a (b) $a/2$
(c) $a/3$ (d) none of these
20. If t, n, t', n' are the lengths of tangent, normal, subtangent & subnormal at a point P (x_1, y_1) on any curve $y = f(x)$ then
- (a) $t^2 + n^2 = t'n'$ (b) $\frac{1}{t^2} + \frac{1}{n^2} = \frac{1}{t'n'}$
(c) $t'n' = tn$ (d) $nt' = n't$
21. Find the shortest distance between $xy = 9$ and $x^2 + y^2 = 1$.
- (a) $3\sqrt{2} + 1$ (b) 2
(c) 4 (d) $3\sqrt{2} - 1$
22. The largest area of a rectangle which has one side on the x -axis and the two vertices on the curve $y = e^{-x^2}$ is
- (a) $\sqrt{2}e^{-1/2}$ (b) $2e^{-1/2}$
(c) $e^{-1/2}$ (d) none
23. If $(x-a)^{2n}(x-b)^{2m+1}$, where m and n are positive integers and $a > b$, is the derivative of a function f , then
- (a) $x = a$ gives neither a maximum nor a minimum
(b) $x = a$ gives a maximum
(c) $x = b$ gives neither a maximum nor a minimum
(d) none of these
24. Let (h, k) be a fixed point, where $h > 0, k > 0$. A straight line passing through this point cuts the positive direction of the coordinate axes at the points P and Q. The minimum area of the ΔOPQ , O being the origin, is
- (a) $2kh$ (b) kh
(c) $4kh$ (d) none of these



25. The set of all values of the parameters a for which the points of local minimum of the function $y = 1 + a^2 x - x^3$ satisfy the inequality $\frac{x^2 + x + 2}{x^2 + 5x + 6} \leq 0$ is

- (a) an empty set
(b) $(-3\sqrt{3}, -2\sqrt{3})$
(c) $(2\sqrt{3}, 3\sqrt{3})$
(d) $(-3\sqrt{3}, -2\sqrt{3}) \cup (2\sqrt{3}, 3\sqrt{3})$

26. The volume of the largest possible right circular cylinder that can be inscribed in a sphere of radius $= \sqrt{3}$ is:

- (a) $\frac{4}{3}\sqrt{3}\pi$ (b) $\frac{8}{3}\sqrt{3}\pi$
(c) 4π (d) 2π

27. Tangent of acute angle between the curves $y = |x^2 - 1|$ and $y = \sqrt{7 - x^2}$ at their points of intersection is

- (a) $\frac{5\sqrt{3}}{2}$ (b) $\frac{3\sqrt{5}}{2}$
(c) $\frac{5\sqrt{3}}{4}$ (d) $\frac{3\sqrt{5}}{4}$

28. A tangent to the curve $y = 1 - x^2$ is drawn so that the abscissa x_0 of the point of tangency belongs to the interval $[0, 1]$. The tangent at x_0 meets the x -axis and y -axis at A & B respectively. The minimum area of the triangle OAB, where O is the origin is

- (a) $\frac{2\sqrt{3}}{9}$ (b) $\frac{4\sqrt{3}}{9}$
(c) $\frac{2\sqrt{2}}{9}$ (d) none

29. If the polynomial equation

$a_n x^n + a_{n-1} x^{n-1} + \dots + a_2 x^2 + a_1 x + a_0 = 0$, n positive integer, has two different real roots α and β , then between α and β , the equation

$na_n x^{n-1} + (n-1)a_{n-1} x^{n-2} + \dots + a_1 = 0$ has

- (a) exactly one root (b) atmost one root
(c) atleast one root (d) no root

30. If $f(x) = \begin{vmatrix} \sin x & \sin a & \sin b \\ \cos x & \cos a & \cos b \\ \tan x & \tan a & \tan b \end{vmatrix}$, where $0 < a < b < \frac{\pi}{2}$,

then the equation $f'(x) = 0$ has, in the interval (a, b)

- (a) atleast one root (b) atmost one root
(c) no root (d) none of these

31. If $f(x) = \frac{x^2}{2 - 2\cos x}$; $g(x) = \frac{x^2}{6x - 6\sin x}$ where $0 < x < 1$,

then :

- (a) both ' f ' and ' g ' are increasing functions
(b) ' f ' is decreasing and ' g ' is increasing function
(c) ' f ' is increasing and ' g ' is decreasing function
(d) both ' f ' and ' g ' are decreasing function

32. For $x \in \left(0, \tan^{-1} \sqrt{\frac{5}{2}}\right)$, the function

$$f(x) = \cot^{-1} \left(\frac{\sqrt{2} \sin x + \sqrt{5} \cos x}{\sqrt{7}} \right)$$

- (a) increases in $\left(0, \tan^{-1} \sqrt{\frac{5}{2}}\right)$

- (b) decreases in $\left(0, \tan^{-1} \sqrt{\frac{5}{2}}\right)$

- (c) increases in $\left(0, \tan^{-1} \sqrt{\frac{2}{5}}\right)$ and decreases in

$$\left(\tan^{-1} \sqrt{\frac{2}{5}}, \tan^{-1} \sqrt{\frac{5}{2}}\right)$$

- (d) increases in $\left(\tan^{-1} \sqrt{\frac{2}{5}}, \tan^{-1} \sqrt{\frac{5}{2}}\right)$ and decreases in

$$\left(0, \tan^{-1} \sqrt{\frac{2}{5}}\right)$$



33. If $f(x) = a^{\{x\} \operatorname{sgn} x}$, $g(x) = a^{\lfloor x \rfloor \operatorname{sgn} x}$ for $a > 1$ and $x \in \mathbb{R}$, where $\{ \}$ & $\lfloor \rfloor$ denote the fractional part and integral part functions respectively, then which of the following statements hold good for the function $h(x)$, where $(\ln a) h(x) = (\ln f(x) + \ln g(x))$.
- 'h' is even and increasing
 - 'h' is odd and decreasing
 - 'h' is even and decreasing
 - 'h' is odd and increasing
34. The sum of tangent and sub-tangent at any point of the curve $y = a \log(x^2 - a^2)$ varies as
- abscissa
 - product of the coordinates
 - ordinate
 - none of these
35. For the curve $x^{m+n} = a^{m-n} y^{2n}$, where a is a positive constant and m, n are positive integers
- (sub-tangent) $^m \propto$ (sub-normal) n
 - (sub-normal) $^m \propto$ (sub-tangent) n
 - the ratio of subtangent and subnormal is constant
 - none of the above
36. $|\sin 2x| - |x| - a = 0$ does not have solution if a lies in
- $\left(\frac{3\sqrt{3}-\pi}{6}, \infty\right)$
 - $\left(\frac{3\sqrt{3}+\pi}{6}, \infty\right)$
 - $(1, \infty)$
 - None of these
37. Let $f(x) = \begin{cases} -x^2, & \text{for } x < 0 \\ x^2 + 8, & \text{for } x \geq 0 \end{cases}$. Then the x-intercept of the line that is tangent to both portions of the graph of $y = f(x)$ is
- zero
 - 1
 - 3
 - 4
38. The least area of a circle circumscribing any right triangle of area S is :
- πS
 - $2\pi S$
 - $\sqrt{2} \pi S$
 - $4\pi S$
39. The minimum value of $a \tan^2 x + b \cot^2 x$ equals the maximum value of $a \sin^2 \theta + b \cos^2 \theta$ where $a > b > 0$, when
- $a = b$
 - $a = 2b$
 - $a = 3b$
 - $a = 4b$
40. A function f such that $f'(a) = f''(a) = \dots = f^{2n}(a) = 0$ and f has a local maximum value b at $x = a$, if $f(x)$ is
- $(x-a)^{2n+2}$
 - $b - 1 - (x+1-a)^{2n+1}$
 - $b - (x-a)^{2n+2}$
 - $(x-a)^{2n+2} - b$
41. A truck is to be driven 300 km on a highway at a constant speed of x kmph. Speed rules of the highway required that $30 \leq x \leq 60$. The fuel costs Rs. 10 per litre and is consumed at the rate of $2 + \frac{x^2}{600}$ liters per hour. The wages of the driver are Rs. 200 per hour. The most economical speed to drive the truck, in kmph, is
- 30
 - 60
 - $30\sqrt{3.3}$
 - $20\sqrt{3.3}$
42. The curve $y = \frac{2x}{1+x^2}$ has
- exactly three points of inflection separated by a point of maximum and a point of minimum
 - exactly two points of inflection with a point of maximum lying between them
 - exactly two points of inflection with a point of minimum lying between them
 - exactly three points of inflection separated by two points of maximum
43. Let $f(x) = \begin{cases} x^3 + x^2 + 3x + \sin x & \left(3 + \sin \frac{1}{x}\right), & x \neq 0 \\ 0 & , & x = 0 \end{cases}$ then number of points (where $f(x)$ attains its minimum value) is
- 1
 - 2
 - 3
 - infinite many
44. The number of points with integral coordinates where tangent exists in the curve $y = \sin^{-1} 2x \sqrt{1-x^2}$ is
- 0
 - 1
 - 3
 - None



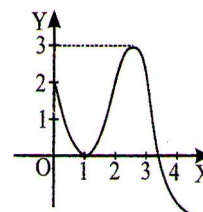
Objective Questions II [One or more than one correct option]

45. The abscissa of a point on the curve $xy = (a + x)^2$, the tangent at which cuts off equal intercepts on the coordinate axes is
 (a) $-a/\sqrt{2}$ (b) $\sqrt{2} a$
 (c) $\sqrt{2} a/2$ (d) $-\sqrt{2} a$
46. If f is an even function then
 (a) f^2 increases on (a, b)
 (b) f cannot be monotonic
 (c) f^2 need not increase on (a, b)
 (d) f has inverse
47. The function $y = \frac{2x-1}{x-2}$ ($x \neq 2$) with codomain $= \mathbb{R} - \{2\}$
 (a) is its own inverse
 (b) decreases at all values of x in the domain
 (c) has a graph entirely above x -axis
 (d) is bound for all x .
48. Let $g'(x) > 0$ and $f'(x) < 0$, $\forall x \in \mathbb{R}$, then
 (a) $g(f(x+1)) > g(f(x-1))$
 (b) $f(g(x-1)) > f(g(x+1))$
 (c) $g(f(x+1)) < g(f(x-1))$
 (d) $g(g(x+1)) < g(g(x-1))$
49. If $f(x) = x^3 - x^2 + 100x + 1001$, then
 (a) $f(2000) > f(2001)$
 (b) $f\left(\frac{1}{1999}\right) > f\left(\frac{1}{2000}\right)$
 (c) $f(x+1) > f(x-1)$
 (d) $f(3x-5) > f(3x)$
50. An extremum of the function,

$$f(x) = \frac{2-x}{\pi} \cos \pi(x+3) + \frac{1}{\pi^2} \sin \pi(x+3) \quad 0 < x < 4$$
 occurs at :
 (a) $x=1$ (b) $x=2$
 (c) $x=3$ (d) $x=\pi$
51. The length of the perpendicular from the origin to the normal of curve $x = a(\cos \theta + \theta \sin \theta)$, $y = a(\sin \theta - \theta \cos \theta)$ at a point θ is 'a', if $\theta =$
 (a) $\pi/4$ (b) $\pi/3$
 (c) $\pi/2$ (d) $\pi/6$
52. The points on the curve $y = x\sqrt{1-x^2}$, $-1 \leq x \leq 1$ at which the tangent line is vertical are
 (a) $(-1, 0)$ (b) $\left(-\frac{1}{\sqrt{2}}, -\frac{1}{2}\right)$
 (c) $(1, 0)$ (d) $\left(\frac{1}{\sqrt{2}}, \frac{1}{2}\right)$
53. Let the parabolas $y = x(c-x)$ and $y = -x^2 - ax + b$ touch each other at the point $(1, 0)$, then
 (a) $a+b+c=0$ (b) $a+b=2$
 (c) $b+c=1$ (d) $a-c=-2$
54. The value of parameter a so that the line $(3-a)x + ay + (a^2-1) = 0$ is normal to the curve $xy=1$, may lie in the interval
 (a) $(-\infty, 0)$ (b) $(1, 3)$
 (c) $(0, 3)$ (d) $(3, \infty)$
55. Which of the following pair (s) of curves is/are orthogonal.
 (a) $y^2 = 4ax$; $y = e^{-x/2a}$
 (b) $y^2 = 4ax$; $x^2 = 4ay$ at $(0, 0)$
 (c) $xy = a^2$; $x^2 - y^2 = b^2$
 (d) $y = ax$; $x^2 + y^2 = c^2$
56. If $\phi(x) = f(x) + f(2a-x)$ and $f''(x) > 0$, $a > 0$, $0 \leq x \leq 2a$ then
 (a) $\phi(x)$ increases in $[a, 2a]$
 (b) $\phi(x)$ increases in $[0, a]$
 (c) $\phi(x)$ decreases in $[0, a]$
 (d) $\phi(x)$ decreases in $[a, 2a]$
57. Let $f(x) = x^{m/n}$ for $x \in \mathbb{R}$ where m and n are integers, m even and n odd and $0 < m < n$. Then
 (a) $f(x)$ decreases on $(-\infty, 0]$
 (b) $f(x)$ increases on $[0, \infty)$
 (c) $f(x)$ increases on $(-\infty, 0]$
 (d) $f(x)$ decreases on $[0, \infty)$



58. For function $f(x) = \frac{\ln x}{x}$, which of the following statements are true.
- $f(x)$ has horizontal tangent at $x = e$
 - $f(x)$ cuts the x -axis only at one point
 - $f(x)$ is many – one function
 - $f(x)$ has one vertical tangent
59. If $f(x) = \frac{x}{1+x \tan x}$, $x \in \left(0, \frac{\pi}{2}\right)$, then
- $f(x)$ has exactly one point of minimum
 - $f(x)$ has exactly one point of maximum
 - $f(x)$ is increasing in $\left(0, \frac{\pi}{2}\right)$
 - maximum occurs at x_0 where $x_0 = \cos x_0$
60. Let $f(x) = (x-1)^4(x-2)^n$, $n \in \mathbb{N}$. then $f(x)$ has
- local minimum at $x = 2$ if n is even
 - local minimum at $x = 1$ if n is odd
 - local maximum at $x = 1$ if n is odd
 - local minimum at $x = 1$ if n is even
61. The angle between the tangent at any point P and the line joining P to the origin, where P is a point on the curve $\ln(x^2 + y^2) = c \tan^{-1} \frac{y}{x}$, c is a constant, is
- independent of x and y
 - dependent on c
 - independent of c but dependent on x
 - none of these
62. The point on the curve $xy^2 = 1$, which is nearest to the origin is
- $(2^{1/3}, 2^{1/6})$
 - $(2^{-1/3}, 2^{1/6})$
 - $(2^{-1/3}, -2^{1/6})$
 - $(-2^{-1/3}, 2^{1/6})$
63. Let $g(x) = -\frac{f(-1)}{2} x^2(x-1) - f(0)(x^2-1) + \frac{f(1)}{2} x^2(x+1) - f'(0)x(x-1)(x+1)$ where f is a thrice differentiable function. Then the correct statements are
- there exists $x \in (-1, 0)$ such that $f'(x) = g'(x)$
 - there exists $x \in (0, 1)$ such that $f'''(x) = g'''(x)$
 - there exists $x \in (-1, 1)$ such that $f'''(x) = g'''(x)$
 - there exists $x \in (-1, 1)$ such that $f'''(x) = 3f(1) - 3f(-1) - 6f'(0)$
64. If $f: [-1, 1] \rightarrow \mathbb{R}$ is a continuously differentiable function such that $f(1) > f(-1)$ and $|f'(y)| \leq 1$ for all $y \in [-1, 1]$ then
- there exists an $x \in [-1, 1]$ such that $f'(x) > 0$
 - there exists an $x \in [-1, 1]$ such that $f'(x) < 0$
 - $f(1) \leq f(-1) + 2$
 - $f(-1) \cdot f(1) < 0$
65. In a triangle ABC
- $\sin A \sin B \sin C \leq \frac{3\sqrt{3}}{8}$
 - $\sin^2 A + \sin^2 B + \sin^2 C \leq \frac{9}{4}$
 - $\sin A \sin B \sin C$ is always positive
 - $\sin^2 A + \sin^2 B = 1 + \cos C$
66. The diagram shows the graph of the derivative of a function $f(x)$ for $0 \leq x \leq 4$ with $f(0) = 0$. Which of the following could be correct statements for $y = f(x)$?



- Tangent line to $y = f(x)$ at $x = 0$ makes an angle of $\sec^{-1} \sqrt{5}$ with the x -axis.
- f is strictly increasing in $(0, 3)$
- $x = 1$ is both an inflection point as well as point of local extremum.
- Number of critical point on $y = f(x)$ is two.



Numerical Value Type Questions

67. If A is the area of the triangle formed by positive x -axis and the normal and the tangent to the circle $x^2 + y^2 = 4$ at $(1, \sqrt{3})$ then $A/\sqrt{3}$ is equal to
68. A cylindrical vessel of volume $25\frac{1}{7}$ cu metres, open at the top is to be manufactured from a sheet of metal. (The value of π is taken as $22/7$). If r and h are the radius and height of the vessel so that amount of metal is used in the least possible then rh is equal to
69. Let α be the angle in radians between $\frac{x^2}{36} + \frac{y^2}{4} = 1$ and the circle $x^2 + y^2 = 12$ at their points of intersection. If $\alpha = \tan^{-1} \frac{k}{2\sqrt{3}}$, then find the value of k^2 .
70. If α is an integer satisfying $|\alpha| \leq 5 - [x]$, where x is a real number for which $2x \tan^{-1} x$ is greater than or equal to $\ln(1 + x^2)$, then find the number of maximum possible values of α . (where $[\cdot]$ represents the greatest integer function)
71. The circle $x^2 + y^2 = 1$ cuts the x -axis at P and Q . Another circle with centre at Q and variable radius intersects the first circle at R above the x -axis and the line segment PQ at S . If A is the maximum area of the triangle QSR then $3\sqrt{3}A$ is equal to _____.
72. If $f(x)$ is a twice differentiable function such that $f(a) = 0, f(b) = 2, f(c) = -1, f(d) = 2, f(e) = 0$, where $a < b < c < d < e$, find the minimum number of zeroes of $g(x) = (f'(x))^2 + f''(x)f(x)$ in the interval $[a, e]$.
73. If the length of the interval of ' a ' such that the inequality $3 - x^2 > |x - a|$ has atleast one negative solution is k then find $4k$.
74. If k is a positive integer, such that
- (i) $\cos^2 x \sin x > -\frac{7}{k}$, for all x
- (ii) $\cos^2 x \sin x < -\frac{7}{k+1}$ for some x , then k must be equal to

Assertion & Reason

- (A) If ASSERTION is true, REASON is true, REASON is a correct explanation for ASSERTION.
- (B) If ASSERTION is true, REASON is true, REASON is not a correct explanation for ASSERTION.
- (C) If ASSERTION is true, REASON is false.
- (D) If ASSERTION is false, REASON is true.

75. **Assertion :** Let $f(x) = 5 - 4(x - 2)^{2/3}$, then at $x = 2$ the function $f(x)$ attains neither least value nor greatest value.

Reason : $x = 2$ is the only critical point of $f(x)$.

- (a) A (b) B
(c) C (d) D

76. **Assertion :** for any triangle ABC

$$\sin \left(\frac{A+B+C}{3} \right) \geq \frac{\sin A + \sin B + \sin C}{3}$$

Reason : $y = \sin x$ is concave downward for $x \in (0, \pi]$.

- (a) A (b) B
(c) C (d) D

77. **Assertion :** The minimum distance of the fixed point

$(0, y_0)$, where $0 \leq y_0 \leq \frac{1}{2}$, from the curve $y = x^2$ is y_0 .

Reason : Maxima and minima of a function is always a root of the equation $f'(x) = 0$.

- (a) A (b) B
(c) C (d) D

78. **Assertion :** The equation $3x^2 + 4ax + b = 0$ has at least one root in $(0, 1)$, if $3 + 4a = 0$.

Reason : $f(x) = 3x^2 + 4ax + b$ is continuous and differentiable in the interval $(0, 1)$.

- (a) A (b) B
(c) C (d) D

79. **Assertion :** Let $f: [0, \infty) \rightarrow [0, \infty)$ and $g: [0, \infty) \rightarrow [0, \infty)$ be non-increasing and non-decreasing functions respectively and $h(x) = g(f(x))$. If f and g are differentiable for all points in their respective domains and $h(0) = 0$ then $h(x)$ is constant function.

Reason : $g(x) \in [0, \infty) \Rightarrow h(x) \geq 0$ and $h'(x) \leq 0$.

- (a) A (b) B
(c) C (d) D



80. **Assertion :** The ratio of length of tangent to length of normal is directly proportional to the ordinate of the point of tangency at the curve $y^2 = 4ax$.

Reason : Length of normal & tangent to a curve

$$y = f(x) \text{ is } \left| y\sqrt{1+m^2} \right| \text{ and } \left| \frac{y\sqrt{1+m^2}}{m} \right|, \text{ where } m = \frac{dy}{dx}.$$

- (a) A (b) B
(c) C (d) D

81. **Assertion :** Among all the rectangles of given perimeter, the square has the largest area. Also among all the rectangles of given area, the square has the least perimeter.

Reason : For $x > 0, y > 0$, if $x + y = \text{const}$, then xy will be maximum for $y = x$ and if $xy = \text{const}$, then $x + y$ will be minimum for $y = x$.

- (a) A (b) B
(c) C (d) D

82. **Assertion :** If $g(x)$ is a differentiable function $g(1) \neq 0, g(-1) \neq 0$ and Rolles theorem is not applicable to

$$f(x) = \frac{x^2 - 1}{g(x)} \text{ in } [-1, 1], \text{ then } g(x) \text{ has atleast one root in } (-1, 1)$$

Reason : If $f(a) = f(b)$, then Rolles theorem is applicable for $x \in (a, b)$

- (a) A (b) B
(c) C (d) D

83. **Assertion :** The tangent at $x = 1$ to the curve $y = x^3 - x^2 - x + 2$ again meets the curve at $x = -2$.

Reason : When a equation of a tangent solved with the curve, repeated roots are obtained at point of tangency.

- (a) A (b) B
(c) C (d) D

84. **Assertion :** Tangent drawn at the point $(0, 1)$ to the curve $y = x^3 - 3x + 1$ meets the curve thrice at one point only.

Reason : Tangent drawn at the point $(1, -1)$ to the curve $y = x^3 - 3x + 1$ meets the curve at 1 point only.

- (a) A (b) B
(c) C (d) D

85. **Assertion :** Shortest distance between

$$|x| + |y| = 2 \text{ \& } x^2 + y^2 = 16 \text{ is } 4 - \sqrt{2}$$

Reason : Shortest distance between the two non intersecting differentiable curves lies along the common normal.

- (a) A (b) B
(c) C (d) D

86. **Assertion :** If $f(x)$ is increasing function with concavity upwards, then concavity of $f^{-1}(x)$ is also upwards.

Reason : If $f(x)$ is decreasing function with concavity upwards, then concavity of $f^{-1}(x)$ is also upwards.

- (a) A (b) B
(c) C (d) D

87. **Assertion :** The largest term in the sequence

$$a_n = \frac{n^2}{n^3 + 200}, n \in \mathbb{N} \text{ is } \frac{(400)^{2/3}}{600}.$$

Reason : $f(x) = \frac{x^2}{x^3 + 200}, x > 0$, then at $x = (400)^{1/3}$, $f(x)$ is maximum.

- (a) A (b) B
(c) C (d) D

Match the Following

Each question has two columns. Four options are given representing matching of elements from Column-I and Column-II. Only one of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.

- | Column-I | Column-II |
|--|-------------------|
| (A) Circular plate is expanded by heat from radius 5 cm to 5.06 cm. Approximate increase in area is | (P) 4 |
| (B) If an edge of a cube increases by 1% then percentage increase in volume is | (Q) 0.6π |
| (C) If the rate of decrease of $\frac{x^2}{2} - 2x + 5$ is twice the rate of decrease of x , then x is equal to (rate of decrease is non-zero) | (R) 3 |
| (D) Rate of increase in area of equilateral triangle of side 15cm, when each side is increasing at the rate of 0.1 cm/sec; is | (S) $3\sqrt{3}/4$ |

The correct matching is :

- (a) (A-Q; B-R; C-P; D-S)
(b) (A-R; B-P; C-Q; D-S)
(c) (A-S; B-Q; C-P; D-S)
(d) (A-P; B-Q; C-R; D-S)



89. **Column-I** **Column-II**
- (A) If portion of the tangent at any point on the curve $x = at^3$, $y = at^4$ between the axes is divided by the abscissa of the point of contact in the ratio $m : n$ externally, then $|n + m|$ is equal to (m and n are coprime) **(P)** 0
- (B) The area of triangle formed by normal at the point (1, 0) on the curve $x = e^{\sin y}$ with axes is **(Q)** $1/2$
- (C) If the angle between curves $x^2 y = 1$ and $y = e^{2(1-x)}$ at the point (1, 1) is θ then $\tan \theta$ is equal to **(R)** 7
- (D) The length of sub-tangent at any point on the curve $y = be^{x/3}$ is equal to **(S)** 3

The correct matching is :

- (a) (A-R; B-Q; C-P; D-S)
 (b) (A-Q; B-R; C-P; D-S)
 (c) (A-P; B-Q; C-R; D-S)
 (d) (A-S; B-P; C-Q; D-S)

90. **Column - I** **Column - II**
- (A) The dimensions of the rectangle of perimeter 36 cm, which sweeps out the largest volume when revolved about one of its sides, are **(P)** 6
- (B) Let A (-1, 2) and B (2, 3) be two fixed points, A point P lying on $y = x$ such that perimeter of triangle PAB is minimum, then sum of the abscissa and ordinate of point P, is **(Q)** 12
- (C) If x_1 and x_2 are abscissae of two points on the curve $f(x) = x - x^2$ in the interval $[0, 1]$, then maximum value of expression $(x_1 + x_2) - (x_1^2 + x_2^2)$ is **(R)** 4

- (D) The number of non-zero integral values of 'a' for which the function

$$f(x) = x^4 + ax^3 + \frac{3x^2}{2} + 1 \text{ is concave}$$

upward along the entire real line is

(T) 2

The correct matching is :

- (a) (A-R; B-P; C-S; D-Q)
 (b) (A-S; B-R; C-P; D-Q)
 (c) (A-P; Q; B-R; C-S; D-R)
 (d) (A-Q; B-S; C-P; D-R)

91. **Column - I** **Column - II**

- (A) The equation $x \log x = 3 - x$ has at least one root in **(P)** (0, 1)

- (B) If $27a + 9b + 3c + d = 0$, then the equation $4ax^3 + 3bx^2 + 2cx + d = 0$ has at least one root in **(Q)** (1, 3)

- (C) If $c = \sqrt{3}$ & $f(x) = x + \frac{1}{x}$ then interval of x in which LMVT is applicable for $f(x)$, is **(R)** (0, 3)

- (D) If $c = \frac{1}{2}$ & $f(x) = 2x - x^2$, then interval of x in which LMVT is applicable for $f(x)$, is **(S)** (-1, 1)

The correct matching is :

- (a) (A-P; B-R; C-Q; D-P)
 (b) (A-R; B-S; C-Q; D-P)
 (c) (A-Q; B-S; C-R; D-P)
 (d) (A-R; B-S; C-P; D-P)



92. Column - I Column - II

(A) If x is real, then the greatest and least value of the expression

$$\frac{x+2}{2x^2+3x+6} \text{ is}$$

(B) If $a+b=1$; $a>0$, $b>0$, then the minimum value of

$$\sqrt{\left(1+\frac{1}{a}\right)\left(1+\frac{1}{b}\right)} \text{ is}$$

(C) The maximum value attained by $y=10-|x-10|$, $-1 \leq x \leq 3$, is

(D) If $P(t^2, 2t)$, $t \in [0, 2]$ is an arbitrary point on parabola $y^2=4x$. Q is foot of perpendicular from focus S on the tangent at P , then maximum area of triangle PQS is

(P) 3

(Q) $\frac{1}{3}$

(R) 5

(S) $-\frac{1}{13}$

The correct matching is :

- (a) (A-S; B-P; C-P; D-R)
(b) (A-Q; B-S; C-P; D-R)
(c) (A-R; B-Q; C-P; D-S)
(d) (A-S; B-R; C-P; D-Q)

Paragraph Type Questions

Using the following, solve Q.93 to Q.95

Passage

If $y = \int_{u(x)}^{v(x)} f(t) dt$, let us define $\frac{dy}{dx}$ in a different manner

as $\frac{dy}{dx} = v'(x)f^2(v(x)) - u'(x)f^2(u(x))$ and the equation of the tangent at (a, b) as

$$y-b = \left(\frac{dy}{dx}\right)_{(a,b)} (x-a)$$

93. If $y = \int_x^{x^2} t^2 dt$, then equation of tangent at $x=1$ is

- (a) $y=x+1$ (b) $x+y=1$
(c) $y=x-1$ (d) $y=x$

94. If $F(x) = \int_1^x e^{t^2/2} (1-t^2) dt$, then $\frac{d}{dx} F(x)$ at $x=1$ is

- (a) 0 (b) 1
(c) 2 (d) -1

95. If $y = \int_{x^3}^{x^4} \ln t dt$, then $\lim_{x \rightarrow 0^+} \frac{dy}{dx}$ is

- (a) 0 (b) 1
(c) 2 (d) -1

Using the following passage, solve Q.96 to Q.98

Passage

Consider a function $f(x) = \left(\alpha - \frac{1}{\alpha} - x\right)(4-3x^2)$ where

' α ' is a positive parameter

96. Number of points of extrema of $f(x)$ for a given value of α is

- (a) 0 (b) 1
(c) 2 (d) 3

97. Absolute difference between local maximum and local minimum values of $f(x)$ in terms of α is

- (a) $\frac{4}{9} \left(\alpha + \frac{1}{\alpha}\right)^3$ (b) $\frac{2}{9} \left(\alpha + \frac{1}{\alpha}\right)^3$
(c) $\left(\alpha + \frac{1}{\alpha}\right)^3$ (d) independent of α

98. Least possible value of the absolute difference between local maximum and local minimum values of $f(x)$ is

- (a) $\frac{32}{9}$ (b) $\frac{16}{9}$
(c) $\frac{8}{9}$ (d) $\frac{1}{9}$



Using the following passage, solve Q.99 to Q.101

Passage

Consider the function $f(x) = \max \{x^2, (1-x)^2, 2x(1-x)\}$ where $0 \leq x \leq 1$.

99. The interval in which $f(x)$ is increasing is

- (a) $\left(\frac{1}{3}, \frac{2}{3}\right)$ (b) $\left(\frac{1}{3}, \frac{1}{2}\right)$
(c) $\left(\frac{1}{3}, \frac{1}{2}\right) \cup \left(\frac{1}{2}, \frac{2}{3}\right)$ (d) $\left(\frac{1}{3}, \frac{1}{2}\right) \cup \left(\frac{2}{3}, 1\right)$

100. The interval in which $f(x)$ is decreasing is

- (a) $\left(\frac{1}{3}, \frac{2}{3}\right)$ (b) $\left(\frac{1}{3}, \frac{1}{2}\right)$
(c) $\left(0, \frac{1}{3}\right) \cup \left(\frac{1}{2}, \frac{2}{3}\right)$ (d) $\left(0, \frac{1}{2}\right) \cup \left(\frac{2}{3}, 1\right)$

101. Let RMVT is applicable for $f(x)$ on (a, b) then $a + b + c$ is (where c is point such that $f'(c) = 0$)

- (a) $\frac{2}{3}$ (b) $\frac{1}{3}$
(c) $\frac{1}{2}$ (d) $\frac{3}{2}$

Using the following passage, solve Q.102 to Q.104

Passage

Let $y = a\sqrt{x} + bx$ be curve, $(2x - y) + \lambda(2x + y - 4) = 0$ be family of lines.

102. If curve has slope $-\frac{1}{2}$ at $(9, 0)$ then a tangent belonging to family of lines is

- (a) $x + 2y - 5 = 0$ (b) $x - 2y + 3 = 0$
(c) $3x - y - 1 = 0$ (d) $3x + y - 5 = 0$

103. A line of the family cutting positive intercepts on axes and forming triangle with coordinate axes, then minimum length of the line segment between axes is

- (a) $(2^{2/3} - 1)^{3/2}$ (b) $(2^{2/3} + 1)^{3/2}$
(c) $7^{3/2}$ (d) 27

104. Two perpendicular chords of curve $y^2 - 4x - 4y + 4 = 0$ belonging to family of lines form diagonals of a quadrilateral. Minimum area of quadrilateral is

- (a) 16 (b) 32
(c) 64 (d) 50

Using the following passage, solve Q.105 to Q.107

Passage

If $y = f(x)$ is a curve and if there exists two points $A(x_1, f(x_1))$ and $B(x_2, f(x_2))$ on it such that

$$f'(x_1) = -\frac{1}{f'(x_2)} = \frac{f(x_2) - f(x_1)}{x_2 - x_1} \text{ then the tangent}$$

at x_1 is normal at x_2 for that curve.

105. Number of such lines on the curve $y = \sin x$ is

- (a) 1 (b) 0
(c) 2 (d) infinite

106. Number of such lines on the curve $y = |\ln x|$ is

- (a) 1 (b) 2
(c) 0 (d) infinite

107. Number of such line on the curve $y^2 = x^3$ is

- (a) 1 (b) 2
(c) 3 (d) 0

Using the following passage, solve Q.108 to Q.110

Passage

$$\text{Let } f'(\sin x) < 0 \text{ and } f''(\sin x) > 0 \quad \forall x \in \left(0, \frac{\pi}{2}\right)$$

Now consider a function $g(x) = f(\sin x) + f(\cos x)$

108. $g(x)$ decreases if x belongs to

- (a) $\left(0, \frac{\pi}{4}\right)$ (b) $\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$
(c) $\left(\frac{\pi}{6}, \frac{\pi}{3}\right)$ (d) none of these

109. $g(x)$ increase if x belongs to

- (a) $\left(0, \frac{\pi}{4}\right)$ (b) $\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$
(c) $\left(\frac{\pi}{8}, \frac{\pi}{3}\right)$ (d) $\left(\frac{\pi}{6}, \frac{\pi}{3}\right)$

110. The set of critical points of $g(x)$ is

- (a) $\left\{\frac{\pi}{8}, \frac{\pi}{6}\right\}$ (b) $\left\{\frac{\pi}{8}, \frac{\pi}{6}, \frac{\pi}{3}\right\}$
(c) $\left\{\frac{\pi}{8}, \frac{\pi}{6}, \frac{\pi}{4}\right\}$ (d) none of these



EXERCISE - 4 : PREVIOUS YEAR JEE ADVANCED QUESTIONS

Objective Questions I [Only one correct option]

1. For all $x \in (0, 1)$ (2000)

- (a) $e^x < 1 + x$ (b) $\log_e(1+x) < x$
(c) $\sin x > x$ (d) $\log_e x > x$

2. Let $f(x) = \int e^x(x-1)(x-2)dx$. Then f decreases in the interval (2000)

- (a) $(-\infty, -2)$ (b) $(-2, -1)$
(c) $(1, 2)$ (d) $(2, \infty)$

3. Let $f(x) = \begin{cases} |x|, & \text{for } 0 < |x| \leq 2 \\ 1, & \text{for } x = 0 \end{cases}$. Then, at $x = 0$, f has (2000)

- (a) a local maximum (b) no local maximum
(c) a local minimum (d) no extremum

4. If the normal to the curve, $y = f(x)$ at the point $(3, 4)$ makes an angle $3\pi/4$ with the positive x -axis, then $f'(3)$ is equal to (2000)

- (a) -1 (b) $-3/4$
(c) $4/3$ (d) 1

5. If $f(x) = xe^{x(1-x)}$, then $f(x)$ is (2001)

- (a) increasing in $\left[-\frac{1}{2}, 1\right]$ (b) decreasing in $\mathbb{R}$
(c) increasing in $\mathbb{R}$ (d) decreasing in $\left[-\frac{1}{2}, 1\right]$

6. The maximum value of $(\cos \alpha_1) \cdot (\cos \alpha_2) \cdot \dots \cdot (\cos \alpha_n)$, under the restrictions $0 \leq \alpha_1, \alpha_2, \dots, \alpha_n \leq \frac{\pi}{2}$ and $(\cot \alpha_1) \cdot (\cot \alpha_2) \cdot \dots \cdot (\cot \alpha_n) = 1$ is (2001)

- (a) $\frac{1}{2^{n/2}}$ (b) $\frac{1}{2^n}$
(c) $\frac{1}{2n}$ (d) 1

7. The length of a longest interval in which the function $3\sin x - 4\sin^3 x$ is increasing, is (2002)

- (a) $\frac{\pi}{3}$ (b) $\frac{\pi}{2}$
(c) $\frac{3\pi}{2}$ (d) π

8. The point(s) on the curve $y^3 + 3x^2 = 12y$ where the tangent is vertical, is (are) (2002)

- (a) $\left(\pm\frac{4}{\sqrt{3}}, -2\right)$ (b) $\left(\pm\sqrt{\frac{11}{3}}, 0\right)$
(c) $(0, 0)$ (d) $\left(\pm\frac{4}{\sqrt{3}}, 2\right)$

9. The equation of the common tangent to the curves $y^2 = 8x$ and $xy = -1$ is (2002)

- (a) $3y = 9x + 2$ (b) $y = 2x + 1$
(c) $2y = x + 8$ (d) $y = x + 2$

10. If $f(x) = x^2 + 2bx + 2c^2$ and $g(x) = -x^2 - 2cx + b^2$ such that $\min f(x) > \max g(x)$, then the relation between b and c , is – (2003)

- (a) no real value of b & c (b) $0 < c < b\sqrt{2}$
(c) $|c| < |b|\sqrt{2}$ (d) $|c| > |b|\sqrt{2}$

11. If $f(x) = x^3 + bx^2 + cx + d$ and $0 < b^2 < c$, then in $(-\infty, \infty)$ (2004)

- (a) $f(x)$ is strictly increasing function
(b) $f(x)$ has a local maxima
(c) $f(x)$ is strictly decreasing function
(d) $f(x)$ is bounded.

12. If $f(x)$ is differentiable and strictly increasing function, then the value of $\lim_{x \rightarrow 0} \frac{f(x^2) - f(x)}{f(x) - f(0)}$ is (2004)

- (a) 1 (b) 0
(c) -1 (d) 2



13. Tangents are drawn to the ellipse $x^2 + 2y^2 = 2$, then the locus of the mid point of the intercept made by the tangents between the coordinate axes is (2004)
- (a) $\frac{1}{2x^2} + \frac{1}{4y^2} = 1$ (b) $\frac{1}{4x^2} + \frac{1}{2y^2} = 1$
- (c) $\frac{x^2}{2} + \frac{y^2}{4} = 1$ (d) $\frac{x^2}{4} + \frac{y^2}{2} = 1$
14. The angle between the tangents drawn from the point (1, 4) to the parabola $y^2 = 4x$ is (2004)
- (a) $\pi/6$ (b) $\pi/4$
- (c) $\pi/3$ (d) $\pi/2$
15. The second degree polynomial $f(x)$, satisfying $f(0) = 0$, $f(1) = 1$, $f'(x) > 0$ for all $x \in (0, 1)$: (2005)
- (a) $f(x) = \phi$
- (b) $f(x) = ax + (1-a)x^2; \forall a \in (0, \infty)$
- (c) $f(x) = ax + (1-a)x^2; a \in (0, 2)$
- (d) No such polynomial
16. The tangent at (1, 7) to the curve $x^2 = y - 6$ touches the circle $x^2 + y^2 + 16x + 12y + c = 0$ at (2005)
- (a) (6, 7) (b) (-6, 7)
- (c) (6, -7) (d) (-6, -7)
17. The tangent to the curve $y = e^x$ drawn at the point (c, e^c) intersects the line joining the points $(c-1, e^{c-1})$ and $(c+1, e^{c+1})$ (2007)
- (a) on the left of $x = c$ (b) on the right of $x = c$
- (c) at no point (d) at all points
18. Let the function $g : (-\infty, \infty) \rightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ be given by $g(u) = 2 \tan^{-1}(e^u) - \frac{\pi}{2}$. Then, g is (2008)
- (a) even and is strictly increasing in $(0, \infty)$
- (b) odd and is strictly decreasing in $(-\infty, \infty)$
- (c) odd and is strictly increasing in $(-\infty, \infty)$
- (d) neither even nor odd, but is strictly increasing in $(-\infty, \infty)$
19. The total number of local maxima and local minima of the function $f(x) = \begin{cases} (2+x)^3, & -3 < x \leq -1 \\ x^{\frac{2}{3}}, & -1 < x < 2 \end{cases}$ is (2008)
- (a) 0 (b) 1
- (c) 2 (d) 3
20. Let f, g and h be real-valued functions defined on the interval $[0, 1]$ by $f(x) = e^{x^2} + e^{-x^2}$, $g(x) = xe^{x^2} + e^{-x^2}$ and $h(x) = x^2e^{x^2} + e^{-x^2}$. If a, b and c denote respectively, the absolute maximum of f, g and h on $[0, 1]$, then (2010)
- (a) $a = b$ and $c \neq b$ (b) $a = c$ and $a \neq b$
- (c) $a \neq b$ and $c \neq b$ (d) $a = b = c$
21. The number of points in $(-\infty, \infty)$, for which $x^2 - x \sin x - \cos x = 0$, is (2013)
- (a) 6 (b) 4
- (c) 2 (d) 0
22. Consider all rectangles lying in the region $\left\{ (x, y) \in R \times R : 0 \leq x \leq \frac{\pi}{2} \text{ and } 0 \leq y \leq 2 \sin(2x) \right\}$ and having one side on the x -axis. The area of the rectangle which has the maximum perimeter among all such rectangles, is (2020)
- (a) $\frac{3\pi}{2}$ (b) π
- (c) $\frac{\pi}{2\sqrt{3}}$ (d) $\frac{\pi\sqrt{3}}{2}$
- Objective Questions II [One or more than one correct option]**
23. If $f(x)$ is cubic polynomial which has local maximum at $x = -1$. If $f(2) = 18, f(1) = -1$ and $f'(x)$ has local minimum at $x = 0$, then (2006)
- (a) the distance between $(-1, 2)$ and $(a, f(a))$ where $x = a$ is the point of local minima, is $2\sqrt{5}$.
- (b) $f(x)$ is increasing for $x \in [1, 2\sqrt{5}]$
- (c) $f(x)$ has local minima at $x = 1$
- (d) the value of $f(0) = 5$



24. If $f(x) = \begin{cases} e^x, & 0 \leq x \leq 1 \\ 2 - e^{x-1}, & 1 < x \leq 2 \\ x - e, & 2 < x \leq 3 \end{cases}$

and $g(x) = \int_0^x f(t) dt$, $x \in [1, 3]$, then (2006)

- (a) $g(x)$ has local maxima at $x = 1 + \log_e 2$ and local minima at $x = e$
 (b) $f(x)$ has local maxima at $x = 1$ and local minima at $x = 2$
 (c) $g(x)$ has no local minima
 (d) $f(x)$ has no local maxima

25. For the function $f(x) = x \cos \frac{1}{x}$, $x \geq 1$. (2009)

- (a) for at least one x in the interval $[1, \infty)$, $f(x+2) - f(x) < 2$
 (b) $\lim_{x \rightarrow \infty} f'(x) = 1$
 (c) for all x in the interval $[1, \infty)$, $f(x+2) - f(x) > 2$
 (d) $f'(x)$ is strictly decreasing in the interval $[1, \infty)$

26. Let f be a real-valued function defined on the interval

$(0, \infty)$, by $f(x) = \ln x + \int_0^x \sqrt{1 + \sin t} dt$. Then which of the

following statement(s) is (are) true? (2010)

- (a) $f'''(x)$ exists for all $x \in (0, \infty)$
 (b) $f'(x)$ exists for all $x \in (0, \infty)$ and f' is continuous on $(0, \infty)$, but not differentiable on $(0, \infty)$
 (c) there exists $\alpha > 1$ such that $|f''(x)| < |f(x)|$ for all $x \in (\alpha, \infty)$
 (d) there exists $\beta > 0$ such that $|f(x)| + |f'(x)| \leq \beta$ from all $x \in (0, \infty)$

27. A rectangular sheet of fixed perimeter with sides having their lengths in the ratio 8 : 15 is converted into an open rectangular box by folding after removing squares of equal area from all four corners. If the total area of removed squares is 100, the resulting box has maximum volume. The lengths of the sides of the rectangular sheet are

(2013)

- (a) 24 (b) 32
 (c) 45 (d) 60

28. The function $f(x) = 2|x| + |x+2| - ||x+2| - 2|x||$ has a local minimum or a local maximum at x is equal to

(2013)

- (a) -2 (b) $-\frac{2}{3}$
 (c) 2 (d) $\frac{2}{3}$

29. Let $a \in \mathbb{R}$ and let $f: \mathbb{R} \rightarrow \mathbb{R}$ be given by

$$f(x) = x^5 - 5x + a.$$

Then

(2014)

- (a) $f(x)$ has three real roots if $a > 4$
 (b) $f(x)$ has only one real root if $a > 4$
 (c) $f(x)$ has three real roots if $a < -4$
 (d) $f(x)$ has three real roots if $-4 < a < 4$

30. Let $f: \mathbb{R} \rightarrow (0, \infty)$ and $g: \mathbb{R} \rightarrow \mathbb{R}$, be twice differentiable functions such that f'' and g'' are continuous functions on $\mathbb{R}$. Suppose $f'(2) = g(2) = 0$, $f''(2) \neq 0$ and $g'(2) \neq 0$. If then (2016)

- (a) f has a local minimum at $x = 2$
 (b) f has a local maximum at $x = 2$
 (c) $f''(2) = f(2)$
 (d) $f(x) - f''(x) = 0$ for at least one $x \in \mathbb{R}$

31. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be given by

$$f(x) = \begin{cases} x^5 + 5x^4 + 10x^3 + 10x^2 + 3x + 1 & x < 0 \\ x^2 - x + 1 & 0 \leq x < 1 \\ (2/3)x^3 - 4x^2 + 7x - (8/3) & 1 \leq x < 3 \\ (x-2)\ln(x-2) - x + (10/3) & x \geq 3 \end{cases}$$

Then which of the following options is/are correct?

(2019)

- (a) f' is not differentiable at $x=1$
 (b) f is increasing on $(-\infty, 0)$
 (c) f is onto
 (d) f' has a local maximum at $x=1$



32. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be given by $f(x) = (x-1)(x-2)(x-5)$. Define $f(x) = \int_0^x f(t) dt$, $x > 0$. Then which of the following options is/are correct? (2019)
- (a) $f(x)$ has a local maximum at $x = 2$
 (b) $f(x)$ has a local minimum at $x = 1$
 (c) $f(x)$ has two local maxima and one local minimum in $(0, \infty)$
 (d) $f(x) \neq 0$, for all $x \in (0, 5)$
33. Let $f(x) = \frac{\sin \pi x}{x^2}$, $x > 0$.
 Let $x_1 < x_2 < x_3, \dots, x_n < \dots$ be all points of local maximum of f and $y_1 < y_2 < y_3 < \dots, y_n < \dots$ be all the points of local minimum of f . Then which of the following options is/are correct? (2019)
- (a) $|x_n - y_n| > 1$ for every n
 (b) $x_1 < y_1$
 (c) $x_{n+1} - x_n > 2$ for every n
 (d) $x_n \in \left(2n, 2n + \frac{1}{2}\right)$ for every n
34. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \frac{x^2 - 3x - 6}{x^2 + 2x + 4}$.
 Then which of the following statements is (are) TRUE? (2021)
- (a) f is decreasing in the interval $(-2, -1)$
 (b) f is increasing in the interval $(1, 2)$
 (c) f is onto
 (d) Range of f is $\left[-\frac{3}{2}, 2\right]$
35. A straight line L with negative slope passes through the point $(8, 2)$ and cuts the positive coordinate axes at points P and Q . Find the absolute minimum value of $OP + OQ$, as L varies, where O is the origin. (2002)
36. Find a point on the curve $x^2 + 2y^2 = 6$ whose distance from the line $x + y = 7$, is minimum. (2003)
37. For the circle $x^2 + y^2 = r^2$, find the value of r for which the area enclosed by the tangents drawn from the point $P(6, 8)$ to the circle and the chord of contact is maximum. (2003)
38. If $f(x)$ is twice differentiable function such that $f(a) = 0, f(b) = 2, f(c) = -1, f(d) = 2, f(e) = 0$, where $a < b < c < d < e$, then the minimum number of zeroes of $g(x) = \{f'(x)\}^2 + f''(x) \cdot f(x)$ in the interval $[a, e]$ is? (2006)
39. The maximum value of the function $f(x) = 2x^3 - 15x^2 + 36x - 48$ on the set $A = \{x | x^2 + 20 \leq 9x\}$ is (2009)
40. The maximum value of the expression $\frac{1}{\sin^2 \theta + 3 \sin \theta \cos \theta + 5 \cos^2 \theta}$ is (2010)
41. Let f be a function defined on $\mathbb{R}$ (the set of all real numbers) such that $f'(x) = 2010(x - 2009)(x - 2010)^2(x - 2011)^3(x - 2012)^4$, for all $x \in \mathbb{R}$. If g is a function defined on $\mathbb{R}$ with values in the interval $(0, \infty)$ such that $f(x) = \ln(g(x))$, for all $x \in \mathbb{R}$, then the number of points in $\mathbb{R}$ at which g has a local maximum is ... (2010)
42. The number of distinct real roots of $x^4 - 4x^3 + 12x^2 + x - 1 = 0$ is (2011)
43. Let $p(x)$ be a real polynomial of least degree which has a local maximum at $x = 1$ and a local minimum at $x = 3$. If $p(1) = 6$ and $p(3) = 2$, then $p'(0)$ is (2012)
44. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = |x| + |x^2 - 1|$. The total number of points at which f attains either a local maximum or a local minimum is (2012)
45. A vertical line passing through the point $(h, 0)$ intersects the ellipse $\frac{x^2}{4} + \frac{y^2}{3} = 1$ at the points P and Q . Let the tangents to the ellipse at P and Q meet at the point R . If $\Delta(h) = \text{area of the } \Delta PQR$, $\Delta_1 = \max_{1/2 \leq h \leq 1} \Delta(h)$ and $\Delta_2 = \min_{1/2 \leq h \leq 1} \Delta(h)$, then $\frac{8}{\sqrt{5}} \Delta_1 - 8\Delta_2$ is equal to (2013)
46. The slope of the tangent to the curve $(y - x^5)^2 = x(1 + x^2)^2$ at the point $(1, 3)$ is (2014)
47. For a polynomial $g(x)$ with real coefficient, let m_g denote the number of distinct real roots of $g(x)$. Suppose S is the set of polynomials with real coefficient defined by $S = \{(x^2 - 1)^2(a_0 + a_1x + a_2x^2 + a_3x^3) : a_0, a_1, a_2, a_3 \in \mathbb{R}\}$.
 For a polynomial f , let f' and f'' denote its first and second order derivatives, respectively. Then the minimum possible value of $(m_{f'} + m_{f''})$, where $f \in S$, is (2020)



48. Let the function $f: (0, \pi) \rightarrow \mathbb{R}$ be defined by
- $$f(\theta) = (\sin \theta + \cos \theta)^2 + (\sin \theta - \cos \theta)^4$$
- Suppose the function g has a local minimum at θ precisely when $\theta \in \{\lambda_1 \pi, \dots, \lambda_r \pi\}$
- where $0 < \lambda_1 < \dots < \lambda_r < 1$. Then the value of $\lambda_1 + \dots + \lambda_r$ is (2020)

Assertion & Reason

49. Consider the following statement S and R :
- S : Both $\sin x$ & $\cos x$ are decreasing functions in the interval $(\pi/2, \pi)$.
- R : If a differentiable function decreases in an interval (a, b) , then its derivative also decreases in (a, b) .
- Which of the following is true ? (2000)
- (a) both S and R are wrong
- (b) both S and R are correct, but R is not the correct explanation for S.
- (c) S is correct and R is the correct explanation for S
- (d) S is correct and R is wrong.

Match the Following

Each question has two columns. Four options are given representing matching of elements from Column-I and Column-II. Only one of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.

50. Let the functions defined in Column I have domain $(-\pi/2, \pi/2)$
- | | |
|------------------|---------------------------------------|
| Column I | Column II |
| (A) $x + \sin x$ | (p) increasing |
| (B) $\sec x$ | (q) decreasing |
| | (r) neither increasing nor decreasing |
- (2008)

Paragraph Type Questions

Using the following passage, solve Q.51 to Q.53

Passage

Consider the function $f: (-\infty, \infty) \rightarrow (-\infty, \infty)$ defined by

$$f(x) = \frac{x^2 - ax + 1}{x^2 + ax + 1}; 0 < a < 2. \quad (2008)$$

51. Which of the following is true ?
- (a) $(2+a)^2 f''(1) + (2-a)^2 f''(-1) = 0$
- (b) $(2-a)^2 f''(1) - (2+a)^2 f''(-1) = 0$
- (c) $f'(1)f'(-1) = (2-a)^2$
- (d) $f'(1)f'(-1) = -(2+a)^2$

52. Which of the following is true ?
- (a) $f(x)$ is decreasing on $(-1, 1)$ and has a local minimum at $x = 1$.
- (b) $f(x)$ is increasing on $(-1, 1)$ and has a local maximum at $x = 1$.
- (c) $f(x)$ is increasing on $(-1, 1)$ but has neither a local maximum nor a local minimum at $x = 1$.
- (d) $f(x)$ is decreasing on $(-1, 1)$ but has neither a local maximum nor a local minimum at $x = 1$.
53. Let $g(x) = \int_0^{e^x} \frac{f'(t)}{1+t^2} dt$. Which of the following is true ?
- (a) $g'(x)$ is positive on $(-\infty, 0)$ and negative on $(0, \infty)$
- (b) $g'(x)$ is negative on $(-\infty, 0)$ and positive on $(0, \infty)$
- (c) $g'(x)$ changes sign on both $(-\infty, 0)$ and $(0, \infty)$
- (d) $g'(x)$ does not change sign $(-\infty, \infty)$

Using the following passage, solve Q.54 to Q.56

Passage

Consider the polynomial $f(x) = 1 + 2x + 3x^2 + 4x^3$. Let s be the sum of all distinct real roots of $f(x)$ and let $t = |s|$

(2010)

54. The real numbers s lies in the interval
- (a) $\left(-\frac{1}{4}, 0\right)$ (b) $\left(-11, -\frac{3}{4}\right)$
- (c) $\left(-\frac{3}{4}, -\frac{1}{2}\right)$ (d) $\left(0, \frac{1}{4}\right)$
55. The area bounded by the curve $y = f(x)$ and the lines $x = 0$, $y = 0$ and $x = t$, lies in the interval
- (a) $\left(\frac{3}{4}, 3\right)$ (b) $\left(\frac{21}{64}, \frac{11}{16}\right)$
- (c) $(9, 10)$ (d) $\left(0, \frac{21}{64}\right)$
56. The function $f'(x)$ is
- (a) increasing in $\left(-t, -\frac{1}{4}\right)$ and decreasing in $\left(-\frac{1}{4}, t\right)$
- (b) decreasing in $\left(-t, -\frac{1}{4}\right)$ and increasing in $\left(-\frac{1}{4}, t\right)$
- (c) increasing in $(-t, t)$
- (d) decreasing in $(-t, t)$



Using the following passage, solve Q.57 and Q.58

Passage

Let $f(x) = (1-x)^2 \sin^2 x + x^2$ for all $x \in \mathbb{R}$ and let

$$g(x) = \int_1^x \left(\frac{2(t-1)}{t+1} - \ln t \right) f(t) dt \text{ for all } x \in (1, \infty)$$

57. Which of the following is true ? (2012)

- (a) g is increasing on $(1, \infty)$
- (b) g is decreasing on $(1, \infty)$
- (c) g is increasing on $(1, 2)$ and decreasing on $(2, \infty)$
- (d) g is decreasing on $(1, 2)$ and increasing on $(2, \infty)$

58. Consider the statements

P : There exists some $x \in \mathbb{R}$ such that

$$f(x) + 2x = 2(1 + x^2)$$

Q : There exists some $x \in \mathbb{R}$ such that

$$2f(x) + 1 = 2x(1 + x)$$

Then,

- (a) Both P and Q are true
- (b) P is true and Q is false
- (c) P is false and Q is true
- (d) Both P and Q are false

Using the following passage, solve Q.59 and Q.60

Passage

Let $f: [0, 1] \rightarrow \mathbb{R}$ (the set of all real numbers) be a function. Suppose the function f is twice differentiable, $f(0) = f(1) = 0$ and satisfies $f''(x) - 2f'(x) + f(x) \geq e^x$, $x \in [0, 1]$.

59. Which of the following is true for $0 < x < 1$? (2013)

(a) $0 < f(x) < \infty$ (b) $-\frac{1}{2} < f(x) < \frac{1}{2}$

(c) $-\frac{1}{4} < f(x) < 1$ (d) $-\infty < f(x) < 0$

60. If the function $e^{-x} f(x)$ assumes its minimum in the interval

$[0, 1]$ at $x = \frac{1}{4}$, which of the following is true ? (2013)

(a) $f'(x) < f(x)$, $\frac{1}{4} < x < \frac{3}{4}$

(b) $f'(x) > f(x)$, $0 < x < \frac{1}{4}$

(c) $f'(x) < f(x)$, $0 < x < \frac{1}{4}$

(d) $f'(x) < f(x)$, $\frac{3}{4} < x < 1$

Using the following passage, solve Q.61 to Q.63

Passage

Let $f(x) = x + \log_e x - x \log_e x$, $x \in (0, \infty)$.

Column 1 contains information about zeros of $f(x)$, $f'(x)$ and $f''(x)$.

Column 2 contains information about the limiting behavior of $f(x)$, $f'(x)$ and $f''(x)$ at infinity.

Column 3 contains information about increasing/decreasing nature of $f(x)$ and $f'(x)$.

Column 1

Column 2

Column 3

(I) $f(x) = 0$ for some $x \in (1, e^2)$ (i) $\lim_{x \rightarrow \infty} f(x) = 0$ (P) f is increasing in $(0, 1)$

(II) $f'(x) = 0$ for some $x \in (1, e)$ (ii) $\lim_{x \rightarrow \infty} f'(x) = -\infty$ (Q) f is decreasing in (e, e^2)

(III) $f'(x) = 0$ for some $x \in (0, 1)$ (iii) $\lim_{x \rightarrow \infty} f'(x) = -\infty$ (R) f' is increasing in $(0, 1)$

(IV) $f''(x) = 0$ for some $x \in (1, e)$ (iv) $\lim_{x \rightarrow \infty} f''(x) = 0$ (S) f' is decreasing in (e, e^2)

(2017)

61. Which of the following options is the only CORRECT combination ?

(a) (I) (ii) (R) (b) (IV) (i) (S)

(c) (III) (iv) (P) (d) (II) (iii) (S)

62. Which of the following options is the only CORRECT combination ?

(a) (I) (i) (P) (b) (II) (ii) (Q)

(c) (III) (iii) (R) (d) (IV) (iv) (S)

63. Which of the following options is the only INCORRECT combination ?

(a) (II) (iii) (P) (b) (I) (iii) (P)

(c) (III) (i) (R) (d) (II) (iv) (Q)



Using the following passage, solve Q.64 and Q.65

Passage

Let $f_1 : (0, \infty) \rightarrow \mathbb{R}$ and $f_2 : (0, \infty) \rightarrow \mathbb{R}$ be defined by

$$f_1(x) = \int_0^x \prod_{j=1}^{21} (t-j)^j dt, \quad x > 0$$

$$\text{and } f_2(x) = 98(x-1)^{50} - 600(x-1)^{49} + 2450, \quad x > 0,$$

Where, for any positive integer n and real numbers

$a_1, a_2, \dots, a_n, \prod_{i=1}^n a_i$ denotes the product of

$a_1, a_2, \dots, a_n$. Let m_i and n_i , respectively, denote the number of points of local minima and the number of points of local maxima of function $f_i, i = 1, 2$, in the interval $(0, \infty)$

(2021)

64. The value of $2m_1 + 3n_1 + m_1n_1$ is-----.

65. The value of $6m_2 + 4n_2 + 8m_2n_2$ is-----.

Text

66. Let $-1 \leq p \leq 1$. Show that the equation $4x^3 - 3x - p = 0$ has a unique root in the interval $\left[\frac{1}{2}, 1\right]$ and identify it.

(2001)

67. If $P(1) = 0$ and $\frac{dP(x)}{dx} > P(x)$ for all $x \geq 1$, then prove that $P(x) > 0$ for all $x > 1$.

(2003)

68. Using the relation $2(1 - \cos x) < x^2, x \neq 0$ or otherwise, prove that $\sin(\tan x) \geq x, \forall x \in \left[0, \frac{\pi}{4}\right]$.

(2003)

69. Prove that $\sin x + 2x \geq \frac{3x \cdot (x+1)}{\pi} \forall x \in \left[0, \frac{\pi}{2}\right]$.

(Justify the inequality, if any used).

(2004)

70. If $|f(x_1) - f(x_2)| \leq (x_1 - x_2)^2$, for all $x_1, x_2 \in \mathbb{R}$. Find the equation of tangent to the curve $y = f(x)$ at the point $(1, 2)$.

(2005)

Find Answer Key and Detailed Solutions at the end of this book



APPLICATIONS OF DERIVATIVES



Please share your valuable feedback by scanning the QR code.

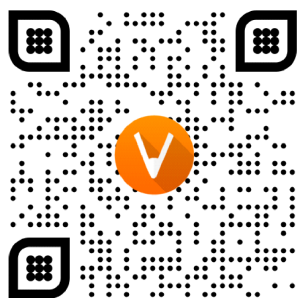


Answer Key



CHAPTER -1 | DETERMINANTS & MATRICES

EXERCISE - 1: BASIC OBJECTIVE QUESTIONS



DIRECTION TO USE -

Scan the QR code and check detailed solutions.

- | | | | | |
|----------|---------|---------|---------|---------|
| 1. (d) | 2. (b) | 3. (d) | 4. (b) | 5. (a) |
| 6. (a) | 7. (d) | 8. (b) | 9. (d) | 10. (b) |
| 11. (d) | 12. (a) | 13. (c) | 14. (c) | 15. (b) |
| 16. (a) | 17. (c) | 18. (b) | 19. (a) | 20. (b) |
| 21. (b) | 22. (c) | 23. (a) | 24. (a) | 25. (a) |
| 26. (b) | 27. (b) | 28. (a) | 29. (c) | 30. (b) |
| 31. (d) | 32. (d) | 33. (b) | 34. (c) | 35. (d) |
| 36. (b) | 37. (d) | 38. (a) | 39. (c) | 40. (b) |
| 41. (c) | 42. (c) | 43. (c) | 44. (d) | 45. (a) |
| 46. (a) | 47. (c) | 48. (b) | 49. (b) | 50. (c) |
| 51. (c) | 52. (a) | 53. (b) | 54. (a) | 55. (d) |
| 56. (b) | 57. (b) | 58. (b) | 59. (a) | 60. (b) |
| 61. (1) | 62. (2) | 63. (0) | 64. (0) | 65. (4) |
| 66. (6) | 67. (0) | 68. (3) | 69. (3) | 70. (4) |
| 71. (16) | 72. (2) | 73. (1) | 74. (2) | 75. (2) |

EXERCISE - 2: PREVIOUS YEAR JEE MAIN QUESTIONS



DIRECTION TO USE -

Scan the QR code and check detailed solutions.

- | | | | | |
|----------------|---------------|--------------|---------------|-------------|
| 1. (c) | 2. (a) | 3. (d) | 4. (c) | 5. (c) |
| 6. (d) | 7. (d) | 8. (a) | 9. (d) | 10. (d) |
| 11. (b) | 12. (b) | 13. (b) | 14. (c) | 15. (d) |
| 16. (d) | 17. (d) | 18. (d) | 19. (a) | 20. (b) |
| 21. (c) | 22. (a) | 23. (b) | 24. (c) | 25. (b) |
| 26. (b) | 27. (d) | 28. (b) | 29. (d) | 30. (d) |
| 31. (b) | 32. (b) | 33. (c) | 34. (a) | 35. (d) |
| 36. (c) | 37. (c) | 38. (a) | 39. (8.00) | 40. (b) |
| 41. (a) | 42. (b) | 43. (b) | 44. (d) | 45. (c) |
| 46. (a) | 47. (d) | 48. (a) | 49. (d) | 50. (d) |
| 51. (c) | 52. (d) | 53. (c) | 54. (10.00) | 55. (c) |
| 56. (8.00) | 57. (d) | 58. (5.00) | 59. (a) | 60. (b) |
| 61. (b) | 62. (b) | 63. (a) | 64. (c) | 65. (b) |
| 66. (a) | 67. (b) | 68. (3.00) | 69. (c) | 70. (c) |
| 71. (a) | 72. (13.00) | 73. (d) | 74. (672.00) | |
| 75. (b) | 76. (a) | 77. (b) | 78. (c) | 79. (c) |
| 80. (b) | 81. (b) | 82. (d) | 83. (1.00) | |
| 84. (910.00) | | 85. (d) | 86. (108.00) | |
| 87. (11.00) | 88. (d) | 89. (6.00) | 90. (5.00) | 91. (b) |
| 92. (a) | 93. (2020.00) | | 94. (3125.00) | |
| 95. (b) | 96. (d) | 97. (a) | 98. (b) | 99. (c) |
| 100. (b) | 101. (a) | 102. (4.00) | 103. (c) | 104. (d) |
| 105. (5.00) | 106. (d) | 107. (c) | 108. (d) | 109. (a) |
| 110. (c) | 111. (8.00) | 112. (d) | 113. (c) | 114. (c) |
| 115. (4.00) | 116. (b) | 117. (c) | 118. (13.00) | 119. (7.00) |
| 120. (21.00) | | 121. (b) | 122. (c) | 123. (c) |
| 124. (a) | 125. (d) | 126. (d) | 127. (540.00) | |
| 128. (17.00) | | 129. (a) | 130. (c) | 131. (c) |
| 132. (d) | 133. (b) | 134. (6.00) | 135. (d) | |
| 136. (766.00) | | 137. (36.00) | | 138. (a) |
| 139. (1.00) | 140. (b) | 141. (a) | 142. (16.00) | 143. (d) |
| 144. (2020.00) | | 145. (2.00) | | |

CHAPTER -1 | DETERMINANTS & MATRICES

EXERCISE - 3 :
ADVANCED OBJECTIVE QUESTIONS

DIRECTION TO USE -

Scan the QR code and check detailed solutions.

1. (b) 2. (d) 3. (a) 4. (c) 5. (d)
 6. (b) 7. (d) 8. (d) 9. (a) 10. (a)
 11. (b) 12. (b) 13. (a) 14. (a) 15. (a)
 16. (b) 17. (a) 18. (c) 19. (c) 20. (d)
 21. (a) 22. (a) 23. (d) 24. (c) 25. (d)
 26. (d) 27. (a,b,c,d) 28. (a,c)
 29. (a,b,c,d) 30. (a,b,c) 31. (b,d) 32. (a,d)
 33. (a,c) 34. (a,b,c,d) 35. (b,c,d)
 36. (a,b,c) 37. (a,b,c) 38. (a,b,d) 39. (a,b,c)
 40. (a,b,c) 41. (a,b,c) 42. (a,b,c) 43. (a,b) 44. (a)
 45. (c) 46. (b,d) 47. (b,c) 48. (81) 49. (0)
 50. (3) 51. (4) 52. (a) 53. (d) 54. (a)
 55. (a) 56. (b) 57. (b) 58. (c) 59. (b)
 60. (d) 61. (d) 62. (a) 63. (c) 64. (b)
 65. (d) 66. (c) 67. (a) 68. (b) 69. (c)
 70. (a)

EXERCISE - 4 :
PREVIOUS YEAR JEE ADVANCED QUESTIONS

DIRECTION TO USE -

Scan the QR code and check detailed solutions.

1. (c) 2. (b) 3. (d) 4. (a) 5. (b)
 6. (c) 7. (b) 8. (b) 9. (a) 10. (a)
 11. (b) 12. (b) 13. (c) 14. (a,d) 15. (c,d)
 16. (a,b) 17. (c,d) 18. (b,c) 19. (c,d) 20. (b,c)
 21. (a,b) 22. (a,c,d) 23. (b,c,d) 24. (b,d)
 25. (a,b,c) 26. (b,c,d) 27. (a,b,d)
 28. (a,b,c) 29. (4) 30. (4) 31. (3) 32. (1)
 33. (9) 34. (1) 35. (2) 36. (1)
 37. (5.00) 38. (1.00) 39. (a) 40. (a)
 41. (b) 42. (a) 43. (d) 44. (c) 45. (d)

Answer Key



CHAPTER - 2 | RELATIONS, FUNCTIONS & INVERSE TRIGONOMETRIC FUNCTIONS

EXERCISE - 1: BASIC OBJECTIVE QUESTIONS



DIRECTION TO USE -

Scan the QR code and check detailed solutions.

- | | | | | |
|----------|-----------|----------|-----------|----------|
| 1. (d) | 2. (a) | 3. (b) | 4. (c) | 5. (d) |
| 6. (b) | 7. (b) | 8. (c) | 9. (b) | 10. (c) |
| 11. (d) | 12. (a) | 13. (d) | 14. (c) | 15. (b) |
| 16. (c) | 17. (d) | 18. (b) | 19. (d) | 20. (d) |
| 21. (b) | 22. (b) | 23. (c) | 24. (b) | 25. (b) |
| 26. (c) | 27. (d) | 28. (d) | 29. (a) | 30. (d) |
| 31. (b) | 32. (a) | 33. (b) | 34. (d) | 35. (a) |
| 36. (b) | 37. (b) | 38. (a) | 39. (d) | 40. (a) |
| 41. (a) | 42. (c) | 43. (a) | 44. (b) | 45. (c) |
| 46. (a) | 47. (a) | 48. (b) | 49. (c) | 50. (b) |
| 51. (a) | 52. (a) | 53. (d) | 54. (b) | 55. (d) |
| 56. (a) | 57. (c) | 58. (a) | 59. (c) | 60. (d) |
| 61. (d) | 62. (c) | 63. (a) | 64. (360) | 65. (2) |
| 66. (8) | 67. (-1) | 68. (-1) | 69. (0) | 70. (1) |
| 71. (0) | 72. (500) | 73. (10) | 74. (3) | 75. (15) |
| 76. (20) | 77. (1) | 78. (1) | 79. (1) | 80. (1) |

EXERCISE - 2: PREVIOUS YEAR JEE MAIN QUESTIONS



DIRECTION TO USE -

Scan the QR code and check detailed solutions.

- | | | | | |
|--------------|-----------|------------|----------|---------|
| 1. (d) | 2. (b) | 3. (b) | 4. (c) | 5. (d) |
| 6. (a) | 7. (a) | 8. (d) | 9. (a) | 10. (c) |
| 11. (b) | 12. (b) | 13. (b) | 14. (b) | 15. (d) |
| 16. (a) | 17. 10.00 | 18. 3.00 | 19. (c) | 20. (b) |
| 21. (a) | 22. (b) | 23. (b) | 24. (a) | 25. (a) |
| 26. (d) | 27. (a) | 28. (b) | 29. (c) | 30. (c) |
| 31. (b) | 32. (b) | 33. (d) | 34. (19) | 35. (b) |
| 36. (b) | 37. (b) | 38. (b) | 39. (a) | 40. (c) |
| 41. (a) | 42. (b) | 43. (2.00) | 44. (a) | 45. (c) |
| 46. (d) | 47. (b) | 48. (c) | 49. (a) | |
| 50. (481.00) | 51. (c) | 52. (c) | 53. (b) | 54. (b) |
| 55. (490.00) | 56. (a) | 57. (c) | 58. (c) | 59. (c) |
| 60. (26.00) | 61. (a) | 62. (b) | 63. (a) | 64. (b) |
| 65. (d) | 66. (b) | 67. (2.00) | 68. (a) | 69. (b) |
| 70. (d) | 71. (d) | 72. (b) | 73. (c) | 74. (a) |
| 75. (a) | 76. (a) | 77. (b) | 78. (b) | 79. (b) |
| 80. (a) | | | | |

CHAPTER - 2 | RELATIONS, FUNCTIONS & INVERSE TRIGONOMETRIC FUNCTIONS

EXERCISE - 3 :
ADVANCED OBJECTIVE QUESTIONS

DIRECTION TO USE -

Scan the QR code and check detailed solutions.

1. (c) 2. (b) 3. (d) 4. (b) 5. (b)
6. (a) 7. (a) 8. (a) 9. (b) 10. (b)
11. (c) 12. (b) 13. (b) 14. (c) 15. (a)
16. (a) 17. (a) 18. (a) 19. (b) 20. (d)
21. (c) 22. (b) 23. (a) 24. (d) 25. (b)
26. (c) 27. (b) 28. (b) 29. (b) 30. (d)
31. (b) 32. (d) 33. (b) 34. (a) 35. (b)
36. (a) 37. (a) 38. (c) 39. (d) 40. (d)
41. (b) 42. (b) 43. (b) 44. (b) 45. (a)
46. (d) 47. (c) 48. (a) 49. (b) 50. (b)
51. (a,b) 52. (a,b,c) 53. (b,c) 54. (a,b,c) 55. (c)
56. (a,b,c,d) 57. (a,b) 58. (b,d)
59. (7) 60. (3) 61. (6) 62. (4) 63. (d)
64. (d) 65. (a) 66. (c) 67. (b) 68. (d)
69. (a) 70. (b) 71. (a) 72. (a) 73. (a)
74. (c) 75. (c) 76. (d) 77. (d) 78. (a)

79. (c) 80. $t \in \left[-\frac{\pi}{2}, -\frac{\pi}{10}\right] \cup \left[\frac{3\pi}{10}, \frac{\pi}{2}\right]$

EXERCISE - 4 :
PREVIOUS YEAR JEE ADVANCED QUESTIONS

DIRECTION TO USE -

Scan the QR code and check detailed solutions.

1. (d) 2. (b) 3. (a) 4. (d) 5. (d)
6. (a) 7. (d) 8. (d) 9. (a) 10. (b)
11. (c) 12. (a) 13. (b) 14. (b) 15. (d)
16. (d) 17. (a) 18. (b) 19. (d) 20. (c)
21. (a,b) 22. (a,b,c) 23. (b,c,d) 24. (a,b) 25. (2)
26. (3) 27. (0) 28. (19) 29. (b) 30. (a)
31. (b) 32. (a) 33. (b) 34. (a) 35. (a)

Answer Key

CHAPTER -3 | CONTINUITY, DIFFERENTIABILITY & DIFFERENTIATION

EXERCISE - 1: BASIC OBJECTIVE QUESTIONS



DIRECTION TO USE -

Scan the QR code and check detailed solutions.

- | | | | | |
|-------------|-------------|------------|------------|------------|
| 1. (d) | 2. (c) | 3. (c) | 4. (d) | 5. (c) |
| 6. (b) | 7. (d) | 8. (b) | 9. (b) | 10. (b) |
| 11. (b) | 12. (a) | 13. (b) | 14. (d) | 15. (b) |
| 16. (c) | 17. (b) | 18. (c) | 19. (c) | 20. (d) |
| 21. (b) | 22. (c) | 23. (a) | 24. (c) | 25. (c) |
| 26. (a) | 27. (c) | 28. (c) | 29. (d) | 30. (a) |
| 31. (b) | 32. (d) | 33. (b) | 34. (b) | 35. (c) |
| 36. (b) | 37. (b) | 38. (a) | 39. (b) | 40. (a) |
| 41. (d) | 42. (d) | 43. (d) | 44. (b) | 45. (d) |
| 46. (a) | 47. (b) | 48. (b) | 49. (a) | 50. (-0.5) |
| 51. (2.24) | 52. (-1) | 53. (0.31) | 54. (0.33) | 55. (0.67) |
| 56. (0.707) | 57. (-0.72) | 58. (0.37) | 59. (2) | 60. (0.37) |
| 61. (1) | 62. (1.6) | 63. (2) | 64. (0.5) | 65. (0.13) |

EXERCISE - 2: PREVIOUS YEAR JEE MAIN QUESTIONS



DIRECTION TO USE -

Scan the QR code and check detailed solutions.

- | | | | | |
|-------------|-------------|-------------|-------------|------------|
| 1. (c) | 2. (a) | 3. (d) | 4. (b) | 5. (b) |
| 6. (c) | 7. (b) | 8. (c) | 9. (b) | 10. (c) |
| 11. (a) | 12. (b) | 13. (a) | 14. (d) | 15. (c) |
| 16. (c) | 17. (b) | 18. (d) | 19. (d) | 20. (a) |
| 21. (b) | 22. (b) | 23. (c) | 24. (c) | 25. (b) |
| 26. (a) | 27. (a) | 28. (d) | 29. (91.00) | 30. (d) |
| 31. (1) | 32. (10.00) | 33. (c) | 34. (d) | 35. (8.00) |
| 36. (d) | 37. (5) | 38. (a) | 39. (c) | 40. (c) |
| 41. (3.00) | 42. (a) | 43. (5.00) | 44. (c) | 45. (d) |
| 46. (a) | 47. (d) | 48. (2.00) | 49. (5.00) | 50. (d) |
| 51. (d) | 52. (c) | 53. (d) | 54. (1.00) | 55. (6.00) |
| 56. (c) | 57. (c) | 58. (b) | 59. (1.00) | 60. (4.00) |
| 61. (c) | 62. (39.00) | 63. (a) | 64. (b) | |
| 65. (5.00) | 66. (b) | 67. (40.00) | | |
| 68. (14.00) | 69. (d) | 70. (17.00) | 71. (d) | 72. (b) |
| 73. (a) | 74. (2.00) | 75. (a) | | |

CHAPTER -3 | CONTINUITY, DIFFERENTIABILITY & DIFFERENTIATION

EXERCISE - 3 :
ADVANCED OBJECTIVE QUESTIONS

DIRECTION TO USE -

Scan the QR code and check detailed solutions.

1. (a) 2. (a) 3. (a) 4. (c) 5. (c)
6. (a) 7. (b) 8. (a) 9. (c) 10. (a)
11. (d) 12. (b) 13. (b) 14. (d) 15. (a)
16. (b) 17. (a) 18. (b) 19. (b) 20. (a)
21. (c) 22. (c) 23. (b) 24. (b) 25. (c)
26. (a) 27. (a) 28. (c) 29. (c) 30. (c)
31. (c) 32. (c) 33. (b) 34. (c) 35. (b)
36. (c) 37. (b) 38. (b) 39. (c) 40. (c)
41. (b) 42. (a) 43. (d) 44. (a) 45. (c)
46. (a,c) 47. (a,b) 48. (a,b,d) 49. (a,b)
50. (b,c) 51. (a,b,c) 52. (b,d) 53. (2000) 54. (1004)
55. (512) 56. (2) 57. (100) 58. (2) 59. (96)
60. (a) 61. (a) 62. (d) 63. (d) 64. (a)
65. (i) $\sec x$ (ii) $e^{x \sin x} (x \cos x + \sin x)$

EXERCISE - 4 :
PREVIOUS YEAR JEE ADVANCED QUESTIONS

DIRECTION TO USE -

Scan the QR code and check detailed solutions.

1. (c) 2. (d) 3. (a) 4. (d) 5. (d)
6. (a) 7. (b) 8. (d) 9. (d) 10. (a)
11. (b) 12. (b) 13. (d) 14. (a,b,c) 15. (b,c)
16. (a,b,c,d) 17. (b,d) 18. (a,c) 19. (a,d)
20. (a,b) 21. (b,c) 22. (a,c) 23. (a,b,d) 24. (0)
25. (3) 26. (4.00) 27. (a) 29. $a = 1$ 30. $1 - \frac{2}{\pi}$

Answer Key



CHAPTER - 4 | APPLICATION OF DERIVATIVE

EXERCISE - 1: BASIC OBJECTIVE QUESTIONS



DIRECTION TO USE -

Scan the QR code and check detailed solutions.

- | | | | | |
|--------------|-------------|--------------|------------|---------|
| 1. (a) | 2. (d) | 3. (a) | 4. (c) | 5. (d) |
| 6. (c) | 7. (a) | 8. (d) | 9. (c) | 10. (c) |
| 11. (c) | 12. (c) | 13. (a) | 14. (d) | 15. (a) |
| 16. (b) | 17. (b) | 18. (a) | 19. (a) | 20. (c) |
| 21. (b) | 22. (c) | 23. (b) | 24. (b) | 25. (a) |
| 26. (c) | 27. (a) | 28. (d) | 29. (b) | 30. (a) |
| 31. (a) | 32. (a) | 33. (d) | 34. (a) | 35. (c) |
| 36. (d) | 37. (b) | 38. (b) | 39. (c) | 40. (a) |
| 41. (a) | 42. (a) | 43. (c) | 44. (b) | 45. (b) |
| 46. (d) | 47. (d) | 48. (c) | 49. (b) | 50. (c) |
| 51. (b) | 52. (b) | 53. (a) | 54. (d) | 55. (c) |
| 56. (a) | 57. (a) | 58. (b) | 59. (d) | 60. (b) |
| 61. (c) | 62. (b) | 63. (c) | 64. (c) | |
| 65. (169.65) | 66. (1.5) | 67. (-3) | 68. (5.2) | |
| 69. (12.57) | 70. (17.32) | 71. (502.65) | 72. (1) | |
| 73. (45) | 74. (17.32) | 75. (-42) | 76. (0.07) | |
| 77. (-3) | 78. (1) | 79. (2.12) | 80. (0.1) | |

EXERCISE - 2: PREVIOUS YEAR JEE MAIN QUESTIONS



DIRECTION TO USE -

Scan the QR code and check detailed solutions.

- | | | | | |
|--------------|-------------|--------------|------------|-----------|
| 1. (b) | 2. (a) | 3. (c) | 4. (a) | 5. (b) |
| 6. (a) | 7. (a) | 8. (c) | 9. (b) | 10. (c) |
| 11. (d) | 12. (a) | 13. (a) | 14. (a) | 15. (d) |
| 16. (a) | 17. (b) | 18. (c) | 19. (c) | 20. (c) |
| 21. (c) | 22. (d) | 23. (d) | 24. 3.00 | 25. (d) |
| 26. (b) | 27. (a) | 28. (a) | 29. (b) | 30. (b) |
| 31. (d) | 32. (b) | 33. (a) | 34. (b) | 35. (c) |
| 36. 122.00 | 37. (a) | 38. (c) | 39. (d) | 40. (b) |
| 41. (c) | 42. (c) | 43. (d) | 44. (c) | 45. (d) |
| 46. (a) | 47. (d) | 48. (a) | 49. (c) | 50. (a) |
| 51. 5 | 52. (a) | 53. (d) | 54. (c) | 55. (b) |
| 56. (d) | 57. (a) | 58. (d) | 59. (b) | 60. 4.00 |
| 61. (b) | 62. (3.00) | 63. (d) | 64. (c) | 65. (c) |
| 66. (c) | 67. (9.00) | 68. (c) | 69. (b) | 70. (a) |
| 71. (d) | 72. (a) | 73. (144.00) | 74. (a) | |
| 75. (4.00) | 76. (d) | 77. (d) | 78. (2.00) | 79. (c) |
| 80. (b) | 81. (b) | 82. (5.00) | 83. (b) | 84. (d) |
| 85. (c) | 86. (a) | 87. (c) | 88. (d) | 89. (3.0) |
| 90. (a) | 91. (36.00) | 92. (a) | 93. (a) | 94. (4.0) |
| 95. (a) | 96. (b) | 97. (2.00) | 98. (d) | 99. (c) |
| 100. (22.00) | | | | |

CHAPTER - 4 | APPLICATION OF DERIVATIVE

EXERCISE - 3 :
ADVANCED OBJECTIVE QUESTIONS

DIRECTION TO USE -

Scan the QR code and check detailed solutions.

1. (c) 2. (d) 3. (a) 4. (c) 5. (c)
6. (b) 7. (d) 8. (c) 9. (a) 10. (a)
11. (d) 12. (b) 13. (d) 14. (d) 15. (b)
16. (b) 17. (a) 18. (c) 19. (a) 20. (b)
21. (d) 22. (a) 23. (a) 24. (a) 25. (d)
26. (c) 27. (c) 28. (b) 29. (c) 30. (a)
31. (c) 32. (d) 33. (d) 34. (b) 35. (a)
36. (a) 37. (b) 38. (a) 39. (d) 40. (c)
41. (b) 42. (a) 43. (a) 44. (c) 45. (a,c)
46. (b,c) 47. (a,b) 48. (b,c) 49. (b,c) 50. (b,d)
51. (a,b,c,d) 52. (a,c) 53. (a,c,d) 54. (a,d)
55. (a,b,c,d) 56. (a,c) 57. (a,b)
58. (a,b,c) 59. (a,c) 60. (a,c,d) 61. (a,b) 62. (b,c)
63. (a,b,c,d) 64. (a,c) 65. (a,b,c)
66. (a,b,d) 67. (2) 68. (4) 69. (16) 70. (11)
71. (4) 72. (6) 73. (25) 74. (18) 75. (d)
76. (a) 77. (c) 78. (d) 79. (a) 80. (a)
81. (a) 82. (c) 83. (d) 84. (c) 85. (d)
86. (d) 87. (d) 88. (a) 89. (a) 90. (c)
91. (a) 92. (a) 93. (c) 94. (a) 95. (a)
96. (d) 97. (a) 98. (a) 99. (d) 100. (c)
101. (d) 102. (b) 103. (b) 104. (b) 105. (b)
106. (c) 107. (b) 108. (b) 109. (b) 110. (d)

EXERCISE - 4 :
PREVIOUS YEAR JEE ADVANCED QUESTIONS

DIRECTION TO USE -

Scan the QR code and check detailed solutions.

1. (b) 2. (c) 3. (a) 4. (d) 5. (a)
6. (a) 7. (a) 8. (d) 9. (d) 10. (d)
11. (a) 12. (c) 13. (c) 14. (c) 15. (c)
16. (d) 17. (a) 18. (c) 19. (c) 20. (d)
21. (c) 22. (c) 23. (b,c) 24. (a,b)
25. (b,c,d) 26. (b,c) 27. (a,c) 28. (a,b) 29. (b,d)
30. (a,d) 31. (a,c,d) 32. (a,b,d) 33. (a,c,d) 34. (a,b)
35. (18) 36. (2,1) 37. (5 unit) 38. (6)
39. (7) 40. (2) 41. (1) 42. (2) 43. (2)
44. (5) 45. (9) 46. (8) 47. (5.00)
48. (0.50) 49. (d) 50. (A-p; B-r)
51. (a) 52. (a) 53. (b) 54. (c) 55. (a)
56. (b) 57. (b) 58. (c) 59. (d) 60. (c)
61. (d) 62. (b) 63. (c) 64. (57.00)
65. (6.00) 66. $\cos\left(\frac{1}{3}\cos^{-1}p\right)$ 70. $y-2=0$

MASTER INDEX

VOLUME 1:

Determinants & Matrices

Relations , Functions & Inverse Trigonometric Functions

Continuity, Differentiability & Differentiation

Applications of Derivatives

VOLUME 2:

Indefinite Integration

Definite Integration, Area under curves

Differential Equations

VOLUME 3:

Vectors & 3-Dimensional Geometry

Probability

Creating Impact at Scale



65 Million+

Monthly
Youtube Views



49 Million+

Monthly
Web + App Visitors



40+ Countries

Where Students
Take Live Classes



25 Million+

Monthly Active
Users



26 Million+

Hours of
Live Learning



25

MILLION+

Parents
Trust
Vedantu



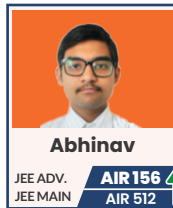
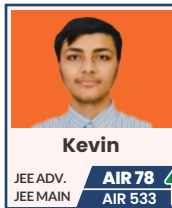
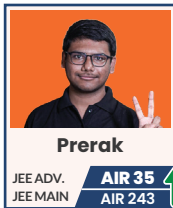
43

MILLION+

Happy
Students

Our Achievers of 2021

JEE Advanced 2021



1500+

Vedantu students ace
JEE Advanced 2021



6%

of IITs' upcoming batch
will be from Vedantu

JEE Main 2021

903

students in
Top 10,000

78

students scored
99.9+ PERCENTILE

869

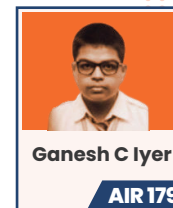
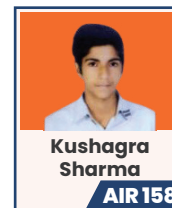
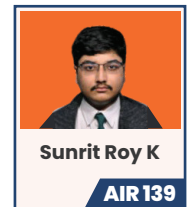
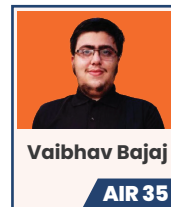
students scored
99.9+ PERCENTILE

68

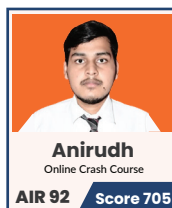
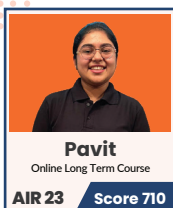
students in
Top AIR 1000

444

students in
Top AIR 5000



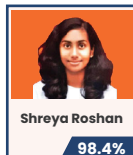
NEET (UG) 2021



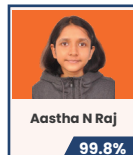
1172

NEET
Qualifiers

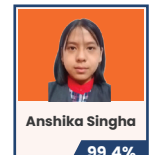
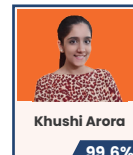
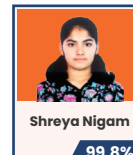
CBSE Class 12



ISC Class 12



CBSE Class 10



ICSE Class 10

